

- a) incomplete angio: (p. 1423)
 - must see both PICA origins (1–2% of aneurysms occur here)
 - need to cross-fill through the ACoA (p. 1423)
- b) degradation of images due to
 - poor patient cooperation (e.g., from agitation). Either sedate patient (use caution in non-intubated patients) or repeat the study at a later time when patient is more cooperative
 - poor-quality equipment providing substandard images
- 2. obliteration of aneurysm by the hemorrhage
- 3. thrombosis of the aneurysm after SAH (p. 1456)
- 4. aneurysm too small to be visualized⁸⁷: although “microaneurysms” may be a source of SAH, their natural history and optimal treatment are unknown
- 5. lack of filling of aneurysm due to vasospasm (of parent artery or of aneurysmal orifice)

► **Nonaneurysmal SAH from source that fails to show up on angiography.** See for etiologies of SAH other than aneurysm (p. 1417) (many of which may not be demonstrated on angiography), including:

1. angiographically occult (or cryptic) vascular malformation (p. 1524)
2. pretruncal nonaneurysmal SAH: see below

89.8.2 Risk of rebleeding

Overall bleed rate is 0.5%/yr, which is lower than with aneurysmal SAH or rebleeding from AVMs. There is also a smaller risk of delayed cerebral ischemia (vasospasm). Neurological outcome is likewise better.

89.8.3 Management

General measures

These patients are still at risk for the same complications of SAH as with aneurysmal SAH: vasospasm, hydrocephalus, hyponatremia, rebleeding, etc. (p. 1432) and should be managed as any SAH (p. 1432). Some subgroups may be at lower risk for complications and may be managed accordingly (e.g., below).

Repeat angiography

Yield of positive second angiogram after technically adequate negative study: 1.8–9.8%⁸⁸ in early (pre-CT) studies, 2–24% quoted more recently.^{87,89,90} CT scan findings are helpful in the decision to repeat angiography.⁹¹ 70% of cases with diffuse SAH and thick layering of blood in the anterior interhemispheric fissure were associated with an ACoA aneurysm that showed up on repeat angiography.⁸⁵ The absence of blood on CT (performed within 4 days of SAH), or thick blood in the perimesencephalic cisterns alone (see below) were unlikely to be associated with a missed aneurysm.

Recommendations regarding repeat angio:

1. repeat angio after ≈ 10–14 days (allows vasospasm & some clot to resolve; **note**: between 5 and 10 days there is decreased chance of seeing an aneurysm because of vasospasm; angiography at ≈ day 10 permits surgery to be done if needed ≈ at day 14, which is about the earliest time after the “no-op” window of day 3–12)
 - a) technically adequate 4-vessel angiogram is negative, and evidence for SAH is strong
 - b) original angio was incomplete or if there are suspicious findings
2. if CT localizes blood clot to particular area, place special attention to this area on repeat angio
3. do not repeat angio for classic pretruncal SAH (see below) or if no blood on CT
4. patients are usually kept in the hospital 10–14 days while waiting for repeat angio (to watch for and manage complication of SAH or rebleeding)

Third arteriogram:

If the first two arteriograms are negative, and the history is suggestive of aneurysmal SAH, a third arteriogram 3–6 months after SAH has ≈ 1% chance of showing a source of bleeding.

Other studies

1. imaging studies of the brain: MRI (with MRA if available) or CT (with angio-CT if available). This may visualize an aneurysm that fails to show up on angiography, and may identify other sources of SAH such as angiographically occult vascular malformation (p. 1524), tumor...
2. tests to rule out spinal AVM: a rare cause of intracerebral SAH (p. 1395)
 - a) spinal MRI: cervical, thoracic, and lumbar
 - b) spinal angiography: too difficult and risky to be justified in most cases of angio-negative SAH. Consider in cases with high suspicion of spinal source

Surgical exploration

Advocated by some for cases of SAH with CT findings compatible with an aneurysmal source in which a suspicious area is demonstrated angiographically⁸⁷ with careful explanation to the patient and family of the possibility of negative operative findings.

89.9 Pretruncal nonaneurysmal SAH (PNSAH)

89.9.1 General information

Formerly perimesencephalic nonaneurysmal SAH.⁹² The suggestion to change the name to pretruncal nonaneurysmal SAH was proposed because neuroimaging has shown the true anatomic localization of the blood to be in front of the brainstem (truncus cerebri) centered in front of the pons rather than perimesencephalic.⁹³ The existing literature on PNSAH is somewhat limited by the lack of a rigorous anatomic definition, with criteria of blood pattern differing amongst studies. Blood often extends into the interpeduncular or premedullary cisterns. It has also sometimes been referred to as the "Dutch disease" due to the initial profusion of information in that literature.

A distinct entity considered to be a benign condition with good outcome and less risk of rebleeding and vasospasm than in other patients with SAH of unknown etiology⁹⁴ (no rebleeding occurred in 37 patients with PNSAH and 45 months mean follow-up,⁹⁵ nor in 169 patients with 8–51 months follow-up⁹⁰; vasospasm has been reported in only 3 patients and may have been related to cerebral angiography rather than the PNSAH, and although it is low, the incidence of angiographic vasospasm may be higher than originally thought⁹⁶).

The actual etiology has yet to be determined (there are 3 case reports of patients explored surgically with no abnormal findings⁹⁰ and one case where a pontine abnormality resembling a capillary telangiectasia was demonstrated on MRI⁹⁷), but it may be secondary to rupture of a small perimesencephalic vein or capillary.⁹⁶ Studies have shown an association with abnormal venous anatomy, including primitive variants of the basal vein of Rosenthal,^{98,99} which some authors have hypothesized results in hemorrhage secondary to central cerebral venous hypertension.^{100,101} Other proposed etiologies include a ruptured perforating artery, cavernous malformation, intraluminal basilar dissection, and capillary telangiectasia.¹⁰²

89.9.2 Presentation

Patients may present with severe paroxysmal H/A, meningismus, photophobia, and nausea. Loss of consciousness is rare. These patients are usually not critically ill (all were grade 1 or 2 [H&H or WFNS grading scale]); however, complications such as hyponatremia or cardiac abnormalities may still occur. Preretinal hemorrhages and sentinel H/A have not occurred. CT and/or MRI demonstrate characteristic findings (see below) although it may initially be missed on CT,⁹⁶ and LP may yield bloody CSF. By definition, all have negative angiography.

89.9.3 Epidemiology

PNSAH has been reported to comprise 20–68% of cases of angiogram-negative SAH^{94,103} (depending on the timing of CT, adequacy of angiography, and the definition of PNSAH). However, the true incidence is probably more in the range of 50–75%.⁹⁰

The reported age range is 3–70 years (mean: 50 yrs),⁹⁰ 52–59% are male, and pre-existing HTN was present in 3–20% of patients.

89.9.4 Relevant anatomy

Posterior fossa cisterns:

The perimesencephalic cisterns include: interpeduncular, crural, ambient, and quadrigeminal cisterns. The prepontine cistern lies immediately anterior to the pons.

Liliequist's membrane (LM)¹⁰⁴: Basically considered to separate the interpeduncular cistern from the chiasmatic cistern¹⁰⁵ (it forms a competent barrier in only 10–30% of the population). In further detail, the superior leaflet of LM (diencephalic membrane) separates the interpeduncular cistern from the chiasmatic cistern medially and from the carotid cisterns laterally.^{106,107} The inferior leaflet (the mesencephalic membrane) separates the interpeduncular from the prepontine cistern.

The diencephalic membrane is thicker and is more often competent, effectively isolating the chiasmatic cistern. However, the carotid cisterns often communicate with the crural cisterns and in turn with the interpeduncular cistern.¹⁰⁷

Thus, blood in the carotid or prepontine cistern is compatible with a low-pressure pretruncal source of bleeding; however, blood in the chiasmatic cistern should raise concern about aneurysmal rupture.

89.9.5 Diagnostic criteria

Without knowledge of the actual substrate of PNSAH, the following suggested diagnostic criteria must be viewed as empiric (adapted⁹⁰): (an example of a typical CT scan is shown in ► Fig. 89.2)

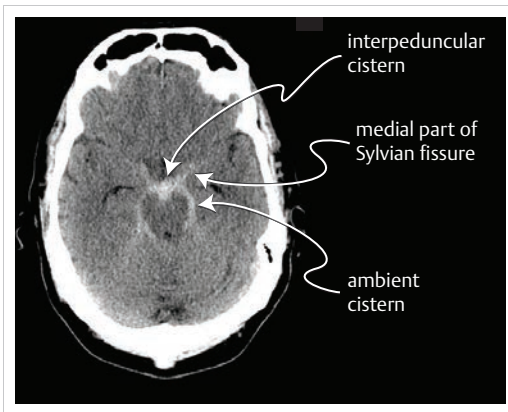


Fig. 89.2 Pretruncal (perimesencephalic) nonaneurysmal SAH in a 67-year-old male. Image: axial noncontrast CT scan of the brain.

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1. CT or MRI scan performed ≤ 2 days from ictus meeting the criteria shown in ► Table 89.6 (later scans render the diagnosis unreliable, e.g., washout could cause an aneurysmal SAH to fit the criteria). These criteria imply that blood should be contained inferior to Liliequist's membrane (LM) (i.e., perimesencephalic and/or prepontine cisterns). Extension into the suprasellar cistern is common. Significant amounts of blood penetrating LM to the chiasmatic, Sylvian, or interhemispheric cisterns should be viewed with suspicion
2. a negative high-quality 4-vessel cerebral angiogram¹⁰⁸ (radiographic vasospasm is common, and does not preclude the diagnosis nor does it mandate repeat angiography). NB: $\approx 3\%$ of patients with a ruptured basilar bifurcation aneurysm meet the criteria of ► Table 89.6,¹⁰⁹ and therefore an *initial* arteriogram is mandatory

Table 89.6 CT or MRI criteria for PNSAH^{96,110}

1. epicenter of hemorrhage immediately anterior to brainstem (interpeduncular or prepontine cistern)
2. there may be extension into anterior part of ambient cistern or basal part of the Sylvian fissure
3. absence of complete filling of anterior interhemispheric fissure
4. no more than minute amounts of blood in lateral portion of Sylvian fissure
5. absence of frank intraventricular hemorrhage (small amounts of blood sedimenting in the occipital horns of the lateral ventricles is permissible)

- appropriate clinical picture: no loss of consciousness, no sentinel H/A, SAH grade 1 or 2 (H&H or WFNS grading scale) (p. 1424), and absence of drug use. Variance from this should raise suspicion of alternate pathogenesis

A more stringent set of anatomic criteria (► Table 89.7) showed excellent inter-observer agreement (97.2%) in PNSAH. Additionally, no aneurysms were identified on catheter angiography when the anatomical criteria were met.¹¹¹

Table 89.7 Alternative anatomic CT criteria for PNSAH¹¹¹

- epicenter of bleeding located immediately anterior to and in contact with the brainstem in the prepontine, interpeduncular, or posterior suprasellar cistern
- blood limited to the prepontine, interpeduncular, suprasellar, crural, ambient, and/or quadrigeminal cisterns and/or cisterna magna
- NO extension of blood into the Sylvian or interhemispheric fissures
- intraventricular blood limited to incomplete filling of the fourth ventricle and occipital horns of the lateral ventricles
- no intraparenchymal blood

89.9.6 Repeat angiography

Controversial. Angiography carries ≈ 0.2 – 0.5% risk of permanent neurologic deficit in this population.⁹⁰ Most experts agree that repeat angiography is not indicated in patients meeting the criteria of PNSAH^{89,108} (although others recommend repeat angiography in all surgical candidates^{87,112}). One should probably repeat the study if any uncertainty exists or if there is a history of a condition associated with increased risk of cerebral aneurysms.⁹⁶

89.9.7 Treatment

Optimal treatment is not known with certainty. The low risk of rebleeding and delayed ischemia suggests that extreme measures are not indicated. The following recommendations are made^{90,96} (period not specified):

1. symptomatic treatment
2. cardiac monitoring
3. electrolyte monitoring for hyponatremia
4. follow patient clinically (and if appropriate, with repeat imaging studies) to rule out hydrocephalus (transient ventricular enlargement is common; however, hydrocephalus requiring shunting is rare (only $\approx 1\%$)⁹⁰)
5. ✖ *not* recommended
 - a) hyperdynamic therapy
 - b) calcium channel blockers: use has not been investigated in PNSAH, but is probably not warranted due to low incidence of vasospasm and should be discontinued when normal angiographic findings are documented⁹⁶
 - c) activity restrictions (except in cases of increasing H/A with mobilization)
 - d) antiseizure medications
 - e) reduction of blood pressure below normal
 - f) surgical exploration

89.9.8 Long-term implications

In one study,¹¹³ 160 patients with PNSAH were contacted for “long term” follow-up (mean 7.5 years, range 1–23). There were no new episodes of SAH in this cohort. Furthermore, the mortality was not elevated above that of the general population. Based on this, the authors concluded that no restrictions should be placed on these patients by physicians or health or life insurance providers.

Absent any long-term prospective study, there is no consensus. Routine repeated vascular workup is not the standard of care. If it is desired to reassess the patient for some specific reason, catheter angiography is almost certainly not indicated (due to the risks and the low yield); here, CTA might be considered. Avoiding HTN, as for the general population, seems reasonable.

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Part XVIII

Vascular Malformations

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XVIII

90 Vascular Malformations

90.1 Vascular malformations – general information and classification

This designation encompasses a number of non-neoplastic vascular lesions of the CNS. The four types originally described by McCormick in 1966 are shown in ► Table 90.1.¹

Possible additional categories:

1. direct fistula AKA arteriovenous fistula (AV- fistula, *not* AVM). Single or multiple dilated arterioles that connect directly to a vein *without* a nidus. These are high-flow, high-pressure. Low incidence of hemorrhage. Usually amenable to interventional neuroradiological procedures.

Examples include:

- a) vein of Galen malformation (aneurysm) (p. 1518)
- b) dural AVM (p. 1514)
- c) carotid-cavernous fistula (p. 1519)

2. mixed or unclassified angiomas: 11% of AOVMS²

Table 90.1 4 classic types of vascular malformation

Type	Prevalence (%)
arteriovenous malformation (AVM) ^a	44–60
cavernous malformation (p. 1525)	19–31
capillary telangiectasia (p. 1525)	4–12
developmental venous anomaly (DVA) (p. 1512)—formerly venous angioma	9–10

^aSometimes referred to as “pial AVM” to distinguish it e.g., from dural AVM

90.2 Arteriovenous malformation (AVM)

90.2.1 General information

Key concepts

- dilated arteries and veins with dysplastic vessels. Arterial blood flows directly between them with no capillary bed and no intervening neural parenchyma (brain) in the nidus
- may be congenital, many are acquired
- AVMs are medium-to-high pressure and high-flow
- usually presents with hemorrhage, less often with seizures
- risk of 1st-time hemorrhage is $\approx 1\%$ /yr. Prognostic factors that alter this risk are uncertain
- risk of recurrent ICH is 5%/year. Increasing age, deep venous drainage, feeding artery aneurysms and female gender may elevate this risk
- 5-year risk of 1st seizure is 8% for unruptured AVM. 5-year risk of epilepsy after a 1st seizure is 58%
- aneurysms on feeding arteries are common and are often the source of the hemorrhage
- demonstrable on angiography, MRI, or CT (especially with contrast)
- main treatment options (when expectant management not elected):
 - surgical excision: often curative. Surgical risk can be estimated by Spetzler-Martin grade (p. 1509)
 - stereotactic radiosurgery (SRS): obliterates 70–80% of AVMs over 1–3 years. Usually reserved for deep lesions < 3 cm dia
 - endovascular embolization: curative in select cases. Usually used as an adjunct to surgery or SRS, for high-risk angiographic entities, or to palliate high-flow symptoms

90.2.2 Description

A collection of dysplastic dilated blood vessels (neither arteries nor veins) wherein arterial blood flows directly into draining veins without the normal interposed capillary beds. AVMs are usually congenital, tend to enlarge somewhat with age, and often progress from low-flow juvenile lesions at birth to medium-to-high-flow high-pressure lesions in adulthood. AVMs appear grossly as a “tangle” of vessels, often with a fairly well-circumscribed center (the nidus) which contains no brain parenchyma, with draining “red veins” (usually on the surface) containing oxygenated blood under higher pressure than in normal veins.

May be classified as:

1. parenchymal AVMs (discussed below). Subclassified as:
 - a) pial
 - b) subcortical
 - c) paraventricular
 - d) combined
2. pure dural AVM (p. 1514)
3. mixed parenchymal and dural (rare)

90.2.3 Epidemiology

Prevalence: usually quoted as 0.14%. Asymptomatic prevalence on brain MRIs: 0.05%.³

Incidence of detected asymptomatic or symptomatic AVMs: ≈ 1.3 per 100,000 person-years.³

Slight male preponderance.

A few types are known to be hereditary. Other types are part of known hereditary syndromes: 15–20% of patients with Osler-Weber-Rendu syndrome (hereditary hemorrhagic telangiectasia) (p. 1524) have cerebral AVMs (the most common genetic cause of AVMs).

Comparison to aneurysms: the average age of patients diagnosed with AVMs is ≈ 33 yrs, which is ≈ 10 yrs younger than for aneurysms.⁴ 64% of AVMs are diagnosed before age 40 (cf. 26% for aneurysms).

90.2.4 Presentation

1. hemorrhage (most common): 58%³ (compared to 92% for aneurysms) (see below)
2. seizures: 34%³
3. remainder: 8%
 - a) mass effect: e.g., trigeminal neuralgia due to CPA AVM
 - b) ischemia: by steal
 - c) H/A: rare. AVMs may occasionally be associated with migraines. Occipital AVMs may present with visual disturbance (typically hemianopsia or quadrantanopsia) and H/A that are indistinguishable from migraine⁵
 - d) bruit: especially with dural AVMs (p. 1514)
 - e) increased ICP
 - f) findings limited almost exclusively to peds, usually with large midline AVMs that drain into an enlarged vein of Galen (“vein of Galen malformation” (p. 1518)):
 - hydrocephalus with macrocephaly: due to compression of Sylvian aqueduct by enlarged vein of Galen or to increased venous pressure
 - congestive heart failure with cardiomegaly
 - prominence of forehead veins (due to increased venous pressure)

90.2.5 Hemorrhage

General information

Most data are from observational research studies, so the true natural history in unselected patients is not known.⁶ Peak age for hemorrhage is between 15 and 20 yrs.⁷ See Pregnancy & intracranial hemorrhage (p. 1425) for a discussion of that topic.

Morbidity and mortality from AVM hemorrhage: data varies widely. With each bleed, an estimate is 10% mortality, 30–50% morbidity⁸ (neurological deficit). ICH from AVMs tend to be more benign than primary ICH.³

Hemorrhage location with AVMs

1. intraparenchymal (ICH): 82% (the most common site of bleeding)⁹

2. intraventricular hemorrhage:
 - a) usually accompanied by ICH as the result of rupture of the ICH into the ventricle
 - b) pure IVH (with no ICH) may indicate an intraventricular AVM
3. subarachnoid: SAH may also be due to rupture of an aneurysm on a feeding artery (common with AVMs)
4. subdural: uncommon. May be the source of a spontaneous SDH (p. 1088)

Annual risk of hemorrhage

A subject of much study and conflicting results.

The oft-quoted annual risk is 2%¹⁰ to 4%.⁴ Based on the largest meta-analysis, the unselected annual risk of hemorrhage is 2.3% per year.¹¹ Factors that may influence the annual risk include the following.

- ▶ **Previous rupture.** The strongest prognosticator for future hemorrhage. 1.3% per year for unruptured AVMs, and 4.3% for (repeat) hemorrhage for ruptured AVMs at diagnosis.¹¹ The risk of rebleeding is highest within the first year (historically, results are conflicting).
- ▶ **Age at diagnosis.** 30% increase in risk per decade¹¹ (historically, results are conflicting).
- ▶ **Drainage pattern.** AVMs with exclusively deep venous drainage: 1.6–2.4-fold increase in annual risk.¹¹
- ▶ **AVM size.** Conflicting information—an association is not found in larger cohorts. The contention was that small AVMs tend to present more often as hemorrhage than do large ones.^{12,13} Small AVMs were thought to have higher pressure in the feeding arteries.¹³
- ▶ **Hemorrhage rate related to Spetzler-Martin grade.** Controversial. The S-M scale was devised for assessing *surgical risk* (not hemorrhage). Some studies show increased risk of hemorrhage with Spetzler-Martin (S-M) grade (p. 1509) 4–5 AVMs¹⁴ (high grade), others studies show the opposite effect.
- ▶ **Combined risk factors.** A bingo sheet showing risk of bleeding based on history of prior bleeding, venous drainage pattern, and nidus location using data from Stapf et al is shown in ▶ Table 90.2 as an illustration (some results are conflicting).

Table 90.2 Annual average hemorrhage rates for various AVM subgroups¹⁵

Venous drainage	Nidus location	No prior hemorrhage	Prior hemorrhage
No deep venous drainage	Not deep	0.9%	4.5%
	Deep	3.1%	14.8%
Deep venous drainage	Not deep	2.4%	11.4%
	Deep	8.0%	34.4%

Lifetime risk of hemorrhage/recurrent hemorrhage

The risk of bleeding over the remainder of one's life is given by Eq (90.1); (this analysis assumes a constant risk of bleeding, ignoring influences of a previous bleed, AVM location, changes in risk with aging or pregnancy, etc.).

$$\text{risk of bleeding (at least once)} = 1 - (\text{annual risk of not bleeding})^{\text{expected years of remaining life}} \quad (90.1)$$

where the annual risk of not bleeding is equal to 1 – the annual risk of bleeding. For example, if a 3% annual risk of bleeding is used as an average, and the remaining life expectancy is 25 years, the result is as illustrated in Eq (90.2).

$$\text{risk of bleeding (at least once in 25 years)} = 1 - 0.97^{25} = 0.53 = 53\% \quad (90.2)$$

A simple first approximation to Eq (90.1) for an annual 3% risk of bleeding is given by Eq (90.3).

$$\text{risk of bleeding (at least once)} \approx 105 - \text{age in years} \quad (90.3)$$

► Table 90.3 shows the risk for various ages using Eq (90.1) (longevity is taken from insurance life-tables).

Table 90.3 Lifetime risk of hemorrhage^a

Age at presentation	Estimated years to live ^b	Lifetime risk of hemorrhage		
		For 1% annual risk ^c	For 2% annual risk	For 3% annual risk
0	76	53%	78%	90%
15	62	46%	71%	85%
25	52	41%	65%	79%
35	43	35%	58%	73%
45	34	29%	50%	64%
55	25	22%	40%	53%
65	18	16%	30%	42%
75	11	10%	20%	28%
85	6	5.8%	11%	17%

^amodified from reference¹⁶

^bbased on 1992 Preliminary Life tables prepared by Metropolitan Life Insurance Company

^c1% annual risk is also presented because it may be applicable for incidental aneurysms (p. 1486)

90.2.6 Seizures

The mechanism of seizure generation is unknown. The younger the patient at the time of diagnosis, the higher the risk of developing convulsions (► Table 90.4). AVM features with increased risk of seizures: temporal location, cortical involvement, nidus diameter > 3 cm.³ Patients presenting with hemorrhage have 22% risk of developing epilepsy in 20 yrs.

90

Table 90.4 Risk of seizures with AVMs

Age at diagnosis (years)	20-year risk of developing seizures
10–19	44%
20–29	31%
30–60	6%

90.2.7 AVMs and aneurysms

7% of patients with AVMs have aneurysms. 75% of these are located on a major feeding artery (probably from increased flow).¹² These aneurysms may be classified into 1 of 5 types shown in ► Table 90.5. Aneurysms also may form within the nidus or on draining veins. When treating tandem AVMs and aneurysms, the symptomatic one is usually treated first (when feasible, both may be treated at the same operation).¹⁷ If it is not clear which bled, the odds are that it was the aneurysm. Although a significant number (≈ 66%) of related aneurysms will regress following removal of the AVM, this does not always occur. In one series, none of the 9 associated aneurysms ruptured or enlarged following AVM removal.¹⁷

Table 90.5 Categories of aneurysms associated with AVMs^a 17

Type	Aneurysm location
I	proximal on ipsilateral major artery feeding AVM
IA	proximal on major artery related but contralateral to AVM
II	distal on superficial feeding artery
III	proximal or distal on deep feeding artery (“bizarre”)
IV	on artery unrelated to AVM

^aexcludes intranidal and venous aneurysms

90.2.8 Evaluation

General information

Various imaging modalities provide information that are additive when analyzed in combination.

Definitive diagnosis and aspects of treatment planning are obtained with catheter angiography.

Cross-section-based modalities (CT, CTA, MRI, MRA) provide important information about adjacent brain that is not obtained with catheter angiography.

MRI and MRA approach CT/CTA in detecting AVMs in the setting of ICH, but MRI/MRA may not show important angiogenic features demonstrable on CTA.

CT

Unenhanced brain CT is the best study to rule out acute hemorrhage (>90% sensitivity). It can miss some AVMs, but can demonstrate calcifications within the lesion or increased density of the nidus. Adding a contrast CT will show enhancement within the vessels, and can delineate the nidus (dense central area of an AVM).

CT angiography (CTA)

Pros: excellent spatial resolution, fast, minimally invasive, fewer contraindications than MRI/MRA.

Cons: involves ionizing radiation, iodine carries risk of allergic reaction in sensitized individuals and risk of renal injury.

In patients with ICH, sensitivity for detecting AVM is 84–100% and specificity is 77–100%^{3,18} (comparable to catheter angiography).

MRI

Characteristics of AVM on MRI:

1. flow void on T1WI or T2WI within the AVM
2. feeding arteries
3. draining veins
4. increased intensity on partial flip-angle (to differentiate signal dropout on T1WI or T2WI from calcium)
5. significant edema around the lesion may indicate a tumor that has bled rather than an AVM
6. T2* GRE (gradient echo sequences) or susceptibility-weighted imaging (SWI) help demonstrate surrounding hemosiderin, which suggests a previous significant hemorrhage
7. a complete ring of low density (due to hemosiderin) surrounding the lesion suggests AVM over neoplasm

Magnetic resonance angiography (MRA)

Cons: both time-of-flight and contrast bolus MRA are more limited in detecting small vessels (<1 mm dia), aneurysms, small AVMs (nidus <10 mm), and venous outflow anatomy, all of which assume greater importance in treatment planning.

Catheter angiography

Characteristics of AVM on angiography:

1. tangle of vessels
2. large feeding artery
3. large draining veins
4. draining veins are visualized in the same images as arteries (arterial phase)

Most but not all AVMs show up on angiography. Few cavernous malformations and venous angiomas do (see Angiographically occult vascular malformations (p.1524)).

AVM grading scales

Surgical grading scales for AVMs

This section presents AVM grading scales (Spetzler-Martin and Lawton-Young) for use with surgical treatment. For treatment with stereotactic radiosurgery, Virginia Radiosurgery AVM Scale (VRAS) (► Table 115.3) is preferable.

Spetzler-Martin (S-M) grade of AVMs

The Spetzler-Martin grading scale is the most widely used and highly validated classification for estimating the risk of surgical resection of an AVM.^{3,19} Often requires catheter angiography and cross-sectional imaging (CT or MRI) to determine the grade.

S-M Grade = sum of points from ► Table 90.6, ranges from 1 to 5. A separate grade 6 is reserved for untreatable lesions (by any means: surgery, SRS...), resection of these would almost unavoidably be associated with disabling deficit or death. May not be applicable to pediatrics (juvenile AVMs are immature and the margins are less well defined; AVMs mature at ≈ age 18 yrs and tend to become more compact).

Table 90.6 Spetzler-Martin AVM grading system²¹

Graded feature	Criteria	Points	Score
Size ^a	small (<3 cm)	1	(1 - 3)
	medium (3–6 cm)	2	
	large (>6 cm)	3	
Eloquence of adjacent brain ^b	non-eloquent	0	(0 - 1)
	eloquent	1	
Pattern of venous drainage ^c	superficial only	0	(0 - 1)
	deep	1	
Spetzler-Martin grade → TOTAL			(1 - 5)

^alargest diameter of nidus on non-magnified angiogram (is related to and therefore implicitly includes other factors relating to difficulty of AVM excision, e.g., number of feeding arteries, degree of steal, etc.)

^beloquent brain²²: sensorimotor, language, and visual cortex; hypothalamus and thalamus; internal capsule; brainstem; cerebellar peduncles; deep cerebellar nuclei

^cconsidered superficial if all drainage is through cortical venous system; considered deep if any or all is through deep veins (e.g., internal cerebral vein, basal vein, or pre-central cerebellar vein)

Based on the Spetzler-Martin grade: 100 consecutive cases operated by an expert (Spetzler) had the outcomes shown in ► Table 90.7 (no deaths). The majority of AVMs presenting to academic medical centers are S-M grade III.²⁰

Table 90.7 Surgical outcome by Spetzler-Martin grade operated on by Spetzler

Grade	No.	No deficit	Minor deficit ^a	Major deficit ^b	Poor outcome ^c
1	23	23 (100%)	0	0	4%
2	21	20 (95%)	1 (5%)	0	10%
3	25	21 (84%)	3 (12%)	1 (4%)	18%
4	15	11 (73%)	3 (20%)	1 (7%)	31%
5	16	11 (69%)	3 (19%)	2 (12%)	37%

^aminor deficit: mild brainstem deficit, mild aphasia, mild ataxia

^bmajor deficit: hemiparesis, increased aphasia, homonymous hemianopsia

^cpooled analysis published in 2011²³

★ Spetzler has since published this 3-tiered management recommendation scheme²³:

- Class A (S-M Grade I & II): surgical resection
- Class B (S-M Grade III): multimodality treatment (also see Lawton-Young scale below)
- Class C (S-M Grade IV & V): follow clinically and repeat angiogram every 5 years. Treatment only for progressive neurologic deficit, steal-related symptoms, or aneurysms identified on surveillance angiograms

Lawton-Young supplementary grading scale for AVMs

The Lawton-Young (L-Y) supplementary grading scale²⁴ is a validated scale²⁵ that enhances the S-M grade. The full weighted model (combining the S-M & L-Y grade) is shown in the reference.²⁴ The simplified model is shown in ► Table 90.8 and the L-Y score is the sum of the points. The accuracy of the L-Y model was almost as good as the full weighted model, and either one was better than the S-M grade at predicting changes in modified Rankin Scale (mRS) score (► Table 98.5) postoperatively. This model is derived only using operated AVMs, and tended to be more helpful when there was a

Table 90.8 Lawton-Young supplementary grading scale²⁴

Graded feature		Points	Score
Patient age	< 20 years	1	(1 - 3)
	20–40 years	2	
	> 40 years	3	
Presentation	ruptured	0	(0 - 1)
	unruptured	1	
Pattern of venous drainage	compact nidus	0	(0 - 1)
	diffuse nidus	1	
L-Y score → TOTAL			(1 - 5)

mismatch with the S-M grade: a S-M grade III with a low L-Y score behaved more like a S-M II AVM, whereas a S-M grade III AVM with a high L-Y grade AVM behaved more like a S-M grade IV.³

90.2.9 Management

- for patients presenting with nontraumatic ICH with clinical or radiologic suspicion for underlying cause, CTA, MRA, & catheter angiography can be useful to evaluate for AVM (Level II³)
- the initial hemorrhage should be managed in accordance with 2015 AHA ICH management guidelines (p. 1619) (covers HTN, seizures, correction of coagulopathy...)
 - surgical evacuation of an ICH should be undertaken if indicated regardless of whether or not there is an AVM. Small superficial AVMs can be removed at the time of emergency surgery
 - the resection of larger, deep AVMs may be deferred for 2–6 weeks to allow a reduction of brain swelling and better evaluation of the residual AVM angiographically and surgically
- for AVMs that have ruptured:
 - surgical resection of low S-M grade AVMs with low L-Y scores, and some S-M grade IV AVMs with low L-Y scores should be considered
 - surgical resection of high S-M grade AVMs is risky, but if the patient has a fixed neurologic deficit that is unlikely to worsen with surgery, it may be a consideration³
- management of unruptured AVMs has been contentious, and an attempt to determine the best course, the ARUBA (A Randomized trial of Unruptured Brain AVM) study²⁶ has been roundly criticised²⁷ for its design and premature closure due to endpoint satisfaction. Its conclusion that medical management alone was superior to medical management + intervention (surgery, SRS, embolization, or a combination) is not uniformly supported

90.2.10 Treatment

General information

Options and some pros and cons of each include:

- surgery: (see below) the treatment of choice for AVMs. When surgical risk is unacceptably high, alternative procedures (e.g., SRS) may be an option:
 - pros: eliminates risk of bleeding almost immediately. Seizure control improves
 - cons: invasive, risk of surgery, cost (high initial cost of treatment may be offset by effectiveness or may be increased by complications)
 - surgical excision: best suited for S-M grades I & II. S-M grades IV & V are high risk for poor outcome.
- radiation treatment:
 - ✗ conventional radiation: effective in only ≤20% of cases.^{28,29} Therefore not considered effective therapy
 - stereotactic radiosurgery (SRS): accepted for some small (≤2.5–3 cm nidus), deep AVMs. See Stereotactic radiosurgery for AVMs (p. 1906)
 - pros: done as an outpatient, non-invasive, gradual reduction of AVM flow, no recovery period. Previous radiation makes AVM tissue more amenable to resection, reducing surgical morbidity.³⁰ Seizure control improves (usually in patients with reduction or obliteration of nidus)³
 - cons: the therapeutic response takes 1–3 years (the so-called “latency period”). During that time, the risk of bleeding is about the same as the natural history (1–3%/year). Limited to lesions with nidus ≤3 cm. Obliteration rate (70–80%) less than with surgery. Delayed side effects include: radiation necrosis, brain edema, & cyst formation. Permanent neurological changes occur in 2–3%³

3. endovascular techniques: e.g., (AVM embolization (p. 1926))
 - a) pros: facilitates surgery³¹ or (controversial) SRS
 - b) cons: sometimes inadequate by itself to permanently obliterate AVMs, induces acute hemodynamic changes, may require multiple procedures, embolization prior to SRS reduces the obliteration rate from 70% (without embolization) to 47% (with embolization)³²
4. combination techniques: e.g., embolization to shrink nidus, then stereotactic radiosurgery

Considerations to take into account in managing AVMs:

1. associated aneurysms: on feeding vessels, draining veins, or intranidal
2. flow: high or low
3. age of patient
4. history of previous hemorrhage
5. size and compactness of nidus
6. availability of interventional neuroradiologist
7. general medical condition of the patient

Surgical treatment for AVMs

Pre-op medical management

Before direct surgical treatment, patient should ideally be pre-treated with propranolol 20 mg PO QID for 3 days to minimize post-op normal perfusion pressure breakthrough (postulated cause of postoperative bleeding and edema,³³ see below). Labetalol has also been used perioperatively to keep MAP 70–80 mm Hg.³⁴

Booking the case: Craniotomy for AVM

Also see defaults & disclaimers (p. 25).

1. position: (depends on location of AVM), radiolucent headholder
2. pre-op embolization (by neuroendovascular interventionalist): typically 24–48 hours pre-op
3. intraoperative angiography (optional)
4. equipment
 - a) microscope (with ICG capability if used)
 - b) image-guided navigation: primarily for the bone flap placement
5. blood availability: type and cross 2 U PRBC
6. post-op: ICU
7. consent (in lay terms for the patient—not all-inclusive):
 - a) procedure: surgery to open the skull and remove the abnormal tangle of blood vessels in the brain, intraoperative angiography
 - b) alternatives: stereotactic radiosurgery, endovascular techniques (not considered definitive treatment for most AVMs, but often used as an adjunct)
 - c) complications: usual craniotomy complications (p. 25) plus stroke (the main concern), bleeding intra-op (requiring transfusion) and post-op, neurologic deficit related to the area of AVM location, failure to be able to remove entire AVM, recurrence in future

Basic tenets of AVM surgery

1. wide exposure
2. isolate and occlude feeding (terminal) arteries before draining veins (lesions with a single draining vein can become impossible to deal with if premature blockage of the draining vein occurs, e.g., by kinking, coagulation)
3. excision of whole nidus is necessary to protect against rebleeding (occluding feeding arteries is not adequate)
4. identify and spare en passant vessels and adjacent (uninvolved) arteries
5. dissect directly on nidus of AVM, work in sulci and fissures whenever possible
6. in lesions that are high-flow on angiography, consider preoperative embolization
7. lesions with supplies from multiple vascular territories may require staging
8. clip accessible aneurysms on feeding arteries

Delayed postoperative deterioration

May be due to any of the following:

1. normal perfusion pressure breakthrough³³: characterized by post-op swelling or hemorrhage. Thought to be due to loss of autoregulation, although this theory has been challenged.³⁵ Risk may be reduced by pre-op medication (see above)
2. occlusive hyperemia³⁶: in the immediate post-op period probably due to obstruction of venous outflow from adjacent normal brain, in a delayed presentation may be due to delayed thrombosis of draining vein or dural sinus.³⁷ Risk may be elevated by keeping the patient “dry” post-op
3. rebleeding from a retained nidus of AVM
4. seizures

90.2.11 Follow-up of treated AVMs

For AVMs that have been resected:

1. an intraoperative or early post-op angiogram should be done to confirm completed removal of the nidus
2. residual nidus should be treated with re-resection or other treatment
3. suggestion: when satisfactory complete angiographic obliteration of an AVM has been accomplished, follow-up with *catheter angiogram* (not CTA or MRA) at 1 & 5 years post treatment

Following radiosurgery, imaging is required.

1. MRI/MRA is commonly repeated at 6-month intervals through the latency period (CT/CTA is substituted if MRI is contraindicated)
2. if SRS-induced obliteration is demonstrated, a catheter angiogram should be done to confirm
3. long-term MRI or CT follow-up may be used to monitor for delayed effects (necrosis, cysts...)

While catheter angiography (CA) is the gold standard, it carries some risk. Post-treatment MRI may be considered, as it demonstrated an accuracy of 90% for AVMs <2.8 cm³ volume and 70% for AVMs >2.8 cm³ compared to CA.³⁸

90.3 Developmental venous anomaly (DVA) (venous angioma)

90.3.1 General information

Key concepts

- a vascular malformation of the brain with normal intervening brain present
- functions as part of the venous drainage for that region of the brain
- low-flow, low-pressure
- characteristic imaging finding: caput medusae (starburst pattern)
- rarely symptomatic: seizures rare, hemorrhage even less common. Venous infarcts may occur
- commonly associated with a cavernous malformation (p. 1525) (cav mal) which is more likely to be the source of any symptoms
- direct treatment is rarely indicated due to the vital function of venous drainage, the risk of venous infarct with excision, and the rarity of symptoms

Developmental venous anomaly (DVA), formerly (developmental) venous angioma AKA venous malformation. A tuft of medullary veins that converge into an enlarged central trunk (draining vein or “collector vein”) that drains either to a deep ependymal vein or into a dural venous sinus. Thought to be congenital abnormalities that arise as a result of a failure in the normal sequence of cerebral venous development.³⁹ The veins lack large amounts of smooth muscle and elastic. No abnormal arteries are found. There is neural parenchyma between the vessels. Most common in regions supplied by the MCA⁴⁰ or in the region of the vein of Galen. They may be associated with a cavernous malformation (p. 1525). Non-hereditary. No apparent gender predilection. DVAs are low-flow and low-pressure.

90.3.2 Presentation

Most are clinically silent, but rarely seizures and even less frequently hemorrhage may occur. Venous infarcts have been reported rarely and may arise from thrombosis of the draining vein.⁴¹ If symptoms are present, look for an associated cavernous malformation which is more likely to be the source (T2* GRE or SWI MRI images may reveal cavernous malformations that might otherwise be occult).

90.3.3 Imaging

Recommendations

- contrast T1 MRI + MRV
- T2* GRE (gradient echo) or SWI: to look for hemosiderin staining (from cav mal)

General features

Classic finding: the appearance of the tuft of veins has been described as the caput medusae (other descriptive terms include: a hydra, spokes of a wheel, a spider, an umbrella, a mushroom, or a sunburst or starburst).^{42(p 1471)} Typically has a long draining vein (longer than a normal vein).

CT scan

Unenhanced CT: is usually negative. May show up as a hyperdense lesion if there is thrombosis or calcifications (► Fig. 90.1).

Enhanced CT: the wedge of veins comprising the caput medusae may be seen. The enlarged elongated draining vein is usually demonstrated.

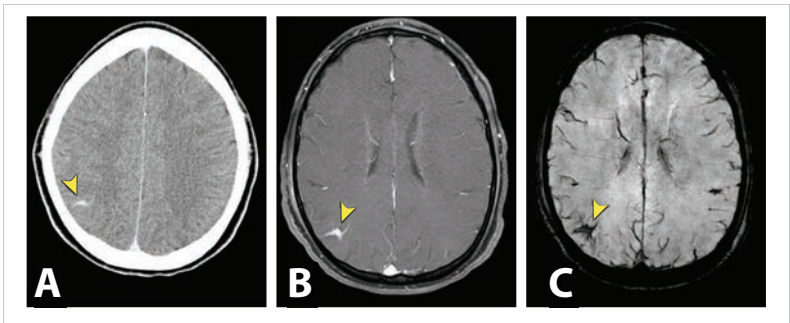


Fig. 90.1 Developmental venous anomaly, right parietal (yellow arrowheads). No hemosiderin staining nor associated cavernous malformation were identified in this patient. The collecting vein drained into the torcular Herophili (not shown). Image: A: CT scan axial noncontrast. B: MRI axial contrast T1. C: MRI axial susceptibility-weighted image (SWI).

MRI/MRV

Noncontrast T1WI & T2WI: if flow is high enough, flow voids may be seen, otherwise may be negative.

T1 contrast images: DVAs enhance intensely (findings as in enhanced CT, see above (► Fig. 90.1)).

T2* GRE or SWI: hemosiderin from previous bleeds (usually from an associated cav mal) will show up as hypointense blooming.

MRV: usually shows the characteristic caput medusae (see above) & draining vein.

Angiogram

Usually not necessary to make the diagnosis. A DVA may occasionally be angiographically occult. Angiographic considerations⁴³ include:

- the caput medusae (see above): can usually be seen in the venous phase

- long draining vein: drains an excessive amount of brain tissue (it is theorized that venous restrictive disease may occur because of the length)
- the arterial phase should show no arteriovenous (AV) shunting (characteristic of an AVM), however, there are isolated reports of an associated AV shunt⁴¹ (these fistulized DVAs—called mixed vascular malformations—may have a more ominous prognosis akin to an AVM)

90.3.4 Treatment

In general, these should not be treated, as they are rarely symptomatic and they represent the venous drainage of the brain in that vicinity. Due to the significant risk of venous infarction, surgery for the angioma itself is reserved only for documented bleeding or for intractable seizures that can definitely be attributed to the lesion. If surgery is indicated for an associated lesion (e.g., cavernous malformation), the angioma should be left alone.

90.4 Dural arteriovenous fistulae (DAVF)

90.4.1 General information

AKA dural AVMs (DAVM). Vascular abnormality in which an arteriovenous shunt is contained within the leaflets of the dura mater, exclusively supplied by branches of the internal/external carotid or vertebral arteries.⁴⁴ Because they are considered acquired rather than congenital lesions, the term fistula is preferred over malformation, although the latter term has also been used in the literature. Multiple fistulas may be found in up to 8% of cases.

Usually found adjacent to dural venous sinuses. Common locations:

1. transverse/sigmoid: the most common⁴⁵ (63% of cases) with a slight left-sided predominance,⁴⁶ with the epicenter of these almost invariably at the junction of the transverse and sigmoid sinuses
2. tentorial/petrosal
3. anterior fossa/ethmoidal
4. middle fossa/Sylvian
5. cavernous sinus (carotid-cavernous fistula (CCF))
6. superior sagittal sinus
7. foramen magnum

90.4.2 Etiology

Evidence suggests that most DAVFs are acquired, idiopathic lesions, and they have a well-recognized association with venous sinus thrombosis, although their exact pathogenesis is not fully understood. Theories include:

1. venous sinus occlusion awakens dormant embryonic dural arteriovenous channels⁴⁵
2. venous hypertension/thrombosis promotes local angiogenesis and the de novo formation of DAVF⁴⁷
3. the DAVF may arise first and itself result in venous sinus thrombosis⁴⁸

90.4.3 Epidemiology

DAVFs comprise 10–15% of all intracranial AVMs.⁴⁶ 61–66% occur in females, and patients are usually in their 40s or 50s. They occur rarely in children, and when they do they tend to be complex, bilateral dural sinus malformations.⁴⁹

90.4.4 Presentation

Common findings are listed in ► Table 90.9. Pulsatile tinnitus is the most common presenting symptom of a DAVF. Cortical venous drainage with resultant venous hypertension can produce IC-HTN, and this is the most common cause of morbidity and mortality and thus the strongest indication for DAVF treatment. DAVFs may also cause global cerebral edema or hydrocephalus due to poor cerebral venous drainage or by impairing the function of the arachnoid granulations, respectively. Other DAVF symptoms/signs include headaches, seizures, cranial nerve palsies, and orbital venous congestion.

Table 90.9 Clinical findings in 27 patients with dural AVMs in the lateral and/or sigmoid sinus⁵⁰

Sign/symptom	No. (%)
pulsatile tinnitus	25 (92%)
occipital bruit	24 (89%)
headache	11 (41%)
visual impairment	9 (33%)
papilledema	7 (26%)

90.4.5 Evaluation

General information

Brain CT or MRI without contrast are often normal. CTA may reveal dilated tortuous vessels corresponding to enlarged arterial feeders or ectatic draining veins. MRA may reveal dilated pial vessels, early prominent venous sinus filling, sinus enlargement or occlusion, and white matter edema related to venous hypertension. Full 6-vessel cerebral angiography (bilateral ICAs, bilateral ECAs, bilateral vertebral arteries) is required to establish the diagnosis and to plan treatment.

Angiographic classification

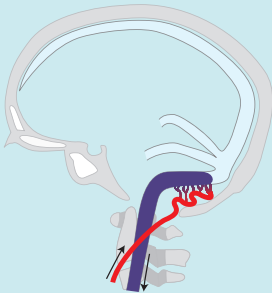
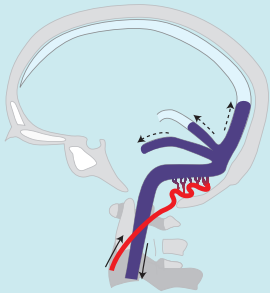
Several classification systems have been described to characterize DAVFs. The Borden⁵¹ (► Table 90.10) and the Cognard⁵² (► Table 90.11) systems have emerged as the most commonly utilized contemporary grading schemes. Cortical venous drainage is the defining angiographic feature that distinguishes benign (low-grade) from aggressive (high-grade) fistulas. (Borden I, Cognard I, and Cognard IIa are low-grade, all others are high-grade.)

Table 90.10 Borden classification

Type	Features
I	DAVF drainage into a dural venous sinus or meningeal veins, with normal antegrade flow. Usually clinically benign.
II	DAVF draining antegrade into dural venous sinus, but with retrograde flow into cortical veins.
III	DAVF with direct retrograde flow from fistula into cortical veins, causing venous hypertension.

Above further subclassified into a: with single hole and b: multiple holes

Table 90.11 Angiographic classification of dural AVMs^{52a}

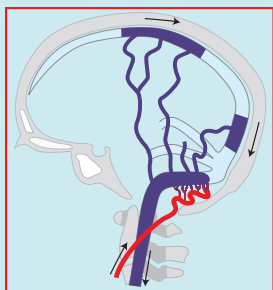
Venous drainage: sinus	
Type I  Normal antegrade flow into a dural venous sinus Course: benign^b	Type IIa  Drainage into a sinus with retrograde flow within the sinus^c Course: sinus reflux caused IC-HTN in 20%

(continued)

Table 90.11 continued

Venous drainage: sinus

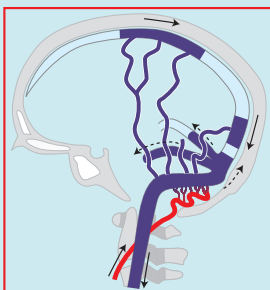
Type IIb



Drainage into a sinus with retrograde flow into cortical vein(s)

Course: reflux into veins induced hemorrhage in 10%

Type IIa + b

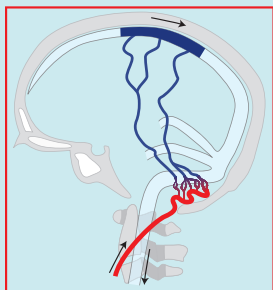


Drainage into a sinus with retrograde flow within the sinus^c and cortical vein(s)

Course: aggressive in 66% with bleeding and/or IC-HTN

Venous drainage: directly into cortical veins^b

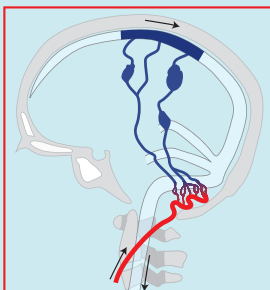
Type III



Direct drainage into a cortical vein without venous ectasia.

Course: hemorrhage occurs in 40%

Type IV

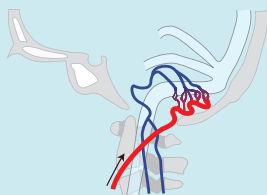


Direct drainage into a cortical vein with venous ectasia

Course: hemorrhage occurs in 65%

Venous drainage: spinal in addition to all of the above

Type V



Direct drainage into spinal perimedullary veins in addition to all of the above

Course: progressive myelopathy in 50%

^athose in red boxes are high risk for bleeding or intracranial hypertension

^bdespite a usually good prognosis, ≈ 2% will progress and therefore follow-up studies may be warranted

^cdashed arrows signify retrograde flow

Borden classification system

The Borden classification⁵¹ is shown in ► Table 90.10.

Cognard angiographic classification

The Cognard system⁵² is shown in ► Table 90.11. This system is generally most applicable to DAVFs involving the transverse sinus.

Cognard found 54% had no cortical venous reflux (Types I and IIa) and usually exhibit benign behavior.

★ Key determinant: in the Cognard system, the pattern of venous drainage is the most critical factor. As a general rule, lesions with retrograde flow in the cortical veins (IIb, IIa + b, III & IV—red frames in ► Table 90.11) are high risk (for bleeding or intracranial hypertension...).

90.4.6 Natural history

The concept of benign vs. aggressive DAVF behavior based on the absence or presence, respectively, of cortical venous drainage was validated by data reported by University of Toronto group. Over a 3-year period, 98% of benign lesions (no cortical venous drainage) remained benign.⁵³ On the other hand, over a 4-year period, annual rates of hemorrhage, non-hemorrhagic neurologic deficit, and mortality were 8.1%, 6.9%, and 10.4%, respectively, for aggressive lesions (with cortical venous drainage).⁵⁴

In a meta-analysis of 377 cases,⁵⁵ three DAVF locations were associated with particularly aggressive behavior (aggressive:benign ratio)—tentorial (31:1), middle fossa/Sylvian (2.5:1), anterior fossa/ethmoidal (2.1:1).

90.4.7 Management

General information

Lesions with cortical venous drainage should generally be treated. Lesions without cortical venous drainage should be followed radiographically and clinically (2% may evolve to develop cortical venous drainage). A change in a bruit (either worsening, or disappearance) should prompt restudy.

Indications for intervention:

1. presence of cortical venous drainage
2. neurologic dysfunction
3. hemorrhage
4. orbital venous congestion
5. refractory symptoms (headache, pulsatile tinnitus)

Manual carotid self-compression

Advocated by some, the thrombosis rate of ≈ 22% and clinical improvement rate of 33%⁵⁶ may mimic the natural course. Patients are advised to compress with the hand that would be affected by ischemia if it were to occur (e.g., with a left-sided DAVF, the right hand should be used to compress the left carotid artery). That way, the hand would fall away if ischemia develops. Recommendations vary, one option: start with 10 minutes once a day, gradually increase frequency and duration.

Endovascular embolization

May be performed transarterial or transvenous. Before the availability of liquid embolic agents (Onyx and NBCA), treatment was directed at the *venous* drainage (unlike pial AVMs) which had higher success, because the coils could be deployed to sacrifice the venous drainage very close to the point of arteriovenous shunting, resulting in thrombosis of the fistula. It is more difficult to deploy coils across the point of arteriovenous shunting from the arterial side, whereas the liquid embolic agents, particularly Onyx, can be injected at somewhat of a distance and pushed forward across the fistulous point. Whether a transarterial, transvenous, or combined approach is utilized depends on the unique angioarchitecture of the fistula.

Surgery

While endovascular approaches have emerged as the primary treatment for most DAVFs, certain fistula types are still best dealt with via open surgery as the first-line strategy.⁵⁷ Furthermore, surgery has been used to successfully treat DAVFs after previous partial, incomplete, or failed endovascular

treatment. Finally, surgery can be used adjunctively in a combined approach to provide direct access for embolization of DAVFs that are inaccessible by a purely endovascular route.

Preoperative embolization may facilitate surgical treatment⁵⁸ by lessening the risk of catastrophic hemorrhage, which may occur simply during the performance of the craniotomy.⁵⁰ The use of the craniotomy is discouraged, as a sinus or venous laceration could produce a fatal hemorrhage. Contingencies for the rapid administration of blood products must be made (large-bore central lines). The scalp incision, craniotomy flap, and dural incision should be planned in a strategic manner to control and sequentially eliminate the blood supply to the lesion at each step, while maximizing the exposure as needed. Surgical options for the treatment of DAVFs include the following techniques³⁷:

1. radical fistula excision
2. sinus skeletonization
3. disconnection of cortical venous drainage
4. ligation of the fistulous point and/or outflow vein
5. sinus packing
6. coagulation of arterial feeders to the lesion

While surgery vs. endovascular treatment can be considered for all DAVF locations, two locations generally remain more favorable for surgery:

1. anterior fossa/ethmoidal
2. tentorial DAVFs

The endovascular approach to these fistulas is difficult, whereas the surgical approach is often straightforward. Surgically-assisted embolization, whereby a craniotomy is performed followed by direct puncture for embolization of the target vessel, may be utilized in select cases.

Stereotactic radiosurgery

May be used post-embolization.⁵⁹ Pan et al⁶⁰ reported a complete obliteration rate of 58% of transverse/sigmoid fistulae treated with only radiosurgery (1650–1900 cGy), or with radiosurgery after surgery/embolization had failed to produce complete obliteration. 71% of the patients were cured of their symptoms.

With the continued improvement of endovascular technology over the past two decades, the role of radiosurgery for DAVF treatment has steadily decreased; however, it remains an option for those difficult lesions in which endovascular/surgical options have been exhausted.

90.5 Vein of Galen malformation

90.5.1 General information

Enlargement of the great cerebral vein of Galen (VOG) may occur in “vein of Galen malformations” (VOGM) (some refer to these as vein of Galen aneurysms). Congenital, develop before the 3-month embryo stage or secondarily to high flow from adjacent deep parenchymatous AVMs or pial fistulae. Most likely consist not of the vein of Galen but rather of the medial vein of the prosencephalon. Parenchymatous AVMs can be distinguished from true VOG malformations by retrograde filling of the internal cerebral vein in the former.⁶¹

True VOG malformations are predictably fed from the medial and lateral choroidal, circumferential, mesencephalic, anterior choroidal, pericallosal, and meningeal arteries.^{61,62} Agenesis of the straight sinus may be an associated finding.

90.5.2 Presentation

Newborns tend to present with congestive heart failure in first few weeks of life (due to high blood flow)⁶³ and a cranial bruit. Hydrocephalus may result from obstruction of the Sylvian aqueduct by the enlarged VOG, or it may be caused by the increased venous pressure (which can also produce prominence of the scalp veins⁶⁴).

Parenchymatous AVMs are usually diagnosed later in life due to neurological manifestations,⁶⁵ including focal neurologic deficit and hemorrhage.

90.5.3 Classification

Classified based on the location of the fistula^{66,67}:

1. pure internal fistulae: single or multiple
2. fistulae between thalamoperforators and the VOG
3. mixed form: the most common
4. plexiform AVMs

90.5.4 Natural history

Untreated VOG malformations have a poor prognosis, with neonates having nearly 100% mortality, and 1–12-month-olds having ≈ 60% mortality, 7% major morbidity, and 21% being normal.⁶⁸

Parenchymatous AVMs behave similarly to other AVMs.

90.5.5 Treatment

Hydrocephalus

Hydrocephalus associated with VOGM is obstructive, due to the varix. Although admonitions about shunting were common due to fear of precipitating a hemorrhage, when hydrocephalus is present the patient needs a shunt.

Vein of Galen malformations

Pediatric patients are often in poor medical condition, limiting the efficacy of operative treatment. Treatment options for these include embolization of the main feeding arteries. Prognosis is poor. Those presenting with hydrocephalus from aqueductal obstruction often do so at the end of the first year of life. Neurosurgical excision may be considered here, and the prognosis is better.

Repeated embolization while monitoring the venous drainage is employed.

Parenchymal AVM with enlarged VOG

The AVM is treated by the same methods as other AVMs (embolization, resection, or radiosurgery).

90

90.6 Carotid-cavernous fistula

90.6.1 General information

Key concepts

- direct (high flow, from ICA) or indirect (low flow, from meningeal branches)
- classic triad (more common with direct CCF): chemosis, pulsatile proptosis, ocular bruit
- risk of SAH is low. Major risk is to vision
- natural history of low-flow CCF is up to 50% spontaneous thrombosis

See also anatomy and venous inflow and outflow of the cavernous sinus (p. 86).

Carotid-cavernous fistula (CCF): divided into direct (Type A) and indirect (Types B–D)⁶⁹:

1. Type A: direct high-flow shunts between the *internal* carotid artery and cavernous sinus:
 - a) traumatic (including iatrogenic): occur in 0.2% of patients with craniocerebral trauma. Iatrogenic: may follow percutaneous trigeminal rhizotomy,⁷⁰ endovascular procedures...
 - b) spontaneous: usually due to ruptured cavernous sinus ICA aneurysm. May also occur in patients with connective tissue disorders
2. indirect (dural): most are shunts: from dural arteries that are branches of the *external* carotid (not from ICA) (exception: Type B) – low flow
 - a) Type B: from meningeal branches of the internal carotid artery (ICA)
 - b) Type C: from meningeal branches of the external carotid artery (ECA)
 - c) Type D: from meningeal branches of both the ICA and ECA

90.6.2 Presentation

1. orbital and/or retro-orbital pain
2. chemosis (arteriolization of conjunctiva)

3. pulsatile proptosis
4. ocular and/or cranial bruit
5. deterioration of visual acuity: may be due to hypoxic retinopathy as a result of reduced arterial pressure, increased venous pressure, and increased intraocular pressure
6. diplopia: abducens (VI) palsy is the most common
7. pupillary dilatation
8. ophthalmoplegia (usually unilateral, but may present initially as bilateral or may progress to bilateral)
9. increased intraocular pressure
10. neo-vascularization of the iris or retina
11. rarely: SAH

Indirect CCFs generally have a more gradual onset and milder presentation than direct.

90.6.3 Evaluation

CT or MRI: usually demonstrates proptosis. Serpiginous and engorged intraocular vessels including the superior ophthalmic vein (best seen on T2WI coronals—helps to differentiate from rectus muscles) and convexity of lateral wall of cavernous sinus.

Angiography: shunting of blood from ICA into cavernous sinus. Rapid opacification of petrosal sinus and/or ophthalmic vein may be seen.

1. Huber maneuver: lateral view, inject VA and manually compress affected carotid. Helps identify upper extent of fistula, multiple fistulous openings, and complete transection of ICA
2. Mehringer-Hieshima maneuver: inject contrast at a rate of 2–3 ml/s into affected carotid while compressing the carotid in the neck (below the catheter tip) to control flow to help demonstrate the fistula

90

90.6.4 Treatment

General information

20–50% of *low-flow* CCF spontaneously thrombose, and therefore one may observe these as long as visual acuity is stable and intra-ocular pressure is ≤ 25 . Symptomatic (e.g., progressive visual deterioration) high-flow CCFs rarely resolve spontaneously, and urgent treatment is usually indicated. Treatment is usually in the form of embolization by an interventional neuroradiologist or trapping between surgically placed clips.

Even if normal ocular motility cannot be achieved in the affected eye, preservation of vision is desirable because:

1. for some ocular motility abnormalities, surgical treatment may reduce diplopia
2. patient may be provided with frosted eyeglass lens which will eliminate diplopia but will maintain peripheral vision
3. in the rare event of injury to contralateral eye (trauma, central retinal artery occlusion...) there would be “reserve” vision in the eye with reduced motility (with loss of the other eye, there would not be diplopia)

Indications for treatment

1. proptosis
2. visual loss
3. cranial nerve VI palsy
4. intractable bruit
5. severely elevated intraocular pressure
6. increased filling of cortical veins on angiography

Endovascular treatment

Options include:

1. electrolytically detachable coils
2. Amplatzer™ Vascular Plug (AVP) (Abbott Laboratories): a self-expanding nitinol wire mesh

Routes available include:

1. transarterial through internal carotid. If this fails (e.g., wide aneurysm neck), the carotid artery may be occluded on either side of fistula to trap it (sacrifices carotid artery, therefore test occlusion must be done first to determine if patient can tolerate this; however, **note:** test occlusion with an open fistula may give a false-positive result because steal through the fistula may reduce CBF and cause neurologic symptoms not related to the occlusion acting alone). The distal occlusion needs to be proximal to the ophthalmic artery
2. transarterial through external carotid: useful only for dural fistulas
3. transvenous:
 - a) traversing heart to enter jugular vein, then through petrosal sinus to cavernous sinus. Lower success rate ($\approx 20\%$) than transarterial route
 - b) via superior ophthalmic vein: entered where supra-optic vein enters orbit to become superior ophthalmic vein. If possible, it is best to wait for the vein to become arterialized by the high flow pressure. Reports of "disasters" due to injury to the fragile vein performed before arterio- lization took place may have been due to more primitive balloon catheters that were standard before current commercially produced versions were available (which are softer than the originals). Must avoid lacerating the vein inside the orbit, and avoid distal ligation of the vein without proximal occlusion (shunts even more blood into eye)

Choice of technique

With indirect fistulas, it is mandatory to place coils on the venous side (otherwise new feeders will be recruited).

Coils or clips may be used to occlude direct fistulas.

90.7 Sigmoid sinus diverticulum

Sigmoid sinus diverticulum (SSD, see ► Fig. 90.2) or sigmoid sinus dehiscence is found in 1.2% of asymptomatic patients.⁷¹ However, these abnormalities may be found ipsilaterally in up to 23% of patients with pulsatile tinnitus, presumably due to turbulent flow which may occur in these abnormalities.⁷² SSD are more common in women.

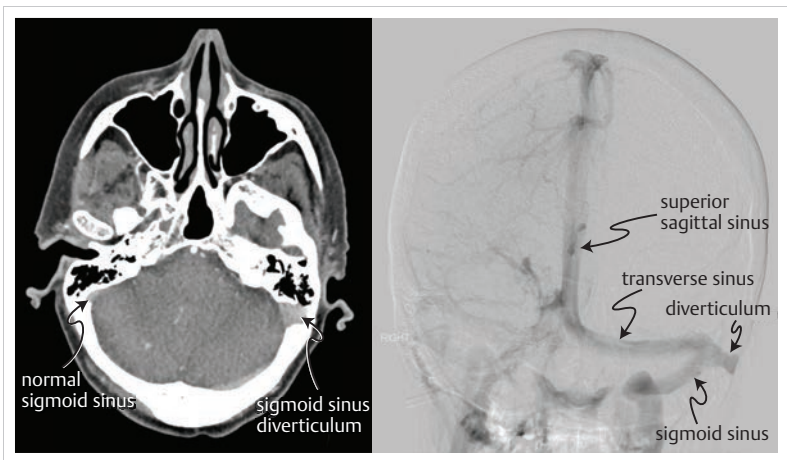


Fig. 90.2 Sigmoid sinus diverticulum.

Image: axial contrast head CT (left) and AP venous-phase angiogram (right) showing sigmoid sinus diverticulum on the patient's left side.

When treatments such as masking noise generators fail, surgical intervention can be considered. Surgical treatment options include:

- Endovascular coiling/stenting
- Transmastoid "resurfacing" (see below)
- Craniectomy with clip reconstruction

Transmastoid "resurfacing" consists of partial mastoidectomy, and subtotal obliteration of the area of the diverticulum (so-called sinus wall resurfacing^{72,73,74}) with e.g., bone chips, fibrin glue, or muscle.

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91 Angiographically Occult Vascular Malformations

91.1 General information

91.1.1 Terminology

Terminology is controversial. The term “cryptic cerebrovascular malformations” was originally applied to angiographically and clinically silent lesions regardless of size.

Recommendation: use the term “**angiographically** occult (or cryptic) vascular malformations” (AOVM) to refer to cerebrovascular malformations that are not demonstrable on technically satisfactory catheter cerebral angiography (i.e., good quality films, with subtraction views, and the following as appropriate: magnification, angiotomography, rapid serial angiograms, or delayed films).¹ Many lesions have large patent vessels at surgery in spite of negative angiography.² Other imaging modalities (i.e., CT, MRI) may be able to reveal these lesions. Although often used interchangeably, the term “occult malformation” (omitting the word “angiographically”) is suggested for use with lesions that also do not appear on these other imaging modalities.

91.1.2 Etiologies

The reasons for a vascular lesion being angiographically cryptic include:

1. lesions that have hemorrhaged
 - a) the bleeding may obliterate the lesion: difficult to substantiate³
 - b) the clot may temporarily compress the lesion³ which may re-open after weeks to months as the clot dissolves
2. sluggish flow
3. small size of the abnormal vessels
4. may require very late angiographic films (i.e., delayed films) to visualize due to late filling

91.1.3 Epidemiology

Incidence of AOVM has been estimated as $\approx 10\%$ of cerebrovascular malformations.⁴ AOVMs were found at necropsy in 21 (4.5%) of 461 patients with spontaneous intracranial hemorrhage (ICH);⁵ but refinements in angiography have occurred since this 1954 report.

The average age at diagnosis in one literature review¹ was 28 yrs.

91.1.4 Presentation

AOVM most often present with seizures or H/A. Less commonly they may present with progressive neurologic symptoms (usually as a result of spontaneous ICH).⁶ They may also be discovered incidentally.

91.2 Osler-Weber-Rendu syndrome

91.2.1 General information

AKA hereditary hemorrhagic telangiectasia (HHT), AKA capillary telangiectasia: slightly enlarged capillaries with low flow. Cannot be imaged on any radiographic study. Usually incidentally found at necropsy without clinical significance (risk of hemorrhage is very low, except possibly in brainstem). Has intervening neural tissue⁴ (unlike cavernous malformations). Usually solitary, but may be multiple when seen as a part of a syndrome: Osler-Weber-Rendu (see below), Louis-Barr (ataxia telangiectasia), Myburn-Mason, Sturge-Weber.

Associated cerebrovascular malformations (CVM) include: telangiectasias, AVMs (the most common CVM, seen in 5–13% of HHT patients⁷), venous angiomas, and aneurysms. Patients are also prone to pulmonary arteriovenous fistulas with associated risk of paradoxical cerebral embolism, which predisposes to embolic stroke and cerebral abscess formation (p.343).

91.2.2 Epidemiology

Rare autosomal dominant genetic disorder of blood vessels affecting ≈ 1 in 5,000 people. 95% have recurrent epistaxis.

91.2.3 Imaging

CT

May show a well-demarcated homogeneous or mottled high density⁶ (high density due to hematoma, calcification, thrombosis, hemosiderin deposition, alterations in BBB, and/or increased blood volume¹) with some form of contrast enhancement (around or within lesion) in 17 of 24 patients.⁶ Surrounding edema or mass effect is rare (except in cases that have recently hemorrhaged).

MRI

May demonstrate previous hemorrhage(s)⁸ (may be important when the presence of multiple occurrences affects therapeutic choices). T2WI finding: reticulated core of increased and decreased intensity, a prominent surrounding rim of reduced intensity may be present (due to hemosiderin-laden macrophages from previous hemorrhages). T2* GRE image demonstrates flow-related enhancement in $\approx 60\%$ of cases, which allows signal dropout from flowing blood on other sequences to be differentiated from that due to calcium (and thus, bone) or air (limitations: hemosiderin causes signal dropout, and slow in-plane flow does not enhance).⁹

91.2.4 Treatment

Surgery is indicated mainly for evacuation of hematoma or diagnosis, especially when favorably located. Also consider surgery for recurrent hemorrhages (rupture has been reported even after normal angiography) or medically intractable seizures. Stereotactic radiosurgery has not had a satisfactorily high enough benefit-to-risk ratio to justify its use.¹⁰

91.3 Cavernous malformation

91.3.1 General information

91

Key concepts

- usually angiographically occult. MRI is the study of choice (open channels $\rightarrow$ flow void on T2WI, previous hemorrhage $\rightarrow$ “popcorn” pattern especially on gradient echo or SWI)
- low-flow. No intervening neural parenchyma. No arteries
- may be associated with developmental venous anomaly (DVA) (p. 1512) which participates in venous outflow and should be preserved
- may be sporadic, hereditary, or may follow XRT
- presentation: usually seizures. Hemorrhage: risk is difficult to predict
- treatment:
 - a) $\blacklozenge$ surgery is the treatment of choice for symptomatic accessible lesions
 - b) $\times$ radiosurgery is controversial, and should be reserved for symptomatic, inaccessible, non-familial CMs

AKA: cavernous hemangioma, cavernoma, cavernous angioma, angioma, hemangioma, and in medical jargon “cav-mal.” A well-circumscribed, benign vascular hamartoma consisting of irregular thick- and thin-walled sinusoidal vascular channels located within the brain but *lacking intervening neural parenchyma*,⁴ large feeding arteries, or large draining veins. Usually 1–5 cm in size. May hemorrhage, calcify, or thrombose. Occur rarely in the spinal cord.¹¹ Caverns are filled with blood in various stages of thrombus formation/organization/dissolution. Frequently associated with venous angiomas (p. 1512). Capillary telangiectasias may be found adjacent to lesions and may represent a precursor. Stain positive for angiogenesis factor.¹² Lesions may arise de novo,¹³ and may grow (although slower than hemangioblastomas), shrink, or remain unchanged with time.¹⁴

91.3.2 Pathology

Gross appearance resembles a mulberry (facetiously dubbed a “hemorrhoid of the brain”). Structural characteristics:

1. comprised of dilated endothelial-lined capillary vessels (caverns) with defective tight junctions
2. no identifiable arterial feeders on angiography

3. often associated with a developmental venous anomaly (DVA)
4. no functional brain tissue with the encapsulated core⁴

Light microscopy: stains for von Willebrand's factor. Smooth muscle layer is absent (except for some tiny portions). EM: shows abnormal gapping of the tight junctions between endothelial cells¹⁵ (may permit leakage of blood) and sparse or poorly characterized subendothelial smooth muscle cells.¹⁵

91.3.3 Epidemiology

Cerebral cavernous malformations (CM) comprise 5–13% of CNS vascular malformations, and develop in 0.02–0.16% of the population (based on large autopsy¹⁶ and MRI^{17,18} series). 48–86% are supratentorial (with 5–10% in the basal ganglia), 4–35% brainstem.¹⁹ Multiple in 23%²⁰ to 50%²¹ of cases, and multiplicity may be more common in hereditary cases.^{17,22}

Spinal CMs: CMs occur rarely in the spinal cord.

CMs may be sporadic, hereditary, or may follow XRT.²³

XRT appears to be a risk factor²⁴ (e.g., following craniospinal XRT²⁵ for medulloblastoma), especially for spinal CMs. 42% of patients with spinal CMs also harbor ≥ 1 intracranial CM.²⁶

91.3.4 Genetics

Many cases of hereditary CMs follow a Mendelian *autosomal dominant* pattern with variable expressivity.²⁷ There appear to be at least 3 gene loci (see ► Table 91.1).

Table 91.1 Subtypes of CCM

	CCM1	CCM2	CCM3
locus	7q11-q22	7p15-13	3q25.2-q27
gene	KRIT1	MGC4607 (malcavernin)	PDCD10
feature	more common in Hispanics		rate of new CM formation: 0.4–2.7 new CMs/year

91.3.5 Presentation/natural history

General information

Presentation^{28,29,30,31}:

1. seizures: 50%³¹
2. hemorrhage: 25%.³¹ Usually intraparenchymal (see box (p. 1527) for definition of hemorrhage)
3. other: 25%. Including focal neurologic deficit without evidence of recent hemorrhage (25%),³¹ hydrocephalus, or as in incidental finding in 20–50%.^{18,32}

Hemorrhage

Accurately determining the risk of hemorrhage has been elusive. Reported rates depend on study design (retrospective vs. prospective, natural history vs. surgical series),³³ the definition of hemorrhage used (clinical vs. radiographic) and whether one assumes the CMs have been present since birth.

A standardized definition of hemorrhage by the Angioma Alliance³⁴ (see box) has been proposed (since essentially all CMs have surrounding hemosiderin indicative of small leaks and CMs are prone to recurrent small hemorrhages that are rarely devastating, and to differentiate a bleed from the mere presence of intralesional thrombus).

Spelling this out in more concrete terms³⁵: a hemorrhagic event is considered to have occurred with:

1. *extralesional* hemorrhage: acute or subacute blood-degradation products outside the CM
2. *intralesional* hemorrhage: acute or subacute blood-degradation products
 - a) in a previously identified CM: accompanied by lesion growth, mass effect, edema or corresponding symptoms
 - b) at first imaging (i.e., in a newly identified CM with no prior imaging): accompanied by corresponding clinical symptoms

✖ A hemorrhage is not considered to have occurred with:

1. only lesion growth without signal change
2. presence of hemosiderin without signs of a recent hemorrhage
3. the appearance of blood-degradation products without lesion growth or edema or clinical symptoms

Σ: Angioma Alliance definition of hemorrhage in cavernous malformations³⁴

Acute or subacute symptoms (any of: headache, seizure, impaired consciousness, or new/worsened focal neurological deficit referable to the anatomic location of the CM) accompanied by radiological, pathological, surgical, or (rarely) only cerebrospinal fluid evidence of recent extra- or intralesional hemorrhage.

This definition does not include either an increase in CM diameter without other evidence of recent hemorrhage, nor the presence of a hemosiderin halo.

Neurologic impairment following initial hemorrhage from a CM tends to be less than with other types of vascular malformations.³⁶ Each hemorrhage carries about 20% risk of morbidity, and 5% mortality.

Risk of hemorrhage in cerebral CMs

- risk of first-time hemorrhage among incidentally discovered CMs is very low (0.08%)³²
- the two factors most consistently associated with increased bleeding risk³¹:
 - CMs initially presenting with hemorrhage: hazard ratio for bleeding = 5.6
 - brainstem CMs: hazard ratio = 4.4
- hemorrhage outside the boundaries of the CM (extralesional hemorrhage) increase the rehemorrhage rate³⁵
- younger age: the hazard ratio for rehemorrhage is increased for younger patients³⁷
- inconsistent findings reported for risk differences with female gender, CM size, and CM multiplicity
- the annual risk of recurrent hemorrhage declines over time^{37,38,39,40}: with rate during the first ≈ 2.5 years that decreases^{33,37,41}
- higher annual ICH rates are reported in familial CMs (4.3–6%) than in sporadic cases⁴² but this could be artifactual³¹
- pregnancy & parturition are not thought to be risk factors for hemorrhage^{27,43,44}
- the risk of hemorrhage from a CM may not be increased by platelet inhibitors³⁹ or anticoagulation,⁴⁵ but this is based on uncontrolled studies that likely avoided treatment of patients with recent hemorrhage
- in a study of 185 Hispanic patients with CCM1 mutation, no significant association was identified with any cardiovascular risk factor including hypertension⁴⁶
- no relation of physical activity to hemorrhage from CMs has been identified³⁹ including aerobic exercise and noncontact sports.⁴⁷ Less is known about contact sports, high-altitude climbing, and scuba diving and spinal cord CMs⁴⁷

► **Hemorrhage as a function of cavernous malformation classification.** The Zabramski Classification⁴² of CMs (with an additional category V denoting extralesional extension added by Nikoubashman et al.³⁵) is shown in ► Table 91.2.

Using the standardized definition of hemorrhage, Nikoubashman et al. found no significant difference in hemorrhage rates for types I, II and V, whereas types III and IV differed significantly.³⁵ This yielded 3 levels of morphology-based risk stratification for CMs shown in ► Table 91.3.

Spinal CMs: data is limited. Annual hemorrhage rate of 2.1% has been reported.⁴⁹ 17% of patients with spinal CMs also have cerebral CMs, and 12% had a family history of CMs.⁴⁹

Seizures

The 5-year risk of first-time seizure was 6% among CM patients presenting with symptoms vs. 4% for those with incidental CMs.⁵⁰

Table 91.2 Zabramski classification of cavernous malformations⁴² (modified from Clinical Radiology, 68, Nikoubashman O, Wiesmann M, Tournier-Lasserre E, et al. Natural history of cerebral dot-like cavernomas. e453–e459, 2013, with permission from Elsevier^{a 35}) - schematic diagrams

Type	T1WI	T2WI	Gradient echo
I			
subacute hemorrhage with a rim of hemosiderin laden macrophages & gliotic brain. AHR ^b = 29.8%	 hyperintense core (methemoglobin)	 hyper- or hypo-intense core (hemosiderin & ferritin) with hypointense halo	
II			
loculated areas of hemorrhage & thrombus of varying age with a rim of gliotic, hemosiderin stained brain; large lesions may have areas of calcifications. AHR ^b = 20.1%	 reticulated core of mixed signal ("popcorn" appearance)	 reticulated core of mixed signal with hypointense rim ("popcorn" appearance)	
III			
chronic resolved hemorrhage with hemosiderin staining within & around the CM. AHR ^b = 3.4%	 iso- or hypo-intense	 hypointense with a hypointense rim that magnifies the size of the lesion	 hypointense with more magnification than T2WI
IV			
(some lesions in this category were found to be telangiectasias). AHR ^b = 1.3%	 poorly seen or not visualized	 poorly seen or not visualized	 punctate hypointense lesions (minute deposits of hemosiderin)
V^a			
presenting with extralesional hemorrhage. AHR ^b = 23.1%	 extralesional hemorrhage	 extralesional hemorrhage	 extralesional hemorrhage

^aType V is not part of the original Zabramski classification and was proposed by Nikoubashman et al.³⁵; only parts of the CM may be visible within the hemorrhage

^b AHR = annual hemorrhage rate found by Nikoubashman et al.³⁵ in patients age < 18 years

Table 91.3 Risk stratification based on CM morphology³⁵

Description	Annual hemorrhage risk	Mean hemorrhage-free interval (months)	Clinical correlates
High risk (Zabramski I, II or V). CMs with acute or subacute blood-degradation products	23.4%	22.63	associated with acute or subacute clinical symptoms
Intermediate risk (Zabramski III). CMs without acute or subacute blood-degradation products	3.4%	27.88	may be symptomatic (especially seizures)
Lowest risk (Zabramski IV). Punctate lesions (visible ≈ only on gradient echo)	1.3%	37.78	asymptomatic unless they are hemorrhagic ⁴⁸

91.3.6 Evaluation

Guidelines

Practice guideline: Imaging recommendations for cavernous malformations

1. **Level I³¹**: brain MRI for the diagnosis and follow-up of known or suspected CMs
2. **Level I³¹**: MRI should include gradient echo or susceptibility-weighted sequences
3. **Level III³¹**: catheter angiography is not recommended unless an AVM is suspected
4. **Level I³¹**: follow-up imaging should be done to assess new or worsened symptoms to guide treatment decisions. There is insufficient data to make recommendations regarding timing of routine surveillance imaging

CT

Not sensitive: CT misses many small lesions, some large ones, and even some that have bled.

Not specific: CT findings may overlap with low-grade tumors, hemorrhages, granulomas. Isolated multifocal calcifications may suggest the diagnosis. Contrast CT may be helpful when MRI cannot be done.

CT scans are usually more readily available than MRI, and can rule out emergent conditions such as significant hematoma, hydrocephalus, and mass effect.

MRI

Diagnostic test of choice. MRI with either *gradient-echo T2WI* or *susceptibility-weighted (SWI)* images is the most sensitive test due to high sensitivity to susceptibility artifact from blood breakdown products within and around CMs. Findings are similar to AOVm in general (mixed-signal core with low-signal rim—sometimes described as “popcorn” pattern; see above); see ► Fig. 91.1. The diagnosis is strongly suggested by finding multiple lesions with these characteristics and a positive family history.²¹ A venous malformation may be seen adjacent to a solitary CM, but not with multiple CMs.⁵¹ Diffusion tensor imaging/white matter tractography⁵² and pre-op 3D-constructive interference in steady-state (CISS) MRI⁵³ may improve localization, approach, and post-op outcomes.

Gadolinium-contrasted MRI may be helpful for identifying possible associated DVAs or capillary telangiectasia or to exclude tumors in questionable cases.

In the absence of spine symptoms, routine imaging of the spine is not indicated in patients with cerebral CMs.³¹

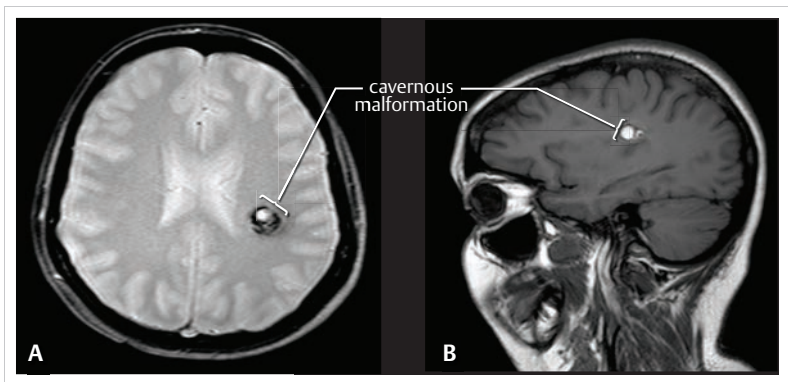


Fig. 91.1 Cavernous malformation of the left hemisphere.
Image: A: T2* GRE axial MRI showing classic “popcorn appearance.”
B: T1 sagittal MRI showing hemosiderin ring around lesion (same patient as A).

Angiography

Does not demonstrate lesion. MRI appearance is nearly pathognomonic, and angiography is not necessary in classically appearing cases. Angiography may be needed to R/O other diagnoses in questionable cases.

Familial considerations

First-degree relatives of patients with more than one family member having a cavernous malformation should have MRI screening and appropriate genetic counseling (consult the guidelines³¹ for details of testing & counseling).

91.3.7 Treatment/management

Overview

Management options:

1. observation
2. surgical excision
3. XRT or stereotactic radiosurgery

Even after many years, there is still conflicting data regarding optimal management of CMs. This has resulted in the somewhat confusing and conflicting treatment guidelines shown in (p. 1530)

Practice guideline: Treatment recommendations for cavernous malformations

1. **Level III³¹:** surgery is not recommended for asymptomatic CMs, especially those that are deep, or in eloquent areas or brainstem, or with multiple CMs
2. **Level II³¹:** consider surgery for: solitary asymptomatic CMs if easily accessible and not in eloquent brain, to prevent future hemorrhage, because of psychological burden, expensive & time-consuming follow-ups, to facilitate lifestyle or career decisions, or in patients who might need to be on anticoagulation
3. **Level II³¹:** consider early CM resection (≤ 6 weeks from hemorrhage) in patients with seizures, especially when medically refractory, if the CM is the likely cause of the seizure
4. **Level II³¹:** consider surgery in symptomatic easily accessible CMs (surgical morbidity & mortality is equivalent to living with the CM for 1–2 years after a first bleed)
5. **Level II³¹:** consider surgery after a second symptomatic bleed in a brainstem CM after reviewing the high risks of early post-op morbidity, mortality, and impact on quality of life
6. **Level II³¹:** consider surgery in deep CMs if symptomatic or after prior hemorrhage (surgical morbidity & mortality is equivalent to living with the CM for 5–10 years after a first bleed)
7. **Level II³¹:** surgical indications are weaker after a single, disabling bleed from a brainstem CM
8. **Level II³¹:** consider radiosurgery in solitary CMs with previous symptomatic hemorrhage if the CM is located in eloquent areas that have an unacceptably high risk with surgery
9. **Level III³¹:** radiosurgery is not indicated for CMs that are asymptomatic, or surgically accessible, or are part of familial CMs because of the concern about precipitating formation of additional CMs

Determining treatment response is difficult since no imaging study can prove elimination of the lesion. Therefore some have suggested that recurrent hemorrhage rate be followed as an endpoint.

No randomized prospective study has been done. No high-quality study has confirmed dramatic benefit or harm from surgical resection, and the few studies that showed some benefit appear to be biased.³¹

Recommendations

Incidental lesions

Asymptomatic, incidentally discovered CMs should be managed expectantly, (especially Zabraski types III & IV) with serial imaging studies for about 2–3 years (to rule out frequent subclinical bleeds); additional studies thereafter based on clinical grounds. Some experts recommend removal for single, easily accessible incidental CMs in non-eloquent brain.⁵⁴ See (p. 1530) for practice guidelines.

✗ Since the radiographic appearance is almost pathognomonic, biopsy or excision solely to verify the diagnosis is rarely appropriate.

Patients presenting with intracerebral or intraventricular hemorrhage

Management should follow evidence-based guidelines (p. 1530) for these entities.

Brainstem CMs

Imaging findings in a brainstem (pontine) cavernous malformation are shown in ► Fig. 91.2.

Surgery is almost never indicated for brainstem CMs that have not bled. Operative management has been suggested for CMs with a history of > 2 prior hemorrhages and “pial/ependymal representa-

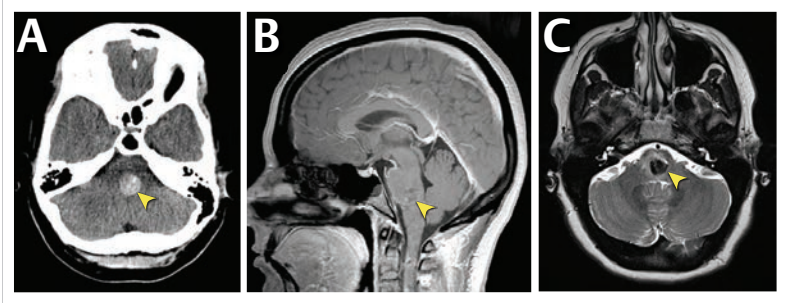


Fig. 91.2 Brainstem cavernous malformation (yellow arrowheads) in a 35-year-old presenting with diplopia and right hand paresthesias.

Image: A: Noncontrast CT scan, B: sagittal T1 MRI with contrast, C: axial T2 MRI.

The admitting CT scan (A) shows acute bleed. The contrast sagittal MRI (B) shows the subtle appearance of the lesion which appears identical on the non-contrast MRI (not shown), i.e., no enhancement. The axial T2 MRI (C) shows the signal dropout from the blood. This lesion and bleed did not present on the surface of the brainstem.

tion” on T1WI MRI.^{55,56} See (p. 1530) for practice guidelines.

Bleeds that do not come to the surface cannot be removed without creating neurologic deficit (worsening of neurologic outcome was 9% vs. 29% in superficial vs. deep brainstem CM resections, respectively⁵⁷). The approach is chosen to expose the site where the bleed comes closest to the surface. CMs are frequently associated with a venous angioma (p. 1512) which must be preserved since it provides critical venous outflow. Outcome was worse with surgery through the floor of the 4th ventricle than with a lateral approach. Significant short-term neurologic deficit is expected in ≈ 50% with brainstem CM resection.⁵⁵

Spinal cord CMs

Managed essentially the same as brainstem CMs.

Cranial nerve CMs

Many case reports and reviews document CMs of cranial nerves (rarely extra-axial) with various presentations.^{58,59,60} Case reports suggest patients may benefit from early surgical decompression from hemorrhagic chiasmal cavernomas since they are at risk for recurrent micro-hemorrhages.⁶¹

Surgery

Indications

Surgical indications for intracranial CMs (see (p. 1530) for practice guidelines):

1. accessible lesions with
 - a) focal deficit
 - b) or symptomatic hemorrhage
 - c) or seizures: seizure control is lower with longer seizure duration⁶²

- less accessible lesions that repeatedly bleed with progressive neurologic deterioration may be considered for excision, even in delicate regions such as the brainstem^{63,64,65} or spinal cord

Surgical risks

The risk of surgery varies with CM location. Overall risk of death or nonfatal stroke after resection of *incidental* CM: 6%.⁶⁶ This risk is greater than the natural history (2.4% over 5 years), but is better than the risk of a recurrent hemorrhage after a first bleed (29.5% over 5 years). The risk for deep CMs (basal ganglia, insula, thalamus) is higher, with 5–18% morbidity and almost 2% mortality.⁶⁷ Brainstem CMs carry even higher risk (morbidity almost 50%) but most improve over time.³¹

Surgical technique

- goal of surgery is complete resection of the lesion. The rate of rehemorrhage is higher with incomplete resection
- stereotactic localization or intraoperative ultrasound are particularly helpful in localizing and increasing the chance for GTR
- choose a surgical corridor along the greatest length of the CM (2-point technique) to minimize injury to adjacent brain
- in comparison to AVMs, a wide exposure is not needed
- initial dissection is directed at separating the lesion from the adjacent brain
- once the dissection is complete, the contents of the CM capsule may be removed piecemeal to minimize the parenchymal opening (especially important in the brainstem)
- with CMs that have bled, one usually encounters a cavity containing the CM and blood degradation products.⁶⁸
- the hemosiderin stained rim is normally not removed, but removal can be considered if seizures are associated with the lesion if it is not located in eloquent brain (controversial). Some improvement in seizure control can occur even with resection limited to the CM alone⁶⁹
- if there is an associated developmental venous anomalies (p. 1512) (DVA) it is not removed as it represents venous drainage of the area, and DVAs often regress following complete removal of the CM
- for deep CMs meeting surgical indications, surgical corridors are discussed in the references for the thalamus⁷⁰ and for the brainstem⁷¹

Brainstem CMs

DTI may be helpful to identify splitting or displacement of critical tracts. The use of retractors is to be avoided; cottonoids and exploitation of hematoma cavity may be used to gain access. Unlike supratentorial CMs, brainstem CMs may be extremely adherent to brain parenchyma⁶⁵. Bipolar cautery: use on low power with constant irrigation to reduce thermal injury. Unlike supratentorial CMs with seizures, the adjacent hemosiderin-stained parenchyma is not removed.

Post-op follow-up

Follow-up MRI ≈ 3 months post-op is recommended. It never looks “normal” but can determine if removal was complete.

Stereotactic radiosurgery (SRS)

See references.^{72,73,74,75,76,77} Controversial: results appear comparable to natural history. Some non-controlled studies have shown a possible reduction in recurrent hemorrhage rate following a 2-year latency period after SRS⁷⁵; however, radiation-induced morbidity was significant.^{78,79} Other series have failed to show reduction.⁸⁰ Findings may reflect the natural history of CMs with temporal clustering of hemorrhagic events with decrease in hemorrhage rates after 28 months.³⁷ SRS may promote development of new CMs in familial cases (controversial). See (p. 1530) for practice guidelines.

91.3.8 Prognosis

When CMs can be completely removed, the risk of further growth or hemorrhage is essentially permanently eliminated⁶⁸ (however, recurrence of symptoms has been reported after partial and even seemingly complete removal^{65,81}).

For CMs treated surgically, patients need to be aware that post-op neurologic worsening is very common, especially with brainstem CMs.⁸² Worsening may be transient,⁸³ but may take months to resolve.

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Part XIX

Stroke and Occlusive Cerebrovascular Disease

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92 Stroke – General Information and Physiology

92.1 Definitions

Acute ischemic stroke (AIS) AKA cerebral infarction. Obsolete term: cerebrovascular accident (CVA).

► **TIA.** (Transient ischemic attack): transient neuronal dysfunction secondary to focal ischemia (of brain, spinal cord, or retina) without (permanent) acute infarction¹ (**note:** obsolete operational definitions used an arbitrary 24-hour cutoff for duration of symptoms).

10–15% of patients with TIA have a stroke within 3 months, 50% of which occur within 48 hours.

► **Stroke.** Permanent (i.e., irreversible) death of neurons caused by inadequate perfusion of a region of the brain or brainstem.

► **Watershed infarct.** Ischemic infarction in a territory located at the periphery of two bordering arterial distributions due to a disturbance in flow in one or both of the arteries.

92.2 Cerebrovascular hemodynamics

92.2.1 Cerebral blood flow (CBF) and oxygen utilization

► Table 92.1 shows typical CBF values and the corresponding neurophysiologic state. CBF < 20 is generally associated with ischemia and if prolonged will produce cell death.² However, this assumes normal metabolic rate and may be more applicable to global cerebral hypoperfusion.³ There is a higher CBF threshold for loss of electrical excitability than that for cell death—this has led to the concept of the ischemic penumbra—nonfunctioning cells that are still viable.²

Table 92.1 Correlates of CBF

CBF (ml per 100 gm tissue/min)	Condition
> 60 (approx)	hyperemia (CBF > tissue demand)
45–60	normal brain at rest
75–80	gray matter
20–30	white matter
< 20: Ischemia	
16–18	EEG becomes flatline
15	physiologic paralysis
12	brainstem auditory evoked response (BAER) changes
10	alterations in cell membrane transport (cell death; stroke)

CBF is related to blood pressure as shown in Eq (92.1).

$$CBF = \frac{CPP}{CVR} = \frac{MAP - ICP}{CVR}$$

(92.1)

where CPP = cerebral perfusion pressure (p. 1036), CVR = cerebrovascular resistance (see below), and MAP = mean arterial pressure.

92.2.2 Cerebrovascular resistance (CVR) and cerebral autoregulation

In the range of CPP ≈ 50–150 mm Hg, CVR (the resistance of the cerebral vascular bed to blood flow) of normal brain tissue varies linearly to maintain an almost constant CBF (► Fig. 92.1). This phenomenon is called (cerebral) autoregulation. It is accomplished by alterations in blood vessel tone via a myogenic mechanism. Cerebral autoregulation can be disrupted in certain pathologic states.

CVR is affected by the PaCO₂ such that there is a linear increase in CBF with increasing PaCO₂ within the range of PaCO₂ = 20–80 mm Hg.

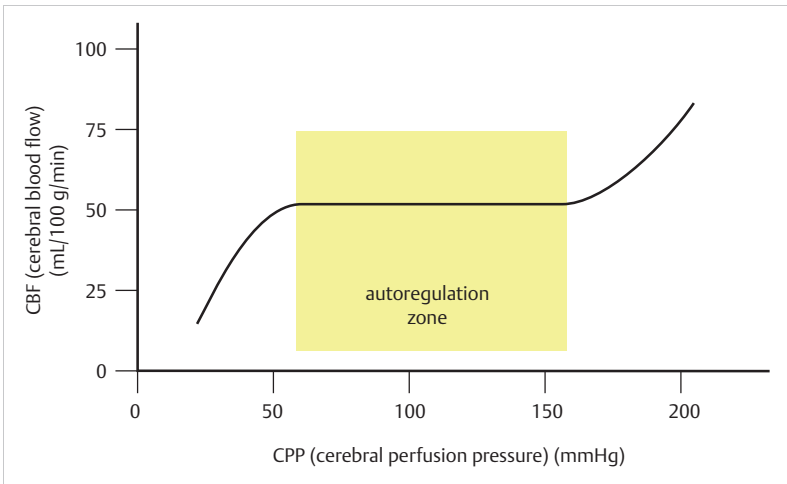


Fig. 92.1 Cerebral autoregulation.

The range of CPP where CBF is relatively constant is indicated by the yellow square.

92.2.3 Cerebral metabolic rate of oxygen consumption (CMRO₂)

CMRO₂ averages 3.0–3.8 ml/100 gm tissue/min. The ratio of CBF to CMRO₂ (the coupling ratio⁴) in the quiescent brain is 14:18. With focal cortical activity, local CBF increases ≈ 30% while CMRO₂ increases ≈ 5%.⁵ CMRO₂ can be manipulated to some degree.

92.2.4 Cerebrovascular reserve and reactivity

May be evaluated with xenon-enhanced CT, CTP (p.239), TCD, SPECT, or MRI.^{6,7,8,9} Response of CBF to a vasodilator challenge with 1000 mg of IV acetazolamide (ACZ) (Diamox®) is classified as^{8,9}:

- Type I: normal baseline CBF with 30–60% increase following ACZ challenge
- Type II: decreased baseline CBF with blunted response of < 10% increase or < 10 ml/100 g/min absolute increase after ACZ challenge
- Type III: decreased baseline CBF with paradoxical decrease of regional CBF following ACZ challenge, suggesting a steal phenomenon in regions with maximally dilated vasculature at baseline

92.3 Collateral circulation

92.3.1 Collateral circulation for ICA stenosis/occlusion

The effects of ICA stenosis/occlusion may be ameliorated by collateral blood flow. Potential alternate routes for blood to reach brain tissue include:

1. flow through the circle of Willis
 - a) from contralateral ICA through anterior communicating a.
 - b) from forward flow through the ipsilateral posterior communicating a.
2. retrograde flow through *ophthalmic a.* parasitizing blood from *both* ECAs via:
 - a) facial a. → angular a. → dorsal nasal a. & medial palpebral a.
 - b) maxillary a.
 - middle meningeal a. → lacrimal a.
 - vidian a. (a. of the pterygoid canal)
 - c) transverse facial a. → lateral palpebral a.
 - d) superficial temporal a. → supraorbital a.
3. proximal maxillary a. → anterior tympanic a. → caroticotympanic branch of ICA
4. cortical-cortical anastomoses
5. dural-leptomeningeal anastomoses

92.3.2 Collateral circulation for vertebrobasilar stenosis/occlusion

Available collaterals depend on the site of occlusion.

Basilar artery occlusion. Collateral flow via:

1. posterior communicating aa.
2. anastomoses between SCA and PICA

Proximal vertebral artery (VA) occlusion. Collateral flow via:

1. ECA → occipital a. → muscular branches of VA → VA
2. thyrocervical trunk → ascending cervical a. → direct connection or spinal radicular aa. → VA
3. contralateral VA and/or ascending cervical a. via spinal radicular branches and anterior spinal artery

92.4 “Occlusion” syndromes

92.4.1 Occlusion of major vessels organized by vascular territories

See ► Fig. 2.1 for the distribution territories of the major cerebral arteries. To indicate lateralization of findings, {CL} = contralateral, {IL} = ipsilateral.

Internal carotid artery and its branches

Risk and extent of stroke is influenced by suddenness of occlusion, location of occlusion, and collateral circulation (see above).

1. statistics:
 - a) acute ICA occlusion (all comers): 26–49% risk of stroke¹⁰ (not all of these strokes are severe)
 - b) annual stroke risk in 1,261 patients with symptomatic ICA occlusion: 7% overall, 5.9% ipsilateral to the occlusion (mean follow-up = 45.5 mos) (even with anticoagulation or antiplatelet drugs) (12 prospective studies¹¹)
 - c) St. Louis Carotid Occlusion Study¹²: 2-year ipsilateral ischemic stroke rate in patients with symptomatic ICA occlusion = 5% in patients with normal O₂ extraction fraction (OEF) by PET scan, and 26% in patients with increased OEF
 - d) stroke risk is less when one includes asymptomatic ICA occlusions (i.e., there are people walking around with ICA occlusion and no symptoms)
 - e) in patients presenting with ICA territory stroke or TIA, complete ICA occlusion is found in 10–15%¹²
2. anterior cerebral artery: {CL} weakness of LE > UE
3. worst-case scenario of total ICA occlusion with no AComA or PComA flow and no collateral rescue: stroke in ACA and MCA territories (► Table 92.2)

Table 92.2 Total ICA occlusion

Deficit ^a	Complete (M1 occlusion)	Superior division	Inferior division
{CL} weakness of UE > LE	X	X	
{CL} weakness of lower face	X	X	
{CL} hemisensory loss (UE & LE)	X	X	
{CL} hemisensory loss of face (all modalities)	X	X	
{CL} neglect ^b	X	X	
{IL} gaze preference	X		
{CL} homonymous hemianopsia	X		X ^c
receptive aphasia ^d (Wernicke's area)	X		X
expressive aphasia ^d (Broca's area)	X	X	

Gerstmann syndrome (p. 99): with dominant parietal lobe infarct

^a{CL} = contralateral, {IL} = ipsilateral. An “X” indicates that the deficit is present

^bwith involvement on side of nondominant hemisphere

^cplus {CL} upper quadrantanopsia

^dwith involvement on the side of the dominant hemisphere

4. posterior cerebral artery
 - a) unilateral occipital lobe infarction → homonymous hemianopsia with macular sparing (visual cortex of the macula receives dual blood supply from MCA and PCA)
 - b) Balint syndrome
 - c) cortical blindness (Anton syndrome)
 - d) Weber syndrome
 - e) alexia without agraphia
 - f) thalamic pain syndrome (Dejerine-Roussy syndrome)
5. recurrent medial striate artery (of Heubner): expressive aphasia + *mild* hemiparesis (UE > LE, proximal muscles weaker than distal) – dense hemiparesis is not typical
6. anterior choroidal artery (AChA) syndrome: first described by Foix et al in 1925. The complete triad consists of {CL} hemiplegia, hemihypesthesia, and homonymous hemianopsia (mnemonic: 3 H's); however, incomplete forms are more common.¹³ Occlusion is usually due to small vessel disease, and CT or MRI usually shows infarct in posterior limb of IC (just above temporal horn of lateral vent)¹⁴ and white matter posterior and lateral to it. Occlusion is usually tolerated fairly well, and ligation of this artery was actually utilized in treatment of Parkinsonism sometimes without ill effect¹⁵ (p 540)—see Surgical treatment of Parkinson's disease (p. 1840)—but internal capsule infarct occurred in ≈ 15%.
7. artery of Percheron (p.82): bilateral thalamic and mesencephalic infarctions¹⁶

Posterior circulation

1. vertebral artery
 - a) medial medullary syndrome (Dejerine syndrome)
 - b) lateral medullary syndrome (Wallenberg syndrome): see below
2. basilar artery
3. AICA: lateral pontine syndrome (Marie-Foix syndrome)
4. PICA: sometimes lateral medullary (Wallenberg) syndrome: see below
5. SCA: infarction of superior cerebellar vermis and superior cerebellum
6. anterior spinal artery

Lateral medullary syndrome (LMS)

AKA Wallenberg's syndrome, AKA PICA syndrome. Classically attributed to PICA occlusion, but in 80–85% of cases the *vertebral artery* is also involved.¹⁷ No cases have been reported arising from brainstem hemorrhage. Onset is usually acute. The findings are listed in ► Table 92.3 (NB: *absence of pyramidal tract findings, and no change in sensorium*). The location of the lesion and medullary structures are shown in ► Fig. 92.2.

Table 92.3 Findings in lateral medullary syndrome¹⁵ (p 547)

GENERALIZED symptoms	Responsible lesion
• vertigo, N/V, nystagmus, diplopia, oscillopsia	vestibular nuclei & connections
• hiccups	?
IPSILATERAL to lesion	Responsible lesion
• facial pain, paresthesias, & impaired sensation	descending tract and nucleus V over half of face
• ataxia of limbs	(restiform body?)
• Horner syndrome	descending sympathetic tract
• dysphagia, diminished gag, hoarseness	exiting fibers of IX & X
• numbness of arm, trunk, or leg	cuneate & gracile nuclei
CONTRALATERAL to lesion	Responsible lesion
• impaired pain & temp sense over half of body	spinothalamic tract

► **Note.** This is essentially the only location where a lesion will produce sensory loss on one side of the face (ipsilateral to the lesion) and contralateral sensory loss in the body. All in the absence of pyramidal tract findings (i.e., overt weakness).

These patients sometimes develop severe cerebellar swelling that responds to neurosurgical decompression (the tissue aspirates easily).

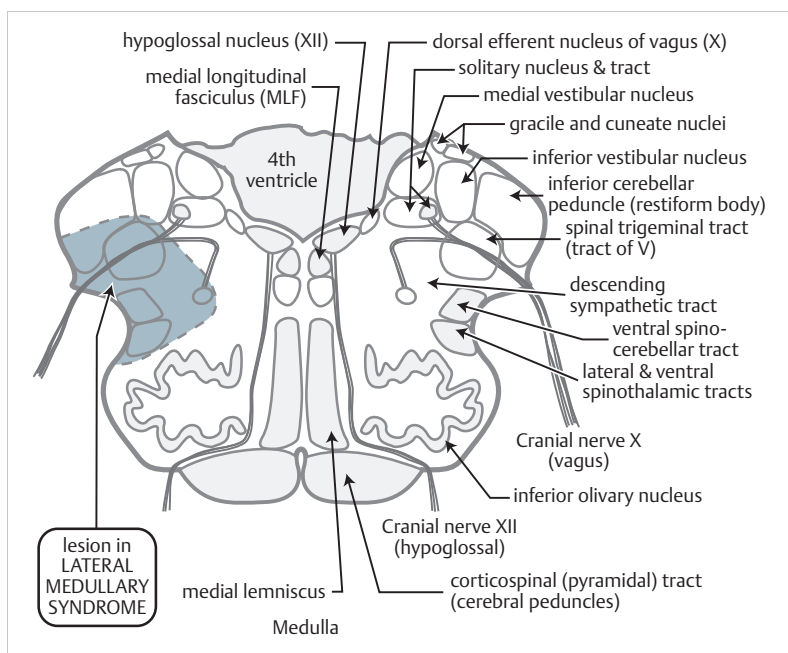


Fig. 92.2 Typical lesion in lateral medullary syndrome. Image: schematic cross-section through medulla (the lesion is indicated as shaded area).

In a patient presenting with LMS, one needs to rule out vertebral dissection (p.1579), since this would be treated with heparin. MRI including fat-suppressed T1WI and MRA would detect dissection in most cases.

Prognosis: 12% of 43 patients died during the acute phase from respiratory and cardiovascular complications and 2 new posterior-fossa strokes occurred.¹⁸ Recurrent vertebrobasilar territory stroke rate was 1.9% per year.¹⁸

92.4.2 Lacunar strokes

General information

Small infarcts in deep noncortical cerebrum or brainstem (► Table 92.4) resulting from occlusion of penetrating branches of cerebral arteries. Size of infarcts ranges from 3–20 mm (CT detects larger ones; better sensitivity in white matter).

Table 92.4 Typical locations for lacunar strokes (in descending frequency)

- putamen
- caudate
- thalamus
- pons
- internal capsule (IC)
- convolutional white matter

Small (3–7 mm) lacunes may be due to lipohyalinosis (vasculopathy due to HTN) of arteries <200 microns (may also be cause of many ICHs); this vasculopathy is indicative of small vessel disease, unlikely to be prevented by carotid endarterectomy.

Clinically, diagnosis is virtually *excluded* by: aphasia, apractagnosia, sensorimotor stroke, monoplegia, homonymous hemianopsia (HH), severe isolated memory impairment, stupor, coma, LOC, or seizures.

L'etat lacunaire: multiple lacunes → chronic progressive neuro decline with one or more episodes of hemiparesis; results in invalidism, dysarthria, small-step gait (*marche à petits pas*), imbalance, incontinence, pseudobulbar signs, dementia. Many signs and symptoms are possibly due to NPH (unrecognized originally).

Lacunar syndromes

Major syndromes (see reference¹⁹ for others):

1. pure sensory stroke or TIA: (the most common lacunar manifestation) usually isolated unilateral numbness of face, arm, and leg. Only 10% of TIA go on to stroke. Lacune in sensory (posteroventral) thalamus → CT detection is poor. Dejerine-Roussy = rare thalamic pain syndrome that may develop late
2. pure motor hemiparesis (PMH): (2nd most common lacunar manifestation) pure unilateral motor deficit of face, arm, and leg without sensory deficit, HH, etc. Lacune in posterior limb of IC, or in lower basis pontis where corticospinal (CS) tracts coalesce, or rarely in mid-cerebral peduncle
3. ataxic hemiparesis: contralateral PMH + cerebellar ataxia of affected limbs (if they can move). Lacune in basis pontis at junction of upper third and lower two-thirds → dysarthria, nystagmus, and unidirectional toppling possible. Differential severity in face, arm, and leg possible because CS fibers are dispersed by nuclei pontis (unlike compact pyramids and peduncle)
 - a) variant: dysarthria-clumsy hand syndrome: lesion in same location or genu of IC. May be mimicked by a cortical infarct, but latter will have numb lips
4. PMH sparing the face: lacune in medullary pyramid; at onset, there may be vertigo and nystagmus (approaching lateral medullary syndrome)
 - a) variant: thalamic dementia: central region of one thalamus + adjacent subthalamus → abulia, memory impairment + partial Horner (miosis + anhydrosis)
5. mesencephalothalamic syndrome: "top o' the basilar syndrome." Usually caused by embolus. Infarct typically butterfly-shaped & bilateral involving rostral brainstem and cerebral hemisphere regions fed by the distal basilar artery. Clinical: III palsy, Parinaud's syndrome (p. 101) & abulia, may have amnesia, hallucinations, and somnolence, usually without significant motor dysfunction
6. Weber's syndrome: Cr. N. III palsy with contralateral PMH (no sensory loss). Usually due to occlusion of interpeduncular branches of basilar artery → central midbrain infarction, disrupting cerebral peduncle and issuing fibers of III. May also be due to aneurysm of basilar bifurcation or BA-SCA junction
7. PMH with crossed VI palsy: lacune in paramedian inferior pons
8. cerebellar ataxia with crossed III palsy (*Claude syndrome*): lacune in dentatorubral tract (superior cerebellar peduncle)
9. hemiballism: classically, infarct or hemorrhage in subthalamic semilunar nucleus of Luys
10. lateral medullary syndrome: see below
11. locked-in syndrome: bilateral PMH from infarct at IC, pons, pyramid, or (rarely) cerebral peduncles

92.5 Stroke in young adults

92.5.1 General information

Only 3% of ischemic strokes occur in patients <40 yrs of age.²⁰ Over 10% of ischemic strokes occur in patients ≤55 yrs.²¹ Incidence: 10 per 100,000 persons age 35–44 yrs,²² 73 per 100,000 for age <55 yrs.²¹

92.5.2 Etiologies

The differential diagnosis is lengthy,²⁰ with *trauma* being the most common cause of strokes (22%) in patients under 45 yrs.²³ Most of the rest are covered by the small number of etiologies listed below (excludes: trauma, post-op stroke, SAH, and intracerebral hemorrhage).

1. atherosclerosis: 20%—less common than in older population (all 18 patients in one series had either ID-DM, or were males > 35 yrs with ≥ 1 risk factor (see below), most had TIAs earlier)
2. embolism with recognized source: 20%

- a) cardiac origin is the most common (above), most have previously known cardiac disease:
 - rheumatic heart disease
 - prosthetic valve
 - endocarditis
 - mitral valve prolapse (MVP): present in 5–10% of young adults, in 20–40% of young adults with stroke (although one series found MVP in only 2% of stroke in young adults²²)
 - a-fib
 - left-atrial myxoma
- b) fat embolism syndrome: neurologic manifestation is usually global neurologic dysfunction; see Fat embolism syndrome (p.1013)
- c) paradoxical embolism: e.g., ASD, pulmonary AVM including Osler-Weber-Rendu syndrome, patent foramen ovale (above)
- d) amniotic fluid embolism: may occur typically in the post-partum period
3. vasculopathy: 10%
 - a) inflammatory
 - Takayasu's
 - infective: TB, syphilis, ophthalmic zoster
 - amphetamine abuse
 - herpes zoster ophthalmicus (HZO): usually presents with delayed contralateral hemiplegia with a mean of $\approx$ 8 weeks following HZO²⁴
 - mucormycosis: a nasal and orbital fungal infection primarily in diabetics and immunocompromised patients that causes an arteritis which may thrombose the orbital veins and ICA or ACA. Produces proptosis, ocular palsy, and hemiplegia
 - associated with systemic disease such as: SLE (lupus) (also see below under Coagulopathy); arteritis (especially periarteritis nodosa (p.208)) when confined to CNS is usually multifocal and progressive, but may mimic stroke early; multiple sclerosis (MS); cancer; rheumatoid arthritis
 - b) non-inflammatory
 - fibromuscular dysplasia (p.209)
 - carotid or vertebral artery dissections (including posttraumatic)
 - moyamoya disease (p.1601)
 - homocystinuria: a genetic defect in methionine metabolism that produces intimal thickening and fibrosis in almost all vessels with associated thromboembolic events (arterial and venous, including dural venous sinuses). Estimated risk of stroke is 10–16%. Patients have a Marfan syndrome-like physical appearance, malar blotches, mental retardation, and elevated levels of urinary homocysteine
 - pseudoxanthoma elasticum
4. coagulopathy: 10%. The following are associated with hypercoagulable states
 - a) SLE: lupus anticoagulant $\rightarrow$ prolonged PTT incompletely corrects with 50/50 mix. Collagen vascular disease only rarely presents initially with stroke
 - b) polycythemia or thrombocytosis
 - c) sickle cell disease
 - d) TTP (thrombotic thrombocytopenic purpura)
 - e) antithrombin III deficiency (controversial—not seen in large series of young adults with stroke)
 - f) protein C or protein S deficiency (familial): protein C attenuates hemostatic reactions, homozygous deficiency is fatal in the neonatal period. Heterozygous deficiency is associated with *thrombotic* strokes. A rare complication during initial therapy with warfarin is a drop in protein C before other coagulation factors resulting in a hypercoagulable state
 - g) antiphospholipid-antibody syndrome (APLAS)^{25,26}: causes venous and/or arterial thrombosis. The two best known antiphospholipid-antibodies are anticardiolipin antibodies (ACLA), and lupus anticoagulant (LAC). Once they become symptomatic, treatment is high-intensity warfarin therapy to an INR $\geq$ 3.²⁷ There is a dramatic increase in thrombotic events after discontinuing warfarin. Aspirin is useless
 - h) following use of the street drug 3,4-methylenedioxymethamphetamine (MDMA, commonly known as ecstasy),²⁸ possibly independent of the hypercoagulable state that occurs with hyperthermia when insufficient fluids are consumed in conjunction with use of the drug
5. peripartum: 5% (usually within 2 wks of parturition)
6. miscellaneous causes: 35%
 - a) uncertain etiology

- b) oral contraceptives (BCP): associated with ninefold increased risk for stroke, many with prior migraine history
- c) cerebral venous thrombosis (CVT) (p. 1594) including dural sinus thrombosis: incidence may be increased with use of BCP
- d) migraine²⁹: widely accepted, but difficult to assess objectively (incidence of stroke in these patients may be same as general population). *Rare*. Usually occurs in women, with a benign long-term course; recurs in < 3%. Possible mechanisms include: vasospasm, platelet dysfunction, and arteriopathy.³⁰ Strokes often occur during a migrainous attack³¹ or shortly thereafter
- e) cocaine abuse³²: stroke may result from vasoconstriction, or from HTN in the presence of aneurysms or AVMs (frank vasculitis occurs³³ but is rare with cocaine, unlike amphetamines); strokes with alkaloidal cocaine (“crack”) are = equally divided between ischemic and hemorrhagic
- f) posterior reversible encephalopathy syndrome (PRES) (p. 202)

92.5.3 Risk factors

In a retrospective “neighborhood control” study of 201 Australian patients aged 15–55 (mean = 45.5) with first-time strokes, the following risk factors were identified²¹:

1. diabetes: odds ratio = 12
2. HTN: odds ratio = 6.8
3. current cigarette smoking: odds ratio = 2.5
4. long-term heavy alcohol consumption: odds ratio = 15 (heavy alcohol ingestion within 24 hrs preceding the stroke was *not* a risk factor)

92.5.4 Evaluation

1. history & physical exam directed at uncovering systemic disease (see above) and modifiable risk factors (see above)
2. cardiology work-up including EKG and echocardiogram
3. bloodwork (include as appropriate):
 - a) routine: electrolytes, CBC, platelet count and/or function, ESR (elevation may suggest SLE, arteritis, atrial myxoma... but a normal ESR does not rule out vasculitis), PT/PTT, VDRL (should be obtained in all young adults with stroke), fasting lipid profile
 - b) for unexplained stroke: ANA, antithrombin III, protein C, protein S, homocysteine, factor V Leiden, PPD, sickle-cell screen, toxicology screen (blood and urine, to R/O drugs such as cocaine), SPEP, lupus anticoagulant, serum amino acid, tissue plasminogen-activator and -inhibitor
4. miscellaneous tests: U/A, CXR, CSF exam when indicated
5. cerebral angiography: not always necessary for patients with obvious systemic disease or strong evidence for cardiac embolism; may occasionally diagnose cerebral embolism if performed within 48 hrs of ictus

92.6 Atherosclerotic carotid artery disease

92.6.1 General information

Atherosclerotic plaques begin to form in the carotid artery at 20 yrs of age. In the extracranial cerebral circulation, plaques typically start on the back wall of the common carotid artery (CCA). As they enlarge, they encroach on the lumen of the ICA. Calcified hard plaques may not change with time. The risk of stroke correlates with the degree of stenosis and with certain types of plaque morphology, and is also increased in hypercoagulable states and with increased blood viscosity.

Plaque morphology

“Vulnerable” plaques are atherosclerotic plaques likely to cause thrombotic complications, or those that tend to progress rapidly. Criteria for vulnerable plaques include: intimal thickening, plaque fissure, lipid/necrotic core with thin fibrous cap, calcification, thrombus, intraplaque hemorrhage, and outward remodeling. Some of these features can be identified with high-resolution MRI.^{34,35,36,37}

92.6.2 Presentation

General information

Carotid artery lesions are considered symptomatic if there are one or more lateralizing ischemic episodes appropriate to the distribution of the lesion. A lesion is considered to be *asymptomatic* if the patient only has non-specific visual complaints, dizziness, or syncope not associated with TIA or stroke.³⁸ The majority (80%) of carotid atherothrombotic strokes occur without warning symptoms.³⁹

Asymptomatic carotid stenosis

Usually discovered as a carotid bruit. Asymptomatic bruit: prevalence increases with age (2.3% in ages 45–54 yrs, 8.2% at ≥ 75).⁴⁰ Accuracy of a bruit in predicting ICA stenosis: 50–83% (depending on cohort, criteria for stenosis...). Sensitivity is as low as 24%.⁴¹

Symptomatic carotid disease

May present as a TIA, RIND, or stroke with any of the following findings; see also ICA occlusion syndromes (p.1538):

1. retinal insufficiency or infarction (central retinal artery is a branch of the ophthalmic artery): ipsilateral monocular blindness
 - a) may be temporary: amaurosis fugax, AKA transient monocular blindness (TMB). Four types:
 - Type I: embolic. Described “like a black curtain coming down” in *one eye*. Complete loss of vision, usually lasts 1–2 minutes
 - Type II: flow-related. Retinal hypoperfusion → desaturation of color, usually described as a graying of vision
 - Type III: vasospastic. May occur with migraines
 - Type IV: miscellaneous. May occur with anticardiolipin antibodies
 - b) blindness may be permanent
2. middle cerebral artery symptoms:
 - a) contralateral motor or sensory TIA (arm and face worse than leg) with hyperreflexia and upgoing toe
 - b) language deficits if dominant hemisphere involved

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92.6.3 Evaluation of the extent of carotid disease

Overview

Symptomatic patients will usually be assessed as part of a stroke/TIA protocol.

Check CBC with platelet count, fibrinogen, PT/PTT/INR (to R/O hypercoagulable state).

Funduscopic exam may show Hollenhorst plaques (cholesterol crystal emboli) in the retina.

Classification of patients based on the hemodynamics and also the embolic propensity of carotid lesions has thus far been too complex to be utilized in large studies. The tests described below place a great deal of emphasis on the greatest degree of stenosis, which is probably an oversimplification. Plaque composition and morphology is probably important.

Recommendations for screening for carotid stenosis

1. the U.S. Preventive Services Task Force (USPSTF) currently recommends against screening for carotid stenosis in the adult general population (grade D recommendation: moderate or high certainty that the service has no net benefit or that the harm outweighs the benefit)⁴²
2. the AHA Primary Prevention of Stroke Guidelines does not recommend screening for asymptomatic carotid stenosis⁴³
3. the American Society of Neuroimaging advised that screening should be considered only for age ≥ 65 years with 3 or more cardiovascular risk factors⁴⁴
4. the Society of Vascular Surgery recommends ultrasonography screening for age ≥ 55 years with cardiovascular risk factors, such as HTN, diabetes, smoking, hypercholesterolemia, or known cardiovascular disease⁴⁵

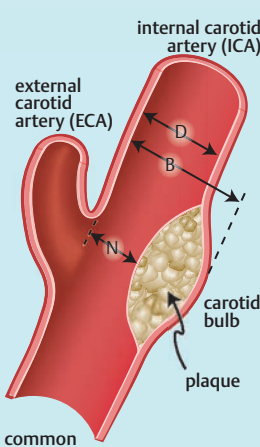
Assessment options

See also recommendations for which tests to use (p.1547).

Angiography

The “gold standard” test is a catheter arteriogram. It cannot be justified as a *screening* test because it is invasive, and too costly and risky (recent data show < 1% risk of transient or permanent deficit (risk is 2–3 times higher in symptomatic patients than in asymptomatic)^{46,47,48} in good hands). Also, unlike duplex Doppler and MRA, it does not provide any information about the thickness of the plaque. Different definitions of the degree of stenosis are employed; ► Table 92.5 compares the definitions used by the NASCET study⁴⁹ to those of the ECST.⁵⁰ For both, N is the linear diameter of the carotid artery at the site of greatest narrowing. The studies differ in the denominator, NASCET uses D (the diameter of the normal artery *distal* to the carotid bulb, taken at the first point at which the arterial walls become parallel), whereas the ECST uses B (the *estimated* carotid bulb diameter).

Table 92.5 Comparison of NASCET and ECST measurements of ICA stenosis^a

	NASCET	ECST
	$1 - \frac{N}{D}$	$1 - \frac{N}{B}$
Approximate equivalent degrees of ICA stenosis based on direct comparison (%)		
30 ^b		65 ^b
40 ^b		70
50		75
60		80
70		85
80		91
90		97

^aadapted from Donnan GA, Davis SM, Chambers BR, et al. Surgery for the prevention of stroke. Lancet 351: 1372, 1998, with permission

^bdegrees of stenosis for which surgery was NOT of clear benefit for symptomatic stenosis (p. 1565)

For example, using the NASCET definition, the degree of stenosis is shown in Eq (92.2).

$$\% \text{ stenosis (NASCET)} = \left(1 - \frac{N}{D}\right) \times 100 \quad (92.2)$$

The relationship between the degree of narrowing based on the NASCET definition vs. that of the ECST has also been estimated by equation⁵¹ as shown in Eq (92.3).

$$\% \text{ stenosis (by ECST)} = 0.6 \times \% \text{ stenosis (by NASCET)} + 40\% \quad (92.3)$$

Angiography also affords the opportunity to perform endovascular intervention if indicated.

Duplex Doppler ultrasound

B-mode image evaluates the artery in cross-sectional plane, and spectrum analysis shows blood flow. Performs poorly with a “string sign.” Cannot scan above the angle of the mandible. Lower frequencies give greater depth of penetration, but signal definition is sacrificed (used in transcranial Doppler). Sensitivity: 88%, specificity: 76%.⁵²

Magnetic resonance angiography (MRA)

May obviate the need for angiography in some cases of carotid stenosis, specifically in symptomatic patients with a focal “gap” of signal intensity loss with distal reappearance of signal.^{53,54} Sometimes overestimates the degree of stenosis.⁵⁵ Sensitivity: 91%, specificity: 88% for extracranial carotid disease.⁵⁶ 2D TOF-MRA is adequate (contrast-MRA shows more, but is not necessary for surgical lesions⁵⁷).

Can be performed at the time as MRI with stroke protocol in TIA/stroke patients, and also detects thrombus or dissection. As with Doppler, has difficulties distinguishing very severe stenosis from occlusion. Less operator-dependent than Doppler, but is more expensive and time-consuming. MRA is more difficult to perform if the patient is critically ill, unable to lie supine, or has claustrophobia, a pacemaker, or ferromagnetic implants. High-resolution MRI may also detect vulnerable plaques (p.1543).

Computed tomography angiography (CTA)

CTA involves ionizing radiation (X-rays) and IV iodinated contrast, limiting its use in patients with dye allergies and renal dysfunction. Results are comparable to MRA and Doppler. CTA can be performed within a few seconds and yields high-resolution images of all vessels from the aortic arch through the intracranial/extracranial vessels as well as the surrounding soft tissues (► Fig. 92.3). In a meta-analysis, sensitivity and specificity for detection of a 70% to 99% stenosis were 85% and 93%, respectively.⁵⁸ CTA is still evolving and may help detect vulnerable plaques (p.1543). Another potential advantage: ability to obtain CT-perfusion (p.239) studies at the same time.

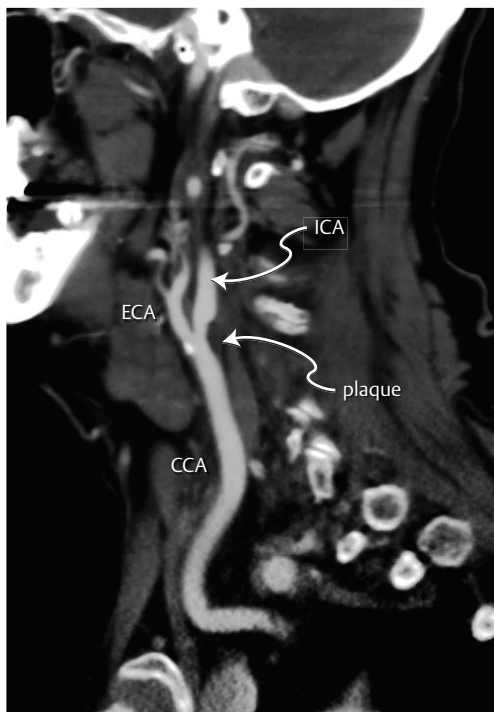


Fig. 92.3 CTA demonstrating soft ICA plaque.

Sagittal CTA through left carotid artery demonstrating soft plaque producing approximately 60% ICA stenosis.

Abbreviations: ICA = internal carotid artery; CCA = common carotid artery; ECA = external carotid artery.

Choice of imaging test/management decisions

Despite a great deal of research on the subject, there are no data to support a particular testing algorithm.¹ Doppler, CTA, or MRA are acceptable initial screening tests. In patients with an abnormal screening test, a common strategy is to obtain a second confirmatory noninvasive test to evaluate the carotid bifurcation before intervention. The combination of carotid ultrasound and MRA has proved cost-effective with good interobserver reliability.⁵⁹ If 2 noninvasive tests are discordant, catheter angiography should be considered before intervention.

92.6.4 Treatment

Treatment alternatives are primarily between the following.

1. “best medical management”: see below
2. carotid endarterectomy (p. 1565)
3. endovascular techniques: combined angioplasty and stenting ($\pm$ distal embolus protection)

Medical treatment

General information

What constitutes “best medical management” has not been precisely determined, and recommendations are constantly changing. Some or all of the following are utilized:

1. antiplatelet therapy (p. 169):
 - a) usually aspirin (ASA): cyclooxygenase inhibitor (see below)
 - b) P2Y₁₂ receptor blockers: clopidogrel, ticlopidine, ticagrelor, prasugrel... either alone or in combination with ASA (see below)
 - c) combination of extended-release dipyridamole and ASA (Aggrenox®) (no benefit from dipyridamole (Persantine®) alone)
2. antihypertensive therapy as appropriate
3. good control of diabetes if present
4. patients with asymptomatic a-fib should be treated with anticoagulation; see Cardiogenic brain embolism (p. 1590)
5. antilipid therapy if needed—statins
6. intervention to help patients to quit smoking

Antiplatelet therapy

Drug info: Aspirin

Irreversibly inhibits cyclooxygenase, preventing synthesis of vascular prostacyclin (a vasodilator and platelet inhibitor) and platelet thromboxane A₂ (a vasoconstrictor and platelet activator). Platelets, lacking cellular organelles, cannot resynthesize cyclooxygenase, whereas the vascular tissues do so rapidly.⁶⁰ NB: < 1000 mg ASA per day probably does not help with high-grade stenosis where there is perfusion failure or flow failure. Some (but not all) studies show less effectiveness in women,⁶¹ and no large study has shown that ASA prevents a second stroke in patients that have already had one.

R: For angina, a bolus dose of 160–325 mg PO is followed by maintenance doses of 80–160 mg/d (lower doses appear to be as effective as higher doses).⁶² Optimal dose for cerebrovascular ischemia continues to be debated. 325 mg PO q d reduces risk of stroke following TIA by 25–30%. Daily doses of 81 or 325 mg when compared to higher doses were associated with a lower rate of stroke, MI, and death (6.2% vs. 8.4%) following carotid endarterectomy.⁶³

Drug info: Aspirin/ER-Dipyridamole (Aggrenox®)

Combination of extended-release dipyridamole and ASA (Aggrenox) is more effective than ASA alone for prevention of TIA, stroke, and myocardial infarction.^{64,65,66} Aggrenox was not superior to clopidogrel, with increased hemorrhage with Aggrenox.⁶⁷ **Side effects:** H/A with initial therapy.

R: 1 capsule PO BID. **Supplied:** fixed-dose capsules of aspirin 25 mg/extended-release with dipyridamole 200 mg.

Drug info: Clopidogrel (Plavix®)

A thienopyridine. Incidence of severe neutropenia (0.04%) is close to that of ASA ($\approx 0.02\%$).⁶⁸ Interferes with platelet membrane function by inhibiting ADP-induced platelet fibrinogen binding and release of platelet granule contents, as well as subsequent platelet-platelet interactions. Produces a time- and dose-dependent irreversible inhibition of platelet aggregation and prolongation of bleeding time. May replace ASA if intolerance or resistance. Used in combination with ASA for some endovascular procedures. Although clopidogrel plus aspirin is recommended over aspirin for acute coronary syndromes, the MATCH⁶⁹ results do not suggest a similar benefit for stroke and TIA. Combination therapy significantly increased risk of hemorrhage.⁶⁹

Pharmacokinetics: Dosed once daily. Requires several days to reach maximal effect ($\therefore$ a loading dose may be used, e.g., after an *acute* event such as an MI, or before stenting). Takes ≈ 5 days off the drug for platelet inhibition to reverse.

R: 75 mg PO q d. Loading dose: 225 mg (3 pills) the first day of therapy. **Supplied:** 75 mg film-coated tablet.

Choice of antiplatelet agents

Individualization is recommended for antiplatelet agents for secondary stroke prevention. ASA is effective, and its low cost may help compliance. A small reduction of vascular events with Aggrenox may justify its expense from a broader healthcare perspective. Clopidogrel is appropriate for those intolerant or resistant to ASA. Clopidogrel plus ASA may be indicated in patients with recent cardiac ischemia or vascular stenting.⁷⁰

Asymptomatic carotid artery stenosis

Key concepts

- natural history: reveals low stroke rate (2%/yr), half of which are not disabling
- large randomized trials have revealed moderate surgical benefit versus medical management for: asymptomatic stenosis $> 60\%$
- treatment selection criteria depend on patient's age, gender, and comorbidities (and therefore life expectancy), and on perioperative complication rate

Practice guideline: Asymptomatic carotid stenosis

- Level I⁷¹: CEA is reasonable in asymptomatic patients with $> 70\%$ ICA stenosis if risk of perioperative stroke, MI, and death is low
- Level II⁷¹: It is reasonable to choose CEA over CAS when revascularization is indicated in older patients especially when the anatomy is unfavorable for endovascular intervention
- Level II⁷¹: It is reasonable to choose CAS over CEA when revascularization is indicated in patients with anatomy unfavorable for surgery
- Level II⁷¹: prophylactic CAS may be considered in highly selected patients with asymptomatic ICA stenosis ($\geq 60\%$ by angiography, $> 70\%$ by validated Doppler ultrasound), but the effectiveness compared to medical therapy alone is not well-established
- Level II⁷¹: In patients with high risk of complications by either CEA or CAS (includes: age > 80 years, NYHA heart failure class II or IV, LVEF $< 30\%$, class III or IV angina pectoris, left main or multivessel CAD, need for cardiac surgery within 30 days, MI within 4 weeks, and severe chronic lung disease), the effectiveness of revascularization over medical therapy alone is not well-established

Abbreviations: CEA = carotid endarterectomy; CAS = carotid artery stenting.

Natural history

Prevalence of carotid stenosis > 50% in men and women > 65 years of age is 5–10%, with 1% having stenosis > 80%.^{72,73,74}

Natural history studies reflect an annual stroke risk of 1–3.4% with asymptomatic carotid artery stenosis of 50–99% at 2–3 years.^{75,76,77,78,79,80} A cohort study found similar cumulative rates of ipsilateral stroke over 10 years (9.3%, or 0.9%/year) and 15 years (16.6%, or 1.1%/year).⁸¹

Attempts to identify subgroups of patients with asymptomatic carotid stenosis at elevated stroke risk suggest that the rate of unheralded stroke ipsilateral to a hemodynamically significant extracranial carotid artery stenosis is 1–2% annually, with some data suggesting that the stroke rate may be higher with progressing stenosis or with more severe stenosis. Asymptomatic carotid stenosis is an important marker of concomitant ischemic cardiac disease.^{75,76,77,80,81} In the REACH Study,⁸² patients with asymptomatic carotid stenosis (n = 3164) had statistically significantly higher age- and sex-adjusted 1-year rates of transient ischemic attack, non-fatal stroke, fatal stroke, and cardiovascular death compared to patients without asymptomatic carotid stenosis (n = 30 329).

Surgery vs. medical management: the studies

ACST

See reference.⁸³

Σ

The ACST⁸³ revealed a moderate benefit for immediate CEA vs. medical management in patients age < 75 with asymptomatic stenosis ≥ 60%.

Details: 3,120 patients with ≥ 60% stenosis by duplex ultrasound were randomized to immediate CEA (50% had CEA within 1 month, 88% within 1 year) or medical therapy at the discretion of the treating physician. Mean follow-up: 3.4 years. Exclusion criteria included: poor surgical risk, prior ipsilateral CEA, and probable cardiac emboli. Surgeons were required to have a perioperative morbidity and mortality rate of < 6%.

Net five-year risk for all stroke or perioperative stroke or death: 6.4% in the CEA group, vs. 11.8% in the medical group (p < 0.0001). Fatal or disabling stroke: 3.5 vs. 6.1%. Fatal stroke alone: 2.1 vs. 4.2%. Although men and women benefited, men benefited more. CEA did not demonstrate a statistically significant benefit for patients over the age of 75. Statistical benefit was not seen in the immediate CEA group until nearly two years after surgery, despite a relatively low perioperative morbidity and mortality rate of 3.1% (in contrast to patients with symptomatic stenosis (NASCET⁸⁴) where benefit was seen much earlier).

ACAS

See reference.⁸⁵

Σ

Large trial that randomized patients in good health with asymptomatic stenosis (calculated in the same manner as the NASCET study) ≥ 60% to CEA plus aspirin, or aspirin alone⁸⁵ found a reduced 5-year risk of ipsilateral stroke if CEA was performed with < 3% perioperative morbidity and mortality and is added to aggressive management of modifiable risk factors.

Details: CEA reduced 5-year stroke risk by 66% in males, 17% in females (not statistically significant), and 53% overall (males & females lumped together). CEA did *not* significantly protect against major stroke or death (P = 0.16) (half of the strokes were not disabling), and was somewhat protective against any stroke or death (P = 0.08). The study group was 95% Caucasian, and 66% were male. Excluded patients (age > 79 yrs, unstable CAD, uncontrolled HTN) may have been higher risk. Surgeons were carefully selected and the surgical morbidity (1.5%) and mortality (0.1%) was very low.

Surprisingly, $\approx$ half of the total morbidity (1.2%) was related to angiography. The implication is that for a generally healthy white male with ACAS > 60%, management with CEA (when performed by a surgeon with a low complication rate, as described) reduces his annual risk of all strokes from 0.5% to 0.17% (the reduction of risk for severe stroke is less). The benefit from CEA is realized within less than one year after the CEA. This is in contrast to the ACST trial (see above) and is most likely due to the lower perioperative event rate. The risk from mortality from other causes (including MI) is $\approx$ 3.9% per year. Combined stroke and death rates in community hospitals,⁸⁶ while improved over the last 20 yrs, remain higher at $\approx$ 6.3% than at centers used in this study.

Veteran's Administration Cooperative Study (VACS)

See reference.⁸⁴

CEA reduces ipsilateral neurologic events, but did not reduce the rate of ipsilateral strokes nor death (most deaths were secondary to MI). This trial did not include women and was not powered to detect differences in outcome subgroups.

CASANOVA Study

See reference.⁸⁷

No difference in outcome between CEA vs. aspirin (new stroke or death), but an unusual protocol lessened its statistical validity.⁸⁸

Mayo Clinic Asymptomatic Carotid Endarterectomy (MACE) Study

See reference.⁸⁹

There were no major strokes or deaths in either the medical or the endarterectomy group. Surgically treated patients were not given aspirin, and 26% had an MI compared to 9% in the aspirin-treated medical arm, reflecting the high incidence of concomitant CAD in patients with an asymptomatic carotid artery stenosis.

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93 Evaluation and Treatment for Acute Ischemic Stroke

93.1 Stroke management – general information (time = brain)

In the complete absence of blood flow, neuronal death occurs within 2–3 minutes from exhaustion of energy stores. However, in most strokes, there is a salvageable penumbra (tissue at risk) that retains viability for a period of time through suboptimal perfusion from collaterals. Local cerebral edema from the stroke results in compromise of these collaterals and progression of ischemic penumbra to infarction if flow is not restored and maintained. Prevention of this secondary neuronal injury drives the treatment of stroke and has led to the creation of designated Primary Stroke Centers that offer appropriate and timely triage and treatment of all potential stroke patients.

This section incorporates recommendations from the 2018 AHA Guidelines for Acute Ischemic Stroke.¹ “Class of Recommendation”: Level I = strong, Level II = moderate to weak.

Key concepts – move quickly (treatment benefits are time-dependent)

Level I.¹

- history/physical exam: include a stroke scale (preferably NIHSS (p. 1554))
- ✓ blood glucose (p. 1557): essential lab to obtain in case IV tPA is indicated
- noncontrast brain CT: the usual initial diagnostic tool of choice (image in ≤ 20 mins)
 - to rule out: hemorrhage (SAH, ICH, EDH, SDH), mass (tumor, abscess...)
 - to calculate ASPECTS (p. 1559) (to identify candidates for thrombectomy)
- CTA for patients with NIHSS score ≥ 10 (correlates with large vessel occlusion [LVO]) to identify candidates for thrombectomy (do not delay IV tPA to get CTA)
- thrombectomy is the standard of care for eligible patients: cerebral ischemia (including infarct) caused by LVO of the ICA or M1 segment of the MCA, 1) when it can be initiated within 6 hours of symptom onset, or 2) if perfusion studies identify viable tissue 6–24 hours from onset
- IV tPA (tissue plasminogen activator, alteplase)
 - within 4.5 hours of onset when thrombectomy not being done *immediately* or for patients who are not thrombectomy candidates
 - goal: “door-to-needle” (DTN) time ≤ 60 minutes

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93.2 Rapid initial evaluation/management

Upon presentation of a patient with symptoms of a potential stroke.

1. **history & physical exam:** key components
 - a) onset or last known well (LKW) time (the last time the patient was seen to be normal): stroke on awakening (“wake-up stroke”) may require additional considerations for management
 - b) current deficit and clinical presentation
 - c) ★ NIH Stroke Scale score (p. 1554) (or Canadian Neurological Scale) assessed and recorded (Level I¹)
2. **laboratories:**
 - a) ★ **blood glucose** (p. 1557) is the only essential lab to get immediately since it affects eligibility for IV tPA (Level I¹)
 - b) see Admitting orders (p. 1556) for subsequent detailed labs (including cardiac troponins...)
3. **imaging:**
 - a) ★ **STAT noncontrast head CT scan:** AHA goal: image the brain in ≤ 20 minutes of arrival in the E/R in ≥ 50% of eligible patients (Level I¹). In most cases this provides the necessary information for management (Level I¹). For details and findings, see section 93.5.1
 - rule out hemorrhage (ICH, SAH, subdural, epidural...) or other lesions (e.g., tumor).
 - determine Alberta stroke program early CT score (ASPECTS) (p. 1559)
 - b) a noninvasive intracranial vascular study (usually a CTA) is obtained in potential candidates for endovascular therapy (EVT) (viz. patients with large vessel occlusion (LVO)) who are best

identified by the NIHSS score²:

- NIHSS score ≥ 10: is 73% sensitive & 74% specific for LVO
- NIHSS score ≥ 6: is 87% sensitive and 52% specific for LVO

This study should not delay IV tPA if indicated; the CTA can be obtained ASAP after IV tPA.

It is reasonable to image the extracranial carotid and vertebral circulations in addition to intracranial vessels in potential candidates for EVT to help determine patient eligibility and to plan the procedure (Level II¹).

4. **intervention:** depending on results of above
- a) candidates for EVT (essentially mechanical thrombectomy or IA tPA) should be taken immediately to angio suite
 - b) if there is going to be a delay, or if the patient is not eligible for EVT, IV tPA is given if indicated

93.3 NIH stroke scale (NIHSS)

Administer in order shown (► Table 93.1). Record initial performance only (do not go back).

Table 93.1 NIH Stroke scale^a

Scale	Finding
1a. Level of consciousness (LOC)	
0	alert; keenly responsive
1	not alert, but arousable by minor stimulation to obey, answer, or respond
2	not alert, requires repeated stimulation to attend, or is obtunded and requires strong painful stimulation to make movements (not stereotyped)
3	comatose: responds only with reflex motor (posturing) or autonomic effects, or totally unresponsive, flaccid, and areflexic
1b. Level-of-consciousness questions	
Patient is asked the month and their age.	
0	answers both questions correctly; must be correct (no credit for being close)
1	answers one question correctly, or cannot answer because of: ET tube, orotracheal trauma, severe dysarthria, language barrier, or any other problem not secondary to aphasia
2	answers neither question correctly, or is: aphasic, stuporous, or does not comprehend the questions
1c. Level-of-consciousness commands	
Patient is asked to open and close the eyes, and then to grip and release the non-paretic hand. Substitute another 1-step command if both hands cannot be used. Credit is given for an unequivocal attempt even if it cannot be completed due to weakness. If there is no response to commands, demonstrate (pantomime) the task. Record only first attempt.	
0	performs both tasks correctly
1	performs one task correctly
2	performs neither task correctly
2. Best gaze	
Test only horizontal eye movement. Use motion to attract attention of aphasic patients.	
0	normal horizontal movements
1	partial gaze palsy (gaze abnormal in one or both eyes, but forced deviation or total gaze paresis are not present) or patient has an isolated cranial nerve III, IV, or VI paresis
2	forced deviation or total gaze paresis not overcome by oculoccephalic (Doll's eyes) maneuver (do not do caloric testing)
3. Visual	
Visual fields (upper and lower quadrants) are tested by confrontation. May be scored as normal if patient looks at side of finger movement. Use ocular threat where consciousness or comprehension limits testing. Then test with double-sided simultaneous stimulation (DSSS).	
0	no visual field deficit
1	partial hemianopia (clear-cut asymmetry), or extinction to DSSS
2	complete hemianopia
3	bilateral hemianopia (blind, including cortical blindness)

(continued)

Table 93.1 continued

Scale	Finding
4. Facial palsy	
Ask patient (or pantomime) to show their teeth, or raise eyebrows and close eyes. Use painful stimulus and grade grimace response in poorly responsive or non-comprehending patients.	
0	normal symmetrical movement
1	minor paralysis (flattened nasolabial fold, asymmetry on smiling)
2	partial paralysis (total or near-total paralysis of lower face)
3	complete paralysis of one or both sides (absent facial movement in upper and lower face)
5. Motor Arm (5a = left, 5b = right)	
Instruct patient to hold the arms outstretched, palms down (at 90° if sitting, or 45° if supine). If consciousness or comprehension impaired, cue patient by actively lifting arms into position while verbally instructing patient to maintain position.	
0	no drift (holds arm at 90° or 45° for full 10 seconds)
1	drift (holds limbs at 90° or 45° position, but drifts before full 10 seconds but does not hit bed or other support)
2	some effort against gravity (cannot get to or hold initial position, drifts down to bed)
3	no effort against gravity, limb falls
4	no movement
9	amputation or joint fusion: explain
6. Motor leg (6a = left, 6b = right)	
While supine, instruct patient to maintain the non-paretic leg at 30°. If consciousness or comprehension impaired, cue patient by actively lifting leg into position while verbally instructing patient to maintain position. Then repeat in paretic leg.	
0	no drift (holds leg at 30° full 5 seconds)
1	drift (leg falls before 5 seconds, but does not hit bed)
2	some effort against gravity (leg falls to bed by 5 seconds)
3	no effort against gravity (leg falls to bed immediately)
4	no movement
9	amputation or joint fusion: explain
7. Limb ataxia	
Looking for unilateral cerebellar lesion. Finger-nose-finger and heel-knee-shin tests are performed on both sides. Ataxia is scored only if clearly out of proportion to weakness. Ataxia is absent in the patient who cannot comprehend or is paralyzed.	
0	absent
1	present in one limb
2	present in two limbs
9	amputation or joint fusion: explain
8. Sensory	
Test with pin. When consciousness or comprehension impaired, score sensation normal unless deficit clearly recognized (e.g., clear-cut asymmetry of grimace or withdrawal). Only hemisensory losses attributed to stroke are counted as abnormal.	
0	normal, no sensory loss
1	mild to moderate sensory loss (pinprick dull or less sharp on the affected side, or loss of superficial pain to pinprick but patient aware of being touched)
2	severe to total (patient unaware of being touched in the face, arm, and leg)
9. Best language	
In addition to judging comprehension of commands in the preceding neurologic exam, the patient is asked to describe a standard picture, to name common items, and to read and interpret the standard text in the box below. The intubated patient should be asked to write:	
• You know how. • Down to earth. • I got home from work. • Near the table in the dining room. • They heard him speak on the radio last night.	

Table 93.1 continued

Scale	Finding
0	normal, no aphasia
1	mild to moderate aphasia (some loss of fluency, word-finding errors, naming errors, paraphasias and/or impairment of communication by either comprehension or expression disability)
2	severe aphasia (great need for inference, questioning, and guessing by listener; limited range of information can be exchanged)
3	mute or global aphasia (no usable speech or auditory comprehension) or patient in coma (item 1a = 3)
10. Dysarthria	
Patient may be graded based on information already gleaned during evaluation. If patient is thought to be normal, have them read (or repeat) the standard text shown in this box.	
• MAMA • TIP-TOP • FIFTY-FIFTY • THANKS • HUCKLEBERRY • BASEBALL PLAYER • CATERPILLAR	
0	normal speech
1	mild to moderate (slurs some words, can be understood with some difficulty)
2	severe (unintelligible slurred speech in the absence of, or out of proportion to, any dysphasia, or is mute/anarthric)
0	intubated or other physical barrier
11. Extinction and inattention (formerly neglect)	
Sufficient information to identify neglect may already be gleaned during evaluation. If the patient has severe visual loss preventing visual DSSS, and the cutaneous stimuli are normal, the score is normal. Scored as abnormal only if present.	
0	normal, no sensory loss
1	visual, tactile, auditory, spatial, or personal inattention or extinction to DSSS in one of the sensory modalities
2	profound hemi-inattention or hemi-inattention to more than one modality. Does not recognize own hand or orients to only one side of space.
A. Distal motor function (not part of NIHSS) (a = left arm, b = right)	
Patient's hand is held up at the forearm by the examiner, and is asked to extend the fingers as much as possible. If patient cannot do so, the examiner does it for them. Do not repeat the command.	
0	normal (no finger flexion after 5 seconds)
1	at least some extension after 5 seconds (any finger movement is scored)
2	no voluntary extension after 5 seconds
^a based on the Cincinnati stroke scale. ³ Free online text version (https://www.ninds.nih.gov/sites/default/files/NIH_Stroke_Scale.pdf) or graphical version (https://www.ninds.nih.gov/sites/default/files/NIH_Stroke_Scale_Booklet.pdf)	

Higher NIHSS scores indicate more deficit and correlate with more proximal vascular lesions (larger vessel occlusion causes more widespread deficit).

93.4 General management for acute ischemic stroke (AIS)

93.4.1 Admitting orders

Orders are for TIA or stroke, but not SAH (p.1432) nor intracerebral hemorrhage (ICH) (p.1619).⁴ Entries from the 2018 AHA guidelines¹ are indicated by the corresponding level of the recommendation.

1. admit to ICU
2. frequent VS with crani checks (q 1 hr × 12 hrs, then if stable, q 2 hrs)
3. monitor cardiac rhythm
4. activity: bed rest

5. labs:
 - a) routine: CBC + platelet count, electrolytes, PT/PTT/INR, U/A, ABG
 - b) ★ only blood glucose (p. 1557) determination must precede initiation of IV tPA in all patients (Level I¹)
 - c) usefulness of CXR in the absence of cardiovascular indications is unclear, and should not delay IV tPA (Level I¹)
 - d) baseline EKG & cardiac troponins are recommended but should not delay IV tPA (Level I¹). EKG changes occur in 5–10% of AIS; acute MIs in 2–3%
 - e) "special" (when appropriate): RPR (to rule out neurosyphilis), ESR (to rule out giant cell arteritis), hepatic profile, cardiac (lipid) profile
 - f) at 24 hrs: CBC, platelet count, cardiac (lipid) profile, EKG
6. oxygenation
 - a) provide supplemental O₂ to maintain O₂ saturation > 94% (Level I¹)
 - b) airway support and ventilator for decreased level of consciousness or for bulbar dysfunction that compromises the patient's ability to protect their airway (Level I¹)
 - c) supplemental O₂ is not recommended in nonhypoxic patients (no benefit¹)
 - d) hyperbaric oxygen is not recommended except for air embolism (no benefit¹)
7. diet: NPO
8. nursing care
 - a) indwelling Foley (urinary) catheter if consciousness impaired or if unable to use urinal or bedpan; intermittent catheterization q 4–6 hrs PRN no void if Foley not used
 - b) accurate I's & O's; notify M.D. for urine output < 20 cc/hr × 2 hrs by Foley, or < 160 cc in 8 hrs if no Foley
9. glucose:
 - a) avoid hyperglycemia in the 1st 24 hours after AIS (worse outcome). Goal: blood sugar 140–180 mg/dL (Level II¹)
Rationale: hyperglycemia may extend ischemic zone (penumbra).⁵
 - b) avoid hypoglycemia < 60 mg/dL (Level I¹)
 - c) hyperglycemia and hypoglycemia may mimic AIS and should be treated if identified (tPA is not indicated for nonvascular conditions) (no benefit¹)
10. IV fluids: NS or 1/2 NS at 75–125 cc/hr for most patients (to eliminate dehydration if present)
 - a) avoid glucose: (see above)
 - b) avoid overhydration in cases of ICH, CHF, or SBP > 180
11. seizures:
 - a) treat recurrent seizures the same as for other neurologic conditions (Level I¹)
 - b) prophylactic seizure medication is not recommended (no benefit¹)
12. treat CHF and arrhythmias. MI or myocardial ischemia may present with neuro deficit
13. blood pressure (BP) management:
 - a) for patients with HTN who are otherwise candidates for IV tPA: carefully lower SBP to < 185 mm Hg, and DBP to < 110 before giving IV-tPA (Level I¹), and maintain < 180/105 for 24 hours after tPA (Level I¹)
Otherwise, for general management, see Hypertension in stroke patients below.
 - b) avoid hypotension and hypovolemia (Level I¹): no specifics provided. Suggestion: for patients presenting with SBP < 110 or DBP < 70:
 - unless contraindicated (viz.: ICH, cerebellar infarct, or decreased cardiac output) give 250 cc NS over 1 hr, then 500 cc over 4 hrs, then 500 cc over 8 hrs
 - if fluid ineffective or contraindicated: consider pressors
 - c) the benefit of drug-induced hypertension is not well-established in AIS (Level II¹)
14. osmotic therapy (mannitol 50 to 100 gm IV over 20 minutes or 3% saline): for clinical deterioration from cerebral edema associated with AIS (Level II¹)
15. patient temperature:
 - a) fever > 38 °C: identify and treat the source, use antipyretics as needed (Level I¹)
 - b) induced hypothermia: use is not well-established and should only be employed within approved clinical trials (Level II¹)
16. other medications
 - a) corticosteroids (including dexamethasone)
 - ★ not recommended for cerebral edema and/or increased ICP complicating stroke (potential harm¹)
 - exceptions: steroid-responsive vasculitis, e.g., giant cell arteritis (temporal arteritis), demonstration of associated brain tumor
 - b) stool softener
 - c) avoid diuretics unless volume overloaded

17. antiplatelet therapy key points (see below (p. 1558) for some additional details)
 - a) ★ aspirin (ASA): recommended within 24–48 hrs after onset of IAS (Level II¹). Doses used: 160–300 mg/d. If IV tPA was given, ASA is usually delayed until 24 hrs later, but can be started earlier for strong indications
 - b) ★ patients with minor stroke: dual antiplatelet therapy (ASA + clopidogrel) for 21 days starting ≤ 24 hrs can reduce secondary stroke for up to 90 days (Level II¹)
 - c) ASA is not a substitute for IV tPA or thrombectomy in eligible patients (no benefit¹)

93.4.2 Hypertension in stroke patients

General information

HTN may actually be needed to maintain CBF in the face of elevated ICP, and it usually resolves spontaneously. Therefore treat HTN cautiously and slowly to avoid rapid reduction and overshooting the target. Avoid treating mild HTN. Indications to treat HTN emergently include:

1. acute LV failure (rare)
2. acute aortic dissection (rare)
3. acute hypertensive renal failure (rare)
4. neurologic complications of HTN
 - a) hypertensive encephalopathy
 - b) conversion of a large pale (ischemic) infarct into a hemorrhagic infarct
 - c) patients with ICH; some HTN is needed to maintain CBF, see Initial management of ICH (p. 1619)

Hypertension treatment algorithm

See reference,⁴ modified.

Recommended lower limits for treatment endpoints are shown in ► Table 93.2.

1. if DBP > 140 (malignant hypertension): ≈ 20–30% reduction is desirable. Cardene infusion or IV

Table 93.2 Guidelines for lower limits of treatment endpoints for HTN in strokes

	No prior history of HTN	Prior history of HTN
do not lower SBP below	160–170 mm Hg	180–185 mm Hg
do not lower DBP below	95–105 mm Hg	105–110 mm Hg

labetalol are agents of choice; arterial-line monitor recommended; sympatholytics (e.g., trimethaphan) contraindicated (they reduce CBF)

2. SBP > 230 or DBP 120–140 × 20 mins: **labetalol** (p. 131) (unless contraindicated): start at 10 mg slow IVP over 2 mins, then double q 10 min (20, 40, 80, then 160 mg slow IVP) until controlled or total of 300 mg given. Maintenance: effective dose (from above) q 6–8 hrs PRN SBP > 180 or DBP > 110
3. SBP 180–230 or DBP 105–120: defer emergency treatment unless there is evidence of LV failure or if readings persist × 60 mins
 - a) oral **labetalol** (p. 131) (unless contraindicated) dosed as follows:
 - for SBP > 210 or DBP > 110: 300 mg PO BID
 - for SBP 180–210 or DBP 100–110: 200 mg PO BID
 - b) if labetalol contraindicated:
 - nifedipine (p. 131)

93.4.3 Antiplatelet drugs in AIS – additional info

- IV tirofiban & eptifibatide: efficacy is not established pending further clinical trials (Level II¹)
- ★ abciximab & other glycoprotein IIb/IIIa receptor antagonists are not recommended (potential harm¹)
- ticagrelor is not recommended over ASA in the acute treatment of minor stroke (no benefit¹)

93.4.4 Emergency surgery

Possible indications:

1. herniation from subdural hematoma

2. suboccipital craniectomy for progressive neurologic deterioration due to brainstem compression from cerebellar hemorrhage or infarction (see below)
3. decompressive craniectomy for malignant MCA territory stroke (see below)
4. carotid endarterectomy for high-grade carotid stenosis ipsilateral to *fluctuating* neuro deficit; see Emergency carotid endarterectomy (p. 1570)

93.5 Imaging in acute ischemic stroke (AIS)

93.5.1 CAT scan findings with acute ischemic stroke (AIS)

Findings at various times after ischemic stroke

NB: These principles do *not* apply to small lacunar infarcts, nor to hemorrhagic strokes (ICH). NB: CT is normal in 8–69% of MCA strokes in the first 24 hours.⁶

► **Hyperacute (<6 hours after stroke).** Early signs of infarction involving large areas of the MCA territory correlate with poor outcome.⁷ Early findings may include⁸:

1. hyperdense artery sign (see below): low sensitivity, but helpful if present
2. focal low attenuation within the gray matter*
3. loss of the gray-white interface*
4. attenuation of the lentiform nucleus
5. mass effect*
 - a) early: effacement of the cerebral sulci (often subtle)⁹
 - b) late: midline shift in large territory infarction
6. loss of the insular ribbon sign (hypodensity involving the insular cortex, susceptible to ischemia due to poor collaterals)
7. enhancement with IV contrast: occurs in only 33%. Stroke becomes isodense (called “masking” effect) or hyperdense with normal brain, and, rarely, may be the only indication of infarction⁹

*These findings are probably due to increased water content resulting from the following: cellular edema arising from altered cell permeability, which produces a shift of sodium and water from the extracellular to the intracellular compartment, which also increases the extracellular osmotic pressure causing transudation of water from capillaries into the interstitium.¹⁰

► **24 hrs.** Most strokes can be identified as low density by this time.

► **1–2 wks.** Strokes are sharply demarcated. In 5–10% there may be a short window (at around day 7–10) where the stroke becomes isodense, called “fogging effect.” IV contrast will usually demonstrate these.

► **3 wks.** Stroke approaches CSF density.

► **Mass effect.** Common between day 1 to 25. Then atrophy is usually seen by ≈ 5 wks (2 wks at the earliest). Serial CT scans have shown that midline shift increases after ischemic stroke and reaches a maximum 2–4 days after the insult.

► **Calcifications.** Over a long period of time (months to years) ≈ 1–2% of strokes calcify (in adults, it is probably a much smaller fraction than this; and in peds it is higher). Therefore, in an adult, calcifications almost rule out a stroke (consider AVM, low-grade tumor...).

Alberta stroke program early CT score (ASPECTS)

The ASPECTS¹¹ is derived from 2 noncontrast axial CT slices: one at the level of the thalamus, the second just rostral to the basal ganglia. The MCA is divided into 10 territories that each get 1 point: 3 subcortical structures (caudate, lentiform nucleus, & internal capsule), and 7 cortical territories: insular cortex (ribbon) and M1 through M6 (as shown in ► Fig. 93.1).

Starting with a normal score of 10, 1 point is subtracted for each of the territories that show signs of early ischemic change: viz. swelling (evidenced by compression of sulci or ventricles) or hypodensity (relative to other areas of the brain) involving ≥ 1/3 of the territory. A score ≤ 7 is associated with a worse outcome from stroke. Despite its limitations (only assesses MCA distribution infarcts, cortical territories are not equally weighted...), it is simple, has reasonably good interrater reliability, and can also be used on MRI.¹²

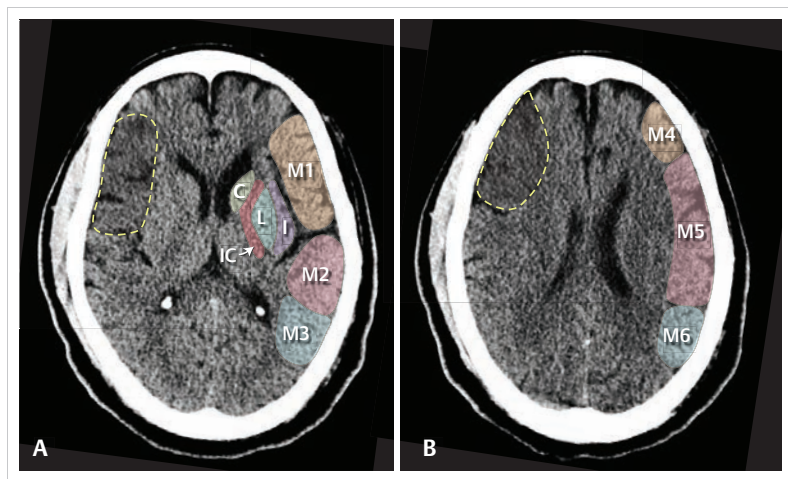


Fig. 93.1 ASPECTS (Alberta stroke program early CT score) with illustration in a 71-year-old male with acute right MCA infarct (demarcated by the yellow broken lines). Image: noncontrast axial head CT, A: at level of the thalamus, B: just above the basal ganglia. The 10 graded areas are shown as colored shapes on the patient's left side: C = caudate (head), IC = internal capsule, L = lentiform nucleus (BG), I = insular cortex, M1 through M6 = convexity cortical territories. ASPECTS here is 8 (10 points minus 1 point for M1, minus 1 point for M4, but no point off for M5 because slightly less than 1/3 of M5 is involved).

Hyperdense artery sign

First described in the MCA in 1983.¹³ The cerebral vessel (usually the MCA) appears as high density on unenhanced CT (► Fig. 93.2), indicating intra-arterial clot (thrombus or embolus).¹⁴ Seen in 12% of 50 patients scanned within 24 hrs of stroke, and in 34% of 23 very early CTs done to R/O hemorrhage. Sensitivity for MCA occlusion is low, but specificity is high (although it may also be seen with carotid dissection, or (usually bilaterally) with calcific atherosclerosis or high hematocrit¹⁴). Does not have independent prognostic significance.¹⁵

Enhancement

CT enhancement with IV contrast in stroke: not routinely used in AIS

1. many enhance by day 6, most by day 10, some will enhance up to 5 wks
2. rule of 2's: 2% enhance at 2 days, 2% enhance at 2 mos
3. gyral enhancement: AKA "ribbon" enhancement. Common. Usually seen by 1 week (gray matter enhances > white). Ddx includes inflammatory infiltrating lesions such as lymphoma, neurosarcooidosis... (due to breakdown of BBB)
4. rule of thumb: there should not be enhancement at the same time that there is mass effect

93.5.2 CT angio (CTA)

CTA (p.238) is useful for assessing the location and extent of vascular occlusion in acute ischemic stroke,¹⁶ and may identify the bleeding source in subarachnoid hemorrhage. Findings can direct treatment towards endovascular options when a proximal or significant large vessel occlusion is seen.

To avoid delays in candidates for endovascular therapy, it is reasonable to proceed with CTA without waiting for serum creatinine if there is no history of renal impairment (Level II¹).

93.5.3 CT perfusion

Theoretically identifies salvageable penumbra as a region of mismatch between CBF and CBV. Assumption: the infarcted core (with no salvageable tissue) has decreased CBF within a region of



Fig. 93.2 Hyperdense artery sign.
Image: noncontrast axial CT scan.
The right middle cerebral artery (MCA) appears hyperdense in this 85-year-old patient with acute right MCA stroke. The left MCA is also slightly hyperdense.

decreased CBV (CBF/CBV match). A mismatched area (decreased CBV *without* a decrease in CBF) represents potentially salvageable penumbra.¹⁷ Implication: thrombolytics and interventional treatment modalities without mismatch will likely increase morbidity and mortality without clinical benefit.

93.5.4 MRI

With newer, faster acquisition times, and with gradient echo sequences that are highly sensitive to hemorrhage, MRI is increasingly being utilized in the hyperacute setting and is at times replacing CT as the initial evaluation. More sensitive than CT (especially DWI-MRI (p.243), and particularly in the first 24 hrs after stroke), and especially with brainstem or cerebellar infarction. More contraindications (p.242) than CT.

Contrast MRI: not often used. 4 enhancement patterns¹⁸:

1. intravascular enhancement: occurs in $\approx 75\%$ of 1–3-day-old cortical infarcts, and is probably due to sluggish flow and vasodilatation (thus, it is not seen with complete occlusion). May indicate areas of brain at risk of infarction
2. meningeal enhancement: especially involving the dura. Seen in 35% of cortical strokes 1–3 days old (not seen in deep cerebral or brainstem strokes). No angiographic nor CT equivalent
3. transitional enhancement: above two types of enhancement coexist with early evidence of BBB breakdown; usually seen on days 3–6
4. parenchymal enhancement: classically appears as a cortical or subcortical gyral ribbon enhancement. May not be apparent for the first 1–2 days, and gradually approaches 100% by 1 week. Enhancement may eliminate “fogging effect” (as on CT) which may obscure some strokes at ≈ 2 weeks on unenhanced T2WI

93.5.5 MRI perfusion

Similar to CT perfusion (p.239), areas of matched DWI and PWI abnormality are thought to represent infarcted tissue. PWI abnormalities that do not have a DWI correlate are thought to represent potentially salvageable penumbra.¹⁹

93.6 Management of TIA or stroke

93.6.1 Treatment options timeline

See ► Fig. 93.3.

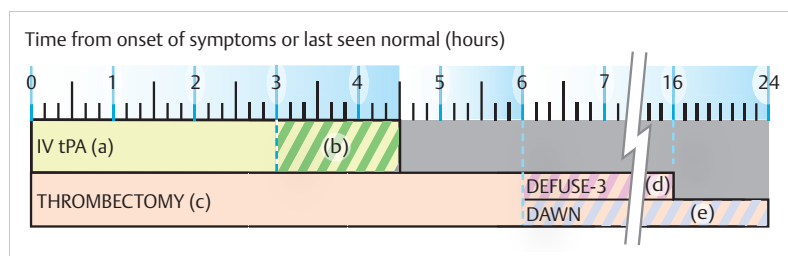


Fig. 93.3 Treatment options timeline.

(a) IV tPA: give to essentially all candidates within 4.5 hours of onset.

(b) 3–4.5 hrs: IV tPA reasonable but not studied in same patient population (see text).

(c) thrombectomy candidates: NIHSS score ≥ 6 , ASPECTS ≥ 6 , large vessel occlusion (LVO), puncture ≤ 6 hrs from onset.

(d) thrombectomy 6–16 hrs in candidates who meet other DAWN or DEFUSE-3 (p. 1562) eligibility (see text).

(e) thrombectomy 16–24 hrs in candidates with anterior circulation LVO who meet other DAWN eligibility criteria.

These times are applicable to anterior circulation strokes. Posterior circulation occlusions may be treated more aggressively.

The American Heart Association has recommended a goal of “door-to-needle” (DTN) time of ≤ 60 minutes in $\geq 50\%$ of acute ischemic stroke patients being treated with tPA (Level I¹), with a DTN of ≤ 45 minutes being a reasonable secondary goal (Level II¹).

93.6.2 Endovascular therapy for stroke

Recent trials favor rapid endovascular intervention in acute ischemic stroke with proximal vessel occlusion, small infarct core, and moderate to good collateral circulation.^{20,21,22,23}

The primary modality is mechanical clot retrieval (thrombectomy), although intra-arterial tPA may be used in select situations. See Endovascular intervention (p. 1932) for details.

Practice guideline: Mechanical thrombectomy for acute ischemic stroke (AIS)

Mechanical thrombectomy with stent retriever is recommended when all criteria met (Level I¹):

- prestroke (modified Rankin Score (mRS), ► Table 98.5) of 0–1
- causative occlusion of ICA or M1 segment of MCA
- age ≥ 18 years
- NIHSS (p. 1554) score ≥ 6
- ASPECTS (p. 1559) ≥ 6
- treatment can be initiated (groin puncture) ≤ 6 hours of onset

Benefits uncertain, but mechanical thrombectomy may be reasonable for (Level II¹)

- carefully selected patients with causative occlusion of M2 or M3 segment of MCA, or anterior cerebral, vertebral, basilar, or posterior cerebral arteries
- or prestroke mRS > 1 , ASPECTS < 6 and causative occlusion of ICA or M1 segment; however, additional randomized trials are needed
- when treatment can be initiated (groin puncture) ≤ 6 hours after onset

Mechanical thrombectomy is recommended for selected patients 6–16 hours from onset with large vessel occlusion (LVO) in the anterior circulation who meet other DAWN^a or DEFUSE-3^a eligibility criteria (Level I¹).

Mechanical thrombectomy is reasonable in selected patients 16–24 hours from onset with anterior circulation LVO who meet other DAWN^a eligibility criteria (Level II¹).

The goal of thrombectomy should be reperfusion to a mTICI 2b/3 angiographic result (► Table 116.2) and to minimize the time to treatment in order to maximize the chances of good functional outcome (Level I¹).

^a DAWN²⁴ & DEFUSE-3²⁵ are the only randomized controlled trials that showed benefit of mechanical thrombectomy > 6 hours from onset. CTP, DW-MRI, or MRI perfusion can aid in patient selection when RCT eligibility criteria are strictly met. Trial eligibility can be found online at <http://stroke.ahajournals.org/lookup/suppl/doi:10.1161/STR.000000000000158/-/DC1>

DEFUSE-3 used perfusion/core mismatch & maximum core size (< 70 ml) to select patients 6–16 hrs from onset. Proprietary software (such as iSchemaView's RAPIDTM) can quickly analyze scans to determine the mismatch.

DAWN: used clinical/imaging mismatch (combination of NIHSS score and imaging findings on CTP or DWI-MRI) to select patients 6–24 hrs from onset.

93.6.3 Thrombolytic therapy

General information

Tissue plasminogen activator (tPA) is an enzyme that is naturally found in vascular endothelial cells. It catalyzes the conversion of plasminogen to the fibrinolytic compound plasmin (the major enzyme involved in thrombolysis). tPA for clinical use is manufactured using recombinant DNA technology (rtPA, or just tPA).

Alteplase (Activase®, Actilyse®) is the primary agent used clinically. It is FDA-approved for the IV treatment of acute ischemic stroke (see below).

Tenecteplase (TNKase®), an alternative tPA, has not shown superiority or inferiority to alteplase,²⁶ and pending further investigation may be considered in patients with minor neurologic deficit and no major intracranial occlusion (Level II¹).

Other agents: benefits are unproven, and their use is not recommended outside of clinical trials (no benefit¹).

IV tissue plasminogen activator (IV tPA)

For this section, tPA refers to alteplase. For *intra-arterial* tPA see endovascular therapy (p. 1932).

The use of tPA administered within the first 3 hours of stroke onset has been shown to be safe and effective in multiple randomized controlled trials and by extensive community experience in numerous countries.¹ This window was extended to 4.5 hours in 2009 after ECASS-3,²⁷ but the study population differed (see Eligibility below).

Guidelines for the administration of IV tPA

Eligibility:

1. age ≥ 18 years (although use in childhood stroke is increasing²⁸)
2. time since last seen normal ≤ 4.5 hrs prior to administration
(NB: the 3–4.5-hr window was not studied in patients ≥ 80 years of age, patients with severe stroke (NIHSS (p. 1554) score > 25), and patients with prior stroke + DM.²⁹ However, it may still be reasonable to treat these patients (Level II¹)
3. tPA is reasonable in otherwise eligible patients with history of ≤ 10 cerebral microbleeds on MRI (Level II¹)
4. AIS in adults with known sickle cell disease

Contraindications¹:

1. intracerebral hemorrhage (ICH): on admitting CT, or history of prior ICH
2. patients with history of > 10 cerebral microbleeds showed an increased risk of ICH and the benefits of tPA are uncertain but treatment may be beneficial if there is substantial potential benefit (Level II¹)
3. clinical presentation of SAH (even with negative CT)

- known intracranial aneurysm or AVM
- active internal bleeding
- known bleeding diathesis, including but not limited to the following:
 - full anticoagulation with LMWH within the previous 24 hours (possible harm¹)
 - platelet count $< 100,000/\text{mm}^3$ (given the low risk of unsuspected abnormal platelet counts or coagulation studies, it is reasonable that urgent tPA should not be delayed to wait for test results (Level II¹))
- serious head trauma, serious stroke, or intracranial surgery within past 3 months
- SBP > 185 mm Hg, or DBP > 110 mm Hg that cannot be controlled despite use of nicardipine infusion or IV labetalol

✗ Cautions:

- seizure witnessed at the time of onset of stroke symptoms
- major surgery within the last 14 days
- arterial puncture at non-compressible site within past 7 days
- recent lumbar puncture
- rapidly improving or minor symptoms. Note: IV tPA should not be delayed to monitor for further improvement (possible harm¹)
- blood glucose > 400 mg/dl or < 50 mg/dl
- history of GI or urinary tract hemorrhage within past 21 days
- post myocardial infarction pericarditis

Additional issues relevant to giving IV tPA:

- hypoattenuation on CT is not a criterion to withhold IV tPA¹
- hyperdense MCA sign is not a criterion to withhold IV tPA (no benefit¹)
- routine use of MRI to exclude cerebral microbleeds before IV tPA is not recommended (no benefit¹)
- use of imaging criteria to select candidates for IV tPA when the time of onset of symptoms is unknown is not recommended (no benefit¹)
- do not delay IV tPA to obtain multimodal CT and/or MRI (possible harm¹)
- for candidates for endovascular therapy, a noninvasive vascular study is recommended during the initial evaluation, but should not delay administration of IV tPA if indicated (Level I¹)
- do not administer abciximab concurrently with tPA (potential harm¹)
- rule out hypoglycemia and hyperglycemia (p.1557)
- be prepared to treat complications of tPA including: bleeding complications, angioedema (may require intubation) (Level I¹)
- hypertension is aggressively controlled
- the use of antithrombotic therapy within the first 24 hours of IV tPA is uncertain. Use may be considered when there is substantial benefit (Level I¹)

Treatment protocol: Also, see Eligibility (p. 1563) & Contraindications (p. 1563) above.

R alteplase (Activase®): initiate < 4.5 hrs from onset of deficit or last known well (see indications/contraindications above).

NINDS protocol: 0.9 mg/kg (up to a maximum of 90 mg total) given as follows: IV bolus 10% of the dose over 1 min, followed *immediately* by the remainder as a constant infusion over 60 minutes (Level I¹).

If there is an indication for anticoagulation, obtain a non-contrast CT 24 hours prior to starting anticoagulation since there is a risk of subclinical intracerebral hemorrhage.

ICH following IV tPA

There is an increased risk of symptomatic intracerebral hemorrhage (ICH) with the use of tPA (NINDS study: 6.4% vs. 0.6% with placebo; ECASS II: 8.8% vs. 3.4%). In spite of this, the NINDS study found that mortality in the tPA group was similar to controls at 3 mos (17% vs. 21%). The following factors were associated with an increased risk of symptomatic ICH (with only a 57% efficiency rate of predicting ICH): severity of NIHSS score, or pre-treatment CT showing brain edema or mass effect. In one study, ICH did not influence outcome except in the rare instance when a massive hematoma occurred.³⁰ Outcomes were still better in the treated group, and the conclusion is that these patients are still reasonable candidates for tPA.³¹ Since then, multicenter analyses have demonstrated that size of infarction and elevated blood sugar are independent risk factors for symptomatic ICH.³²

Management of post-tPA ICH:

- discontinue tPA infusion and obtain STAT head CT

2. send labs: PT, aPTT, platelet count, fibrinogen, and type & cross
3. prepare to administer 6–8 units cryoprecipitate containing Factor VIII
4. prepare to administer 6–8 units of platelets
5. if emergent EVD placement or other interventional procedure is needed, consider the use of recombinant Factor VIIa (40–80 mg/kg) immediately beforehand (NB: this is only a temporizing measure and cryoprecipitate still needs to be given)

93.7 Carotid endarterectomy

93.7.1 Indications

Trials and results

► Table 93.3 shows the status of current studies for the surgical treatment of carotid stenosis (NB: some of the results may be contradictory).

Table 93.3 Summary of study findings for carotid endarterectomy (CEA)^a (modified³³)

Stenosis	Relevant study	Recommendation	Risk reduction ^b
Symptomatic Narrowing			
70–99%	NASCET ³⁴	CEA	16.5 @ 2 yrs
> 60%	ECST ³⁵	CEA	11.6 @ 3 yrs
50–69%	NASCET ³⁶	CEA ^c	10.1 @ 5 yrs
< 30%	NASCET ³⁶	BMM	0.8 @ 5 yrs
< 40%	ECST ³⁷	BMM	CEA worse @ 3 yrs
Asymptomatic Narrowing (p. 1548)			
> 60%	ACST ³⁸	CEA if age < 75 yrs	5.4% @ 5 yrs
> 60%	ACAS, ³⁹ ACST ^d	CEA ^d	6.3 @ 5 yrs
> 50%	VACS	± CEA ^e	
< 90%	CASANOVA	BMM ^e	

^aabbreviations: NASCET = North American Symptomatic Carotid Endarterectomy Trial; ECST = European Carotid Surgery Trial; CASANOVA = Carotid Artery Stenosis with Asymptomatic Narrowing Operation Versus Aspirin;

ACAS = Asymptomatic Carotid Atherosclerosis Study; ACST = Asymptomatic Carotid Surgery Trial; VACS = Veteran's Administration Cooperative Study; CEA = carotid endarterectomy; BMM = best medical management

^breduction in risk of all nonfatal strokes and death from any cause with CEA vs. BMM (e.g., with an absolute risk reduction of 16.5 at 2 yrs, for every 100 patients treated, 16.5 nonfatal strokes or deaths were prevented over a 2-yr period)

^csurgery moderately beneficial (requires low complication rate)

^dthe overall health of the patient is critical

^eresults equivocal

The North American Symptomatic Carotid Endarterectomy Trial³⁴ (NASCET) found that for patients with a hemispheric or retinal TIA or a mild (non-disabling) stroke within 120 days and *ipsilateral high-grade* stenosis (> 70%), that carotid endarterectomy (CEA) reduced the rate of fatal and non-fatal strokes (by 17% at 18 months) and death from any cause (by 7% at 18 months) when compared to best medical management (when surgery was performed with perioperative risk of stroke or death of 5.8%). Results were twice as good for patients with stenosis from 90–99% than for those with 70–79%. Furthermore, with CEA the frequency of major functional impairment was reduced at 2 years.⁴⁰ NB: see differences in techniques for measuring stenosis between NASCET and ECST (► Table 92.5).

See also **asymptomatic** patients (p. 1548).

Unresolved controversies

Include:

1. progressive STROKE (“stroke in evolution”): see Emergency carotid endarterectomy (p. 1570)
2. abrupt occlusion: see Emergency endarterectomy (p. 1570)

3. tandem lesions (e.g., carotid siphon and bifurcation stenosis): although this topic remains controversial, CEA in patients with tandem lesions has not been associated with increased postoperative stroke rates.^{41,42} Recent case series also report success with endovascular treatment
4. progressive retinal ischemia

93.7.2 Timing with respect to acute stroke

For patients with small fixed deficits or small infarcts on CT or MRI, the risk of early CEA is not increased.^{41,43} In the pooled analysis of the three symptomatic CEA studies, patients randomized in the trials within 2 weeks of the last symptomatic event had greater benefit from CEA.⁴⁴ Data from Sundt (see below) indicates that a stroke is a risk factor for a complication only if it occurred within 7 days pre-op.

Since the introduction of tPA for the treatment of acute ischemic stroke, there have been reports on the successful treatment of residual critical ICA stenosis following tPA recanalization as early as 24 hours after administration of tPA in patients with small fixed deficit or small ischemic areas on MRI.^{44,45}

93.7.3 Pre-op risk factors for CEA

Σ: Pre-op risk factors for CEA

The characteristics of patients who are high risk for complications from CEA have not been well defined, despite the perception that this group exists.

Identifying patients at high risk for complications after CEA has proven challenging. Typically, the exclusion criteria from studies are cited, but in most cases these are simply patients who were not included in the study because it was the investigators' perception that these patients might be "high risk." Therefore these risk factors are not validated. They are included here for completeness.

NASCET and ACAS: age > 80 years, prior ipsilateral CEA, prior contralateral CEA within 4 months, prior neck XRT, tandem lesion larger than target lesion, other conditions that could cause symptoms (atrial fibrillation, prior stroke with persistent major deficit, valvular heart disease), major organ failure, uncontrolled hypertension or diabetes mellitus, and significant coronary artery disease.^{46,47}

The SAPHIRE Trial (Stenting and Angioplasty with Protection in Patients at High-Risk for Endarterectomy): patients with clinically significant cardiac disease (CHF, abnormal stress test, or need for open-heart surgery), severe pulmonary disease, contralateral carotid occlusion, contralateral laryngeal-nerve palsy, previous radical neck surgery or neck XRT, recurrent stenosis after endarterectomy, and age > 80 years.⁴⁸ The ARCHER Trial (ACCULINK for Revascularization of Carotids in High-Risk patients) also included patients with tracheostomy, spinal immobility, and dialysis-dependent renal failure.⁴⁹

93.7.4 Carotid endarterectomy—surgical considerations

Perioperative management

Pre-op management (carotid endarterectomy)

1. ASA 325 mg TID for at least 2 days, preferably 5 days pre-op⁵⁰ (NB: patients should be kept on their ASA for surgery, and if not on ASA they should be started, in order to reduce risks of MI and TIA⁵¹)

Post-op management (carotid endarterectomy)

1. patient monitored in ICU with A-line
2. keep patient well hydrated (run IVF ≥ 100 cc/hr for most adults)
3. SBP ideally 110–150 mm Hg (higher pressures are permitted in patients with chronic severe HTN)
 - a) BP frequently labile in first 24 hrs post-op, may be due to "new" pressure in carotid bulb; to prevent rebound hyper- or hypotension, avoid long-acting agents
 - b) hypotension
 - check EKG – R/O cardiogenic shock
 - if mild, start with fluids (crystalloid or colloid)
 - phenylephrine (Neo-Synephrine®) for resistant hypotension

- c) hypertension: nicardipine (Cardene®) (p. 131) is the agent of choice. Avoid rebound hypotension
4. avoid antiplatelet drugs for 24–48 hrs post-op (causes oozing); may start these 24–72 hrs post-op (note: ASA 325 mg + dipyridamole 75 mg TID have been shown *not* to reduce the rate of restenosis after endarterectomy⁵²)
5. optional: reverse half of heparin with protamine 10 minutes after closing arteriotomy

Post-op check (carotid endarterectomy)

In addition to routine, the following should be checked:

1. change in neurologic status due to cerebral dysfunction, including:
 - a) pronator drift (R/O new hemiparesis)
 - b) signs of dysphasia (especially for left-sided surgery)
 - c) mimetic muscle symmetry (assesses facial nerve function)
2. pupil diameter and reaction (R/O stroke, Horner syndrome)
3. severe H/A (especially unilateral) > may indicate hyperperfusion syndrome
4. STA pulses (R/O external carotid occlusion)
5. tongue deviation (R/O hypoglossal nerve injury)
6. symmetry of lips (R/O weakness of lower lip depressors due to retraction of marginal mandibular branch of facial nerve against mandible, usually resolves in 6–12 wks, must differentiate from central VII palsy due to stroke)
7. check for hoarseness (R/O recurrent laryngeal nerve injury)
8. assess for hematoma in operative site: note any tracheal deviation, dysphagia

Post-op complications (carotid endarterectomy)

To justify CEA, the absolute upper limit of (significant) complication rate should be $\leq 3\%$.

1. overall in-hospital mortality: 1%⁵³
2. disruption of arteriotomy closure: rare, but emergent (see below)
 - a) evidenced by:
 - swelling of neck: rupture may produce a pseudoaneurysm
 - tracheal deviation (visible, palpable, or on CXR)
 - symptoms: dysphagia, air hunger or worsening hoarseness, difficulty swallowing
 - b) dangers:
 - asphyxiation: most immediate danger
 - stroke
 - exsanguination (unlikely, unless skin closure is also disrupted)
 - c) late (often delayed weeks to months): false aneurysm.⁵⁴ Risk = 0.33%. Presents as neck mass. Risk is increased with wound infection and possibly with patch graft as compared to endarterectomy alone.^{54,55,56}
3. stroke (cerebral infarction) intra-op or post-op rate⁵⁷: 5%
 - a) embolic (the most common cause of *minor* post-op neurologic deficit): source may be denuded media of endarterectomy
 - b) intracerebral hemorrhagic (ICH) (breakthrough bleeding): occurs in <0.6%.⁵⁸ Related to cerebral hyperperfusion in most^{59,60} (see below). Usually occurs within first 2 weeks, often in basal ganglion 3–4 days post-op with hypertensive episode. Patients at greatest risk are those with severe stenosis and limited hemispheric collateral flow
 - c) post-op ICA occlusion
 - most common cause of *major* post-op stroke, but may be asymptomatic
 - risk is reduced by attention to technical details at surgery⁶¹ (p 249)
 - some may be due to hypercoagulable state induced by heparin (predictable in patients whose platelet count drops while on heparin. No known therapy for this condition⁶¹ (p 249–50))
 - the endarterectomized surface is highly thrombogenic for 4 hrs following endarterectomy (Sundt recommends not reversing heparin)
 - in Sundt's series using patch graft⁶¹ (p 229): 0.8% incidence, associated with major stroke in 33% and minor stroke in 20%
 - occlusion rate with primary closure: 4% in Sundt's experience, 2–5% in literature⁶¹ (p 249)
4. post-op TIAs: most due to ICA occlusion. Some may be due to microemboli. Hyperperfusion syndrome produces a 1% incidence of post-op TIAs⁶¹ (p 229)
5. seizures⁶²: usually focal in onset with possible generalization, most occur late (post-op day 5–13) with an incidence of $\approx 0.4\%$ ⁵⁸ to 1%.⁶³ May be due to cerebral hyperperfusion,⁵⁸ emboli,⁶⁴ and/or intracerebral hemorrhage. Usually difficult to control initially, lorazepam and phenytoin are recommended (p.213)

6. late restenosis: identifiable restenosis occurs in $\approx 25\%$ by 1 yr, and half of these reduce luminal diameters by $> 50\%$.⁶⁵ Restenosis within 2 yrs is usually due to fibrous hyperplasia, after 2 yrs it is typically due to atherosclerosis⁶⁶
7. cerebral hyperperfusion syndrome (AKA normal pressure hyperperfusion breakthrough): classically thought to result from return of blood flow to an area that has lost autoregulation due to chronic cerebral ischemia typically from high-grade stenosis. Controversial.⁶⁰ Usually presents as ipsilateral vascular H/A or eye pain that subsides within several days⁶⁷ or with seizures ($\pm$ PLEDs on EEG, more common with Halothane®, due to petechial hemorrhages⁶⁸). May cause ICH.⁶⁸ Most complications occur several days post-op
8. hoarseness: the most common cause is laryngeal edema and *not* superior nor recurrent laryngeal nerve injury
9. cranial nerve injury: the most common complication after CEA with an incidence of up to 8–10%⁶⁹
 - a) hypoglossal nerve $\rightarrow$ tongue deviation towards the side of injury; incidence $\approx 1\%$ (with mobilizing XII to allow displacement). Unilateral injury may cause speaking, chewing, and swallowing difficulties. Bilateral injuries can cause upper airway obstruction.⁷⁰ The presence of a unilateral palsy is a contraindication to doing contralateral endarterectomy until the first side recovers. May last as long as four months
 - b) vagus or recurrent laryngeal nerve $\rightarrow$ unilateral vocal cord paralysis: 1% risk
 - c) mandibular branch of facial nerve $\rightarrow$ loss of ipsilateral lip depressor
10. headache⁶⁸
11. hypertension^{71,72}: may develop 5–7 days post-op. Longstanding HTN may occur as a result of the loss of the carotid sinus baroreceptor reflex

Complication management

1. post-op TIAs
 - a) if TIA occurs in recovery room, emergency CT (to R/O hemorrhage) and then angiogram recommended to assess for ICA or CCA occlusion (vs. emboli)
 - b) if TIA occurs later, consider emergency ocular pneumoplethysmography; if abnormal $\rightarrow$ emergency surgery (if neurologically intact, pre-op angiogram is appropriate)⁶¹
2. fixed post-op deficit in distribution of endarterectomized carotid
 - a) if deficit occurs immediately post-op (i.e., in PACU), recommend immediate re-exploration without delay for CT or angiogram⁷³ (case reports of no deficit when flow re-established in ≤ 45 mins). For later onset, workup is indicated. Technical considerations for emergency reoperation⁶¹ (p 255):
 - isolate the 3 arteries (CCA, ECA, & ICA)
 - occlude CCA first, then ECA, and ICA last (to minimize emboli)
 - open arteriotomy, check backflow; if none, pass a No. 4 Fogarty catheter into ICA, gently inflate and withdraw (avoid intimal tears)
 - if good backflow established, close with patch graft
 - remove tortuous vessel loops and kinks before closing
 - b) immediate management (unless ICH or SDH are likely) includes
 - fluids (e.g., Plasmanate®) to improve rheology and to elevate BP
 - pressors (e.g., phenylephrine) to elevate SBP to ≈ 180 mm Hg
 - oxygen
 - heparinization (may be controversial)
 - c) theoretical benefits of radiographic evaluation include:
 - CT: identifies ICH or SDH that might require treatment other than re-exploration of the surgical site, elevating BP, etc.
 - angiogram: identifies whether ICA is occluded, or if deficit is from another cause (e.g., emboli from endarterectomy site) that would not benefit from re-exploration or possibly endovascular treatment
3. disruption of arteriotomy closure, management
 - a) OPEN WOUND—if there is any stridor, it is critical to do this before trying to intubate (although ideally performed in O.R., the delay may be decisive). Evacuate clot (start with a sterile gloved finger) and stop bleeding, preferably without traumatizing the artery; a DeBakey clamp is optimal
 - b) INTUBATION—high priority, may be difficult or impossible if trachea is deviated (open wound immediately). Preferably done by anesthesiologist in controlled setting (i.e., O.R.) unless there is acute airway obstruction
 - c) call O.R. and have them prepare set-up for endarterectomy, and take patient to O.R.

93.7.5 Operative technique

Anesthesia and monitoring

Most (but not all) surgeons monitor some parameter of neurologic function during carotid endarterectomy, and will alter technique (e.g., insert a vascular shunt) if there is evidence of hemodynamic intolerance of carotid clamping (only occurs in $\approx 1\text{--}4\%$).

1. local/regional anesthesia: permits "clinical" monitoring of patient's neurologic function.^{74,75} Disadvantages: patient movement during procedure (often exacerbated by sedation and alterations in CBF), lack of cerebral protection from anesthetic and adjunctive agents. The only prospective randomized study found no difference between local and general anesthesia.⁷⁶ The multicenter, randomized controlled General Anesthesia versus Local Anesthesia (GALA) Trial⁷⁷ found no significant differences in the prevention of stroke, MI, or death for either anesthetic technique. Subgroup analysis showed trends (not statistically significant) favoring local anesthesia for perioperative death, event-free survival at 1 year, and patients with contralateral occlusion. Local anesthesia was associated with a significant reduction of shunt insertion.⁷⁷ A Cochrane Database Review found no evidence from randomized trials to favor either anesthetic technique⁷⁸
2. general anesthesia, possibly including barbiturates (thiopental boluses of 125–250 mg until 15–30-second burst suppression on EEG, followed by small bolus injections or constant infusion to maintain burst suppression⁵⁰)
 - a) EEG monitoring
 - b) SSEP monitoring
 - c) measurement of distal stump pressure after CCA occlusion (unreliable), e.g., using a shunt if stump pressure < 25 mm Hg
 - d) transcranial Doppler
 - e) near-infrared spectroscopy

Position and incision

1. supine, neck slightly extended and rotated slightly ($\approx 30^\circ$) away from the operative side
2. the incision curves gently and follows the anterior border of the sternocleidomastoid muscle, and curves posteriorly at the rostral end
3. keep the horizontal portion of the incision ≈ 1 cm away from the mandible to avoid injury to marginal mandibular branch of facial nerve (which lies in the inferior parotid gland and supplies lip depressor) due to retraction against mandible
4. retractors should not be placed deeper than the platysma to avoid injury to recurrent laryngeal nerve, which runs between the esophagus and trachea. Blunt retractors are used to avoid internal jugular vein injury

Dissection

1. the common facial vein (CFV) usually crosses the field over the carotid bifurcation, it is doubly ligated and divided. It leads to the internal jugular vein (IJV)
2. identifying the IJV is key, dissection is carried down between the carotid artery and the IJV
3. the *ansa hypoglossi* runs superficial to the ICA and serves as a useful guide to the *hypoglossal nerve* (XII) which should be identified since it is at greater risk when it is not seen. XII can arise anywhere from the carotid bifurcation to the angle of the mandible, although it is usually in the vicinity of the CFV. Mobilization can be facilitated by dividing the small artery (sternocleidomastoid branch of the ECA) and vein that cross over it⁷⁰
4. the *ansa hypoglossi* can usually be spared, and if mobilized, allows medial retraction of the hypoglossal nerve out of harm's way. If it is necessary to divide the *ansa* it is done close to the hypoglossal nerve to be certain it is not a branch of the vagus and to minimize neurologic deficit (the *ansa* has an anterior cervical limb from the cervical plexus)
5. the superior thyroid artery is the first branch of the ECA, and helps differentiate ECA from the ICA (the ICA is located posterior to the ECA)
6. the carotid bulb may be anesthetized with $\approx 2\text{--}3$ ml of 1% plain lidocaine using a 27 Ga needle. This may be done routinely, or, as some prefer only if hypotension and/or bradycardia occur during dissection (indicating IX nerve stimulation)
7. the ICA must be exposed beyond the extent of the plaque which can be determined by gentle palpation with a moistened finger and by visualization as the area where the artery turns from yellowish to its normal pink color

Occlusion and arteriotomy

1. a vessel loop is placed around the ECA at least 2 cm above the bifurcation
2. a vessel loop is also placed around the ICA but is looped only once
3. umbilical tape with a choke is placed around the CCA 2–3 cm below the bifurcation
4. IV heparin (usually 5,000 IU) is given 1 minute prior to cross clamping
5. a temporary aneurysm clip is placed on the superior thyroid artery
6. the order of *occlusion* of the vessels is as follows (mnemonic: "ICE"):
 - a) ICA (e.g., with temporary aneurysm clip)
 - b) CCA (e.g., with a small DeBakey clamp)
 - c) ECA (e.g., with temporary aneurysm clip)
7. during ICA clamping, mild hypertension is maintained by the anesthesiologist
8. shunt: some surgeons use some form of monitoring (EEG, BSAER, etc.) to determine if a shunt is needed—see Anesthesia and monitoring (p. 1569); yet others routinely use a shunt whenever possible without assessing the need
9. the arteriotomy is begun in the CCA with a #11 scalpel, and once the lumen is entered, Potts' scissors carry the incision through to the ICA beyond the plaque. Stay in the midline to facilitate arteriotomy closure

Plaque removal

1. the plaque usually cannot be completely removed from the CCA, and thus it is usually transected with Potts' scissors, taking care not to inadvertently incise the artery wall and to leave as smooth an edge as possible
2. in the ICA, great care must be made to avoid leaving an intimal flap which could become a nidus for an arterial dissection. If necessary the intima may be tacked down by suturing from the lumen out on both ends (using double-armed suture) and tying the knot outside the vessel

Arteriotomy closure and vessel release

1. arteriotomy may be performed with a running Prolene suture using either
 - a) primary closure
 - b) or a patch graft to increase the caliber of the vessel and reduce the risk of re-stenosis
 - c) limited evidence suggests that carotid patch angioplasty may reduce the risk of perioperative arterial occlusion and re-stenosis. Synthetic patches (Dacron, PTFE) are preferred to autologous vein (risk of aneurysmal dilatation, thrombogenic surface)^{79,80}
2. the order of releasing the vessels (reverse that of the clamping order):
 - a) ECA
 - b) CCA (allows air and debris to be washed into the ECA)
 - c) ICA

93.7.6 Emergency carotid endarterectomy

General information

Emergency CEA indications include crescendo TIAs and stroke in evolution. The treatment paradigm of these conditions has shifted towards the use of interventional methods, such as thrombolysis and stenting, although there are no randomized controlled trial data to support that approach. A recent meta-analysis of emergent CEA has shown that the pooled stroke and stroke/death rates after CEA for crescendo TIA in 176 patients were 6.5% and 9.0%, respectively. For those with stroke in evolution, the overall stroke and stroke/death rates in 114 patients were 16.9% and 20.0%, respectively.⁸¹

After retrospective analysis of 64 emergency endarterectomies⁸² the guidelines given below were suggested. However, the efficacy of immediate surgical removal of obstruction is controversial and unproven. In one early study, over 50% of patients suffered fatal intracranial hemorrhage within 72 hours of emergency carotid endarterectomy.

Initial management of patient presenting with acute neuro deficit

1. obtain history directed at determining presence of previous stroke and other serious medical illness, and to try to differentiate from seizure
2. baseline neurological assessment including evaluation of STA pulses and carotid bruits
3. during evaluation: close control of BP. O₂ per NC. Labs + EKG; see Management of TIA or stroke (p. 1562). Consider hemodilution with LMD

4. CT to R/O ICH or infarction (early stroke will not be visible)
5. when carotid disease is suspected, and CT is negative for ICH or acute infarct, emergency angiography, MRI/MRA, or CTA is performed

Indications for emergency carotid endarterectomy

General information

In patients with acute neurological deficits, the need for rapid decision making often does not allow differentiating between TIA, stroke in evolution and acute stroke, nor in assessing the stability or fluctuating nature of the deficit.

Indications

1. stroke in evolution
2. crescendo TIAs: TIAs that abruptly increase in frequency to $\geq$ several per day
3. following intra-arterial thrombolysis, emergent/urgent CEA is indicated for residual critical carotid stenosis^{45,83}

Contraindications

See also more details (p.1563). Patients with depressed levels of consciousness or acute fixed deficits.

Surgical management

Again, most cases would now be managed initially with endovascular thrombolysis and stenting. Surgery would be considered if this is not an option.

1. for emergency surgery, it is essential that blood pressure be stable
2. in patients with complete occlusion, ICA is not occluded intra-op (to avoid breaking up thrombus, if present)
3. if thrombus is present
 - a) attempt spontaneous extrusion using back pressure
 - b) if this fails, attempt to remove with smoothened suction catheter
 - c) if this fails, pass balloon embolectomy catheter as far as base of skull (caution: avoid injury to distal ICA that could cause CCF)
 - d) obtain intra-op angiogram unless thrombus emerges and backflow is excellent
 - e) plicate ICA (avoid creating a blind pouch at origin) if there is good backflow or if satisfactory angiography cannot be obtained

Surgical results

Highest correlation was with presenting neurologic status (► Table 93.4).

Table 93.4 Surgical results

Presenting deficit	Same or improved	Deaths
intact or mild	92%	0
moderate	80%	1 (7%)
severe	77%	3 (13%)

93.8 Carotid angioplasty/stenting

93.8.1 General information

Σ: Endarterectomy vs. stenting

There are no well-designed studies that convincingly show superiority of angioplasty/stenting over CEA in *average-risk* symptomatic patients, and the recommendation in these patients is to continue with the time-tested technique of CEA.

There is a paucity of randomized control trials^{48,49,84,85,86,87} comparing carotid angioplasty/stenting with CEA, and many nonrandomized registries.^{49,88,89,90,91,92,93,94,95,96}

However, data from multicenter randomized trials showing that carotid angioplasty/stenting is as safe over the short term or as efficacious over the long term as CEA in average-risk symptomatic patients are lacking. Published trials are heterogeneous (clinically and methodologically), too small to provide robust and convincing data, and limited in long-term follow-up. Only the SAPHIRE study⁴⁸ comparing CEA with stenting (using a distal embolic protection device) for moderate to severe carotid stenosis with comorbidities that might increase the risk of CEA (high-risk patients), found that angioplasty/stenting was not inferior (risk within 3%, $P = 0.004$) to CEA (based on a composite primary end point of stroke, death, or MI within 30 days, or death from neurologic causes or ipsilateral stroke between 31 days and 1 year).⁴⁸ However, the study methodology has been criticized.^{97,98,99}

A 2007 Cochrane review concluded that available data on carotid angioplasty/stenting are difficult to interpret and does not support a change in clinical practice away from recommending CEA as the treatment of choice for suitable carotid artery stenosis.¹⁰⁰

93.8.2 Indications for angioplasty/stenting

Carotid stenting performed with adequate procedural quality levels should be considered instead of CEA in the presence of¹⁰¹:

1. severe vascular and cardiac comorbidities:
 - a) congestive heart failure (New York Heart Association class III/IV) and/or known severe left ventricular dysfunction
 - b) open heart surgery needed within 6 weeks
 - c) recent myocardial infarction (< 24 hours and > 4 weeks)
 - d) unstable angina (Canadian Cardiovascular Society class III/IV)
 - e) contralateral carotid occlusion
2. specific conditions:
 - a) contralateral laryngeal nerve palsy
 - b) radiation therapy to the neck
 - c) previous CEA with recurrent restenosis
 - d) high cervical internal carotid/below the clavicle common carotid lesions
 - e) severe tandem lesions
 - f) age > 80 years
 - g) severe pulmonary disease

The 2009 European Society for Vascular Surgery (ESVS) Guidelines state that carotid angioplasty/stenting is indicated in cases of: contralateral laryngeal nerve palsy, previous radical neck dissection or cervical XRT, prior CEA (re-stenosis), high bifurcation or intracranial extension of a carotid lesion, provided that the peri-interventional stroke or death rate is not higher than that accepted for CEA (Class C recommendation).⁷⁹

AHA Guidelines state that angioplasty/stenting might be a reasonable alternative to CEA in *asymptomatic high-risk* patients. However, they stress that it remains uncertain whether this group of patients should have either procedure.¹⁰²

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94 Cerebral Arterial Dissections and Moyamoya Disease

94.1 Cerebral Arterial Dissections

94.1.1 Cerebral arterial dissections – key concepts

Key concepts

- hemorrhage into the tunica media of an artery
- may be spontaneous, posttraumatic, or iatrogenic (e.g., angiography-related), may be intracranial or extracranial
- may present with pain (usually ipsilateral H/A or carotidynia), Horner syndrome (in carotid dissections), TIA/stroke, or SAH
- extracranial dissections are usually treated medically (anticoagulation), intracranial dissections with SAH are treated surgically

This section primarily discusses “spontaneous” dissections. Cerebrovascular arterial dissections following blunt cervical trauma (p. 1028) are much more common.

94.1.2 Nomenclature

Some confusion has arisen because of inconsistent terminology in the literature. Although by no means standard, Yamaura¹ has suggested the following:

- ▶ **Dissection.** Extravasation of blood between the intima and media, creating luminal narrowing or occlusion.
- ▶ **Dissecting aneurysm.** Dissection of blood between the media and adventitia, or at the media, causing aneurysmal dilatation, which may rupture into the subarachnoid space.
- ▶ **Pseudoaneurysm.** Rupture of artery with subsequent encapsulation of the extravascular hematoma, may or may not produce luminal narrowing.

94.1.3 Pathophysiology

The lesion common to all dissections is hemorrhage outside of the vascular lumen due to pathological trans-intimal extravasation of blood from the true lumen into the vessel wall. The hematoma may either dissect the internal elastic membrane from the intima² causing narrowing of the true lumen, or it may dissect into the subadventitial plane producing an adventitial outpouching from the vessel wall (pseudoaneurysm). Rupture through the vessel wall producing SAH occurs occasionally.

Subintimal dissection is more common with intracranial dissections, whereas extracranial vessels (including the aorta) usually dissect at the media or between media and adventitia.

“Spontaneous” dissections have been associated with a large number of conditions, oftentimes the association is unproven. These conditions include:

- fibromuscular dysplasia (FMD): found in ≈ 15% of cases³
- cystic medial necrosis (or degeneration): originally thought to be a common finding, now thought to perhaps be linked to a higher likelihood of *fatal* dissection
- saccular aneurysm
- Marfan syndrome: autosomal dominant inherited disorder of connective tissue. Phenotypic manifestations are due to production of abnormal fibrillin, the main component of extracellular microfibrils, a component in the media of certain blood vessels, encoded by the FBN1 gene on chromosome 15q21
- Ehlers-Danlos syndrome
- atherosclerosis: only rarely implicated as an etiology. More likely to be a factor with subintimal dissection of *extracranial* arteries

- Takayasu's disease
- medial degeneration
- syphilitic arteritis (more common in the past, associated with 60% of dissections before 1950)
- autosomal dominant polycystic kidney disease (p. 1455): associated with a higher incidence of cerebral aneurysms
- variant periarteritis nodosa
- allergic arteritis
- homocystinuria
- moyamoya disease (p. 1581) ⁴
- strenuous physical activity

94.1.4 Epidemiology

Occurs primarily in middle-aged patients, with a mean age of ≈ 45 yrs (average age of traumatic dissections is slightly younger). More frequent in men.^{1,3} Incidence is unknown, since oftentimes the condition produces mild, transient symptoms. Increased awareness of the condition has resulted in an increased rate of diagnosis. ICA dissection accounts for 1–2.5% of first strokes.⁵ However, in middle-aged and young adults it comprises 10–25% of strokes.⁶

94.1.5 Sites of dissection

A review of 260 cases¹ (literature review+new cases) found the incidence by location shown in ► Table 94.1. The vertebral artery was the most common intracranial site. Previously, it was believed that the ICA was the most common site. This change may be due to the recent increased recognition of arterial dissections as a source of SAH (and vertebral dissections most often present as SAH). Multiple dissections occur in $\approx 10\%$ (the most common: bilateral vertebrobasilar lesions).

Table 94.1 Spontaneous intracranial dissections by site¹

Location	Left	Right	Total
vertebral	122	82	204
basilar		35	35
internal carotid	17	13	30
middle cerebral	16	10	26
anterior cerebral	10	3	13
posterior cerebral	7	9	16
PICA	4	10	14
Total	176	127	338

94.1.6 Clinical

Cerebral arterial dissections may cause symptoms by:

1. embolization secondary to:
 - a) platelet aggregation stimulated by the exposed surfaces
 - b) dislodged thrombus (formation of which is enhanced by reduced flow)
2. reduced distal flow secondary to:
 - a) thrombosis due to reduced flow
 - b) occlusion of the true lumen by the expansion of the mural hematoma
3. subarachnoid hemorrhage (atypical presentation, may be more common with posterior circulation dissection than with anterior circulation)⁷

The most common presentation in patients <30 yrs of age was due to internal carotid dissection without SAH. In patients >30 yrs, vertebrobasilar artery (VBA) dissection with SAH was the most common.¹

Headache, usually severe, often predates neurologic deficit by days or weeks (p.1492). See sections ICA (p. 1578) and vertebrobasilar (p. 1579) for specifics.

94.1.7 Evaluation

- ▶ **CT.** More useful for evaluating brain for infarction. Dissection can sometimes be visualized directly.⁸
- ▶ **CT angiogram (CTA).** Often obviates the need for cerebral angiography since CTA scanners with ≥ 16 detectors are equal in predictive value and have an accuracy near 99%.⁹
- ▶ **Angiography.** The definitive diagnostic study. However, diagnosis may be delayed if the dissection is misinterpreted as:
 1. an unusual saccular aneurysm (the most common error)
 2. atherosclerotic lesions: with dissections, the location is unusual, the lesion may be isolated, the age is usually younger, and the stenosis is smooth. Cervical ICA dissection typically spares the carotid bulb whereas cervical ICA atherosclerosis tends to involve the bulb.
 3. vasospasm following SAH: however, the narrowing with vasospasm is delayed in onset vs. the changes with dissection which are present from the beginning.

Angiographic findings may include:

1. luminal stenosis: irregular stenosis over long segments of the artery often with focal areas of near-total stenosis ("string sign")
2. fusiform dilation with proximal or distal narrowing ("string and pearl" sign)
3. occlusion: artery usually tapers to a point
4. intimal flap: when seen, usually found at proximal end of dissection
5. may see proximal beading ("string of beads" configuration, indicative of FMD)
6. "double lumen sign": true vessel lumen and an intramural false lumen with an intimal flap. Usually with retention of contrast within the false lumen well into the venous phase. The only pathognomonic sign
7. wavy "ripple" appearance
8. severe kinking (frequently bilateral). With VBA lesions: dolichoectasia

A characteristic of arterial dissections is that they often change configuration on repeat angiography¹⁰ (some resolve, and some worsen). NB: forceful intra-arterial contrast injection during the performance of angiography carries a potential for worsening the dissection.

- ▶ **MRI.** Probably not as accurate as CTA or angiography. Optimal MRI study is source *T1WI axial images with fat suppression* ("fat sat"), look for loss of visualization over several slices, with good visualization above and below. May visualize intimal flap and distinguish a dissection from a fusiform aneurysm.

Crescent sign: bright signal in wall of ICA (▶ Fig. 61.2) on T2WI axial images (hematoma in vessel wall).

94.1.8 Overall outcome

An early review of the literature found an 83% mortality within a few weeks of presentation with vertebralbasilar artery (VBA) dissection.¹¹ A later report tempered that grim prognosis.¹²

Based on a review of 260 cases,¹ an overall mortality of 26% was found. 70% had a favorable outcome (assessed using Glasgow Outcome scale, ▶ Table 98.4), 5% poor. Mortality was higher in ICA lesions (49%) than VBA lesions (22%). Mortality was 24% in the SAH group, and 29% in non-SAH cases.

94.1.9 Internal carotid dissection

See General information on arterial dissections (p.1576). Posttraumatic ICA dissection (p.1577) is much more common than spontaneous.

Some cases considered "spontaneous" may actually be due to trivial trauma, including violent coughing, nose blowing, and simple neck turning. Usually seen in young women.

In spontaneous dissection, the most common initial symptom is ipsilateral headache. Most of these (60%) are orbital or periorbital, but they may also be auricular or mastoid (39%), frontal (36%), temporal (27%). May also produce sudden onset of severe pain over carotid artery (carotidynia).¹³

Incomplete Horner syndrome (ocul sympathetic palsy): ptosis and miosis without anhidrosis (due to involvement of plexus around the ICA, sparing the ECA plexus which innervates facial sweat

glands) may occur. Bruits may be heard by the examiner or by the patient. These and other clinical features are shown in ► Table 94.2.

May be a cause of some cases of infantile and childhood hemiplegia and hemiparesis.¹⁴

Table 94.2 Clinical features of spontaneous ICA dissection³

Feature	%
focal cerebral ischemia	76
headache	59
oculosympathetic palsy	30
bruit	25
amaurosis fugax	10
neck pain	9
syncope	4
scalp tenderness	2
neck swelling	2

94.1.10 Vertebrobasilar system artery dissection

Vertebral artery dissections

General information

See general information on cerebral arterial dissections (p. 1576). See also posttraumatic vertebrobasilar dissection (p. 1032).

Less common than carotid. Extracranial lesions outnumber intracranial.

Traumatic dissections tend to occur where the VA crosses bony prominences, e.g., at the C1–2 junction or where it enters the foramen transversarium (usually at C6). Spontaneous dissections tend to be intracranial and commonly occur on the *dominant* VA. Unlike cervical ICA dissections, which tend not to propagate intracranially through the carotid canal, high cervical VA dissections can readily propagate intracranially through the foramen magnum.

Spontaneous VA dissections have been associated with FMD, migraine, and oral contraceptives.¹⁵ Unrecognized or forgotten trauma or sudden head motion may have occurred in some cases reported as spontaneous. Commonly occurs in young adults (mean age: 48 yrs). With spontaneous dissections, 36% of patients have dissections at other sites, 21% of cases have bilateral VA dissections.¹⁶

Dissecting aneurysms of the VA (possibly a distinct entity) are also described.^{17,18,19} They tend to be fusiform, and may be amenable to clipping, and were associated with vertebral dissections in 5 of 7 cases reported in one series.²⁰ As of 1984, only ≈ 50 cases of dissecting aneurysms were published.²⁰

Presentation

In spontaneous extradural dissections, neck pain is a prominent early finding in most patients, and is commonly located over the occiput and posterior cervical region. Generalized severe headache is also common. TIAs or stroke (usually lateral medullary syndrome (p. 1539)²¹ or cerebellar infarction, especially in patients with occlusion of the third or fourth portion of the VA²²). None of 5 patients developed new neurologic symptoms after the original stroke in an average of 21 months follow-up.²² In 3 of these 5, VA dissection was bilateral.

Dissecting aneurysms may present with altered consciousness, and may cause SAH (seen in 6 of 30 cases of vertebrobasilar complex dissections).²⁰ Rebleeding occurs in 24–30% of those cases presenting with SAH,¹⁶ making these lesions treacherous, with a very high mortality.^{23,24}

Traumatic extradural dissections or pseudoaneurysms may have a similar presentation, but can also produce massive external hemorrhage or neck hematomas.¹⁶

Evaluation

See section under Cerebral arterial dissections, Evaluation (p. 1578).

► **Angiography.** Diagnosis by angiography may be difficult in many cases (the most common misdiagnosis is ruptured saccular aneurysm of unusual shape²⁵).

In posttraumatic dissections, the most common finding is irregular stenosis of horizontal loops of distal extracranial VAs as they pass behind C1, often bilateral.

In 14 of 15 posttraumatic VA dissections, the lesion was located posterior to the atlas (distal extracranial 3rd segment), the single exception being a patient with direct trauma causing proximal VA involvement. This predilection is possibly explained by the fact that the first and third portions of the VA are movable, whereas the second and fourth are relatively immobilized by bone.

Treatment

Except for cases presenting with hemorrhage or large ischemic stroke, medical therapy should be started emergently. Classically consists of anticoagulation, with heparin acutely, followed by oral agents (e.g., Coumadin) probably for a total of 6 months. Recent preliminary study showed antiplatelet therapy was equally as effective.²⁶

As with traumatic dissections, endovascular techniques are now assuming a more prominent role in management.

► **Indications for intervention.** Surgery or endovascular techniques (mostly stents, but also occlusion, angioplasty¹⁶) are required for dissections presenting with SAH (due to their propensity to rebleed) and is recommended for most intradural dissections. For extradural lesions it is indicated for dissections that progress (angiographically) or for persistent symptoms in spite of adequate medical therapy. Some less malignant lesions may be amenable to endovascular stenting.

► **Endovascular treatment.** Balloon-mounted, self-expandable, or covered stents have been used relatively infrequently to treat dissections of the internal carotid or vertebral arteries, with good technical results and low procedure-related complication rates.²⁷ Given the small number of patients treated and the fact that medical therapy is generally effective, the role of stenting for dissection remains to be defined. It should be reserved for patients in whom medical therapy is ineffective or contraindicated or when the dissection causes symptomatic flow-limiting stenosis.

► **Surgical treatment.** At the time of surgery, the site of dissection may be recognized by fusiform or tubular enlargement of the artery with discoloration due to blood within the arterial wall (the discoloration has been described as black, bluish, purple, purple-red, or brown²⁵).

Surgical treatment of intradural dissection when endovascular techniques are not an option includes the following alternatives:

1. non-clippable aneurysms may be candidates for Hunterian occlusion of the VA proximal to the BA (either by microsurgical technique, or by endovascular techniques which may not be as precise). Some may not tolerate clipping the dominant VA, especially if the contralateral VA is hypoplastic. Conversely, some may tolerate bilateral VA occlusion.²⁸ Balloon test occlusion¹⁶ is recommended
 - a) if the dissection involves the PICA origin, then clip proximal to dissection. PICA then fills from retrograde flow, and the reversal of flow across the site of dissection should push the intima back against the wall
 - b) if the dissection is proximal to PICA and doesn't involve PICA, then trap the aneurysm between clips. PICA fills by retrograde flow
 - c) if the aneurysm begins distal to the PICA origin, occlude the VA⁷ distal to the PICA takeoff²⁹
2. combining VA clipping (non-clippable aneurysms may be candidates for Hunterian occlusion of the VA proximal to the aneurysm) with vascular bypass, options:
 - a) side-to-side PICA-PICA anastomosis
 - b) transplantation of the PICA origin to the VA outside the aneurysm
 - c) occipital artery-to-PICA bypass
3. resection accompanied by autogenous interposition vein graft
4. non-occlusive surgical techniques
 - a) clipping with specially designed clips for fusiform aneurysms (e.g., Sundt-Kees clip)
 - b) wrapping: of dubious benefit

94.1.11 Vertebrobasilar system dissections excluding the VA

Basilar artery dissections tend to present with brainstem infarction and more rarely with SAH.²⁴ The prognosis is generally regarded as poor. Endovascular techniques may be able to treat some.

94.2 Moyamoya disease

94.2.1 General information

Key concepts

- progressive bilateral spontaneous occlusion or stenosis of terminal ICAs with compensatory capillary collaterals that look like a “puff of smoke” (Japanese: moyamoya) on angio
- typical presentation: juvenile form → ischemic infarcts/TIAs (suspect diagnosis in any child presenting with TIAs). Adult form → hemorrhage
- pathology: intimal thickening w/o inflammation, may also involve heart, kidneys. Associated aneurysms may be source of bleeding
- evaluation: cerebral angiography is necessary to delineate degree of stenosis as well as to evaluate potential extracranial donor vessels for revascularization. Also identifies aneurysms
- treatment:
 - a) medical treatment (antiplatelet drugs, anticoagulation, vasodilators...): not shown to be effective, although antiplatelet/anticoagulation is often used
 - b) surgical revascularization: reduces the incidence of strokes and TIAs, but benefit on reducing the rate of hemorrhage is unproven

Progressive spontaneous occlusion of one or usually both ICAs (usually at the level of the carotid siphon) and their major branches, with secondary formation of an anastomotic collateral capillary network at the base of the brain which has been termed “moyamoya,” the Japanese word for something hazy like a “puff of cigarette smoke”³⁰ (which it fancifully resembles on angiography) (first described in 1957,³¹ and named in 1969³⁰). With progression, involvement includes the proximal MCAs and ACAs and on rare occasion the vertebrobasilar system. Associated aneurysms (see below) and rarely AVMs^{32,33} may be observed.

Eventually the dilated capillary (moyamoya) vessels disappear with the development of collaterals from the ECA (meningeal collaterals are called “rete mirabile”).

94.2.2 Pathophysiology

Primary moyamoya disease

The most common pathology is stenosis of the proximal anterior and middle cerebral arteries that is neither atherosclerotic nor inflammatory in origin. Exact etiology is unknown but some studies show elevated basic fibroblast growth factor in the dura and scalp arteries in patients with moyamoya.³⁴ The internal elastic lamina of affected vessels may be thinned or duplicated. Similar vascular changes may also occur in the heart, kidney, and other organs, suggesting it may be a systemic vascular disease.

Secondary moyamoya disease

AKA “quasi-moyamoya disease” or “moyamoya syndrome.”³⁵ Angiographic findings of moyamoya associated with e.g.:

1. Graves' disease/thyroidosis
2. history of cerebral inflammatory disease, including meningitis (especially tubercular (TB) meningitis and leptospirosis)
3. retinitis pigmentosa
4. vascular disorders: atherosclerosis, fibromuscular dysplasia, pseudoxanthoma elasticum
5. congenital disorders: Down syndrome, Marfan syndrome, Turner syndrome, neurofibromatosis type 1, tuberous sclerosis, Apert syndrome
6. hematologic disorders: Fanconi anemia, sickle cell disease (in the U.S. one of the more common associations) and sickle cell trait
7. following radiation therapy for skull base glioma in children³⁶
8. head trauma
9. systemic lupus erythematosus (SLE)

Associated aneurysms

Intracranial *aneurysms* are frequently associated with moyamoya disease (MMD). This may be a result of the increased flow through dilated collaterals, or it may be that patients with moyamoya may also have a congenital defect in the arterial wall that predisposes them to aneurysms. 3 types: 1) usual sites of aneurysms in the Circle of Willis, 2) in peripheral portions of cerebral arteries, e.g., posterior/anterior choroidal, Heubner's, and 3) within moyamoya vessels. The frequency of aneurysms in the vertebrobasilar system is $\approx 62\%$, which is much higher than in the general population.³⁷ Aneurysmal SAH may be the actual cause of some hemorrhages that were erroneously attributed to moyamoya vessels.

94.2.3 Epidemiology

Risk factors

A history of inflammation in the head & neck region has been implicated.

Demographics

Incidence in Japan is higher (0.35/100,000/yr) than in North America. Two peaks (may not be same disease): juvenile (highest peak), age < 10 yrs (mean 3); adult, 3rd & 4th decade. Slight female predominance (1.8:1). Some evidence for familial tendency (some Asian families have an incidence as high as 7%), genetics appears autosomal dominant with low penetrance. Associated with some HLA antigens (B40 in juvenile form; B54(20) in adult) and anti-double-stranded DNA antibody.

94.2.4 Presentation

Juvenile form

Moyamoya is associated with 6% of childhood strokes.³⁴ Ischemic presentation more common (81%); includes TIAs (41%) which may alternate sides (alternating hemiplegia is a suggestive clinical finding), RINDs, or infarct (40%). Neurologic events are often provoked by straining or by hyperventilation (e.g., during crying or blowing a wind instrument), which is thought to produce hypocapnea with reactive vasoconstriction.

Headache is the most common presenting symptom, but seizures, focal neurologic deficits, choreoathetotic movements and hemorrhages can also be presenting symptoms. The risk of hemorrhage is increased in stages 5 & 6 of MMD.

Adult form

Hemorrhage has been described as being more common (60%), but in a Stanford series³⁸ 89% presented with ischemia. Rupture of the fragile moyamoya vessels produces bleeding in the basal ganglia (BG), thalamus, or ventricles (from the ventricular wall) in 70–80% of hemorrhages. SAH may occur, usually due to rupture of associated aneurysms (see above). In the pre-CT era, the most common form of hemorrhage was thought to be SAH from the rupture of moyamoya vessels, but most cases were probably intraventricular blood or SAH from associated aneurysms.³⁹

94.2.5 Natural history

Incidence of disease progression in one study was 20% in adult patients with MMD.⁴⁰ Female patients had a higher risk of disease progression than males.

Prognosis of untreated MMD is poor, with 73% rate of major deficit or death within 2 years of diagnosis in children, and similarly poor outlook in adults.³⁸

94.2.6 Evaluation and diagnosis

Diagnostic criteria

Diagnosis of moyamoya requires bilateral symmetrical stenosis or occlusion of the terminal portion of the ICAs as well as the presence of dilated collateral vessels at the base of the brain.³⁴ (If unilateral, the diagnosis is considered questionable,⁴¹ and these cases may progress to bilateral involvement).

Other characteristic findings include:

1. stenosis/occlusion starting at termination of ICA and at origins of ACA and MCA

2. abnormal vascular network in region of BG (intraparenchymal anastomosis)
3. transdural anastomosis (rete mirabile), AKA “vault moyamoya.” Contributing arteries: anterior facial, middle meningeal, ethmoidal, occipital, tentorial, STA
4. moyamoya collaterals may also form from internal maxillary artery via ethmoid sinus to fore-brain in frontobasal region

CT

Work-up in suspected cases typically begins with a non-enhanced head CT. Up to 40% of ischemic cases have normal CT. Low-density areas (LDAs) may be seen, usually confined to cortical and sub-cortical areas (unlike atherosclerotic disease or acute infantile hemiplegia which tend to have LDAs in basal ganglia as well). LDAs tend to be multiple and bilateral, especially in the PCA distribution (poor collaterals), and are more common in children.

MRI and MRA

MRA usually discloses the stenosis or occlusion of the ICA. Moyamoya vessels appear as flow voids on MRI (especially in basal ganglia) and a fine network of vessels on MRA, and are demonstrated better in children than adults. Parenchymal ischemic changes are commonly shown, usually in watershed areas.

Angiography

In addition to helping to establish the diagnosis, angiography also identifies suitable vessels for revascularization procedures and unearths associated aneurysms. The angiography-related complication rate is higher than with atherosclerotic occlusive disease. Avoid dehydration prior to and hypotension during the procedure. Six angiographic stages of MMD that tend to progress up until adolescence and stabilize by age 20 are described in ► Table 94.3.³⁰

Table 94.3 Six angiographic stages of MMD (Suzuki stages)³⁰

Stage	Finding
1	stenosis of suprasellar ICA, usually bilateral
2	development of moyamoya vessels at base of brain. ACA, MCA, & PCA dilated
3	increasing ICA stenosis & prominence of moyamoya vessels (most cases diagnosed at this stage). Maximal basal moyamoya
4	entire circle of Willis and PCAs occluded, extracranial collaterals start to appear, moyamoya vessels begin to diminish
5	further progression of stage 4 with intensification of ECA collaterals & reduction of moyamoya associated vessels
6	total occlusion of ICA & major cerebral arteries and complete absence of moyamoya vessels

EEG

Non-specific in the adult. Juvenile cases: high-voltage slow waves may be seen at rest, predominantly in the occipital and frontal lobes. Hyperventilation produces a normal buildup of monophasic slow waves (delta-bursts) that return to normal 20–60 seconds after hyperventilation. In >50% of cases, after or sometimes continuous with buildup is a second phase of slow waves (this characteristic finding is called “rebuildup”), which are more irregular and slower than the earlier waves, and usually normalize in ≤10 minutes.⁴²

Cerebral blood flow (CBF) studies

CBF is decreased in children with MMD, but relatively normal in adults. There is a shift of CBF from the frontal to the occipital lobes⁴³ probably reflecting the increasing dependency of CBF on the posterior circulation. Children with MMD have impaired autoregulation of CBF to blood pressure and CO₂ (with more impairment of vasodilatation in response to hypercapnia or hypotension than vasoconstriction in response to hypocapnia or hypertension).⁴⁴

Xenon (Xe-133) CT can identify areas of low perfusion. Repeating the study after an acetazolamide challenge (which causes vasodilatation) evaluates reserve capacity of CBF and can identify areas of “steal” which are at high risk of future infarction.

94.2.7 Treatment

General information

No medical or surgical treatment has been proven effective in reducing the rate of hemorrhage in the adult with MMD. However, multiple large case series have supported the efficacy of cerebral revascularization for reducing the incidence of ischemic strokes and TIAs.³⁵

Asymptomatic moyamoya disease

Guidelines for management of asymptomatic moyamoya disease have not yet been established. A multicenter, nationwide survey in Japan focusing on asymptomatic moyamoya disease provided the following findings⁴⁵: subtle findings of cerebral infarction and disturbed cerebral hemodynamics were detected in 20% and 40% of the involved hemispheres, respectively. Angiographic stage was more advanced in elderly patients. Of 34 medically-treated patients, 7 experienced TIA, ischemic stroke, or hemorrhage during a mean follow-up period of 43.7 months. Cerebral infarction or hemorrhage did not occur in the 6 patients who underwent surgical revascularization.

Medical treatment

Medical treatment with platelet inhibitors, anticoagulants, calcium channel blockers,³⁸ steroids, mannitol, low-molecular-weight dextran, and antibiotics have not proven to be of benefit. Steroids may be considered for involuntary movements and acutely during recurrent TIAs.

Surgical treatment

General information

Patients with mass effect from clot may be candidates for urgent decompression. Revascularization procedures, however, should be performed when the patient is stable under nonemergent conditions.

Perioperative management

During any surgical procedure:

1. avoid hyperventilation: due to increased sensitivity of collaterals, keep PaCO₂ 40–50 mm Hg to avoid ischemic infarction
2. avoid hypotension: maintain BP at normotensive levels
3. avoid alpha-adrenergic agents because of vasoconstrictive effects
4. cerebral protection: mild hypothermia (32–34 °C)⁴⁶ and barbiturates are routinely used
5. papaverine helps prevent vascular spasm

Postoperatively following STA-MAC bypass procedures:

1. avoid hypertension: may cause bleeding at anastomotic site and in areas of increased perfusion within the brain
2. avoid hypotension: may result in graft occlusion
3. aspirin is started on the post-op day #1
4. watch for evidence of CSF leak
5. monitor coag studies and correct abnormalities
6. cerebral arteriogram is recommended 2–6 months post-op

Suggested criteria for revascularization procedures

See reference.³⁵

1. patients presenting with infarction or hemorrhage but are in good neurologic condition
2. infarction < 2 cm maximal diameter on CT, and all previous hemorrhages have completely resolved
3. angiographic stage is II–IV (see ► Table 94.3)
4. timing of operation: ≥ 2 months after most recent attack

Surgical revascularization options

Various methods to revascularize the ischemic brain, used primarily in children, include:

1. direct revascularization procedures:

- a) results are superior to indirect revascularization procedures^{47,48} if a donor and recipient vessel of sufficient caliber (≥ 1 mm outer dia) can be identified (may be difficult in the pediatric age group who are the most likely to benefit⁴⁹). Otherwise, indirect revascularization procedures (see below) are options.
- b) Among direct revascularization procedures, STA-MCA bypass⁵⁰ is the procedure of choice.
2. indirect revascularization procedures: usually reserved for younger patients (suggested cutoff age $\approx < 15$ years). May be combined with STA-MCA bypass. Includes:
 - a) encephalomyosynangiosis (EMS): laying the temporalis muscle on the surface of the brain (may cause problems with muscle contractions during talking and chewing, and neural impulses on surface of brain)
 - b) encephaloduroarteriosynangiosis (EDAS)^{51,52}: suturing the STA with a galeal cuff to a linear defect created in the dura. Variations on this technique include splitting the dura⁵³
 - c) omental transposition⁵⁴: either as a pedicle graft or as a vascularized free flap. Felt to have higher potential to revascularize ischemic tissue than above procedures, but there is greater risk of mass effect from the thickness of the omentum
3. the above indirect revascularization procedures improve blood flow in the MCA distribution, but not ACA circulation. This may be rectified by:
 - a) simple placement of frontal burr holes with opening of the underlying dura and arachnoid⁵⁵
 - b) "ribbon EDAS" where a pedicle of galea is inserted into the interhemispheric fissure on both sides⁵⁶
4. stellate gangliectomy and perivascular sympathectomy: unproven that this increases CBF permanently

Outcome with surgical treatment

Neurologic status at time of treatment generally predicts long-term outcome.³⁴ The mortality rate in adults ($\approx 10\%$) is higher than for juveniles ($\approx 4.3\%$).⁴¹ The cause of death was bleeding in 56% of 9 children and 63% of 30 adults. With treatment the prognosis is good in 58%.³⁹

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95 Other Vascular Occlusive Conditions

95.1 Totally occluded internal carotid artery

95.1.1 General information

10–15% of patients presenting with carotid territory stroke or transient ischemic attacks (TIA) are found to have carotid occlusion. This amounts to an estimated 61,000 first-ever strokes and 19,000 TIAs per year in the United States. Prevention of subsequent stroke in symptomatic patients with carotid artery occlusion remains a difficult challenge. The overall rate of subsequent stroke is 7% per year for all stroke and 5.9% per year for ischemic stroke ipsilateral to the occluded carotid artery.¹ These risks persist even despite treatment with antiaggregants and anticoagulants.² The prevalence of asymptomatic carotid occlusion is not known, and the incidence of ipsilateral stroke in never-symptomatic carotid occlusion is negligible.³

95.1.2 Presentation

3 patterns of stroke with total internal carotid artery occlusion:

1. whole hemisphere stroke
2. watershed infarct
3. **carotid stump syndrome**: continued cerebral or retinal symptoms following complete occlusion of the ipsilateral ICA.⁴ Rare. Both hemodynamic and embolic mechanisms have been implicated⁵:
 - a) hemodynamic: inadequacy of collaterals may lead to recurrent ischemic symptoms
 - b) emboli formed in the turbulent flow at the ostium of the stump can reach MCA-supplied brain tissue via ECA-ICA anastomoses either directly or through reverse flow through the ophthalmic artery. Other pathways have also been described⁶

Treatment options include: stump excision⁷ or endovascular treatment with stenting across the stump.⁸ If there is ipsilateral ECA stenosis, either of these may be combined with ECA endarterectomy or angioplasty ± stenting. Most patients also receive post-treatment dual antiplatelet therapy.

In symptomatic patients⁹: hemiparetic TIA 53%, dysphasic TIA 34%, fixed neuro deficit 21%, crescendo TIAs 21%, amaurosis fugax 17%, acute hemiplegia 6%. 27% were asymptomatic in one series.¹⁰

95.1.3 Natural history

See reference.¹¹

Patients with mild deficit and angiographically proven ICA occlusion have a stroke rate (in two series) of 3 or 5% per year (2 or 3.3% related to occluded side). In patients with acute ICA occlusion and profound neurological deficit, 2–12% make good recovery, 40–69% will have profound deficit, and 16–55% will have died by the time of follow-up.

95.1.4 Endovascular thrombolysis and stenting for acute carotid occlusion

Case reports and series of endovascular treatment of internal carotid artery occlusion have confirmed the feasibility of this technique. Intra-arterial thrombolysis within 6 hours of stroke onset may increase recanalization rates to 37–100% and clinical improvement to 53–94% without significant increase in hemorrhagic transformation when compared with intravenous thrombolytic therapy alone.^{12,13,14,15,16,17} Although results appear promising, randomized controlled trials on cervical carotid thrombolysis and/or stenting are lacking.

95.1.5 Surgery

Options include: endarterectomy, Fogarty balloon catheter embolectomy (utilizing a No. 2 French catheter with 0.2-ml balloon gently passed 10–12 cm up ICA from small arteriotomy made distal to atheromatous plaque¹⁸), extracranial-intracranial bypass. Restored patency rate is inversely related to suspected duration of occlusion. Chronically occluded ICA has poor patency rate and little gain from re-opening.

Determining the exact time of occlusion is frequently impossible. One must often rely on clinical grounds; therefore an occasional chronic occlusion will be included.

Retrograde filling of ICA to petrous or cavernous segment from ECA (e.g., via ophthalmic) or from contralateral ICA is a good sign of operability.⁹

Surgical results⁹

32% (15/47 cases) immediate surgical failures (no or minimal back bleeding), at least 3 deaths. Among immediate successes no strokes and no TIAs. If operated <2 days reported patency rate 70–100%, from 3–7 days 50–100%, 8–14 days 27–58%, 15–30 days 4–61%, over 1 month (2 series) 20–50%.

95.1.6 Guidelines

Emergency operations for acute neuro deficit associated with total occlusion should not be performed after about 2 hrs. Extremely poor neuro status (lethargy/coma) is a contraindication to surgery. Patients without persistent neuro deficit: operate ASAP. If the patient has recurrent TIAs (despite maximal medical therapy) following recent carotid occlusion, and no definite infarct on MRI, consider bypass surgery.

95.2 Cerebellar infarction

95.2.1 General information

Relatively rare (seen on only 0.6% of all CTs obtained for any reason¹⁹). Cerebellar infarcts may be classified as involving the PICA distribution (cerebellar tonsil and/or inferior vermis), superior cerebellar artery distribution (superior hemisphere or superior vermis), or other indeterminate patterns.²⁰ 80% of patients developing signs of brainstem compression will die, usually within hours to days.

95.2.2 Early clinical findings

In most cases the onset is sudden, without premonitory symptoms.²¹ The first 12 hrs after onset were characterized by lack of progression. Early findings are due to the *intrinsic* cerebellar lesion (ischemic infarction or hemorrhage):

1. symptoms
 - a) dizziness or vertigo
 - b) nausea/vomiting
 - c) loss of balance, often with a fall and inability to get up
 - d) headache (infrequent in one series²¹)
2. signs
 - a) truncal and appendicular ataxia
 - b) nystagmus
 - c) dysarthria

95.2.3 Later clinical findings

Patients with cerebellar infarction may subsequently develop increased pressure within the posterior fossa (due to cerebellar edema or mass effect from clot), with brainstem compression (particularly posterior pons). Clinical findings generally increase between 12 and 96 hrs following onset. Compression of the Sylvian aqueduct can cause acute hydrocephalus with attendant increased ICP.

95.2.4 Imaging studies

CT scan: may be normal very early in these patients. There may be subtle findings of a tight posterior fossa: compression or obliteration of basal cisterns or 4th ventricle or hydrocephalus.

MRI: (including DWI) more sensitive for ischemia, especially in the posterior fossa.

95.2.5 Surgical indications

Surgical decompression (see below) should probably be done as soon as any of the following signs develop if there is no response to medical therapy.²² It is important to recognize a lateral medullary syndrome (LMS) (p. 1539) which may often accompany a cerebellar infarct. With LMS, the signs are usually present from the onset (dysphagia, dysarthria, Horner syndrome, ipsilateral facial numbness, crossed sensory loss...), and are not accompanied by a change in sensorium. There is no place for surgical decompression in LMS since it represents primary brainstem ischemia and not compression.

Findings proceed in the approximate following sequence if there is no intervention:

1. abducens (VI) nerve palsy
2. loss of ipsilateral gaze (compression of VI nucleus and lateral gaze center)
3. peripheral facial nerve paresis (compression of facial colliculus)
4. confusion and somnolence (may be partly due to developing hydrocephalus)
5. Babinski sign
6. hemiparesis
7. lethargy
8. small but reactive pupils
9. coma
10. posturing → flaccidity
11. ataxic respirations

95.2.6 Suboccipital craniectomy for cerebellar infarction

Unlike the situation with supratentorial masses causing herniation, there are several reports of patients in deep coma from direct brainstem compression who were operated upon quickly and made useful recovery.^{22,23,24} See also Guidelines for patients with cerebellar hemorrhage (p. 1625).

The operation of choice is a suboccipital decompression to include enlargement of the foramen magnum. The dura is then opened and the infarcted cerebellar tissue usually exudes “like tooth-paste” and is easily aspirated. Avoid using ventricular drainage alone as this may cause upward cerebellar herniation (p. 325) and does not relieve the direct brainstem compression.

95.3 Malignant middle cerebral artery territory infarction

95.3.1 General information

A distinct syndrome that occurs in up to 10% of stroke patients,^{25,26} which carries a mortality of up to 80% (mostly due to severe postischemic cerebral edema → increased ICP → herniation).²⁶

Patients usually present with findings of severe hemispheric stroke (hemiplegia, forced eye and head deviation) often with CT findings of major infarct within the first 12 hours. Most develop drowsiness shortly after admission. There is progressive deterioration during the first 2 days, and subsequent transtentorial herniation usually within 2–4 days of stroke. Fatalities are often associated with: severe drowsiness, dense hemiplegia, age > 45–50 yrs,²⁷ early parenchymal hypodensity involving > 50% of the MCA distribution on CT scan,²⁸ midline shift > 8–10 mm, early sulci effacement, and hyperdense artery sign (p. 1560) ²⁷ in MCA.

Neurosurgeons may become involved in caring for these patients because aggressive therapies in these patients may reduce morbidity and mortality. Options include:

1. conventional measures to control ICP (with or without ICP monitor): mortality is still high in this group, and elevated ICP is not a common cause of initial neurologic deterioration in large hemispheric stroke
2. hemicraniectomy (decompressive craniectomy): see below
3. ✖ to date, the following treatments have not improved outcome: agents to lyse clot, hyperventilation, mannitol, or barbiturate coma

95.3.2 Hemicraniectomy for malignant MCA territory infarction

May reduce mortality to as low as 32% in nondominant hemisphere strokes²⁹ (37% in all comers³⁰) with surprising reduction of hemiplegia, and in dominant hemisphere strokes, with only mild-moderate aphasia (better results occur with early surgery, especially if surgery is performed *before* any changes associated with herniation occur). Meta-analysis³¹ of 3 randomized controlled trials found that hemicraniectomy within 48 hours after stroke onset resulted in decreased mortality and increased the number of patients with a favorable functional outcome.

Indications: No firm indications. Guidelines:

1. age < 70 years
2. more strongly considered in nondominant hemisphere (usually right)
3. clinical & CT evidence of acute, complete ICA or MCA infarcts and direct signs of impending or complete severe hemispheric brain swelling (severe post-admission neurologic deterioration is the usual event that triggers surgical intervention)

See also **Technique** (p. 1766).

95.4 Cardiogenic brain embolism

95.4.1 General information

About one stroke in six is cardioembolic. Emboli may be composed of fibrin-rich thrombi (e.g., mural thrombi due to segmental myocardial hypokinesis following MI or ventricular aneurysm), platelets (e.g., nonbacterial thrombotic endocarditis), calcified material (e.g., in aortic stenosis), or tumor particles (e.g., atrial myxoma).

► **Following acute myocardial infarction (AMI).** 2.5% of patients will have a stroke within 1–2 weeks of an AMI (the period when most emboli occur). The risk is higher with anterior wall MI ($\approx 6\%$) vs. inferior wall MI ($\approx 1\%$).

► **Atrial fibrillation (a-fib).** Nonrheumatic patients with a-fib have a 3–5-fold increased risk of stroke,³² with a 4.5% rate of stroke per year without treatment.³³ The incidence of a-fib in the U.S. is 2.2 million. About 75% of strokes in patients with a-fib are due to left atrial thrombi.³⁴ Independent risk factors for stroke in patients with a-fib are: advanced age, prior embolism (stroke or TIA), HTN, DM, and echocardiographic evidence of left atrial enlargement or left ventricular dysfunction.³²

CHADS₂ scoring system (► Table 95.1) for risk of stroke with a-fib has been widely validated.³⁵ Points are totaled and risk is shown in ► Table 95.2. If CHADS₂ score is ≥ 2 , warfarin therapy is significantly protective for out-of-hospital death or hospitalization for stroke, MI, or hemorrhage (CI = 0.61–0.91).³⁶

Table 95.1 CHADS₂ scoring items

Item	Points
CHF (any history)	1
HTN (prior history)	1
Age > 75 yrs	1
Diabetes mellitus	1
Secondary prevention: in patients with prior ischemic stroke or TIA; most also include systemic embolic events	2

Table 95.2 Risk based on CHADS₂ score

CHADS ₂ score	Annual stroke risk (%/year)
0	1.9
1	2.8
2	4
3	5.9
4	8.5
5	12.5
6	18.2

► **Prosthetic heart valves.** Patients with mechanical prosthetic heart valves on long-term anticoagulation have an embolism rate of 3%/year for mitral and 1.5%/year for aortic valves. With bioprosthetic heart valves and no anticoagulation, the risk is 2–4%/year.

► **Paradoxical embolism.** Paradoxical embolism can occur with a patent foramen ovale which is present in 10–18% of the general population, but in up to 56% of young adults with unexplained stroke.³⁷

► **Endocarditis.** Blood cultures and TEE help evaluate.

95.4.2 Diagnosis of cardiogenic brain embolism

General information

No specific neurologic features can distinguish these patients. The diagnosis is suggested in imaging studies showing multiple intracranial ischemic strokes in different arterial distributions, the

differential diagnosis includes: vasculitis, intracranial atherosclerosis (focal plaques, more common in Asian populations that consume Western diets), and intravascular lymphomatosis.

The diagnosis of cardiogenic brain embolism (CBE) as a cause of a stroke relies on demonstrating a potential cardiac source, the absence of cerebrovascular disease, and non-lacunar stroke.

Large areas of hemorrhagic transformation within an ischemic infarct may be more indicative of CBE due to thrombolysis of the clot and reperfusion of infarcted brain with subsequent hemorrhagic conversion. Hemorrhagic transformation most often occurs within 48 hrs of a CBE stroke, and is more common with larger strokes.

Detection of cardiac source

Most centers rely on echocardiography (without transesophageal ability). Using restricted criteria (i.e., excluding mitral valve prolapse), about 10% of patients with ischemic stroke will have potential cardiac source detected by echo, and most of these patients have other manifestations of cardiac disease. In stroke patients without clinical heart disease, only 1.5% will have a positive echo; the yield is higher in younger patients without cerebrovascular disease.³⁸

EKG may detect atrial fibrillation, which may be seen in 6–24% of ischemic strokes and may be associated with a 5-fold increased risk of stroke (see below).

95.4.3 Treatment

CBE is essentially the only condition for which anticoagulation has been shown to significantly reduce the rate of further strokes.

One must balance the risk of recurrent emboli (12% of patients with a cardioembolic stroke will have a second embolic stroke within 2 weeks) against that of converting a pale infarct into a hemorrhagic one. No study has shown a clear benefit of *early* anticoagulation.

Recommendations for anticoagulation:

1. if anticoagulation is to be used, it should not be instituted within the first 48 hrs of a probable CBE stroke
2. CT should be obtained after 48 hrs following a CBE stroke and before starting anticoagulation (to R/O hemorrhage)
3. anticoagulation should not be used in the face of large infarcts
4. start heparin and warfarin simultaneously. Continue heparin for 3 days into warfarin therapy, see Anticoagulation (p. 163)
5. optimal range of oral anticoagulation to minimize subsequent embolism and/or hemorrhage has not been determined, but pending further data, an INR of 2–3 appears satisfactory
6. patients with asymptomatic a-fib have 66–86% reduction in stroke risk with warfarin (Coumadin®).^{32,39} ASA is only about half as effective, but may be sufficient for those without associated risk factors (p. 1590)³²

95.5 Vertebrobasilar insufficiency

95.5.1 General information

Signs and symptoms resulting from inadequate blood flow through the posterior cerebral circulation (vertebral arteries, basilar artery and their branches).

95.5.2 Symptoms

► Table 95.3 shows a mnemonic of the symptoms of vertebrobasilar insufficiency (VBI). Predicting the site of the lesion based only on clinical evaluation is very unreliable.

Diagnostic criteria for VBI are shown in ► Table 95.4.

Table 95.3 Mnemonic: “The 5 D’s of VBI”

- “drop attack”
- diplopia
- dysarthria
- defect (visual field)
- dizziness

Table 95.4 Criteria for clinical diagnosis of VBI

Clinical diagnosis requires 2 or more of the following

- motor or sensory symptoms or both, occurring bilaterally in the same event
- diplopia: ischemia of upper brainstem (midbrain) near ocular nuclei
- dysarthria: ischemia of lower brainstem
- homonymous hemianopsia: ischemia of occipital cortex (NB: this is *binocular*, in contrast to amaurosis fugax which is monocular)

VBI may also be suspected in a patient with transient episodes of “dizziness” (vertigo that is otherwise unexplained, e.g., absence of orthostatic hypotension or benign positional vertigo) that are initiated by positional changes. VBI may sometimes be due to compression of the VA at the C1–2 level with:

1. head turning (see below)
2. os odontoideum (p. 1175)
3. anterior atlantoaxial subluxation: e.g., in rheumatoid arthritis (p. 1377)
4. with rotatory atlantoaxial subluxation (p. 1158)

95.5.3 Pathophysiology

Atheromatous and stenotic lesions of the posterior circulation occur most frequently at the VA origin.

VBI symptoms may be due to:

1. hemodynamic insufficiency (may be the most common etiology), including:
 - a) subclavian steal: reversed flow in VA due to proximal stenosis of subclavian artery
 - b) stenosis of both VAs or of one VA where the other is hypofunctional (e.g., hypoplastic, occluded, or terminates in PICA) causing reduced distal flow in face of inadequate collaterals (see below)
2. embolism from ulcerations
3. atherosclerotic occlusion of brainstem perforators
4. vertebrobasilar hypoplasia: reported as a possible etiology for cerebellar stroke.

95.5.4 Natural history

No clinical study accurately defines the natural history. The estimated stroke rate is 22–35% over 5 years, or 4.5–7% per year⁴⁰ (one study estimating 35% stroke rate in 5 years did not use angiography).

Risk of stroke after first VBI-TIA has been estimated as 22% for first year.⁴¹

95.5.5 Evaluation

Adequate investigation usually requires selective four-vessel angiography,⁴² sometimes with provocative maneuvers (see e.g., Bow hunter's stroke (p. 1592)). CTA may also be useful.

95.5.6 Treatment

Anticoagulation is the mainstay of medical management. Alternatives include anti-platelet drugs such as ASA (efficacy of either remains unproven^{40,42}).

Surgical treatment includes:

1. vertebral endarterectomy
2. transposition of VA to ICA (with or without carotid endarterectomy, with or without saphenous vein patch graft) or to thyrocervical trunk or to subclavian artery⁴³
3. bypass grafting (e.g., occipital artery to PICA)
4. C1–2 posterior arthrodesis (p. 1778) may prevent potentially life-threatening stroke in cases of os odontoideum (p. 1176)

95.6 Bow hunter's stroke

95.6.1 General information

A special subset of VBI. The term was coined in 1978 by Sorensen.⁴⁴ Bow hunter's stroke (BHS): hemodynamic VBI induced by intermittent VA occlusion resulting from head rotation⁴⁵ (ischemic

sequelae range from TIA (bow hunter's sign) to completed stroke). May occur with forced (e.g., with chiropractic neck manipulation⁴⁶) or voluntary⁴⁷ head rotation.

Occlusion usually involves the VA *contralateral* to the direction of rotation, and usually occurs at the C1–2 junction (due to the immobility of the VA at this location).⁴⁸ However, other sites have also been reported.^{49,50}

VA occlusion does not produce symptoms in most individuals due to collateral flow through the contralateral VA and/or the circle of Willis. Symptomatic occlusion usually involves the dominant VA,⁵¹ but it may also occur with non-dominant VA⁴⁷. Most cases of BHS occur in patients with an isolated posterior circulation (incompetent posterior communicating arteries).

BHS has also been postulated as one possible cause of SIDS.⁵²

95.6.2 Contributing factors

- external VA compression⁵⁰
 - spondylotic bone spurs: particularly in the foramen transversarium⁵³
 - tumors
 - fibrous bands (e.g., proximal to entrance of VA into C6 foramen transversarium⁴⁹)
 - infectious processes
 - trauma
- tethering of the VA
 - at the transverse foramina of C1 & C2
 - along the sulcus arteriosus proximal to where the VA enters the dura
- defect in odontoid process⁵⁴
- atherosclerotic vascular disease

95.6.3 Diagnosis

General information

BHS should be suspected in patient with symptoms of VBI precipitated by head movement. This may be very difficult to differentiate from vertigo and nausea due to vestibular dysfunction (which can also be induced by head movement). (Rotation of the body keeping the head motionless should not cause symptoms from vestibular dysfunction and might help distinguish these conditions.⁵⁵)

Dynamic cerebral angiography (DCA)

✖ NB: significant consequences can be precipitated during DCA in patients with BHS.⁴⁸ The involved VA shows loss of flow as the head is rotated from the neutral position to the contralateral side. Carotid injections demonstrate patency of PComAs and the presence of any persistent fetal anastomoses.

CT angiogram (CTA)

Same precautions as with DCA (see above). Probably not the initial diagnostic study of choice. If the DCA is negative, CTA is not needed. If DCA is positive, CTA may be helpful to demonstrate the arterial relationship to the bony anatomy.

95.6.4 Treatment

Options include:

- anticoagulation⁵⁵
- cervical collar: to remind patient not to turn their head
- for VA compression at C1–2 (see ► Table 95.5 for a comparison):
 - C1–2 fusion (p.1781)
 - VA decompression: C1 “hemilaminectomy” via a posterior approach⁵⁶

Table 95.5 Comparison of surgical treatment for positional VA occlusion at C1–2

Procedure	Advantages	Disadvantages
C1–2 fusion	high success rate in eliminating symptoms	loss of 50–70% of neck rotation with possible discomfort
VA decompression	no loss of motion	33% continue to have symptoms ⁵⁷

4. for compression at other sites: elimination of the source of compression where possible (e.g., sectioning off offending fibrous band,⁴⁹ removal of osteophytic spurs⁵³...)

Management recommendations: For compression at C1–2, it is suggested that VA decompression be performed as the initial treatment. This should be followed by DCA to verify maintenance of patency with head turning. Patients who fail clinically or on DCA should undergo C1–2 fusion.⁴⁸ Patients need to know pros and cons of each option.

95.7 Cerebral venous thrombosis

95.7.1 General information

Cerebral venous thrombosis (CVT) involves the dural sinuses and/or cerebral veins.

It is an uncommon condition that comprises 0.5–1% of strokes⁵⁸ and usually affects young people (78% occur in patients age < 50 years⁵⁹). Estimated incidence: 1.32 per 100,000 person-years, with a higher incidence of 2.78 in women 31–50 years of age.

It has been observed that patients are usually either mildly symptomatic or extremely ill and unstable.

There are 3 types of CVT, any of which may produce venous infarctions:

1. dural sinus thrombosis (DST)
2. cortical venous thrombosis
3. deep venous thrombosis

95.7.2 Etiologies

Partial list of etiologies

Many conditions have been incriminated with CVT, some of which may be due to referral and ascertainment biases.⁶⁰ Some common ones are listed here^{60,61} (p 1301):

1. infection
 - a) usually local, e.g., otitis media^{62,63} (leading to the now obsolete term otitic hydrocephalus), sinusitis, peritonsillar abscess, paranasal sinusitis⁶⁴; in the pre-antibiotic era, CVT was most commonly associated with chronic suppurative infection
 - b) meningitis
2. pregnancy & puerperium: see below
3. oral contraceptives† (birth control pills [BCP])⁶⁵
4. dehydration and cachexia (marantic thrombosis): includes burns and cachexia of neoplastic disease
5. cardiac disease (including CHF)
6. ulcerative colitis (UC): 1% of UC patients have some thrombotic complication (not necessarily all intracranial), and this is the cause of ≈ 33% of deaths (usually pulmonary embolism, PE)
7. periarteritis nodosa
8. sickle cell trait
9. trauma: including closed head injury (see below)
10. iatrogenic: e.g., S/P radical neck surgery,⁶⁶ transvenous pacemaker placement, post-craniotomy, internal jugular vein catheterization
11. malignancy: including myeloproliferative disorders
12. hypercoagulable state (AKA thrombophilia or prothrombotic conditions)
 - a) protein C deficiency† or resistance to activated protein C: hereditary factor V Leiden mutation may produce resistance to activated protein C.⁶⁷ Apparent protein C deficiency may be an artifact of dehydration in some cases
 - b) antithrombin III deficiency†
 - c) protein S deficiency†
 - d) antiphospholipid antibodies[‡]: associated with a variety of clinical syndromes including ischemic stroke, DVTs, thrombocytopenia, and systemic lupus erythematosus (SLE). The best known antibodies include
 - anticardiolipin antibodies[‡]
 - lupus anticoagulant
 - e) paroxysmal nocturnal hemoglobinuria (PNH)
 - f) plasminogen deficiency
 - g) prothrombin G20210A mutation of factor II[‡]: causes a mild elevation of prothrombin
 - h) systemic lupus erythematosus⁶⁸
 - i) factor VIII elevation⁶⁹: may explain some cases of CVT in pregnancy (see below)

13. diabetes mellitus: especially with ketoacidosis
14. (hyper)homocystinuria (p.208)
15. Behçet's syndrome (p.208)⁷⁰
16. rarely associated with lumbar puncture, associated with hereditary activated protein C resistance due to the factor V R506Q mutation (FV Leiden) in one report⁷¹

† Strongly associated with CVT (odds ratio ≥ 5)⁶⁰

Pregnancy/puerperium

Highest risk is in the third trimester and for 6–8 weeks post-partum.⁶⁰ One series⁷² found that no case of CVT occurred later than 16 days post-partum. Incidence $\approx 1/10,000$ births. Etiology may be related to elevation of clotting factors (VII, X, and especially factor VIII⁷³). Hypercoagulability increases post-partum due to volume depletion and trauma.

Trauma

A rare sequelae of closed head injury.⁷⁴ CVT occurs in $\approx 10\%$ of combat injuries involving the brain. May occur in absence of skull fracture. CVT should be suspected in patients with fractures or missiles crossing sinus.

95.7.3 Relative frequency of venous structures involved

The relative frequency of involvement of dural sinuses and other veins with thrombotic issues

1. sinuses
 - a) superior sagittal sinus (SSS) and left transverse sinus (TS) (70% each)
 - b) multiple sinuses in 71%
 - c) isolated inferior sagittal sinus: rare, first case report in 1997⁷⁵
 - d) straight sinus⁷⁶
2. superficial cortical veins
3. deep venous system (e.g., internal cerebral vein)
4. cavernous sinus^{77,78}: rare. Thrombophlebitis of the cavernous sinus may be caused by sphenoid sinusitis. MRI may show enlargement and abnormal enhancement of the cavernous sinus, increased signal of the petrous apex and clivus on T2WI, and narrowing of the cavernous portion of the ICA⁷⁸

95.7.4 Pathophysiology

Venous thrombosis reduces venous outflow from the brain and diminishes effective blood flow to the involved area. This venous engorgement causes white matter edema. The increased venous pressure may also lead to infarction and/or hemorrhage. These processes may all elevate ICP. Thus, clinical findings may be due to elevated ICP, and focal findings may be due to edema and/or hemorrhage. Cerebral infarction due to venous stasis is called venous infarction.

95.7.5 Clinical

Clinical presentations of DST are shown in ► Table 95.6. There are no pathognomonic findings. Many signs and symptoms are due to elevated ICP as a result of impaired venous outflow and may present as a syndrome clinically indistinguishable from pseudotumor cerebri (p.955). Others present as a result of focal brain injury from venous insufficiency or infarction/hemorrhage. Headache is the most common symptom.⁷⁹

There is a high association of concurrent thromboembolic disease in other organs.

The anterior 1/3 of the SSS may occlude without neurologic sequelae. Posterior to this (especially posterior to the vein of Trolard), venous infarction is more likely to develop. Midportion SSS occlusion usually $\rightarrow$ increased muscle tone ranging from spastic hemi- or quadriplegia to decerebration. Posterior SSS thrombosis $\rightarrow$ field cuts or cortical blindness, or massive stroke with cerebral edema and death. Occlusion of the TS may occur without deficit unless the contralateral TS is hypoplastic, in which case presentation is similar to posterior SSS occlusion.

SSS occlusion alone will not cause cranial nerve findings except perhaps for visual obscuration and abducens (VI) nerve palsy from elevated ICP. Thrombosis in the jugular bulb may compress the nerves in the jugular foramen pars nervosa causing hoarseness, aphonia, difficulty swallowing, and breathlessness; see Vernet's syndrome (p. 102).⁸⁰

Table 95.6 Presentation of dural sinus thrombosis^a

Sign/symptom	Series A ^b	Series B ^c
H/A	100%	74%
N/V	75%	–
seizures	70%	29%
hemiparesis	70%	34%
papilledema	70%	45%
blurred vision	60%	–
altered consciousness	35%	26%

^aother symptoms include: focal neurologic deficit, cranial nerve palsy, nystagmus⁸¹^bseries A: 20 young females⁷²^cseries B: 38 cases from France⁸²

95.7.6 Diagnosis of DST

General information

CT (especially CTV) & MRI/MRV are very sensitive and specific for identifying areas of clot.

Catheter angiography is better at demonstrating the presence of residual flow, and can identify areas of reversal of flow, and sometimes is able to demonstrate clot as a filling defect. Angiography is often used as a complementary test⁸³ when the diagnosis is suggested by CT or MRI.

Non-contrast CT

Low sensitivity. A wide range of accuracy is reported: 30–100% sensitivity & 83–100% specificity (a realistic range of sensitivity from reviews is $\approx 30\%$ ⁶⁰ to 75% ⁸⁴ of cases of CVT). False positives may occur with high hematocrit. Hounsfield unit (p.238) > 70 is highly specific for acute CVT.⁸⁴ Findings include:

1. hyperdense sinus (► Fig. 95.1) or cortical vein: high-density clots in cortical veins; has been dubbed the cord sign, which is pathognomonic for cerebral venous thrombosis; seen in only 2/30 patients

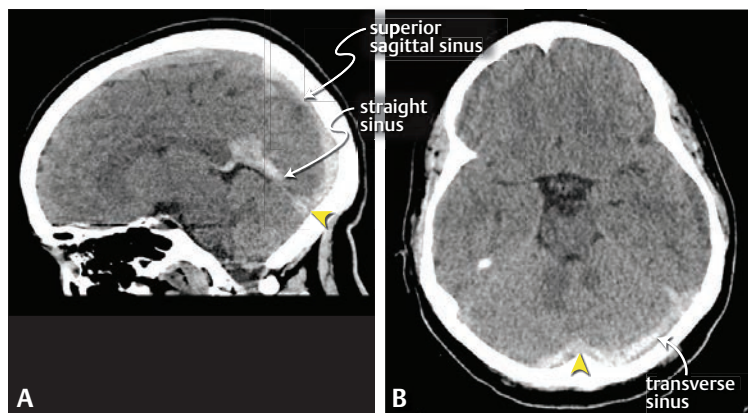


Fig. 95.1 Cerebral venous thrombosis.

Non-contrast CT scan, A: sagittal, B: axial.

Thrombus (white) is seen in the superior sagittal sinus, the straight sinus, the left transverse sinus extending slightly into the right transverse sinus, and in the torcular Herophili (yellow arrowheads).

2. hemorrhage: 9–39% show some form of hemorrhage
 - a) petechial “flame” hemorrhages (intraparenchymal): seen in 20% (suspect sinus thrombosis with intracerebral hemorrhages in unusual locations for aneurysm or “hypertensive” hemorrhage)
 - b) SAH: in < 1% (usually over convexity)
 - c) intraventricular
3. small ventricles: seen in 50%
4. thrombosis of superior sagittal sinus may produce a triangular-shaped high density within the sinus posteriorly near the torcular Herophili on axial CT images (delta sign (► Fig. 95.2, panel A) or “dense/filled delta sign” as opposed to the “empty delta” sign, see below & ► Fig. 95.2, panel B) (there is also confusion when an apparent “empty delta” sign is seen *without* contrast; this may occur when there is blood surrounding the SSS, e.g., following subarachnoid hemorrhage; this has been called a “false delta sign” or pseudodelta sign⁸⁵). Recommendation: avoid the confusion of the variations on “delta signs” and describe the findings
5. venous infarct: infarct that doesn’t follow arterial territories, especially near a dural sinus⁸⁶
6. white matter edema
7. above changes occurring *bilaterally*

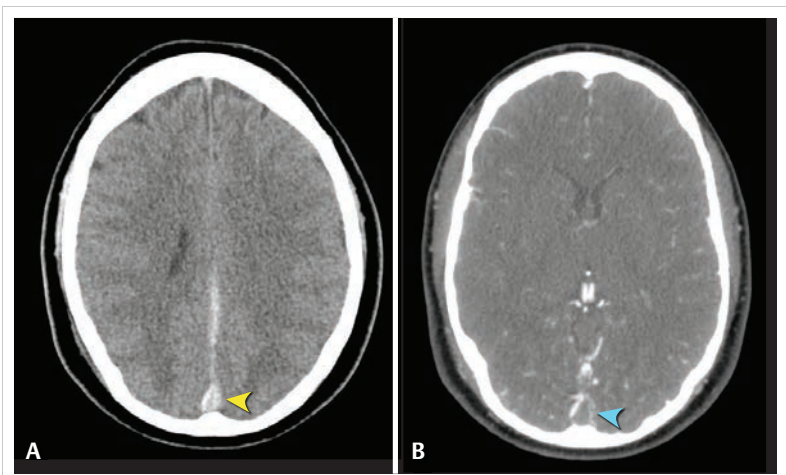


Fig. 95.2 Delta sign on axial CT scan in CVT.

A: non-contrast, demonstrating the “(full) delta sign” (yellow arrowhead) in the torcular Herophili.
B: IV contrast-enhanced, demonstrating the “empty delta sign” (blue arrowhead).

IV contrast CT

Findings of DST include:

1. with *contrast*, the dura around the sinus may enhance and become denser than clot in 35% of cases.⁸⁷ Near the torcular Herophili this produces what has been called the empty delta sign (► Fig. 95.2, panel B),⁸⁸ but sometimes this, too, is called the delta sign
2. gyral enhancement occurs in 32%
3. dense deep (white matter) veins (collateral flow)
4. intense tentorial enhancement (common)

CTV

CT venography is approximately equivalent to MRV in diagnosing CVT.⁶⁰

MRI & MRA/MRV

MRI excels for diagnosis and follow-up. MRI shows absence of flow and clot burden and demonstrates parenchymal changes including venous infarcts (► Fig. 95.3, panel B). Can differentiate

occluded sinus from congenital absence. Shows cerebral edema and non-acute hemorrhagic changes to better advantage than CT. Also may help estimate age of clots (► Table 95.7). Findings include the hyperintense vein sign (► Fig. 95.3, panel A).
MR angiography may increase the utility. MR venography (MRV) tends to overestimate the degree of occlusion.

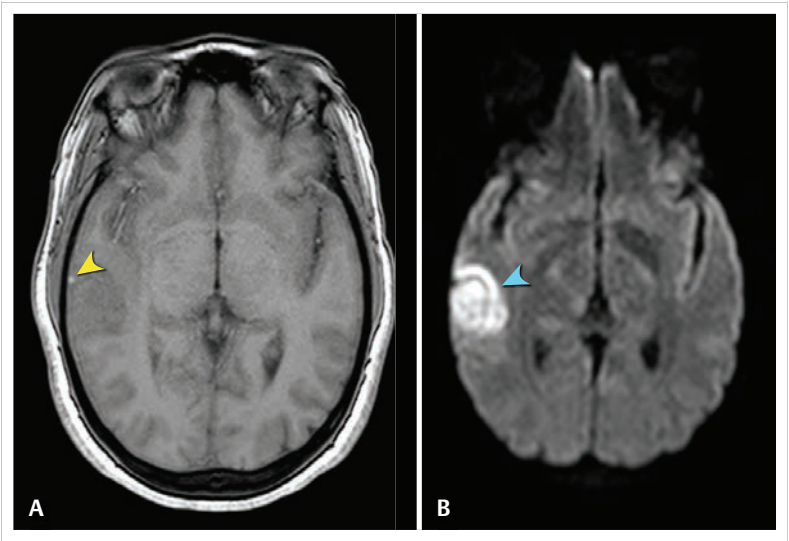


Fig. 95.3 MRI findings in CVT.
A: T1 axial showing hyperintense vein sign (yellow arrowhead).
B: DWI showing venous infarct (blue arrowhead) adjacent to the thrombosed vein.

Table 95.7 MRI appearance of thrombosed sinuses at various stages⁶⁰

Age of clot in sinus	Hemoglobin state	Appearance of clotted sinus	
		T1WI	T2WI
acute (0–7 days)	deoxyhemoglobin	isointense	hypointense (black): can mimic flow void
subacute	methemoglobin	hyperintense (1st)	hyperintense (2nd)
late (recanalized)	paramagnetic products of above	black (flow void)	black (flow void)

Catheter angiography for CVT

Catheter angiography is close to MRI in sensitivity, and some still consider it to be the standard of diagnosis. MRI has some advantages over angiography (e.g., on angiography a hypoplastic transverse sinus may not visualize, or non-opacified blood entering a sinus may mimic a filling defect). Be sure to image all the way to the venous phase (requires extra vigilance).

Findings include:

- 1. non-filling of segments of sinuses, or filling defects in segments that are visualized
- 2. prolonged circulation time: present in 50% of cases (may need delayed films to see veins)
- 3. stumps and abnormal collateral pathways

LP

LP (lumbar puncture) is generally not indicated unless there is suspicion of meningitis. There are no specific CSF abnormalities, so LP is often not helpful. Opening pressure (OP) is increased in > 80%.⁷⁹ CSF may be bloody or xanthochromic.

Bloodwork

Routine blood studies (CBC, chemistry, PT & aPTT) should be performed (Level I⁶⁰).

To detect prothrombotic conditions, useful tests include: protein C and S levels*, antithrombin deficiency*, antiphospholipid antibodies = anticardiolipin antibodies and lupus anticoagulant as well as tests for specific predisposing conditions (Factor II level, serum homocysteine level, paroxysmal nocturnal hemoglobinuria (PNH) panel, leukocyte alkaline phosphatase (Level I⁶⁰)).

* Tests for protein C & protein S, and antithrombin deficiency should be deferred until 2–4 weeks after completing anticoagulation (testing is of limited value on warfarin or in the acute setting since the acute process will cause numerous abnormalities in the clotting system (Level II⁶⁰)).

D-dimer: a fibrin degradation product, showed a sensitivity of 97.1%, specificity of 91.2%, a negative predictive value of 99.6%, and a positive predictive value of 55.7%.⁸⁹ A normal D-dimer level by a sensitive radioimmunoassay or ELISA may help identify patients with a low probability of CVT (Level II⁶⁰). However, if suspicion is high, further evaluation is indicated.

Ultrasound for DST

May be used in diagnosis of superior sagittal sinus thrombosis in the neonate.⁹⁰

95.7.7 Management of CVT

General information

Driving principle: treat the underlying abnormality (if possible).

Management should be aggressive because recoverability of brain is probably greater than with arterial occlusive stroke. Management is challenging because measures that counteract thrombosis (e.g., anticoagulation) tend to increase the risk of hemorrhagic infarct (the risk of which is already increased), and measures that lower ICP tend to increase blood viscosity → increased coagulability.

The following is derived from the 2011 AHA Scientific Statement for the Diagnosis and Management of Cerebral Venous Thrombosis⁶⁰ and the 2018 Report of the Society of NeuroInterventional Surgery on Endovascular Strategies for Cerebral Venous Thrombosis.⁸⁴

Key concepts: Evaluation and treatment of CVT

1. history & bloodwork: identify risk factors (p. 1594) and symptoms (► Table 95.6). Treat underlying conditions
2. initial diagnostic test in cases of suspected CVT: either brain CT + CTV head/neck, or brain MRI + MRV head/neck⁸⁴ (if normal, these tests do not rule out CVT). MRI/MRV is probably more sensitive than CT/CTV
3. catheter angiography may be helpful with inconclusive CTV or MRV when the clinical suspicion is high (Level II⁶⁰). Be sure to obtain late images (venous phase)
4. if CVT is confirmed: treatment
 - a) anticoagulation: start with dose-adjusted unfractionated heparin (UFW) or weight-based LMWH, followed by vitamin K antagonists (e.g., warfarin) (Level II⁶⁰). ICH is not a contraindication
 - b) indications for endovascular treatment (p. 1934) (chemical thrombolysis or thrombectomy):
 1. clinical deterioration despite anticoagulation
 2. contraindications to anticoagulation
 3. coma
 4. deep CVT
 5. intracerebral hemorrhage (ICH)
 - c) decompressive hemicraniectomy indications: large lesions with herniation
 - d) intracranial pressure monitoring and treatment: start with acetazolamide for elevated ICP
5. if CVT is not confirmed: diagnosis of alternative conditions
 - a) brain tumor
 - b) infection
 - c) ischemic stroke
 - d) posterior reversible encephalopathy syndrome (PRES) (p. 202)
 - e) pseudotumor cerebri (p. 955)

Treatment specifics

1. general measures
 - a) patients suspected of having infection should receive appropriate treatment: antibiotics, drainage of purulent collections... (Level I⁶⁰)
 - b) seizures:
 - ASMs for a defined duration are recommended for a single seizure with (Level I⁶⁰) or without (Level II⁶⁰) parenchymal lesions to prevent further seizures
 - ASMs are not recommended in the absence of seizures (potential harm⁶⁰)
 - c) steroids: not recommended unless needed for other underlying disease (potential harm⁶⁰) (reduces fibrinolysis, increases coagulation)
 - d) control HTN
2. anticoagulation
 - a) anticoagulation: start with dose-adjusted unfractionated heparin (UFW) or weight-based LMWH (Level II⁶⁰). See dosing information (p. 172). Numerous studies show a lower mortality rate with heparin than without.^{91,92,93} It remains the best treatment even when there is evidence of intracerebral hemorrhage (ICH) with the attendant risk of increasing the size of the hemorrhage.⁸³ There is no consensus on duration of treatment. Success rate may be higher if administered before patient becomes moribund
 - ★ during pregnancy, full-dose LMWH is recommended over UFH (Level II⁶⁰)
 - b) transition from heparin to vitamin K antagonists (e.g., warfarin)
3. monitor ICP if patient continues to deteriorate: ventriculostomy is preferred, but use caution in placing catheter if patient is on heparin
 - a) hydrate aggressively as ICP tolerates
 - b) measures to lower ICP: differ somewhat from treatment for traumatic ICP elevation
 - acetazolamide (Level II⁶⁰): one of the few measures to treat elevated ICP that do not rely on venous outflow
 - elevate HOB
 - drain CSF
 - pentobarbital coma: also does not depend on venous outflow
 - hyperosmotic and/or loop diuretics: reserve for last because diuretics → hypertonicity → ↑ viscosity → ↑ coagulation. Replace fluid loss with isotonic IV fluids to prevent dehydration; i.e., goal is hypertonic euvoolemia
 - c) monitor patients with increased ICP for progressive visual loss and treat elevated ICP urgently if detected (Level I⁶⁰). Treatment measures include: serial LPs, optic nerve sheath fenestration,⁹⁴ or VP shunt (Level II⁶⁰)
4. endovascular therapy (EVT) (p. 1934) for deterioration despite intensive anticoagulation (Level II⁶⁰). There are no guidelines for how long to wait before declaring medical therapy a failure, decision takes into account how sick the patient is. Also, there is no information when to use DC vs. EVT. Modalities include chemical thrombolysis (direct injection of tPA into the sinus) and mechanical thrombectomy
5. decompressive craniectomy (DC): consider for neurologic deterioration due to mass effect or intracranial hemorrhage causing increased ICP (Level II⁶⁰). Risky, but may be lifesaving. Early surgery (≤ 12 hrs from admission) & younger patients are predictors for a more favorable response to DC. If DC is elected, the options of heparinization and tPA cannot be employed for ≈ 2–3 days
6. long-term anticoagulation after resolution of acute phase with heparin (Level II⁶⁰):
 - a) spontaneous CVT: vitamin K antagonists (VKA) for 6–12 months, target INR = 2–3
 - b) CVT with risk factors that have been eliminated: VKA for 3–6 months, target INR = 2–3
 - c) recurrent CVT, VTE after CVT, or severe thrombophilia (e.g., homozygous prothrombin G20210A, homozygous factor V Leiden, protein C or S or antithrombin deficiency, antiphospholipid syndrome, or combined thrombophilic conditions): indefinite VKA, target INR = 2–3
 - d) consider consultation with a physician with expertise in coagulation
 - e) for CVT during pregnancy: continue full anticoagulation with LMWH during the pregnancy, and LMWH or VKA with a target INR = 2–3 for ≥ 6 weeks post-partum (for a total duration of therapy of 6 months) (Level I⁶⁰)

95.7.8 Prognosis

Morbidity & mortality:

Natural history (without treatment): reported mortality range is 14–40% (based on small sample sizes).⁸⁴

With anticoagulation: ≈ 13% die or remain dependent.⁸⁴

Poor prognosticators:

1. clinical status:
 - a) coma⁹⁵
 - b) rapid neurologic deterioration,⁹⁵ focal signs
2. demographics
 - a) age: extremes of age (infancy or elderly)⁹⁵
 - b) male gender
3. radiographic findings:
 - a) hemorrhages, especially larger hemorrhages
 - b) venous infarcts
4. deep venous involvement

► Future pregnancies in patients with a history of CVT

1. advise the patient that future pregnancy is *not* contraindicated. Consultation with a hematologist and/or maternal fetal medicine specialist is reasonable (Level II⁶⁰)
2. prophylaxis with LMWH during future pregnancies and the post-partum period for women with a history of CVT is recommended (Level II⁶⁰)

95.8 Extracranial-intracranial (EC/IC) bypass

95.8.1 The 1985 international EC/IC bypass study

The EC/IC bypass, pioneered by Donaghy and Yasargil in 1967,⁹⁶ plummeted in popularity⁹⁷ after publication of the international cooperative EC/IC bypass study⁹⁸ in 1985. The EC/IC trial randomized 1,377 patients with symptomatic ICA or MCA stenosis to either STA-MCA bypass or medical therapy with ASA. Despite a graft patency rate of 96%, surgical patients suffered more and earlier fatal and nonfatal strokes. Patients with severe MCA stenosis and those with persistent symptoms following ICA occlusion fared especially worse with bypass.

Critique: inclusion criteria failed to distinguish between hemodynamic vs. thromboembolic causes of stroke.^{2,99,100} Ischemia secondary to thromboembolic events would not be expected to improve with flow augmentation, and inclusion of such patients in the surgical arm could therefore artificially lower the apparent efficacy of the procedure.

95.8.2 Current state of affairs

Imaging technologies introduced since the 1985 EC/IC trial can identify flow-dependent ischemia. Xenon-CT, TCD, SPECT, and MRI and CT perfusion may be used in combination with acetazolamide challenge to evaluate cerebrovascular reserve and reactivity (p.1537).

As cerebral perfusion pressure decreases in severe atherosclerotic occlusive disease, cerebral autoregulation (p.1536) is unable to maintain adequate CBF to meet metabolic demands. In this state of "misery perfusion," oxygen extraction fraction (OEF) of available blood flow will increase.^{101,102} Abnormal OEF, as quantified by PET, is an independent predictor of subsequent stroke.² Patients with abnormal response to acetazolamide challenge (p.239) and/or with elevated OEF are therefore potential candidates for cerebral revascularization.^{2,100,103,104,105}

The Japanese EC-IC Bypass Trial (JET) and Carotid Occlusion Surgery Study (COSS) were designed to stratify candidates according to hemodynamic criteria.

- JET: 169 patients, study period of almost 4 years, 2-year follow-up. A significant reduction of the primary and secondary stroke was observed after surgery.
- COSS: halted after inclusion of 195 patients due to high 30-day event rate and no significant benefits on the overall outcome of the surgical patients. The high event rate (14.4%) was dramatically higher than that of the 1985 EC-IC bypass study. Furthermore, the COSS data demonstrated a clear postoperative reduction of ipsilateral ischemic events beyond the 30-day perioperative period.

Conclusion: EC-IC bypass remains an option in carefully selected patients with ischemic cerebrovascular disease, limited to interdisciplinary and specialized high-volume centers and within the framework of controlled studies.^{106,107,108,109} Recently, the Japanese Adult Moyamoya Trail (JAM) reported the preventative effect of direct bypass against rebleeding in adult patients with moyamoya disease who had experienced intracranial hemorrhage.¹¹⁰

95.8.3 Indications for EC/IC bypass

1. symptomatic patients with misery perfusion (see above)¹¹¹
2. aneurysms: certain aneurysms are not amenable to either direct microsurgical clipping or endovascular coiling due to extreme size, location, calcification or atherosclerosis, dissection, or the incorporation of perforators or major arteries. EC/IC bypass remains a highly viable adjunctive measure in patients requiring Hunterian occlusion of parent vessel or prolonged temporary occlusion for definite treatment.^{112,113,114,115,116} Cerebrovascular reserve and need for bypass can be assessed preoperatively using balloon test occlusion (BTO) with hypotensive challenge
3. tumors encasing or invading major arteries
4. moyamoya disease (p.1581) and moyamoya syndrome

95.8.4 Bypass types

The type of graft used depends on preoperative determination of amount of flow augmentation necessary, the size of the recipient graft, and the availability of donor vessel^{117,118}:

1. pedicled arterial grafts: STA, occipital artery
 - a) low-flow (15–25 ml/min)
 - b) only one anastomosis required
 - c) 95% graft patency in superficial temporal artery-middle cerebral artery (STA-MCA) bypasses
2. radial artery graft
 - a) moderate to high flow (40–70 ml/min)
 - b) advantages: physiological conduit for arterial blood; constant location makes it easy to harvest; lumen size closely approximates that of M2 or P1 and reduces flow mismatch with subsequent flow turbulence and graft thrombosis
 - c) disadvantages: risk of vasospasm (reduced with pressure distension technique)
 - d) >90% graft patency at 5 years
3. saphenous vein graft
 - a) high flow (70–140 ml/min)
 - b) advantages: easy accessibility; longer length
 - c) disadvantages: risk of thrombosis at distal anastomosis due to flow mismatch and turbulence; lower graft patency rates
 - d) 82% graft patency at 5 years

95.8.5 Perioperative complications of EC/IC bypass

In addition to risks of surgery, risks of EC/IC bypass include: risk of stroke from temporary occlusion of a cortical vessel, cerebral hyperperfusion, and “watershed shift” phenomenon with the risk of cerebral hemorrhage from a sharp increase in focal CBF.¹¹⁹

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Part XX

Intracerebral Hemorrhage

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XX

96 Intracerebral Hemorrhage in Older Adults

96.1 Intracerebral hemorrhage – general information

Intracerebral hemorrhage (ICH) is a hemorrhage within the brain parenchyma. In the past it was commonly referred to as “hypertensive hemorrhage,” but hypertension is a debatable etiology in many cases; see *Hypertension as a cause?* (p. 1612). This section deals primarily with spontaneous intracerebral hemorrhage (sICH), i.e., not due to trauma, tumor, etc. Ongoing trials for this condition may be viewed at www.strokecenter.org/trials/.

Key concepts: intracerebral hemorrhage (ICH) in adults

- the second most common form of stroke (15–30% of strokes), but the most deadly
- unlike ischemic infarct: smooth progressive onset over minutes to hours, often with severe headache, vomiting, and alterations in level of consciousness
- unenhanced CT scan of the brain is the initial diagnostic study of choice
- the volume of the hematoma correlates highly with morbidity and mortality
- ICH enlarges in > 33% of cases within the first 3 hours of onset
- worse prognosis for patient on anticoagulants or with elevated cardiac troponins
- additional diagnostics (e.g., CTA, catheter angiogram, MRV...) if suspicion of underlying abnormality (e.g., AVM, dural sinus thrombosis, neoplasm...)
- hypertension (SBP > 150 mm Hg): reduce SBP to 140
- surgery
 - a) supratentorial ICH: still controversial. Surgery does not improve neurologic outcome, but may reduce length of ICU stay and mortality
 - b) cerebellar ICH with neurologic deterioration, or brainstem compression, or obstructive hydrocephalus from intraventricular clot: surgical evacuation ASAP (initial treatment with EVD is *not* recommended)

96.2 Epidemiology

96.2.1 Incidence

ICH is the second most common form of stroke ($\approx$ 15–30% of all strokes) (earlier estimates: 10%¹), and the most deadly. Approximately 12–15 cases per 100,000/yr. Early studies estimated an incidence equal to SAH, but more recent studies in the CT era show approximately twice the incidence as SAH² (pre-CT studies may have misclassified some ICH as ischemic stroke, and some cases of ICH that rupture into the subarachnoid space (occurs in $\approx$ 7%) may have been misclassified as SAH). After a decline in the 1970's, the incidence increased in the 1980s for those age $\geq$ 65 years.³ Onset is usually during activity (rarely during sleep), which may be related to elevation of BP or increased CBF; see *Etiologies* (p. 1610).

The diagnosis and management of ICH is a medical emergency. Over 20% of patients deteriorate by $\geq$ 2 GCS levels between the first EMS assessment and the E/R,⁴ and 15–23% of patients deteriorate further within the first few hours in the hospital.⁵

96.2.2 Risk factors

The following are epidemiologic risk factors; see also others (p. 1610).

1. age: the incidence increases significantly after age 55 years and doubles with each decade of age until age > 80 yrs where incidence is 25 times that during previous decade. Relative risk for age > 70 yrs is 7
2. gender: more common in men
3. race: in the U.S., ICH affects blacks more than whites. May be related to higher prevalence of poorly controlled HTN in blacks. Incidence may also be higher in Asians⁶
4. previous stroke (any type) increases risk to 23:1
5. alcohol consumption^{6,7}:
 - a) recent use: moderate or heavy alcohol consumption both within the 24 hours and the week preceding the ICH were independent risk factors for ICH⁸ as shown in ► Table 96.1

- b) chronic use: one study suggests that consuming > 3 drinks a day increases the risk of ICH by ≈ 7 times^{9(p 15)}
- c) ICH in patients with high ethanol consumption were more commonly lobar than the typical “hypertensive hemorrhages” in the basal ganglia¹⁰
- 6. cigarette smoking: increases the risk of SAH and ischemic infarction but probably does *not* increase the risk of ICH,^{11,12} further clarification is needed
- 7. street drugs: cocaine, amphetamines, phencyclidine¹³
- 8. liver dysfunction: hemostasis may be impaired on the basis of thrombocytopenia, reduced coagulation factors, and hyperfibrinolysis¹⁴ (may be responsible for the increased risk of ICH with chronic EtOH consumption)

Table 96.1 Relative risk of ICH with EtOH consumption

Period prior to ICH	Amount ^a (g EtOH)	Relative risk
24 hours	41–120	4.6
	> 120	11.3
1 week	1–150	2.0
	151–300	4.3
	> 300	6.5

^a1 standard drink = 12 g EtOH

96.3 Locations of hemorrhage within the brain

96.3.1 General information

Common sites of spontaneous intracerebral hemorrhage (sICH) are shown in ► Table 96.2. Common arterial feeders of ICHs:

1. lenticulostriates: the source of putaminal hemorrhages (possibly secondary to microaneurysms of Charcot-Bouchard, see below)
2. thalamoperforators
3. paramedian branches of BA
4. intraventricular hemorrhage: occurs in $\approx 45\%$ of sICH and is an independent risk factor for worse outcome.⁵ May be primary, or more commonly, results from intraventricular extension

Table 96.2 Common sites for ICH (modified¹⁵)

%	Location
50	striate body (basal ganglia); putamen most common; also includes: lenticular nucleus, internal capsule, globus pallidus
15	thalamus
10–15	pons ($\approx 90\%$ of these are genuinely hypertensive)
10	cerebellum
10–20	cerebral white matter
1–6	brainstem

96.3.2 Lobar hemorrhage

This term was popularized in 1980 after a report delineating 4 clinical syndromes associated with ICH in each of the cerebral lobes¹⁶ (occipital, temporal, frontal, and parietal) as in contrast to hemorrhage of deep structures (e.g., basal ganglion, thalamus, and infratentorial structures).¹⁶ Accounts for 10–32% of nontraumatic ICHs.¹⁶ With large hemorrhages, it may be difficult to distinguish between lobar and deep ICH.

Lobar hemorrhages are more likely to be associated with structural abnormalities than deep hemorrhages (see below). They may also be more common in patients with high alcohol consumption (see above). Lobar hemorrhages may also have a more benign outcome than ganglionic-thalamic hemorrhages.¹⁶

Etiologies of lobar hemorrhage: Although many causes of ICH can produce lobar hemorrhages (see below for a detailed list), those that are more likely to produce lobar hemorrhages include:

1. extension of a deep hemorrhage

2. cerebral amyloid angiopathy (p. 1612): the most common cause of lobar ICH in elderly normotensive patients
3. trauma
4. hemorrhagic transformation of an ischemic infarct: see below
5. hemorrhagic tumor (p. 1612). Multiple lobar hemorrhages may occur with metastases
6. cerebrovascular malformation (especially AVM) (p. 1524)
7. rupture of an aneurysm: see below for circumstances likely to produce this
8. idiopathic

96.3.3 Internal capsule hemorrhages

There may be prognostic significance with regard to contralateral motor function if the hemorrhage is medial to and/or extending through the internal capsule (IC), or lateral to the IC and merely compressing it, making the clot more accessible to surgical treatment without damaging the IC.

96.4 Etiologies

1. "hypertension" (debatable as a cause or effect, see below) but is a risk factor
 - a) acute hypertension (HTN): as may occur in eclampsia (see below) or with certain drugs, e.g., cocaine (p. 215)
 - b) chronic HTN: possibly causes degenerative changes within blood vessels
2. possibly associated with acutely increased CBF (globally or focally),¹⁷ especially to areas previously rendered ischemic:
 - a) following carotid endarterectomy^{18,19}
 - b) following repair of congenital heart defects in children²⁰
 - c) previous stroke (embolic²¹ or otherwise): hemorrhagic transformation may occur in up to 43% of strokes during the first month.²² May follow dislodgment or recanalization of an arterial occlusion, although it has been demonstrated with persistent occlusion.²³ May occur as early as ≤ 24 hrs after a stroke in patients with a negative CT done within 6 hours.²⁴ Two types^{22,25}:
 - type 1: diffuse or multifocal. Heterogeneous or mottled appearance within the boundaries of the stroke. Less hyperdense than primary ICH
 - type 2: extensive hematoma. Probably unifocal source. As hyperdense as primary ICH and may extend outside the original stroke boundaries. Unlike type 1, classically associated with anticoagulation therapy, and tends to occur in initial few days after stroke and is often associated with clinical worsening. May be difficult to distinguish from primary ICH, and may be frequently misdiagnosed as such²⁴
 - d) migraine: during²⁶ or following²⁷ a migraine attack (probably an exceedingly rare event)
 - e) following surgery to remove an AVM: "normal perfusion pressure breakthrough." Some cases may be due to incomplete AVM excision
 - f) physical factors: following strenuous physical exertion,²⁸ exposure to cold²⁹...
3. vascular anomalies
 - a) AVM: rupture; see Arteriovenous malformation (p. 1505)
 - b) aneurysm rupture
 - saccular ("berry") aneurysms: (i) **aneurysms of the circle of Willis (COW)**: aneurysms that have become adherent to brain surface by fibrosis as a result of inflammation or previous hemorrhage may produce ICH when they rupture instead of the usual SAH; (ii) **aneurysms distal to the COW** (e.g., MCA aneurysms)
 - microaneurysms of Charcot-Bouchard (p. 1612)
 - c) venous angioma rupture: significant ICH from these common lesions is a very rare event
4. "arteriopathies"
 - a) cerebral amyloid angiopathy (p. 1612): usually $\rightarrow$ repeated lobar hemorrhages (see below)
 - b) fibrinoid necrosis^{1,30} (sometimes seen in cases of amyloid angiopathy)
 - c) lipohyalinosis: subintimal lipid-rich hyaline material³¹
 - d) cerebral arteritis (including necrotizing angiitis)
5. brain tumor (primary or metastatic): see below
6. coagulation or clotting disorders
 - a) leukemia
 - b) thrombocytopenia:
 - thrombotic thrombocytopenic purpura
 - aplastic anemia
 - c) patients on anticoagulation therapy (p. 1613) constitute 12–20% of patients with ICH⁵

- d) patients receiving thrombolytic therapy:
 - for acute ischemic stroke: incidence of *symptomatic* ICH within 36 hrs of treatment with rtPA is 6.4% (vs. 0.6% in the placebo-treated group)³²
 - for acute MI or other thrombosis: incidence is $\approx 0.36\text{--}2\%$.^{33,34,35} Risk is increased with higher doses than the recommended 100 mg of alteplase (Activase®, recombinant tissue plasminogen activator (rt-PA))³⁶ in older patients, in those with anterior MI or higher Killip class, and with bolus administration (vs. infusion).³⁷ When heparin was used adjunctively, higher doses were associated with higher risk of ICH.³⁸ ICH is thought to occur in those patients with some preexisting underlying vascular abnormality.³⁹ Immediate coronary angioplasty is safer than rt-PA when available³⁵
- e) aspirin therapy:
 - one ASA qod was associated with increased risk of ICH,⁴⁰ with a rate of 0.2–0.8% per year⁴¹
 - ASA 100 mg/d did not increase the risk of significant ICH in patients > 60 yrs with mild-to-moderate head injury (GCS ≥ 9)⁴²
- f) Vitamin E supplements⁴³: associated with reduction of 1 ischemic stroke in 476 individuals, and increase of 1 ICH in 1,250 patients taking vitamin E
7. CNS infection:
 - a) especially fungal, which damages blood vessels
 - b) granulomas
 - c) herpes simplex encephalitis: may initially produce low-density lesions that progress to hemorrhagic ones
8. venous or dural sinus thrombosis (p. 1594)
9. drug-related
 - a) substance abuse
 - alcohol: consumption of > 3 drinks/day increases the risk of ICH ≈ 7 -fold (p. 1608)
 - drug abuse: especially sympathomimetics (cocaine,^{44,45} amphetamine⁴⁶)
 - b) drugs that raise BP:
 - alpha-adrenergic agonists (sympathomimetics): including phenylephrine, ephedrine,⁴⁷ and pseudoephedrine^{48,49}
 - ephedra alkaloids: sold as a dietary supplement (ma huang) to suppress appetite and increase energy. Associated in case reports with HTN, SAH, ICH, seizures, and death⁵⁰
10. posttraumatic: often in a delayed fashion^{51,52}; see **Hemorrhagic contusion** (p. 1071)
11. pregnancy-related: the risk of ICH in pregnancy and puerperium (up to 6 weeks postpartum) is ≈ 1 in 9,500 births⁵³
 - a) most commonly associated with eclampsia or preeclampsia: the mortality of eclampsia is $\approx 6\%$ with ICH being the most frequent direct cause⁵⁴; also see Pregnancy & intracranial hemorrhage (p. 1425)
 - b) postpartum ICH (median 8 days, range 3–35 days) in the absence of eclampsia has been reported⁵⁵; when associated with vasculopathy the term **postpartum cerebral angiopathy** has been used
 - c) vascular findings:
 - some cases associated with isolated cerebral vasculopathy in the absence of systemic vasculitis⁵⁶
 - some cases demonstrate vasospasm
 - some cases show findings (e.g., patchy enhancement in occipital lobes) suggestive of cerebrovascular dysautoregulation (p. 1536)
 - some cases show no vascular-related abnormalities
12. postoperative:
 - a) following carotid endarterectomy (see above)
 - b) following craniotomy:
 - at site of craniotomy⁵⁷: risk factors identified: within residual astrocytoma after subtotal resection, following craniotomy for AVM (see above)
 - at site remote from craniotomy. In a series of 37 patients, unlike hematomas at the craniotomy site, the following were identified as *not* being related to risk of hemorrhage: HTN, coagulopathy, CSF drainage, underlying occult lesion following drainage of chronic SDH (p. 1082), and cerebellar hemorrhage following pterional craniotomy⁵⁸ (this author incriminated possibly rapid overdrainage of CSF), or following temporal lobectomy⁵⁹
13. idiopathic¹⁶

96.4.1 Cerebellar hemorrhage etiologies

Etiologies are similar to ICH of any location; however, some nuances:

1. HTN is a factor in up to two-thirds of cerebellar hemorrhages

2. AVM is a consideration, aneurysm is very rare (possibly AICA aneurysm, but usually only in association with other high-flow lesions, e.g., AVM⁶⁰)
3. may be related to recent previous spinal or supratentorial surgery

96.4.2 Hypertension as a cause?

Hypertension (HTN) is controversial as cause of ICH since the incidence of both ICH and HTN increases with age (66% of patients > 65 yrs have HTN). The relative risk for ICH with HTN is 3.9–5.4, depending on the definition of HTN used.⁶¹ Many patients with ICH are dramatically hypertensive on presentation; however, acute elevations of ICP from the hemorrhage may actually precipitate HTN (part of Cushing's triad, see ► Table 62.2). HTN is probably a risk factor primarily for pontine/cerebellar ICH and is probably not a factor in at least 35% of basal ganglion hemorrhages.

96.4.3 Microaneurysms of Charcot-Bouchard

AKA miliary aneurysms.⁶² Occur primarily at bifurcation of small (< 300 μ m) perforating branches of lateral lenticulostriate arteries in basal ganglia (found in 46% of hypertensive patients over age 66, but only in 7% of controls⁶³). Possibly the origin of some "hypertensive" ganglionic (putaminal) hemorrhages,⁶⁴ but this is controversial.

96.4.4 (Cerebral) amyloid angiopathy

Cerebral amyloid angiopathy (CAA) AKA congophilic angiopathy. Pathologic deposition of beta amyloid protein (appears as birefringent "apple-green" under polarized light when stained with congo red) within the media of small meningeal and cortical vessels (especially those in white matter) without evidence of systemic amyloidosis.⁶⁵ Some vessels may show fibrinoid necrosis of vessel wall.^{66,67} There is a treatable subset of CAA with inflammation (see below).

CAA should be suspected in patients with *recurrent* hemorrhages (uncommon with "hypertensive hemorrhages" (p. 1608)⁶⁸) that are *lobar* in location. Gradient-echo MRI may identify petechial hemorrhages (microbleeds) or hemosiderin deposits from small cortical hemorrhages.⁶⁹ Less likely in the case of basal ganglion or brainstem hemorrhages.¹⁶

Incidence increases with age: CAA is present in $\approx$ 50% of those over 70 years of age,⁷⁰ but most do not hemorrhage. CAA is probably responsible for $\approx$ 10% of cases of ICH. May be associated with genetic factors (including the apolipoprotein E ϵ 4 allele⁷¹), and may be more prevalent in patients with Down syndrome. Although they are distinct diseases, there is some overlap between CAA and Alzheimer's disease; the amyloid in CAA is identical to that found in senile plaques of Alzheimer's disease. CAA may increase the risk of ICH by potentiating plasminogen⁷² (may be of special relevance to patients receiving tissue plasminogen activator (t-PA) to treat MI or stroke).

Patients with CAA may present with a TIA-like prodrome (see below).

Among patients with lobar hemorrhage, those with the apoE ϵ 4 allele typically have their first hemorrhage > 5 yrs earlier than noncarriers (73 $\pm$ 8 yrs vs. 79 $\pm$ 7 yrs).⁷¹

Diagnostic tests are useful mainly to rule out other conditions. The definitive diagnosis of CAA requires pathologic evaluation of brain tissue. Criteria for the diagnosis of CAA are shown in ► Table 96.3.⁷³

A subset of CAA patients treatable with immunosuppressants have CAA-related inflammation (CAA-ri) as a result of an autoimmune response to the amyloid protein. These patients present with H/A, subacute cognitive impairment or seizures, with vasogenic edema on imaging. Anti- β -amyloid antibodies appear in the CSF, but currently no clinical test is available. To avoid resorting to brain biopsy, MRI criteria for probable CAA-ri have been developed: uni- or multi-focal asymmetric white matter hyperintensities that extend to subcortical in addition to the typical CAA findings of lobar-ICH, microbleeds, superficial siderosis... (82% sensitive, 97% specific).⁷⁴

96.4.5 Hemorrhagic brain tumors

Although any brain tumor can hemorrhage, tumoral ICH is usually associated with malignancies. Tumors on occasion can also produce SAH or subdural hematomas.

Malignant tumors most commonly associated with ICH:

1. glioblastoma
2. lymphoma
3. metastatic tumors
 - a) melanoma^{75,76}; $\approx$ 40% hemorrhage
 - b) choriocarcinoma^{75,77,78}; $\approx$ 60% hemorrhage

Table 96.3 Criteria for the diagnosis of cerebral amyloid angiopathy (CAA)⁷³

Diagnosis	Criteria
definite CAA	full postmortem exam showing all 3 of the following: a) lobar, cortical, or corticosubcortical hemorrhage b) severe CAA c) absence of another diagnostic lesion
probable CAA with supporting pathological evidence	clinical data & pathological tissue showing all 3 of the following: a) lobar, cortical, or corticosubcortical hemorrhage b) some degree of vascular amyloid deposition in specimen c) absence of another diagnostic lesion
probable CAA	clinical data and MRI findings showing all 3 of the following: a) age $\geq$ 60 yrs b) multiple hemorrhages restricted to the lobar, cortical, or corticosubcortical region c) absence of another cause of hemorrhage ^a
possible CAA	clinical data and MRI findings: a) age $\geq$ 60 yrs b) single lobar, cortical, or corticosubcortical hemorrhage without another cause ^a , or multiple hemorrhages with a possible but not a definite cause ^a , or with some hemorrhages in an atypical location (e.g., brainstem)

^ae.g., excessive anticoagulation (INR > 3.0), head trauma, ischemic stroke, CNS tumor, cerebrovascular malformation, vasculitis, or blood dyscrasia

- c) renal cell carcinoma
- d) bronchogenic carcinoma: although only $\approx$ 9% hemorrhage, this tumor is such a frequent source of cerebral mets that it therefore is a more common source of tumoral ICH

Malignant tumors that hemorrhage less commonly include:

1. medulloblastoma^{79,80,81,82} (most commonly in children)
2. gliomas^{83,84}

Some *benign* brain tumors that have been associated with ICH include:

1. meningiomas have been associated with intratumoral, subdural, and nearby parenchymal hemorrhage.^{85,86,87,88} Tendency to bleed is similar for angioblastic variety as for other highly vascular meningiomas
2. PitNET/adenoma: see Pituitary apoplexy (p.865)
3. oligodendroglioma (relatively benign): rarely presents with hemorrhage,⁸⁹ classically after years of causing seizures
4. hemangioblastoma⁹⁰
5. vestibular schwannoma^{91,92,93}
6. cerebellar astrocytoma⁹⁴

96.4.6 Anticoagulation preceding ICH

10% of patients on warfarin (Coumadin®) develop a significant bleeding complication per year (not all are intracranial), including ICH (65% mortality in this group). The risk of ICH in patients treated with warfarin for a-fib varies between 0–0.3% per year⁴¹ (historically, this was as high as $\approx$ 1.8% in older studies⁹⁵ from the 1960s and 1970s), but when an elderly subgroup (mean age 80 yrs) was analyzed, this rate was 1.8% per year.⁴¹ ICH was the only cause of fatal bleeding complications of warfarin therapy in one series where the cumulative risk of a fatal hemorrhage was 1% at 1 year and 2% at 3 yrs.⁹⁶

The risk of hemorrhagic complications was increased with the length and also the variability of the PT, and during the first three months of anticoagulation.⁹⁶ Patients with cerebral amyloid angiopathy (CAA) (see above) are also at increased risk of ICH following administration of antiplatelet drugs or anticoagulants.⁷³

96.5 Clinical

96.5.1 General information

In general, the neurologic deficit with ICH is characterized by a smooth progressive onset over minutes to hours, unlike embolic/ischemic stroke where deficit is maximal at onset. With ICH, severe

headache, vomiting, and alterations in level of consciousness may be more common (H/A may not be more prevalent than in embolic stroke, but it is often a first and prominent symptom¹⁶).

96.5.2 Prodrome

TIA-like symptoms may precede lobar hemorrhages^{97,98} in patients with CAA, and may occur in up to $\approx 50\%$ of patients for whom a complete history is obtainable. Unlike typical TIAs, these usually consist of numbness, tingling, or weakness (corresponding to the area where the hemorrhage will subsequently occur) that gradually spreads in a manner reminiscent of a Jacksonian-march and may spill-over vascular territories (probably an electrical phenomenon rather than an ischemic event). This is suggestive of but not pathognomonic for the subsequent development of lobar ICH.

96.5.3 Concomitants of specific lesions in ICH

Putaminal hemorrhage

The most common site for ICH. Smooth gradual deterioration in 62% (maximal deficit at onset in 30%); never fluctuating. Contralateral hemiparesis, may progress to hemiplegia or even coma or death. H/A in 14% at onset. No H/A at any time in 72%. Papilledema and subhyaloid preretinal hemorrhage are rare.

Thalamic hemorrhage

Classically, contralateral hemisensory loss. Also hemiparesis when the internal capsule is involved. Extension into upper brainstem $\rightarrow$ vertical gaze palsy, retraction nystagmus, skew deviation, loss of convergence, ptosis, miosis, anisocoria, $\pm$ unreactive pupils. H/A in 20–40%. Motor deficit similar to putaminal hemorrhage, but contralateral sensory deficit widespread and striking. Hydrocephalus may occur from compression of CSF pathways.

In 41 patients, when hemorrhage > 3.3 cm on CT, all died. Smaller hematomas usually caused permanent disability.

Cerebellar hemorrhage

See ► Fig. 96.1. May include any combination of the following:

1. symptoms of increased ICP (lethargy, N/V, HTN with bradycardia...) due to hydrocephalus, which may occur as a result of:
 - a) compression of the 4th ventricle $\rightarrow$ obstruction of CSF
 - b) extension of the hemorrhage into the ventricular system
2. direct compression of brainstem may produce:
 - a) facial palsy: due to pressure on the facial colliculus
 - b) these patients classically become comatose without first having hemiparesis, unlike many supratentorial etiologies

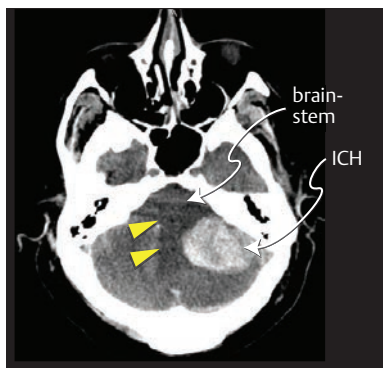


Fig. 96.1 Spontaneous left cerebellar hemisphere ICH in 73-year-old male.

Axial CT showing ICH producing brainstem compression and obliteration of the 4th ventricle. The patient had hydrocephalus. There is also edema (low-density) in the cerebellar vermis and brainstem (yellow arrow-heads) most likely due to ischemia.

Lobar hemorrhage

Syndromes associated with hemorrhage in the 4 cerebral lobes¹⁶ ($\approx$ 50% have H/A as a first and prominent symptom):

1. frontal lobe (the most distinctive of the syndromes): frontal H/A with contralateral hemiparesis, usually in the arm with mild leg and facial weakness
2. parietal lobe: contralateral hemisensory deficit and mild hemiparesis
3. occipital lobe: ipsilateral eye pain and contralateral homonymous hemianopsia, some may spare superior quadrant
4. temporal lobe: on dominant side, produces fluent dysphasia with poor auditory comprehension but relatively good repetition

96.5.4 Delayed deterioration

General information

Deterioration after the initial hemorrhage is usually due to any combination of the following:

1. rebleeding: see below
2. edema: see below
3. hydrocephalus: higher risk with intraventricular extension or posterior fossa ICH
4. seizures
5. increased ICP

Rebleeding or extension of bleed

Early rebleeding: Rebleeding (more so in basal ganglion hemorrhages than in lobar hemorrhages) has been documented during the first hour by “ultra-early” scanning and repeating CT scans. Rebleeding is usually accompanied by clinical deterioration.⁹⁹ The incidence of hematoma enlargement decreases with time, 33–38% in 1–3 hrs,¹⁰⁰ 16% in 3–6 hrs, and 14% between 24 hrs of onset and a second CT within 24 hrs of the first.¹⁰¹ Risk of early rebleeding is increased in patients with a “spot sign” on CTA (p.1616). Patients with enlarging hematomas were more likely to have larger hematomas and/or coagulopathy, and had a worse outcome.¹⁰¹ Rebleeding may still occur following surgical evacuation of clot even with satisfactory intraoperative hemostasis. Hemostatic agents (p.1619) (e.g., NovoSeven®) may reduce this risk.

Late rebleeding: Quoted rates for late rebleeding from ICH range from 1.8–5.3% (depending on length of follow-up).¹⁰² Diastolic BP was significantly higher in the group with recurrent hemorrhage, with a 10%/yr risk for DBP > 90 mm Hg vs. < 1.5% for DBP $\leq$ 90 (mean F/U of 67 months).¹⁰² Other risk factors include diabetes and tobacco and alcohol abuse.¹⁰³ Recurrent hemorrhages may indicate underlying vascular malformations or amyloid angiopathy (lobar rebleeding is likely to be due to amyloid angiopathy¹⁰³).

Edema

Edema and ischemic necrosis around the hemorrhage may cause delayed deterioration.¹ Although necrosis from mass effect of the clot contributes a small part to the edema, experiments indicate that by itself, the mass effect is insufficient to account for the amount of edema that occurs. It is believed that an edemogenic toxin is released from the clot. Experiments with various components of blood clots have disclosed that thrombin in concentrations that could be released from the clot causes increased permeability of the blood-brain barrier, and is also a potent vasoconstrictor. This is the leading suspect as the major cause of delayed edema and deterioration. Also see Cerebral edema (p.90).

96.6 Evaluation

96.6.1 Overview

Practice guideline: Initial diagnosis & assessment in spontaneous ICH

Level 1⁵:

- obtain a baseline severity score
- rapid imaging with noncontrast CT (or MRI) to differentiate from ischemic stroke

Level II⁵:

- consider CTA & contrast CT to identify patients at risk for hematoma expansion
- consider CT venogram, contrast CT, contrast MRI, MRA, and/or catheter angiogram as appropriate when clinical or imaging suspicion of underlying abnormality (vascular or neoplastic)

96.6.2 CT scan

General information

Noncontrast head CT scan is the usual initial imaging procedure of choice. It is rapid, has few contraindications, and easily demonstrates blood as high-density within the brain parenchyma immediately after hemorrhage.

Although mass effect is common, the tendency for the hemorrhage to dissect through brain tissue often results in less mass effect than would be anticipated from the size of the clot.

Volume measurements on CT scan

ICH volume carries prognostic significance. It can be measured volumetrically using computer algorithms available on some CT scanners, or it can very simply be approximated by the ellipsoid method¹⁰⁴ as shown in Eq (96.1), where AP, LAT, and HT are the *diameters* of the clot in each of the 3 dimensions (anteroposterior, lateral, and height). This estimate was originally developed for AVMs, based on the principle that the volume of an ellipsoid is approximately half of that of a parallelepiped into which it is placed¹⁰⁵ and is simpler than other slightly more accurate estimation methods.¹⁰⁶ An illustration of the measurements is shown in ► Fig. 96.2. To estimate the height of a lesion when only axial images are available (as on some CTs), count the number of images on which the lesion is seen, and multiply by the slice thickness of the CT cuts^{104,106,107} (this information is usually indicated on the images), or subtract the table position of the highest cut that shows the clot from the table position of the lowest cut showing the clot.

$$\text{ellipsoid volume} \approx \frac{\text{AP} \times \text{LAT} \times \text{HT}}{2} \quad (96.1)$$

On average, the size of the clot decreases ≈ 0.75 mm/day, and the density decreases by ≈ 2 CT units/day, with little change for the first 2 wks.

96.6.3 CT angiography (CTA)

Not necessary to identify ICH. However, CTA is very sensitive and specific in detecting vascular lesions (e.g., AVM or aneurysm). May also demonstrate the “spot sign”¹⁰⁸: a small enhancing focus within an acute ICH (see ► Fig. 96.2 panel C) which correlated with increased risk of hematoma expansion.

CTA has supplanted catheter angiography in most situations. Indications for CTA include:

- subarachnoid hemorrhage
- findings suggestive of cerebral venous thrombosis (p. 1594) (CVT): increased attenuation in dural sinuses or cortical veins
- enlarged vessels/flow voids or rim calcifications suggestive of AVM
- also see entry under Management guidelines (p. 1620) below

96.6.4 MRI

Usually *not* the procedure of choice for initial study. Disadvantages compared to CT scan:

1. does not show blood well within the first few hours (gradient echo and SWI are the best sequences)
2. difficult to ventilate or access patient during the study
3. slower and more expensive, not available in as many E/Rs
4. contraindications e.g., claustrophobia in awake patients, as well as others that may not be known at the time of initial evaluation (e.g., shrapnel in the eye, non-MRI-compatible pacemaker or spinal cord stimulator...)

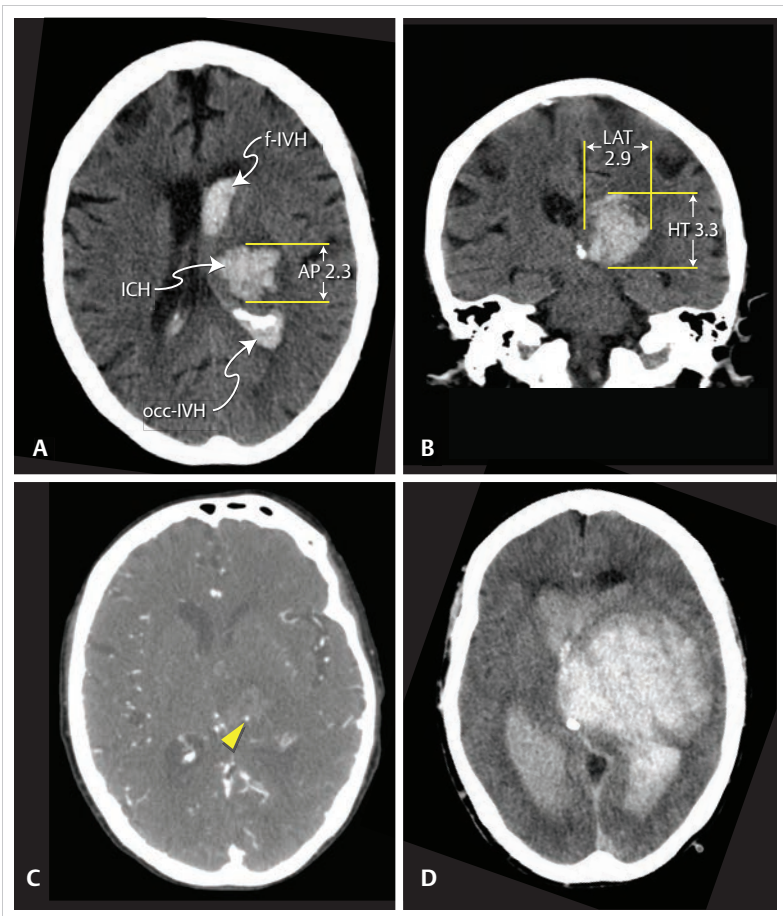


Fig. 96.2 Spontaneous left thalamic ICH in 71-year-old female.

Image: A: axial CT showing ICH with ventricular extension (f-IVH = intraventricular hemorrhage in frontal horn of left lateral ventricle, occ-IVH = IVH in occipital horn).

B: coronal CT.

Estimated ICH volume = $(2.3 \text{ cm} \times 2.9 \text{ cm} \times 3.3 \text{ cm})/2 = 11 \text{ cc}$.

C: CTA done immediately after CT shown in A & B, demonstrating a "spot sign" (yellow arrowhead) within the ICH.

D: Repeat CT scan done 1 hour later showing massive enlargement of ICH.

AP = AP dimension of ICH in cm, LAT = mediolateral dimension, HT = height.

May be useful later, e.g., to help diagnose cerebral amyloid angiopathy (CAA) (p. 1612).

The appearance of ICH on MRI is very complicated. It is highly dependent on the age of the clot¹⁰⁹ as summarized in (► Table 96.4).

96.6.5 Catheter cerebral angiography

For diagnosing ICH itself, angiography cannot reliably differentiate the mass effect due to an ICH from that due to an ischemic infarct or tumor.¹¹⁰ May demonstrate AVMs and aneurysms when they are associated with the ICH. The yield may be increased by delaying the study.¹⁶ May demonstrate vascular blush in some cases of tumor. Normal arteriography cannot eliminate cerebral amyloid

Table 96.4 Variation of brain MRI signal characteristics of intraparenchymal blood over time (modified¹⁰⁹)

Phase	Approximate time after onset	Hemoglobin (Hgb) state	T1 MRI ^a	T2 MRI ^a	Mnemonic ^b
hyperacute	0–6 hrs ^c	oxy-Hgb (intracellular)	I	B (slight ↑)	I be
acute	6–72 hrs	deoxy-Hgb (intracellular)	I (or slight ↑)	D	iddy
early subacute	3–7 d	met-Hgb (intracellular)	B	D	biddy
late subacute	7–14 d	met-Hgb (extracellular ^d)	B	B	baby
chronic	> 2 weeks ^e	hemosiderin (intracellular)	D (slight ↑)	D	doodoo

^aB = bright (hyperintense compared to brain), D = dark (hypointense), I = isointense
^bsilly/easy-to-remember mnemonic made from I, B, or D in the preceding columns
^csome authors consider up to about 24 hrs as hyperacute
^dwhen RBCs lyse, the Hgb becomes extracellular
^ethe center of the clot may be isointense on T1 and slightly hyperintense on T2

angiopathy as the etiology of ICH in the elderly.¹¹¹ CTA has supplanted most indications for catheter angiography.

Indications for CTA/catheter angiography in ICH: see entry under Nonsurgical management guidelines (p.1620).

96.6.6 ICH score

A caveat about overreliance on grading systems, especially early in the course, bears discussion. In an effort to apply evidence-based methodology to the care of the ICH patient, a number of prognostic models have been developed. While they are helpful, their development may have been biased by self-fulfilling prophecies based on withdrawal of medical support in the face of apparently devastating neurologic injury. The American Heart Association statement on “Palliative and End-of-Life Care in Stroke”¹¹² discusses palliative care, and it is recommended that in the absence of known preexisting DNR wishes, the decision to withdraw support not be made until at least the second full day of hospitalization after ICH (Level II³).

The widely used ICH score¹¹³ assigns points based on 5 features as indicated in ► Table 96.5. The points are then summed for the “ICH score.” The associated 30-day mortality is tabulated in ► Table 96.6.

Table 96.5 ICH Score¹¹³

Graded feature	Criteria	Points	Score
GCS (Glasgow coma scale score; ► Table 18.1)	3–4	2	(0 - 2)
	5–12	1	
	13–15	0	
Age ^a	≥ 80 yrs	1	(0 - 1)
	< 80 yrs	0	
Location	infratentorial	1	(0 - 1)
	supratentorial	0	
ICH volume see Eq (96.1)	≥ 30 cc	1	(0 - 1)
	< 30 cc	0	
Intraventricular blood	yes	1	(0 - 1)
	no	0	
ICH Score → TOTAL			(0 - 6)

^apossible bias since treatment decisions in elderly patients may have differed from younger patients

Table 96.6 Mortality based on ICH Score

ICH Score ^a	30-day mortality	N ^b
0	0%	26
1	13%	32
2	26%	27
3	72%	32
4	97%	29
5	100%	6
6	? 100% ^c	0

^a from ► Table 96.5^b N is the number of patients with that score^c no patient in the study had a score of 6, but "it is expected this would be associated with a high rate of mortality"¹¹³

96.7 Initial management of ICH

96.7.1 History checklist

The following checklist is presented to assist in gathering historical information important in evaluating & managing the adult with spontaneous ICH (modified)⁵:

1. time course of symptoms: time of initial onset (or when last seen normal)
2. initial symptoms and progression
3. seizure
4. hypertension history
5. drugs:
 - a) sympathomimetics:
 - amphetamines, cocaine
 - appetite suppressants or nasal decongestants (pseudoephedrine)
 - b) dietary supplements: especially ephedra alkaloids (ma huang)
 - c) anticoagulants: warfarin (Coumadin®), dabigatran (Pradaxa®), apixaban (Eliquis®)...
 - d) antiplatelet drugs: aspirin (patients often forget taking low-dose 81 mg), Plavix, NSAIDS
 - e) oral contraceptive (birth control pills): questionable association
 - f) history of alcohol abuse
6. past medical history
 - a) coagulopathies
 - b) history of dementia: ICH may be associated with cerebral amyloid angiopathy (p. 1612)
 - c) liver disease: may be associated with coagulopathy
 - d) previous stroke
 - e) history of known vascular abnormalities (AVM, venous angioma...)
 - f) tumor: known history of cancer, especially those that tend to go to brain (lung, breast, GI, renal, melanoma...) or associated with coagulopathy (leukemia)
7. recent surgery: especially carotid stenting or endarterectomy, procedures requiring heparin...
8. recent childbirth and/or eclampsia or preeclampsia
9. history of recent trauma

96.7.2 Initial laboratory tests

1. CBC (including platelet count), electrolytes (including BUN/creatinine and glucose): hyperglycemia is associated with worse prognosis
2. PT/INR & aPTT
3. cardiac-specific troponins: elevation is associated with worse outcome
4. toxicology screen (urine or serum): mainly for cocaine and/or sympathomimetics
5. pregnancy test in women of childbearing potential

96.7.3 Nonsurgical management outline

The following assumes the diagnosis has already been made, usually on CT scan. There is not uniform agreement on many management aspects. The following is offered as a guide.

1. patients should initially be managed in an ICU or dedicated stroke unit by personnel experienced in critical care (Level I⁵)

2. hypertension (HTN) may contribute to further bleeding, especially within the first hour.⁹⁹ However, some HTN may be needed to maintain perfusion. See practice guideline (p. 1621)
3. intubate if stuporous or comatose
4. maintain normothermia: treatment of fever after ICH may be reasonable (Level II⁵) (NB: a causal relation between fever and worse outcome and/or increase in ICH is not established; mild hypothermia is considered investigational)
5. antiepileptic medications
 - a) seizures (clinical and subclinical) are treated with appropriate ASMs
 - b) ✖ prophylactic ASMs: not recommended
 - c) ASM options
 - Keppra has a very favorable therapeutic/toxic profile. Dose 500 mg BID
 - OR phenytoin (p.488) (Dilantin®): load with 17 mg/kg slow IV over 1 hour, follow with 100 mg q 8 hrs
6. hemostatic issues:
 - a) INR (or PT), PTT & platelet count (PC), platelet function assay (PFA)
 - correct coagulopathies, see Correction of coagulopathies or reversal of anticoagulants (p. 174)
 - platelets: correct thrombocytopenia or platelet-inhibiting drugs as discussed below
 - b) ✖ bleeding time: obsolete, not generally helpful
 - c) for hemostasis management issues, see practice guideline (p. 1621)
7. steroids: ✖ not recommended for ICH⁵ (exception: if underlying tumor is identified)
8. treat intracranial hypertension presumptively: mannitol and/or furosemide as tolerated, also helps with HTN; for more, see Treatment measures for elevated ICP (p. 1046). If significant problems from suspected increased ICP, consider ICP monitor
9. external ventricular drain (EVD): for hydrocephalus (✖ but not for hydrocephalus associated with cerebellar hemorrhage – see below), some cases of intraventricular blood, or to manage ICP (see below). ✖ R/O coagulopathy before placing
10. follow electrolytes and osmolarity:
 - a) monitor for hyponatremia (possible SIADH (p.118))
 - b) maintain euglycemia: avoid hyper- or hypoglycemia (Level I⁵)
11. **cardiac:** systematic screening for myocardial ischemia or infarction is recommended. EKG & cardiac enzymes should be checked routinely (Level II⁵)
12. swallowing: formal screening for dysphagia before initiating oral intake is recommended (Level I⁵)
13. **CTA or catheter angiography:** primarily to R/O underlying vascular malformation, aneurysm (a less common cause of ICH) & tumor (which is usually better diagnosed on contrast CT or MRI)
 - a) if urgent surgery is indicated (e.g., for herniation), the delay in obtaining an angiogram may be detrimental and it may be best deferred to post-op
 - b) ★ **indications:** angiography is not routinely recommended for patients > 45 yrs of age with preexisting hypertension *and* ICH in thalamus, putamen, or posterior fossa because there was a 0% yield out of 29 patients in this group¹¹⁴ and low yield in all patients with isolated deep ICH.¹¹⁵ It may be considered for:
 - patients > 45 yrs with a history of HTN and a *lobar* ICH: angiography had a 10% yield,¹¹⁴ with the ratio of AVM:aneurysm ≈ 4.3:1
 - patients with intraventricular hemorrhage (without parenchymal hematoma): the yield of angiography was ≈ 65%,¹¹⁴ primarily AVM
 - c) an underlying lesion may be obliterated by ICH, especially acutely. If initial angio is negative, repeat after CT shows resorption of clot (≈ in 2–3 mos). If still negative, follow CT or MRI q 4–6 mos for ≈ 1 year to R/O tumor.¹ Delaying the initial angiogram for several weeks may increase the yield and is also an option¹⁶
 - d) MRI/MRA has only ≈ 90% sensitivity for detecting structural abnormalities in this setting, and so a negative study cannot completely exclude this possibility¹¹⁴
 - e) the yield of angiography in ICH would be expected to be lower in patients at increased risk of ICH: patients on warfarin (Coumadin®), chronic alcoholics, patients with amyloid angiopathy...

Practice guideline: Blood pressure management in patients with ICH

1. patients with SBP 150–200 mm Hg and no contraindication to acute BP management: lowering SBP to 140^a is safe (Level I⁵) and improves functional outcome (Level II⁵)
2. patients with SBP > 200 mm Hg: it is reasonable to consider aggressive reduction of BP with continuous IV infusion and monitoring of BP (Level II⁵)

^a since the publication of these guidelines,⁵ the INTERACT-II¹¹⁶ & ATACH-2¹¹⁷ trials have shown that rebleeding occurs despite blood pressure control, and that reducing SBP < 140 mm Hg is associated with increased incidence of adverse renal events¹¹⁷ probably from hypoperfusion

Practice guideline: Seizure & ASM management in patients with ICH

1. treat clinical seizures, or seizures on EEG in patients with altered mental status (Level I⁵)
2. continuous EEG monitoring is probably indicated when depression of mental status is out of proportion to the degree of brain injury (Level II⁵)
3. prophylactic ASMs are not recommended (Level III⁵)

Practice guideline: Hemostasis & coagulopathy in patients with ICH

1. for severe coagulation factor deficiency or severe thrombocytopenia: replace deficient factors or administer platelets (Level I⁵)
2. patients on vitamin K antagonists (VKA) (e.g., warfarin) with elevated INR:
 - a) withhold VKA (Level I⁵)
 - b) replace vitamin K-dependent clotting factors (Level I⁵)
 - c) correct the INR (Level I⁵)
 - consider prothrombin complex concentrate (PCC) (p. 174) over FFP because PCC may have fewer complications, and corrects INR faster & closer to normal (Level II⁵)
 - ✖ not recommended: rFVIIa (doesn't replace all clotting factors & may not restore clotting in vivo despite normalization of INR, & thromboembolic complications also occur) (Level III⁵)
 - d) administer IV vitamin K (Level I⁵)
3. patients on dabigatran (Pradaxa®)^a, rivaroxaban (Xarelto®)^a, or apixaban (Eliquis®)^a: consider treatment with activated PCC factor eight bypassing activity (FEIBA), other PCCs or vFVIIa (Level II⁵); consider dialysis for dabigatran (Level II⁵)
4. patients on heparin: consider reversal with protamine sulfate (Level II⁵)
5. patients on antiplatelet drugs: platelet transfusion is of uncertain benefit (Level II⁵)
6. ✖ not recommended: rFVIIa in unselected ICH patients (no clear clinical benefit) (Level III⁵)

^a subsequent to publication of these guidelines,⁵ the following reversal agents have become available and may be considered (not part of guidelines): idarucizumab (p. 173) for dabigatran, andexanet alfa (p. 175) for rivaroxaban or apixaban

Practice guideline: Prophylaxis & treatment of thromboembolism in patients with ICH

1. intermittent pneumatic compression device beginning the day of admission to prevent DVT (Level I⁵). ✖ not recommended: graduated compression device to prevent DVT or improve outcome (Level III⁵)

2. after documentation of cessation of bleeding: consider low-dose sub-Q heparin (LMW or unfractionated) to prevent DVT in patients with lack of mobility after 1–4 days from ICH (Level II⁵)
3. for patients with symptomatic DVT or PE: systemic anticoagulation or IVC filter placement is probably indicated (Level II⁵). The choice of modality should consider the time from ICH, ICH stability, cause of ICH, & overall patient condition (Level II⁵)

► Thrombocytopenia or platelet inhibiting drugs

1. thrombocytopenia: in general situations platelet transfusions are recommended only for PC < 50 K, but ICH is so serious that it is suggested to keep PC > 100K (if this is difficult to attain, aim for platelet count > 75K)
2. benefit of platelet transfusion in patients receiving platelet-inhibiting drugs (e.g., aspirin or Plavix®) is uncertain
3. when needed: start with 6 units of platelets; see Platelets (p. 161)

► **NovoSeven® (recombinant-activated coagulation factor VII [rFVIIa]).** ✖ Use in ICH is not recommended.⁵ FDA-approved for various bleeding diatheses (including hemophiliacs with antibodies to factor VIII or IX). Initial results appeared promising for ICH¹¹⁸; however, further study could not show clear benefit in noncoagulopathic patients, and thromboembolic complications also occurred.

96.7.4 Anticoagulation following ICH

Practice guideline: Resumption of anticoagulation in patients following ICH

1. for nonvalvular a-fib: long-term anticoagulation with warfarin should probably be avoided after warfarin-associated lobar ICH because of relatively high risk of recurrence (Level II⁵)
2. for strong indications, consider anticoagulation after nonlobar sICH or antiplatelet monotherapy after any sICH (Level II⁵)
3. optimal timing to resume treatment is uncertain
 - avoidance of oral anticoagulant (OAC) use for at least 4 weeks after OAC-associated ICH in patients without mechanical heart valves may decrease the risk of recurrent ICH (Level II⁵)
 - if indicated, aspirin monotherapy can probably be resumed in the days following ICH although optimal timing is uncertain (Level II⁵)
4. substitution of dabigatran, rivaroxaban, or apixaban for warfarin in patients with a-fib in an effort to reduce the risk of recurrent ICH is uncertain (Level II⁵)
5. data regarding use of statins after ICH is insufficient to make recommendations (Level II⁵)

Patients with ICH who subsequently require anticoagulation (e.g., for embolic ischemic stroke or for mechanical heart valve) pose a management dilemma. In the case of embolic disease, the fear of converting an ischemic infarct to a hematoma or increasing the size of a small ICH with continued anticoagulation has traditionally outweighed the possible benefit of protection from further embolization. However, an anecdotal (retrospective uncontrolled) report of 12 such patients found no incidence of increased intracranial bleeding with either continued anticoagulation (6 patients) or resumption of anticoagulation after a hiatus (several days in 4 patients, 5 days in 1, and 14 days in 1).¹¹⁹ In another study,¹²⁰ none of 35 patients who had resumption of warfarin had recurrent intracranial hemorrhage (ICH, SAH, or subdural hematoma). While this does not prove that anticoagulation is safe after ICH, it does demonstrate that if there is a strong indication for anticoagulation, and if there is not an acceptable alternative (e.g., Greenfield filter for DVT (p. 179)), that anticoagulation in this setting is not always met with disastrous results.

The probability of having an ischemic stroke at 30 days following cessation of warfarin for a median of 10 days using Kaplan-Meier survival estimates are approximately 2.9% for patients who had originally been treated with warfarin for prosthetic heart valves, 2.6% for those treated for atrial fibrillation, and 4.8% for those treated for cardioembolic stroke.¹²⁰ These numbers may be gross underestimates as many patients died within 2 weeks, and follow-up imaging was scant¹²¹; another study¹²² showed a much higher rate of 20%; see Cardiogenic brain embolism (p. 1590) for more details.

Antiplatelet therapy after ICH is not associated with a substantially increased risk of recurrent ICH¹²³ (prospective cohort study).

Recommendations

A-fib: long-term anticoagulation should be *avoided* after ICH.¹²⁴

Mechanical heart valves: 1–2 weeks off anticoagulation (to observe ICH, or to evacuate an SDH or clip an aneurysm).^{120,125} Patients with deep hemispheric ICH at high risk for thromboembolic stroke may benefit from resumption of long-term anticoagulation.¹²⁴

Patients requiring hemodialysis after ICH: heparin-free dialysis may be used.

96.8 Ventriculostomy (IVC) with ICH

Indications for Ventriculostomy (IVC) AKA external ventricular drainage (EVD) (also see guidelines below)

1. acute hydrocephalus:
 - a) due to compression of CSF pathways: e.g., thalamic ICH compressing the 3rd ventricle (not recommended for cerebellar ICH (Level III⁵), see (p. 1623))
 - b) associated with intraventricular blood: clot may cause obstructive hydrocephalus, and intraventricular blood alone is a risk factor for communicating hydrocephalus due to toxic effect on CSF resorptive mechanisms. IVCs may get blocked by clot, and intraventricular rtPA has been used to help maintain patency, but the overall benefit is uncertain⁵
2. for ICP management

Practice guideline: ICP monitoring & treatment in patients with ICH

1. EVD is reasonable for patients with hydrocephalus, especially with reduced level of consciousness (Level II⁵) (✗ not for cerebellar ICH – see text)
2. patients with GCS ≤ 8, or evidence of transtentorial herniation or significant IVH or hydrocephalus: consider ICP monitoring and treatment. A goal of CPP 50–70 mm Hg may be reasonable depending on the status of cerebral autoregulation (Level II⁵)
3. ✗ corticosteroids are not recommended for treatment of increased ICP (Level III⁵)
4. intraventricular rtPA appears to have a low complication rate, but the overall safety and benefit are uncertain (Level III⁵)

96.9 Surgical treatment

96.9.1 General information

The first successful evacuation of an intracerebral hematoma was reported by MacEwan in 1888.¹²⁶ The patient recovered completely from an upper-extremity monoplegia. Ever since then, surgical treatment of ICH has been fraught with controversy. See (p. 1623).

Practice guideline: Surgical treatment in patients with ICH

1. cerebellar hemorrhage with neurologic deterioration, or brainstem compression and/or obstructive hydrocephalus from intraventricular clot:
 - a) surgical removal of the clot should be done ASAP (Level I⁵); see Cerebellar hemorrhage (p. 1614) for details
 - b) ✗ initial treatment with EVD instead of surgical evacuation is not recommended (Level III⁵)
2. supratentorial ICH: the usefulness of surgery is not well-established (Level II⁵). Potential exceptions & additional considerations:
 - a) early ICH evacuation is not clearly superior to evacuation when the patient deteriorates (Level II⁵)
 - b) deteriorating patients: ICH evacuation may be considered as a life-saving measure (Level II⁵)
 - c) patients in coma, or large ICH with significant midline shift, or elevated ICP refractory to medical management: decompressive craniectomy (DC) with or without ICH evacuation may reduce mortality (Level II⁵)
 - d) minimally invasive ICH evacuation with stereotactic or endoscopic aspiration with or without thrombolytics is of uncertain effectiveness (Level II⁵)

Booking the case: Craniotomy for ICH

Also see defaults & disclaimers (p. 25).

1. position: (depends on location of bleed)
2. equipment:
 - a) microscope (not used for all cases)
 - b) image-guided navigation (typically used for minimally invasive ICH removal)
3. post-op: ICU
4. consent (in lay terms for the patient—not all-inclusive):
 - a) procedure: surgery through the skull to remove blood clot, stop any bleeding identified, possible placement of external (ventricular) drain
 - b) alternatives: nonsurgical management
 - c) complications: usual craniotomy complications (p. 25) plus further bleeding which may cause problems (especially in patients taking blood thinners, antiplatelet drugs including aspirin, or those with coagulation abnormalities or previous bleeds) and may require further surgery, areas of the brain that have already been damaged by the bleeding are not likely to recover, hydrocephalus (buildup of fluid in the brain) may occur with or without surgery and may require a temporary drainage tube or a permanent shunt

96.9.2 Indications for surgery

General information

Considerable controversy persists regarding indications for surgery. Surgery may lower morbidity from rebleeding (especially if an aneurysm or AVM is identified as the cause of the ICH), edema, or necrosis from mass effect of hematoma (unproven), but rarely causes neurologic improvement. Meta-analyses^{127,128} yield inconclusive or conflicting results and could not identify whether there was a favorable effect of surgery, the types of ICH and patients that are likely to benefit, and the relative effectiveness of the various available surgical options.

Randomized prospective studies (RPS) in the current CT/surgical era

One RPS¹²⁹ found lower mortality for patients with GCS 7–10 treated surgically (**note:** only 20% of these patients were operated on <8 hrs from the bleed, and the mean time for all patients to operation was 14.5 hours (range: 6–48 hrs), which may be long). However, survivors in this group were all severely disabled (none were independent).

Another¹³⁰ found no benefit from surgery for *putaminal* hemorrhages, also with poor outcomes in all patients.

International STICH¹³¹: enrolled 1,033 patients. Study shortcomings: possible selection bias (the responsible neurosurgeon had to be uncertain of the benefits of medical vs. surgical treatment), “early surgery” had a somewhat long median time to treatment of 30 hours, and 26% of medically treated patients crossed over and had surgery at a mean of 60 hours (late). Given these limitations, the conclusion was that for supratentorial ICH there was no benefit of early surgery (although there may have been some benefit in the subgroup with a hematoma within 1 cm of the cortical surface). This trial may be more accurately considered to be a comparison of early vs. delayed surgery in patients subjectively judged to need surgery by the investigator.

Conclusion

The decision to operate therefore must be individualized based on patient’s neurologic condition, size and location of hematoma, patient’s age, and the patient’s expressed preferences (e.g., by a “living will”) and the family’s wishes concerning “heroic” measures in the face of catastrophic illness. Also, see (p. 1623).

Guidelines for considering surgery vs. medical management

(For separate indications for surgery for *cerebellar* hemorrhage, see below.)

1. NON-SURGICAL: factors that may favor medical management
 - a) minimally symptomatic lesions: e.g., alert patient with subtle hemiparesis (especially patients with GCS > 10¹²⁹)

- b) situations with little chance of good outcome
 - high ICH score (p. 1618), which overlaps with the following
 - massive hemorrhage with significant neuronal destruction (see below)
 - large hemorrhage in dominant hemisphere
 - poor neurologic condition: e.g., comatose with posturing (i.e., GCS ≤ 5), loss of brainstem function (fixed pupils, posturing...)
 - $\approx$ age > 75 yrs: do not do well with surgery for this
 - c) severe coagulopathy or other significant underlying medical disorder(s): in the event of herniation, rapid decompression surgery may be considered in spite of the risks
 - d) basal ganglion (putaminal) or thalamic hemorrhage: surgery is no better than medical management, and both have little to offer^{130,132} (see below)
2. SURGICAL: factors that may favor rapid surgical removal of the blood clot
- a) lesions with marked mass effect, edema, or midline shift on imaging (removal is considered due to the potential for herniation)
 - b) lesions where the symptoms (e.g., hemiparesis/plegia, aphasia, or sometimes just confusion or agitation...) appear to be due to increased ICP or to mass effect (i.e., compression) from the clot or surrounding edema. Symptoms attributable directly to brain injury from the hemorrhage are unlikely to be reversed by surgical evacuation
 - c) volume: surgery for *moderate volume* hematomas (i.e., ≈ 10 –30 cc, Eq (96.1)) may be more appropriate than with:
 - ✖ small clot (< 10 cc): mass effect from clot + edema is usually not significant enough to require surgery
 - ✖ large clot (> 30 cc): associated with poor outcome (only 1 of 71 patients could function independently at 30 days¹³³)
 - ✖ massive hemorrhage > 60 cc with GCS ≤ 8 : 91% 30-day mortality¹³³
 - ✖ massive hemorrhage > 85 cc (the volume of a sphere with a diameter of 5.5 cm): no patient survived, regardless of treatment in one series¹³⁴
 - d) persistent elevated ICP in spite of therapy (failure of medical management). Evacuating clot definitely lowers ICP, but the effect on outcome is uncertain
 - e) rapid deterioration (especially with signs of brainstem compression) regardless of location in a patient considered to be salvageable
 - f) favorable location, for example:
 - lobar (as opposed to deep hemispheric): in spite of optimistic results in a non-randomized study done in 1983 indicating good outcomes in patients with deep hemorrhages treated with early surgery,⁶⁴ a later randomized study failed to confirm this benefit¹³⁰
 - cerebellar: see below
 - external capsule
 - non-dominant hemisphere
 - g) young patient (especially age ≤ 50 yrs): they tolerate surgery better than elderly patients, and, unlike elderly patients with brain atrophy, they also have less room in the head to accommodate the mass effect of clot + edema
 - h) early intervention following hemorrhage: surgery after 24 hrs from onset of symptoms or deterioration may be of less benefit¹²⁹

Management of cerebellar hemorrhage

Recommendations⁵:

1. cerebellar hematoma ≤ 3 cm diameter: treat conservatively
2. cerebellar hematoma > 3 cm with either hydrocephalus or clinical signs of brainstem compression: surgical evacuation ASAP (level I⁵)
3. patients with absent brainstem reflexes and flaccid quadriplegia: intensive therapy is not indicated. **Note:** some authors contend that the loss of brainstem reflexes from direct compression may not be irreversible,¹³⁵ and that cerebellar hemorrhage represents a surgical emergency (and that the above criteria would thus deny potentially helpful surgery to some, see discussion of cerebellar infarction and decompression (p. 1588))
4. patients with hydrocephalus from ventricular obstruction: ✖ initial treatment with an intraventricular catheter (IVC) without ICH evacuation is not recommended. IVC can only be inserted if there is no coagulopathy. Risk of upward cerebellar herniation (p. 325).

Endoscopic treatment of intraventricular hemorrhage

Endoscopic treatment of intraventricular hemorrhage is of uncertain efficacy (Level II⁵).

96.9.3 Surgical considerations

General recommendations

1. send specimens (hematoma, abnormal-looking tangle of blood vessels if present, and possibly biopsy walls of hematoma cavity) to pathology for analysis¹³⁶ (to rule out tumor, AVM, amyloid angiopathy...)
2. surgical options:
 - a) "standard approach": craniotomy with evacuation of the clot under direct vision (with or without microscope)
 - b) stereotactic aspiration with thrombolytic agents has also been used; see **Stereotactic surgery**, evacuation of intracerebral hemorrhage (p. 1838), including surgery using an "exoscope" through a tube
 - c) endoscopic surgery¹³⁷

Surgical techniques for cerebellar hemorrhage

1. position: lateral oblique (p. 1738) with the involved side up
2. if rapidity is crucial, a midline skin incision is preferred because it can be taken down quickly with little fear of encountering a vertebral artery
3. craniectomy (without bone replacement) is preferred over craniotomy to accommodate post-op swelling
4. a prophylactic Frazier burr hole is recommended to allow rapid treatment if post-op hydrocephalus develops—see placement (p. 1742) and use (p. 1745), or a ventricular catheter may be placed to monitor ICP and allow CSF drainage post-op
5. in cases where there has been rupture into the ventricular system, the surgical microscope should be used to follow the clot to the fourth ventricle, which is then cleared of clot

Intraventricular tissue plasminogen activator (tPA)

Intraventricular tPA may help lyse clot and maintain intraventricular catheter (IVC) patency or reopen a clotted IVC. However, the CLEAR III trial¹³⁸ showed no benefit of using 1 mg tPA vs. saline for the primary endpoint of mRS score (≥ 3 at 6 mos for patients with < 30 ml clot (lower mortality at 6 mos was offset by a larger number of survivors with mRS = 5)). ✖ In cases of suspected aneurysm, AVM, or other vascular malformation, it cannot be used until the source of bleeding has been neutralized.^{139,140}

R: 2–5 mg of tPA^{139,141,142} in NS is administered through an intraventricular catheter (IVC) every 8 hours for up to 4 days. The IVC is closed for 2 hours after injection.¹⁴²

96.10 Outcome

Thalamic hemorrhages that tend to destroy the internal capsule (IC) are more likely to produce hemiplegia than hemorrhages lateral to the IC that compress but do not disrupt the IC.

Mortality: The chief cause of death (in a series testing the effects of dexamethasone) is cerebral herniation,¹⁴³ occurring mainly during the first week and mostly in patients with initial Glasgow Coma Scale scores ≤ 7 . The in-hospital death rate decreased overall during the 1980s but increased for patients ≥ 65 years of age.³

Quoted mortality rates vary widely, and depend on size and location of clot, age, and medical condition of the patient, and etiology of the hemorrhage. Overall, the 30-day mortality rate is $\approx 44\%$ for ICH,² which is similar to that for SAH ($\approx 46\%$). Patients with lobar hemorrhages (p. 1614) tend to fare better than deep ICH (basal ganglion, thalamus...) with only $\approx 11\%$ mortality in 26 patients.¹⁶ Patients on anticoagulation for a-fib fare worse than those who are not.¹⁴⁴

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97 ICH in Young Adults and Pediatrics

97.1 ICH in young adults

97.1.1 General information

In a review of 72 patients age 15–45 yrs suffering nontraumatic ICH,¹ a presumed cause was found in 76% (► Table 97.1).

AVM: lobar hemorrhages in this age group are highly suggestive of AVM. Of 40 lobar hemorrhages, 37.5% were determined to be from AVMs.¹

Pregnancy/postpartum: 3 patients had labor or postpartum hemorrhages (p. 1611); see also Pregnancy & intracranial hemorrhage (p. 1425).

Herpes simplex encephalitis: may produce hemorrhagic changes on CT, especially in the temporal lobes; see Herpes simplex encephalitis (p. 397).

Drug abuse: especially with sympathomimetics such as cocaine (p. 1611) should also be considered in young adults.

Leukemia: ICH may be the initial presentation of leukemia in a young adult (may be due to metastases (chloroma) or to thrombocytopenia).

Table 97.1 Causes of spontaneous ICH in young adults¹

Etiology	%
ruptured AVM	29.1
arterial hypertension	15.3
ruptured saccular aneurysm	9.7
sympathomimetic drug abuse	6.9
tumor ^a	4.2
acute EtOH intoxication	2.8
preeclampsia/eclampsia	2.8
superior sagittal sinus thrombosis	1.4
moyamoya	1.4
cryoglobulinemia	1.4
undetermined	23.6
^a hemangioma, ependymoma, metastatic choriocarcinoma...see Hemorrhagic brain tumors (p. 1612)	

97.1.2 Outcome

Overall in-hospital survival (including those treated medically) was 87.5%.

97.2 Intracerebral hemorrhage in the newborn

97.2.1 General information

Occurs primarily in premature infants. Alternate terms: subependymal hemorrhage (SEH), germinal matrix hemorrhage (GMH), periventricular-intraventricular hemorrhage (PIVH). Intraventricular hemorrhage (IVH) arises from extension of SEH through ependymal lining of ventricle and occurs in 80% of cases of SEH.²

97.2.2 Etiology

The highly vascular germinal matrix is part of the primordial tissue of the developing brain and is the source of future neurons and glial cells. It is located just beneath the ependymal lining of the lateral ventricles, and undergoes progressive involution until 36 weeks gestational age (GA). Thus, the matrix may persist out of utero in premature infants. A disproportionate amount of the total CBF perfuses the periventricular circulation through these capillaries which are immature and fragile and have impaired autoregulation.^{3,4} The site of hemorrhage is age-dependent. Between 24–28

weeks GA they occur over the body of the caudate nucleus and at 29 weeks GA or greater they arise over the head of the caudate nucleus.⁵

97.2.3 Pathogenesis of PIVH in the pre-term infant

The metabolically active GM is susceptible to hypotension and hypoperfusion, which can lead to infarction. The GM is a vulnerable watershed zone supplied by Heubner's artery (from the anterior cerebral artery), terminal branches of the lateral striate arteries (off the middle cerebral artery), and the anterior choroidal artery (off the internal carotid or middle cerebral artery).

1. postnatal hypoxia due to respiratory distress syndrome related to hyaline membrane disease, pneumothorax, and/or anemia can deprive the metabolically active GM of oxygen. This ischemia to the endothelial cells lining the capillaries makes them vulnerable to infarction and then disruption
2. hypercapnia maximally dilates the thin-walled vessels of the GM. If this is followed by sudden increases in perfusion the result can be rupture of the vessels
3. increased venous pressure from any cause (labor and delivery, positive pressure ventilation, stimulation, endotracheal suctioning, myocardial failure from ischemia) can result in increased venous pressure in the GM leading to hemorrhage
4. dehydration followed by rapid resuscitation with hyperosmolar solutions increases the intravascular volume by osmotically encouraging the movement of fluid from tissues into the intravascular space. With associated increases in systemic blood pressure the GM capillaries are at increased risk of rupture

97.2.4 Risk factors for PIVH

Increased cerebral perfusion pressure (CPP) with the associated increased cerebral blood flow (CBF) and hypoxia are the common denominators for most risk factors for PIVH. The elevated pressure may cause the hemorrhage by rupturing the fragile vessels of the germinal matrix, possibly already damaged by previous insults of high or fluctuating CBF and hypoxia.

Risk factors for PIVH include⁶:

1. those associated primarily with increased CBF or CPP:
 - a) asphyxia: including hypercapnia (see above)
 - b) rapid volume expansion
 - c) seizures
 - d) pneumothorax
 - e) cyanotic heart disease (including PDA)
 - f) infants being mechanically ventilated having RDS and fluctuating CBF velocity documented by Doppler flow meter⁷
 - g) anemia
 - h) decreased blood glucose
 - i) arterial catheterization
 - j) blood pressure fluctuations
2. younger gestational age (GA)
3. low birth weight
4. acute amnionitis
5. failure to give antenatal steroids (p. 1632) during the 48 hours prior to pre-term delivery⁸ (i.e., to women at risk of delivering low-birth-weight infants)
6. APGAR's <4 at 1 minute and <8 at 5 minutes
7. acidosis
8. coagulopathies
9. general anesthesia for C-section
10. extracorporeal membrane oxygenation (ECMO): due to heparinization in addition to increased CPP
11. maternal cocaine abuse⁹
12. maternal aspirin use

97.2.5 Epidemiology

Incidence

Depends on the method used for detection (many PIVHs are asymptomatic) and the population being evaluated. 540,000 pre-term infants are born in the United States annually. 85,000 are very

pre-term (<32 weeks GA) and 385,000 are late pre-term (34–36 weeks GA). 63,000 very-low-birth-weight (<1500 grams) infants are born each year. Of the preemies weighing <1500 gm birth weight, 20–25% will suffer from a PIVH.^{10,11}

In a 1978 study, PIVH was found by CT in 43% (20/46) of preemies with birth-weight <1500 gm.¹² Mortality in infants with PIVH was 55%, compared to 23% in those without PIVH.¹² Ultrasound (U/S) detected PIVH in 90% of 113 preemies <34 weeks gestation¹³ (49% were grade III or IV, see ► Table 97.2 for grading).

Table 97.2 Grading subependymal hemorrhage¹²

Grade	Description
I	subependymal
II	IVH without ventricular dilatation
III	IVH with ventricular dilatation
IV	IVH with parenchymal hemorrhage
Abbreviation: IVH = intraventricular hemorrhage.	

Timing

The timing of PIVH has a bimodal distribution. A substantial number occur within 6 hours of birth with 50% occurring within 12 hours of birth.^{14,15} At postnatal days 3–4, a second peak occurs. Only 5% of bleeds will develop after postnatal day 4. Progression of hemorrhage has been documented in 10–20% of infants.¹⁵ Early-onset PIVH is more likely to progress and has a higher mortality.¹⁶

97.2.6 Prevention

Numerous studies have been conducted to find a method of directly reducing the incidence of PIVH among premature infants. Many are controversial. Optimal resuscitation and neonatal care, with an emphasis on measures which minimize cerebral blood flow fluctuations, are key.

1. good prenatal care and avoiding pre-term labor
2. antenatal corticosteroids: administration of one course of antenatal corticosteroids to women at risk of having premature birth infants reduces neonatal mortality, respiratory distress syndrome, and PIVH.¹⁷ Multiple courses of antenatal corticosteroids did not improve outcomes and were associated with decreased head circumference, weight, and length at birth¹⁸
3. indomethacin: results in cerebral vasoconstriction and reduces the responsiveness of CBF to changes in CO₂, lowers CBF and increases arterial oxygenation reducing patent ductus arteriosus (PDA). However, use is possibly associated with increased risk of intestinal perforation
4. antenatal vitamin K given IM >4 hrs prior to delivery decreases PIVH from 33% to 5%
5. sluicing umbilical cord blood and delaying umbilical cord clamping by 30–120 seconds in *prema-ture* babies increased hematocrit and decreased PIVH in 5 of 7 studies¹⁹
6. using surfactant to reduce RDS
7. minimizing external stimulation (some centers use fentanyl drips)
8. steroids to stabilize the GM vessels

97.2.7 Clinical

Grading

The most commonly used grading system of Papile et al (► Table 97.2) is based on MRI, CT, or U/S findings, as illustrated in ► Fig. 97.1.

There is a direct correlation between younger gestational age (GA) and the severity of PIVH. In infants 24–26 weeks GA, 32% will have a Grade III PIVH and 19% will have a Grade IV PIVH, whereas in infants 31–32 weeks GA, 11% will have a Grade III PIVH and 5% will have a Grade IV PIVH.²⁰

Presentation

Asymptomatic PIVH

Most PIVHs are clinically unsuspected (especially with smaller hemorrhages) and are discovered incidentally on surveillance U/S. Symptomatic PIVH may present acutely or subacutely (see below).

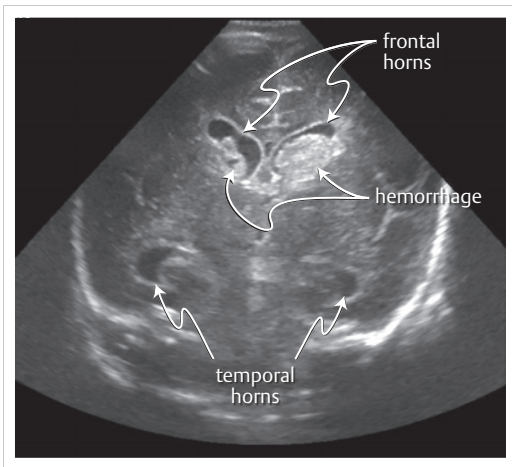


Fig. 97.1 Grade IV bilateral germinal matrix hemorrhages in premature infant 27 weeks gestational age. Temporal horns of lateral ventricles are slightly dilated. Coronal ultrasound through anterior fontanelle.

Retrospectively, these PIVHs may have been suggested by a fall in Hct or delays in neurologic development. Silent PIVHs have a 78% 6-month survival, vs. 20% for PIVH showing signs.

Subacute presentation

Usually smaller or more slowly developing hemorrhages. Clinically may present as irritability, reduced motor activity, or abnormal eye movements.

Acute presentation

1. changes in muscle tone or activity: usually decerebrate or decorticate posturing, sometimes flaccid paralysis
2. seizures: often subclinical
3. tense fontanelle
4. hypotension
5. respiratory and cardiac irregularities: apnea & bradycardia ("A's and B's")
6. unreactive pupils and/or loss of extraocular muscle movements
7. Hct drop > 10%

Hydrocephalus

General information

20–50% of infants with PIVH will develop either transient or progressive hydrocephalus (HCP). Grades III and IV are more often associated with progressive ventricular dilatation than are lower grades (however, HCP may develop even after low-grade PIVH²¹). Younger gestational age infants may be at lower risk.

Post-PIVH hydrocephalus usually occurs 1–3 weeks after the hemorrhage. Probably caused by cellular debris and/or the toxic effects of blood breakdown products on the arachnoid granulations (communicating HCP), or by an adhesive arachnoiditis in the posterior fossa, or, rarely, by compression or blockage of critical pathways, e.g., at the Sylvian aqueduct (obstructive HCP). In a case of HCP following intra-uterine PIVH, aqueductal gliosis was found at autopsy.²²

Differential diagnosis of ventriculomegaly in PIVH

When ventriculomegaly is detected, it needs to be differentiated from the following:

1. transient ventriculomegaly: occurs in the first few days after PIVH. This may not cause elevated ICP. As implied, it is self-limited
2. progressive ventriculomegaly: occurs in 20–50% of cases (true hydrocephalus)
3. "hydrocephalus ex vacuo": due to loss of brain tissue or maldevelopment. Is not progressive on serial U/S. OFCs may fall below normal due to lack of growing brain as stimulus for head growth

Possible presentations

Abnormally increasing OFC (crossing percentile curves faster than body weight), lethargy, apnea and bradycardia, vomiting. There is progressive dilatation of the ventricular system on serial U/S or CT or MRI evaluations.

97.2.8 Pathophysiologic effects of PIVH

Deleterious effects of PIVH on the brain are due to²³:

1. destruction of the germinal matrix and glial precursors
2. direct injury to neural tissue from hematoma: once hemorrhage resorbs may leave patient with porencephaly or cystic lesions
3. pressure of hematoma on nearby brain tissue reducing CBF even to parts of the same hemisphere distant from the hemorrhage²⁴
4. diffuse decreased CBF following the hemorrhage²⁵ due to elevated ICP
5. injury from the same hypoxic event that precipitated the PIVH
6. decreased CPP leads to periventricular leukomalacia (PVL) and cerebral infarction
7. periventricular hemorrhagic infarction
8. hydrocephalus (see above): numerous deleterious effects on the CNS
9. seizures: repeated or prolonged seizures may be deleterious to neuronal function

97.2.9 Diagnosis

Ultrasound (U/S)

Performed through the open fontanelles.¹³ Accuracy $\approx$ 88% (91% sensitivity, 85% specificity).²⁶ U/S is invaluable because:

1. it demonstrates the size of the ventricles, the location and size of the hematoma, and the thickness of the cortical mantle
2. it may be brought to the infant's bedside (obviating transportation)
3. it is non-invasive
4. it is not adversely affected by occasional infant movements (eliminating the need for sedation)
5. there is no exposure to ionizing radiation (radiation from diagnostic imaging in children has long-term risks for cancer²⁷ and damage to the lens)
6. it may be followed serially with relative ease

CT scan

Sometimes necessary when U/S is not readily available, or in complicated cases where anatomy is difficult to deduce from U/S images. Many ICUs have portable CT scans available which obviates need for patient transport.

Rapid-sequence MRI

Pros: Eliminates the risk of ionizing radiation associated with CT scan.

Cons: Requires moving the infant from the neonatal ICU to the radiology suite.

97.2.10 Treatment

General measures

General measures are directed at optimizing CPP without further excessive elevation of CBF by carefully maintaining normal MAP and normalizing pCO_2 , and by treating active hydrocephalus as needed (see above).

While daily LPs can control the deleterious effects of posthemorrhagic HCP, they do not reduce the frequency of long-term HCP (requiring permanent shunting). Ventricular size must be monitored with serial U/S.

Medical treatment

1. not very effective. Treated patients fared worse in several studies
2. osmotic agents: isosorbide, glycerol. Effects are short-lived
3. ✖ diuretic therapy: has been used, but a large study showed increased nephrocalcinosis and biochemical abnormalities, resulting in a borderline increase in the risk for motor impairment at

one year.²⁸ The results were so compelling, the data-monitoring committee terminated the study prematurely. Furosemide and acetazolamide therapy was deemed neither safe nor effective in treating post-hemorrhagic ventricular dilatation and cannot therefore be recommended.²⁹

Surgical/interventional treatment for the clot

Due to poor operative results, surgical evacuation of an intracerebral hemorrhage in the newborn is not indicated with the possible exception of a posterior fossa hemorrhage causing brainstem compression that does not respond to medical treatment.³⁰ Supportive measures are usually in order.

Intervention for intraventricular blood

General information

34% of infants < 1500 g require shunt/reservoir drainage after failed medical management. Grade III and IV PIVH: > 70% of cases develop progressive ventricular dilatation, and 32–47% of this subset will ultimately require shunting.³¹

Indications for intervention

Intervention for intraventricular blood is indicated in the setting of progressive ventriculomegaly with the OFC crossing percentile curves and clinical evidence of increased ICP (split sutures, tense fontanelle...).

Serial lumbar punctures

Used at many facilities for hemorrhages with intraventricular extension and communicating hydrocephalus (the usual type of HCP that occurs with PIVH).³²

This should be undertaken with the knowledge that meta-analysis³³ showed that sequential lumbar or ventricular taps of ≈ 10 ml/kg/tap for prophylaxis or treatment of progressive hydrocephalus offers no clear benefit over conservative treatment, and had an infection rate of 5–9%. In rare cases, LPs may succeed in temporizing progressive HCP for a few weeks until the infant is large enough for shunt placement.

Infants < 800 gm may not tolerate LPs because of desaturation when lying on their side, or the LP itself may be difficult. In these patients, consider 1–2 ventricular taps to at least obtain fluid for analysis (in some cases nothing further needs to be done).

Serial ventricular taps

May be a viable short-term option for those infants who cannot tolerate LPs or in whom there is obstruction to CSF flow in the lumbar subarachnoid space (e.g., due to spinal subdural hematoma from previous LP). However, it is not desirable for long-term use because of repeated trauma to brain (risk of porencephaly) and risk of intracerebral, intraventricular, or subdural hemorrhage.

If continued taps are likely (i.e., large hemorrhage, or rapid recurrence of intracranial hypertension as determined by palpation of fullness of anterior fontanelle (AF) following several taps), the acceptable options include:

1. continuing serial LPs (see below)
2. percutaneous ventricular taps: not recommended for more than a few treatments as it causes porencephaly
3. placement of a temporary ventricular access device (TVAD)—a *ventricular catheter* connected to a subgaleal *reservoir* (e.g., a Rickham reservoir, or a low profile McComb reservoir³⁴). These can be inserted safely at the bedside, obviating the need for transport to the O.R.³⁵
 - a) temporary ventricular access: the reservoir can be used for serial percutaneous taps. Usually tapped QD or QOD (see below). Use a 27 Ga butterfly needle, clean with at least 3 betadine stick swabs, withdraw ≈ 10 ml and send for culture. Reported infection rate: 8–12%³⁶
 - b) ventricular-subgaleal shunt: the side-port of the reservoir is left uncapped. A subgaleal pocket must be created at the time of surgery. Fluid is reabsorbed from this potential space. First performed in 1893 by Mikulicz-Radecki (1850–1905). Use has been reported up to 35 days.³⁷ Infection rate: $\approx 6\%$
 - c) the reservoir may be converted to VP shunt if and when appropriate. Not recommended in infants < 1100 gms due to very high infection rate
4. external ventricular drainage (EVD): similar to reservoir placement, but with possibility of inadvertent dislodgment (13%) and comparable infection rate (6%)

5. early VP shunting: high infection rate, peritoneal cavity not suitable in many cases, e.g., due to necrotizing enterocolitis (NEC), paucity of subcutaneous tissue through which to pass shunt tube... Not recommended for infants < 2000 gms

Temporary ventricular access device (TVAD)

Advantages of TVAD

1. avoids shunt in unhealthy children at risk of infection, skin breakdown, or other operative/anesthetic complications
2. clears protein and cellular debris (more favorable for subsequent shunting)
3. avoids repeated penetration of brain with risk of porencephaly
4. provides port for infusion of medication (e.g., antibiotics) PRN
5. avoids cumbersome, easily dislodged EVD with infection risk 6% on average of 13 days of EVD
6. up to 25% of patients will recover and avoid permanent shunt placement^{38,39}

Disadvantages of TVAD

1. requires services of a neurosurgeon (not always available)
2. increases risk of infection of subsequent permanent shunt from 5% to 13%⁴⁰
3. inherent risks of surgery including hemorrhage, infection, ventriculitis, meningitis, CSF leak
4. risks of overdrainage including subdural hematoma, impaired skull growth

Technical considerations for serial taps (via ventricular reservoir or LP)

8–20 cc of fluid are removed initially, and this is repeated daily (or more often if AF become very tense before 24 hours elapse) for several days, and then usually varies from 5–20 cc qod to 15 cc TID depending on response. The frequency and volume of the taps are modified based on:

1. fullness of AF: attempt to keep AF from becoming tense
2. appearance of ventricles on serial U/S: strive to prevent progressive enlargement, reduction in size can usually be achieved
3. follow OFC: should not cross percentile curves (need to differentiate from the so-called “catch-up phase” of brain growth which may occur once the infant overcomes their overall medical problems and is able to adequately utilize nutrition^{41,42}; serial U/S will show rapid brain growth without progressive ventriculomegaly in cases of catch-up brain growth)
4. CSF protein concentration: controversial. Diminishes with serial taps. Some feel that as long as it is ≥ 100 mg/dl it is unlikely that significant spontaneous resorption will occur and continued serial taps will probably be needed
5. NB: removal of this volume of fluid may cause electrolyte disturbances, primarily hyponatremia;
 - ∴ follow serum electrolytes on a regular basis

Follow with serial U/S on day 3–5, and then weekly for several weeks, and then bi-weekly. A baseline CT scan is often obtained prior to placement of a permanent shunt.

Insertion of VP shunt or conversion of sub-Q reservoir to VP shunt

Indications and requirements:

1. symptomatic hydrocephalus (p. 1633) and/or progressive ventriculomegaly
2. infant is extubated (and thus off ventilator)
3. infant weighs ≥ 2000 grams (some prefer ≥ 2500 grams)
4. no evidence of NEC (might create problems with peritoneal end of catheter)
5. CSF protein ideally < 100 mg/dl (because of concerns about plugging of the shunt, or causing ileus or malabsorption of the fluid—which was not seen with high-protein fluid shunted from the subdural space⁴³—and also to see if patient will start reabsorbing CSF on their own)

Technical recommendations:

1. do not tap reservoir for at least 24 hrs before inserting a new ventricular catheter (allows ventricles to expand to facilitate catheterization)
2. obtain U/S the day prior to conversion
3. use a low- or very-low-pressure system (if CSF protein is high, consider a valveless system), upgrade later in infancy if necessary
4. avoid placing shunt hardware in areas on which these debilitated infants tend to lie (to prevent skin breakdown with hardware exposure)

97.2.11 Outcome

Short-term

Preemies with PIVH have higher mortality than matched preemies without PIVH.

The incidence of mortality and progression of hemorrhage is higher the earlier the hemorrhage occurs. The more severe the hemorrhage, the higher the mortality and the higher the risk of HCP (► Table 97.3).

Table 97.3 Short-term outcome of PIVH ($\approx$ 250 cases²)

Severity of hemorrhage	Deaths (%)	Progressive hydrocephalus (%)
mild	0	0–10
moderate	5–15	15–25
severe	50–65	65–100

Long-term

The effect of low-grade PIVH on long-term neurodevelopment has not been studied well. Most investigators feel that higher grades of PIVH are associated with greater degrees of handicaps than matched controls.

In one study of 12 infants with Grade II PIVH treated with serial LPs and in the 7 with progressive ventriculomegaly with VP shunt followed for a mean of 4.5 years, all were ambulatory and 75% had IQ within normal range.⁴⁴

A recent study of very-low-birth-weight infants showed that children 18–22 months of age with severe PIVH and shunts had significantly lower scores on the Bayley Scales of Infant Development IIR compared with children with no PIVH and children with equal grades of PIVH who did not require a shunt.⁴⁵

97.3 Other causes of intracerebral hemorrhage in the newborn

1. birth trauma may result in subdural hemorrhage, tentorial hemorrhage, parenchymal hemorrhage, and/or subarachnoid blood. This is usually detected by imaging (U/S or CT) when an infant develops seizures, apnea, bradycardia, or, rarely, focal neurological deficits. It rarely requires surgical intervention
2. choroid plexus hemorrhage can result in IVH. In some cases HCP can develop and require shunt placement
3. hemorrhagic stroke has been identified in 6.2 per 100,000 live births.⁴⁶ The usual presentation was with encephalopathy (100%) and seizures (65%). 75% of the strokes were idiopathic. Other identified etiologies were thrombocytopenia and a single case of a cavernous malformation. Risk factors for perinatal hemorrhagic stroke include: male gender, fetal distress, emergent C-section, prematurity, and post-maturity
4. tumors in the neonate can present with hemorrhage
5. vascular malformations of any form can present in the neonate with hemorrhage, although this is uncommon. Vein of Galen malformations are diagnosed in the neonate in about 40% of cases.⁴⁷ Most of these infants present with fulminant congestive heart failure and 50% have ventriculomegaly

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Part XXI

Outcome Assessment

98 Outcome Assessment

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XXI

98 Outcome Assessment

98.1 Cancer

► **Karnofsky performance scale (KPS).** ► Table 98.1 (after David A. Karnofsky), often used for grading functional status in patients with cancer. A KPS score < 70 (particularly with brain tumors) often identifies patients with a worse prognosis for any given treatment.

Table 98.1 Karnofsky performance status scale (modified^{1,2})

Score	Criteria	General category
100	normal: no complaints, no evidence of disease	Able to carry on normal activity and work. No special care is needed
90	able to carry on normal activity: minor signs or symptoms	
80	normal activity with effort: some signs or symptoms	
70	cares for self: unable to carry on normal activity or to do active work	Unable to work. Able to live at home, care for most personal needs. Variable assistance is required
60	requires occasional assistance: cares for most of needs	
50	requires considerable assistance and frequent care	
40	disabled: requires special care and assistance	Unable to care for self. Requires equivalent of institutional or hospital care. Disease may be rapidly progressing
30	severely disabled: hospitalized; death not imminent	
20	very sick: hospitalized; active supportive care needed	
10	moribund: fatal processes are progressing rapidly	
0	dead	

WHO performance score. ► Table 98.2, the WHO Performance score³ (AKA The Eastern Cooperative Oncology Group (ECOG) score, AKA Zubrod score [after C. Gordon Zubrod]), ranges from 0 to 5, with 0 indicating perfect health and 5 death. The advantage over the Karnofsky scale is its simplicity.

Table 98.2 WHO performance scale

Grade	Description
0	Fully active, no performance restriction as a result of disease.
1	Restricted in physically strenuous activity. Ambulatory. Able to do light work, e.g., light house work, desk work.
2	Unable to perform any work activities. Ambulatory. Up & about > 50% of waking hours.
3	Only able to perform limited self-care. Wheelchair-confined > 50% of waking hours.
4	Completely disabled. Unable to perform self-care. Totally bed- or chair-confined.
5	Dead

98.2 Head injury

The **Rancho Los Amigos scale** (► Table 98.3) is often used in rating *disability* following head injury. The Glasgow outcome scale⁴ and the augmented Glasgow Outcome Scale - Extended (GOSE)⁵ (► Table 98.4) are frequently employed in *outcome* assessment. For the GOSE, the use of a structured interview is recommended to enhance the consistency of ratings.

98.3 Cerebrovascular events

98.3.1 General information

Several outcome grading scales have come to be favored for use following strokes or SAH. Each emphasizes different aspects of outcome. The Barthel Index (► Table 98.6) places weight on activities

Level	Meaning
I	no response to pain, touch, sight, or sound
II	generalized reflex responses to pain
III	Localized response blinks to strong light, turns towards/away from sound, responds to physical discomfort, inconsistent responses to commands
IV	Confused—Agitated alert, very active, agitated, aggressive, or bizarre behaviors. Performs motor activities but behavior is non-purposeful, extremely short attention span
V	Confused—Non-agitated gross attention to environment, easily distracted, requires continual redirection, difficulty learning new tasks, agitated by excess stimulation. May converse socially but with inappropriate verbalizations
VI	Confused—Appropriate inconsistent orientation to time and place. Retention span and recent memory impaired. Begins to recall past, consistently follows simple commands, goal-directed behavior with assistance
VII	Automatic—Appropriate performs daily routine in highly familiar environment in a non-confused but automatic “robot-like” fashion. Skills deteriorate in unfamiliar environment. Lacks realistic planning for future
VIII	Purposeful—Appropriate

Glasgow Outcome Scale - Extended (GOSE) ⁵		Score	Glasgow Outcome Scale (original) ⁴
death		1	death—most deaths ascribable to primary head injury occur within 48 hrs
vegetative		2	persistent vegetative state—unresponsive & speechless. After 2–3 weeks, may open eyes & have sleep/wake cycles
severe disability	low severe disability: dependent for daily support for physical and/or mental disability, but can be left alone at home > 8 hrs	3	severe disability (conscious but disabled)—dependent for daily support (may be institutionalized, but this is not a criterion)
	upper severe disability: as with #3, but cannot be left alone at home > 8 hrs	4	moderate disability (disabled but independent)—travel by public transportation, can work in sheltered setting (exceeds mere ability to perform “activities of daily living”)
moderate disability	low moderate disability: some disability (e.g., aphasia, hemiparesis, seizures, memory deficits...). Independent at home, but dependent outside. Unable to return to work	5	good recovery—resumption of normal life despite minor deficits (“return to work” not reliable)
	upper moderate disability: as with #5 but able to return to work even if special arrangement is needed	6	—
good recovery	lower good recovery: resumption of normal life with capacity to work, even if pre-injury status is not achieved, but no disabling deficits	7	—
	upper good recovery: as with #7, but with disabling deficits	8	—

of daily living (ADLs), while others, such as the modified Rankin Scale⁶ (► Table 98.5) assess levels of independence, include a comparison to previous activity levels, and show fairly good interobserver consistency.⁷ While it does measure functional status, the modified Rankin is not sensitive to subtle neurologic deficits such as dysphasia or visual field defects.

Table 98.5 The modified* Rankin Scale

Grade	Description
0	no symptoms at all
1	no significant disability despite symptoms: able to carry out all usual duties & activities
2	slight disability: unable to carry out all previous activities. Able to look after own affairs without assistance
3	moderate disability: requiring some help, but able to walk without assistance
4	moderately severe disability: unable to walk without assistance, and unable to attend to own bodily needs without assistance
5	severe disability: bedridden, incontinent, and requiring constant nursing care and attention

*the original Rankin scale⁸: did not have Grade 0, Grade 1 did not include the words “despite symptoms” and “& activities,” and it defined Grade 2 as “unable to carry out some of previous activities...”

Table 98.6 The Barthel index

Item	Original Barthel Index			Modified Barthel Index				
	Unable to perform task	Needs assistance	Fully independent	CODE 1 Unable to perform task	CODE 2 Attempts task but unsafe	CODE 3 Moderate help required	CODE 4 Minimal help required	CODE 5 Fully independent
Personal hygiene	0	0	5	0	1	3	4	5
Self-bathing	0	0	5	0	1	3	4	5
Feeding	0	5	10	0	2	5	8	10
Toilet	0	5	10	0	2	5	8	10
Stair climbing	0	5	10	0	2	5	8	10
Dressing	0	5	10	0	2	5	8	10
Bowel control	0	5	10	0	2	5	8	10
Bladder control	0	5	10	0	2	5	8	10
Ambulation	0	5–10	15	0	3	8	12	15
Wheelchair ^a	0	0	5	0	1	3	4	5
Chair/bed transfers	0	5–10	15	0	3	8	12	15
Total (range)	0	→→	100	0	→→→→			100

^ascore only if unable to walk and patient trained in wheelchair management

98.3.2 Scales

► Modified Rankin scale (► Table 98.5)

► **Barthel index** (► Table 98.6). The original Barthel index^{9,10} assigns one of three scores to 10 ratable ADLs, and then the individual scores are summed. The modified Barthel index (MBI) with a 5-step scoring system appears to have greater sensitivity.¹¹ The total ranges from 0 to 100 (a score of 100 implies functional independence, not necessarily normality).

Of all the factors, independence in bathing was the most difficult. Abilities on the Barthel index tend to return in a fairly consistent order, and so most patients with the same score will have similar patterns of disability.

98.4 Spinal cord injury

► **Functional Independence Measure™ (FIM™).**^{12,13,14} Developed to provide uniform evaluation of disability for spinal cord injuries. Rates 18 items shown in ► Table 98.7 (13 motor, 5 cognitive) on the 7-level scale shown in ► Table 98.8.

The FIM™ has high internal consistency and is a good indicator of burden of care.^{15,16}

Table 98.7 The Functional Independence Measure™ (FIM)

Classification	Item	Score*
Motor		
self-care	eating	(1 - 7)
	grooming	(1 - 7)
	bathing	(1 - 7)
	dressing—upper body	(1 - 7)
	dressing—lower body	(1 - 7)
	toileting	(1 - 7)
sphincter control	bladder management	(1 - 7)
	bowel management	(1 - 7)
mobility	bed, chair, wheelchair	(1 - 7)
	toilet	(1 - 7)
	tub, shower	(1 - 7)
locomotion	walk or wheelchair	(1 - 7)
	stairs	(1 - 7)
Cognitive		
communication	comprehension	(1 - 7)
	expression	(1 - 7)
social cognition	social interaction	(1 - 7)
	problem solving	(1 - 7)
	memory	(1 - 7)
FIM™ → TOTAL		(18 - 126)

* Score 1–7 as shown in ► Table 98.8

Table 98.8 The 7 FIM™ rating levels of disability

Degree of dependency	Level of function	Score
no helper	complete independence	7
	modified independence	6
modified dependence on a helper	supervision	5
	minimal assist (≥ 75% independent)	4
	moderate assist (≥ 50% independent)	3
complete dependence on a helper	maximal assist (≥ 25% independence)	2
	total assist (< 25% independence)	1

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Part XXII

Differential Diagnosis

- | | | |
|-----|--|------|
| 99 | Differential Diagnosis by Location or Radiographic Finding – Intracranial | 1646 |
| 100 | Differential Diagnosis by Location or Radiographic Finding – Spine | 1677 |
| 101 | Differential Diagnosis (DDx) by Signs and Symptoms – Primarily Intracranial | 1683 |
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99 Differential Diagnosis by Location or Radiographic Finding – Intracranial

99.1 Diagnoses covered outside this chapter

See ► Table 99.1.

Table 99.1 Differential diagnoses by location or radiographic finding, intracranial—covered outside this chapter

DDx
chordomas (p. 825)
extra-axial fluid (peds) (p. 1087)
gyral enhancement (p. 1560)
hydrocephalus (p. 433)
pineal region tumors (p. 758)
pneumocephalus (p. 1067)
schizencephaly (p. 309)

99.2 Posterior fossa lesions

99.2.1 Cerebellar lesions

General information

The following addresses *intra*-axial p-fossa abnormalities (for extra-axial lesions, see below).

Adult

Single lesion

► **Note.** Rule of thumb: “the differential diagnosis of a solitary intraparenchymal lesion in an adult p-fossa is metastasis, metastasis, metastasis, until proven otherwise.”

- tumors:
 - metastasis
 - hemangioblastoma (p. 822): the most common PRIMARY intra-axial p-fossa tumor in adults (7–12% of p-fossa tumors). Very vascular nodule, often has cyst. Almost all p-fossa tumors are relatively avascular on angiography *except* these (look for serpentine signal voids especially in the periphery of the lesion on MRI,¹ much less common in cavernous hemangioma)
 - cerebellar (pilocytic) astrocytoma (p. 693): may be solid or cystic, tends to occur in younger adults
 - brainstem glioma: an isolated glioblastoma in the posterior fossa of an adult is a reportable rarity
 - choroid plexus tumor: usually infratentorial in adults (p. 1665)
 - cerebellar liponeurocytoma (p. 721)
- infectious: abscess
- vascular
 - cavernous hemangioma
 - hemorrhage
 - infarction: cerebellar stroke may be associated with H/A and/or pain in suboccipital region or upper neck
 - embolic
 - thrombotic/plaque-related
 - vertebral artery dissection: much less common than carotid dissection (p. 1579)
 - vertebrobasilar hypoplasia (p. 1592)
- Lhermitte-Duclos (p. 716): Focal or diffuse. Nonenhancing. Characteristic tiger stripes. Widens folia (cf. most neoplasms which destroy folial pattern)

Multiple lesions

- metastases
- hemangioblastoma (possibly as part of von Hippel-Lindau) (p. 646)

3. abscesses
4. cavernous hemangiomas

Pediatric

Also see Pediatric brain tumors (p.621).

Early data: 67% of childhood brain tumors occur in p-fossa, and astrocytomas were the most common there. Currently: p-fossa tumors comprise 54–60% of childhood brain tumors (breakdown listed below). 4 types account for $\approx$ 95% of intratentorial tumors in patients $\leq$ 18 yrs of age.² The 3 most common are equal in incidence (expressed as percent of p-fossa tumors pooled from 1,350 pediatric brain tumors³):

1. medulloblastoma (p.744): 27%
 - a) most start in *roof* of 4th ventricle (fastigium), and most are solid
 - b) differentiating medulloblastoma (MB) from ependymoma:
 - 4th ventricle drapes around medulloblastoma (“banana sign”) from the anterior aspect, cf. ependymoma which tends to grow into 4th ventricle from floor. Ependymoma may grow through foramen of Luschka and/or Magendie
 - ependymomas tend to be inhomogeneous on T1WI MRI (unlike MB)
 - the exophytic component of ependymomas tends to be high signal on T2WI MRI (with MB this is only mildly hyperintense)
 - calcifications: common in ependymomas, but only in $<$ 10% of MB
2. cerebellar (pilocytic) astrocytoma (p.693): 27%. Most start in cerebellar hemisphere. Often cystic with enhancing mural nodule
3. brainstem gliomas (p.695): 28%. Usually present with multiple cranial nerve palsies and long tract findings
4. ependymoma (p.724): usually arise in *floor* of 4th ventricle
5. choroid plexus papilloma (p.739): majority of patients are $<$ 2 yrs old
6. atypical teratoid/rhabdoid tumor (AT/RT) (p.754)
7. metastasis: neuroblastoma, rhabdomyosarcoma, Wilms tumor...
8. PHACES syndrome: acronym for a group of findings including Posterior fossa malformations, cervicofacial Hemangioma, Arterial anomalies of the head and neck, Coarctation of the aorta and cardiac defects, Eye anomalies, and Sternal cleft. Ratio girls:boys = 9:1. Thought to begin during gestation weeks 8–10

99.2.2 Cerebellopontine angle (CPA) lesions

Lesions in general

Vestibular schwannoma, meningioma, and epidermoid account for most. For those lesions that may be cystic, see below.

1. vestibular schwannoma: (80–90% of CPA lesions); see below for differentiating from meningioma (p.1648)
2. meningioma: (5–10%); see below for differentiating from vestibular schwannoma (p.1648)
3. ectodermal inclusion tumors (p.937)
 - a) epidermoid (cholesteatoma): 5–7%. High signal on DWI MRI (p.937). Tumor passing from the posterior fossa to the middle fossa though the incisura is highly suggestive of epidermoid
 - b) dermoid
4. metastases
5. neuroma from cranial nerves other than VIII (also see below for some differentiating features)
 - a) trigeminal neuroma: expands towards Meckel's cave
 - b) facial nerve neuroma⁴: may arise in any portion of the VII nerve, with a predilection for the geniculate ganglion.⁵ Even in these tumors, hearing loss tends to precede facial paresis. Hearing loss may be sensorineural from VIII nerve compression from tumors arising in the proximal portion of VII (cisternal or internal auditory canal (IAC) segment), or it may be conductive from erosion of the ossicles by tumors arising in the second (tympanic, or horizontal) segment of VII. Facial palsy (peripheral) (p.607) may also develop, usually late⁴
 - c) neurinoma of lowest 4 cranial nerves (IX, X, XI, XII)
6. arachnoid cyst (p.260)
7. neurenteric cyst (p.313): rare.⁶ May secrete mucin
8. cholesterol granuloma (distinct from epidermoid) (p.937)
9. lipoma
10. aneurysm: PICA, AICA, vertebrobasilar
11. dolichobasilar ectasia

12. cysticercosis
13. extensions of:
 - a) brainstem or cerebellar glioma
 - b) PitNET/adenoma
 - c) craniopharyngioma
 - d) chordoma & tumors of skull base
 - e) 4th ventricle tumors (ependymoma, medulloblastoma)
 - f) choroid plexus papilloma: from 4th ventricle through foramen of Luschka
 - g) glomus tumor
 - glomus jugulare
 - glomus tympanicum
 - h) primary tumors of temporal bone (e.g., sarcoma or carcinoma)

Cystic lesions of the CPA

CPA lesions from the above list that may be cystic or have a cystic component⁶:

1. arachnoid cyst: same intensity as CSF on all MRI sequences, homogeneous
2. epidermoid cyst (► Fig. 56.1): ★ high signal on DWI MRI differentiates this from arachnoid cyst
3. dermoid cyst: high-intensity areas on T1WI similar to fat; usually midline
4. cystic schwannoma
5. cholesterol granuloma: ★ ≈ only lesion that is high signal on T1WI (due to blood breakdown products; exception: the rare “white” epidermoid). Also high signal on T2WI. Usually extradural, especially near petrous apex. Bone destruction is common
6. neurenteric cyst: nonenhancing. Low intensity on DWI MRI
7. choroidal cyst
8. cysticercosis: enhancing nodule (scolex)

Differentiating neuromas of V, VII, and VIII cranial nerves

All 3 of these tumors may present in the CPA and may cross from posterior fossa to middle fossa, but they tend to do so in different manners. Vestibular schwannomas show “translational” extension by passing through the tentorial hiatus medially. Most trigeminal neuromas show “transapicopetrosal” extension by crossing into the middle fossa via the petrous apex (although some show translational extension). When facial neuromas cross, they tend to spread across the midpetrosal bone, which is characteristic for facial neuromas.⁴ When a facial neuroma enlarges the IAC, unlike a vestibular schwannoma, it tends to erode the anterosuperior aspect of the IAC.

Differentiating vestibular schwannoma from CPA meningioma

1. vestibular schwannoma (VS) (AKA acoustic neuroma):
 - a) clinical: progressive unilateral hearing loss, usually with tinnitus. Progression results in unsteadiness, with true vertigo being rare. The facial nerve is more resistant to stretching, and thus facial nerve signs and symptoms occur late. Trigeminal nerve involvement may occur with tumors > 3 cm (check corneal reflex), with tic douloureux-like symptoms being unusual
 - b) imaging: often heterogeneous signal and nonuniform enhancement. Medium-size tumors look like ice cream in a cone (IAC is the cone). Rarely calcified. Except for very small tumors, IAC is frequently enlarged. Look for an acute angle between the tumor and the petrous bone (meningiomas usually have an obtuse angle)
2. meningiomas: may mimic VSs with these differences:
 - a) clinical: since they often arise from the superior anterior edge of the IAC, early facial nerve involvement is more common, and hearing loss is usually *late*. Trigeminal neuralgia-like pain is more common than with VSs
 - b) imaging: homogeneous signal and enhancement. The tumor may enter the IAC but it tends not to enlarge it. IAC often eccentric in tumor. Tumor is flat against petrous bone with an obtuse angle to the bone. *Calcification and bony hypertrophy* may occur (which occasionally *narrows* the IAC).

99.2.3 Petrous apex lesions

1. infection/inflammatory:
 - a) osteomyelitis: may produce Gradenigo's syndrome (p.601)
 - b) cholesterol granuloma (bright on T1WI; epidermoid cysts, ► Fig. 56.1 are bright on DWI, neither enhance)

2. vascular lesions: aneurysm
3. neoplastic:
 - a) squamous cell cancer
 - b) glomus tumor
 - c) chondrosarcoma: will displace the carotid from medial to lateral (almost every other tumor in this region encases the carotid)

99.2.4 Foramen magnum lesions

Differential diagnosis

See Foramen magnum lesions (p.1649) for *nonneoplastic* lesions. Most foramen magnum (FM) region tumors are extra-axial. This includes:

1. extra-axial tumors
 - a) meningioma: the anterior lip of the foramen magnum is the second most common site of origin of posterior fossa meningiomas. Meningiomas (p.817) comprise 38–46% of FM tumors^{7,8} and most are intradural
 - b) Chordoma (p.825): a mass behind the dens compressing the spinal cord is a chordoma until proven otherwise
 - c) neurilemmoma
 - d) epidermoid
 - e) chondroma
 - f) chondrosarcoma
 - g) metastases
2. exophytic component of a brainstem tumor
3. non-neoplastic lesions
 - a) aneurysms or ectasia of the vertebral artery
 - b) odontoid process in cases of basilar invagination (p.228)
 - c) pannus from involvement of the odontoid with rheumatoid arthritis or old nonunion of fracture
 - d) synovial cyst of the quadrate ligament of the odontoid⁹

Presentation

In the pre-imaging era (i.e., before CT & MRI) these lesions were often diagnosed relatively late due to the unusual associated clinical syndromes and the rarity of visualizing this region on myelography.

Clinical findings

Symptoms:

1. sensory
 - a) craniocervical pain: usually an early symptom, commonly in neck and occiput. Aching in nature. ↑ with head movement
 - b) sensory findings: usually occur later. Numbness and tingling of the fingers
2. motor
 - a) spastic weakness of the extremities: weakness usually starts in the ipsilateral UE, then the ipsilateral LE, then the contralateral LE, and finally the contralateral UE (“rotating paralysis”)

Signs:

1. sensory
 - a) dissociated sensory loss: loss of pain and temperature contralateral to lesion with preservation of tactile sensation
 - b) loss of position and vibratory sense, greater in the upper than the lower extremities
2. motor
 - a) spastic weakness of the extremities
 - b) atrophy of the intrinsic hand muscles: a lower motor nerve finding
 - c) cerebellar findings may rarely be present with extensive intracranial extension
3. long tract findings
 - a) brisk muscle stretch reflexes (hyperreflexia, spasticity)
 - b) loss of abdominal cutaneous reflexes
 - c) neurogenic bladder: usually a very late finding

4. ipsilateral Horner syndrome: due to compression of cervical sympathetics
5. nystagmus: classically downbeat nystagmus (p.586), but other types can occur

It had been postulated that long tract findings were due to direct compression at the cervicomedullary junction, and that lower motor nerve findings in the upper extremities were due to central necrosis of the gray matter as a result of compression of arterial blood supply. Anatomic study suggests that it is actually *venous* infarction at lower cervical levels (C8–1) that is responsible for the lower motor neuron findings.

99.3 Multiple intracranial lesions on CT or MRI

1. neoplastic
 - a) primary
 - multicentric gliomas; ≈ 6% of gliomas are multicentric, more common in neurofibromatosis, see Multiple gliomas (p.667)
 - tuberous sclerosis (including giant cell astrocytomas); (usually periventricular)
 - multiple meningiomas
 - lymphoma
 - PNET
 - multiple neuromas (usually in neurofibromatosis, including bilateral vestibular schwannomas)
 - b) metastatic: usually cortical or subcortical, surrounded by prominent vasogenic edema (p.911). More common tumors include:
 - lung
 - breast
 - melanoma: may be higher density than brain on unenhanced CT
 - renal cell
 - gastrointestinal tumors
 - genitourinary tract tumors
 - choriocarcinoma
 - testicular
 - atrial myxoma
 - leukemia
2. infection: mostly abscess or cerebritis. Most commonly due to:
 - a) pyogenic bacteria
 - b) toxoplasmosis: common in AIDS patients (p.404)
 - c) fungal
 - cryptococcus
 - mycoplasma
 - coccidiomycosis
 - aspergillosis
 - candidiasis
 - d) echinococcus
 - e) schistosomiasis
 - f) paragonimiasis
 - g) herpes simplex encephalitis (HSE): usually temporal lobe (p.397)
3. inflammatory
 - a) demyelinating disease
 - MS: usually in white matter, periventricular, with little mass effect, margins are usually very sharp. Ring-enhancing lesions can occur with tumefactive demyelinating lesions (p.187)
 - progressive multifocal leukoencephalopathy (PML): primarily in white matter. No enhancement. Patients are usually very sick
 - b) gummas
 - c) granulomas
 - d) amyloidosis
 - e) sarcoidosis
 - f) vasculitis or arteritis
 - g) collagen vascular disease, including:
 - periarteritis nodosa (PAN) (p.208)
 - systemic lupus erythematosus (SLE)
 - granulomatous arteritis
4. vascular
 - a) multiple aneurysms (congenital or atherosclerotic)

- b) multiple hemorrhages, e.g., associated with DIC or other coagulopathies (including anticoagulant therapy)
 - c) venous infarctions, especially in dural sinus thrombosis (p. 1594)
 - d) moyamoya disease (p. 1581)
 - e) subacute hypertension (as in malignant HTN, eclampsia...) → symmetric confluent lesions with mild mass effect and patchy enhancement, usually in occipital subcortical white matter
 - f) multiple strokes
 - lacunar strokes (l'état lacunaire)
 - multiple emboli (e.g., in atrial fibrillation, mitral valve prolapse, SBE, air emboli)
 - sickle cell disease
 - vasculitis
 - intravascular lymphomatosis (p. 842)
 - 5. hematomas and contusions
 - a) traumatic (multiple hemorrhagic contusions, multiple SDH)
 - b) multiple "hypertensive" hemorrhages (amyloid angiopathy, etc.)
 - 6. intracranial calcifications (p. 1665)
 - 7. miscellaneous
 - a) radiation necrosis
 - b) foreign bodies (e.g., post gunshot wound)
 - c) periventricular low densities
 - Binswanger's disease
 - transependymal absorption of CSF (e.g., in active hydrocephalus)
- **Evaluation.** Deciding which of the following tests are needed to evaluate a patient with multiple intracranial lesions must be individualized for the appropriate clinical setting.
1. cardiac echo: to R/O SBE that could shed septic emboli
 2. "metastatic workup" (p. 914) including:
 - a) CT of chest/abdomen/pelvis with and without contrast: has become a relatively standard part of the metastatic workup. It has largely supplanted CXR, lower GI (barium enema), and IVP.

Rationale:

 - Chest: R/O primary bronchogenic Ca or pulmonary metastases of another Ca. Can demonstrate mediastinal lymphadenopathy. Also to R/O pulmonary abscess that could shed septic emboli
 - Assesses for possible primary lesions: e.g., kidneys, GI, prostate
 - Evaluates for metastases to liver, adrenal, and even spine
 - b) mammogram in women
 - c) PSA in men

99.4 Ring-enhancing lesions on CT/MRI

99.4.1 Abscess vs. tumor

See ► Fig. 99.1 and ► Fig. 99.2. Tumor: the enhancing ring may be incomplete and irregular. Abscess: ring is usually complete, often thinner and smoother than with tumor. Abscess: usually brighter than tumor on DWI MRI.

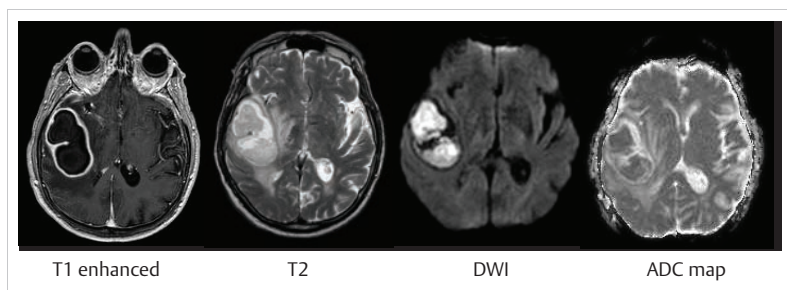


Fig. 99.1 Cerebral abscess, right side. Image: MRI. NB: contents are bright on DWI.

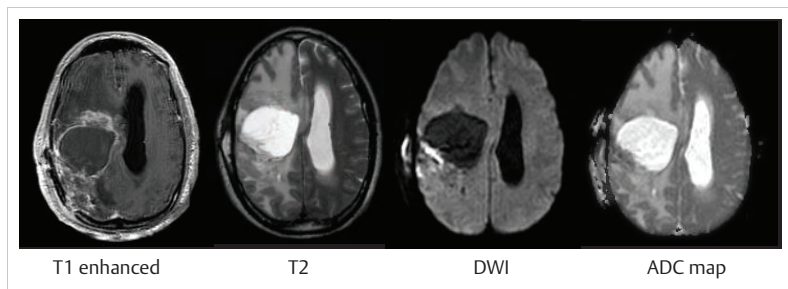


Fig. 99.2 Glioblastoma, right side. Image: MRI. NB: dark on DWI.

MR spectroscopy (MRS) (p.244) should theoretically be ideal for differentiating tumor from abscess (abscess should show reduced NAA, Cr and choline, and “atypical peaks” may be present), but in practice is often not conclusive.

99.4.2 Short list

Multiple lesions: metastases or abscess are much more likely than astrocytoma.

In adults, the main differential (short list) is:

1. high-grade glioma (glioblastoma)
2. metastasis
3. abscess
4. lymphoma should also always be tacked on as a possibility

99.4.3 Long list

Mnemonic: “Magic Dr” (metastasis (including lymphoma), abscess, glioma, infarct, contusion, demyelination, radiation).

1. astrocytoma: usually glioblastoma (GBM)
2. metastases (p.908): especially lung
3. abscess (p.343):
 - a) may see visible growth over several days on serial imaging
 - b) pyogenic abscesses are often (but not always) associated with fever and rapidly progressing neurologic deficit
 - c) Nocardia abscesses (p.366) are often *multiloculated* and are usually associated with a lung lesion
4. others
 - a) lymphoma (primary brain lymphoma or metastatic systemic lymphoma): wall is thicker than abscess.¹⁰ Incidence is increasing (p.843)
 - b) radiation necrosis
 - c) resolving intracerebral hematoma: on T1 gradient echo sequence, a continuous ring suggests hematoma, an interrupted ring suggests malignancy
 - d) cystic lesions with enhancing wall or mural nodule (see also intracranial cysts):
 - cysticercosis cyst, see Neurocysticercosis (p.404)
 - hemangioblastoma
 - pilocytic astrocytoma
 - cystic acoustic neuroma
 - e) trauma
 - f) recent infarct
 - g) thrombosed giant aneurysm

99.5 White matter lesions

99.5.1 Leukoencephalopathy

Disease largely confined to the white matter. Most of these lesions are caused by demyelinating disease.

Appear as white matter low density on CT or low signal on T1WI MRI, and high-intensity on T2WI. Usually does not enhance. Unlike a stroke, changes tend to spare the cortex. Conditions such as metabolic derangements, leuko-araiosis, etc. tend to produce fairly symmetric findings.

Differential diagnosis:

1. anoxia/ischemia
2. demyelinating disease
 - a) MS
 - b) ADEM (p. 190)
3. intoxication: cyanide, organic solvents, carbon monoxide
4. vitamin deficiencies: B12 with subacute combined degeneration
5. infectious, especially viral:
 - a) progressive multifocal leukoencephalopathy (PML) (p. 354)
 - b) herpes varicella-zoster leukoencephalitis (p. 399)
 - c) HIV infection (AIDS): perivascular pattern of demyelination
 - d) cytomegalovirus infection
 - e) Creutzfeldt-Jakob disease: small and perivascular demyelination
6. metabolic derangements: hyponatremia (p. 114), excessively rapid correction of hyponatremia (causing osmotic myelinolysis)
7. hereditary: metachromatic leukodystrophy, adult-onset Schilder's disease
8. leuko-araiosis (p. 1670)
9. multiple myeloma (p. 928)
10. low-grade (WHO grade 2 infiltrating) glioma

99.5.2 Corpus callosum lesions

1. lymphoma
2. MS plaque
3. tumefactive demyelinating lesions (p. 190)
4. lipoma
5. diffuse axonal injury from trauma

99.6 Sellar, suprasellar, and parasellar lesions

99.6.1 General information

May enlarge, erode, or destroy the sella turcica. Considerations in adults (adenoma is the most common enhancing pituitary lesion) are different than for children (adenomas are rare, craniopharyngioma and germinoma are more common). Includes (modified¹¹):

99.6.2 Tumors/pseudotumors

Tumors having epicenter within the sella

► Pituitary tumor:

1. adenohypophyseal tumors
 - a) pituitary neuroendocrine tumor (p. 854) (PitNET) AKA pituitary adenoma
 - microadenoma: < 1 cm diameter
 - macroadenoma: ≥ 1 cm diameter
 - invasive adenoma (p. 855): Includes aggressive tumors of Nelson's syndrome (p. 869)
 - b) pituitary carcinoma (p. 855)
2. neurohypophyseal tumors (tumors of the posterior pituitary)
 - a) metastases: the most common tumor found in the posterior pituitary (presumably due to rich blood supply); breast and lung are most common primaries¹²
 - b) pituitaryoma (p. 853): the most common tumor arising from neurohypophysis/pituitary stalk (i.e., primary)
 - c) astrocytoma: arising from stalk or posterior pituitary

► Pituitary "pseudotumor":

1. hyperplasia (enlargement)
 - a) thyrotroph hyperplasia due to primary hypothyroidism¹³ (see ► Table 52.2) causing chronic pituitary stimulation by TRH. Typically: free T4 is low or normal, TSH ↑ ↑, symmetrical sellar mass on MRI

- b) gonadotroph hyperplasia: due to primary hypogonadism
- c) somatotroph hyperplasia: due to ectopic GH-RH secretion
- d) pregnancy: during pregnancy, the number of lactotrophs increases by 40% in response to elevated estrogen levels (lactotroph hyperplasia). The mean height of the gland on MRI during pregnancy is 9.6–10 mm, immediately postpartum it is 10–12 mm, and it returns to its pre-gravid size $\approx$ 6 months postpartum¹⁴
- 2. pituitary enlargement may occur in intracranial hypotension (p.421)
- 3. the pituitary gland of young women of childbearing potential is normally slightly enlarged

Juxtaseellar or suprasellar tumors or masses: any of these lesions may extend into the sella

- 1. craniopharyngioma (p.849): in this region, these account for 20% of tumors in adult, 54% in peds
- 2. Rathke cleft cyst (p.949)
- 3. meningioma (parasellar, tuberculum sellae, or diaphragma sellae): to differentiate tuberculum sellae meningioma from pituitary macroadenoma on MRI ($\blacktriangleright$ Fig. 99.3), 3 characteristics of meningioma are (1) bright homogeneous enhancement with gadolinium (cf. heterogeneous, poor enhancement with macroadenoma), (2) suprasellar epicenter (vs. sellar), and (3) tapered extension of intracranial dural base¹⁵ (*dural tail*). Also, the *sella is usually not enlarged*, and even large suprasellar meningiomas rarely produce endocrine disturbances.¹⁶ The pituitary stalk is sometimes seen being pushed posteriorly by a meningioma. Tuberculum sellae meningiomas may be associated with sphenoid pneumosinus dilatans¹⁷ (enlargement of the underlying sphenoid sinus without bone erosion) as shown in $\blacktriangleright$ Fig. 46.7

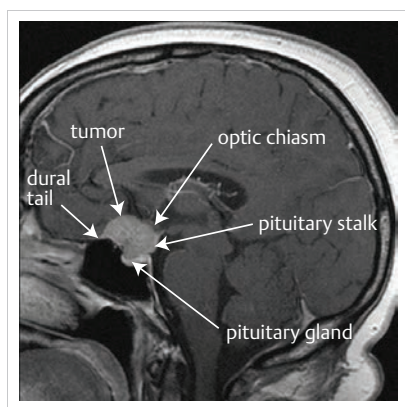


Fig. 99.3 Tuberculum sellae meningioma that could be mistaken for a PitNET/adenoma. Image: contrast-enhanced T1WI sagittal MRI.

- 4. pituitary tumor (mostly adenomas) with extrasellar extension: tends to push carotids laterally (unlike meningioma which may encase carotid), more symmetric than meningioma
- 5. germ cell tumors (GCT) (p.831): choriocarcinoma, germinoma, teratoma, embryonal carcinoma, endodermal sinus tumor. In females, suprasellar GCTs are more common; in males pineal region is more common
 - a) *suprasellar GCT*: triad of diabetes insipidus, visual deficit, and panhypopituitarism.¹⁸ May also present with obstructive hydrocephalus
 - b) simultaneous suprasellar and pineal lesions is diagnostic of GCT (so-called synchronous germ cell tumors (p.831))
- 6. glioma
- 7. hypothalamic glioma
- 8. optic nerve or chiasm (optic glioma) (p.694)
- 9. metastasis
- 10. chordoma
- 11. parasitic infections: cysticercosis
- 12. epidermoid cyst
- 13. suprasellar arachnoid cyst: see Arachnoid cysts (p.280)

14. sarcoidosis (p.198): hypothalamic involvement is a more likely site as a cause of anterior and/or posterior pituitary insufficiency
15. bone abnormalities
 - a) giant cell tumor (p.992)
 - b) chondromyxoid fibroma
 - c) brown tumor of hyperparathyroidism
 - d) bone spur
 - e) extramedullary hematopoiesis¹⁹

99.6.3 Vascular lesions

- a) aneurysm: ACoA, ICA (p.1612) (cavernous carotid or suprasellar variant of superior hypophyseal artery aneurysm), ophthalmic, basilar bifurcation. Giant aneurysms may produce mass effect, and some may be mistaken for a PitNET/adenoma (see ► Fig. 99.4)
- b) carotid-cavernous fistula (CCF) (p.1519)

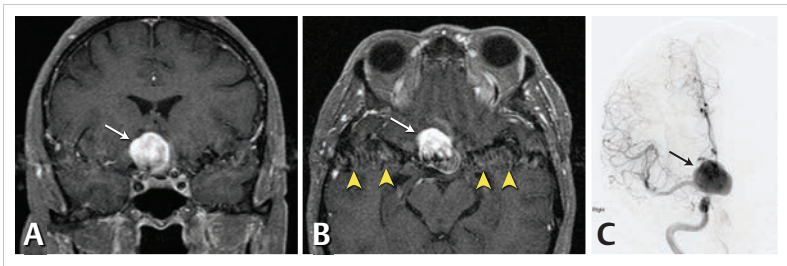


Fig. 99.4 Giant right ICA aneurysm that could be mistaken for a PitNET/adenoma.

Image: MRI T1 + contrast, A: coronal, B: axial; C: AP R ICA arteriogram.

The aneurysm is indicated by the long arrows. Pulsation artifact (yellow arrowheads in B) may be a tip-off that this is an aneurysm.

99.6.4 Inflammatory

- a) (autoimmune) hypophysitis (see below):
 - distinguishing imaging characteristics are shown in ► Table 99.2
 - the most important clinical feature: pregnancy
 - the most important laboratory feature: diabetes insipidus (if DI is present, it is unlikely to be an adenoma)
- b) pituitary granuloma²⁰

Table 99.2 Imaging characteristics of hypophysitis vs. adenoma²¹

Feature	Hypophysitis	Adenoma
Enlargement	symmetric	asymmetric
Pituitary stalk	thickened, nontapering	not thickened, tapering, deviated
Sellar floor ^a	spared	may be eroded
Enhancement	intense, may be heterogeneous	less intense, usually homogeneous
Mean size at time of presentation	3 cm ³	10 cm ³
Posterior pituitary bright spot ^b	lost	preserved in 97%

^aon CT scan

^b the normal hyperintensity of the posterior pituitary on T1WI MRI (p. 882) ²²

99.6.5 Empty sella syndrome

1. primary (p.952)
2. secondary: following pituitary tumor resection (p.952)

99.6.6 Hypophysitis

AKA autoimmune hypophysitis (AH)

Two main forms:

1. lymphocytic (adeno)hypophysitis²¹ AKA lymphoid adenohypophysitis: the more commonly encountered form. Inflammation of the pituitary stalk with lymphocytic infiltrate. Well-established autoimmune etiology, although the antigens have not been identified. Primarily affects women in late pregnancy or early postpartum period, and classically presents with sudden onset of diabetes insipidus
2. granulomatous hypophysitis: more aggressive. No gender bias. No association with pregnancy. May be autoimmune, but pathogenesis not definitely known

Because AH often mimics a nonsecretory pituitary macroadenoma (enhancing sellar mass on imaging, with negative endocrine tests), many of these lesions undergo surgical resection instead of what may be more appropriate medical therapy (e.g., steroids,²³ or discontinuing possible offending agents such as ipilimumab²⁴).

For distinguishing imaging characteristics, see ► Table 99.2.

99.7 Intracranial cysts

99.7.1 In general

Modified²⁵:

1. arachnoid cysts (p.260): Typically lined with meningotheial cells
2. suprasellar cyst from dilated 3rd ventricle
3. interhemispheric cyst from porencephaly
4. neuroectodermal cysts (gliependymal cysts): intraparenchymal, located near ventricles
5. old infarct: if it communicates with a ventricle it is called a porencephalic cyst
6. tumor cysts (the solid portion may sometimes be isodense to brain on CT):
 - a) ganglioglioma (p.707): usually solid but may appear cystic on CT
 - b) pilocytic astrocytoma (p.691): usually has enhancing mural nodule
 - c) neurilemmomas may be cystic
 - d) supratentorial ependymomas (p.724) are often cystic
7. infectious
 - a) abscess
 - b) cysticercosis: see Neurocysticercosis (p.404)
 - c) hydatid cyst: see Echinococcosis (p.408)
8. pineal cysts (p.948)
9. colloid cyst (p.944)
10. Rathke's cleft cyst (p.949)
11. giant aneurysm
12. on CT, a low-density non-enhancing tumor can mimic a cyst
13. chronic subdural hematoma or hygroma may mimic a cyst
14. posterior fossa: (for cysts of the CPA) (p.1648). Includes:
 - a) cyst associated with Dandy-Walker malformation (p.270). Cerebellar vermis is usually hypoplastic (or, less commonly, completely absent). Hydrocephalus often present
 - b) epidermoid (p.936)
 - c) enlarged (mega) cisterna magna (p.271) may mimic a cyst. No mass effect on cerebellum
 - d) cerebellar hemangioblastoma (p.648): often has an enhancing mural nodule
 - e) arachnoid cyst of posterior fossa: usually associated with mass effect on cerebellum
 - f) neurenteric cyst (p.313)
 - g) pilocytic astrocytoma of the cerebellum (p.689): usually has an enhancing mural nodule
15. intraventricular cysts include: arachnoid cysts (p.260), cysticercosis cyst

99.7.2 Midline cavities

Three potential supratentorial midline cavities in the center of the brain and differentiating features are shown in ► Table 99.3.

Table 99.3 Features of midline brain cavities²⁶

Cavity	Anatomy	Frequency	Clinical significance
cavum septum pellucidum (CSP) (see text)	located between leaflets of septum pellucidum	100% of preemies, 97% of newborns, 10% of adults	may be a normal variant, but has been linked to several neurodevelopmental & psychiatric disorders
cavum vergae (see text)	directly posterior to, and often communicating with CSP. Bounded posteriorly by the splenium of the corpus callosum	relatively uncommon	possible association with neurologic abnormalities ^a
cavum velum interpositum	due to separation of crura of fornix between thalami above the 3rd ventricle	present in 60% of children < 1 yr of age, and in 30% between 1 and 10 yrs	no known association with pathologic conditions

^aincluding developmental delay, macrocephaly, Apert's syndrome, abnormal EE

99.7.3 Cavum septi pellucidi (CSP) and cavum vergae (CV)

The septum pellucidum (Latin: translucent wall) is a membrane that separates the two lateral ventricles. It is absent in septo-optic dysplasia (p.276). Embryologically it starts as two thin parallel leaves (the septi pellucidi) with fluid between them (the cavum septi pellucidi or cavity of the septum pellucidum) (CSP) (A, ► Fig. 99.5) along with its posterior extension, the cavum vergae (CV) (B, ► Fig. 99.5) (first described by Andrea Verga). Archaic terms are “5th ventricle” for the CSP and “6th ventricle” for the CV.

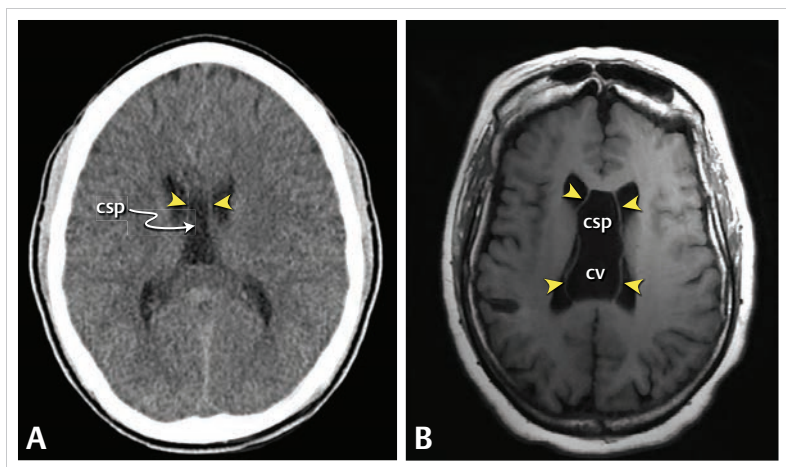


Fig. 99.5 Cavum septi pellucidi and cavum vergae (in 2 different patients).

Image: A: Axial noncontrast CT shows a cavum septi pellucidi (csp) by itself (which was not previously present but developed in a 16-year-old female following a gunshot wound to the brain).

B: Axial T1 MRI demonstrating a cavum vergae (cv), situated posterior to a csp in a 40-year-old woman. The leaflets of the septi pellucidi are indicated by the yellow arrowheads.

The leaflets comprising the walls of the CSP and the CV begin to fuse together at $\approx$ 24 weeks gestation and both are usually completely obliterated by the time of a full-term birth or shortly thereafter. As a result, CSP is present in $\approx$ all preemies. The CSP and the CV fuse from posterior to anterior, therefore persistence of the CV is almost always accompanied by CSP (together called “cavum septi pellucidi et vergae”).

Persistence of the CSP is found in up to 10% of the adult population, and can be a normal variant representing an asymptomatic developmental anomaly. However, sometimes it is associated with neurodevelopmental disruption. It has been *loosely* associated with psychiatric conditions including bipolar disorder, Tourette's syndrome, obsessive-compulsive disorder, and schizophrenia, among others.²⁷ CSP may also result from trauma possibly as a result of tears in the membrane, and has been associated with TBIs especially in professional boxers²⁸ suffering from chronic traumatic encephalopathy (p.1112), and motor vehicle trauma.²⁷

The compartment is usually isolated, although some communicate with the 3rd ventricle.

99.8 Orbital lesions

99.8.1 General information

4 compartments of the orbit:

1. ocular (AKA globe, AKA bulbar)
2. optic nerve sheath
3. intraconal
4. extraconal

CT remains a strong imaging modality within the orbit (less susceptible to motion artifact than MRI because of the speed, images bony structures to good advantage). MRI adds info for related extra-orbital structures (optic chiasm, hypothalamus...).

99.8.2 Orbital lesions in adults

Orbital pseudotumor is the most common.

1. neoplastic
 - a) discrete tumors that may occur adjacent to but not envelop the optic nerve sheath
 - cavernous hemangioma: the most common *benign* primary intraorbital neoplasm. Choroidal hemangioma is seen in Sturge-Weber syndrome
 - fibrohistiocytoma
 - hemangiopericytoma
 - b) capillary hemangioma: produces infantile proptosis. Regresses spontaneously
 - c) lymphangioma: produces infantile proptosis. Does *not* regress
 - d) melanoma: the most common primary ocular malignancy of adulthood
 - e) retinoblastoma: congenital, malignant primary retinal tumor. 40% are bilateral, 90% are calcified (often a key differentiating feature; does not portend benignity as with other lesions). CT may show retinal detachment
 - f) lymphoma of the orbit: causes painless proptosis. The 3rd most common cause of proptosis
 - g) intraorbital meningioma
 - h) primary tumors of the optic nerve & its coverings
 - optic pathway glioma (p.694): fusiform nerve enlargement usually > 1 cm in length
 - optic nerve sheath tumor (schwannoma)
2. congenital
 - a) Coats disease: telangiectatic vascular malformation of retina which leaks a lipid exudate causing retinal detachment. May mimic retinoblastoma. Vitreous is hyperintense on MRI on both T1WI and T2WI due to lipid
 - b) persistent hyperplastic primary vitreous
 - c) retinopathy of prematurity (retrolental fibroplasia)
3. infectious
 - a) toxocara endophthalmitis
4. inflammatory/collagen vascular disease: usually bilateral
 - a) scleritis
 - b) pseudotumor of the orbit: the most common intraconal lesion. Usually unilateral (p.600)
 - c) sarcoidosis: usually affects the conjunctiva and lacrimal gland and spares connective tissues and intraorbital muscles
 - d) Sjögren's syndrome
5. vascular
 - a) enlargement of the superior orbital vein: may occur in thrombosis of cavernous sinus or in carotid-cavernous fistula
 - b) dural AVM

6. miscellaneous

- a) drusen: degenerated subretinal epithelial pigmented deposits in the posterior globe that may resemble calcified masses on CT. Often associated with age-related macular degeneration (AMD)
- b) thyroid ophthalmopathy: Graves' disease (hyperthyroidism & swelling of EOMs → *painless* proptosis). 80% of cases are bilateral. The ophthalmopathy is independent of the level of thyroid hormone (possibly an autoimmune process). NB: a swollen inferior rectus muscle may resemble an orbital tumor if seen only on lower CT cut through the orbit
- c) EOM enlargement can also occur with steroid use or occasionally with obesity
- d) fibrous dysplasia (p.975)
- e) optic nerve sheath dilatation: this and other intraorbital findings (p.964) may occur with increased intracranial pressure

99.8.3 Orbital tumors in pediatrics

1. dermoid cyst: 37%. The most common lesion in children
2. hemangioma: 12%. Most regress spontaneously without surgery
3. rhabdomyosarcoma: 9%. The most common malignant tumor of the orbit
4. optic nerve glioma: 6%
5. lymphangioma: < 7%. Imaging resembles hemangioma. But will not regress spontaneously, requires surgery. Proptosis may worsen after a URI. May bleed into itself (chocolate cysts)

99.9 Dural sinus lesions

This section considers lesions within the dural sinuses (e.g., superior sagittal sinus, transverse sinus...) and not external compression of the sinus by a mass.

1. dural sinus thrombus (p.1594): acute thrombus is hyperdense on CT, and hyperintense on T1 MRI
2. tumor: especially parasagittal meningiomas (p.808) (up to 50% invade the superior sagittal sinus)
3. focal sinus stenosis: e.g., transverse sinus stenosis (p.964) is common in pseudotumor cerebri (bilateral stenosis was seen in 93%). This may be causative in some cases of pseudotumor, but it may be a result of increased CSF pressure in some. Transverse sinuses are normally asymmetric, with the right usually larger/dominant
4. giant arachnoid granulation: most common in the transverse sinus in the middle or lateral third with a slight left-sided prevalence, the superior sagittal sinus is the next most common location (► Fig. 99.6).²⁹ Typically > 1 cm diameter. May erode into the inner table of the skull. The arachnoid granulation does not enhance (but there is often a linear central enhancing vessel, presumably a vein) and does not show diffusion restriction on DWI MRI

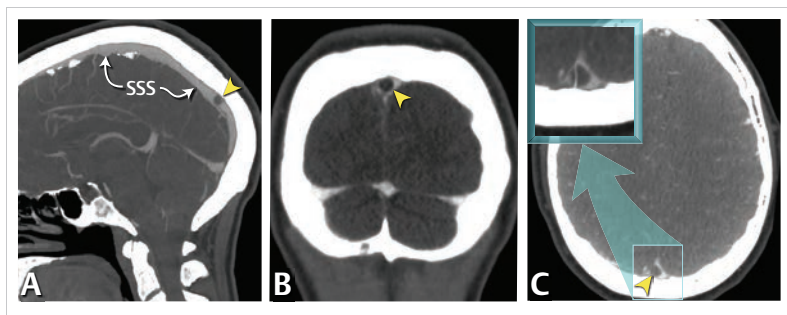


Fig. 99.6 Giant arachnoid granulation (yellow arrowheads) in the posterior portion of the superior sagittal sinus (SSS) appearing as a filling defect in the SSS.

Image: A: sagittal CTV, B: coronal CTV, C: axial CTA (inset shows an enlargement of the lesion in the SSS which demonstrates the enhancing central vessel [retouched photo to accentuate the visibility of the vessel]). This was an incidental finding in a patient who had a contraindication to MRI.

99.10 Cavernous sinus lesions

Modified³⁰:

1. primary tumors (rare)
 - a) meningiomas³¹
 - b) neurinomas
2. tumors from adjacent areas that may extend into cavernous sinus (head and neck cancers may track intracranially along cranial nerves, especially V)
 - a) meningiomas
 - b) neurinomas
 - c) chordomas
 - d) chondromas
 - e) chondrosarcomas
 - f) pituitary tumors³²
 - g) nasopharyngeal carcinomas
 - h) olfactory neuroblastoma (p.935)
 - i) nasopharyngeal angiofibromas
 - j) metastatic tumors
3. inflammation: e.g., Tolosa-Hunt (p.600)
4. infection: mucormycosis (phycomycosis) (p.599). Usually in diabetics
5. vascular
 - a) cavernous carotid aneurysm
 - b) carotid-cavernous fistula (p.1519)
 - c) cavernous sinus thrombosis

99.11 Meckel's cave abnormalities

99.11.1 Etiologies

Also see anatomy of Meckel's cave (p.1858). Some of these may rarely cause trigeminal neuralgia or atypical facial pain (p.1857) that may mimic trigeminal neuralgia (e.g., trigeminal neuroma,³³ arachnoid cyst³⁴).

► Enlargement of Meckel's cave

1. neoplasms
 - a) nerve sheath tumors: trigeminal neuroma is the most common neoplasm
 - b) meningioma
 - c) tumors of the head and neck with perineural extension along the trigeminal nerve
 - d) extension of PitNET/adenoma
 - e) epidermoid³⁵
 - f) metastases: including lymphoma
 - g) invasive pituitary macroadenomas invading through cavernous sinus into Meckel's cave
2. non-neoplastic
 - a) enlargement of Meckel's cave: (► Fig. 57.1) may be seen with intracranial hypertension (see below) (with or without spontaneous CSF leak³⁶)
 - b) arachnoid cyst within Meckel's cave (► Fig. 99.7): a rare cause of trigeminal neuralgia³⁴
 - c) vascular
 1. persistent trigeminal artery
 2. aneurysms of the internal carotid artery
 3. trigeminal AVM
 - d) infectious/inflammatory: sarcoid, herpes, petrous apex mucocele, Tolosa-Hunt (p.600),³⁷ idiopathic inflammatory sensory neuropathy (IIHS) (which can mimic a schwannoma)³⁸

► **Compression of Meckel's cave.** Increased ICP can distend Meckel's cave if the porus trigeminus (p.1858) is enlarged, but it may compress Meckel's cave if it isn't enlarged. It can also be compressed by tumors or aneurysms external to the cave.

99.11.2 Evaluation

► **MRI.** Recommended MRI protocol: imaging in 3 planes with T1, T2 short-tau inversion recovery (STIR) and gadolinium enhanced T1 images with fat suppression. The contrast study rules out most neoplasms that are potential causes.

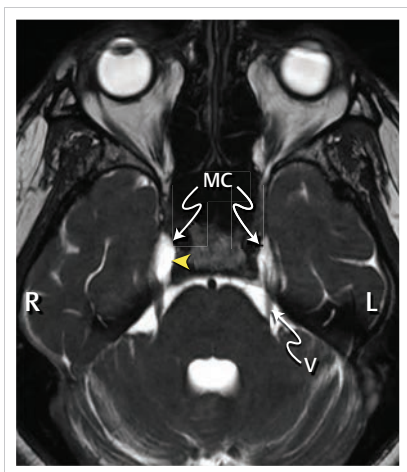


Fig. 99.7 Arachnoid cyst of Meckel's cave.

Image: axial FIESTA MRI in a 44-year-old patient with bilateral dilatation of Meckel's cave, with trigeminal symptoms on the right. The trigeminal nerve can be seen trifurcating within Meckel's cave on the left, whereas on the right the nerve is compressed against the lateral wall of the cave by what appears to be an arachnoid cyst (yellow arrowhead).

Abbreviations: MC = Meckel's cave; V = trigeminal nerve (Cr.N. V) as it crosses the CPA cistern.

Other MRI findings of increased ICP (p.963) should be sought, and include: empty sella (► Fig. 56.6, ± infundibulum sign), dilated optic nerve sheath (► Fig. 33.1), flattening of the posterior globe, empty sella, dilated optic nerve sheath, flattening of the posterior globes.

► **Ophthalmology consultation.** If increased intracranial pressure is suspected, ophthalmology consultation should be obtained to rule out papilledema, visual field deficit and Cr.N. VI palsy. If any of these findings are present in the absence of a mass, hydrocephalus or evidence of infection, a workup for pseudotumor cerebri (p.962) may need to be undertaken (including lumbar puncture).

► **Percutaneous biopsy through the foramen ovale.** In questionable cases, a percutaneous biopsy through the foramen ovale may be performed.^{38,39}

99.12 Skull lesions

99.12.1 General information

The most common *benign* tumors of skull are osteoma and hemangioma. Osteogenic sarcoma is the most common *malignancy*. See also specific skull tumors (p.972).

Evaluating roentgenographic skull lucencies

There is enough overlap of features to prevent any systematic means of determining the etiology of all or even most radiographic skull lucencies. The following features should be noted for any lucency; some are more helpful than others (modified⁴⁰):

1. multiplicity (single or multiple?): except for multiple venous lakes, the presence of 6 or more defects is usually indicative of a malignancy
2. origin (intradiploic, full-thickness, inner or outer table only):
 - a) most vault lesions originate intradiploically, so limitation to this space may merely signify early recognition of a lesion
 - b) expansion of the diploë with bulging of one or both tables almost always signifies a benign lesion
 - c) full-thickness lesions affecting both tables congruently usually indicates malignancy, whereas non-congruent erosion is more common with benign lesions
3. edges (smooth or ragged):
 - a) smooth edges, whether regular, distinct, or indistinct: no predictive value
 - b) irregular margins (especially ragged undermined edges): more suggestive of infection (osteomyelitis) or malignancy
 - c) sharply demarcated, full-thickness punched-out defects: suggest myeloma

4. presence of peripheral sclerosis: circumferential bony sclerosis suggests benignity (may indicate slow expansion and longstanding nature). The ring of sclerosis is generally narrow except in fibrous dysplasia
5. presence or absence of peripheral vascular channels: presence is highly suggestive of benign lesions (seen in $\approx 66\%$ of venous lakes and $\approx 50\%$ of hemangiomas)
6. pattern within the lucency:
 - a) ★ hemangiomas classically show *honeycomb* or *trabecular* pattern (seen in $\approx 50\%$ of cases) or *sunburst* pattern (seen in $\approx 11\%$ of cases)
 - b) fibrous dysplasia (p.975) may show well-defined islands of bone, or a grossly mottled appearance with randomly arranged cystic and dense areas
7. location on the cranial vault (high vs. low): poor correlation with benign vs. malignant lesions
8. pain: Langerhans cell histiocytosis lesions are often *tender*

NB: keep in mind that skull lesions may have an intracranial component. CT scanning is good for assessing bone (MRI is poor for this); however, CT may miss small intracranial lesions tucked within the convexity of the calvaria due to bone-hardening artifact (MRI has better sensitivity in this setting).

Nuclear bone scan may be a helpful adjunctive test (see specific lesion for findings).

Biopsy: indicated for questionable skull lesions. If the bone has not been destroyed by soft tissue, biopsy may be accomplished with a Craig needle, and the specimen may need decalcification by the pathologist before histologic evaluation can be completed.

99.12.2 Radiolucent lesion or bone defect in skull (AKA lytic lesions)

1. congenital or developmental
 - a) epidermoid (cholesteatoma): *sclerotic* edge
 - b) congenital: encephalocele, meningoencephalocele, dermal sinus
 - c) fibrous dysplasia (p.974). A benign condition in which normal bone is replaced by fibrous connective tissue. Tends to occur higher in calvaria. 3 types:
 - cystic: widening of the diploë usually with thinning of the outer table and little involvement of the inner table. Typically involves calvaria
 - sclerotic: usually involves skull base (especially sphenoid bone) and facial bones
 - mixed: appearance is similar to cystic type with patches of increased density within the lucent lesions
 - d) hemangioma or AVM of bone or scalp
 - e) pacchionian depression: arachnoid granulations (older terms: Pacchioni's granulation (after Italian anatomist Antonio Pacchioni) or pacchionian bodies) resorb CSF into vascular system and occasionally cause a bony lucency, usually near the superior sagittal sinus
 - f) Albright's syndrome
 - g) congenital foramina: "holes" in skull traversed by emissary veins
 - h) parietal thinning: usually a bilateral process
 - i) frontal fenestrae
 - j) venous lakes
 - k) cerebral herniations: AKA occipital pacchionian granulations
2. traumatic
 - a) surgical defect: burr hole, craniectomy
 - b) fracture
 - c) posttraumatic leptomeningeal cyst (p. 1100)
 - d) following trauma in children⁴¹
3. inflammatory
 - a) osteomyelitis: including tuberculosis⁴²
 - b) sarcoidosis
 - c) syphilis
4. neoplastic
 - a) hemangioma: fine, honeycombed matrix. Classic X-ray finding: "starburst" appearance due to radiating bone spicules (may occur in as few as $\approx 11\%$ of cases⁴⁰)
 - b) intracranial tumor with erosion
 - c) lymphoma, leukemia
 - d) meningioma
 - e) metastasis: usually hot on bone scan
 - f) multiple myeloma, plasmacytoma (p.929). Usually cool on bone scan
 - g) sarcoma or fibrosarcoma of bone

- h) skin tumor with invasion (rodent ulcer)
- i) neuroblastoma
- j) lipoma
- k) epidermoid (may also be considered congenital, thus also see above)
- 5. miscellaneous
 - a) Langerhans cell histiocytosis (p. 846). Perfectly round *non-sclerotic* punched-out lesion, may be single (formerly called eosinophilic granuloma) or multiple, tender
 - b) Paget's disease (when seen as a zone of osteolysis without osteoblastic sclerosis on skull films, this is defined as osteoporosis circumscripta). Usually "hot" on bone scan
 - c) aneurysmal bone cyst: rare. Arises in diploë and expands both tables, which become thin but remain intact
 - d) brown tumor of hyperparathyroidism

99.12.3 Diffuse demineralization or destruction of the skull

Includes "salt and pepper skull."

- 1. common
 - a) hyperparathyroidism, primary or secondary
 - b) metastatic carcinoma or neuroblastoma
 - c) multiple myeloma
 - d) osteoporosis
- 2. uncommon
 - a) Paget's disease (osteoporosis circumscripta)

99.12.4 "Hair-On-End" appearance in skull

- 1. common
 - a) congenital hemolytic anemia (e.g., thalassemia, sickle cell, hereditary spherocytosis, pyruvate kinase deficiency)
- 2. uncommon
 - a) hemangioma
 - b) cyanotic congenital heart disease (with secondary polycythemia)
 - c) iron deficiency anemia
 - d) metastases: especially neuroblastoma, thyroid carcinoma
 - e) multiple myeloma
 - f) meningioma
 - g) osteosarcoma
 - h) polycythemia vera

99.12.5 Diffuse increased density, hyperostosis, or calvarial thickening

- 1. common
 - a) anemia (sickle cell, iron deficiency, thalassemia, hereditary spherocytosis)
 - b) fibrous dysplasia
 - leontiasis ossea ("lion-like facies"): a form of polyostotic fibrous dysplasia
 - c) hyperostosis interna generalisata
 - d) osteoblastic metastases (especially prostate, breast)
 - e) Paget's disease (begins with lytic zone and diploic thickening)
 - f) treated hydrocephalus
- 2. uncommon
 - a) chronic phenytoin therapy
 - b) Engelman's disease (progressive diaphyseal dysplasia)
 - c) fluorosis
 - d) hypervitaminosis D
 - e) hypoparathyroidism, pseudohypoparathyroidism
 - f) meningioma
 - g) osteogenesis imperfecta
 - h) osteopetrosis (p. 1689)
 - i) secondary polycythemia
 - j) syphilitic osteitis
 - k) tuberous sclerosis

99.12.6 Focal increased density of skull base

1. common
 - a) fibrous dysplasia
 - b) meningioma
2. uncommon
 - a) mastoiditis
 - b) nasopharyngeal carcinoma
 - c) osteoblastic metastasis
 - d) osteoma of the outer table or diploë
 - e) chondroma
 - f) sarcoma of bone (e.g., osteosarcoma, chondrosarcoma)
 - g) sphenoid sinusitis

99.12.7 Generalized increased density of skull base

1. common
 - a) fibrous dysplasia
 - b) Paget's disease
2. uncommon
 - a) severe anemia (e.g., thalassemia, sickle cell)
 - b) Engelman's disease (progressive diaphyseal dysplasia)
 - c) fluorosis
 - d) hyperparathyroidism, primary or secondary (treated)
 - e) hypervitaminosis D
 - f) idiopathic hypercalcemia
 - g) meningioma
 - h) osteopetrosis (p. 1689)

99.12.8 Localized increased density or hyperostosis of the calvaria

1. common
 - a) anatomic variation (e.g., sutural sclerosis)
 - b) fibrous dysplasia
 - c) osteoma (p. 972)
 - d) meningioma
 - e) hyperostosis frontalis interna (p. 974)
 - f) osteoblastic metastases (especially: prostate, breast)
 - g) Paget's disease (begins with lytic zone and diploic thickening)
 - h) cephalohematoma
 - i) depressed skull fracture
2. uncommon
 - a) osteosarcoma
 - b) chronic osteomyelitis, tuberculosis
 - c) tuberous sclerosis
 - d) osteoid osteomas: radiolucent nidus with surrounding zone of dense sclerosis
 - e) osteoblastoma
 - f) ossifying fibromas: predilection for frontotemporal region
 - g) radiation necrosis

99.13 Other abnormalities of the skull

99.13.1 Pneumocele

Pneumocele: enlargement of an air sinus often with bone erosion. Pneumosinus dilatans, in contrast, generally denotes enlargement of an air sinus *without* bone erosion (► Fig. 46.7), as may occur with tuberculum sellae (p. 809) or planum sphenoidale (p. 809) meningiomas.

Pneumoceles occur primarily in the maxillary antrum. Etiology is unknown, and may involve a trap-valve mechanism, ruptured mucocele, or possibly congenital.

Presentation of pneumocele or pneumosinus dilatans:

1. headache
2. neuralgia

3. facial asymmetry
4. frontal bossing (with frontal pneumosinus dilatans)
5. exophthalmous
6. CSF fistula (leak)
7. treatment for maxillary pneumocele: opening the sinus into the nasal cavity via endoscopic approach. Watch for encephalocele.

99.13.2 Frontal bossing

Protuberance of the frontal bone. Forehead appears prominent. May be associated with an exaggerated brow ridge.

Etiologies include:

1. hydrocephalus (p.427) in a young child
2. acromegaly (p.870)
3. fibrous dysplasia (p.975)
4. extramedullary hematopoiesis (p.179)
5. achondroplasia
6. Hurler syndrome

99.14 Combined intracranial/extracranial lesions

Lesion causing mass outside skull with intracranial component.

1. intra-axial: rule of thumb—"there is no intra-axial lesion that grows out of skull"; however, untreated fungating malignant gliomas may do this
2. extra-axial:
 - a) meningioma
 - may arise in diploë, grows outward and inward
 - intracranial meningioma can grow through bone by destroying it
 - intracranial meningioma can induce hyperostosis that causes extracranial mass
 - b) metastatic disease (e.g., GI carcinoma, and especially prostate Ca)
 - c) bone (skull) lesion:
 - hemangioma
 - epidermoid
 - fibrous dysplasia (rare)
 - giant cell tumor (rare)
 - Ewing's sarcoma (rare in skull)
 - aneurysmal bone cyst (5% occur in skull, occipital bone most common)

99.15 Intracranial hyperdensities

Differential diagnosis of an intracranial structure that is hyperdense with respect to brain (i.e., appear "whiter" than brain) on non-contrast CT:

1. acute blood
2. calcium
3. vessels with low flow
4. melanoma: may be slightly hyperdense to brain due to melanin

99.16 Intracranial calcifications

99.16.1 Single intracranial calcifications

1. benign ("physiologic")
 - a) choroid plexus: (► Fig. 99.8) calcifications usually bilateral (see below)
 - b) arachnoid granulation
 - c) diaphragma sellae
 - d) dural: common locations include falx (falx) (► Fig. 99.8), tentorial, sagittal sinus
 - e) habenular commissure
 - f) petroclinoid or interclinoid ligaments
 - g) pineal: (► Fig. 99.8) 55% of patients > 20 yrs of age have a calcified pineal gland visible on plain skull X-ray

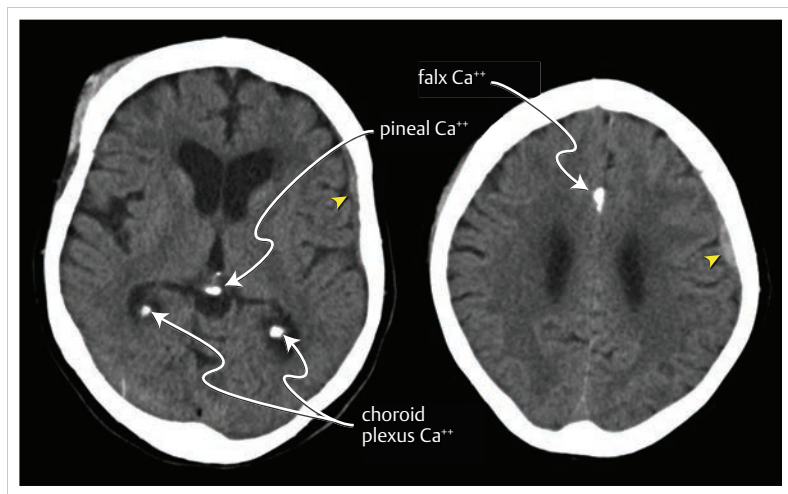


Fig. 99.8 Benign intracranial calcifications in an 81-year-old woman who also has small subdural hematomas (yellow arrowheads).

Abbreviations: Ca^{++} = calcifications.

Image: axial noncontrast CT scans at two different positions.

2. infection
 - a) cysticercosis cyst: single or multiple, see Neurocysticercosis (p.404)
 - b) encephalitis, meningitis, cerebral abscess (acute and healed)
 - c) granuloma (torulosis and other fungi)
 - d) hydatid cyst
 - e) tuberculoma
 - f) paragonimiasis
 - g) rubella
 - h) syphilitic gumma
3. vascular
 - a) aneurysm, including:
 - vein of Galen aneurysm
 - giant aneurysm
 - b) arteriosclerosis (especially carotid artery in siphon region)
 - c) hemangioma, AVM, Sturge-Weber syndrome
4. neoplastic: calcifications usually suggest a more benign process
 - a) meningioma (p.803)
 - b) craniopharyngioma
 - c) choroid plexus papilloma
 - d) ependymoma
 - e) glioma (especially oligodendroglioma, also astrocytoma)
 - f) ganglioglioma
 - g) lipoma of corpus callosum
 - h) pinealoma
 - i) hamartoma of tuber cinereum
5. miscellaneous
 - a) hematoma: ICH, EDH, or SDH. Calcifications usually only when chronic
 - b) idiopathic
 - c) tuberous sclerosis (p.644)

99.16.2 Multiple intracranial calcifications

1. common
 - a) choroid plexus: the most common site for physiologic calcification (in lateral ventricles where it is usually bilateral and symmetric; rare in 3rd & 4th ventricles). Increases in frequency and extent with age (prevalence: 75% by 5th decade). Rare under age 3. Under age 10, consider possible choroid plexus papilloma. Involvement in the temporal horns is often associated with neurofibromatosis
 - b) basal ganglia (BG): slight bilateral BG calcifications on CT are common, especially in the elderly. Considered a normal radiographic variant by some. They may be idiopathic, secondary to conditions such as hypoparathyroidism or long-term antiepileptic medication use, or part of rare conditions such as Fahr's disease (see below). BG calcifications > 0.5 cm dia are possibly associated with cognitive impairment and a high prevalence of psychiatric symptoms (including bipolar and obsessive-compulsive disorders, but no patients had schizophreniform disorders)⁴³
2. uncommon
 - a) Fahr's disease: progressive idiopathic calcification of medial portions of basal ganglia, sulcal depths of cerebral cortex, and dentate nuclei⁴⁴
 - b) hemangioma, AVM, Sturge-Weber syndrome, von Hippel-Lindau disease
 - c) basal cell nevus syndrome (fals, tentorium)
 - d) Gorlin's syndrome. Associated findings: mandibular cysts, rib and vertebral deformities, short metacarpals. Medulloblastoma seen in several patients
 - e) deposition of calcium in the media of medium-sized blood vessels without compromise of the lumen. Usually asymptomatic. May become symptomatic by the time the involvement is significant enough to be visible on plain X-ray in a young person
 - f) cytomegalic inclusion disease
 - g) encephalitis (e.g., measles, chickenpox, neonatal herpes simplex)
 - h) hematomas (SDH or EDH, chronic)
 - i) neurofibromatosis (choroid plexi)
 - j) toxoplasmosis
 - k) tuberculomas; tuberculous meningitis (treated)
 - l) tuberculous sclerosis
 - m) hypoparathyroidism (including post-thyroidectomy cases⁴⁵) and pseudohypoparathyroidism
 - n) multiple tumors (e.g., meningiomas, gliomas, metastases)
 - o) cysticercosis cyst: may be single or multiple, see Neurocysticercosis (p.404)

99.17 Intraventricular lesions

99.17.1 General information

Intraventricular tumors represent only ≈ 10% of CNS neoplasms. A clue to differentiating a tumor located within the ventricle from an intraparenchymal tumor invaginating into the ventricle is a "cap" of CSF surrounding an intraventricular tumor on CT or MRI.

99.17.2 Differential diagnosis

(Percentages quoted here are from a series of 73 patients with an intraventricular lesion on CT seen at UCSF⁴⁶).

1. astrocytoma: (20%) the most common lesion. Hydrocephalus (HCP) is present in 73%. Hyperdense on non-contrast CT (NCCT) in 77%.
Locations in descending order of frequency:
 - a) frontal horn
 - b) 3rd ventricle
 - c) atrium (AKA trigone)
 - d) 4th ventricle
2. colloid cyst (p.944): (14%) essentially seen only in anterior 3rd ventricle at foramen of Monro (other sites have been described but are exceedingly rare). 50% are hyperdense on NCCT. MRI appearance is variable, and may occasionally be missed. Little or no enhancement on CT/MRI. DDx includes xanthogranuloma

3. meningioma: (12%) most in atrium, rarely in frontal horn. All hyperdense with dense uniform enhancement. May be calcified. Most have dense tumor blush on angiogram, most supplied from anterior choroidal artery, posterior choroidal less common. Thought to arise from arachnoid cells within the choroid plexus
4. ependymoma (p.724): (10%) most in 4th ventricle, may occur in body of lateral ventricle. Often hyperdense on CT because of high cellularity
5. craniopharyngioma: (7%) primarily in 3rd ventricle. Most have punctate calcification. Squamous epithelial rests in region of lamina terminalis are felt to give rise to this uncommon variety of craniopharyngioma
6. medulloblastoma: (5%) often fill 4th ventricle. Hyperdense on CT with homogeneous enhancement
7. cysticercosis: (5%) may involve any ventricle or may be panventricular (NB: incidence related to geographic location)
8. choroid plexus papilloma: (5%) most common in lateral ventricle (may be bilateral), but also may be seen in 4th and occasionally in 3rd. Non-obstructive HCP may occur (possible CSF overproduction). Intense blush on angiogram
9. epidermoid: (4%) mostly in 4th ventricle. Hypodense on CT with no enhancement (tend to follow CSF signal). The most common 4th ventricular low-density lesion in the U.S.
10. dermoid: (3%) common in 4th ventricle. May see free-floating fat in ventricles suggestive of cyst rupture. Tendency to form in midline
11. choroid plexus carcinoma: (3%) common in atrium of lateral ventricle. May extend into adjacent brain parenchyma with edema and shift. Intense blush on angio. NB: very rare lesion
12. subependymoma (p.735): (3%) 4th ventricle or frontal horn. Typically isodense on CT with *minimal or no enhancement* (► Fig. 41.12). May have calcification or cystic degeneration (more common in ependymoma). Most commonly in floor of 4th ventricle near obex
13. ependymal cyst: (3%) common in lateral ventricle. Absence of communication demonstrated by water-soluble contrast cisternography
14. arachnoid cyst: (1%) lateral ventricle. Absence of communication demonstrated by water-soluble contrast cisternography
15. arteriovenous malformation (AVM): (3%)
16. teratoma: (1%) Located in anterior 3rd ventricle. Partially calcified with foci of fat density. Marked enhancement
17. central neurocytoma (p.719)
18. metastases: breast and lung reported⁴⁷
19. chordoid glioma of the 3rd ventricle⁴⁸

Additional lesions not included in this list: intraventricular arachnoid cyst (p.260) (uncommon), cysticercosis

99.17.3 Features to help identify type of intraventricular lesions

By location within ventricular system

► Table 99.4 shows the breakdown of lesion type by location within the ventricular system.

By location and age within lateral ventricle

See reference.⁴⁹

See ► Table 99.5. This study excluded tumors that were clearly arising in the 3rd ventricle or were predominantly parenchymal with intraventricular extension.

The teratoma and both PNETs occurred in age < 1 year, and all showed calcifications. Only one CPP occurred above age 5 years.

In adults > 30 years of age, the only tumors found in the trigone were meningiomas. Subependymomas (p.735) were the *only* nonenhancing intraventricular tumor in this age group.

By location within 3rd ventricle

1. anterior 3rd ventricle
 - a) colloid cyst
 - b) sellar mass
 - c) sarcoidosis
 - d) aneurysm

Table 99.4 Type of intraventricular lesions by location⁴⁶ (numbers are patients out of 73^a)

3rd ventricle		4th ventricle		Lateral ventricle					
				Atrium		Body		Frontal horn	
colloid cyst	10	medulloblast.	4	meningioma	8	ependymoma	3	astrocytoma	7
craniopharyng.	5	ependymoma	4	astrocytoma	3	ch. plexus papil. ^b	1	meningioma	1
astrocytoma	4	epidermoid	3	ch. plexus papil.	1	ch. plexus carc.	1	subependym.	1
teratoma	1	cysticercosis	2	ch. plexus carc.	1	ependym. cyst	1	dermoid	1
ch. plexus papil.	1	astrocytoma	1	arachnoid cyst (p.260)	1	AVM	1		
cysticercosis	1	subependym.	1	ependym. cyst	1				
dermoid	1								
ch. plexus carc.	1								
AVM	1								

^a1 patient had cysticercosis diffusely throughout ventricles^b1 patient with bilateral lateral ventricle papillomas**Table 99.5** Lateral ventricle tumor type by location & age

Age (yrs)	Location within lateral ventricle ^a		
	Foramen of Monro region	Trigone	Body
0–5	0	8 CPP	2 PNETs 1 teratoma
6–30	5 SEGAs 2 pilocytic astrocytomas 1 CPP 1 meningioma 1 oligodendroglioma	1 ependymoma 1 oligodendroglioma	1 mixed glioma 1 ependymoma 1 pilocytic astrocytoma
> 30	2 metastases	8 meningiomas	2 glioblastomas 1 lymphoma 1 metastasis 6 subependymomas

^aabbreviations: CPP = choroid plexus papillomas, PNET = primitive neuroectodermal tumor, SEGA = subependymal giant cell astrocytoma

- e) hypothalamic glioma
- f) histiocytosis
- g) meningioma
- h) optic glioma
- 2. posterior third ventricle
 - a) pinealoma (dysgerminoma)
 - b) meningioma
 - c) arachnoid cyst (p.260)
 - d) vein of Galen aneurysm

By enhancement

All lesions enhanced except: cysts (ependymal and arachnoid), dermoids, and epidermoids.

Subependymomas (p.735): there are differences of opinion of the tendency for these to enhance. Jelínek et al⁴⁹ found that they were the only consistently nonenhancing tumors at the foramen of Monro or in the body of the ventricles (3rd ventricular colloid cysts were excluded).

By multiplicity

Multiple lesions are more suggestive of: neurocysticercosis, metastases, or ruptured epidermoid cyst.

99.18 Periventricular lesions

99.18.1 Periventricular solid enhancing lesions (in decreasing frequency)

1. lymphoma; CNS involvement from systemic, or rarely primary brain (p.840): must be included in differential diagnosis of any solid enhancing periventricular brain tumor. Very radiosensitive.
2. ependymoma (usually invaginates)
3. metastatic Ca: especially malignant melanoma or choriocarcinoma
4. ventriculitis
5. medulloblastoma (in peds), AKA cerebellar sarcoma in adults
6. pineal tumor (dysgerminoma type): usually midline, young patient
7. occasionally, glioblastoma can present like this

99.18.2 Periventricular low density on CT, or high signal on T2WI MRI

1. increased extracellular or intracellular water content (edema)
 - a) in hydrocephalus: transependymal CSF absorption (p.431)
 - b) necrosis from infarction
 - c) edema from tumor
2. uncommon late variants of adrenoleukodystrophy
3. vascular disorders
 - a) subacute arteriosclerotic encephalopathy (Binswanger's disease)^{50,51}
 - b) cerebral embolism
 - c) vasculitis
 - d) amyloid angiopathy
 - e) low flow states
4. demyelination: including multiple sclerosis
5. leukoaraiosis⁵²: a term coined to describe white matter disease with symmetric (or nearly so) periventricular white matter changes on CT or MRI. May be asymptomatic or may present with findings including dementia. May be related to:
 - a) Binswanger's encephalopathy
 - b) watershed infarction⁵³
 - c) normal aging⁵⁴: increases each decade after age 60, usually patchy
 - d) hypoxia
 - e) hypoglycemia⁵⁵
6. heterotopias: islands of gray matter in abnormal locations
7. following radiation therapy (XRT)

99.19 Meningeal thickening/enhancement

99.19.1 Dural enhancement (intracranial)

Visible beneath the inner table of the skull. Unlike leptomeningeal enhancement, does not follow the gyral convolutions. May be either focal or diffuse⁵⁶:

1. focal
 - a) adjacent to meningioma: so called "dural tail" (► Fig. 46.4 panel D)
 - b) pleomorphic xanthoastrocytoma (PXA): also can have "dural tail"
2. diffuse dural enhancement⁵⁷: associated with extraaxial neoplastic processes in ≈ 65%. Clinically: H/A, multiple cranial nerve palsies, seizures
 - a) intracranial hypotension (p.421): diffuse pachymeningeal enhancement on cerebral MRI
 - b) infection: bacterial meningitis
 - c) primary CNS tumors: medulloblastoma, malignant meningioma
 - d) sarcoidosis
 - e) following craniotomy
 - f) metastases (mostly carcinomas). May cause dural thickening as well:
 - dural metastases

- leptomeningeal carcinomatosis (carcinomatous meningitis) (p.920)
- lymphoma⁵⁸
- g) neurosarcoidosis
- h) following subdural hemorrhage⁵⁹
- i) idiopathic

99.19.2 Leptomeningeal enhancement

See reference.⁵⁶

1. thin linear enhancement that closely follows the gyri
2. small nodules attached to the brain

99.20 Ependymal and subependymal enhancement

Some overlap with periventricular enhancement. Ependymal enhancement often heralds a serious condition.⁶⁰ Main DDx is tumor vs. infectious process.

1. ventriculitis or ependymitis: ependymal enhancement occurs in 64% of cases of pyogenic ventriculitis⁶¹
 - a) infection may occur in the following settings:
 - following shunt surgery
 - after intraventricular surgery
 - with indwelling prosthetic devices (e.g., Ommaya reservoir)
 - with use of intrathecal chemotherapy
 - with meningitis
 - with viral ependymitis
 - in some cases of CMV encephalitis in immunocompromised patients
 - granulomatous involvement: esp. in immunocompromised patients; e.g., tuberculosis, *Mycobacterium*, syphilis
 - b) infections may be⁶⁰
 - bacterial (pyogenic) ventriculitis
 - tuberculous ventriculitis
 - cystic lesions suggest cysticercosis
2. carcinomatous meningitis: typically also produces meningeal enhancement (p.920)
3. multiple sclerosis: usually more *periventricular* (in the white matter)
4. tumors
 - a) lymphoproliferative disorders
 - CNS lymphoma (p.840)
 - leukemia
 - b) ependymoma
 - with tumor spread
 - transient enhancement reported in a child with ependymoma in the absence of tumor spread⁶²
 - c) metastasis
 - d) germ cell tumors
5. tuberous sclerosis: subependymal hamartomas appear as nodules which occasionally enhance (p.644). These gradually calcify with age
6. in the presence of appropriate constitutional symptoms: rare causes of linear enhancement include: neurosarcoidosis, Whipple's disease, metastatic multiple myeloma (usually nodular)

In immunocompromised patients, the enhancement pattern may help distinguish between the following (which tend to occur in this population⁶⁰):

1. thin linear enhancement: suggests virus (CMV or varicella-zoster)
2. nodular enhancement: suggests CNS lymphoma
3. band enhancement: less specific (may occur with virus, lymphoma, or tuberculosis (TB))

99.21 Intraventricular hemorrhage

Etiologies:

1. most occur as a result of extension of intraparenchymal hemorrhages
 - a) in the adult:
 - spontaneous ICH: especially thalamic or putaminal hemorrhages (p.1614)
 - associated with AVM

- b) in newborns: extension of subependymal hemorrhage (p. 1630)
- 2. *pure* intraventricular hemorrhage (IVH) is usually the result of a rupture of
 - a) aneurysm: accounts for $\approx 25\%$ of IVH in adults, and is second only to extension of intracerebral hemorrhage as the most common cause. IVH occurs in 13–28% of ruptured aneurysms in clinical series.⁶³ More common with the following aneurysms: ACoMA, distal basilar artery or carotid terminus, VA or distal PICA (for patterns) (p. 1454)
 - b) vertebral artery dissection (p. 1577) (or dissecting aneurysms)
 - c) intraventricular AVM
 - d) intraventricular tumor
 - e) SAH outside the ventricles refluxing into foramina of Luschka and/or Magendie

99.22 Medial temporal lobe lesions

May be responsible for seizures, especially “uncal fits” (temporal lobe seizures).

1. hamartoma: nonenhancing
2. mesial temporal sclerosis (p. 482): should see atrophy of the parenchyma in this area with dilatation of the temporal horn of the lateral ventricle
3. glioma: look for mass effect. Usually unilateral. Enhancement suggests higher grade than low grade
4. (autoimmune) limbic encephalitis: an inflammatory disease of the medial temporal lobes that may present with temporal lobe seizures or rapid onset of neuropsychiatric symptoms that may respond to immunotherapy.⁶⁴ May be a paraneoplastic condition, however, disease-causing antibodies are identified in many cases. Diagnostic criteria⁶⁵:
 - a) rapid onset (< 3 months) of short-term memory deficit, seizures, or psychiatric symptoms consistent with limbic system involvement (e.g., anxiety, depression, behavioral changes)
 - b) *bilateral* abnormalities highly restricted to medial temporal lobes on T2 FLAIR MRI ($\blacktriangleright$ Fig. 99.9) or 18-fluorodeoxyglucose (^{18}F -FDG) PET scan
 - c) one or both of:
 1. CSF pleocytosis: $\text{WBCs} > 5/\text{mm}^3$
 2. EEG: epileptic or slow-wave activity involving the temporal lobes
 - d) reasonable exclusion of alternative causes
 - e) if one of the first 3 criteria is not met, the diagnosis may only be made with identification of antibodies against cell-surface, synaptic, or onconeural proteins
5. acute ischemic stroke involving the posterior cerebral artery
6. encephalitis: including
 - a) *herpes simplex encephalitis* (p. 397): comprises 25% of cases of encephalitis with temporal lobe involvement. Differentiating features: bilateral involvement is characteristic (but may be *unilateral*), often hemorrhagic, crosses into insula (“transylvian sign”), spares basal ganglia

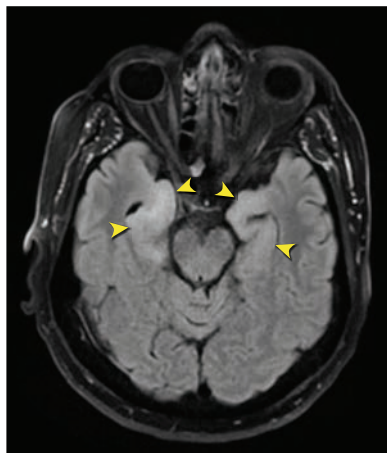


Fig. 99.9 Limbic encephalitis.

Image: axial T2 FLAIR MRI. Finding: increased signal in bilateral medial temporal lobes (yellow arrowheads).

- b) varicella-zoster
- c) TB
- d) neurosyphilis

99.23 Basal ganglion abnormalities

1. generally symmetric abnormalities
 - a) calcification (p.1667)
 - b) Wilson's disease (hepatolenticular degeneration): autosomal recessive disease causing accumulation of copper in tissues
 - c) Huntington's disease (or chorea): caused by > 40 trinucleotide CAG repeats in the Huntington gene (4p16.3) which leads to the production of the protein huntingtin. Cell loss in caudate nucleus can be seen on CT or MRI
 - d) manganese (p.185): symmetrical high-signal abnormalities on T1WI primarily in the globus pallidus with essentially no findings on T2WI or T2* GRE (almost pathognomonic)
 - e) globus pallidus (low density on CT):
 - severe carbon monoxide intoxication
 - cyanide poisoning
 - hypoxia
 - f) putamen
 - hypoglycemia: affects corpus striatum (caudate and putamen)
2. stroke

99.24 Thalamic lesions

Astrocytomas are the most common tumors.

1. common neoplasms
 - a) adults
 - astrocytoma, IDH-mutant, grade 3
 - glioblastoma, IDH-wildtype
 - metastasis
 - primary CNS lymphoma
 - b) pediatrics
 - astrocytoma, IDH-mutant, grade 2
 - astrocytoma, IDH-mutant, grade 3
 - glioblastoma, IDH-wildtype
 - pilocytic astrocytoma
2. uncommon neoplasms
 - a) adults
 - astrocytoma, IDH-mutant, grade 2
 - neurocytoma
 - oligodendroglioma
 - pilocytic astrocytoma
 - hamartoma
 - b) pediatrics
 - germinoma
 - oligodendroglioma
 - PNET
 - subependymal giant cell tumor
3. non-neoplastic (pediatric and adult)
 - a) cavernous angioma
 - b) granuloma
 - c) heterotopias
 - d) AVM
 - e) infarct

99.25 Intranasal/intracranial lesions

Lesions within the nose that may communicate with the intracranial cavity:

1. infectious
 - a) tuberculosis

- b) syphilis
 - c) Hansen's disease (leprosy)
 - d) fungal infections, especially:
 - aspergillosis
 - mucormycosis: seen primarily in diabetics or immunocompromised patients (p.599)
 - *Sporothrix schenckii*
 - *Coccidioides*
 - e) Wegener's granulomatosis (p.207): necrotizing granulomatous vasculitis of the upper and lower respiratory tracts with glomerulonephritis and nasal destruction⁶⁶
 - f) lethal midline granuloma (p.207): a locally destructive lymphomatoid infiltrative disease that may not have true granulomas, and may also cause local nasal destruction. However, renal and tracheal involvement do not occur as in Wegener's granulomatosis
 - g) polymorphic reticulosis: may be a nasal lymphoma. Possibly the same disease as lethal midline granuloma (see above)
2. mucocele: a retention cyst of an air sinus that results from an occluded ostium and may cause expansive erosion of the involved sinus. Often enhances with IV contrast (MRI or CT), and may contain mucus or pus
 3. neoplasms
 - a) carcinoma of the nasal sinus
 - squamous cell
 - glandular
 - nasopharyngeal carcinomas: may be related to Epstein-Barr Virus (EBV) infection
 - sinonasal undifferentiated carcinoma (SNUC)⁶⁷: distinct from lymphoepithelioma (less keratinizing). Rare, aggressive carcinoma (more lethal variant of squamous cell carcinoma) with poor prognosis. Incidence may be higher with prior XRT and in woodworkers and nickel factory workers. May invade adjacent structures, those relevant to neurosurgeons: frontal fossa and cavernous sinus. No relation to EBV. Treatment: tri-modal therapy (XRT, chemotherapy, and salvage surgery)
 - b) olfactory neuroblastoma (p.935) (formerly esthesioneuroblastoma). A very rare malignant tumor arising from crest cells of the nasal vault, often with intracranial invasion. Presents with epistaxis (76%), nasal obstruction (71%), tearing (14%), pain (11%), diplopia, proptosis, anosmia, and endocrinopathies⁶⁸
 - c) metastatic tumors: very rare, possibly with renal cell carcinoma
 - d) benign tumors
 - frontal meningioma: rarely erodes into nasal cavity
 - rhabdomyoma
 - benign hemangiopericytoma
 - cholesteatoma
 - chordoma
 4. congenital lesions
 - a) **encephalocele**: a nasal polypoid mass in a *newborn* should be considered an encephalocele until proven otherwise. Classifications:
 - cranial vault
 - frontal ethmoidal
 - basal
 - posterior fossa
 - b) nasal glioma: non-neoplastic glial tissue located within the nose, often conceptually and diagnostically confused with an encephalocele (► Table 99.6). The term "glioma" is a misnomer, and nasal glial heterotopia is preferred. Does not communicate with the subarachnoid space

Table 99.6 Encephalocele vs. nasal glioma

Finding	Encephalocele	Nasal glioma
pulsatile?	frequently (may not be if small)	no
changes with Valsalva maneuver	swells (Furstenberg sign)	no change
presence of hypertelorism	suggests encephalocele	does not correlate
attachment to CNS	stalk	none, or minimal
probe	can be passed lateral	cannot be passed lateral

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100 Differential Diagnosis by Location or Radiographic Finding – Spine

100.1 Diagnoses covered outside this chapter

See ► Table 100.1.

Table 100.1 Differential diagnoses by location or radiographic finding, spine—covered outside this chapter

DDx
chordomas (p. 825)
lateral disc herniation (p. 1264)
spinal cord tumors (p. 979)
spinal epidural abscess (p. 382)
spinal stenosis
• lumbar (p. 1332)
synovial cyst (spinal) (p. 1341)
thoracic outlet syndrome (p. 581)

100.2 Atlantoaxial subluxation

- incompetence of the transverse atlantal ligament (TAL): results in *increased* atlanto-dental interval (ADI) (p. 223)
 - rheumatoid arthritis: erosion of insertion points of the TAL (p. 1377)
 - traumatic
 - disruption (tear) of the TAL (rare)
 - avulsion of the insertion points of the TAL (as in comminuted C1 fx)
 - congenital laxity of the TAL:
 - Down syndrome: 20% incidence (p. 1381) ¹
 - may be associated with neurofibromatosis
 - retropharyngeal infections: chronic tonsillitis (p. 1158), Grisel syndrome
 - chronic steroid use
- incompetence of the odontoid process: ADI is *normal*
 - odontoid fractures (p. 1171)
 - os odontoides (p. 1175)
 - erosion of the odontoid due to rheumatoid arthritis (RA) (p. 1376)
 - neoplastic erosion of the odontoid:
 - metastases to the upper cervical spine (p. 922)
 - other tumors of the axis
 - Morquio syndrome: hypoplasia of the dens (p. 1308)
 - congenital absence/dysplasia of the odontoid
 - following transoral odontoidectomy: creates severe ligamentous instability (p. 1771)
 - local infection

► **Note.** Chronic atlantoaxial subluxation (AAS) seen in conditions such as rheumatoid arthritis or Down syndrome may be significant yet asymptomatic. Treatment decisions in this group are difficult. Acute AAS is more commonly symptomatic and may be life-threatening.

100.3 Abnormalities in vertebral bodies

See also lesions unique to the craniocervical junction & upper cervical spine (p. 1308).

For abnormalities unique to the axis (C2), see below.

- neoplasms; see more extensive list (p. 570)
 - metastases: prostate, breast, lung, renal cell, thyroid, lymphoma, & myeloma commonly go to bone. Four patterns (≈ all are *low intensity* on T1WI):
 - focal lytic (most common): T1WI = hypointense, T2WI = hyperintense

- focal sclerotic: hypointense on T1WI and T2WI
- diffuse homogeneous: T1WI = hypointense, T2WI = hyperintense or heterogeneous
- diffuse heterogeneous: mixed signal intensities on T1WI & T2WI
- b) primary bone tumors; see more extensive discussion (p.989)
 - vertebral hemangioma
 - osteoblastoma
- 2. infection: osteomyelitis/discitis
- 3. fatty infiltrate or replacement of bone marrow: with age, hematopoietic red marrow of VBs is gradually replaced by yellow marrow in a patchy pattern at a slower rate than in many other locations, e.g., distal appendicular bones.² T1WI: yellow marrow (MRI characteristics similar to subcutaneous fat) is hyperintense to red marrow (caution: bright areas on T1WI may be fat, or may be a normal area next to a low-intensity met). T2WI: yellow marrow is bright
- 4. degenerative changes (Modic changes), see ► Table 81.2
- 5. metabolic
 - a) Paget's disease: plain X-rays → enlargement of VBs with cortical thickening usually involving **several contiguous levels** (p.1363)
 - b) osteoporosis: reduced bone density. Vertebral compression fractures may be seen
 - c) ankylosing spondylitis (p.1365): osteoporotic VBs, calcified intervertebral discs (sparing the nucleus pulposus), and ossified ligaments → square VBs with bridging syndesmophytes ("bamboo spine"). Starts in sacroiliac joints & lumbar spine

100.4 Axis (C2) vertebra lesions

1. tumors: rare. Possibilities include those that involve the spine at any location. Some factors pertinent to this location³:
 - a) primary bone
 - chondroma
 - chondrosarcoma: rare in the craniovertebral junction. Lobulated tumors with calcified areas
 - chordoma: slow-growing radioresistant malignancy (p.827)
 - osteochondroma (chondroma)
 - osteoblastoma (p.990)
 - osteoid osteoma (p.990): more common in posterior elements than VB⁴
 - giant-cell tumors of bone: typically arise in adolescence. Lytic with bony collapse⁵
 - b) metastatic: including
 - typical metastases that spread hematogenously to bone, including: breast cancer, prostate cancer, malignant melanoma, paraganglioma, renal cell carcinoma
 - extension of regional tumors: nasopharyngeal tumors, craniopharyngioma
 - c) meningioma
 - d) neurofibroma
 - e) miscellaneous
 - plasmacytoma
 - multiple myeloma
 - Langerhans cell histiocytosis: osteolytic defect with progressive vertebral collapse. Occasionally occur in C2⁶
 - Ewing's sarcoma: malignant. Peak incidence during 2nd decade of life
 - aneurysmal bone cyst⁷
2. infection: osteomyelitis of the axis
3. pannus from old nonunion of fracture or from rheumatoid arthritis (RA)
4. erosive changes in the odontoid process with RA (p.1377)

100.5 Mass posterior to the odontoid process

1. RA pannus. Involvement of the c-spine in RA typically begins early in the disease process and often parallels the extent of peripheral disease correlating with the degree of hand and wrist erosion
2. retro-odontoid pseudotumor (ROP)⁸ in patients without RA. AAS may play a role in developing ROP. ROP unassociated with AAS can be seen in patients without RA but is rare. All cases reported were seen in the elderly and most have severe OA changes at the atlanto-axial joint
3. calcium pyrophosphate deposition (CPPD) disease (formerly referred to as pseudogout)
4. psoriatic arthritis
5. chondrocalcinosis

100.6 Pathologic fractures of the spine

100.6.1 General information

Fractures due to metastatic involvement are hypointense on T1WI and hyperintense on T2WI. Benign VB collapse should be isointense to normal VBs on all sequences^{9,10} and the VB should look homogeneous. On T2WI or STIR images, the cortex of the VB (which should be a dark border around the VB due to low water content of cortical bone) should be intact.

100.6.2 Etiologies

1. osteoporosis
2. neoplasm: short list
 - a) metastases: common sources of spine mets: lung, breast, prostate, myeloma
 - b) Langerhans cell histiocytosis (p. 846): may cause vertebra plana (see below)
 - c) lymphoma
 - d) hemangioma (p. 992)
3. infection
4. avascular necrosis of the vertebral body
 - a) Calve-Kummel-Verneuil disease (see below)
 - b) with steroid use

100.6.3 Vertebra plana

Criteria:

1. uniform collapse of vertebral body into flat thin disc
2. increased density of vertebra
3. spares neural arches
4. normal disc and intervertebral disc space
5. intervertebral vacuum cleft sign (pathognomonic)
6. no kyphosis

Etiologies include:

1. Langerhans cell histiocytosis
2. Calve-Kummel-Verneuil disease: avascular necrosis of the vertebral body. Occurs in 2–15-year-olds
3. hemangioma

100.7 Spinal epidural masses

See items marked with a dagger (†) under Myelopathy (p. 1696).

100.8 Destructive lesions of the spine

100.8.1 Etiologies

1. neoplastic; see Differential diagnosis: spine & spinal cord **tumors** (p. 979) for more:
 - a) metastatic tumors with a predilection for bone: prostate, breast, renal cell, lymphoma, thyroid, lung...; see Spinal epidural metastases (p. 921)
 - b) primary bone tumors: chordomas (p. 825), osteoid osteoma (p. 990), hemangioma (p. 992)
2. infection:
 - a) vertebral osteomyelitis: occurs mostly in IV drug abusers, patients with diabetes mellitus, and hemodialysis patients. May have associated spinal epidural abscess. Also see Vertebral osteomyelitis (p. 386)
 - b) discitis (p. 390)
3. chronic renal failure: some patients develop a destructive spondyloarthropathy that resembles infection^{11,12}
4. ankylosing spondylitis (p. 1365): bamboo spine (square VBs with bridging syndesmophytes)
5. lesions producing *posterior* scalloping of VB (mnemonic: AMEN)

Acromegaly or achondroplasia
Marfan syndrome or mucopolysaccharidosis

Ehlers-Danlos
Neurofibromatosis
also: dural ectasia

6. lesions producing *anterior* scalloping of VB
 - a) aortic aneurysm
 - b) lymphoma
 - c) spinal TB

100.8.2 Differentiating factors

Of the many lytic or destructive lesions that involve the vertebra, destruction of the disc space is highly suggestive of *infection*, which often involves at least two adjacent vertebral levels. Although tumors may involve adjacent vertebral levels and cause collapse of disc height, the disc space is usually not destroyed¹³ (possible exceptions include: some vertebral plasmacytomas, a reported metastatic cervical carcinoma, and there may occasionally be destruction of the disc in ankylosing spondylitis¹⁴). Unlike pyogenic infections, the disc may be relatively resistant to tuberculous involvement in Pott's disease (► Fig. 21.1).¹⁵ Also, since metastatic tumor involvement usually produces widespread bony involvement, it is less likely with involvement of a single bone.

100.9 Vertebral hyperostosis

1. Paget's disease (p.1362): classic "ivory bone" with cortical thickening ("picture frame" appearance on plain X-rays). Consider Paget's with a *dense* vertebra on X-ray in an older patient, commonly involving several contiguous vertebrae
2. osteoblastic metastases
 - a) in men: prostate
 - b) in women: breast
 - c) lymphoma
3. osteoid osteoma and osteoblastoma (p.990)
4. bone island (AKA enostosis)¹⁶: cortical bone (histologically normal) within cancellous bone (abnormal location). Some consider it a hamartoma. Usually asymptomatic, most common in long bones and pelvis. Low signal on MRI (like cortical bone). High density on CT and plain radiographs. Usually cold on bone scan

100.10 Sacral lesions

100.10.1 Tumors

1. metastases: the most common sacral neoplasm
2. primary neoplasms of the sacrum are uncommon and include:
 - a) giant cell tumor (p.1701)
 - b) chordoma
 - c) teratoma:
 - adults: pre-sacral or sacro-coccygeal teratomas may arise from cells sequestered from Hensen's node in the caudal embryo. Rarely cause neurologic involvement (distinguishing this from chordoma). Sacrum may be normal in up to 50% (abnormal in almost all chordomas). Treatment is complete removal, usually by general surgeon
 - peds: malignant pre-sacral teratoma is a rare tumor seen primarily in female children

100.10.2 Infection

Most infections of the sacrum or sacroiliac joint are due to contiguous spread from a suppurative focus.

100.10.3 Arthritic disorders

1. ankylosing spondylitis (p.1367): involves SI joint almost by definition
2. osteoarthritis

100.10.4 Sacral fractures

May be due to:

1. trauma
2. repetitive stress
3. sacral insufficiency (p. 1705)

100.10.5 Congenital

Sacral agenesis (caudal regression syndrome): rare (prevalence: 0.005–0.01%; higher (0.1–0.2%) in children of diabetic mothers (16–20% of children with sacral agenesis have diabetic mothers)). Increased incidence of associated spinal abnormalities including: syrinx, tethered cord, lipoma, and lipomyelomeningocele.

1. Four types:
 - a) Type 1: partial unilateral agenesis, localized to the sacrum or coccyx
 - b) Type 2: partial bilaterally symmetric defects in the sacrum. Iliac bones articulate with S1, and distal segments of the sacrum and coccyx fail to develop
 - c) Type 3: total sacral agenesis + iliac bones articulate with the lowest segment of the lumbar spine present
 - d) Type 4: total sacral agenesis + iliac bones fused posteriorly along the midline
2. in cases of total sacral agenesis (types 3 & 4), MR findings include: absence of the sacrum and coccyx and variable absence of a portion of the lumbar spine, with a characteristic club-shaped configuration of the conus medullaris

100.10.6 Miscellaneous

Osteitis condensans ilii: increased density in ilium, usually asymptomatic (incidental) finding. Occasionally may produce low back pain or tenderness.

100.11 Enhancing nerve roots

1. tumor
 - a) meningeal carcinomatosis
 - b) lymphoma
2. infection: especially CMV (often seen in AIDS patients)
3. inflammatory
 - a) Guillain-Barré
 - b) arachnoiditis
 - c) sarcoid

100.12 Nodular enhancing lesions in the spinal canal

1. neurofibromatosis (NFT)
2. tumor
 - a) drop mets
 - b) neurofibroma
 - c) schwannoma

100.13 Intraspinal cysts

1. spinal meningeal cysts (p. 1400)
2. cystic neurofibroma
3. ependymoma: may be cystic. In filum terminale: myxopapillary ependymoma (p. 985)
4. syringomyelia (p. 1405)
5. dilated central canal (p. 1405)

100.14 Diffuse enhancement of nerve roots/cauda equina

(As distinct from nodular enhancement, see above)

1. Guillain-Barré (p. 193)

2. meningitis
3. cytomegalovirus (CMV) (especially in AIDS)
4. lymphoma
5. sarcoid (look for hilar adenopathy)

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101 Differential Diagnosis (DDx) by Signs and Symptoms – Primarily Intracranial

101

101.1 Diagnoses covered outside this chapter

See ► Table 101.1.

Table 101.1 Differential diagnoses by signs and symptoms, primarily intracranial—covered outside this chapter

DDx

abducens palsy (p. 598)
anisocoria (p. 591)
chordomas (p. 825)
chronic meningitis (p. 341)
coma (p. 319)
Creutzfeldt-Jakob disease (p. 403)
diabetes insipidus (p. 124)
dizziness (p. 603)
facial nerve palsy (Bell's palsy) (p. 607)
giant cell arteritis (p. 203)
gyral enhancement (p. 1560)
hemiplegia/hemiparesis—see spine section (p. 1703)
internuclear ophthalmoplegia (p. 596)
Meniere disease (p. 604)
multiple sclerosis (p. 188)
ophthalmoplegia
• painful (p. 599)
• painless (p. 600)
papilledema (p. 586)
Parinaud's syndrome (p. 101)
Parkinson's disease (p. 185)
pneumocephalus (p. 1067)
prolactin elevation (► Table 52.4)
pseudotumor cerebri (p. 961)
retinal hemorrhage (p. 1104)
sarcoidosis (p. 199)
seizures
• new onset, adult (p. 503)
• new onset, peds (p. 504)
• nonepileptic (p. 507)
• status epilepticus (p. 511)
schizencephaly (p. 309)
torticollis (p. 1844)
trigeminal neuralgia (p. 1859)
vertigo (p. 603)

101.2 Encephalopathy

Many etiologies are similar to that for coma (p. 319). EEG may be helpful in distinguishing some etiologies (p. 249).

1. a rare cause may be (spontaneous) intracranial hypotension (p. 421)
2. hypertensive encephalopathy from malignant hypertension

101.3 Syncope and apoplexy

101.3.1 General information

Syncope may be defined as one or more episodes of brief loss of consciousness (LOC) with prompt recovery (this term is considered by many to signify a vasovagal episode). The uncommonly used term lipothymia may be less likely to imply an etiology. Prevalence may be as high as $\approx 50\%$ (higher in the elderly). Apoplexy is traditionally considered a form of hemorrhage, usually intracerebral. The recovery from apoplexy would therefore usually be slower than for syncope.

101.3.2 Etiologies

(Adapted.^{1,2}) NB: in a large number of cases no cause can be determined.

1. vascular: a few myotonic jerks may be seen in cerebral ischemia
 - a) cerebrovascular
 - subarachnoid hemorrhage (most commonly aneurysmal)
 - intracerebral hemorrhage
 - brainstem infarction
 - pituitary apoplexy (p.865) (rare)
 - vertebrobasilar insufficiency (VBI) (p. 1591)
 - rarely with migraine
 - b) cardiovascular
 - Stokes-Adams attacks: disorder of AV-node conduction in the heart resulting in syncope with bradycardia
 - carotid sinus syncope: minimal stimulation (e.g., tight shirt collar, syncope while shaving...) causes reflex bradycardia with hypotension, more common in patients with carotid vascular disease. Bedside carotid massage with ECG and BP monitor may diagnose²
 - cardiac standstill: seen rarely in patients with glossopharyngeal neuralgia (p. 1873)
 - vasodepressor syncope (the common faint), AKA vasovagal response, and recently AKA neurocardiogenic syncope³: the most common cause of transient LOC. Hypotension usually with any of the following autonomic manifestations: pallor, nausea, heavy perspiration, pupillary dilatation, bradycardia, hyperventilation, salivation. Usually benign. Most common in age < 35 yrs
 - orthostatic hypotension: drop in SBP ≥ 25 mm Hg on standing
 - triggered syncope: includes micturition syncope, tussive syncope, weight-lifting syncope... (most involve elevation of intrathoracic pressure)
2. infectious
 - a) meningitis
 - b) encephalitis
3. seizure (p.480): in general, there are involuntary movements and confusion afterwards, lasts at least several minutes. Todd's paralysis may follow and usually resolves slowly over a period of a few hours. There may be irritative special-sense phenomena (visual, auditory, or olfactory hallucinations)
 - a) generalized
 - b) complex partial
 - c) akinetic seizure
 - d) drop attack (loss of posture without LOC): seen in Lennox-Gastaut
4. metabolic: hypoglycemia (may produce seizure, usually generalized)
5. miscellaneous
 - a) intermittent ventricular obstruction: the classic example is a colloid cyst of the 3rd ventricle (p.944), but this mechanism is questionable
 - b) narcoleptic cataplexy: narcolepsy is characterized by somnolence and sudden attacks of weakness (cataplexy) when awake. Easy arousal and lack of post-ictal drowsiness distinguishes cataplexy from a seizure. The somnolence is treated with CNS stimulants (such as amphetamines or modafinil (Provigil®) 200 mg PO q AM), and cataplexy is treated with antidepressants
 - c) psychogenic
6. intracranial hypotension: typically when assuming upright position. May be spontaneous (p.421), or may follow LP, CSF shunt, spinal surgery with CSF leak
7. unknown: in $\approx 40\%$ of cases no cause can be diagnosed

101.3.3 Practical approach to syncope

Introduction

The core of diagnosis and management are the H&P, orthostatic vital signs, and the ECG, with a combined diagnostic yield of 50%⁴ covering:

1. reflex-mediated such as vasovagal or Valsalva/stress-induced: 36–62%
2. cardiac valvular etiology or arrhythmia: 10–30%
3. orthostatic due to autonomic dysregulation, dehydration, or polypharmacy: 2–24%
4. cerebrovascular due to stroke: ≈ 1%
5. seizure

Evaluation

1. history: includes
 - a) medication list: look for drugs that may cause orthostatic hypotension, especially blood pressure medication, beta-blockers
 - b) precipitating factors: e.g., change in position, sensitivity to tight collars...
 - c) premonitory factors: e.g., sweating and tremulousness may signify hypoglycemia, bradycardia is associated with vasovagal events, tonic-clonic movements may occur with a seizure
 - d) post-ictal emergence: usually rapid after a simple faint, slower after seizure which may also exhibit Todd's paralysis (p.480)
2. cardiovascular etiologies: Testing is also guided by H&P, vital signs & ECG:
 - a) cardiac arrhythmia evaluation: 12-lead ECG & 24-hour Holter monitor, and may lead to electrophysiologic (EP) testing/intervention^{4,5}
 - b) abnormal orthostatics warrant a formal tilt-table test
 - c) history of cardiomyopathy or CAD merits an echocardiogram and formal stress testing. These results determine the need for cardiac catheterization
3. neurologic etiologies: comprise < 1% of cases.⁶ In the absence of clinical evidence of a neurologic etiology, neurodiagnostic testing (EEG, CT scan, MRI/MRA, carotid Doppler) has a diagnostic yield of 2–6%. ∴ These tests are warranted only when clinically indicated⁵ (seizures, altered consciousness, gradually resolving Todd's paralysis, known history of cerebrovascular compromise). Tests include:
 - a) unenhanced brain CT: rules out most acute neurosurgical etiologies (bleed, hydrocephalus, edema that may be associated with tumor)
 - b) MRI without and with enhancement in cases with unexplained CT findings, or with a negative CT but high suspicion of a CNS etiology
 - c) seizure evaluation: when symptoms suggest possible seizure:
 - EEG: usually a sleep-deprived EEG. Not very sensitive
 - 24-hour video EEG monitoring: in cases with high index of suspicion of seizures or nonepileptic seizures

Management

Admission and inpatient management are warranted for patients with diagnosed cardiac or neurologic syncope, either by suggestive history (family history of sudden death, syncope during exertion, witnessed seizure) or diagnostic testing (arrhythmia, severe orthostatic changes, hemodynamic instability).^{4,7}

101.4 Transient neurologic deficit

For apoplexy, etc., see Syncope and apoplexy (p.1684).

The first three etiologies listed below cover most cases of transient neurologic deficit:

1. transient ischemic attack (TIA) (p.1536): temporary neurologic dysfunction as a result of ischemia. Maximum deficit usually at onset. Most resolve in < 20 mins
2. migraine: unlike TIA, tends to progress in a march-like fashion over several minutes. May or may not be followed by headache; see Migraine (p.183)
3. seizure: may be followed by Todd's paralysis (p.480)
4. TIA-like syndrome
 - a) "tumor TIA": a transient deficit in a patient with a tumor, may be clinically indistinguishable from an ischemic TIA. Intravascular lymphomatosis may mimic TIAs (p.842)
 - b) TIA-like symptoms may occur as a prodrome (p.1614) to a lobar intracerebral hemorrhage^{8,9} in cases of cerebral amyloid angiopathy (CAA). Unlike typical TIAs, these usually consist of

numbness, tingling, or weakness that gradually spreads in a manner reminiscent of a Jacksonian march and may cross over vascular territories. Caution: antiplatelet drugs and anticoagulation may increase the risk of hemorrhage in patients with CAA (p. 1612)

- c) chronic subdural hematoma: may cause recurrent TIA-like symptoms of the involved hemisphere¹⁰ (including transient aphasia with dominant hemisphere involvement, hemisensory or motor abnormalities). The duration of symptoms tends to be longer than the typical TIA.¹⁰ Postulated mechanisms include:
 - electrical basis: the possibility of epileptic activity (e.g., due to irritation of the cortex by blood breakdown products) has not been supported in the literature; however, spreading depression of Leao has been considered¹¹
 - impairment of venous outflow by compression of surface veins
 - compromised regional cerebral perfusion by indirect shifting of the anterior and posterior cerebral arteries¹²
 - transient ICP elevations → variations in cerebral perfusion pressure

101.5 Ataxia/balance difficulties

1. cerebellar origin: usually with involvement of UEs in addition to LEs
 - a) cerebellar tumors
 - b) cerebellar hemorrhage
 - c) acute cerebellar ataxia: usually follows viral infection in a child < 3 years. Usually self-limited with good prognosis for complete recovery
2. spinal cord: usually worse with eyes closed (loss of proprioceptive input)
 - a) spinal stenosis
 - b) neoplastic cord compression
 - c) syringomyelia (may be part of Chiari malformation)
3. degenerative
 - a) ataxia-telangiectasia syndrome
 - b) ataxia oculomotor apraxia
 - c) Friedreich's ataxia
 - d) spinocerebellar degeneration
4. metabolic/nutritional
 - a) vitamin B₁₂ deficiency
 - b) drugs
 - ASMs (especially phenytoin or carbamazepine)
 - alcohol: acutely with intoxication and chronic
 - heavy metal poisoning: primarily lead (wrist drop is more common)
5. conditions that may mimic ataxia
 - a) weakness
 - b) peripheral neuropathy
 - c) dizziness: including orthostatic hypotension; see Dizziness and vertigo (p. 603)
6. peripheral neuropathy:
 - a) ataxia can occur with Guillain-Barré syndrome (p. 193), especially Miller Fisher variant (p. 194)
 - b) balance difficulties are common with chronic inflammatory demyelinating polyneuropathy (CIDP) (p. 195)

101.6 Wide-based gait

Definition: gait with feet spread > 2–4 inches apart.

This symptom may have intracranial or spinal etiology (myelopathy).

1. idiopathic normal pressure hydrocephalus (iNPH) (p. 438). Often accompanied by magnetic gait (p. 439). Memory disturbance is common
2. cerebellar ataxia. Poor placement of feet. Look for other cerebellar signs: dysidiadochokinesia (impairment of rapid alternating movements), dysmetria, nystagmus...
3. sensorimotor peripheral neuropathy: due to loss of input of joint position sense. Worsens with loss of visual input (closing eyes, as in Romberg's test)
4. vestibular ataxia: Meniere disease, labyrinthitis. Associated with N/V, nystagmus, hearing loss...
5. myelopathy: due to cervical or thoracic spinal cord dysfunction (compression, or intrinsic spinal cord disease, see myelopathy (p. 1696)). Usually accompanied by other long tract signs: reflex abnormalities (Babinski, clonus, hyperreflexia), decreased vibratory and/or joint position sense

(posterior columns). With cervical spinal cord involvement, upper extremity findings may also occur (typically numb/clumsy hands). Asymmetric spinal cord involvement may produce Brown-Sequard syndrome (p. 1135): dissociated sensory loss (contralateral loss of pain and temperature with ipsilateral loss of vibratory sense and proprioception) with ipsilateral weakness

6. ✖ Parkinson disease (p. 184): wide-based gait is *not* typical. Classically: short shuffling steps, intermittent freezing. Associated with resting tremor, bradykinesia, cogwheeling, reduced arm swing...

101.7 Diplopia

1. cranial nerve palsy of any one or combination of III, IV (rare), or VI
 - a) multiple cranial nerve palsies (p. 1687)
 - b) VI palsy (p. 598): can occur with increased intracranial pressure, e.g., in pseudotumor cerebri (p. 955), sphenoid sinusitis...; see other causes of abducens palsy (p. 344)
 - c) isolated muscle paresis of III suggests nuclear lesion or myasthenia gravis
2. intraorbital mass compressing extraocular muscles
 - a) orbital pseudotumor (p. 600)
 - b) meningioma
3. Graves' disease: hyperthyroidism + ophthalmopathy (p. 1658)
4. myasthenia gravis
5. giant cell arteritis (p. 204)
6. botulism: due to toxin from *Clostridium botulinum* (in adults: ingested or in wound). N/V, abdominal cramps, and diarrhea often precede neurologic symptoms. Neurologic involvement is typically symmetric. Dry mouth & cranial nerve palsies (diplopia, ptosis, loss of accommodation, and pupillary light reflex) are followed by descending weakness. Bulbar paresis (dysarthria, dysphagia, dysphonia, flaccid facial muscles) follows. Muscles of the trunk/extremities and respiration progressively weaken in a descending fashion. Sensory disturbances are absent. Sensorium usually remains clear
7. following head trauma: includes injury to EOMs, orbital hematoma, VI palsy from increased ICP

101.8 Anosmia

1. abrupt onset of anosmia
 - a) severe upper respiratory infection with damage to the neuroepithelium: the most common cause
 - b) head trauma: second most common cause. Anosmia occurs in 7–15% of patients with significant head trauma
2. gradual onset of anosmia
 - a) allergic rhinitis and sinus disease¹³: third most common cause of anosmia (anosmia in this setting may be intermittent)
 - b) intracranial neoplasms: olfactory groove meningioma (see Foster Kennedy syndrome (p. 100)), olfactory neuroblastoma (p. 935)
 - c) may also be associated with Alzheimer's disease
 - d) olfactory sense diminishes with age: ~ 50% of patients 65–85 years of age have some loss of sense of smell
 - e) metabolic abnormalities: vitamin deficiency
 - f) physical blockage of nasal passages: nasal polyps...
 - g) endocrine abnormalities: diabetes...
 - h) chemical: alcohol abuse, exposure to solvents,¹⁴ cocaine (ischemic infarction of olfactory mucosa from vasoconstriction)
3. congenital anosmia: Kallmann syndrome (anosmia with hypogonadotropic hypogonadism¹⁵)

101.9 Multiple cranial nerve palsies (cranial neuropathies)

101.9.1 Framework

The differential diagnosis is legion. The following is a framework (modified¹⁶):

1. congenital
 - a) Möbius syndrome: AKA congenital facial diplegia. Facial plegia is complete in ~ 35% (in rest, affects upper face more than lower face, unlike central or peripheral facial palsy), associated

with abducens palsy in 70%, external ophthalmoplegia in 25%, ptosis in 10%, lingual palsy in 18%

- b) congenital facial diplegia may be part of facioscapulohumeral or myotonic muscular dystrophy
2. infectious
 - a) chronic meningitis:
 - spirochetal, fungal, mycoplasma, viral (including AIDS)
 - mycobacterial AKA tuberculous (TB) meningitis: 6th nerve involved first and most frequently. CSF shows lymphocytic pleocytosis and hypoglycorrhachia. Smears are usually negative and multiple cultures are needed to diagnose
 - b) stage II Lyme disease (p.607). Facial nerve weakness is common, sometimes bilateral (Lyme disease is the most common cause of facial diplegia in endemic areas). Other cranial nerve involvement is rare
 - c) neurosyphilis: rare nowadays except with AIDS. Diagnosed by serologic testing
 - d) fungal infection
 - cryptococcal meningitis (p.409): CSF analysis for cryptococcal antigen and India ink prep can detect
 - aspergillosis: may extend to the orbit from sinuses and involve cranial nerves
 - mucormycosis (phycomycosis) (p.1541): produces cavernous sinus syndrome, usually occurs in diabetics
 - e) cysticercosis: especially with basal form; see Neurocysticercosis (p.404)
3. traumatic: especially with basal skull fractures. Lower cranial nerve palsies may occur (sometimes delayed in onset) with occipital condyle fractures (p.1156) or atlantooccipital dislocation (p.1153)
4. neoplastic (brainstem compression and intrinsic lesions usually also produce long tract findings early). Also see Jugular foramen syndromes (p.102)
 - a) chordoma (p.827)
 - b) sphenoid-ridge meningioma
 - c) neoplasms of the temporal bone (often in conjunction with chronic otitis media and otalgia): adenoid cystic carcinoma, adenocarcinoma, mucoepidermoid carcinoma
 - d) glomus jugulare tumors: often affects nerves IX, X, and XI. May cause pulsatile tinnitus; see Paraganglioma (p.939)
 - e) carcinomatous or lymphomatous meningitis (p.920): CSF pleocytosis and elevated protein. Palsies are painless or associated with diffuse headache. Sensory palsies are common, resulting in deafness and blindness
 - f) invasive PitNET/adenomas involving the cavernous sinus (p.864): extraocular cranial neuropathies tend to develop after visual field deficits in these tumors, and are less common when compared to other intracavernous solid tumors¹⁷
 - g) primary CNS lymphoma (p.843)
 - h) multiple myeloma involving the skull base (p.929)
 - i) intrinsic brainstem tumors: gliomas, ependymoma, metastases...
5. vascular
 - a) aneurysm: intracranial or cavernous sinus (p.1489)
 - b) brainstem stroke: usually also produces long tract findings (p.101)
 - Weber's syndrome: Cr N III (usually pupil-sparing) + contralateral hemiparesis
 - Millard-Gubler syndrome: Cr N VI + VII + contralateral hemiparesis
 - c) vasculitis: Wegener's granulomatosis usually affects eighth nerve in addition to others
6. granulomatous
 - a) sarcoidosis: ~ 5% have CNS involvement, usually as fluctuating single or multiple cranial neuropathies (facial nerve is most common, and may be indistinguishable from Bell's palsy). CSF pleocytosis (p.208) is common
7. inflammatory
8. neuropathies
 - a) Guillain-Barré syndrome (GBS) (p.193): cranial nerve involvement includes facial diplegia, oropharyngeal paresis. Peripheral neuropathy usually presents with ascending weakness, proximal muscle weakness > distal, and absent deep tendon reflexes
 - b) Miller Fisher variant GBS: ataxia, areflexia & ophthalmoplegia. Serum marker: anti-GQ1b antibodies
 - c) idiopathic cranial polyneuropathy: subacute onset of constant facial pain, usually retro-orbital. Frequently precedes sudden onset of cranial nerve palsies, usually involving III, IV, & VI, less frequently V, VII, and lower nerves (IX through XII). Olfactory and auditory nerves are

usually spared. Acute and chronic inflammation of unknown etiology similar to Tolosa-Hunt and orbital pseudotumor. Steroids reduce pain and expedite recovery

9. entrapment in abnormal bone
 - a) hyperostosis cranialis interna: a rare autosomal dominant abnormality of the bone of the base of the skull causing recurrent facial palsy and other cranial nerve palsies¹⁸
 - b) osteopetrosis: see below
 - c) Pager's disease (p.1362) involving the skull: 8th nerve involvement (deafness) is most common. Optic nerve atrophy, and palsies of oculomotor, facial, IX, XI, olfactory nerves and others may also occur¹⁹
 - d) fibrous dysplasia (p.974)

101.9.2 Specific syndromes

Facial diplegia

Items culled from the above list that have facial diplegia (p.607) as a prominent finding:

1. congenital: Möbius syndrome, congenital facial diplegia
2. infectious: Lyme disease
3. neuropathies: Guillain-Barré syndrome
4. isolated 4th ventricle (p.607): compression at the facial colliculus
5. granulomatous: sarcoidosis

Cavernous sinus syndrome

Multiple cranial nerve palsies (involving any of the cavernous sinus cranial nerves: III, IV, V1, V2, V6) that primarily produce diplopia (due to ophthalmoplegia). Classically the third nerve palsy (e.g., from an enlarging cavernous carotid artery aneurysm) will *not* produce a dilated pupil because the sympathetic which dilate the pupil are also paralyzed.²⁰ (p.1492) Facial pain or altered facial sensation may occur.

See list of lesions that may produce cavernous sinus syndrome (p.1660).

Osteopetrosis

AKA "marble bone disease" (there is also some confusion with the term osteosclerosis; osteosclerosis fragilis generalisata is the obsolete term for osteopetrosis). A rare group of genetic disorders of defective osteoclastic resorption of bone resulting in increased bone density, may be transmitted either as autosomal dominant or recessive.²¹ The dominant form is usually benign and is seen in adults and adolescents. The recessive ("malignant") form is often associated with consanguinity, and is similar to hyperostosis cranialis interna (see above), but in addition to the proclivity for the skull, it also involves ribs, clavicles, long bones, and pelvis (long-bone involvement results in destruction of marrow and subsequent anemia). Cranial nerves involved primarily include optic (optic atrophy and blindness are the most common neurologic manifestation), facial, and vestibulo-acoustic (with deafness); trigeminal nerve may also be involved. There may also be extensive intracranial calcifications, hydrocephalus, intracranial hemorrhage, and seizures.

Bilateral optic nerve decompression via a supraorbital approach may improve or stabilize vision.²¹

101.10 Binocular blindness

► Bilateral occipital lobe dysfunction

1. bilateral posterior cerebral artery flow impairment
 - a) top of the basilar syndrome
 - b) increased intracranial pressure
 - hydrocephalus with shunt malfunction
 - pseudotumor cerebri (p.956)
 - cryptococcal meningitis: decreased visual acuity (p.409)
2. trauma: bilateral occipital lobe injury (e.g., contrecoup injury)

► Other etiologies

1. seizures: epileptic blindness
2. migraine: cortical spreading depression

3. posterior ischemic optic neuropathy (p. 1261): usually in the setting of shock, rarely following spine surgery in the prone position
4. bilateral vitreous hemorrhage: e.g., with SAH (Terson syndrome (p. 1419))
5. functional: conversion reaction, hysterical blindness...

101.11 Monocular blindness

Due to a lesion anterior to the optic chiasm.

1. Amaurosis fugax: often described as a "shade coming down" over one eye
 - a) TIA: usually due to occlusion of the retinal artery (p. 1544)
 - b) giant cell arteritis (GCA) (p. 203): usually due to ischemia of optic nerve or tracts (less commonly due to retinal artery occlusion)²²
2. trauma: optic nerve injury
3. ruptured carotid cavernous aneurysm: resultant carotid-cavernous fistula increases intraocular pressure by impeding venous return
4. intraorbital pathology: tumors
5. injury within the globe: retinal detachment, ocular trauma
6. unilateral vitreous hemorrhage: e.g., with SAH (Terson's syndrome)

101.12 Exophthalmos

101.12.1 General information

Alternate spelling: exophthalmus.

Definition: abnormal protrusion of the eyeball. Some authors reserve the term exophthalmos for cases due to endocrinopathies and use proptosis (of the eye) for other causes, but these terms are widely used interchangeably.

Clinical criteria (there are other criteria that may be used): anterior displacement of the globe > 18 mm (Hertel exophthalmometry can be used to measure clinically – requires intact lateral orbital bone).

CT/MRI criteria: proptosis is diagnosed when > 2/3 of the globe lies anterior to a line drawn from lateral orbit to medial canthus. For most accurate results, eyes should be open and fixated on a point in the primary gaze position. NB: most routine CT scans do not include the orbits to reduce the radiation to the eye.

101.12.2 Pulsatile

1. carotid-cavernous fistula (CCF) (p. 1519)
2. transmitted intracranial pulsation due to defect in orbital roof
 - a) seen unilaterally e.g., in neurofibromatosis type 1 (p. 638)
 - b) post-op following procedures that remove orbital roof or wall
3. vascular tumors

101.12.3 Non-pulsatile

1. tumor
 - a) intraorbital tumor: may be due to mass effect from tumor or to compromised venous drainage from the orbit
 - optic glioma (p. 694)
 - optic sheath neuroma
 - lymphoma
 - optic sheath meningioma²³
 - orbital involvement with multiple myeloma (p. 929)
 - orbital invasion by invasive PitNET/adenoma (p. 963)
 - in peds: metastatic neuroblastoma
 - in peds: Langerhans cell histiocytosis (p. 846) as part of Hand-Schüller-Christian (triad: DI, exophthalmos, and lytic bone lesions (particularly of cranium))
 - b) due to hyperostosis from a sphenoid ridge meningioma
2. Graves' disease (hyperthyroidism + exophthalmos) (p. 1658): even though the exophthalmos is usually bilateral with this (80%), thyroid disease is still the most common cause of *unilateral* proptosis²⁴
3. enlargement of periorbital fat²⁵

4. infection: orbital cellulitis (usually has concomitant sinusitis)
5. inflammatory: orbital pseudotumor. Usually unilateral (p.600)
6. hemorrhage
 - a) traumatic
 - b) spontaneous
7. 3rd nerve palsy: can cause up to 3-mm proptosis from relaxation of the rectus muscles
8. cavernous sinus occlusion (may affect both eyes)
 - a) cavernous sinus thrombosis (p.1595)
 - b) cavernous sinus tumor obstructing venous outflow
9. pseudo-exophthalmos
 - a) congenital macropthalmos (bull's eye)
 - b) lid retraction: e.g., in Graves' disease (p.1691)
 - c) coronal craniosynostosis can cause a "relative" proptosis (p.266)

101.13 Ptosis

AKA blepharoptosis. Drooping of the upper eyelid.

Distinguished from pseudoptosis (lid droop not resulting from weakness of levator palpebrae superioris (LPS)), which can be due to enophthalmos (globe displaced posteriorly, e.g., with orbital floor blow-out fracture), microphthalmia, blepharospasm, Duane syndrome.

Etiologies of ptosis:

1. congenital: most are simple (autosomal dominant inheritance), complicated ptosis is associated with other findings (e.g., ptosis with ophthalmoplegia)
2. traumatic: injury to eyelid, orbital roof fracture...
3. neurogenic:
 - a) third nerve palsy (p.596)
 - involvement of main trunk of third nerve: can occur intradurally or within cavernous sinus. Ptosis may be an early sign of pituitary tumor expansion (apoplexy) (p.865)
 - involvement of the superior division of the third nerve within the orbit
 - b) Horner syndrome (p.594): ptosis here is partial (may be a pseudoptosis since weakness is in tarsal muscles, not LPS), and the lower eyelid will be higher than the uninvolved contralateral lower eyelid
4. myogenic ptosis
 - a) botulinum toxin injection (e.g., Botox®)
 - b) myasthenia gravis
5. mechanical ptosis
 - a) tumors: neurofibroma, hemangioma, malignant melanoma, mets...
 - b) extension of mucocele of frontal sinus
6. pharmacologic (drugs). Partial list:
 - a) corticosteroids: including topical
 - b) alcohol
 - c) opium

101.14 Pathologic lid retraction

1. hyperthyroidism (p.1659)
2. psychiatric: schizophrenia...
3. steroids
4. Parinaud's syndrome (p.101)

101.15 Macrocephaly

Macrocephaly means increased size of the head.²⁶ Although sometimes used synonymously, the term macrocrania by convention refers to a head circumference > 98th percentile.²⁷ (p.203) Also, not to be confused with macrencephaly AKA megalencephaly (see below). In a pediatric practice the 3 most common etiologies in decreasing order of frequency: familial (parents have big heads), benign subdural fluid collections of infancy (p.1087), and hydrocephalus.

1. with ventricular enlargement
 - a) (hydrostatic) hydrocephalus (HCP), see for etiologies (p.426)
 - communicating
 - obstructive

- b) hydranencephaly (p.309)
- c) constitutional ventriculomegaly: ventricular enlargement of no known etiology with normal neurologic function
- d) hydrocephalus ex vacuo: loss of cerebral tissue (more often associated with *microcephaly*, e.g., with TORCH infections)
- e) vein of Galen aneurysms: see below
- 2. with normal or mildly enlarged ventricles
 - a) "external hydrocephalus": prominent subarachnoid spaces and basal cisterns; see External hydrocephalus (AKA benign external hydrocephalus) (p.433)
 - b) subdural fluid
 - hematoma
 - hygroma
 - effusion benign and symptomatic
 - benign subdural collections of infancy (p.1087)
 - c) cerebral edema: some consider this to be a form of pseudotumor cerebri²⁶
 - toxic: e.g., lead encephalopathy (from chronic lead poisoning)
 - endocrine: hypoparathyroidism, galactosemia, hypophosphatasia, hypervitaminosis A, adrenal insufficiency...
 - d) familial (hereditary) macrocrania: parents also have large heads, the brains eventually "catch up"
 - e) idiopathic
 - f) megalencephaly (AKA macrencephaly): an enlarged brain (p.312)
 - g) neurocutaneous disorders (p.637): usually due to increased volume of brain tissue (megalencephaly, see above).²⁶ Seen especially in neurofibromatosis and congenital hypermelanosis (Ito's syndrome). Less common in tuberous sclerosis and Sturge-Weber. Also seen in the rare hemimegalencephaly syndrome
 - h) arachnoid cyst (AKA subependymal or subarachnoid cyst)²⁶: a duplication of the ependyma or arachnoid layer filled with CSF. Usually reach maximal size by 1 month of age and do not enlarge further. Treatment is required in $\approx 30\%$ due to rapid enlargement or growth beyond first month. Cyst may be shunted or fenestrated. Prognosis with true arachnoid cyst is generally good (unlike porencephalic cyst) if no increased ICP or progressive macrocephaly during 1st year of life
 - i) arteriovenous malformation: especially vein of Galen "aneurysm" (p.1518). Auscultate for cranial bruit. With vein of Galen aneurysms, macrocephaly may be due to HCP from obstruction of the Sylvian aqueduct.²⁶ With other malformations, macrocrania may be due to increased pressure in venous system without HCP
 - j) brain tumors without hydrocephalus: brain tumors are rare in infancy, and most cause obstructive HCP. Tumors that occasionally present without HCP include astrocytomas. May also be seen in the rare diencephalic syndrome, see tumor of anterior hypothalamus (p.695)
 - k) "gigantism syndromes"
 - Soto's syndrome: associated with advanced bone age on X-ray, and multiple dysplastic features of face, skin, and bones
 - exomphalomalaglossia-gigantism (EMG) syndrome: hypoglycemia (from abnormalities in islets of Langerhans), large birth weight, large umbilicus or umbilical hernia, and macroglossia
 - l) "craniocerebral disproportion" (p.1087)²⁶: may be the same as benign extra-axial fluid of infancy
 - m) achondroplastic dwarf: cranial structures are enlarged but the skull base is small, giving rise to a prominent forehead and an OFC ≥ 97 th percentile for age, hypoplasia of midface, and stenosis at foramen magnum. Head growth follows different curve than normal (OFC ≥ 97 th percentile for age is not unusual and does not necessitate shunting)
 - n) Canavan's disease: AKA spongy degeneration of the brain, an autosomal recessive disease of infancy prevalent among Ashkenazi Jews. Produces symmetrical low attenuation of hemispheric white matter on CT²⁸ and macrocephaly
 - o) neurometabolic diseases: usually due to deposition of metabolic substances in the brain. Seen in Tay-Sachs gangliosidosis, Krabbe disease...
- 3. due to thickening of the skull
 - a) anemia: e.g., thalassemia
 - b) skull dysplasia: e.g., osteopetrosis (p.1689)

101.16 Tinnitus

101.16.1 General information

May be either subjective (heard only by patient) or objective (e.g., cranial bruit, can be heard by examiner as well, usually with a stethoscope placed over the cranium, orbit, or carotid arteries in the neck). Objective tinnitus is almost always due to vascular turbulence (from increased flow or partial obstruction).

101.16.2 Pulsatile tinnitus

Most cases are due to vascular lesions.

1. pulse-synchronous:
 - a) carotid-cavernous fistula (p.1519)
 - b) AVM:
 - cerebral (pial) AVM
 - dural AVM (p.1514)
 - c) glomus jugulare tumor (p.941)
 - d) cerebral aneurysm: (rare) possibly with turbulent flow in giant aneurysm
 - e) hypertension
 - f) hyperthyroidism
 - g) pseudotumor cerebri (p.955)
 - h) transmitted bruit: from heart (e.g., aortic stenosis), carotid artery stenosis (especially external carotid)
 - i) dehiscent jugular bulb or high-riding jugular bulb: normal venous variant
 - j) rarely with posterior fossa tumors: CP-angle tumors e.g., vestibular schwannoma or meningioma, vascular intraparenchymal tumors e.g., hemangioblastoma (especially in CPA)
 - k) lesions that can present with a red tympanic membrane
 - aberrant carotid artery in middle ear
 - persistent stapedia artery: rare. Arises from aberrant ICA or from junction of horizontal and vertical petrous ICA. Foramen spinosum is absent on the affected side. Enlargement of anterior tympanic segment of seventh nerve canal
 - glomus tympanicum tumor (p.941)
 - l) sigmoid sinus diverticulum
2. non-pulse-synchronous: asymmetrical enlargement of sigmoid sinus and jugular vein may produce a low-grade hum

Workup for pulsatile tinnitus:

1. MRI without and with enhancement: to look for tumors, e.g., glomus jugulare
2. angiogram: include internal and external carotid injections
3. tests that are usually *not* helpful and should not be ordered routinely:
 - a) carotid ultrasound: nonspecific, not sensitive
 - b) MRI/MRV: may miss small dural fistulas and do not give details needed for treatment for large ones

101.16.3 Non-pulsatile tinnitus

1. occlusion of external ear: cerumen, foreign body
2. middle ear infection (otitis media)
3. otosclerosis
4. stapedia muscle spasms: as occurs in hemifacial spasm
5. CP-angle tumors: including vestibular schwannoma (p.777)
6. Meniere disease (p.604)
7. labyrinthitis
8. endolymphatic sac tumors: e.g., as in von Hippel-Lindau disease (p.648)
9. drugs
 - a) salicylates: aspirin, bismuth subsalicylate (Pepto Bismol®)
 - b) quinine
 - c) aminoglycoside toxicity: streptomycin, tobramycin (tinnitus precedes hearing loss)

101.17 Facial sensory changes

1. circumoral paresthesias
 - a) hypocalcemia
 - b) syringobulbia
2. unilateral facial sensory changes
 - a) trigeminal nerve neuroma
 - b) vestibular schwannoma (VS): to involve Cr. N. V a VS has to be > 2 cm in diameter; see symptoms from 5th nerve compression under vestibular schwannoma (p. 778)
 - c) compression of the spinal trigeminal tract (large compressive lesions may cause bilateral alteration of facial sensation) that chiefly manifests in diminution of pain and temperature sense with little effect on touch sense.²⁹ The tract usually extends down into the spinal cord as far as ≈ the C2 vertebral level (although it may occasionally extend down to C4)

101.18 Language disturbance

1. aphasia:
 - a) injury to speech areas of brain: the classic nosology is likely an oversimplification in view of more recent models of speech (e.g., dual stream model)
 - Wernicke's aphasia (p. 90): classically produces *fluent* aphasia (normal sentence length & intonation, devoid of meaning)
 - Broca's aphasia (p. 90): haltering, dysarthric
 - conduction aphasia (p. 90): fluent spontaneous speech and paraphasias, but patients understand spoken or written words, and are aware of their deficit
 - b) transitory aphasia following a seizure; see Todd's paralysis (p. 480)
 - c) primary progressive aphasia of adulthood: idiopathic & degenerative
2. akinetic mutism: seen with bilateral frontal lobe dysfunction (e.g., with bilateral ACA distribution infarction due to vasospasm from AComA aneurysm rupture or with large bilateral frontal lesions; may actually be abulia) or with bilateral cingulate gyrus lesions
3. cerebellar mutism (p. 100) (AKA muteness of cerebellar origin)^{30,31}
4. following transcallosal surgery: as a result of bilateral cingulate gyrus retraction or thalamic injury together with section of the midportion of the corpus callosum³²

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102 Differential Diagnosis (DDx) by Signs and Symptoms – Primarily Spine and Other

102

102.1 Diagnoses covered outside this chapter

See ► Table 102.1.

Table 102.1 Differential diagnoses by signs and symptoms, spine and other—covered outside this chapter

DDx
ankylosing spondylitis (p. 1367)
bladder dysfunction (p. 93)
brachial plexopathy (p. 570)
carpal tunnel syndrome (p. 547)
cervical stenosis (p. 1300)
lateral disc herniation (p. 1265)
meralgia paresthetica (p. 561)
myopathy
spinal cord tumors (p. 979)
spinal epidural abscess (p. 381)
spinal stenosis
• lumbar (p. 1332)
synovial cyst (spinal) (p. 1341)
thoracic outlet syndrome (p. 581)
torticollis (p. 1844)
urinary retention (p. 95)

102.2 Myelopathy

Items marked with a dagger (†) may present as a *spinal epidural mass*.

► Congenital

1. (Arnold)–Chiari malformation (p. 303): Type I often presents in early adulthood
2. tethered cord: often may not present until after some trauma
3. syringomyelia: may be congenital or posttraumatic in quadriplegics, usually presents with a central cord syndrome—see Syringomyelia (p. 1405)—or progressive myelopathy
4. neurenteric cyst (p. 313)
5. cord compression that occurs with some mucopolysaccharidoses: e.g., Morquio syndrome (due to atlantoaxial subluxation), Hurler syndrome
6. hereditary spastic paraplegia: family history is key. Diagnosis of exclusion¹

► Acquired

1. cervical or thoracic spinal stenosis: often degenerative disease superimposed on congenitally narrow canal (congenital narrowing is frequent in achondroplasia)
2. traumatic: including spinal shock, hematomyelia, spinal epidural hematoma (see vascular below), barotrauma, electrical injuries, compression by bone fracture†. May follow minor trauma in the setting of spinal stenosis
3. herniated intervertebral disc†: myelopathy more common in thoracic region, radiculopathy more common in cervical region (long tract signs are rare with herniated cervical disc)
4. kyphosis
5. extramedullary hematopoiesis† (p. 179): hypertrophy of marrow → cord compression. Primarily in chronic anemias (e.g., thalassemia major)
6. bony compression secondary to incompetence of odontoid process or transverse atlantal ligament†. May be congenital, traumatic (p. 1171), neoplastic, or inflammatory (especially rheumatoid arthritis)

7. epidural lipomatosis† (p. 1381): hypertrophy of epidural fat most often due to years of exogenous steroid therapy.² Rarely symptomatic
8. ossification of the posterior longitudinal ligament (OPLL) (p. 1370)³
9. vertebral Paget's disease† (p. 1362)
10. idiopathic spinal cord herniation (p. 1404)^{4,5}: rare. Thoracic spinal cord herniates through an anterior dural defect frequently producing a Brown-Séquard syndrome or spastic paraparesis
11. spinal arachnoid cyst (p. 1401)
12. spinal arachnoid web (p. 1402)
13. Hirayama disease (p. 1403): a rare condition characterized by insidious, usually asymmetric painless wasting of the distal UE muscles typically affecting males in the age range of 15–20 years. May be caused by asymmetric compression of the spinal cord by the dura separating from the posterior bony spinal canal in flexion
14. superficial siderosis: rare. Hemosiderin deposition on the surface of the brain, cranial nerves, and spinal cord causes neurodegeneration. Triad: hearing loss, ataxia, and myelopathy. May follow trauma, surgical procedures, dural laceration, CNS tumors. Some success reported in treating with deferiprone (Ferriprox®), a lipid-soluble iron chelator that crosses the BBB⁶
15. arachnoiditis ossificans: a rare condition involving calcification of the arachnoid membrane.⁷ In the T-spine, may occur as ossified plaques or in a cylindrical form surrounding the spinal cord. May be difficult to detect on MRI and myelography. Plain unenhanced CT may be optimal for diagnosis

► Neoplastic

1. spine/spinal cord tumors† (p. 979) (for details)
 - a) extradural (55%):
 - primary tumors (rare) include: neurofibromas, chordomas, osteoid osteoma, aneurysmal bone cyst, vertebral hemangioma⁸
 - if age > 40 yrs, suspect extradural lymphoma (primary or secondary) or leukemic deposits (chloroma), especially with pre-existing diagnosis of hematopoietic or lymphatic disorder
 - epidural metastases (p. 921) become increasingly common after age 50 yrs. Occurs in up to 10% of cancer patients. 5–10% of malignancies present initially with cord compression
 - b) intradural-extramedullary (40%): meningiomas, neurofibromas
 - c) intradural-intramedullary: primary cord tumors (p. 980) (ependymoma, astrocytoma) and, rarely, intramedullary mets
2. carcinomatous meningitis (p. 920): neurologic deficit usually cannot be localized to a single level
3. paraneoplastic syndrome (p. 569): including effects on spinal cord or on peripheral nerves

► Vascular

1. hematoma/hemorrhage
 - a) spinal epidural hematoma† (p. 1383): usually associated with anticoagulation therapy⁹
 - traumatic: following LP or epidural anesthesia (p. 1383)
 - spontaneous¹⁰: rare. Includes hemorrhage from spinal cord AVM (p. 1383) or from vertebral hemangioma (p. 1395)
 - b) spinal subarachnoid hemorrhage: as is the case with spinal epidural hematoma (p. 1383), this may also be posttraumatic (e.g., following LP^{11,12}) or secondary to spinal cord AVM
 - c) spinal subdural hematoma
 - d) hematomyelia
2. spinal cord infarction (p. 1385): uncommon. Usually involves the anterior spinal artery, sparing posterior columns. Usually ≈ T4 level (watershed zone).
 - a) atherosclerosis of radicular artery in elderly patient with hypotension
 - b) clamping aorta during surgery (e.g., for abdominal aortic aneurysm)
 - c) hypotension (relative or absolute) during surgery in the sitting position in the presence of spinal stenosis¹³
 - d) aortic dissection: especially thoracic spinal levels
 - e) vertebral artery dissection: especially cervical spinal levels
 - f) embolization of spinal arteries
3. spinal cord vascular malformations† (p. 1395): 10–20% present as sudden onset of myelopathy, usually in patients < 30 yrs,¹⁴ myelopathy may be secondary to:
 - a) mass effect from AVM: spinal AVMs account for < 5% of lesions presenting as cord “tumors”
 - b) rupture → SAH, hematomyelia, or epidural hematoma
 - c) watershed infarction due to “steal”
 - d) spontaneous thrombosis (necrotizing myelopathy of Foix-Alajouanine disease (p. 1398)¹⁵): presents as spastic → flaccid paraplegia, with ascending sensory level
4. radiation myelopathy: due to microvascular occlusion (p. 1901)

5. secondary to iodinated contrast material used for mesenteric or aortic angiography. Especially when angiogrammed in presence of hypotension, where cardiac output is shunted away from viscera and into spinal radicular arteries. Treatment: place patient sitting, remove ≥ 100 ml of CSF via LP and replace with equal amount of saline over 30 mins¹⁶

► Autoimmune

1. post-viral (or post-vaccination): may actually be etiology of autoimmune process (i.e., transverse myelitis). Viral prodrome present in $\approx 37\%$ of cases of ATM. Viral infection is usually most damaging to gray matter (e.g., poliomyelitis)

► Demyelinating

1. acute (idiopathic) transverse myelitis (ATM) (p. 195). Peak incidence during first 2 decades of life. Abrupt onset of LE weakness, sensory loss, back pain, and sphincter disturbance indistinguishable from spinal cord compression. Thoracic region most common. CT and myelogram are normal. MRI may demonstrate. CSF $\rightarrow$ pleocytosis and hyperproteinemia
2. multiple sclerosis (MS): diagnosed in only 7% of patients presenting as acute transverse myelopathy. Although more common in young adults, MS can occur at any time in life. Myelopathy of MS is usually insidious, and is usually incomplete (i.e., some sparing). Affects myelin, thus sparing gray matter. Abdominal cutaneous reflexes are almost always absent in MS
3. **neuromyelitis optica (NMO)** (AKA Devic syndrome): an idiopathic inflammatory, demyelinating process characterized by acute bilateral optic neuritis and transverse myelitis extending ≥ 3 vertebral levels¹⁷ (► Fig. 102.1). These processes evolve separately over a period of months to years. Brain lesions are uncommon but can involve the brainstem, thalamus, and corpus callosum.¹⁸ Callosal lesions are longitudinally oriented (unlike in MS where they are ovoid and horizontally oriented).¹⁹ Spinal cord edema may become so severe as to cause complete block on myelography.

NMO is no longer considered a variant of MS, supported by the discovery of circulating antibodies (NMO-IgG)²⁰ against aquaporin-4²¹ (the dominant water channel of the CNS primarily located on the foot processes of astrocytes²²). Mean age of onset: 41.1 years.²³ Strong female predominance. Prevalence in Caucasians is 0.4 per 100,000, and is higher in those of Asian or Indian descent.²⁴ Seropositivity is not required for the diagnosis, but is associated with a higher incidence of co-existing autoimmune disorder and more severe clinical attacks. Seronegative patients have a higher incidence of simultaneous optic neuritis and transverse myelitis with a shorter time to

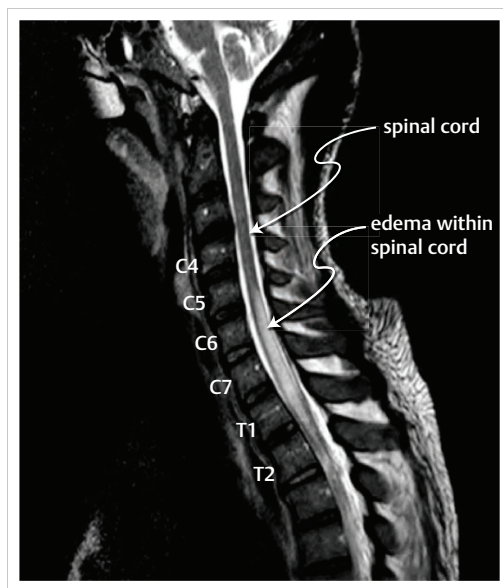


Fig. 102.1 Neuromyelitis optica. Image: T2 sagittal cervical MRI. Spinal cord edema extends from C5 to T2.

diagnosis.²⁵ Unlike MS, only partial recovery occurs after acute attacks. Relapse is common: 60% after 1 year, 90% after 3 years.²⁶
Acute treatment includes high-dose IV methylprednisolone or rescue plasma exchange (7 total exchanges)

► Metabolic/toxic

1. (subacute) combined system disease (CSD) (AKA subacute combined columnar degeneration): due to vitamin B₁₂ (cyanocobalamin) deficiency
Vitamin B₁₂ is a water-soluble vitamin found in meats and dairy products. Its absorption in the distal duodenum is facilitated by intrinsic factor, a small polypeptide secreted by gastric parietal cells.²⁷ B₁₂ is involved in neurologic function, production of RBCs, DNA synthesis, conversion of methylmalonic acid to succinyl coenzyme A, and conversion of homocysteine to methionine²⁸
 - a) etiologies of vitamin B₁₂ deficiency:
 - dietary deficiency of B₁₂: non meat-eaters may have insufficient intake.
 - chronic autoimmune gastritis: reduces available intrinsic factor which leads to intestinal malabsorption of B₁₂ in the distal ileum
 - other gastric disorders: low gastric pH, e.g., in Zollinger-Ellison syndrome, can inhibit attachment of intrinsic factor to ileal receptors
 - chronic use of metformin (> 4 months), proton pump inhibitors (> 12 months), or histamine H₂ blockers (> 12 months)
 - b) clinical manifestations of B₁₂ deficiency include:
 - pernicious anemia: a macrocytic megaloblastic anemia
 - combined system disease (CSD): myelopathy involving primarily the posterior columns. Onset is gradual and uniform. Begins with symmetrical paresthesias in feet or hands (posterior column involvement) → leg stiffness, weakness, and proprioceptive deficits with unsteadiness that is worse in the dark → spasticity → paraplegia → bowel and bladder dysfunction. Dementia (confusion, memory impairment, irritability...) occurs in advanced cases due to cerebral white matter changes. Visual disturbances with or without optic atrophy may be due to optic nerve demyelination
 - c) labs:
 - CBC: most (but not all) patients will have a *macrocytic* (megalocytic) anemia (folic acid deficiency also produces megaloblastic anemia. Folic acid corrects the anemia, even with CSD, but *not* the neurologic deficits which may actually worsen)
 - serum vitamin B₁₂ (cobalamin): levels < 150 pg/ml are diagnostic. However, normal B₁₂ levels do not R/O B₁₂ deficiency²⁸
 - if B₁₂ levels are low, or if they are normal but the patient has macrocytic anemia or neurologic symptoms, then check for elevated serum methylmalonic acid (also check homocysteine to R/O folate deficiency)
 - **Schilling test**: determines the cause of the B₁₂ deficiency even if B₁₂ injections have already been given (radiolabeled cyanocobalamin is given orally, followed by a parenteral flushing dose of nonradioactive vitamin, and the percentage of radioactivity is measured in the urine over 24 hours, performed once without and then once with added intrinsic factor, and then once following antibiotic therapy)
 - d) imaging: T2WI MRI may demonstrate increased signal within the white matter of the spinal cord, predominantly in the posterior columns but may also be seen in spinothalamic tracts
 - e) treatment: B₁₂ injections q 1–3 months or large doses of oral preparations²⁹ (other transport systems independent of intrinsic factor result in absorption of ≈ 1% of orally administered B₁₂, doses of 300–100,000 mcg result in absorption of more than the daily requirement of 1–2.5 mcg)
 2. toxins: e.g., local anesthetics used for spinal anesthesia rarely cause myelopathy
- #### ► Infectious
1. (para) spinal abscess, AKA **spinal epidural abscess** or epidural empyema (p.381) †: often history of *Staphylococcus* infection, usually a skin furuncle. Vertebral osteomyelitis often accompanies.³⁰ Produces local tenderness, back pain, fever, elevated ESR
 2. vertebral osteitis/osteomyelitis (p.386) †
 3. pyogenic discitis†: spontaneous or following procedures (p.390)
 4. HIV- or AIDS-related myelopathy: similar to B₁₂ deficiency. Spastic weakness & ataxia. Can cause vacuolization of spinal cord. “Tropical (spastic) paraparesis of AIDS” also seen in HTLV-I infection³¹
 5. tuberculosis: Pott’s disease, see Tuberculous vertebral osteomyelitis (p.387)
 6. spinal meningitis with pachymeningitis
 7. viral:
 - a) herpes varicella-zoster: rarely causes necrotizing myelopathy

- b) herpes simplex type 2: may cause ascending myelitis
- c) cytomegalovirus: may cause transverse myelitis
- 8. syphilitic involvement: may cause tabes dorsalis, syphilitic meningomyelitis, or spinal vascular syphilis. Diagnosed by serum and CSF serology
- 9. parasitic cysts†
- 10. some forms of Creutzfeldt-Jakob disease (CJD) (p. 399) with predominant initial muscle wasting may mimic spinal cord disease or ALS

► Peripheral neuromuscular disorder

- 1. Guillain-Barré syndrome (GBS) (p. 193): rapidly ascending weakness (mimics cord compression) with areflexia and near-normal sensation
- 2. chronic dysimmune neuropathies: presumed to be immune-mediated³²
 - a) chronic inflammatory demyelinating polyneuropathy (CIDP) (p. 194): similar to GBS but can progress over a longer period of time
 - b) multifocal motor neuropathy (MMN): a rare almost purely motor neuropathy affecting only lower motor neurons (unlike ALS which also affects upper motor neurons) characterized by asymmetric muscle wasting, cramping & LE twitching. May mimic ALS, but is treatable (with IVIg or immunosuppression)
- 3. myopathies: including steroid myopathy (usually affects proximal > distal muscles)

► Motor neuron diseases

- 1. amyotrophic lateral sclerosis (ALS) (p. 191): upper and lower motor neuron disease. *Slight* spasticity of LEs (extreme spasticity is rare), atrophic weakness of the hands and forearms, fasciculations in the UE, absence of sensory changes (including lack of pain), sphincter control usually preserved
- 2. primary lateral sclerosis: age > 50. No LMN signs. Slower progression than ALS (years to decades). Pseudobulbar palsy (p. 186) is common³³

† Items with dagger may also present as a spinal epidural mass.

102.3 Sciatica

102.3.1 General information

Definition: pain in the distribution of the sciatic nerve.

Anatomy: the sciatic nerve is comprised of components of nerve roots of L4–S3 containing fibers from the anterior and posterior divisions of the lumbosacral plexus. The nerve passes out of the pelvis through the greater sciatic foramen along the back of the thigh. In the lower third of the thigh it divides into the tibial and common peroneal nerves.

102.3.2 Etiologies

The most common cause of sciatica is *radiculopathy* due to a herniated lumbar disc.³⁴ The differential diagnosis is similar to that for myelopathy (see above) but also includes:

- 1. congenital:
 - a) meningeal cyst (perineural cyst); see Spinal meningeal cysts (p. 1400)
 - b) conjoined nerve root (p. 293): initially dismissed as a possible cause of radiculopathy, but current thinking recognizes that these may be symptomatic possibly by tethering
- 2. acquired:
 - a) spinal stenosis/spondylosis/spondylolysis/spondylolisthesis
 - b) juxtafacet cyst: includes synovial cyst and ganglion cyst (p. 1340)³⁵: detection is increasing with the use of MRI
 - c) nerve root sheath cyst: may be congenital or acquired. May arise near axilla of nerve root and cause compression of adjacent roots. Treatment: excise cyst and oversew the ostium
 - d) arachnoiditis ossificans (p. 1697): rare. In the lumbar region may occur as columnar, cylindrical, or irregularly shaped masses.³⁶ May produce low back pain, radiculopathy, or cauda equina syndrome
 - e) heterotopic ossification around the hip³⁷
 - f) injection injuries from misplaced IM injections
 - g) compartment syndrome of the posterior thigh
 - h) injury complicating total hip arthroplasty³⁸
 - i) radiation injury following treatment of nearby tumors

3. infectious:
 - a) discitis (p. 390): usually causes excruciating pain with any movement
 - b) Lyme disease (p. 364)
 - c) herpes zoster: a rare cause of radiculopathy.³⁹ Lumbosacral dermatomes are involved in ≈ 10–15% of zoster cases. Pain is usually independent of position. Typical herpetic skin lesions usually follow onset of pain by 3–5 days. 1–5% develop motor weakness (usually in arms or trunk). Sacral zoster can cause detrusor paralysis, producing urinary retention. 55% of those with motor symptoms have good recovery, 30% have fair to good recovery
4. neoplastic:
 - a) spine tumors: multiple myeloma (p. 928), metastases (p. 928)
 - b) bone or soft-tissue tumors along the course of the sciatic nerve: may result in erroneous laminectomy for herniated lumbar disc.⁴⁰ Pain is usually *insidious* in onset, and *not positional* (see below)
 - intra-abdominal or pelvic neoplasm
 - tumors of the thigh
 - tumors in the popliteal fossa or calf
5. inflammatory:
 - a) trochanteric bursitis (p. 1333): may produce pseudoradiculopathy. Rarely extends to the posterior thigh or as far distally as the knee
 - b) myositis ossificans of the biceps femoris muscle⁴¹
6. vascular:
 - a) sciatica may be mimicked by intermittent (i.e., vascular) claudication
 - b) psoas hematoma: usually in patient on anticoagulant. Sometimes drainage is required
7. referred pain of nonspinal origin: not dermatomal. Nerve root tension signs (p. 1333) are usually negative. Includes:
 - a) pyelonephritis
 - b) renolithiasis, including ureteral obstruction
 - c) cholecystitis
 - d) appendicitis
 - e) endometritis/endometriosis
 - f) posterior perforating duodenal ulcer
 - g) inguinal hernia, especially if incarcerated
 - h) aortic dissection (p. 1705)
8. piriformis syndrome (PS): controversial. Piriformis muscle originates on anterior S2–4 VBs, sacrotuberous ligament and passes through the greater sciatic notch to attach to the greater trochanter of the femur. It is innervated by L5–1. It is the principal *external* rotator of the extended hip. It may irritate or compress the sciatic nerve (AKA pseudosciatica, can mimic symptoms of a herniated disc). The superior gluteal nerve is spared as it has a take-off proximal to the muscle. Conversely, PS may occur secondary to lower lumbar radiculopathy. Produces pain in the sciatic distribution and weakness of external rotation and abduction of the hip. Signs: Freiberg test (pain with forced internal rotation of the hip with thigh extension) or the Pace test (pain on resisted abduction/external rotation of the hip). No well-designed studies of treatments. Advocated therapies include: PT, stretching, injection of the muscle localized by digital rectal exam taking care not to inject the sciatic nerve itself & piriformis muscle section. Sometimes long-lasting relief can follow injection with local anesthetic. Use of botulinum toxin (Botox®) injections has been described
9. more peripheral involvement (i.e., neuropathy) that may be confused with radiculopathy. Including:
 - a) femoral neuropathy mistaken for L4 radiculopathy (see below)
 - b) proximal sacral plexus lesion mistaken for S1 radiculopathy (see below)
 - c) diabetic neuropathy (p. 572) including diabetic amyotrophy
 - d) tumors (see below)

102.3.3 Extraspinal tumors causing sciatica

★ Pain characteristics: pain is almost always insidious in onset.⁴⁰ It may be intermittent initially, but eventually all patients develop pain that is constant, progressive, and unaffected by position or rest.⁴⁰ Significant night pain is described in ≈ 80%.

Straight leg raising was positive in most, but in more than half the pain was localized to a specific point along the course of the nerve, distal to the sciatic notch.⁴⁰ Conservative treatment brings either no or only temporary relief.

Approximately 20% will have a previous history of tumor (usually neurofibromatosis or previous malignancy). Malignancies include⁴⁰: metastatic lesions, primary bone sarcomas (chondrosarcoma...), soft-tissue sarcomas (liposarcoma...). Benign tumors include: lipoma, neurofibroma, schwannoma, aneurysmal bone cyst of the sacrum, giant cell tumor of the sacrum, tenosynovial giant cell tumor.

In two-thirds of cases, a detailed medical history and physical exam allowed localization and even determined the nature (bone tumor vs. soft-tissue) of the lesion.⁴⁰ Radiographs that show the entire pelvis and the proximal femur will demonstrate almost all tumors in these locations.^{40,42}

102.3.4 Features differentiating radiculopathy in sciatica

General information

Sciatica may result from nerve root involvement within the spinal canal (e.g., with lumbar disc herniation). Clinically this produces a nerve root syndrome; see Nerve root syndromes (p. 1253). Spinal imaging studies (MRI, myelogram/CT) will usually detect nerve root compression here.

L4 involvement

Femoral neuropathy is often mistakenly identified as an L4 radiculopathy. Distinguishing features are shown in ► Table 102.2.

Table 102.2 Distinguishing femoral neuropathy from L4 radiculopathy

Feature	Femoral neuropathy	L4 radiculopathy
Sensory loss		
distribution (► Fig. 1.16)	anterior thigh	dermatome from ≈ knee to medial malleolus, spares anterior thigh
Muscle weakness		
iliopsoas	weak	normal
thigh adductors	normal (innervated by obturator nerve)	may be weak
quadriceps	weak	weak

L5 involvement

Peroneal nerve palsy may be mistaken for L5 radiculopathy (p. 1706).

S1 involvement

Outside the spinal canal, S1 can also be involved as it enters the sacral plexus, e.g., by a pelvic tumor. In plexus lesions, EMG will show sparing of the paraspinal muscles (nerves to paraspinal muscles exit in the region of the neural foramen) and the gluteus maximus and medius (superior and inferior gluteal nerves take off just distal to the paraspinal nerves).

102.4 Acute paraplegia or quadriplegia

102.4.1 General information

Entities causing spinal cord compression usually present as: paraplegia or -paresis (or quadriplegia/ paresis), urinary retention (may require bladder ultrasound or checking a post-void residual to detect), and impaired sensation below level of compression. May develop over hours or days. Reflexes may be hyper- or hypoactive. There may or may not be a Babinski sign. Excluding trauma, the most common cause is compression by tumor or bone.

102.4.2 Etiologies

Some overlap with myelopathy.

- in infancy (may produce "floppy infant syndrome")
 - spinal muscular atrophy (the most severe form is called Werdnig-Hoffmann disease and is usually fatal within months): a rare autosomal recessive congenital disease of childhood with

degeneration of anterior horn cells. Only rarely evident at birth (where it presents as a paucity of movement), produces weakness, areflexia, muscle and tongue fasciculations with normal sensation. Usually starts in proximal muscles and muscles of respiration. Severe cases progress over the first year or two to quadriplegia.

Disease progression was halted in 60% of cases associated with mutation of the SMN1 gene on chromosome 5 (which codes for survival motor neuron (SMN) protein) with intrathecal administration of Spinraza™ (nusinersen),⁴³ an orphan drug that costs \$125,000 U.S. per injection or \$750,000 per year.

- b) spinal cord injury during parturition: a rare sequela of breech delivery
- c) congenital myopathies: e.g., infantile acid maltase deficiency (Pompe disease)
- d) infantile botulism: ileus, hypotonia, weakness, mydriasis, *Clostridium botulinum* bacteria and toxin in feces
2. traumatic spinal cord injury
 - a) major trauma: diagnosis is usually evident
 - b) minor trauma: may cause cord injury in setting of spinal stenosis which may → central cord syndrome (p. 1132)
 - c) atlantoaxial dislocation: from major trauma or due to instability from tumor or rheumatoid arthritis
3. congenital
 - a) extradural spinal cord compression by bone secondary to cervical hemivertebra (symptoms not present at birth, may develop decades later, occasionally after minor trauma)
 - b) cervical stenosis (p. 1301) (usually with superimposed spondylosis): quadriplegia or central cord syndrome may follow minor trauma
 - c) achondroplasia: spinal stenosis (animal model: dachshund)
 - d) syringomyelia: usually presents with central cord syndrome
4. metabolic
 - a) combined system disease* (p. 1699)
 - b) thallium poisoning: usually causes sensory and autonomic symptoms, quadriplegia and dysarthria may be seen in severe cases
 - c) central pontine myelinolysis (p. 119)
5. infectious
 - a) epidural spinal infection (abscess or empyema)*
 - b) post-viral (or post-vaccination): may be a transverse myelitis*
6. peripheral neuromuscular disorder*
 - a) Guillain-Barré syndrome (p. 194): classically an *ascending* paralysis, but paraparesis mimicking a spinal cord lesion is an unusual variant⁴⁴
 - b) myopathies
7. neoplastic*: spinal cord tumors
8. autoimmune*
9. vascular
 - a) acute pontomedullary infarction: age usually > 50 yrs. Patient is quadriplegic, alert, with bulbar palsies (eye movement abnormalities, impaired gag and speech)
 - b) spinal cord infarction*: including AVM, radiation myelopathy...
10. miscellaneous compressive*: including epidural hematoma, bony compression, epidural lipomatosis
11. functional: hysteria, malingering
12. bilateral cerebral hemisphere lesion (involving both motor strips): e.g., post-cerebral irradiation or parasagittal lesion. Will not have sensory *level*

* For items with asterisk, see Myelopathy (p. 1696) for details.

102.5 Hemiparesis or hemiplegia

102.5.1 General information

May be produced by anything that interrupts the corticospinal tract from its origin in the pyramidal cells of Betz in the motor strip (p. 56) down to the cervical spine. This results in upper motor neuron paralysis (see ► Table 30.4), which should also produce long tract findings, including Babinski sign ipsilateral to hemiplegia.

102.5.2 Etiologies

► **Lesions of the cerebral hemisphere in the region of the contralateral motor strip.** Large lesions may also involve the sensory cortex producing reduced sensation ipsilateral to the hemiparesis

1. tumor (neoplasm): primary or metastatic
2. traumatic: epidural or subdural hematoma, hemorrhagic contusion of the brain, compression by depressed skull fracture
3. vascular:
 - a) infarction
 - ischemic: embolic, low flow (due to atherosclerosis, arterial dissection...)
 - hemorrhagic: intracerebral hemorrhage, aneurysmal SAH...
 - b) TIA (p. 1536)
4. infection: cerebritis, abscess

► **Lesions of the contralateral internal capsule.** Produces pure motor hemiplegia without sensory loss. Most common etiology is ischemic lacunar infarct

► **Lesions of the brainstem.** Ischemic infarct, hemorrhage, tumor

► **Lesions of the cervicomedullary junction.** Foramen magnum lesions (p. 1649)

► **Unilateral spinal cord lesions.** Above $\approx$ C5 ipsilateral to the weakness producing a Brown-Séquard syndrome (p. 1135) with contralateral sensory loss to pain and temperature. See etiologies (p. 1135)

► **Hypoglycemia.** Can sometimes be associated with hemiparesis that clears after administration of glucose

► **Note.** In a patient with unexplained hemiparesis/hemiplegia, especially after trauma, consider carotid dissection

102.6 Ascending paralysis

1. Guillain-Barré syndrome (p. 193)
2. tick-borne paralysis: usually asymmetrical
3. spinal infection: e.g., spinal epidural abscess (p. 381)
4. in a patient with prior spinal cord injury below
 - a) early after spinal cord injury (≤ 4 weeks): ascending hematomyelia (blood within spinal cord), spinal cord edema. An argument for surgical decompression and rigid fusion early after spinal cord injury was to reduce the risk of acute ascending myelopathy⁴⁵
 - b) delayed after spinal cord injury: posttraumatic syringomyelia (p. 1411), arachnoiditis

102.7 Descending paralysis

1. botulism: key features in addition to descending paralysis are oculomotor weakness (ophthalmoplegia), miosis (pupillary constriction), and constipation
2. myasthenia gravis: can also produce ophthalmoplegia

102.8 Low back pain

102.8.1 General information

The following considers primarily low back pain (LBP) *without* radiculopathy or myelopathy, although some overlap occurs. Trauma is usually obvious and is not discussed here. See Sciatica (p. 1700) for differential diagnosis of that condition and also Low back pain and radiculopathy (p. 1226) for evaluation.

102.8.2 Acute low back pain

Similar to list for myelopathy (p. 1696). Most cases are non-specific (e.g., lumbosacral sprain), only 10–20% can be given a precise pathoanatomical diagnosis⁴⁶:

► **Patients writhing in pain.** Should be evaluated for an intra-abdominal or vascular condition (e.g., pain of aortic dissection is typically described as a “tearing” pain): patients with neurogenic LBP tend to remain as still as possible, possibly needing to change positions at intervals

► **Unrelenting pain at rest:**

1. spinal tumor (intradural or extradural) (p. 1235)
 - a) primary or metastatic spine tumor: suspected in patients with pain duration > 1 month, unrelieved by bed rest, failure to improve with conservative therapy, unexplained weight loss, age > 50 yrs⁴⁷
 - b) nocturnal back pain relieved by aspirin is suggestive of osteoid osteoma or benign osteoblastoma (p. 990)⁴⁸
2. infection (especially in IV drug abusers, diabetics, post-spinal surgery, immunosuppressed patients, or those with pyelonephritis or UTI post-GU surgery). Fever is somewhat insensitive for spinal infections. Spine tenderness to percussion has 86% sensitivity with bacterial infections, but a low specificity of 60%.⁴⁷ Types of infections include:
 - a) discitis
 - b) spinal epidural abscess: should be considered in patients with back pain, fever, spine tenderness, or skin infection (furuncle)
 - c) vertebral osteomyelitis
3. inflammatory
4. sacroiliitis: may produce pain and tenderness over one or both SI joints. Pelvic X-rays may show sclerosis of one or both sacroiliac joints.
 - a) bilateral & symmetric
 - ankylosing spondylitis (p. 1365): morning back stiffness, no relief at rest, improvement with exercise.⁴⁹ Usually seen in males with symptom onset before age 40 yrs. Positive Patrick’s test (p. 1252) and pain on compressing the pelvis with the patient in the lateral decubitus position
 - Reiter syndrome (after Hans Reiter, a German bacteriologist): a reactive arthritis (usually 1–3 weeks following certain bacterial infections) with involvement of at least one other non-joint area (urethritis, uveitis/conjunctivitis, skin lesions, mucosal ulcerations...). 75% are HLA-B27–positive
 - may occur in Crohn’s disease
 - b) bilateral & asymmetric
 - psoriatic arthritis
 - rheumatoid arthritis: adult & juvenile forms
 - c) unilateral
 - gout
 - osteoarthritis
 - infection

► **Evolving neurologic deficit.** (**Cauda equina syndrome:** perineal anesthesia, urinary incontinence or urgency or retention, progressive weakness) all require emergent diagnostic evaluation to rule out treatable conditions such as:

1. spinal epidural abscess (p. 381)
2. spinal epidural hematoma (p. 1383)
3. spinal tumor (intradural or extradural) (p. 979)
4. massive central disc herniation (p. 1254)

► **Pathologic fracture.** Acute pain in patients at risk for osteoporosis or with known Ca should prompt evaluation for pathologic fractures

1. lumbar compression fracture: see Osteoporotic spine fractures (p. 1208)
2. sacral insufficiency fracture⁵⁰: especially in rheumatoid arthritis patients on chronic steroids, often with no antecedent history of trauma. May cause back pain and/or radiculopathy. Often missed on plain films, best seen on CT, but may also be detected on bone scan

► **Coccydynia** (p. 1241). Pain and tenderness around the coccyx

► **Tears in the anulus fibrosus.** (“Anular tears”)⁵¹ (NB: also present in 40% of asymptomatic patients between 50 and 60 yrs of age, and 75% between 60 and 70 yrs⁵²)

► **Rarely following subarachnoid hemorrhage.** (SAH) due to irritation of lumbar nerve roots and dura: usually accompanied by other signs of SAH (p. 1417)

► **Myalgia.** May be a side effect of “statins” (drugs used to lower serum concentration of LDL cholesterol) with or without elevation of serum creatinine phosphokinase, sometimes with accompanying weakness and rarely with severe rhabdomyolysis and myoglobinuria leading to renal failure (risk may be increased with renal or hepatic dysfunction, advanced age, hypothyroidism, or serious infection)⁵³

► **Drug-induced:**

1. statins: see above under Myalgia
2. phosphodiesterase type 5 (PDE5) inhibitors used for erectile dysfunction: all may be associated with LBP, but the incidence is higher with tadalafil,⁵⁴ etiology unknown. Usually occurs 12–24 hours post-dose and resolves by 48 hours. Most respond to simple analgesics

102.8.3 Subacute low back pain

10% of patients with LBP have symptoms that persist > 6 weeks.

Differential diagnosis includes causes of acute LBP (above) and also:

1. continued pain at rest should prompt evaluation for spinal osteomyelitis (especially with fever and elevated ESR) or neoplasm if not already done
2. plain spine X-rays may show possibly causative conditions, although many or all of the following may also be seen in *asymptomatic* patients:
 - a) spondylolisthesis (p. 1337)
 - b) spinal osteophytes
 - c) lumbar stenosis
 - d) **Schmorl's node or nodule** (p. 1266): disc herniation through cartilaginous endplate into vertebral body (NB: may also be seen in 19% of asymptomatic patients⁵⁵)

Chronic low back pain

After 3 months, only ~ 5% of patients with LBP will continue to have persistent symptoms. A structural diagnosis is possible in only ~ 50% of these patients. These patients account for 85% of the cost in lost work and compensation.⁴⁶

Differential diagnosis includes causes of acute and subacute LBP listed above, as well as:

1. degenerative conditions
 - a) degenerative spondylolisthesis (p. 1340)
 - b) spinal stenosis (affecting the spinal canal)
 - c) lateral recess syndrome
2. spondyloarthropathies
 - a) ankylosing spondylitis: look for erosive changes adjacent to SI joint and positive test for HLA-B27 antigen
 - b) Paget's disease of the spine: vertebral involvement is very common in patients with Paget's disease
3. osteitis condensans ilii: increased density in ilium, usually asymptomatic (incidental) finding. Occasionally may produce low back pain or tenderness. Usually found in women who have been pregnant
4. psychological overlay: including secondary gain (financial, emotional...)

102.9 Foot drop

102.9.1 General information

Key concepts

- weak anterior tibialis (foot extension) innervated by deep peroneal nerve (L4, 5)
- most common etiologies: L4/L5 radiculopathy, common peroneal nerve palsy
- in a patient with foot drop, check posterior tibialis (foot inversion) and gluteus medius (internal rotation of flexed hip)—both are spared in peroneal nerve palsy and both should be involved with L4/L5 radiculopathy
- EMG can assist in localization and prognostication

Definition: weakness of anterior tibialis (primarily L4 and to a lesser extent L5), often accompanied by a weak extensor digitorum longus and extensor hallucis longus (primarily L5 with some S1 contribution), all of which are innervated by the *deep peroneal nerve*.

102.9.2 Underlying substrates of foot drop

The most common dilemma is to distinguish foot drop due to radiculopathy from that due to peroneal nerve palsy (usually common peroneal nerve). With common peroneal nerve (CPN) palsy, there is *sparing* of posterior tibialis (foot inversion, innervated by posterior tibial nerve) and gluteus medius (internal rotation of the thigh with the hip flexed, innervated by superior gluteal nerve, primarily L5 with some L4, the takeoff is shortly after the roots exit from neural foramen). With L4 or L5 root lesions these muscles will also be weak; see ► Fig. 102.2.

Flail foot results from paralysis of dorsiflexors *plus* plantarflexors, e.g., in sciatic nerve dysfunction as can occur during surgery for hip fracture-dislocation⁵⁶ or injection injuries (IM injections should be given superiorly and laterally to a line drawn between the posterior superior iliac spine and the greater trochanter of the hip). NB: the peroneal division of the sciatic nerve tends to be more vulnerable to injury than the tibial division.

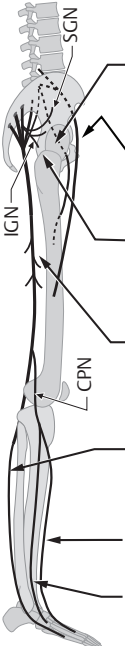
	Muscle to check	Rationale	Finding: exam
	Thigh adductors (adductors longus, brevis, and magnus) (L2, L3): innervated by the obturator nerve	Involvement indicates that the lesion includes more than sciatic nerve/L5 root (e.g., paravertebral mass, cauda equina lesion if bilateral findings)	Weakness when adducting thigh while supine with knee extended
	Quadriceps femoris innervated by the femoral nerve	(Same as above)	Weakness of knee extension (L2, L3, L4)
	Muscles innervated by L5 branches that exit the lumbar plexus very close to the neural foramina	Involvement indicates very proximal lesion (e.g., nerve root or very proximal (paravertebral) lumbar plexus) ^a	Weakness of: 1. gluteus medius (L4, L5, S1): internally rotate thigh 2. gluteus maximus (L5, S1, 2): dig heel into bed while supine
	L5 innervated muscles (via sciatic nerve) proximal to the take off of the common peroneal nerve (CPN) ^b	If muscles listed above are intact, involvement of muscles to the right localizes the lesion to sciatic nerve above mid thigh (e.g., injury to sciatic nerve at the greater sciatic notch)	1. Slight weakness of biceps femoris (lateral hamstrings) (L5, S1, 2): flex the knee (with thigh flexed) 2. Gastrocnemius weakness (foot plantarflexion) unless injury only to peroneal division of sciatic nerve ^b
	Muscles innervated by (posterior) tibial nerve	Sparing of these with foot drop indicates lesion distal to take off of common peroneal nerve (foot drop with weak foot inversion may be L4 or 5 radiculopathy)	Weakness of tibialis posterior (L4, L5): invert foot (foot should be plantarflexed to eliminate anterior tibialis)
	Anterior tibialis innervated by deep peroneal nerve	Involvement does not narrow down etiology until other muscles are examined	Weakness of ankle dorsiflexion (foot drop)
	Muscles innervated by superficial peroneal nerve	Preservation of these with foot drop localizes the lesion to the deep peroneal nerve	Weakness of the peroneus longus and brevis (L5, S1): evert the foot
Abbreviations: CPN = common peroneal nerve ICN = inferior gluteal nerve SGN = superior gluteal nerve		^a Note: EMG can differentiate root lesion from plexus by detecting involvement of paraspinal muscles, which occurs in root but not in plexus lesions because dorsal rami exit proximal to plexus ^b The peroneal division of the sciatic nerve is more vulnerable to injury than the tibial division for several reasons. It is thus not unusual to see isolated peroneal nerve injuries above the knee, e.g. from hip dislocation or fractures, stab wounds, injection injuries...	

Fig. 102.2 Physical exam to localize the lesion in a patient with LE weakness.

Table 102.3 Localization of lesion with foot drop

Lesion	Motor deficit ^a					Sensory changes
	anterior tibialis (L4, 5 ankle dorsiflexion)	peroneus longus/brevis (L5, S1 foot eversion)	tibialis posterior (L4, 5 foot inversion)	biceps femoris (L5, S1, 2 knee flexion)	gastrocnemius (S1, 2 plantar-flexion)	
deep peroneal nerve	x					minimal, or great toe web space
superficial peroneal nerve		x				lateral distal leg and dorsum of foot
common peroneal nerve (CPN)	x	x				all of the above
L4 or L5 radiculopathy	x	x	x			dermatomal (► Fig. 1.16)
peroneal division of sciatic nerve ^b	x	x	x	x		as with common peroneal
main trunk of sciatic nerve	x	x	x	x	x	lateral distal leg and entire foot

^ax denotes that the indicated muscle is involved (i.e., weak)

^bsee footnote (b) under ► Fig. 102.2

102.9.3 Etiologies of foot drop

Three major categories: 1) muscular, 2) neurologic, 3) anatomic.

1. peripheral nerve palsies (more common). See ► Table 102.3 and ► Fig. 102.2.

a) peroneal nerve injury (also, see Common peroneal nerve palsy (p.563) for details including etiologies). Branches that may be involved:

- deep peroneal nerve: isolated foot drop with minimal sensory loss (except possibly in great toe web space)
- superficial peroneal nerve: weakness of peroneus longus and brevis (foot eversion) with *no* foot drop. Sensory loss: lateral aspect of lower half of leg and foot
- common peroneal nerve: combination of above, i.e., foot drop + weak foot eversion, with sparing of tibialis posterior (foot inversion). Sensory loss: lateral aspect of lower half of leg and foot

b) L5 radiculopathy (or, less commonly, L4). The most common cause is HLD at L4–5, other etiologies include: lumbar spinal stenosis at L4–5, sacral ala fracture (p.1215)

- results in pain and/or sensory changes in L5 (or L4) dermatome
- weakness with radiculopathy tends to be more pronounced in distal muscles (e.g., anterior tibialis) than in proximal (e.g., gluteus maximus)
- *painless* foot drop is unlikely to be due to radiculopathy; consider peroneal neuropathy, diabetic neuropathy, lesion anywhere along pyramidal tract, motor neuron disease...

c) lumbar plexus injury

d) lumbosacral plexus neuropathy (p.571)

e) injury to lateral trunk of sciatic nerve

f) peripheral neuropathy: weakness tends to be greater distally, producing wrist or foot drop. Classic example: Charcot-Marie-Tooth (p.568) (CMT), findings tend to be rather dramatic in spite of the fact that it often doesn't seem to bother the patient very much

g) early in the course of motor neuron disease (ALS)

h) heavy metal poisoning

2. central nervous system causes (foot drop here is usually painless)

a) cortical lesion (UMN): parasagittal lesions in region of motor strip (see ► Fig. 1.3) (sensation will be spared if the lesion does not extend posteriorly to the sensory cortex).⁵⁷ There may be a Babinski sign or hyperactive Achilles reflex (so-called "spastic foot drop"). Usually painless

b) spinal cord injury: including cervical spinal myelopathy

3. non-neurogenic causes

a) muscular dystrophy

b) lead toxicity: in children may cause foot drop with no sensory loss

c) anterior compartment syndrome

102.9.4 Clinical

Loss of dorsiflexion causes foot slap with the front of the foot when the heel strikes the ground while walking. Also during the swing phase of gait the front of the foot may snag the ground (especially on uneven surfaces) which may cause tripping; thus patients develop steppage gait (exaggerated thigh & knee flexion) on the affected side. Associated weakness of tibialis posterior, when present (e.g., with L5 radiculopathy) destabilizes the ankle, permitting eversion which also predisposes to falls and to ankle fractures. Chronic foot drop may produce Achilles tendon contracture with talipes equinus.

Wasting of the extensor digitorum brevis may be seen.

102.9.5 Evaluation

1. bloodwork: glucose, ESR
2. EMG: can help differentiate L5 radiculopathy from peroneal nerve palsy, plexus lesion (► Fig. 102.2), or motor neuron disease (p. 191) (for details). EMG is not reliable until symptoms have been present at least ≈ 3 weeks
3. For suspected radiculopathy: MRI (or CT/myelogram if MRI is not possible)

102.10 Weakness/atrophy of the hands/UEs

102.10.1 Hand/UE weakness or atrophy with relatively preserved function in the LEs

1. cervical spondylosis (p. 1296): often causes sensory disturbance
2. cervical radiculopathy (p. 1280)
3. amyotrophic lateral sclerosis (ALS): no sensory involvement. One of the few causes of clinically prominent *fasciculations*. See details of ALS (p. 1700), other distinguishing features (p. 1700), fibrillations (p. 531)
4. spinal cord pathology
 - a) central cord syndrome (p. 1132): typically causes more impairment (weakness, sensory disturbance) in the UE than the LE
 - b) syringomyelia (p. 1405): usually burning dysesthesias of the hands with dissociated sensory loss
5. brachial plexus injury (p. 579)
6. brachial plexus neuropathy (p. 570) (includes Parsonage-Turner syndrome (p. 570))
7. peripheral nerve problems, including:
 - a) carpal tunnel syndrome (p. 546)
 - b) ulnar neuropathy (p. 553)
 - c) other peripheral nerve entrapment syndromes (p. 541)
8. foramen magnum lesions (p. 1649): can cause (Bell's) cruciate paralysis⁵⁸ due to compression above the pyramidal decussation (located in the inferior medulla oblongata), which produces bilateral UE weakness and possibly atrophy of the hands with sparing of the LEs⁵⁹ (in the differential diagnosis for central cord syndrome). Compression on only one side may produce the similarly named but clinically different hemiplegia cruciata (spastic palsy of one UE and the contralateral LE)⁵⁹
9. thoracic outlet syndrome (p. 581)
10. botulism (p. 1687)
11. Hirayama disease (p. 1403): due to asymmetric compression of the spinal cord by the dura in flexion
12. pharyngeal-cervical-brachial variant of Guillain-Barré syndrome (p. 194)

102.10.2 Atrophy of the first dorsal interosseous muscle

Etiologies: either C8/T1 nerve root or ulnar nerve involvement (may be either focal or diffuse). There are 4 main differential diagnoses:

1. ulnar neuropathy: check median nerve to see if findings extend to a nearby but separate nerve
 - a) at the elbow (p. 558)
 - b) at Guyon's canal (p. 558)
2. nerve root involvement:
 - a) cervical radiculopathy: C8 or T1

- b) nerve root avulsion (p.255): weakness + sensory loss with normal SNAP on EMG, usually with a history of precipitating trauma
- 3. lower brachial plexus involvement
 - a) thoracic outlet syndrome (p.581)
 - b) Pancoast tumor (p.570)
- 4. neurodegenerative disorders
 - a) amyotrophic lateral sclerosis (ALS) (p.1301) (AKA motor neuron disease)
 - b) multifocal motor neuropathy (MMN) (p.1700): a chronic dysimmune motor neuropathy with asymmetric muscle wasting, cramping, & LE twitching

102.11 Radiculopathy, upper extremity (cervical)

See Weakness/atrophy of the hands/UEs (p.1709). In addition to those items:

1. primary shoulder pathology: characteristically, pain is aggravated by active and/or passive shoulder movement. In general, shoulder pathology does not produce pain referred to the neck
 - a) rotator cuff tear
 - b) bicipital tendonitis: tenderness over biceps tendon
 - c) subacromial bursitis: there may be tenderness over the AC joint
 - d) adhesive capsulitis
 - e) impingement syndrome: the “empty can test” is usually positive (each arm held out in front, 30° lateral to straight forward, thumbs pointing down, as in emptying out a soda can. Examiner pushes down on the patient’s hands while the patient resists. Test is positive if it reproduces pain)
2. shoulder pain is very common in polymyalgia rheumatica (p.206), typically worsens with movement
3. interscapular pain: a common location for referred pain with cervical radiculopathy, may also occur with cholecystitis or some shoulder pathologies
4. MI: some cases of cervical radiculopathy (especially left C6) may present with symptoms that are suggestive of an acute myocardial infarction
5. complex regional pain syndrome AKA reflex sympathetic dystrophy (p.525): may be difficult to distinguish from cervical radiculopathy. Stellate ganglion blocks may help.⁶⁰

102.12 Neck pain (cervical pain)

This section deals primarily with axial neck-pain without radicular features. For radicular features, see Radiculopathy, upper extremity (cervical) above.

1. cervical spondylosis (including facet arthritis)
2. cervical sprain: including whiplash-associated disorder
3. fracture of the cervical spine: with upper cervical spine fractures (e.g., odontoid), patients characteristically hold their head in their hands, especially when going from recumbent to upright position
 - a) traumatic
 - b) pathologic (tumor invasion, rheumatoid arthritis)
4. occipital neuralgia (p.541)
5. herniated cervical disc:
 - a) lateral herniated disc: if symptomatic, tends to produce more radicular symptoms in the UE than actual neck pain
 - b) central disc herniation: if symptomatic, tends to produce myelopathy, does not produce any neck pain whatsoever in many cases
6. abnormalities of the craniocervical junction:
 - a) Chiari type 1 malformation (p.295)
 - b) atlantoaxial subluxation
7. **fibromyalgia**: idiopathic chronic pain syndrome characterized by widespread nonarticular musculoskeletal pain, nodularity, and stiffness^{61,62} without pathologic inflammation. Possible link to neuroendocrine dysfunction.⁶³ Afflicts 2% of the population.⁶² female:male ratio is 7:1. No diagnostic laboratory study. May be associated with psychiatric illness and multiple non-specific somatic complaints including malaise, fatigue, impaired sleep, GI complaints, and cognitive impairment
8. **Eagle syndrome** (first described by otolaryngologist Watt Weems Eagle in 1937): AKA stylohyoid syndrome. Elongation of the styloid process (defined as length > 3 cm⁶⁴) (► Fig. 102.3) and/or calcification of the stylohyoid ligament. Symptoms may be due to compression of adjacent nerves

(e.g., IX, V, chorda tympani, or sympathetic plexus on carotid sheath), or pain from fracture or inflammation.⁶⁴ Palpation over the styloid process may reproduce symptoms. Surgical resection can ameliorate the pain. Two variants:

- a) typical variant: history of tonsillectomy. Pharyngeal pain, dysphagia, and otalgia
- b) second variant: AKA carotid artery-styloid process syndrome. Carotidynia radiating into ipsilateral eye and vertex



Fig. 102.3 Eagle syndrome.

Image: CT bone window showing elongated left styloid process (yellow arrowheads). A: coronal, B: sagittal.

9. crystal deposition diseases: gout, pseudogout, hydroxyapatite (HA), or calcium pyrophosphate dihydrate (CPPD) crystal deposition diseases. May appear as a crown-like density encompassing the odontoid process (crowned-dens syndrome)⁶⁵ representing calcifications in the transverse ligament, seen best on cervical CT. May be treated with a short course of prednisolone (e.g., 15 mg/d) followed by NSAID

102.13 Burning hands/feet

1. spinal cord syndromes:
 - a) central cord syndrome (CCS) (p.1132)
 - b) burning hands syndrome (p.1126): a possible variant of CCS, described in football-related cervical spine injury
 - c) numb-clumsy hand syndrome (p.1299): seen in cervical myelopathy
2. complex regional pain syndrome (CRPS) (p.525) AKA reflex sympathetic dystrophy
3. peripheral neuropathy
 - a) diabetic amyotrophy (p.572) AKA Bruns-Garland syndrome
4. erythromelgia AKA erythromelalgia: a rare disorder characterized by erythema, edema, increased skin temperature, and burning pain of the hands and/or feet. Usually refractory to medical management, although some success is reported with epidural bupivacaine,⁶⁶ lidocaine patches,⁶⁷ or cold soaks
 - a) primary erythromelgia: etiology is idiopathic
 - b) secondary erythromelgia: associated with autoimmune and rheumatologic factors
5. vascular:
 - a) occlusive arterial disease: atherosclerosis, Raynaud's syndrome
 - b) venous insufficiency

102.14 Glove/stocking sensory disturbance

Sensory loss in a stocking distribution is more common than a glove or glove-stockings sensory loss.

Sensory level: there is a subjective and often an objective sensory level in a limb distal to which there is impairment. This differs from a *dermatomal* sensory disturbance which produces issues that follow one or more dermatomes (areas of sensory disturbance oriented along the long axis of the limb, as shown in the blue and white portions of ► Fig. 1.16) or a mononeuropathy (i.e., involvement of a single peripheral nerve) which produces patches of sensory loss corresponding to the distribution of a peripheral nerve (as shown in the colored portions of ► Fig. 1.16), neither of which is circumferential.

Symptoms may include any combination of: numbness, tingling, burning dysesthesias, hyperesthesia and aching sensation.

Etiologies

1. vascular insufficiency: this usually involves the lower extremities only
2. polyneuropathy: most cases start in the feet and gradually progress upwards
 - a) large fiber polyneuropathy: can be diagnosed with EMG
 - b) small fiber polyneuropathy: does not show up on an EMG. The small nerves involved do not produce imbalance or muscle weakness. May be associated with diabetes mellitus (the most common), autoimmune disorders (e.g., Sjögren or sarcoidosis), paraproteinemia, amyloid angiopathies, sodium channelopathies. It may present with somatic or autonomic symptoms.⁶⁸

If it is felt to be necessary, skin biopsy can aid in diagnosis

102.15 Muscle pain/tenderness

1. fibromyalgia: see above
2. myopathy
3. statin-induced myopathy: can vary from mild (with symptoms of muscle aches, symptoms usually abate rapidly after discontinuation of the statin, although occasionally up to 2 months may be required) to severe (with rhabdomyolysis which can → nephropathy)
4. diffuse severe sensitivity to light touch is often a marker of nonorganic pain⁶⁹

102.16 Lhermitte's sign

102.16.1 General information

Technically, a symptom (not a sign). Electrical shock-like sensation radiating down the spine usually provoked by neck flexion (shocks radiating up the spine are sometimes referred to as reverse Lhermitte's sign). Classically attributed to MS, but may occur in any pathology involving primarily the posterior columns of the spinal cord (see below).

102.16.2 Etiologies

1. multiple sclerosis (MS) (p. 187)
2. cervical spondylosis (cervical spondylotic myelopathy (p. 1297))
3. subacute combined degeneration (p. 1699): check for vitamin B₁₂ deficiency
4. cervical cord tumor
5. cervical disc herniation
6. radiation myelopathy (p. 1901)
7. Chiari type I malformation (p. 295)
8. central cord syndrome (p. 1132)
9. SCIWORA (spinal cord injury without radiographic abnormality) (p. 1196)

102.17 Swallowing difficulties

1. mechanical: the term globus describes a sensation of a lump in the throat. Odynophagia is pain on swallowing, which may occur without dysphagia
 - a) ossification of the anterior longitudinal ligament (OALL) (p. 1372)
 - b) diffuse idiopathic skeletal hyperostosis (DISH) (AKA Forestier disease) (p. 1373): An enthesopathy
 - c) post-op following ACDF
 - it is normal to have a little swelling and fullness early post-op
 - may be increased with multiple levels and with anterior plating
 - as a complication from post-op hematoma
2. neurologic

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Part XXIII

Procedures, Interventions, Operations

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103 Intraoperative Dyes, O.R. Equipment, Surgical Hemostasis, and Bone Extenders

103.1 Introduction

This section provides information useful in the O.R. that applies to a number of different topics. Some items that are pertinent to only one topic will be found in their respective sections instead (e.g., transsphenoidal tumor removal is found in the section on pituitary tumors).

REMEMBER: before performing any invasive procedure, know the patient's coagulation status (history of anticoagulant and/or antiplatelet function medications, and if indicated: PT, PTT, INR, platelet count and platelet function assay, FDP...).

103.2 Intraoperative dyes

This section covers visible dyes that may be useful in the O.R. For radio-opaque dyes, see Contrast agents in neuroradiology (p.230). There is little information available in the literature regarding the intrathecal (IT) use of the following agents:

Indigo carmine: a blue dye which has been used intrathecally to locate CSF leaks. There are few published reports, and no accounts of adverse effects. In 1933 it was reported¹ that IT injection of 5 ml of 0.6% indigo carmine solution produced blue-green discoloration of the CSF draining through a fistula into the nose within 15 minutes, lasting for 5 hours, with no indication of toxicity. It is excreted in the urine (and not in mucous membranes). The consensus is that it should be relatively safe for IT use, but the manufacturer did not recommend this application.

✖ **Methylene blue:** methylene blue is probably *cytotoxic* and appears to become fixed to neural tissue. It should therefore *not* be used as a stain in neurosurgical operations or diagnostic tests. CNS damage (some permanent) occurred in 14 patients given an IT injection of a 1% solution. Symptoms included: paraparesis, quadriplegia, multiple cranial nerve involvement (including anosmia and optic atrophy), dementia, and hydrocephalus.²

Fluorescein: although intrathecal injection (e.g., to look for CSF leak) has been used by ENT surgeons with apparently acceptable results, there is a risk of seizures. 2.5% fluorescein is diluted 1:10 with CSF or saline and ~ 6 ml is injected into the spinal subarachnoid space (or 0.5 ml of 5% fluorescein mixed with 5–10 ml of CSF³).

Fluorescein has also been used IV (adult dose: 1 amp IV) to help mark areas of the brain where there is breakdown of the blood-brain barrier (BBB), e.g., in tumors (p.628); however, fluorescein is eventually excreted in mucus, urine, etc., and just about everything turns orange. It has also been used to perform intraoperative "visible angiograms" during the removal of AVMs.

Indigocyanine green (ICG): used for intraoperative angiography (p.1466).

103.3 Operating room equipment

103.3.1 Operating microscope—observer's eyepiece

For spine cases, the ideal location of the observer eyepiece is usually directly opposite the surgeon. For intracranial work, the observer's (assistant's) eyepiece is placed to the *right* of the operator's, except in the following cases where it is placed to the left:

1. transsphenoidal surgery (when the surgeon stands to the patient's right)
2. *right* posterior fossa craniotomy in the lateral oblique (suboccipital) position

103.3.2 Head stabilization

General information

Options include:

1. non-pin head stabilization: when absolute head fixation is not indicated, the following options avoid some of the complications associated with pin-based head fixation (p.1717):
 - a) horse-shoe head rest
 - b) doughnut: gel type, or one can be fashioned using a stockinette
 - c) Prone-view® for prone position e.g., during posterior spine surgery

2. pin-based head-fixation: the Mayfield head-holder or head frame is the most common, devised by Dr. Frank Henderson Mayfield (1908–1991)

Pin-based head-fixation

Indications for pin stabilization:

1. ✖ not recommended for use in age < 3 years, use pediatric pins with care in ages 3–10 years (this age cutoff is not based on scientific evidence, most reported complications occur in age > 3 years)
2. craniotomies:
 - a) most intracranial vascular operations: a radiolucent head-holder should be used if intraoperative angiogram is to be performed
 - b) often for tumor operations, especially if there is need for a self-retaining retractor system that attaches to the Mayfield head-holder (e.g., Budde halo)
 - c) when intraoperative image-guidance (IG) systems are used (an alternative here is to use mask-based or strap-on registration array holders)
3. cervical spine: often used for posterior cervical operations (laminectomies, instrumentation, fusions...)

Application of pin head holder:

1. “skull block” (p. 1733) (blockade of scalp innervation) may be administered prior to pin placement. Critical for awake craniotomies (for the wake-up), and for vascular cases and cases with increased ICP where the block blunts the precipitous increase in blood pressure^{4,5} that may otherwise accompany pinning, which may → increased ICP. For technique, see Typical sequence of anesthesia (p. 1733).
2. plan pin placement
 - a) the manufacturer recommends that the pins be placed within a band-like area similar to a “sweatband” worn just above the orbits and the pinna
 - b) avoid placing pins on the thin temporal squamosa, and use with caution over the frontal sinus⁶
 - c) the single pin is typically placed anteriorly when the patient is in the supine position (► Fig. 106.1); for a posterior fossa approach in the prone position, if the craniotomy is on one side, the single pin is placed on that side (► Fig. 105.2)
 - d) most current head clamps have a dual pin “rocker” opposite the single pin, and this should be placed equidistant from the centerline for maximal stability
3. appropriately sized sterile pins (usually coated with antibiotic ointment) are placed in the head-holder (disposable pins have largely superseded reusable pins):
 - a) ✖ children < 3 yrs of age: increased risk of calvarial penetration or depressed skull fractures. A padded cerebellar head rest should be used instead^{6,7}
 - b) children ≥ 3 yrs of age and < 10 yrs of age: pediatric skull pins should be employed. These pins have a “shoulder” (► Fig. 103.1) to ensure shorter depth of skull penetration. NB: with a thick scalp these pins may not adequately contact the skull
4. NB: before doing the following, make sure that the single pin that has the spring loaded pressure adjustment is backed out enough so that there is room to tighten it after this step is completed). The ends of the Mayfield clamp are squeezed together, allowing the linear ratchet and pawl to slide, until the pins are initially seated in the skull
5. the knob housing the tension spring and gauge is tightened (each ring = 20 lbs)
 - a) adults: tighten until the third ring (60 lbs) is visible (► Fig. 103.1); up to 80 lbs has been described
 - b) pediatrics: less pressure should be applied, 30–40 lbs has been suggested.⁶ Even with pediatric pins & decreased pressure, complications may occur, consider horseshoe headrest

Complications of pin head-fixation:

1. malposition of pins:
 - a) through unintended anatomic structures: pinna, orbit, superficial temporal artery, shunt hardware, prior craniotomy defect
 - b) poor fixation by not properly placing pins close to “equator” resulting in movement during surgery (with risk of cervical spine injury, injury to structures being operated on due to the sudden movement, loss of image-guided registration) and possible skin laceration
2. skin penetration by pins: can cause injury to intracranial structures, or infection including delayed abscess, epidural hematoma⁶...
 - a) overtightening of pins

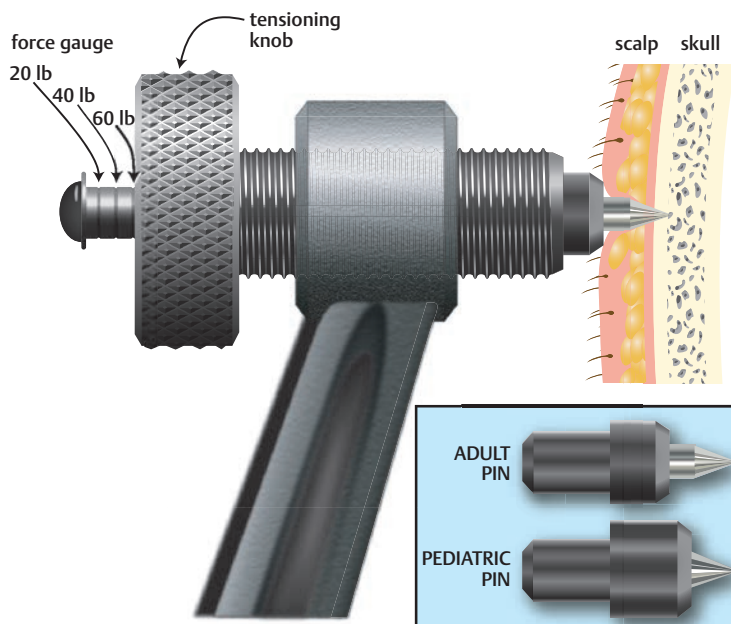


Fig. 103.1 Mayfield head-holder.

Image: detail of the tension knob of a Mayfield-head holder that has been tightened to 60 lbs of force (3 rings showing).

Inset shows disposable pins, highlighting the shorter exposed metal points of the “pediatric pin” (for ages approximately 3–10 years) compared to the adult pin.

- b) incorrect pin selection: see above
- c) soft skull: in elderly patients, poorly calcified skulls, pediatric patients⁶
- 3. skin necrosis: especially with pediatric pins due to the “shoulder” of the pins
- 4. skull fracture: including “ping-pong ball” fractures in young children
- 5. slippage of any of the joints or connections to the O.R. table
- 6. clamp breakage^{8,9}: inspect the head-holder for cracks prior to each application, store properly and maintain per the manufacturer’s specifications
- 7. bleeding from pin site: usually when the head-holder is removed. If bleeding does not stop after a minute or so of pressure, a suture or surgical staple may be used

103.4 Surgical hemostasis

103.4.1 Basic options

1. thermocoagulation
 - a) electrical coagulation:
 - monopolar (Bovie) cautery: electric current passes through the patient to a grounding pad. Because of possible transmission through electrically and thermally sensitive neural structures, this modality is not used directly on the brain or in proximity to named nerves (including cranial nerves) & nerve roots
 - bipolar cautery: current passes only between the tips of the cautery device. Used for precise coagulation. When used directly on or near to the brain or nerves, the current setting is typically reduced from that employed for general use

- b) thermal units: e.g., AccuTemp® disposable eye cautery units (particularly useful to coagulate dura when inserting a ventriculostomy in the ICU)
- c) laser: especially neodymium:yttrium-aluminum-garnet (Nd:YAG) laser
- 2. mechanical
 - a) bone wax: originated by Sir Victor Horsely. Stops bone bleeding. NB: *Inhibits bone formation* so its use is limited in spinal fusions
 - b) ligature: less commonly used in neurosurgery than other specialties
 - c) “silver clips” (e.g., HemoClips®)
- 3. chemical hemostasis: see below

103.4.2 Chemical hemostasis

See review¹⁰ for more information. Some key points:

1. gelatin sponge (Gelfoam®): no intrinsic coagulating effect. Absorbs 45 times its weight in blood which causes it to expand and tamponade bleeding. Absorbable. May be combined with thrombin as patties or as powder (e.g., FLOSEAL®, SurgiFlo®)
2. oxidized cellulose (Oxycel®) and oxidized regenerated cellulose (Surgicel®): absorbable. Acidic material that reacts with blood to form a reddish brown “pseudoclot.” Bactericidal to over 20 different organisms. May retard bone growth. Oxycel® interferes with epithelialization more than Surgicel®
3. microfibrillar collagen (Avitene®): promotes adhesion and aggregation of platelets. Loses effectiveness in severe thrombocytopenia (< 10,000/ml). May be used on bone bleeding. Remove excess material to reduce risk of infection
4. thrombin (Thrombostat®): does not depend on any intermediate physiological agent. Caution: although thrombin may cause significant edema when placed on brain where the pia has been disrupted, practical experience indicates this is uncommon

103.5 Dural substitutes

Indications: incompetent dura due to any of a number of causes: trauma, tumor, infection, surgery....

Dura consists primarily of collagen and elastin.¹¹

Options:

- autologous grafts (from the patient). Common donor sites include:
 - a) fascia lata
 - b) galea/pericranium (when doing a craniotomy)
- allograft: some are designed to be absorbable, some are non-absorbable
 - a) reconstituted materials: processed collagen...
 - b) processed intact natural materials: e.g., ovine (sheep) or bovine (cattle) pericardium, porcine intestinal mucosa...
 - c) synthetic materials: e.g., Polyesterurethane, PLGA (Polyglactin 910) and PDO (Polydioxanone)...
 - d) biosynthetic materials: e.g., cellulose produced by *Gluconacetobacter xylinus*

Examples of some commercially available dural substitutes are shown in ► Table 103.1.

103.6 Localizing levels in spine surgery

Identifying the correct level in spine surgery may be extremely challenging in certain situations. With the proliferation of minimally invasive spine techniques and the associated reduction in the structures that are directly visualized, the reliance on intraoperative imaging or navigation to determine the spinal level has increased.

Potential pitfalls which increase the chances of error, including:

1. pre-op, pathology is usually identified on MRI, and there can be challenges in translating MRI images to imaging available in the O.R.
 - a) thoracic lesions: pre-op MRI usually counts from the top (C2) down, and in surgery it is often necessary to count from the bottom (S1) up, the count could be off if there are not 5 lumbar vertebrae and 12 pairs of ribs
 - b) lumbar spine: a well-developed S1–2 disc or an L5 vertebra fused to the sacrum can confuse the count (see below)
2. not all patients have 12 ribs, or 5 “lumbar” vertebrae. In the modal (most common) human spine, there are 24 presacral vertebrae; however, some individuals have 23 and others have 25

Table 103.1 Characteristics of some commercially available dural allograft substitutes

Trade name (manufacturer)	Material	Abs ^a	Sut ^b	Seal ^c	Description
Biodesign Dural Graft (Cook Medical)	collagen (porcine small intestinal submucosa derived)	+	+	*	thin; lower amount of elastin than dermis derived products
Biodesign Duraplasty Graft (Cook Medical)	collagen (porcine small intestinal submucosa derived)	+	–	*	
Cerafix® (Acera Surgical)	PLGA (polyglactin 910) and PDO (polydioxanone)	+	–	–	porous, fully synthetic, fully resorbable material; mimics architecture of extracellular matrix
DuraFlex™ Suturable Graft (Integra)	bovine pericardium	+	–	*	non-crosslinked collagen; manufactured using the Tutoplast® Tissue Sterilization Process
DuraGen® Plus (Integra)	collagen (bovine derived)	+	–	+	porous monolayer crosslinked collagen
DuraGen® Classic Matrix (Integra)	collagen (bovine derived)	+	–	–	
DuraGen® Secure (Integra)	collagen (bovine derived)	+	–	–	has additional cellulose microlayer used to keep graft in place; applied dry
DuraGen® Suturable (Integra)	collagen (bovine derived)	+	+	–	thick, porous bilayer collagen
Durepair® Regeneration Matrix (Medtronic)	collagen (bovine derived)	+	+	–	thin; uniquely uses type III fetal bovine tissue in addition to type I collagen without crosslinking
DuraMatrix® Only (Stryker)	collagen (bovine dermis derived)	+	+	*	thin; highly conformable; lower liquid permeability than other similar products
DuraMatrix® Only PLUS (Stryker)	collagen (bovine dermis derived)	+	–	*	thick; non-porous, multi-layered with lower liquid permeability than other similar products
DuraMatrix® Suturable (Stryker)	collagen (bovine dermis derived)	+	+	*	thick; bovine collagen reinforced with additional crosslinking
Durepair® Regeneration Matrix (Medtronic)	collagen (bovine derived)	+	+	–	thin; uniquely uses type III fetal bovine tissue in addition to type I collagen without crosslinking
Tutopatch® (Tutogen Medical)	bovine pericardium	+	+	+	thin; retains a structure similar to non-processed collagen
Neuro-Patch® (B. Braun)	polyester urethane	–	+	+	thin; microporous fleece with high liquid tightness
SYNTHACEL® Dura Repair (DePuy Synthes)	bionanocellulose	–	–	–	thin; porous; made of biosynthesized cellulose and water; entirely non-animal derived

^a Abs: + indicates absorbable, – indicates non-absorbable.

^b Sut: + indicates suturable. Most other allografts may be **anchored** using tensionless sutures; a tapered (not cutting) needle is recommended.

^c Seal: – means the manufacturer states dural sealants should not be used with this product, * means no information is provided regarding sealants.

(variations include: 11 or 13 rib-bearing vertebrae, or a lumbosacral transitional vertebra; the terminology of a “lumbarized S1 vertebra” or a “sacralized L5 vertebra” is imprecise and confusing). An HLD at the ultimate disc space (usually L5–1) most often impinges on the 25th nerve root (however, in the variant cases, it may actually impinge on the 24th or 26th root)¹²

3. patients may have variant or ambiguous anatomy (e.g., a well-developed S1–2 disc space, an enlarged L1 transverse process mimicking a 12th rib...)
4. some “landmarks” used for localizing levels are unreliable or changeable (e.g., mobile spondylolisthesis)
5. plain radiographs/fluoroscopy have difficulty imaging the upper thoracic and sometimes the lower cervical spine
 - a) on lateral imaging, the shoulders often obscure lower cervical/upper thoracic levels
 - b) on AP imaging, the pronounced kyphosis of this region requires cranio-caudal angulation of the X-ray beam, which throws off imaging at other levels
6. spinous processes of lumbar and especially thoracic levels are below the corresponding VB
7. changes may occur between the time of the pre-op imaging and the surgery

Aids in determining spinal levels:

1. image-guided systems, when available
2. built-in or portable intra-op MRI or CT (e.g., Airo® CT by Mobius marketed by BrainLab), or CT-like modalities (e.g., O-arm™ by Medtronic, ARCADIS® C-arm by Siemens), or image-guided spine technology
3. ★ pre-op plain X-rays:
 - a) for lumbar pathology: lumbar AP & lateral
 - b) for thoracic pathology: lumbar + thoracic AP & lateral (to verify there are 12 thoracic vertebrae and 5 lumbar vertebrae)
4. on lateral lumbar spine X-rays, the top of the iliac crests are even with the L4 spinous process or the L4–5 interspinous space
5. on sagittal MRI there are generally no numbering cues, but on axial MRI, the sacral ala are reliably identifiable and this can be used to identify L5–1 disc space
6. counting methods (if possible, using more than 1 method is highly recommended)
 - a) counting up from T12 or L5 on fluoro: be sure there are 12 ribs. You can “bridge” from lumbar or lower thoracic spine to higher thoracic levels using an instrument on the patient as a marker, or you can count up to a given level (e.g., T9) and then while a hemostat is placed at this level on the fluoro screen, under real-time fluoro the machine is slowly moved up the spine and the hemostat is moved along with T9 (to count up from L5, verify there are 5 lumbar (i.e., non-rib-bearing) presacral vertebrae). Radiation safety: avoid live fluoro as much as possible
 - b) AP view: starting at T12 (lowest rib) or from L5
 - c) lateral view: starting at the L5 and counting up
 - d) counting down from T1 (first rib) on AP fluoro: the fluoro machine may need to be angled caudally from the anterior position because of the thoracic kyphosis. Sometimes counting pedicles helps
 - e) by palpation: with thoracotomy, in the upper T-spine you can palpate the ribs from the inside from T1 and count down. The rib inserts at the upper end of the thoracic vertebra near the junction with the VB above (e.g., the T5 rib joins T5 close to the T4–5 disc space)

103.7 Bone graft

103.7.1 Use of bone graft extenders/substitutes as an adjunct to fusion

Practice guideline: Bone graft extenders and substitutes

Level I¹³: autologous bone or recombinant human bone morphogenetic protein (rhBMP-2) bone graft substitute is recommended in the setting of an ALIF in conjunction with a threaded titanium cage

Level III¹³:

- rhBMP-2 in conjunction with hydroxyapatite and tricalcium phosphate may be substituted for autograft in some cases of posterolateral fusion
- calcium phosphate is recommended as a bone graft extender, especially when combined with autologous bone

103.7.2 Assessing surgical lumbar fusion

See **Practice guideline: Radiographic assessment of fusion** (p.1722).

Practice guideline: Radiographic assessment of fusion

Level I¹⁴: static X-rays alone are *not* recommended

Level II¹⁴:

- in the *absence* of rigid instrumentation, lack of motion between vertebrae on lateral flexion/extension X-rays is highly suggestive of successful fusion
- ✗ technetium-99 bone scanning is *not* recommended

Level III¹⁴: radiographic techniques, often in combination, may be used when failed lumbar fusion is suspected, including: static and flexion/extension X-rays, CT scan

Practice guideline: Correlation between fusion and outcome

Level III¹⁵: the correlation between fusion and clinical outcome is not strong, and in any given situation fusion status may be unrelated to outcome

103.7.3 Bone graft properties

General information

For spine fusions, components of bone graft that are important for fusion:

1. osteoinduction: recruitment of mesenchymal cells and the stimulation of these cells to develop into osteoblasts and osteoclasts
2. osteogenesis: formation of new bone by host or graft mesenchymal stem cells transformed into osteoblasts
3. osteoconduction: the structure of the graft that acts as a scaffold upon which new bone and blood vessels form
4. mechanical stability: the structural anatomical biomechanical support provided, e.g., following discectomy, corpectomy, or resection of vertebral tumor

► Table 103.2 summarizes the properties of various bone graft materials (adapted^{16,17,18}). See the sections that follow for more details.

Table 103.2 Characteristics of bone graft materials^a (see text for details)

Material	Mechanical stability	Osteogenic	Osteo-inductive	Osteo-conductive
Cancellous autograft	±	++++	++	+++
Cortical autograft	+++	+	+	+
Vascularized autograft	+++	+++	++	+++
Allograft	+	–	±	+
Bone marrow aspirate	–	+	±	+
Demineralized bone matrix (DBM)	–	–	+	+
Bone morphogenetic protein (BMP)	–	–	++++	–
Collagen	–	–	–	–
Ceramics	+	–	–	+++

^a Key: – no effect, ± minimal or no effect, + mild, ++ moderate, +++ strong, ++++ very strong effect

Autograft

Common donor sites: iliac crest, rib,¹⁹ fibula, bone removed during decompression. Characteristics:

1. PROS: no histocompatibility or disease transmission issues
2. CONS:
 - a) persistent post-op donor site pain: occurs in as many as 34% of patients (the severity of which was graded as “unacceptable” in 3%)²⁰
 - b) increased surgical risks of:
 - blood loss
 - wound infection
 - fracture
 - cosmetic deformity
 - increased operative time to procure
 - numbness from nerve injury (e.g., cluneal nerves see below)
 - hematoma
3. subtypes
 - a) cancellous bone: provides all graft components except mechanical stability
 - b) cortical bone:
 - provides superior and immediate mechanical strength
 - has diminished osteoinduction and osteoconduction capacity
 - c) corticocancellous bone: e.g., tricortical iliac crest wedge. Contains all bone graft components
 - d) vascularized autograft:
 - technically challenging
 - best suited for areas that are scarred, irradiated, or that span long segments
 - e) autologous bone marrow:
 - source of osteoprogenitor cells and osteoinductive substrates
 - diminished donor site risks
 - no osteoconductive nor structural properties

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Allograft

Acquired through organ procurement agencies. Primarily frozen or freeze-dried. Donor sites include: ilium, tibia, fibula, femur, rib.

1. PROS: eliminates risks associated with harvesting autograft
2. CONS:
 - a) small but real risk of disease transmission
 - b) provides only osteoconduction (lacks osteoinduction and osteogenesis)
 - c) availability may vary from time to time
3. subtypes
 - a) tricortical block, bicortical plug, or unicortical dowel
 - b) corticocancellous: matchsticks, crushed
 - c) cancellous: cubes, block, crushed, bone powder
4. uses: allografts are acceptable for structural grafts such as in anterior spinal interbody fusion, where compressive forces are applied to the graft. However, for onlay grafts such as for posterior cervical fusion, the lack of osteoinductive and osteogenetic properties is a critical shortcoming

Demineralized bone matrix (DBM)

Prepared by acid extraction, reducing antigenicity, but preserving some osteoconductive and variable osteoinductive properties.

1. available as putty, gel, chips, granules, or powder
2. primarily used as an adjunct to other grafting materials
3. CONS:
 - a) increased cost
 - b) variable efficacy between preparations and batches of the same preparation
 - c) no mechanical or structural properties

Bone morphogenetic proteins (BMP)

Biological compounds that are a group of cytokines which induce the transformation of mesenchymal stem cells into osteoblasts (osteoinduction) with the potential to induce ectopic bone formation. There are ≈ 20 different proteins from the transforming growth factor- β family. Produced using recombinant DNA technology.

1. a carrier matrix is required to retain the soluble factor at the graft site (i.e., to prevent the BMP from diffusing into adjacent tissues, thereby reducing the desired effect and possibly inducing bone growth at undesired foci)
2. FDA approved in U.S. only for ALIF. Other uses are “off label”
3. available preparations: rhBMP-2 (Infuse® by Medtronic)
4. PROS: increases fusion rates
5. CONS:
 - a) expensive
 - b) ectopic bone formation, bone resorption (so-called osteolysis) or remodelling at the graft site²¹
 - c) in anterior cervical spine surgery: neck swelling with airway compromise, hematoma, painful seroma²¹
6. contraindications include:
 - a) patients with active malignancy or those undergoing treatment for a malignancy
 - b) should not be used in the vicinity of a resected tumor
 - c) patients who are skeletally immature

Collagen

Used primarily as a carrier for other osteoinductive, osteoconductive, or osteogenetic materials and as a composite with other graft extenders

1. PROS: contributes to vascular ingrowth, mineral deposition, and growth factor binding
2. CONS:
 - a) minimal structural support
 - b) potential immunogenicity

Ceramics

Includes tricalcium phosphate, calcium carbonate, & hydroxyapatite.

1. PROS: no risk of disease transmission
2. CONS: only recommended for use as bone graft extenders (i.e., must be combined with autograft, bone marrow aspirate, BMP...)

103.7.4 Bone growth stimulators

Introduction

Electrical current, electromagnetic fields, and ultrasound have been shown to influence osteogenesis (p.1722) and thereby bone growth and healing, possibly as a result of affecting some combination of the following: calcium influx through voltage-gated channels,²² modulation of gene expression in connective tissues with a resultant increase in production of various bone morphogenetic proteins (BMP) (p.1723).²³

There are a number of commercially available bone growth stimulators (BGS) for spine fusions that can be implanted internally (at the time of surgery), or, more commonly, worn externally after surgery in accordance to specific protocols established by the manufacturer for each device.

Since there is controversy regarding the criteria for determination of spinal fusion, and with the correlation of fusion with good outcome (some patients without fusion have good outcome), the efficacy and cost-effectiveness of BGS are unknown.

Available technologies

BGS technologies typically employed in spine fusions include:

- DCS: direct current stimulation using electrodes implanted at the time of surgery
- CCS: capacitance coupling stimulation using 2 electrodes placed on the skin over a fusion site
- PEMFS: pulsed electromagnetic field stimulation using coils typically embedded in a brace

Indications & contraindications

According to the National Coverage Determination (NCD),²⁴ electrical Osteogenic Stimulators (BGS) are effective in increasing the fusion rate as an adjunct to spinal fusion surgery in patients at high-risk for pseudoarthrosis shown in the practice guideline below.

Technologies recommended for payer coverage²⁵ for spine fusion include: DCS, CCS, & PEMFS.

Technologies with insufficient information to recommend payer coverage for spine fusion include: LIPUS (low-intensity pulsed ultrasound) and CMF (combined magnetic field = DC field & AC field).

Practice guideline: Nationally covered indications for bone growth stimulators in spine fusions

- previously failed spinal fusion at the same site
- fusion involving 3 or more vertebrae

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Other factors that the NASS Coverage Policy²⁵ recommended for consideration for BGS include the presence of one or more of the following: diabetes, inflammatory arthritis requiring long-term corticosteroids, immunocompromise (e.g., chemotherapy and radiation therapy to the spine, hypogammaglobulinemia, granulocytopenia, acquired immune deficiency syndrome, chronic granulomatous disease), systemic vascular disease, osteopenia or osteoporosis, cigarette smokers who cannot stop smoking in preparation for surgery.

In addition to those not qualifying for BGS based on the above, BGS is *not* indicated: in patients with malignancy, as an adjunct for primary bone healing for spinal fractures, or as a nonsurgical treatment for established pseudarthrosis.

The safety of BGS in the following situations is not fully known: pregnancy, infection, patients with cardiac pacemakers or defibrillators (consult a cardiologist), skeletally immature patients (children).

With implanted BGS: MRI procedures must follow specific guidelines related to magnet strength and spatial gradient (confer with your MRI facility).

Specific recommendations for the type of BGS in the *lumbar* spine²⁵:

- DCS & CCS: for posterolateral fusion using autograft + extender
- PEMFS: for lumbar interbody fusion

103.7.5 Bone graft procurement

Iliac crest

Anterior iliac bone graft

Should be obtained at least ~ 3–4 cm lateral to the anterior superior iliac spine (ASIS) to avoid the lateral femoral cutaneous nerve and to reduce the risk of avulsion fractures of the remaining ilium. When a tricortical graft is taken, keep the dissection in the subperiosteal plane and avoid electrocautery on the medial (inner) surface when detaching the iliacus muscle to avoid injury to the ilioinguinal, iliohypogastric, and lateral femoral cutaneous nerves.

Posterior iliac crest bone graft

May be used to obtain corticocancellous strips or plates for onlay bone grafts, or tricortical grafts which may be used as strut grafts or for C1–2 arthrodesis.

They are taken from the medial 6–8 cm of the iliac crest (► Fig. 103.2) to avoid the superior cluneal nerves (which cross the posterior iliac crest ~ 8 cm lateral to the posterior superior iliac spine) with resultant buttock numbness or the development of painful neuromas. A vertical incision just medial to the posterior superior iliac spine usually works well.

The spine may sometimes be found on corpulent patients by locating the “dimple of Venus” (foresae lumbales laterales—indentation sometimes visible superior to the gluteal cleft, directly superficial to the sacroiliac joint) and incising slightly lateral to it. Avoid mistaking the sacrum for the iliac spine.

The gluteus maximus is dissected off the lateral surface subperiosteally. To avoid fractures extending into the iliac crest, a wide osteotome should be used to create a “stop cut”; alternatively, a sagittal saw may be used. Avoid penetration through the inner cortical surface of the crest so as not to enter the pelvis and possibly cause an intra-abdominal hematoma. Another potential complication is fracture extension into the greater sciatic notch with possible injury to the gluteal arteries and sciatic nerve among others. Once the graft is removed and cancellous bone is gouged out, the exposed bone surfaces should be waxed and closed system drainage should be used to reduce the risk of local hematoma formation.

Fibula

Autogenous fibular graft provides a high arthrodesis rate,²⁶ but may be associated with significant morbidity, and so may be best reserved for salvage procedures.²⁷ Preserve the proximal fibular head to avoid injury to the peroneal nerve. At least 7 cm of distal fibula should be maintained to preserve ankle stability.¹⁹

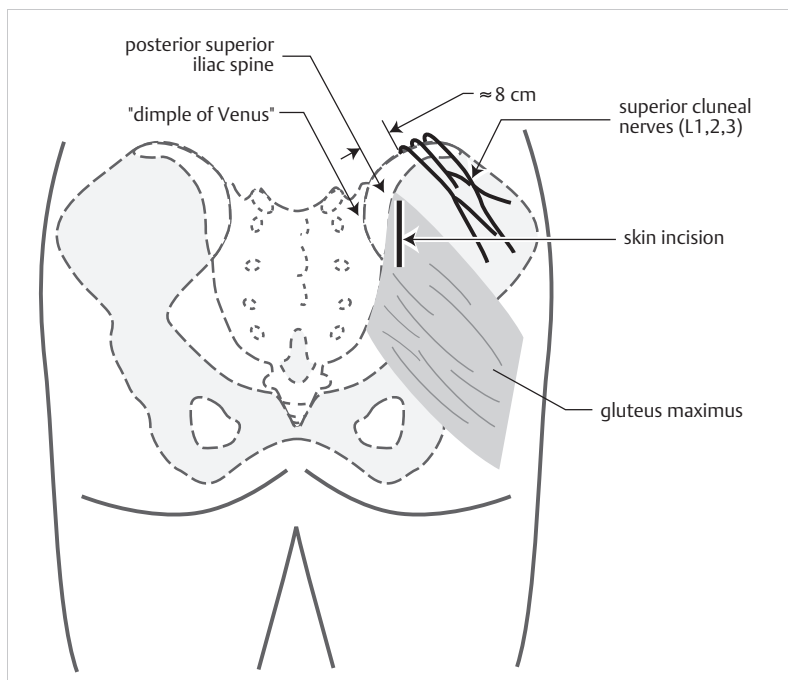


Fig. 103.2 Posterior iliac crest bone donor site.

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104 Craniotomies – General Information and Cortical Mapping

104.1 Craniotomy – general information

104.1.1 Cranial perforators

Perforators are one means used to create burr holes which enable the surgeon to access the brain for intracranial procedures. Many brands of perforators are designed with a slip clutch that disengages the drill and stops the outer shaft from spinning and entering the skull once the central part of the drill penetrates the inner table. While these clutches are generally reliable and immensely helpful, there is a possibility of malfunction such that the outer shaft continues to rotate and if it penetrates the inner table, the entire drill can plunge into the brain. In 8 months of 2005 the FDA received reports of 200 injuries as a result of the drill failing to disengage.¹ The FDA released a number of recommendations to reduce the risk of injury,¹ excerpts of which are shown here:

- select the appropriate perforator based on the skull thickness (pediatric vs. adult)
- keep the perforator perpendicular to the skull throughout the drilling process
- do not rock, rotate, or change the angle of the device during drilling
- avoid using excessive pressure on the drill. Brace the hand holding the drill on the other hand, which should rest on the patient's skull, and thereby prevent plunging if the drill completely penetrates the skull
- use caution when:
 - drilling through areas of irregular bone contour, curvature, or variations in thickness
 - drilling on skulls of infants/children, the elderly, or any other patient who might have softer bone consistency (including osteogenesis imperfecta...)
 - perforating bone in an area where the bone might be diseased or incompetent or have loose fragments

104.1.2 Intraparenchymal cyst aspiration

When a cystic tumor or intracerebral hemorrhage is being operated on, an attempt should be made to insert a ventricular needle into the lesion and aspirate some but not all of the cyst contents. This often produces significant decompression. Avoid evacuating all of the contents; otherwise the lesion might be difficult to find. The needle may then be left in place to allow localization of the lesion (or the needle track can be followed, which may occasionally be difficult).

104.1.3 Intraoperative brain swelling

Background

Under certain circumstances during surgery, the brain may start to severely swell out of the craniotomy wound. Etiologies of this emergency situation include:

1. extraparenchymal bleeding: from a vessel or intraoperative aneurysm rupture, remotely situated epidural/subdural hematoma
2. intracerebral hemorrhage
3. venous outflow obstruction
4. vasodilatation induced by hypercarbia
5. severe diffuse cerebral edema following stroke or traumatic brain injury (TBI)

Management

First efforts should be aimed at ruling out and correcting the aforementioned causes as well as some adjunctive measures. Most maneuvers are similar to those used in controlling an ICP crisis. During the process, it is critical to try to avoid having the brain compress itself against the craniotomy bone edges, which can lacerate the cortex and can also further compromise cortical veins, which impairs venous outflow, causing more brain edema and swelling, which accelerates the vicious cycle.

1. elevate the head of the patient (e.g., with reverse Trendelenburg of the O.R. table)
2. make sure the jugular veins are not kinked: this may require rotating the head by loosening the pivot that connects the table adapter to the Mayfield head-holder and rotating the head to a more neutral position

3. rule out hypercarbia: make sure the endotracheal tube is not kinked, check the patient's end-tidal $p\text{CO}_2$
4. measures to lower ICP and protect the brain
 - a) give mannitol 1 gm/kg IV bolus
 - b) drain CSF if an option: from adjacent cistern or lumbar drain
 - c) tap and drain a ventricle e.g., using an EVD or a ventricular needle
 - d) have anesthesiologist hyperventilate to a $p\text{CO}_2$ of 30–35 mm Hg
 - e) have anesthesiologist induce burst suppression (e.g., with barbiturates, or propofol...)
 - f) apply sterile ice water on the brain
5. emergently intubate patients who are undergoing awake craniotomy
6. consider intraoperative CT or ultrasound if rapidly available to rule out hematoma (intracerebral, EDH, SDH) which could potentially be immediately evacuated
7. during the above steps, place a moist sponge on the surface of the brain and gently but firmly apply evenly distributed pressure to push the brain back into the wound
8. if all else is failing, the craniotomy flap can be enlarged as much as possible to create a decompressive craniotomy. Enlarging the skin incision to do so is preferable to having too small a bony opening, which risks brain compression/laceration against the edges. The skin is closed without the bone flap and without dural closure as in a decompressive craniectomy (p. 1765)
9. a last-ditch life-saving measure for continued uncontrollable swelling which is to be taken under advisement with eloquent cortex: use a gloved hand to sweep the herniating brain out of the wound (i.e., remove it from the patient)

104.1.4 Craniotomy pre- and post-op management

Risks

Many risks cannot be generalized for all craniotomies and are specific to various tumors, aneurysms, etc. General information:

1. postoperative hemorrhage
 - a) overall risk of postoperative hemorrhage^{2,3}: 0.8–1.1%. The most common indication for craniotomy in these series was for meningioma, followed by trauma, aneurysm, and then intrinsic supratentorial tumors. 43–60% of the hematomas were intraparenchymal, 28–33% epidural, 5–7% subdural, 5% intrasellar, 8% mixed, 11% confined to superficial wound. Overall mortality was 32%
 - b) hematoma may occur at the surgical site or in remote locations, e.g., intracerebellar hemorrhage after pterional⁴ and temporal⁵ craniotomies
2. in craniotomy for brain tumor⁶:
 - a) risk of anesthetic complications: 0.2%
 - b) increased neurologic deficit in first 24 hours post-op: $\approx 10\%$
 - c) wound infection: 2%
3. postoperative headache (p. 1731)

Pre-op orders

1. for tumor: if patient on steroids, give $\approx 50\%$ higher dose 6 hrs before and on-call to O.R. (stress doses); if not on steroids give dexamethasone 10 mg PO 6 hrs before and on-call to O.R. (in A.M., give with a sip of water)
2. antiepileptic medication
 - a) if there is a history of seizures:
 - if already on antiseizure medications (ASMs) continue same doses
 - if not on ASMs, load with Keppra 500 mg or oral PHT (may give 300 mg PO q 4 hrs $\times$ 3 doses (total 900 mg) to load orally)
 - b) if no history of seizures
 - if surgery does not require a cortical incision (e.g., aneurysm) then ASMs are generally not used
 - if a cortical incision is anticipated, option to load with ASMs as above
3. prophylactic antibiotics: (optional) ideally 30–60 minutes before incision. For most antibiotics, it is given in the O.R. before the skin incision. For antibiotics that take a long time to infuse (e.g., vancomycin) it may help to order it to be given “on call to O.R.”
4. DVT prophylaxis: pneumatic compression boots or knee-high TED® hose

Post-op orders

Guidelines (individualize as appropriate) for patient to be extubated

1. admit PACU, transfer to ICU (neuro unit if available) when stable
2. VS: q 15 min x 4 hrs, then q 1 hr. Temperature q 4 hrs x 3 d, then q 8 hrs. Neuro check q 1 hr
3. activity: bed rest (BR) with HOB elevated 20–30°
4. knee-high TED hose or pneumatic compression boots
5. I & O q 1 hr (if no Foley: straight cath q 4 hrs PRN bladder distension)
6. incentive spirometry q 2 hrs while awake (*do not use following transsphenoidal surgery*)
7. diet: NPO except minimal ice chips and meds as ordered
8. IVF: NS + 20 mEq KCl/L @ 90 ml/hr
9. O₂: 2 L per NC
10. meds:
 - a) dexamethasone (Decadron®): if not on chronic steroids, give 4 mg IV q 6 hrs. Otherwise give stress doses based on patient's current dose and length of treatment
 - b) H₂ antagonist, e.g., ranitidine 50 mg IVPB q 8 hrs
 - c) Anti-epileptic drug (ASM) especially when cerebral cortex is violated: typically Keppra® (levetiracetam): 500 mg PO or IV q 12 hours. If there is no prior history of seizure, typically discontinue after ≈ 1 week
 - d) Cardene® drip: titrate to keep SBP < 160 mm Hg and/or DBP < 100 mm Hg (use cuff pressures, may use A-line pressures if they correlate with cuff pressures)
 - e) codeine 30–60 mg IM q 3–4 hrs PRN H/A
 - f) acetaminophen (Tylenol®) 650 mg PO/PR q 4 hrs PRN temperature > 100.5 °F (38 °C)
 - g) continue prophylactic antibiotics if used: (e.g., cefazolin (Kefzol®) 500–1000 mg IVPB q 6 hrs x 24 hrs, then D/C)
11. labs:
 - a) CBC once stabilized in ICU and q d thereafter
 - b) renal profile once stabilized in ICU and q 12 hrs thereafter
 - c) ABG once stabilized in ICU and q 12 hrs x 2 days, then D/C (also check ABG after any ventilator change if patient on ventilator)
12. call M.D. if any deterioration in crani checks, for T > 101 °F (38.5 °C), sudden increase in SBP, SBP < 120, U.O. < 60 ml/2-hrs
13. post-op CT: non-contrast post-op head CT is performed if the patient does not return to baseline neurologic function within a reasonable amount of time; also, performed routinely at many institutions following all craniotomies

104.1.5 Postoperative deterioration

General information

When the postoperative neurologic status is worse than pre-op, especially in a patient who deteriorates after initially doing well following surgery, emergency evaluation and treatment is indicated.

Possible etiologies:

1. hematoma (p. 1729)
 - a) intracerebral hemorrhage (ICH): at or remote from surgical site
 - b) epidural hematoma
 - c) subdural hematoma
2. cerebral infarction
 - a) arterial
 - b) venous infarction: especially with surgery on or around the venous sinuses (p. 1761)
3. postoperative seizure: may be due to inadequate antiseizure medication levels, and may be exacerbated by any of the above (see below for management)
4. acute hydrocephalus
5. pneumocephalus; also see Pneumocephalus (p. 1067):
 - a) tension pneumocephalus: see Tension pneumocephalus (p. 1068)
 - b) simple pneumocephalus: the simple presence of air in the cranium can cause neurologic symptoms even if not under tension (as would commonly occur following the now outdated pneumoencephalogram). Symptoms include: lethargy, confusion, severe headache, nausea & vomiting, seizures. Air may be located over the cerebral convexities, in the p-fossa, and/or in the ventricles and usually resorbs with symptomatic improvement in 1–3 days

6. edema: may improve with steroids
 - a) worsening of cerebral edema: moderate post-op worsening of cortical function of immediately adjacent brain is not unexpected in many operations, and is usually transient. However, reversible etiologies (such as subdural hematoma (SDH)) must be ruled out
 - b) traction or manipulation of cranial nerves may cause dysfunction that may be temporary. Division of cranial nerves can cause permanent dysfunction
7. persistent anesthetic effect (including paralytics): unlikely in a patient who deteriorates after initially doing well post-op. Consider reversing medication given during surgery (caution re: hypertension and agitation), e.g., naloxone, flumazenil (p. 320), or reversal of pharmacologic muscle block (p. 143)
8. vasospasm: following SAH or may be due to manipulation of blood vessels

Postoperative seizure management

1. intubate if patient does not rapidly regain consciousness, is not protecting airway, or has labored respirations
2. CT scan: rule out hematoma (intracerebral or extra-axial) or hydrocephalus
3. antiseizure medications:
 - a) draw blood for appropriate antiseizure medication level
 - b) bolus with additional antiseizure medications: do not wait for levels

104.1.6 Postoperative headache

General information

Also see the “syndrome of the trephined” (p. 1763).

Persistent headache (H/A) is well described following posterior fossa craniectomy (incidence range: 0–83%⁷). The time course in one series⁸ was: 23% at 3 mos, 16% at 1 yr, and 9% at 2 yrs.

Persistent H/A may also be observed following supratentorial craniotomy⁹ (prevalence 1 year after anterior temporal lobectomy for seizures: 12%⁹).

These H/A have been attributed to traction on the dura when the bone is not replaced, tension on the dura due to tight dural closure, temporalis or nuchal muscle dissection, nerve entrapment in the closing sutures or in the healing scar (especially the greater occipital nerve following suboccipital craniotomies), intradural blood and/or bone dust, and CSF leak.⁹

Prevention

No single method or group of methods has been successful in completely eliminating the complaint of post-op H/A.^{10,11} Until further research can further advance the understanding of the cause and prevention of these H/A, it seems reasonable to employ the following measures as much as possible in an attempt to minimize these debilitating symptoms: restoring function of the temporalis or suboccipital musculature, rigid fixation of bone flaps, cranioplasty for large craniectomies, meticulous tension-free dural closure (using duraplasty when necessary), and keeping intradural blood clot and bone dust to the minimum possible.¹² Cranioplasty following posterior fossa surgery for vestibular schwannoma reduced the incidence of post-op H/A from 17% to 4%.¹³

Treatment

Initially, symptomatic treatment is indicated. Referral to a H/A specialist may be appropriate when it becomes apparent that the H/A are not resolving spontaneously after $\approx$ 3 months.¹²

104.2 Intraoperative cortical mapping (brain mapping)

104.2.1 General information

Indications: typically used to locate the motor strip, sensory cortex, or speech centers intraoperatively for surgery in and around these eloquent areas. Localization of these areas based on visible anatomy alone is unreliable. These techniques are typically employed in seizure surgery as well as in treating lesions in areas of eloquent brain.

Some techniques require an awake patient, with the surgery done under local anesthesia with sedation. Motor and sensory cortex can also be localized in anesthetized patients using SSEPs (see below).

104.2.2 Phase reversal method for localizing primary sensory and motor cortex

General information

Utilizes intraoperative SSEPs to localize primary sensory and motor cortex in patients under general anesthesia (as opposed to using brain mapping techniques in awake patients).^{14,15}

Technique

See anesthesia requirements for intraoperative EP monitoring (p.112). A strip grid is placed on the surface of the brain perpendicular to the anticipated orientation of the central sulcus. SSEP stimulation is performed while recording through the strip grid. Phase reversal of the N20/P20 peak between a pair of electrodes in the strip grid indicates that those electrodes straddle the central sulcus (► Fig. 104.1) with primary motor cortex located anteriorly, and sensory cortex posteriorly. The grid is then repositioned and the test is run again to verify the findings.

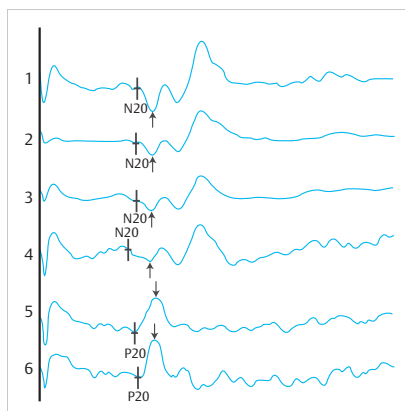


Fig. 104.1 Phase reversal.

Intra-op 6-electrode recording strip placed on the brain during SSEP recording. Phase reversal of the negative N20 peak (arrows) to a positive P20 peak between electrodes #4 & 5 indicates that electrodes #4 & 5 straddle the central sulcus.

104.2.3 Awake craniotomy

General information

Usually employed for brain mapping, especially for speech areas. Numerous techniques and protocols have been described. Typically, the patient is temporarily anesthetized with short-acting agents (inhalational and/or injectable). This is supplemented with local anesthetic. The craniotomy is then performed and the patient is allowed to wake up while the brain is exposed to permit neurophysiologic testing during surgery. If (short-acting) paralytics are used, it is critical to reverse these agents 15–30 minutes prior to applying the electrical stimulation and that a train-of-four muscle twitch can be elicited.

Booking the case: Awake craniotomy

Also see defaults & disclaimers (p.25) and pre-op counselling (see below).

1. position: depends on lesion location, with pin headholder (for image-guided navigation if used)
2. equipment:
 - a) microscope if needed e.g., for tumor dissection
 - b) image-guided navigation system (if used)
 - c) ultrasonic aspirator (for tumors)
3. anesthesia: pre-op consult for “awake craniotomy” & skull block
4. consult neurology or neuropsychology to be available during surgery for intra-op neurologic testing for “awake craniotomy”

5. EEG techs to perform intra-op EEG and provide brain stimulator
6. post-op: ICU
7. consent (in lay terms for the patient—not all-inclusive):
 - a) procedure: surgery on the brain to be performed with periods where the patient will be woken up for testing (plus whatever else is planned, e.g., removal of tumor, removal of seizure source...)
 - b) alternatives: the same surgery under general anesthesia, nonsurgical management (for some diagnoses, e.g., tumor, radiation therapy)
 - c) complications: (usual craniotomy complications (p. 25): stroke, bleeding, coma, death, infection, seizures), difficulty accurately mapping the desired areas of the brain

Indications

1. surgery in eloquent brain (near motor strip (Brodmann's area 4 in ► Fig. 1.1) or speech/language centers or thalamus) including tumors and epileptic foci
2. removal of brainstem tumors
3. some seizure surgery to look for seizure focus

Contraindications to awake craniotomy

1. patients unlikely to be able to cooperate: very young or very elderly patients, confused patients, those with significant speech deficits already present or language barrier

Patient counselling pre-op

Patients need to be aware of what the sequence of events will be and what will be expected of them. It may be helpful to have them practice reading some typical material that will be used in the O.R. Patients over age ≈ 40 usually need reading glasses to see written material, and they should have their own available in the O.R., although the temples (earpieces) usually can't be accommodated. The patient should be advised that there may be some pain involved.

Patient positioning for surgery

Significantly more time must be spent on patient positioning to ensure that they will be as comfortable as possible without moving. Extra padding is employed. Access to the patient's face is necessary for the anesthesiologist and the neurophysiologist.

Typical sequence for anesthesia

See reference.¹⁶

1. in the pre-op holding area, load with Precedex® (dexmedetomidine) 0.5 mcg/kg IV over 20 minutes followed by intra-op infusion at 0.4–1.0 mcg/kg/hr
2. induction of anesthesia utilizes propofol 3 mg/kg IV followed by laryngeal mask airway (LMA) placement
3. skull block¹⁷: injection of local anesthetic (e.g., 30 ml of 0.5% bupivacaine) to permit the skin incision and also rigid head fixation with pins (as required for image navigation devices, and situations where no head movement can be tolerated during surgery) without pain at the time of the wake-up. Injection at 4 regions on each side as shown in ► Fig. 104.2:
 - ① supraorbital & supratrochlear nerves: 2 ml injected 1.5 cm above the supraorbital foramen above the medial third of the orbit. NB: if you are going to use surface matching to register the patient for image guidance (e.g., BrainLab or Stealth), injecting here may deform the skin and affect the registration accuracy. Consider injecting a lower volume of higher concentration agent (e.g., 2% lidocaine)
 - ② auriculotemporal nerve: 5 ml injected 1.5 cm anterior to the tragus. ✗ Caution: to avoid anesthetizing the facial nerve, inject just deep to the subcutaneous tissue
 - ③ postauricular branches of the greater auricular nerve: 2 ml 1.5 cm posterior to the antitragus
 - ④ greater, lesser, & third occipital nerves: inject 5 ml with a 22 gauge spinal needle at the mastoid process and proceed along the nuchal ridge until the midline is reached
4. start inhalational anesthesia with 0.5 MAC desflurane with the patient breathing spontaneously while the scalp incision, craniotomy, and dural opening are performed (the dura is pain-sensitive, the brain is not)

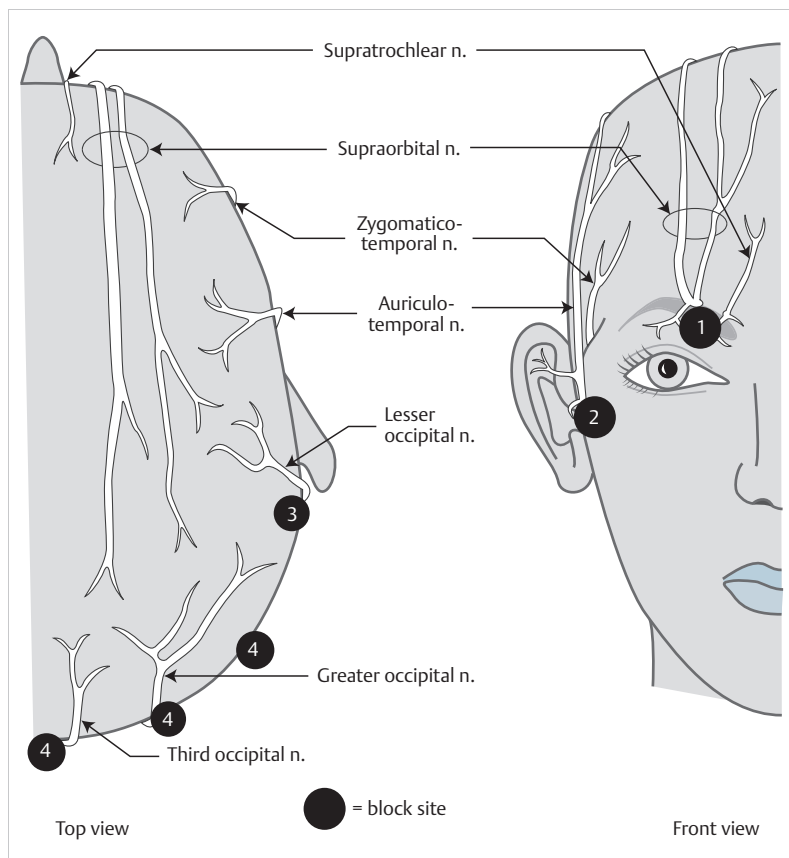


Fig. 104.2 Infiltration sites for skull block.

5. as the dural opening is begun, the desflurane is turned off and a remifentanyl infusion of 0.1–0.2 mcg/kg/min IV is started
6. by the time the dural opening is completed, the desflurane has usually worn off and the LMA can be removed
7. remifentanyl is then titrated for pain control
8. neurophysiologic testing can usually be performed at this time (e.g., below)
9. the operation may often be carried to completion with the patient awake, although once the intracranial part of the operation is completed, more pain relief may be desired and general anesthesia may be needed for pain control or agitation (LMA may suffice here)

104.2.4 Speech mapping

General information

Typical settings for a constant current generator using a bipolar electrode are shown in ► Table 104.1. If a voltage-based unit is used, start at 1 volt and increase.

Table 104.1 Settings for constant current generator

Control	Setting ^a
frequency	50–60 Hz
waveform	biphasic square wave
duration	2–4 mS peak-to-peak
mode	repeat
polarity	normal
current	varies between 2–16 mA

^anot all settings are present on all models

Techniques for language mapping

There are numerous methodologies. One protocol for cortical mapping:

1. requires awake craniotomy
2. once the temporal lobe is exposed, a recording electrode strip is placed on the brain surface
3. using a bipolar stimulator, start with a low current (e.g., 2 mA) and begin stimulating an area of the cortex for 3–5 seconds, and observe for afterdischarges (akin to a focal seizure) on the recording strip. If no afterdischarges, increase the current in 2-mA increments up to a maximum of ≈ 10 mA. If afterdischarges occur, back off by 1–2 mA and then test that area for speech changes as follows
4. stimulate cortex while patient names objects shown on picture cards (automatic verbalization, such as counting, is robust and may persist). Observe for effects including¹⁸:
 - a) total speech arrest¹⁸: on the dominant hemisphere typically in the pars opercularis or precentral gyrus, but also in frontal operculum and temporoparietal region. On the non-dominant hemisphere, this occurred only in the pars opercularis.
 - b) able to speak but unable to name^{18,19} (dysnomia): in dominant hemisphere, typically in posterior inferior frontal gyrus, and posterior temporal and inferior parietal regions
 - c) semantic errors: posterior middle temporal gyrus, anterior supramarginal gyrus, and inferior frontal gyrus
 - d) phonological paraphrasias, neologisms, and circumlocution: superior temporal sulcus
 - e) NB: subcortical fiber mapping may identify white matter tracts that participate in language processing (see reference¹⁹)
5. repeat the above steps at the next area (first finding threshold for afterdischarges and then stimulating while testing)

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105 Posterior Fossa Craniotomies

105.1 Indications

To gain access to the cerebellum, cerebellopontine angle (CPA), to one vertebral artery, posterior brainstem, 4th ventricle, pineal region, or using extreme lateral posterior fossa approach to the anterolateral brainstem. See paramedian (p. 1740) and midline (p. 1742) suboccipital craniectomies for details.

105.2 Position

105.2.1 Options

Position options include:

1. sitting position (p. 1737)
2. lateral oblique (p. 1738) ("park bench"): patient three-quarters oblique (almost prone). Often used for access to cerebellar hemisphere lesion
3. semi-sitting
4. supine with shoulder roll, head almost horizontal
5. prone
6. Concorde position: often used for access to midline (e.g., 4th ventricle). Prone, thorax elevated 20°, neck flexed and tilted slightly away from the side on which the surgeon will be standing

105.2.2 Sitting position

Used less frequently than in the past because of associated complications and acceptable alternative positions (except for some specific circumstances). However, some experts feel that the risks of the sitting position have been greatly overstated.¹ The presence of a ventricular CSF shunt is considered to be a contraindication to the use of the sitting position by some.

Advantages

1. improved drainage of blood and CSF out of surgical site
2. enhanced venous drainage which helps reduce venous bleeding and also ICP
3. easy ventilation due to unencumbered chest
4. patient's head may be kept exactly midline, aiding operator orientation and reducing risk of kinking of vertebral arteries

Disadvantages/risks

1. possible air embolism (see below)
2. fatigue of operator's hands
3. increased surgical risks from placement of CVP catheter (required to treat possible AE): e.g., pneumothorax with subclavian vein catheterization, thrombosis
4. risk of post-op hematoma at the operative site may be increased because potential venous bleeders may remain occult while the patient is sitting, but may manifest when patient returns to a horizontal position post-op. However, one study found no such increased incidence²
5. risk of post-op subdural hematoma: 1.3% of p-fossa cases³
6. possible brachial plexus injury: prevent this by not allowing patient's arms to hang at the side. Instead, fold them across abdomen
7. midcervical quadriplegia^{4,5}: presumably due to flexion myelopathy.^{6,7,8} The combination of the sitting position with hypotension⁹ or neck flexion with possible compression of the anterior spinal artery, ± cervical bar, and elevation of the head thus reducing the arterial pressure may all contribute
8. sciatic nerve injury (piriformis syndrome)¹⁰: prevent this by flexing patient's knees (reduces tension on sciatic nerve)
9. extent of post-op pneumocephalus is more pronounced, and may increase the risk of tension pneumocephalus¹¹; see Pneumocephalus (p. 1067)
10. venous pooling of blood in the LEs under anesthesia may cause relative hypovolemia and should be counteracted by binding the LEs prior to positioning
11. decreased cerebral blood flow due to lower hemodynamic arterial pressure¹²

► **Air embolism (AE).** A potentially fatal complication of any operation when an opening to air occurs in a non-collapsible vein (e.g., diploic vein or a dural sinus) when there is a negative pressure in the vein (e.g., when the head is elevated above the heart).¹³ Air is entrained in the vein and can become trapped in the right atrium of the heart, which may impair venous return, causing hypotension. AE has also been described with endovascular procedures.¹⁴ AE may also produce cardiac arrhythmias. Paradoxical air embolism can occur in the presence of a patent foramen ovale¹⁵ or pulmonary AV fistula, and may produce ischemic cerebral infarction.

Greater negative pressures occur in the sitting position due to the extreme elevation of the head, but AE can occur in any operation with the head elevated higher than the heart. Incidence: a wide range has been quoted in the literature, and depends on the monitoring method used: $\approx 7\text{--}25\%$ incidence with the sitting position using Doppler monitoring is an estimate.³

For operations with a *significant* risk of AE, a right atrial CVP line is recommended (to aspirate air), and monitoring for air embolism; options include: transesophageal echo (the most sensitive), precordial Doppler monitoring. (Although technically the risk of air embolism includes *any* case where the head is higher than the right cardiac atrium, practically it is limited to cases where the head of the bed is $\approx > 30^\circ$ which is mostly limited to the sitting position for posterior fossa tumors.)

Diagnosis and treatment:

AE should be suspected in any operative case in which the surgical site is higher than the heart when there is any unexplained hypotension or decrease in EtCO_2 .¹⁸

- transesophageal echocardiography (TEE). Bubbles can be seen on the 2D echo display
pros: considered the most sensitive monitoring modality
cons: significant false-positive rate, expensive, invasive, requires experience and vigilance
- precordial Doppler U/S: probe may be placed over 2nd to 4th intercostal space either to right or left of sternum, or posteriorly between the scapula and spine. AE is heralded by a change in sonic intensity and character at first by a superimposed irregular high-pitched swishing sound, and then as more air is entrained so called “mill wheel” or machinery sounds dominate
pros: the most sensitive of the non-invasive techniques
cons: difficult in morbidly obese patients and in certain patient positions (e.g., prone or lateral), interference from other sounds in the O.R., requires vigilance

The earliest indication of AE may be a rise in the end-tidal nitrogen (requires mass-spectrometer on monitor), then a fall in the end-tidal pCO_2 occurs. Machinery sounds in the precordial Doppler also suggest AE. Hypotension may develop. Measures shown in ► Table 105.1 should be immediately instituted.

Table 105.1 Treatment for air embolism

1. find and occlude site of air entry, or else rapidly pack wound with sopping wet sponges/laps and wax bone edges
2. Durant’s maneuver¹⁶: lower patient’s head if at all possible (Trendelenburg) and rotate patient *LEFT* side down (attempt to trap air in right atrium)
3. jugular venous compression (bilateral best; second choice: right only)
4. aspirate air from right atrium via CVP catheter
5. ventilate patient with 100% O_2
6. discontinue nitrous oxide if used (may expand AE)¹⁷
7. use pressors and volume expanders to maintain BP
8. PEEP is *ineffective* in preventing or treating AE; may increase the risk of paradoxical AE¹³

105.2.3 Lateral oblique position

AKA “park bench” position.

- axillary roll for the down-side arm (see ► Fig. 105.1) (or position the patient so that the down-side arm extends over the edge of the table and is held in place by a sling formed by the Mayfield table attachment with copious padding)
- upper arm supported on pillows or towels (avoid using a Mayo stand, which restricts the ability to laterally tilt the O.R. table during surgery)
- adhesive tape to gently pull down on the upper shoulder
- bring the patient’s back as close to the side edge of the table as possible (usually limited by the travel of the head clamp) to bring the patient closer to the surgeon
- elevate thorax $10\text{--}15^\circ$
- see Head positioning (p. 1739)
- optional spinal drainage (usually for large tumors)

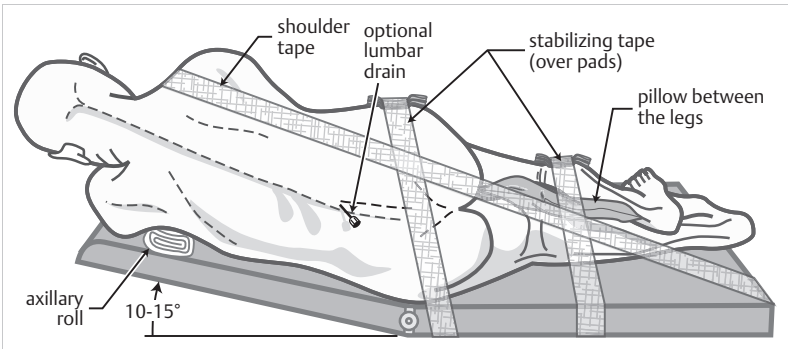


Fig. 105.1 Lateral oblique ("park bench") position.

- some surgeons prefer the LE that contacts the table to be slightly flexed at the hip & knee to stabilize the patient, others prefer the uppermost LE to be flexed to allow the patient to "fall forward" a little more at the hips. Place a pillow between the legs
- secure patient with adhesive tape over pads so the table can be "airplaned" (rolled) during the operation

For access to the porus acusticus or more caudally

(E.g., for vestibular schwannomas; not necessary for microvascular decompression for trigeminal neuralgia.)

Get the shoulders out of the way by flexing the neck as much as possible while maintaining patient airway (aided by use of non-kinking wire-reinforced ET tube, so-called "armored tube"). The upper shoulder is retracted caudally by adhesive tape (avoid excess traction which may injure brachial plexus).

Head positioning

A Mayfield head-holder (p. 1717) is placed with the single pin on the side of the lesion, slightly anterior to a true-lateral on the skull (► Fig. 105.2). In an adult, it is tightened until the third ring (60 lbs) is just visible on the spring-loaded indicator.

Tip the vertex of the head slightly towards the floor by bringing the downside ear towards the ipsilateral shoulder.

The head is rotated 20–30° face-down from the horizontal (► Fig. 105.2).

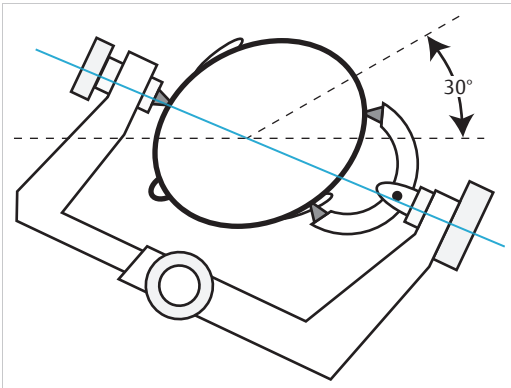


Fig. 105.2 Head-holder placement and rotation of head for a right suboccipital craniectomy (looking down on top of patient's head).

The neck is flexed to a slight degree for cerebellar hemisphere lesions. For access down to the foramen magnum, more aggressive neck flexion is used (maintain a distance of at least 2 finger-breadths between the mandible and the chest), and avoid kinking the endotracheal tube (some surgeons request an "armored" tube which is reinforced with a wire).

105.3 Paramedian suboccipital craniectomy

105.3.1 Indications

1. access to the cerebellopontine angle (CPA)
 - a) CPA tumors, including:
 - vestibular schwannoma
 - CPA meningioma
 - epidermoid
 - b) microvascular decompression
 - trigeminal neuralgia
 - hemifacial spasm
 - miscellaneous: geniculate neuralgia, glossopharyngeal neuralgia
2. lesions of one cerebellar hemisphere (typically lesions < 2.5 cm diameter):
 - a) tumors: metastases, hemangioblastomas...
 - b) hemorrhage within cerebellar hemisphere
3. access to vertebral artery
 - a) aneurysms: PICA, vertebralbasilar junction
 - b) vertebral endarterectomy
4. access to anterolateral brainstem tumors (extreme lateral p-fossa approach)
 - a) foramen magnum tumors, including: chordomas, meningiomas

105.3.2 Position, skin incision, craniectomy, approach...

See list of alternatives (p.1737). See lateral oblique position (p.1738).

105.3.3 Skin incision

Linear (paramedian) incisions

A vertical linear paramedian incision provides adequate exposure for lesions < 2.5 cm diameter and involves less trauma to overlying muscles, and may be easier to get watertight closure than with midline incision.

For lesions in the cerebellar hemisphere: a linear vertical incision approximately midway between the midline and the mastoid notch may be used.

Access to CPA (for microvascular decompressions and *small* CPA tumors): a slightly curved retro-mastoid incision placed 5 mm medial to the mastoid notch (a palpable landmark) is used (► Fig. 105.3):

1. "5–6–4" incision (incision placed 5 mm medial to mastoid notch, extending from 6 cm above notch to 4 cm below). High enough to expose transverse sinus:
 - a) for approach to fifth nerve: microvascular decompression for trigeminal neuralgia
2. "5–5–5" incision (5 mm medial, extending 5 cm up to 5 cm down), used for approach to seventh/eighth nerve complex:
 - a) microvascular decompression for hemifacial spasm
 - b) small vestibular schwannoma
3. "5–4–6" incision (5 mm medial, extending 4 cm up to 6 cm down): used for approach to lower cranial nerves:
 - a) glossopharyngeal neuralgia

"Hockey-stick" incision

Useful for lesions > 2.5 cm in the cerebellar hemisphere or CPA where getting the muscles out of the way will facilitate maneuvering instruments about the posterior fossa.

Incision is made in the midline starting at ≈ C2 spinous process, proceeding superiorly to just above theinion, and then laterally to just beyond the mastoid tip (► Fig. 105.4). A short optional caudal curve may be made laterally to further remove the muscle from the operative field.

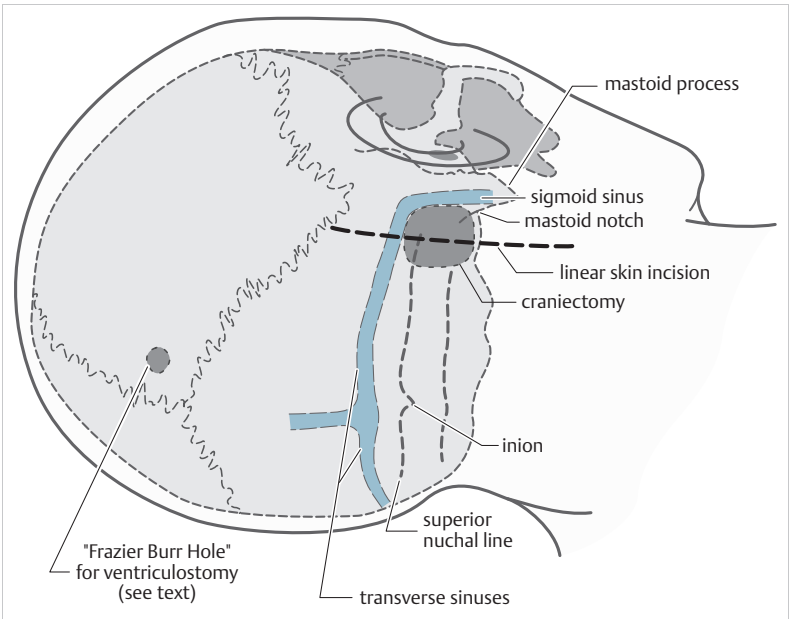


Fig. 105.3 Retromastoid suboccipital craniectomy.

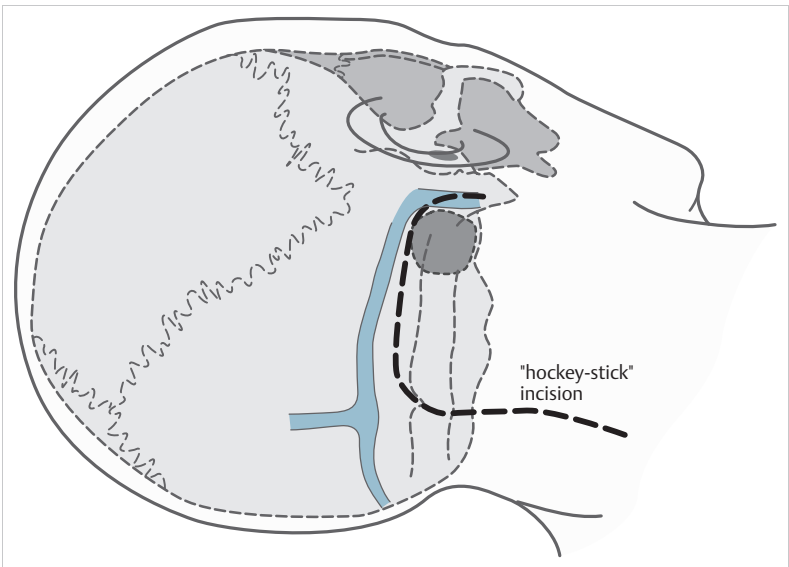


Fig. 105.4 "Hockey-stick" skin incision.

105.3.4 Craniectomy

Landmarks

The location of the inferior margin of the transverse sinus is quite accurately estimated at two finger-breadths above the upper limit of the mastoid notch (usually just above the superior nuchal line). This should be the upper limit of the skull opening.

For microvascular decompression

Craniectomy $\approx$ 2 cm diameter placed in the angle between transverse and sigmoid sinuses.

For small tumors (<2.5 cm)

Craniectomy $\approx$ 4 cm diameter placed in the angle between transverse and sigmoid sinuses.

For large tumors

A larger craniectomy may be needed, the size of which is limited by:

1. transverse sinus superiorly
2. foramen magnum inferiorly (which may be opened as prophylaxis against tonsillar herniation in the event of p-fossa edema post-op)
3. sigmoid sinus laterally (opening mastoid air cells is acceptable, but to prevent CSF leak, these must be packed with bone wax and muscle (or bone dust from craniectomy¹⁹), and may be covered with reflected dura or fascia)
4. midline medially (unless the tumor extends across the midline)

For approach to lower cranial nerves

(e.g., for glossopharyngeal neuralgia).

Craniectomy is extended inferiorly to $\approx$ 1/2 cm above foramen magnum.

Burr hole for emergency ventriculostomy

Optionally placed prophylactic occipital burr hole (**Frazier burr hole**) usually for intraparenchymal cerebellar tumors or any situation where post-op swelling or hydrocephalus is likely (not commonly used for microvascular decompression or small vestibular schwannomas).

Location: 3–4 cm from midline. In adults, 6–7 cm above theinion²⁰; in pediatrics, 2–3 cm above the transverse sinus²¹ (p 429) (i.e., $\approx$ 3–4 cm above theinion).

See Post-op management (p. 1745) for use.

105.3.5 Approach to the cerebellopontine angle (CPA)

The angle of approach determines which portion of the posterior fossa is visualized.

1. retracting the cerebellum *inferiorly* (working in the junction of the tentorium and petrous bone) gives access to the region of the trigeminal nerve, e.g., for microvascular decompression for trigeminal neuralgia
2. *medial* retraction gains access to the region of the porus acusticus, e.g., for vestibular schwannomas
3. *superior* retraction gains access to the lower cranial nerves, e.g., for geniculate neuralgia

105.4 Midline suboccipital craniectomy

105.4.1 Indications

Access to the midline or both sides of the posterior fossa

1. midline posterior fossa lesions
 - a) cerebellar vermal and paravermal lesions, including: vermian AVM, cerebellar astrocytoma near the midline
 - b) tumors of the 4th ventricle: ependymoma, medulloblastoma
 - c) pineal region tumors
 - d) brainstem lesions: brainstem vascular lesions (e.g., cavernous angioma)
2. decompressive craniectomies: e.g., for Chiari malformation
3. cerebellar hemisphere tumors: metastases, hemangioblastoma, pilocytic astrocytoma...

105.4.2 Position

See positioning (p. 1737). For midline lesions the Concorde position (p. 1737) is usually used.

105.4.3 Skin/fascia incision

Midline incision from $\approx$ 6 cm aboveinion to $\approx$ C2 spinous process. Take the incision a little higher if a Frazier burr hole is to be done (can then utilize the same skin incision). The skin incision should leave the muscles and fascia intact. It is often difficult to place Raney clips on the skin in this region. To facilitate water-tight closure, the fascia is "T'd" at the top, leaving a cuff of tissue on the occiput just above the superior nuchal line.

105.4.4 Craniectomy

Craniectomy implies removal of bone (often piecemeal) with no intention of replacing it. The advantage to not replacing the bone is that if there is post-op swelling, the inelastic bone flap may cause more pressure to be transmitted to the brainstem. A down-side to not replacing the bone is that local pain and/or "syndrome of the trephined" (p. 1763) may be more common. The bone opening is usually taken down to foramen magnum.

C1 removal: for cerebellar hemisphere tumors, many remove the posterior arch of C1 (caution re: vertebral arteries on superior aspect of C1). For 4th ventricular tumors, consider removing the arch if the tonsils extend below the foramen magnum.²²

105.4.5 Approach

A "Y"-shaped drotomy is often used. If the lesion has a cystic component, aspiration through a ventricular needle is used to partially decompress it.

105.4.6 Approaches to the 4th ventricle

Applied anatomy²³:

- 9 lobules of the cerebellar vermis: lingula, central lobule, culmen, declive, folium, tuber, pyramid, uvula, and nodule
- the tonsils are part of the cerebellar hemispheres. The 2 tonsils generally meet in the midline and must be separated to access the inferior 4th ventricle
- roof of the 4th ventricle
 - fastigium: the apex of the roof of the 4th ventricle and the dividing point between the superior and inferior portion of the 4th ventricle
 - the roof of the superior portion of the 4th ventricle is comprised primarily of the superior medullary velum and the medial surface of the superior cerebellar peduncles
 - the roof of the inferior portion of the 4th ventricle is formed by the tela choroidea and the inferior medullary velum. Key point: the roof of the inferior 4th ventricle is devoid of any known functional neural structures
- floor of the 4th ventricle
 - rostral two-thirds is the posterior aspect of the pons. Anatomical feature: the facial colliculus – a slight protuberance due to the internal genu (p. 607) of the facial nerve (Cr. N. VII)
 - caudal one-third is the posterior medulla oblongata
 - at the apex of the floor is the inferior terminus of the Sylvian aqueduct
 - the obex (where the 4th ventricle becomes the central spinal canal) is at the caudal end (deep and anterior to the foramen of Magendie)

General information:

- position, skin incision, craniectomy: as in Midline suboccipital craniectomy (p. 1742) using the Concorde position (p. 1737)
- the posterior arch of C1 does not need to be removed unless the tonsils extend inferior to the foramen magnum²²
- options:
 - neuromonitoring: SSEP/MEP, BAER
 - temporary pacemaker in case of bradycardia due to brainstem manipulation
 - image-guided navigation: if used, fiducials placed before pre-op imaging and kept in place until surgery usually helps with registration

- complications:
 - hydrocephalus: incidence as high as 30%^{24,25}; average is probably lower²²
 - cerebellar mutism (p.100): develops in up to 30%^{24,25,26}
 - other complications²⁶: dysarthria: 30%, dysphagia: 33%

Two main approaches to the 4th ventricle²³:

1. transvermian approach (splitting the cerebellar vermis in the midline)
 - overview: the inferior vermis is incised and the two halves are retracted to opposite sides
 - split the vermis to the smallest extent possible (usually up to the fastigium, but not into the superior medullary velum)
 - the nodule, tela choroidea, and inferior medullary velum also need to be incised to access the 4th ventricle
 - superior exposure is limited by the superior medullary velum
 - PROS: wider and slightly more rostral exposure than telovelar approach
 - CONS: risk of caudal vermis syndrome (truncal ataxia, dysequilibrium, oscillation of head and trunk, nystagmus), cerebellar mutism, injury to dentate nucleus (more severe dysequilibrium)
2. telovelar approach (TVA)^{22,23}
 - overview: exposes the 4th ventricle through the cerebello-medullary fissure without incising the vermis or cerebellar hemisphere
 - PROS: no functioning nerve tissue is harmed in the approach – which may reduce the risk of cerebellar dysfunction, including cerebellar mutism (p.100); improved access to the lateral recess of the 4th ventricle
 - CONS: narrower corridor than with a widely split vermis; limited access to deep or large tumors involving the rostral third of the 4th ventricle; limited access to contralateral floor of the 4th ventricle (TVA can be done bilaterally to circumvent this limitation²³)
 - technique:
 - the uvula is exposed by separating the two tonsils
 - on the side where the approach is planned, dissect between the tonsil and the adjacent uvula
 - the tonsil is retracted to the side of the approach and the uvula is retracted to the opposite side to expose the tela choroidea and inferior medullary velum
 - PICA is usually seen clearly at this time & should be protected (along with its branches)
 - the inferior aspect of large tumors can often be seen protruding through the foramen of Magendie under the thin inferior medullary velum and tela choroidea
 - the tela choroidea and inferior medullary velum are opened to one side of the vermis and the vermis is retracted as a whole
 - the floor of the 4th ventricle (brainstem) is protected by sliding a cottonoid up along the floor

105.5 Extreme lateral posterior fossa approach

Allows access to anterolateral region of brainstem. Differs from above in that the skin incision is designed to get the bulk of the skin and muscle flap out of the way.

Key: remove the lip of the foramen magnum as far laterally as possible, best done with a diamond drill.

105.6 Cranioplasty for suboccipital craniectomy

Methylmethacrylate cranioplasty as part of the closure following suboccipital crani for vestibular schwannoma reduced the incidence of post-op H/A from 17% to 4%.²⁷

105.7 Post-op considerations for p-fossa craniotomies

105.7.1 Post-op check

In addition to routine, the following should be checked:

1. respirations: rate, pattern (see Intubation below)
2. follow closely for hypertension (see below)
3. evidence of CSF leak through wound

105.7.2 Post-op management

Intubation

Post-op intubation for 24–48 hours is sometimes maintained on a precautionary basis: many complications often have respiratory arrest as the initial manifestation (see below), and the patient may deteriorate precipitously from this point. There is a trade-off as the stimulus of the endotracheal tube may exacerbate hypertension and patient agitation, and so sedation is often required, which may obscure the neuro exam and depress respirations. If the patient wakes up extremely well from an uncomplicated p-fossa crani and it is not late at night, most surgeons will extubate.

Hypertension

Hypertension should be avoided at all costs to prevent bleeding from tenuous vessels (e.g., nicardipine or clevidipine should be prepared prior to termination of the operation, and should be hanging and ready to titrate to keep SBP $\geq$ 160 mm Hg during the reversal of anesthesia and post-op).

Physician should be called for any sudden changes in BP post-op (may indicate elevated pressure in posterior fossa, see below).

105.7.3 Post-op complications

Posterior fossa edema and/or hematoma

In the posterior fossa, a small amount of mass effect can be rapidly fatal due to the paucity of room and the immediate transmission of pressure directly to the brainstem. It can also occlude CSF circulation through the aqueduct and cause *acute hydrocephalus* with the attendant risk of tonsillar herniation. Increased pressure in the p-fossa is usually heralded by sudden increases in BP or changes in respiratory pattern (pupillary reflexes, level of consciousness, and ICP are *not* affected until late). See (p. 1745) for emergency treatment measures.

Emergency treatment for p-fossa swelling

★ Rapid intubation, ventricular tap (through previously placed burr hole, if possible, see text), and reoperation are indicated. The wound should be opened immediately wherever patient is (recovery room, ICU, floor...). CT scanning may cost valuable minutes; it is rarely appropriate to delay treatment for this (must be judged on an individual basis).

To expedite ventricular taps, a prophylactic occipital burr hole (Frazier burr hole) is often placed during posterior fossa surgery to permit drainage of CSF from the lateral ventricles in the event of acute hydrocephalus from blockage of the 4th ventricle or aqueduct. If acute hydrocephalus develops (e.g., from a hematoma), an emergent percutaneous ventricular tap with ventricular needle (or, if not available, spinal needle) is performed, passing the needle through the burr hole aiming for the middle of the forehead. In the presence of acute hydrocephalus, CSF should be encountered at a depth of 3–5 cm. A final insertion depth of 10 cm is usually used. NB: this maneuver may provide a few more minutes while preparing for the definitive treatment of re-opening the wound; however, hydrocephalus may not initially be present since it takes some time to develop.

Suboccipital pseudomeningocele

An “internal” CSF fistula. Incidence following suboccipital craniectomy: 8²⁸–28%.²⁹

May be asymptomatic, but also may be associated with H/A, nausea/vomiting, local pain/tenderness. Some are soft and compressible, others may be tense.

Indications for operation:

1. external leak (CSF fistula, see below)
2. threatening integrity of incision
3. cosmetic deformity
4. causing symptoms

Treatment options (up to 67% require permanent CSF drainage³⁰):

1. noninvasive measures: expectant management, fluid restriction, head wrapping, keeping HOB elevated, acetazolamide. Steroids may be used if aseptic meningitis is suspected

2. percutaneous aspiration: "tap and wrap."^{21 (p 436),31} Risks introducing bacteria, causing infection
3. direct surgical exploration with multilayer re-closure^{21 (p 436)}
4. lumbar drain: effective only if pseudomeningocele communicates with the subarachnoid space.
 ✖ May produce acute posterior fossa syndrome (H/A, nausea, vomiting, ataxia...) ²⁸ especially if the pseudomeningocele doesn't communicate. Symptoms usually resolve with prompt discontinuation of lumbar drainage.^{28,29} Other potential complications: vagal nerve palsy, tonsillar herniation, subdural hematoma, kinking of PCA → stroke. Drainage options:
 - a) external drain (temporary)
 - b) lumboperitoneal shunt (permanent)
5. ventricular drainage
 - a) EVD (temporary)
 - b) shunt (permanent)

CSF fistula

Occurs in 5–17% of cases. A potential source of meningitis, thus CSF leak must be treated immediately.

Etiologies: controversial. May include:

1. abnormal CSF hydrodynamics (i.e., hydrocephalus). Maneuvers to stem the leak will likely fail until the CSF is shunted or hydrodynamics normalize
2. poor wound closure: probably blamed more often than it is the actual cause
3. subarachnoid scarring

May be associated with meningitis (aseptic or infectious), multiple operations. Formation may be facilitated by coughing/sneezing, postural changes, one-way ball-valve mechanism due to a tissue flap.

An external CSF leak may occur through:

1. the skin incision
2. via the eustachian tube; see possible routes of egress following suboccipital vestibular schwannoma removal (p.794):
 - a) through the nose (CSF rhinorrhea)
 - b) down the back of the throat
3. the ear (CSF otorrhea) in cases with perforated TM

Treatment

Initial treatment measures to temporize in the hope that CSF hydrodynamics will normalize and/or that the leak site will scar closed within a few days:

1. elevate the HOB
2. lumbar subarachnoid drainage
3. if the leak occurs through the skin incision:
 - a) reinforce the incision with sutures, e.g., running locked 3–0 nylon after preparation of the skin with antimicrobial and local anesthetic
 - b) alternatively, the incision may be painted with several coats of collodion

If persistent, a CSF fistula requires surgical correction, see CSF fistula (cranial); for general information (p.415), see CSF fistula following suboccipital removal of vestibular schwannoma (p.796).

Fifth or seventh nerve injuries

Causes diminished corneal reflex with potential corneal ulceration; initially managed with isotonic eye drops (e.g., Natural Tears®) q 2–4 hrs & PRN, or with a moisturizing insert (e.g., Lacricert®) q day, and at night with an eye patch or taping eyelid shut.

Miscellaneous

Supratentorial intracerebral hemorrhage has been described, and may result from transient hypertension.³²

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106 Supratentorial Craniotomies

106.1 Pterional craniotomy

106.1.1 Indications

1. aneurysms
 - a) all aneurysms of anterior circulation
 - b) basilar tip aneurysms
2. direct surgical approach to cavernous sinus
3. suprasellar tumors
 - a) PitNET/adenoma (when there is a large suprasellar component)
 - b) craniopharyngioma

106.1.2 Technique

Position, skin incision, craniectomy, approach...

1. supine, ipsilateral shoulder roll if head turned $> 30^\circ$ (see below)
2. elevate thorax $10\text{--}15^\circ$: reduces venous distension
3. flex knees
4. Mayfield 3 pin head-holder: applied between true AP and true lateral (so that it is $\approx$ horizontal when head is rotated to the necessary position, see ► Fig. 106.1)
5. neck extended 15° : allows gravity to retract frontal lobe away from skull base
6. head rotated from vertical as shown in ► Fig. 106.1

Room arrangement

1. microscope: observer tube to operator's *right* for either right or left pterional crani

Skin incision

See ► Fig. 106.2. From zygomatic arch 1 cm in front of tragus (to avoid frontalis branch of facial nerve and frontal branch of superficial temporal artery), curving slightly anteriorly, staying behind hairline to widow's peak, optional additional curve beyond midline to aid in skin retraction. Over temporalis muscle, incise skin down to but not through temporalis fascia.

The temporalis muscle may be incised caudal to the skin incision (i.e., closer to zygomatic arch); this minimizes the muscle mass that needs to be retracted inferiorly and yet keeps the scar behind the hairline (note: there is a greater risk of frontalis weakness with this technique than if the temporalis muscle is incised in-line with the skin incision).

Burr holes

Two burr holes are sufficient; made as far caudally as possible to minimize the amount of bone to be rongeur off to gain access to the floor of the middle cranial fossa. One burr hole is made at the posterior insertion of the zygomatic arch ("A" in ► Fig. 106.3); this burr hole may be placed slightly forward when exposure is centered over structures around the ACoA (e.g., suprasellar tumor). The second burr hole ("Z") is made at the intersection of the zygomatic bone (near the frontozygomatic suture), the superior temporal line, and the supraorbital ridge. The hole should be as low as possible on the orbit; aim the drill slightly superiorly to avoid actually entering the orbit. The dura is dissected off the inner table with a Penfield #3 dissector.

Craniotomy

The resulting bone flap is centered over the depression of the sphenoid ridge. Approximately 33% of the craniotomy is anterior to the anterior margin of temporalis muscle insertion, $\approx 66\%$ is posterior.

With the craniotome, starting at the frontal burr hole the craniotomy is taken anteriorly across the anterior margin of the superior temporal line, staying as low as possible on the orbit (to obviate having to rongeur bone, which is unsightly on the forehead). The distance "B" from the medial extent of the craniotomy to the frontal burr hole is 3 cm for anterior circulation aneurysms. For the approaches to skull base (e.g., Dolenc approach), distance "B" is larger and takes the opening to $\approx$ the

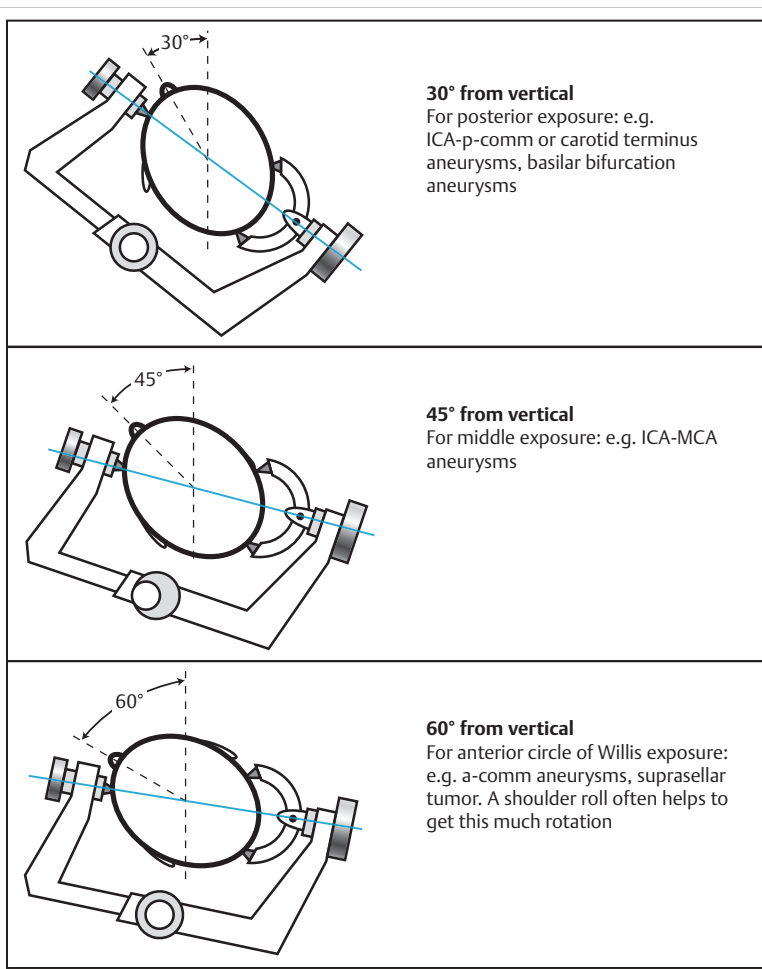


Fig. 106.1 Head position for pterional craniotomy depending on exposure required. The blue line indicates the approximate center line.

mid-orbit. Then from point "B," a sharp superior turn is made and the opening is taken back to point "A." The height ("H") of the craniotomy needs to be only ≈ 3 cm for aneurysms of the Circle of Willis, and slightly larger (≈ 5 cm) for middle cerebral artery aneurysms. Minimal exposure of temporal cortex is necessary for aneurysms of the skull base region. For large flaps (e.g., for tumors), "H" is made larger to expose more temporal lobe.

From the frontal burr hole, the craniotomy is then taken posteriorly towards the depression corresponding to the sphenoid wing until the drill hangs up.

The craniotomy from the posterior burr hole is taken forward towards the depression corresponding to the sphenoid wing until the drill hangs up.

The bone between the two points where the drill hangs up is scored with the craniotome, and then the bone is fractured at this point. A rongeur is used to remove as much sphenoid wing as possible.

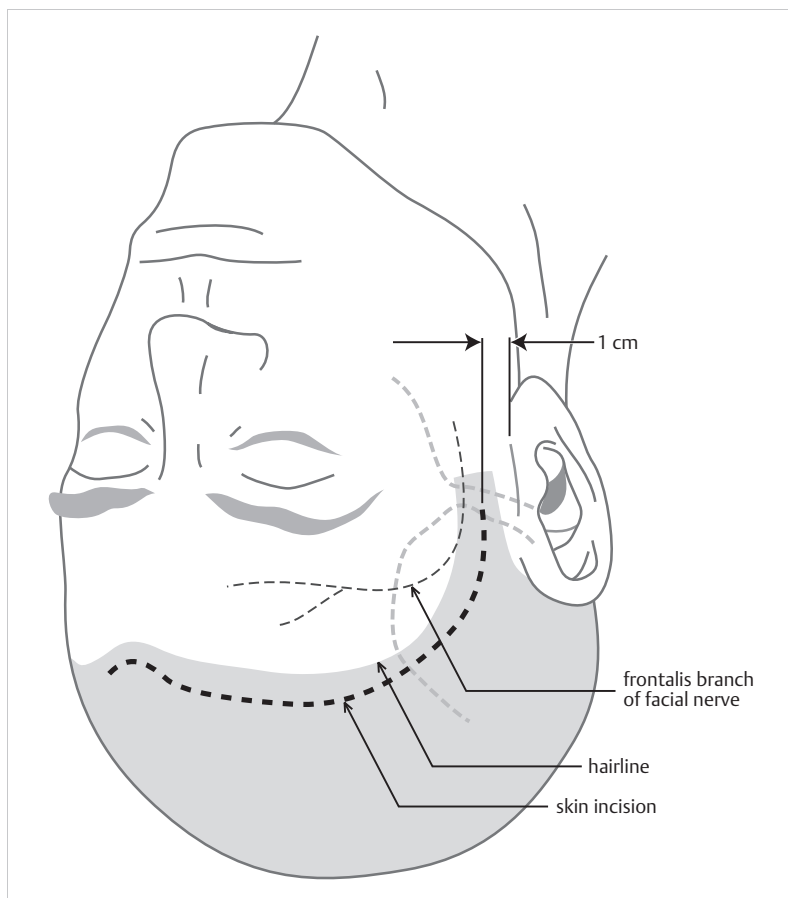


Fig. 106.2 Skin incision for pterional craniotomy.

Dural flap

Curvilinear, centered over sphenoid wing, retracted inferiorly with dural stitch.

Dissection

For some anterior circulation aneurysms (e.g., MCA aneurysms) and for the Yasargil approach to basilar tip aneurysms, the Sylvian fissure needs to be split. This can be accomplished by working from the lateral aspect of the fissure medially, or by starting at the point where the carotid artery penetrates the fissure and working laterally. The latter method may be easier when prolific veins overlie the junction of the frontal and temporal lobe. There are no arteries that cross the Sylvian fissure, and so if the correct plane is maintained, no arteries need to be sacrificed.

► Fig. 106.4 shows a theoretical exposure of the circle of Willis possible through a pterional craniotomy. This diagram is semi-schematic, and in reality dissection would be directed either anteriorly (e.g., to expose ACoA) or posteriorly (e.g., for basilar tip aneurysms) but not both.

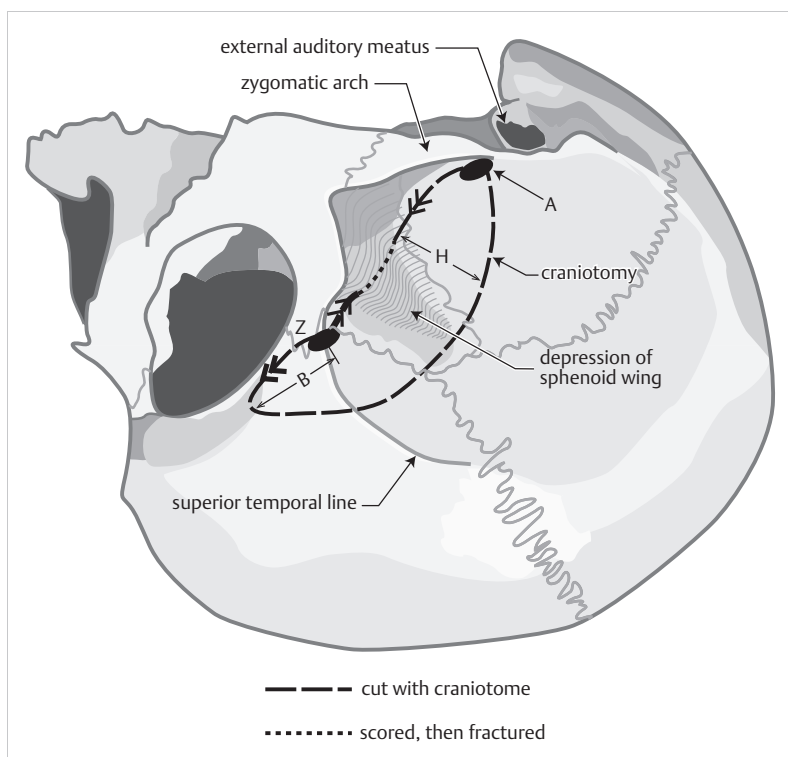


Fig. 106.3 Skull landmarks for right pterional craniotomy.

106.2 Temporal craniotomy

106.2.1 Indications

1. temporal lobe biopsy: herpes simplex encephalitis
2. temporal lobectomy: for resection of seizure focus, decompression post-trauma...
3. hematoma (epidural or subdural) overlying temporal lobe
4. tumors of the temporal lobe
5. small, laterally located vestibular schwannomas¹
6. access to the floor of the middle cranial fossa (including foramen ovale/Meckel's cave, the labyrinthine and upper tympanic portion of the facial nerve)
7. access to medial temporal lobe e.g., for amygdalo-hippocampectomy (p. 1893) or for mesial temporal sclerosis (p. 482)

106.2.2 Technique

See ► Fig. 106.5. Two basic methods for temporal craniotomy:

1. small craniotomy or craniectomy through a linear skin incision: good for cortical biopsy or draining chronic subdural hematoma. Also permits access to floor of middle fossa. Simple quick closure
2. question-mark skin incision with standard craniotomy flap: useful for temporal lobe exposure for tumor or acute hematoma

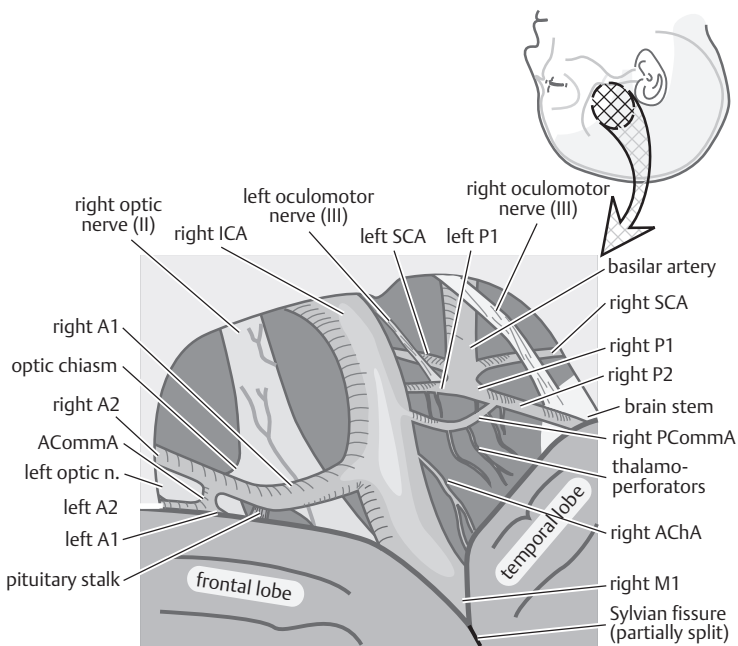


Fig. 106.4 Right-sided Sylvian fissure dissection, surgical view.

106.2.3 Position, skin incision, craniectomy, approach...

1. patient supine with shoulder roll (to assist in rotating neck to get head almost horizontal)
2. elevate thorax 10–15°: reduces venous distension
3. flex knees slightly
4. Mayfield 3 pin head-holder: true AP with single pin anteriorly
5. head rotated almost horizontal to floor: avoid over-extending to prevent kinking neck veins

106.2.4 Craniotomies

Small craniectomy

Linear skin incision completely within the extent of the temporalis muscle. To access the temporal tip: place the incision midway between the lateral canthus and external auditory canal (EAC); extend it from the zygomatic arch upward for ≈ 6 cm. For small, laterally located vestibular schwannomas, the incision is made 0.5 cm anterior to the EAC, extending ≈ 7–8 cm above the zygomatic arch.¹ To drain a subdural, place the incision just anterior to the tragus and start it 1–2 cm above the zygomatic arch for ≈ 6 cm (modified based on the location of the epicenter of the subdural). Take the incision down to temporalis fascia with the knife, and incise the fascia and muscle with Bovie cautery. Spread with self-retaining retractors, and make a burr hole. Enlarge with rongeurs and/or Kerrison punches.

Standard craniotomy

Question-mark skin incision

See ► Fig. 106.5. Used for access to the temporal lobe including tip (a reverse question mark incision may be used to gain access to the middle and posterior temporal lobe).

1. the pinna is either sutured inferiorly out of the way before draping, or it can be folded under the drapes which may be stapled to the skin

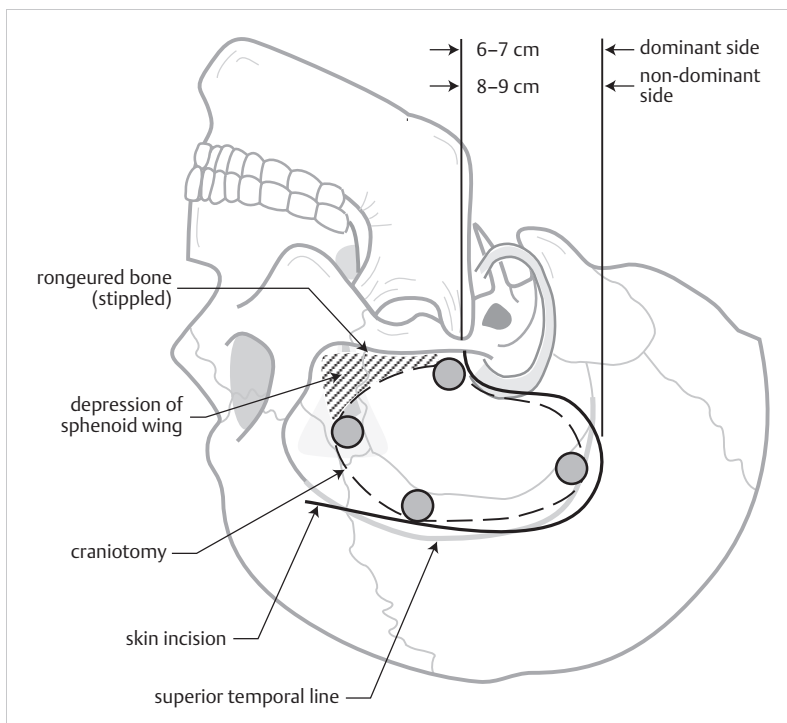


Fig. 106.5 Temporal craniotomy (exposing entire temporal lobe).

- the lower limb extends from the zygomatic arch just anterior to the tragus (to avoid the superficial temporal artery)
- curve as far posteriorly as $\approx 6-7$ cm on the dominant side, or $\approx 8-9$ cm on the non-dominant side at the level of the top of the pinna (these dimensions allow access to the “safe” area of temporal tip for lobectomy)
- then superiorly to the level of the superior temporal line
- then anteriorly towards the forehead, stopping at the hairline

Burr hole placement

- at the posterior insertion of the zygomatic arch
- at upper anterior junction of zygomatic arch
- one or two burr holes along posterior and superior aspect of the skin incision

Craniotomy

Connect the burr holes with the craniotome, keeping as low as possible in the middle fossa to minimize the amount of bone that must be rongeured. The remaining bone is rongeured down to the floor of the middle fossa (cross-hatched area in ► Fig. 106.5).

Temporal lobectomy

✗ Danger points:

- dominant hemisphere: Wernicke’s speech area. Although variable (see Temporal lobectomy (p. 1893)), one can usually safely resect up to 4–5 cm from temporal tip without use of mapping techniques to localize speech

2. non-dominant hemisphere: one can resect up to 6–7 cm before running the risk of injuring the optic radiation
3. Sylvian fissure (middle cerebral artery): it is best to amputate the temporal lobe backward from the tip for the extent of the desired resection, and then work deep
4. medially, the incisura should be identified to avoid injury to the brainstem which lies just medial to this

106.3 Frontal craniotomy

106.3.1 Indications

1. access to frontal lobe: e.g., for infiltrating tumor
2. approach to 3rd ventricle or to sellar region tumors in some situations, including craniopharyngiomas, planum sphenoidale meningiomas
3. repair of ethmoidal CSF fistula

106.3.2 ✕ Danger points

1. anterior cerebral arteries in the midline (deep)
2. superior sagittal sinus (SSS) in the midline (note: the SSS may be sacrificed in its anterior third without engendering venous infarction in most cases, whereas venous infarction will almost always occur with division of the SSS posterior to that)
3. avoid inadvertently crossing the midline into the contralateral hemisphere through the corpus callosum
4. dominant hemisphere: Broca's (motor speech) area is located in the inferior frontal gyrus

106.3.3 Technique

Craniotomy options

Two basic choices for craniotomy:

1. unilateral craniotomy through a curved skin incision taken anteriorly up to the hairline: used when one does not need to be low in the frontal fossa in the midline (otherwise the skin incision would have to be taken far into forehead) and when there is no need to cross the midline
2. large bifrontal skin incision from "ear-to-ear" (souttar skin incision²) allowing low approach to one or both frontal fossa

Unilateral frontal craniotomy

► Fig. 106.6. Skin incision starts < 1 cm anterior to the tragus, and does not need to go all the way down to the zygomatic arch. It curves superiorly and slightly posteriorly before being taken to the midline frontally.

Burr holes

1. at the junction of the superior temporal line and the orbital rim
2. just posterior to the depression of the sphenoid wing (behind the pterion)
3. anteriorly just behind the hairline to avoid having a burr hole under the forehead (which causes an unsightly depression)
4. superiorly

Bilateral frontal craniotomy

1. "ear-to-ear" or souttar skin incision
 - a) just behind hairline with a slight widow's peak at the front
 - b) does not need to go all the way to the zygomatic arch, it just needs to be ≈ as low as the orbital roof
 - c) unlike pterional craniotomy, usually do not need to incise the temporalis muscle and fascia. Dissect the flap off the muscle/fascia
 - d) if a periosteal flap is likely to be needed, it sometimes helps not to incise the periosteum at the same time as the skin incision. Then, the periosteum can be incised behind the skin incision to yield a longer periosteal graft than would have otherwise been obtained

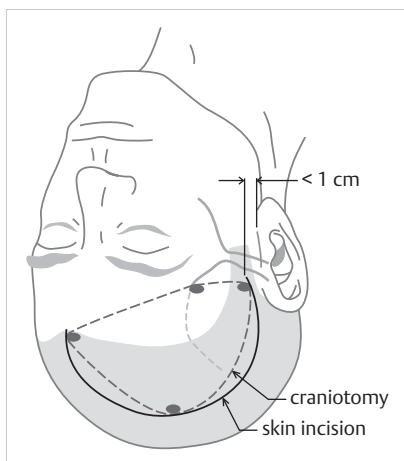


Fig. 106.6 Unilateral frontal craniotomy.

2. burr holes: to avoid burr hole defects on the forehead, the bone flap can be created with two burr holes straddling the SSS close to the skin incision, and two burr holes laterally
3. the SSS may be divided low, near the orbital roof, with little risk
4. if the frontal sinus is entered, it is dealt with as outlined under Frontal sinus fractures (p. 1066)

106.4 Petrosal craniotomy

106.4.1 Indications

1. lesions of the petrous apex (e.g., petroclival meningiomas)
2. lesions of the clivus (e.g., chordomas) with both posterior fossa and supratentorial components

106.4.2 Advantages

Sparses sinus and otologic apparatuses. Minimizes cerebellar and temporal lobe retraction.

106.4.3 Technique

See reference.³

Position

1. patient supine, ipsilateral shoulder roll
2. elevate thorax 10°: reduces venous distension
3. flex knees
4. Mayfield 3 pin head-holder: close to true AP with single pin on forehead
5. head positioned to place petrous base at highest point of field:
 - a) head rotated 40–60° from vertical
 - b) head abducted towards contralateral shoulder
 - c) neck extended 15°: allows gravity to retract frontal lobe away from skull base

Skin incision

Reverse question mark starting from zygomatic arch 1 cm anterior to tragus, arcing posteriorly over ear, descending to 0.5–1 cm medial to mastoid notch.

Temporalis muscle and periosteum reflected anteriorly and inferiorly.

Craniotomy

Four burr holes are utilized, two on each side of the sinuses (near the junction of the transverse and sigmoid sinuses).

106.5 Occipital craniotomy

106.5.1 Indications

Occipital lobe tumors including posterior falx meningiomas or tentorial meningiomas with only supratentorial component. Occipital lobe intracerebral hemorrhages.

106.5.2 Positions

Supine

Shoulder roll on affected side; elevate thorax 15°. Mayfield 3 pin head-holder with single pin in forehead off to the side of the crani, double pin just over midline on opposite side.

Lateral oblique

1. affected side up, can operate either
 - a) from behind patient similar to p-fossa crani for CPA lesion
OR
 - b) from top of table
2. alternative approach: affected side down. Useful in lesions adjacent to the falx; see **Interhemispheric approach** (p. 1762)

References

- | | |
|---|---|
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|---|---|

107 Approaches to the Lateral and Third Ventricles, Decompressive Craniectomies, and Cranioplasty

107.1 Approaches to the lateral ventricle

Surgical approaches to the trigone¹

Classic review² (p 561–74) summarized:

1. atrium (AKA trigone); numerous approaches include¹:
 - a) middle temporal gyrus: through the dilated temporal horn
 - b) lateral temporal parietal
 - c) superior parietal occipital
 - d) transcallosal (see below)
 - e) transtemporal horn: access to temporal horn is via lobectomy of the temporal tip
 - f) occipital lobe incision or occipital lobectomy: recommended only if patient has homonymous hemianopsia pre-op
2. frontal horn
 - a) middle frontal gyrus
3. midventricular body
 - a) transcallosal
 - b) middle frontal gyrus: usually prevents access to vascular supply until most of the tumor is removed (especially for tumors supplied primarily by posterior choroidal artery)
4. temporal horn
 - a) middle temporal gyrus
 - b) transtemporal horn

107.2 Approaches to the third ventricle

107.2.1 General information

Classic references review the microsurgical anatomy³ and surgical approaches,⁴ and are briefly summarized below.

Alternative approaches for lesions of the anterior 3rd ventricle⁵:

1. transcortical: approach is through the lateral ventricle and is feasible only in the presence of hydrocephalus; especially useful if the tumor extends from the 3rd ventricle into one of the lateral ventricles. Risk of seizures is 5% (higher than with transcallosal) (p. 1762).
2. transcallosal: may be preferable in the absence of hydrocephalus (see below)
 - a) anterior transcallosal: good visualization of both walls of 3rd ventricle; risk of bilateral fornix damage
 - b) posterior transcallosal: allows approach to quadrigeminal plate or pineal region; risk of damage to deep veins
3. subfrontal: allows four different approaches
 - a) subchiasmatic: between optic nerve and optic chiasm
 - b) optico-carotid: through the triangular space bordered by optic nerve medially, carotid artery laterally, and ACA posteriorly
 - c) lamina terminalis: above the optic chiasm⁶
 - d) transphenoidal: requires removal of tuberculum sellae, planum sphenoidale, and anterior wall of the sella turcica
4. transphenoidal
5. subtemporal
6. stereotactic: may be useful for aspiration of colloid cysts; see Stereotactic drainage of colloid cysts (p. 947)

107.2.2 General principles of tumor removal

Summarized.⁴ During the approach, deep veins should be preserved at all costs, even if it means stretching them to the point that they may rupture.

It is helpful to place a suture through the tumor capsule to act as a tether.

The tumor should first be removed from within the capsule; techniques include aspiration, and then opening the capsule and debulking from within. The capsule may then be collapsed and

dissected from adherent structures. If the capsule adhesions seem unyielding, the most likely cause is incomplete intracapsular evacuation.

Vessels on the surface of the tumor should be presumed to be supplying normal brain, and should be dissected off the capsule once it is completely emptied.

107.2.3 Transcallosal approach to lateral or third ventricle

General information

Performed through an interhemispheric approach to the corpus callosum (CC) via a parietal craniotomy, usually right-sided in a left-hemisphere-dominant patient.

Indications

Primarily for tumors or lesions of the lateral or 3rd ventricle, including:

1. colloid cysts
2. craniopharyngiomas
3. cysticercosis cysts
4. thalamic glioma
5. AVM

Booking the case: Transcallosal surgery

Also see defaults & disclaimers (p. 25).

1. position: supine with pin head-holder
2. equipment:
 - a) microscope
 - b) image-guided navigation system
3. post-op: ICU
4. consent (in lay terms for the patient—not all-inclusive):
 - a) procedure: operation between the two halves of the brain to remove lesion
 - b) alternatives: non-surgical management, surgery through the surface of the brain (transcortical), radiation therapy for some diagnoses
 - c) complications: stroke, “disconnection syndrome” (uncommon) (p. 1893), hydrocephalus with possible need for a shunt, memory deficits

Technique

See references.^{3,4,7}

General information

► Fig. 107.1. Image-guided navigation is very helpful in ascertaining the correct trajectory which permits minimizing the size of the callosotomy, and helps distinguish the corpus callosum from the cingulate gyri.

Position

Supine with neck flexed. Thorax elevated 20°. Spinal drain *not* used. Keep the head perfectly vertical to minimize disorientation that can easily occur with this approach. Alternatively, gravity retraction may be employed either by tilting the head slightly to the right (causing the right hemisphere to fall away) or by using the lateral position.

Skin incision

Either of the following may be used:

1. inverted “U” with the top just left of midline, extending from 6 cm anterior to the coronal suture to 2 cm behind the coronal suture, taking the sides for 7–8 cm
2. scuttar skin incision

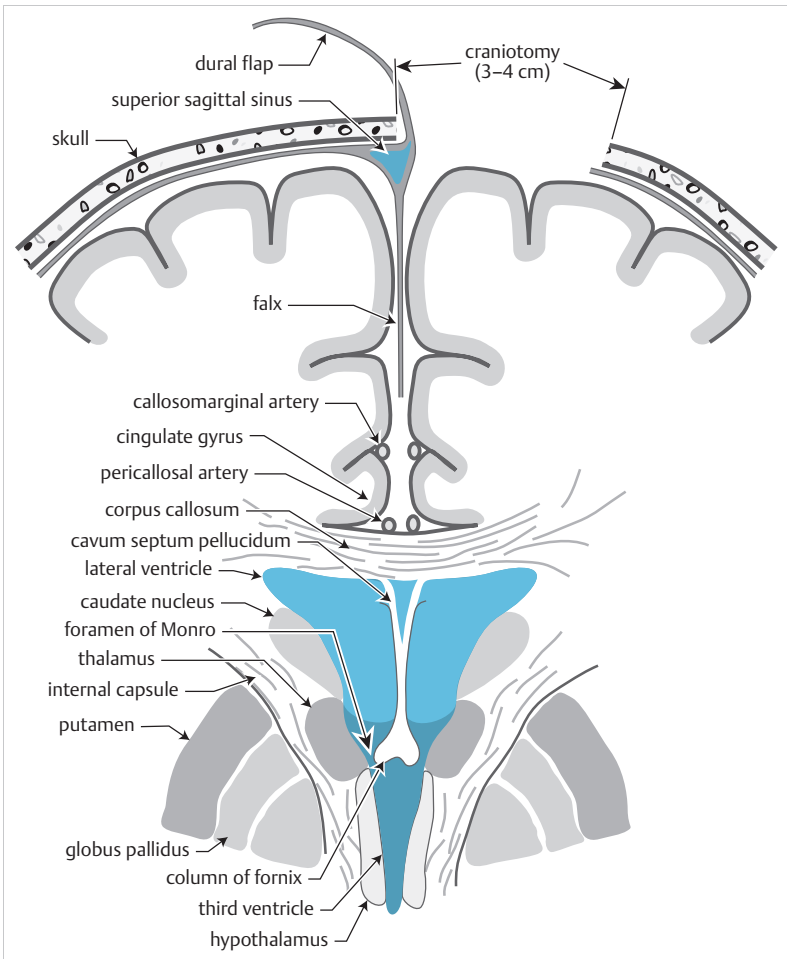


Fig. 107.1 Transcallosal approach to the third ventricle: frontal view.

Craniotomy

Pre-op angiography is recommended to plan the position of the flap to avoid sacrificing large cortical veins. MRI may also suffice for this.⁸ There tend to be fewer veins bridging from the cortex to the superior sagittal sinus (SSS) anterior to the coronal suture; therefore, this is often a good location to enter the interhemispheric fissure. The bone flap is either trapezoidal or triangular in shape, for adequate exposure it is *critical* to go all the way to the SSS. Several techniques may be used. NB: the SSS is often to the right of the sagittal suture (p.61).

1. to expose the SSS, straddle the SSS with paired burr holes anteriorly and posteriorly, dissect the dura from the inner table between pairs, and make the longitudinal cut on the *left* of midline. Disadvantage: removing the midline bone puts the SSS at greater risk of injury and makes it more difficult to control lacerations
2. one can make the long cut well to the right of midline, and then under direct vision rongeur off the bone to the SSS. Safe, but leaves a large bone gap that may need to be filled (e.g., with methyl methacrylate) and is time-consuming

- most risky for causing a sagittal sinus laceration is to make the long cut just right of the midline (over the edge of the SSS, which may lacerate it)

To stay away from the motor strip (p.56) and to keep the sagittal sinus exposure as anterior as possible, 2/3 of the opening should lie anterior to coronal suture, 1/3 posterior (generally: 6 cm total with 4 cm anterior and 2 cm posterior). The craniotomy extends laterally to 3–4 cm to right of midline. The last cut with the craniotome should connect the burr holes along the sinus (midline); leaving this cut for last permits rapid access to the sinus in case it is torn. The dural flap is based towards the sagittal sinus.

Approach to corpus callosum

None, or at most, only one bridging vein from the cortex to the sagittal sinus may be sacrificed (and then, only if it is not a large draining vein). Gently retract the right hemisphere. Avoid retractors on the sagittal sinus to prevent injury to the SSS which may lead to sinus thrombosis (once CSF is released (with the callosotomy) retraction will be easier). Enter the interhemispheric fissure and follow the falx deep. Open the arachnoid membrane beyond the deep edge of the falx.

The two cingulate gyri may be adherent in the midline, and can easily be mistaken for the corpus callosum (CC). This error may be compounded by mistaking the callosomarginal arteries for the pericallosal arteries. Erroneously entering the cingulate gyrus disorients the surgeon and could cause injury to the pericallosal arteries. To differentiate: the CC is a pure white structure, is usually deeper than one anticipates, and is appreciated beneath the paired pericallosal arteries. Image-guided surgery or measuring the depth to the CC on the pre-op MRI may help.

Callosotomy

The callosotomy is usually performed between the two pericallosal arteries. Some arterial branches may cross the midline, and occasionally it is necessary to sacrifice some. Trajectory: a line drawn from the coronal suture (in the midline) to the external auditory canal (the foramen of Monro lies along this line); this helps avoid the tendency to tunnel posteriorly through the CC. Either the bipolar cautery, suction and sharp knife, or the laser is used to make the callosotomy. In hydrocephalus, the callosum will be thin. Entering the lateral ventricle releases CSF, which aids retraction. When the foramen of Monro is occluded (e.g., with colloid cyst), it helps to fenestrate the septum pellucidum to prevent it from bulging into the ventricle in which one is operating (otherwise, as CSF is aspirated from the ipsilateral lateral ventricle, it cannot escape from the other).

Disconnection syndrome (p.1893): more common with posterior callosotomy (near the splenium) where more visual information crosses. The risk is reduced by creating a callosotomy <2.5 cm in length, extending posteriorly from a point 1–2 cm behind the tip of the genu.⁹ For an interforaminal approach, the callosotomy must be perfectly midline.

Approach to third ventricle

Usually, the callosotomy will not be exactly midline, and one of the lateral ventricles will be entered. Great care must be taken to correctly identify which lateral ventricle has been entered, another potentially disorienting pitfall. For orientation (► Fig. 107.2), the choroid plexus passes forward in the choroidal fissure to the foramen of Monro (which is medial) where it converges with the thalamostriate vein approaching from a more lateral position in the groove between the thalamus and caudate. The septal and caudate veins approach the foramen from anterior. With colloid cysts, the foramen of Monro may be hard to recognize initially as it will be plugged with the cyst, which can resemble the ependymal lining of the ventricle, but on close inspection is usually slightly grayer (the choroid plexus enters the posterior aspect of the foramen).

Another possible pitfall upon incising the CC is entering a cavum septum pellucidum (p.1657). The give-away here is that *no* landmarks will be visible.

Alternative approaches to third ventricle

- interforaminal⁹: go *above* body of fornix, approaches *roof* of 3rd ventricle. Well suited for lesions of the mid- and posterior 3rd ventricle. Callosotomy should be as close to midline as possible
- from lateral ventricle through the foramen of Monro: with hydrocephalus, the foramen of Monro is usually dilated. If the foramen is too small for adequate access to the 3rd ventricle, one can
 - return to the interforaminal approach (see above) or
 - enlarge the foramen of Monro only if absolutely necessary. Either by:
 - opening the foramen laterally

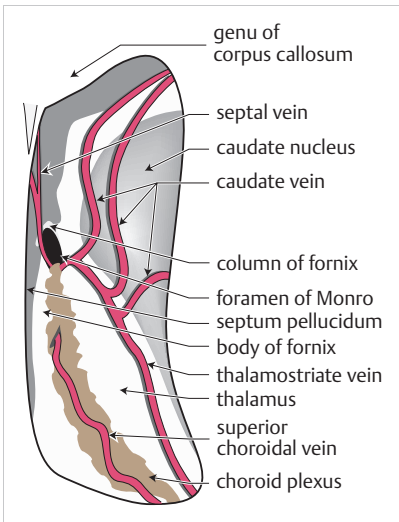


Fig. 107.2 Right foramen of Monro viewed from above through right lateral ventricle.³

- the “subchoroidal” approach, making the incision posteriorly (sacrificing the thalamostriate vein) which is reportedly well tolerated^{4,10}
- last resort: incising the anterosuperior margin of the foramen through the column of one fornix.⁹ Caution: if the other fornix is non-functional for any reason, this would produce a bilateral forniceal lesion and may (but not definitely⁹) result in loss of short-term memory and ability for new learning

Colloid cyst removal

It is critical to debulk and empty a cystic lesion, such as a colloid cyst, before delivering it from the 3rd ventricle through the foramen of Monro. This will minimize the retraction and manipulation of the fornix. Inserting a needle and aspirating may work. The partially emptied cyst is grasped with a micro-pituitary and is delivered into the lateral ventricle through the foramen of Monro. One should only attempt to deliver the empty capsule through the foramen of Monro (p. 1757). There is usually a stalk attaching the lesion to the roof of the 3rd ventricle, this is coagulated with bipolar cautery and divided.

For other tumors, if the tumor is too large to fit through the foramen of Monro, it should be gutted from within.

Complications

1. venous infarction, may be due to:
 - a) sacrifice of critical cortical draining veins: plan the flap to avoid this with preoperative angiography, or with sagittal T2WI MRI images¹¹
 - b) superior sagittal sinus (SSS) thrombosis.¹² Factors that may contribute to sinus injury include⁸:
 - injury from retractor: avoid placing retractor on sinus (deformation of midline should not exceed 5 mm)
 - over-retraction of the dural sinus flap or on SSS itself (lateral deformation should be < 2 cm)
 - injury during the opening of the bone in the region of the sinus
 - over-use of bipolar coagulation in the region of the SSS
 - hypercoagulable state of the patient, including dehydration
2. transient mutism as a result of bilateral cingulate gyrus retraction or thalamic injury in conjunction with section of the midportion of the callosum¹¹

107.2.4 Transcortical approach to lateral or third ventricle

Indications

In the absence of hydrocephalus, it is difficult to navigate through the ventricular system. Thus, with normal sized ventricles, the 3rd ventricle and region of the foramen of Monro are better approached transcallosally (p. 1758).

1. tumors of the atrium of the lateral ventricle
2. tumors of the roof of the 3rd ventricle
3. 3rd ventricular tumors with significant extension into one lateral ventricle

Approaches

1. posterior parietal
2. middle temporal gyrus: useful when temporal horn of lateral ventricle is dilated due to hydrocephalus caused by the tumor; access is through the temporal horn
3. middle frontal gyrus approach: a 4-cm incision is made parallel to the axis of the middle frontal gyrus, above and anterior to the expressive speech center (Broca's area) and anterior to the motor strip⁴; about the same point as used for frontal ventriculostomy, see Kocher's point (p. 1821)

107.3 Interhemispheric approach

107.3.1 Indications

For lesions abutting on midline, deep to surface, but superficial to corpus callosum (lesions that can "fall away" from midline). Similar to transcallosal approach above, except that the pathology can be placed on the down-side, which allows gravity to retract the hemisphere and thus minimizes pressure necrosis injury from mechanical retractors.

107.3.2 Technique

Position

True lateral (prevents getting lost from unusual angles). Head tilted slightly up.

Approach

Similar to transcallosal (p. 1758). Need to be sure that lateral portion of craniotomy extends at least 4 cm from midline to minimize the necessity of retraction of brain against bone.

107.4 Cranioplasty

107.4.1 Indications/contraindications

General indications

1. cosmetic restoration of external skull appearance and symmetry
2. relief of symptoms due to craniotomy defect (p. 1731)
 - a) pain or tenderness: especially at the bone edges
 - b) syndrome of the trephined: *nonfocal* (see below)
 - c) focal deficit related to the defect: e.g., sinking skin syndrome (see below)
 - d) seizures originating in the brain beneath the defect
3. protection from trauma (blunt or penetrating) in area of post-craniotomy or posttraumatic skull defect
4. reduction of irritation of the brain as a result of pressure on and deformity of the surface of the brain. Reducing this irritation may improve seizure control if this is an issue
5. cognitive deficits: may improve following cranioplasty (especially with large defects)¹³

Onset of symptoms may be delayed months to years after the craniectomy.

Contraindications

1. infection: especially at the craniectomy site, but also at distal sites because of the fear of seeding the cranioplasty flap
2. brain protruding beyond the confines of the skull
3. untreated hydrocephalus: this is complicated because treating hydrocephalus may require repairing the bone defect, usually in combination with a shunt

Syndrome of the trephined

Symptoms first described in the French literature during World War I. The term “syndrome of the trephined” was coined by Grant & Norcross,¹⁴ at that time the description consisted of nonfocal symptoms including: headache (present in 54%¹⁵) and sometimes pulsatile pain (usually localized to the area of the skull defect), vertigo (24%), amnesia, inability to concentrate, insomnia, fatigue, depression... Similar in many ways to postconcussive syndrome (p. 1111).

Since then the definition has been liberally expanded to include delayed focal neurologic symptoms,¹⁶ tinnitus,¹⁷ and even “any symptoms reversible with cranioplasty.”¹⁸
The symptoms sometimes change with posture.¹⁹

Other syndromes related to the bone defect

For those purists who want to restrict the use of the term “syndrome of the trephined” to nonfocal symptoms described above, the following syndromes have been proposed.

Sinking skin flap syndrome²⁰: neurologic deficit related to brain immediately underlying the skull defect when the scalp is displaced inward by atmospheric pressure. Deficits may include contralateral hemiparesis (more common in UE than LE due to location of UE motor neurons over the convexity; see ► Fig. 1.3), contralateral visual and somatosensory deficit, speech/language deficit (with dominant hemisphere defects). Not all patients with a sunken flap develop a deficit.

Motor trephine syndrome¹⁶: limited to contralateral hemiparesis. Mean delay to onset: 5 months.

107.4.2 Etiology of symptoms

Possible explanation for symptoms related to the bone defect:

1. atmospheric pressure exerted directly on the surface of the brain including compression of cortical veins
 - changes in cerebral blood flow (CBF)
 - impaired cerebral glucose metabolism
 - local alteration in CSF flow
2. globally altered CSF flow dynamics
3. scar tissue adhering to the dura and/or brain
4. acute angulation of scalp at bone edges stimulating pain fibers causing local pain
5. pulsations (CSF and/or blood) causing irritation of (pain-sensitive) dura

The effects of barometric pressure are modified by:

1. scalp elasticity
2. size of the bone defect
3. amount of skull curvature at defect site
4. changes in CSF pressure with: postural changes (incorrectly referred to as “siphoning,” in reality is a reduction in pressure of a fluid column (hydrostatics)), coughing/straining, hydrocephalus

107.4.3 Timing of cranioplasty

- brain swelling: enough time needs to have lapsed so that any brain swelling that was present has subsided to the point that the brain is no longer protruding beyond the normal confines of the intracranial space (you don’t want to be “pushing brain” back into the skull with the cranioplasty flap). This sometimes requires shunting hydrocephalus if present
- contamination status of wound
 - skull defect from clean wounds (following decompressive craniectomy, post-craniotomy brain swelling...): recent studies found no significant difference in *infection* between early and late (variously defined) cranioplasty (contaminated cases were not included).^{21,22} However, one series reported an increased risk of infection for repair of a decompressive craniectomy within 14 days.²³

Clean surgical cases without brain swelling (e.g., repair of defect after removing skull hemangioma); there appears to be little risk of infection with immediate cranioplasty

- contaminated wound (open fracture, fractures that traverse the nasal sinuses, penetrating trauma, infection...): classic teaching was to delay cranioplasty using allograft (a “foreign body” e.g., PEEK implant) at least 6 months to reduce the risk of infection. Some authors even recommended waiting > 1 year.²⁴ More contemporary practice seems to be to wait ≈ 3–6 months after the craniectomy if there are no signs of infection
- extenuating circumstances: consideration may be made to shorten the delay before cranioplasty e.g., if the patient is having significant symptoms related to the absent area of bone (e.g., “syndrome of the trephined” or “sinking skin flap” syndrome)

107.4.4 Material

Options for material include:

1. the patient's own bone. Bone that is removed at the time of craniectomy that has been preserved for future implantation. This is generally not employed in contaminated cases (penetrating trauma, infection...). Storage options:
 - a) in a “pocket” created in the patient's subcutaneous abdominal fat
 - b) in preservative (e.g., RPMI (p. 1766)) and stored in ultralow-temperature freezer
2. materials that can be formed by the surgeon
 - a) polymethylmethacrylate (PMMA): created by mixing methylmethacrylate powder with a liquid methylmethacrylate monomer in the O.R., which is molded to desired shape, and allowed to set (harden) before being attached to the skull with plates, sutures, or wire. The setting reaction is exothermic, and to prevent heat injury to the underlying brain, insulate the brain with wet surgical sponges and irrigate copiously during the setting process, or preferably, once the material is reasonably firm it can be removed from the operative site to complete the setting
 - b) mesh: may be made of titanium or tantalum. Options include:
 - standard flat mesh. Can be molded to a limited degree. Usually better for smaller defects, e.g., < 5 cm diameter
 - preformed mesh (e.g., SmartMesh by KLS Martin, or CranioCurve® by Zimmer Biomet): mesh implants that are contoured to average skull shapes and common craniectomy configurations
3. pre-fabricated custom bone flaps: commercially manufactured using 3-D printing based on thin-cut CT scans of the defect and, if available, utilizing a “mirror image” of the intact contralateral side as a model for the desired contour
 - a) polymethylmethacrylate (PMMA)
 - b) PEEK (poly-ether-ether-ketone)
 - c) titanium
 - d) tantalum
 - e) acrylic
4. split-thickness calvaria

When synthetic material is used, the flap should be perforated with a dozen or so drill holes to prevent the accumulation of fluid (either underneath the flap, or between the flap and the skull).

107.4.5 Complications

Complications include:

- a) infection: ≈ 8% risk²¹
- b) hematoma: under the cranioplasty flap (epidural or subdural)
- c) seizures
- d) brain injury
- e) bone flap resorption (BFR): when using autologous bone (the patient's own bone). Reported range: 3–51%.²⁵ When BFR occurs, the rate of resorption is about 3%/year.²⁵ Risk factors have been postulated, including length of time to reimplantation²⁶ but so far only younger age has been shown to correlate with increased incidence of BFR²⁵
- f) hydrocephalus
- g) implant dislodgement

Risk of complication is increased with bifrontal bone defects.^{27,28,29}

107.4.6 Technique

General information

The following pertains primarily to cranioplasty following a decompressive craniotomy.

Surgical objectives

- separate the temporalis muscle from where it has scarred onto the dura
- avoid CSF leak by not violating the dura (or pseudo-dura) or by closing any opening that is identified
- repair the bone defect with a bone flap (see above for material options)
- replace the temporalis muscle outside the bone graft and, if necessary, tack it into position

Risks

- infection
- post-op hematoma: epidural more likely than subdural
- CSF leak

Surgical details

The following pertains in particular to decompressive craniectomy defect which extends from the parietal and frontal regions to include varying amounts of bone overlying the middle fossa.

- re-incise the previous skin incision, being careful to stay on bone where possible and, where the incision is not over bone, by using, e.g., a hemostat under the skin to prevent the scalpel from intracranial penetration
- starting at a point near the superior-most aspect of the defect, begin to separate the scalp flap from where it is scarred to the dura or pseudodura for a short distance (a couple of centimeters or so) inside the bone edges. This is usually easier if a barrier (e.g., silastic sheet) was placed at the time of the craniectomy. A Langenbeck periosteal elevator may work in areas where the planes separate easily, a #10 scalpel used with the sharp side of the blade pointing up may be used where scarring is more tenacious
- work around the defect in both directions towards the base of the flap, which is where the caudal aspect of the temporalis muscle crosses the edge of the defect to the outside of the skull
- as long as the tissue is thin (i.e., scar only, no temporalis muscle) use monopolar (Bovie) cautery along the bone as close as possible to the bone edge to expose the bone
- when you get to the point where the anterior and posterior aspect of the temporalis muscle pedicle is identified, you can begin to separate the muscle from where it is scarred to the dura/pseudodura and lift it off the dura along with the scalp
- during the process, some or all of the temporalis muscle will be detached from the dura and the overlying scalp (some surgeons intentionally detach it completely). Later in the case, the muscle may be tacked down to the bone flap (e.g., through perforation holes) or to the underside of the scalp
- the cup end of a Penfield #1 dissector may be used to free the scar tissue off the bone edge around the entire defect. The dissector need only expose down to the deep edge of the inner table, without separating the scar from the inner table (which could facilitate epidural bleeding/hematoma formation)
- if the flap is not already perforated, on the back table, the flap is multiply drilled to provide a route of drainage for epidural blood
- the bone flap is placed in the defect. If it is riding up at any point, the soft tissue and any irregularities may be corrected with a drill and/or rongeur
- the flap is secured in position, usually with titanium plates and screws
- a subgaleal drain is brought out through a separate stab incision and closure is performed in the usual manner

107.5 Decompressive craniectomy

107.5.1 Indications

Indications (controversial) include:

- malignant middle cerebral artery occlusion syndrome (p.1589) primarily for nondominant hemisphere. Use on dominant side is more controversial

2. traumatic intracranial hypertension
 - a) as an adjunct for persistent intracranial hypertension when other ICP control measures fail (p.1100)³⁰
 - b) early in the management: may be considered for patients undergoing emergent surgery (for fracture, EDH, SDH...)³¹
3. uncontrollable brain swelling during craniotomy (p.1729)
4. reported in children with refractory nontraumatic intracranial hypertension³² (e.g., infection, infarction, Reye's syndrome...)

107.5.2 Potential complications

1. bleeding
2. herniation of the brain through the opening, compressing and lacerating the brain on the bone edges (risk may be reduced by making generous craniectomy)
3. post-op injury to the brain from inadvertent external pressure applied to the now relatively less protected brain
4. post-op fluid collections: hygromas or hematomas at the operative site, on the contralateral side, or interhemispheric

107.5.3 Techniques

General considerations

1. it is necessary to open the dura
2. options for the removed bone flap
 - a) discard it: this may be the best option when the bone flap has been contaminated as a result of an open traumatic scalp laceration
 - b) place it in a separate subcutaneous pouch in the patient's abdomen for later retrieval and reimplantation into the skull. This is especially helpful if the patient's own skull is preferred and the patient is not at the facility where the cranioplasty to replace the bone is likely to be performed (the patient takes the flap with them)
 - c) store it externally for future reimplantation: saturate with sterile preservative solution (e.g., RPMI medium 1640; www.invitrogen.com/GIBCO) and then place within sterile storage (e.g., intestinal bags which are then placed in a sterile plastic container) and store in an ultralow-temperature freezer (approximately -80 °C)
 - d) for non-contaminated situations (e.g., stroke): reimplantation can be considered after 6–12 weeks
3. bone openings need to be large (e.g., > 12 cm diameter,³³ often > 15 cm)

Hemicraniectomy

1. some prefer use of a Mayfield head-holder placed low (► Fig. 107.3) to give greater access³¹ (not feasible with severe comminuted skull fractures)
2. AP axis of head is placed horizontal to floor (unless C-spine not cleared or if neck too immobile—one may compensate for this by rotating table)
3. skin incision: two options
 - a) ► Fig. 107.4 A starts at widow's peak, similar to trauma flap (p.1015), but with increased exposure by taking it posteriorly close to theinion, then turning sharply anteriorly and hugging the ear to preserve blood supply

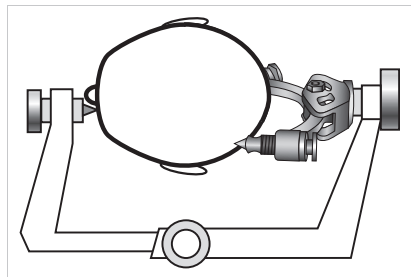


Fig. 107.3 Position of head and head-holder for right hemicraniectomy (looking down on top of patient's head).

- b) ► Fig. 107.4 B “T” incision. Less risk of flap ischemia. The “T” joins the midline incision behind the coronal suture to preserve the STA³¹
- c) burr holes (► Fig. 107.5): a burr hole is made just above the posterior root of the zygomatic arch, a second one may be made just behind the frontal insertion of the zygomatic arch, inferior to the superior temporal line
- d) bone flap: proceed posteriorly from the posterior zygomatic arch using the footplated craniotome. Posteriorly, stay ≈ 1 cm superior to asterion to avoid the transverse sinus. The flap is taken 1 cm beyond the lambdoid suture, and then up towards the sagittal suture, crossing the lambdoid suture again (this leaves a small amount of bone posteriorly on which the head can rest post-op). An anterior turn is made 1 cm short of the sagittal suture to avoid the superior sagittal sinus, and the sagittal suture is paralleled. The coronal suture is crossed and the drill is taken as low as possible in the frontal fossa near the midline. Staying as low as possible, the orbital roof is followed posteriorly towards the second burr hole. The burr holes are then connected

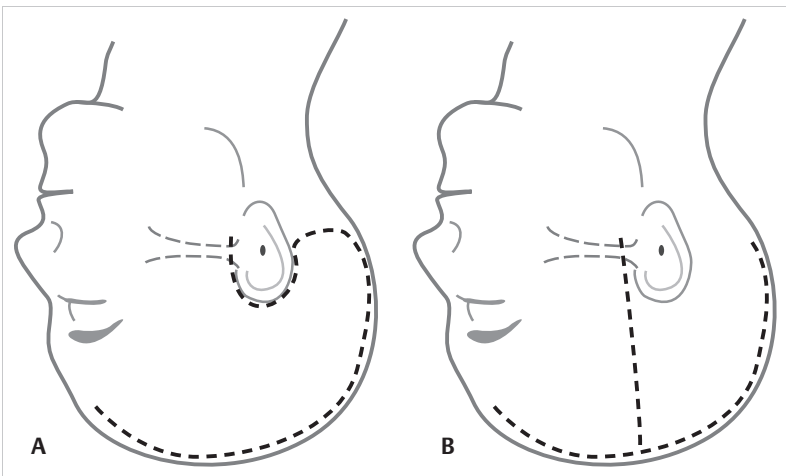


Fig. 107.4 Two options for skin incision for hemicraniectomy (see text).

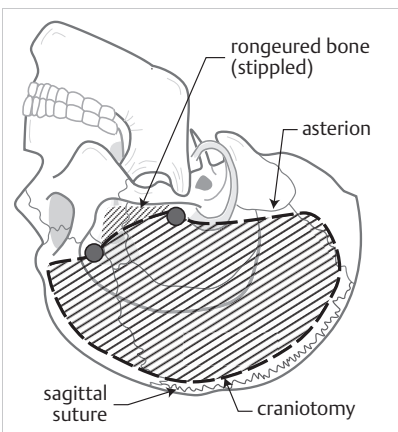


Fig. 107.5 Hemicraniectomy bone flap.

- e) some bone may need to be rongeured to expose the floor of the middle fossa (stippled area in ► Fig. 107.5)
- f) dural opening: based inferiorly, taken to 1 cm short of the craniotomy edge. Dural releasing incisions may be made at intervals up to the bone margin to avoid strangulation of the brain on the dural edge
- g) duraplasty
 - onlay: 2-cm-wide strips of dural substitute can then be placed partway under the dural edge around the periphery to isolate the brain from the undersurface of the skin flap where there will be a gap in the dura
 - some authors suture a dural graft in place
- h) the dural flap is then replaced on top of the brain and dural substitute strips, and is not sutured

Bilateral craniectomy

The above procedure can be performed bilaterally; however, it is difficult to position the head to do this. Alternatively, a bifrontal craniectomy can be performed.

1. skin incision: bicoronal, posterior to the coronal suture (► Fig. 107.6)
2. burr holes: may use the same ones as for hemicraniectomy (see above) bilaterally. Additional burr holes to straddle the superior sagittal sinus may be made if a large single bone flap is planned
3. bone flap (► Fig. 107.7): two options, both are taken back to the coronal suture:
 - a) a single large bone flap³⁴ extending back to the coronal sutures, or
 - b) two frontal flaps leaving a thin strip of bone in the midline overlying the superior sagittal sinus (if this strip is too wide, it can damage the brain)
4. dural opening: bilateral, based against the midline (superior sagittal sinus)

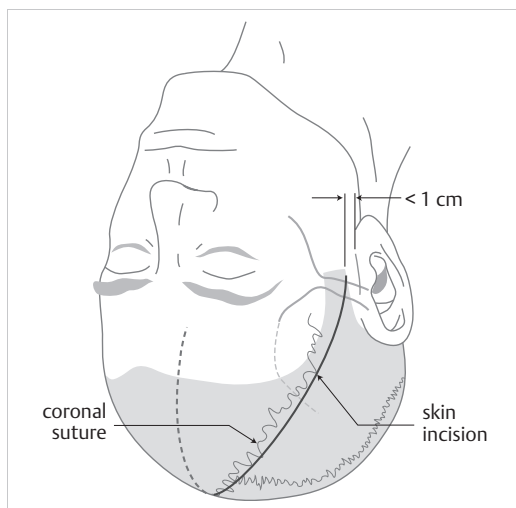


Fig. 107.6 Bilateral craniectomy skin incision.

Posterior fossa decompressive craniectomy

1. skin incision: midline skin incision from aboveinion to ≈ C2 spinous process
2. bone opening: laterally to sigmoid sinuses, superiorly to transverse sinus. C1 laminectomy is typically performed as well³²
3. dural opening: "Y"-shaped incision

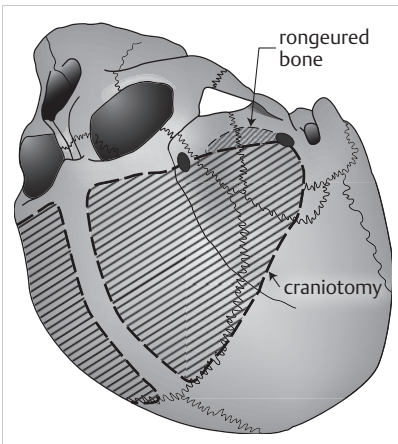


Fig. 107.7 Bilateral craniectomy skull flap shown with 2 separate frontal flaps (the preserved midline bone strip over the superior sagittal sinus is optional).

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108 Spine, Cervical

108.1 Anterior approaches to the cervical spine

1. anterior odontoid screw (p. 1775)
2. C1–3 (upper cervical spine):
 - a) transoral approach: including odontoidectomy (p. 1771)
 - b) extrapharyngeal approaches: use nasotracheal intubation (so that the mandible can be completely closed) through the contralateral nares. The head is slightly extended and is rotated 15° to the contralateral side. Avoid any oral tubes
 - medial extrapharyngeal approach: medial to carotid sheath. Provides a more anterior position than the lateral retropharyngeal approach. Structures encountered: branches of external carotid artery, upper laryngeal nerves, hypoglossal nerve
 - lateral retropharyngeal approach: only the spinal accessory nerve is encountered
3. C3–7: standard anterior cervical discectomy approach
 - a) For 1 or 2 level ACDF or 1 level corpectomy, a horizontal incision is usually employed
 - b) For more levels, a vertically oriented incision may be preferred to facilitate access

108.2 Transoral approach to anterior craniocervical junction

108

108.2.1 General information

Primarily useful for midline *extradural* lesions (approach to intradural lesions has been described,¹ but the use has been extremely limited because of difficulties obtaining watertight closure and increased risk of meningitis). Refinements in techniques and equipment (e.g., flexible reinforced oral endotracheal tube, McGarver or Crockard retractor, operating microscope, and suturing transnasal red-rubber catheters to the uvula to aid in retraction) allow access from as high as the inferior third of the clivus to as low as C3 (and sometimes C4²) vertebral body without need for tracheostomy or splitting of the tongue. Additional access can be achieved with use of extended techniques, including splitting of the hard & soft palate, tongue splitting, and transmandibular approach.

108.2.2 Transoral odontoidectomy

Indications

Anterior extradural compression of the cervicomedullary junction as with pannus from rheumatoid arthritis, irreducible basilar invagination, tumors of C2, infection.

Stabilization

75% of patients undergoing transoral removal of the odontoid process required posterior fusion afterwards³ due to ligamentous instability.^{4,5} While the stabilization intuitively seems like it should be done first, it is often done following the decompression at the same sitting or at a soon-to-follow date. Some reasons for decompressing before stabilization:

1. positioning the patient for fusion may cause neurologic compromise if there is cord compression
2. a post-op MRI can be done to determine if enough decompression was achieved from the odontoidectomy. If not, a laminectomy can be done at the same time as the posterior stabilization
3. the amount of destabilization may not be known until after the odontoidectomy—in some cases a C1–2 fusion may suffice⁵

Stabilization usually entails posterior occipitocervical fusion. Occasionally fusion may be limited to C1–2 or C1–3 without the occiput. It is also possible to place an anterior strut between the body of C2 and the clivus, or between C2 and C1. Fibula is recommended. Metal instrumentation should be avoided.

Pre-op preparation

1. make sure that the patient can open the mouth at least 25 mm. If not, then other approaches such as translabiomandibular should be considered

2. for conditions resulting in malalignment or basilar invagination, cervical traction for 1 or more days is sometimes required
3. radiographic evaluation
 - a) cervical MRI without and with contrast to define the soft tissue pathology
 - b) CT of the craniocervical junction with sagittal and coronal reconstruction
 - c) CTA to assess the position and involvement of the vertebral arteries. Measuring the distance between the VAs provides useful information

Booking the case: Transoral approach

All see defaults & disclaimers (p. 25).

1. position: supine with pin head-holder
2. equipment
 - a) microscope
 - b) high-speed drill with long bits
 - c) C-arm
 - d) image-guided navigation system (if used)
3. instruments
 - a) transoral set (usually includes oral retractor such as Crockard, Dingman, Dickman-Sonntag...)
 - b) long instruments: microdiscectomy instruments often work
4. anesthesia: intubation using video laryngoscopy, or second choice, asleep fiberoptic bronchoscopy⁶
5. some surgeons use ENT to perform the approach and closure and for follow-up
6. consent (in lay terms for the patient—not all-inclusive):
 - a) procedure: transoral resection of odontoid, placement of halo-vest immobilization, MEP monitoring (MEP should be consented specifically due to risk of seizures). Need for posterior stabilization at the same setting or in the immediate future
 - b) alternatives: nonsurgical management, radiation therapy for some diagnoses
 - c) complications: CSF leak with possible meningitis, spinal cord injury, wound breakdown, breathing problems (may require tracheostomy), seizures with MEP, swallowing difficulties (may require PEG tube, usually temporary). The patient should not undergo the operation if they would not be willing to have a PEG tube placed if it is indicated in the opinion of the operating surgeon and/or their consultants

Technical considerations

Some key points (see references for details^{2,3,7}):

Video laryngoscopy, or second choice, asleep fiberoptic bronchoscopy are now the intubation methods of choice (awake fiberoptic intubation is rarely used).⁶ Nasotracheal (NT) is used by some, but at the narrow upper part of the exposure the NT tube tends to get in the way.

SSEP and MEP monitoring are used in appropriate cases.

Positioning: 3-point fixation with a Mayfield head-holder is typically used. The patient is supine with no *neck rotation* (distorts the anatomical relationships and may bring one VA closer to the midline). Tilt the whole patient or the table towards the surgeon. 10–15° neck extension improves the exposure. Alternatively, the surgeon can stand above the patient who is kept perfectly supine.

A specialized retractor (e.g., Crockard transoral retractor) or a conventional Dingman retractor is placed. Verify that the tongue is not being compressed against the teeth.

Landmark: the tubercle of the atlas can be palpated through the posterior pharynx to locate the midline and for craniocaudal orientation.

The mucosa of the posterior pharynx is infiltrated with 1% lidocaine with epinephrine. Some authors culture the oropharynx to obtain drug sensitivities of organisms for use in the event of infection. Some advocate liberal topical use of 1% hydrocortisone ointment to the mucosa of the oropharynx and posterior tongue at the beginning and also during the operation to reduce intra-op and post-op swelling. Others feel it has no effect, and IV Decadron is used by some.

A 3-cm-long vertical midline incision is made.

To reduce the risk of C1 spreading and allowing basilar invagination, C1 ring-sparing surgery may be attempted by removing only the inferior half to two-thirds of the anterior C1. When C1 ring-sparing is not done, the central 3 cm of the atlas is removed with a high-speed drill.

There is $\approx$ about 20–25 mm working distance between the two vertebral arteries at their point of closest approximation where they enter the foramen transversarium at the inferior aspect of the lateral mass of C2.

The odontoid is hollowed out (“like a canoe”) using a high-speed drill, checking progress on lateral fluoroscopy at frequent intervals. Once the bone has been reduced to a thin shell, it can be fractured in towards the hollowed-out portion using curettes. The superior tip of the odontoid is particularly challenging due to the apical ligament.

Closure: a two-layer closure is preferred by some. Others recommend a single-layer closure incorporating deep muscle, superficial muscle, and mucosa.² If the dura has been violated, a fascial patch is secured with tissue adhesive and a lumbar subarachnoid drain is placed in the O.R. and maintained at low pressure for 3–4 days. An NG tube is placed under direct visualization to avoid injury or penetration of the mucosal closure.

Posterior stabilization

Transoral odontoidectomy produces instability in most cases (sometimes delayed).^{4,5}

For basilar invagination or occipitocervical instability, an occipitocervical fusion (p.1773) is recommended.⁷

For C1–2 instability alone, a posterior C1–2 arthrodesis may be performed (p.1778).⁷

Possible complications

1. dural tear with CSF leak and risk of meningitis
2. vertebral artery injury
3. spinal cord injury

Post-op care

1. NG feeding or IV hyperalimentation is used initially (in anticipation of oropharyngeal swelling (lasts 2–3 days) and to avoid disruption of the mucosal closure)
2. intubation is maintained until the swelling subsides. Initial removal of the endotracheal tube over a tube changer facilitates reintubation if needed; the tube changer can be removed if no problem develops after 1 hour⁷
3. if the NG tube comes out, it should only be replaced under direct vision (usually by ENT physician) to avoid injury/penetration of the mucosal incision
4. halo-vest immobilization is maintained until the posterior fusion is performed
5. for staged procedures when the fusion is being done at a later date, a post-op MRI should be done to assess the degree of decompression. If further decompression is needed, then a laminectomy can be added to the posterior fusion

108.3 Occipitocervical fusion

The patient will lose about 30% of neck flexion mobility with an occipital to C1 fusion.

Indications for occipitocervical fusion⁸:

1. traumatic occipitocervical dislocation
2. absence of a complete arch of C1. **Note:** Alternatively, C1–2 lateral mass fusion (with or without lateral mass screws) (p.1779) may be used in cases b and c if only the posterior arch of C1 is compromised
 - a) congenital
 - b) post-decompression
 - c) posttraumatic: “bursting” C1 fracture (bilateral or multiple C1 ring fractures). NB: some feel that this may be satisfactorily treated with halo immobilization until the atlas fracture heals (as they almost all do) followed by C1 to C2 wiring/fusion.⁹
3. congenital anomalies of the occipitocervical joints
4. upward migration of the odontoid into the foramen magnum
5. marked irreducible shifts of C1 or C2

Disadvantages of occipitocervical fusion:

1. loss of movement at the occipitocervical junction further reduces the range of motion as follows¹⁰:
 - a) flexion/extension: reduced by $\approx$ 30% (13° occurs at occiput–C1 junction)
 - b) lateral rotation: 10° is lost
 - c) lateral bending: 8° is lost
2. non-union rate is higher than with C1–2 fusion alone¹¹

Options:

1. keel plate (placed centrally over the thickest portion of the occipital bone) connected via rods to cervical screws (C2 pedicle screws and C3 lateral mass screws): reduced range of motion (ROM) to 17% of normal in a cadaver study¹² (for technique, see below)
2. occipital condyle (OC)–C1 polyaxial screws¹³; see below
3. occipital–C1 (AKA atlantooccipital) transarticular screws (see below)
4. looped rod wired to the occiput via wire cables placed through holes drilled in the occiput. Reduced ROM only to 31% of normal¹²

108.3.1 Keel plate occipital-cervical fusion

Pre-op planning:

1. CT scan through C2
 - a) to rule out aberrant position of foramen transversarium
 - b) to measure diameter of pedicles (may be best done on coronal sections due to the fact that the axial images are not usually oriented along path of pedicle) and estimate length of screws to be used
 - c) to verify trajectory of screws
2. measure thickness of occipital bone to determine screw-length for occipital screws

Technique:

1. occipital keel screws/plate
 - a) a drill, tap, and screwdriver with flexible shafts or universal joints are usually needed because of interference from patient's skin
 - b) midline holes are preferred since occipital bone is thickest here
 - c) drill with drill guide set to 8 mm, check depth with probe, if the inner cortex has not been breached, increase the drill depth by 2 mm, drill to the new depth, check depth again, and continue the process increasing the drill length by 2 mm at a time until the inner cortex is breached or 14–16 mm is reached. Use a screw length equal to the depth that was drilled
 - d) **SCREWS** occipital screws are cortical screws (narrow pitch) and the distal tip is **blunt** (to avoid dural injury).
Typical dimensions: 4.5 mm diameter, 8–12 mm length
2. C2 pedicle screws (p. 1784)
3. C3 lateral mass screws (p. 1784) (if used)

108.3.2 Occipital condyle to C1 polyaxial screw fusion

See references.^{13,14}

Utilizes polyaxial screws placed in the occipital condyles that are then connected to screws placed at lower levels (see below) via connecting rods.

1. PROS compared to occipital plate/keel instrumentation:
 - a) circumvents problem of poor occipital bone purchase which may occur with keel plates
 - b) can be used even following posterior fossa craniectomy
 - c) greater surface area for fusion
 - d) avoids risk of intracranial injury from occipital screws
2. CONS: due to condylar variability, not all patients are candidates
3. biomechanics: compared to occipital plate, similar stiffness in flexion-extension and axial rotation, increased stiffness to lateral bending¹⁵

Pre-op planning: CT scan occiput through C2.

Technique:

✖ Structures to avoid include: hypoglossal nerve in hypoglossal canal (just above the occipital condyles (OC)), carotid and vertebral arteries, jugular bulb. Image guidance can be helpful.

1. occipital condyle screws
 - a) **ENTRY** 4–5 mm lateral to the foramen magnum, 1–2 mm rostral to the atlantooccipital joint (no need to expose the entire condyle—it is only necessary to expose as far laterally as the posterior condylar emissary vein which is best left alone but if necessary may be safely coagulated in most cases¹³)
 - b) **TRAJ** 12–22° medial (mean: 17°), 5° maximal superior angulation
 - c) **SCREWS** 3.5 mm diameter polyaxial screws; bicortical purchase obtained using 20–24 mm length (mean: 22 mm)

2. condyle screws are connected with 3 mm diameter rods to either:
 - a) C1 lateral mass screws (p. 1781) and C2 pars screws (p. 1784), or
 - b) C1–2 transarticular screws (p. 1779)

108.3.3 Occipital–C1 (AKA atlantooccipital) transarticular screws

See references.^{16,17}

1. PROS: no compromise of C1–2 joint
2. CONS: steep trajectory requires additional incision at level of C–T junction
3. **ENTRY** midpoint of posterior C1 lateral mass
4. **TRAJ** 10–20° medially, aiming cranially to **TARGET** middle of occipital condyle
5. **SCREWS** 28–32 mm cannulated lag screws
6. biomechanics: ≈ equal to occipital plate–C1 lateral mass fusion¹⁸
7. clinical data: 2 cases reported, 2 year F/U, no complication

108.3.4 Post-op immobilization/bracing

1. for severe C1 fractures, or those with impaired bone healing capacity (elderly or unreliable patients, smokers...) a halo-vest is recommended × 8–12 weeks
2. otherwise, if C1 is not badly damaged, a collar that limits flexion (e.g., Miami-J collar) suffices × 8–12 weeks

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108.4 Anterior odontoid screw fixation (OSF)

108.4.1 Introduction

50% of axial rotation of the head occurs at the C1–2 complex. Stability of the C1–2 joint depends primarily on the integrity of the odontoid process and the atlantal transverse ligament (p. 69) (which is the most important structure holding the odontoid process in position against the anterior arch of C1).

Odontoid screw fixation (OSF) treats odontoid fractures by restoring the structural integrity of the odontoid process (osteosynthesis) without sacrificing normal mobility. Other treatment methods, e.g., C1–2 fusion, significantly reduce rotational mobility (although subaxial articulations will compensate to some degree over time).

108.4.2 Evaluation

A full set of C-spine X-rays is needed (including an open-mouth odontoid view in addition to AP, lateral, and flexion/extension views). MRI is recommended to rule out disruption of atlantal transverse ligament. Cervical axial CT with coronal and sagittal reconstructions are also recommended to demonstrate the orientation of the fracture path and to verify the integrity of the posterior elements.

108.4.3 Indications

See guidelines for management of isolated odontoid fractures. (p. 1173)

In general, a reducible odontoid Type II fracture (and Type III fractures where the fracture line is in the cephalad portion of the body of C2 in an elderly patient who may not fuse as well with immobilization as a younger patient¹⁹). Absolute requirement: the transverse ligament *must be* intact.

108.4.4 Contraindications

1. fractures of the C2 vertebral *body* (except cephalad Type III fracture)
2. disruption of transverse atlantal ligament (TAL): see Atlantoaxial transverse ligament (TAL) injuries (p. 1778). May be directly demonstrated on MRI. Indirect evidence: if the sum of the overhang of the lateral masses of C1 on C2 exceeds 7 mm (rule of Spence (p. 1161) - NB: this rule is inaccurate and should not be used as the sole determinant)
3. large odontoid fracture gap
4. irreducible fracture
5. age of fracture: controversial. Fusion rates in fracture > 18 months old was 25%.²⁰ Fracture < 6 months old have ≈ 90% fusion rate²⁰

6. patients with short, thick necks and/or barrel chest: makes it difficult to achieve the proper angle. May be circumvented by the instrumentation distributed by Richard-Nephew which utilizes a cannulated flexible drill, tap, and screwdriver
7. pathologic odontoid fracture
8. fracture line in oblique orientation to frontal plane (shearing forces can cause malalignment during screw tightening)

Booking the case: Odontoid screw fixation

See defaults & disclaimers (p. 25).

1. position: supine head on horseshoe headrest, halter traction
2. anesthesia: video laryngoscopy, or second choice, asleep fiberoptic bronchoscopy are the intubation methods of choice (awake fiberoptic intubation is rarely used).⁶ Do NOT use a wire-reinforced endotracheal tube
3. equipment: **2 C-arms** for biplane fluoro, or image guidance system
4. instrumentation:
 - a) some surgeons use specialized instrumentation (e.g., Apfelbaum set). Alternatively the following may be used
 - b) ACDF surgical set
 - c) retractor: a tubular retractor (e.g., METRx® by Medtronic) works well, a radiolucent tube may be advantageous
 - d) consider having posterior cervical fusion instrumentation available (e.g., for C1–2 fusion (p. 1778) ± occipital plate as appropriate) in case the odontoid screw has to be aborted
5. implants: 4 mm diameter cannulated screws, lag screws & conventional screws, 40–50 mm lengths will cover most situations. These may be found in cannulated screw sets e.g., as used by orthopedic surgeons for minor fractures or in specialty sets such as Medtronic UCSS (Universal Cannulated Screw Set)
6. consent (in lay terms for the patient—not all-inclusive):
 - a) procedure: surgery to place screw(s) from the front of the neck across the fractured odontoid bone. Possible posterior approach in case the anterior approach cannot be completed
 - b) alternatives: nonsurgical management in a collar (which has a low chance of success in certain situations, e.g., age ≥ 50, displacement ≥ 5 mm), fusion (C1–2 or occiput–C2 or other, as appropriate) which will result in some loss of motion
 - c) complications: screw breakage/pullout, failure to fuse which might require additional surgery (which will reduce neck motion), some of the usual risks of anterior cervical spine approaches (p. 1286) (swallowing difficulties, injury to esophagus, trachea, carotid artery, recurrent laryngeal nerve palsy..., but generally not including spinal cord, nerve root, or vertebral artery injury). The patient should not undergo the operation if they would not be willing to have a PEG tube placed if it is indicated in the opinion of the operating surgeon and/or their consultants
 - d) consider consenting patient for posterior C1–2 fusion in case the odontoid screw has to be aborted

108.4.5 Technique summary

Preparation

Various instrumentation systems have been developed to facilitate the procedure. The following describes some of the basic elements that are not specific to any one instrumentation (see reference by Apfelbaum²¹ for details of his instrumentation distributed by Aesculap Instrument Corporation, South San Francisco, CA).

Two C-arm fluoroscopy machines are mandatory for bi-plane imaging (simultaneous AP and lateral views). Some surgeons prefer placing 2 screws if there is enough bone to accommodate them; however, this may also reduce the amount of bone surface that can heal and the fusion rate appears to be the same.²²

Anesthetic considerations

The anesthesiologist is positioned at the *foot* of the table. Video laryngoscopy, or second choice, asleep fiberoptic bronchoscopy are now the intubation methods of choice (awake fiberoptic intubation is rarely used).⁶ Do NOT use a wire-reinforced endotracheal tube since the wire interferes with the AP imaging.

Position

Supine. The neck is placed in *extension* (critical to performing the procedure) either with Holter traction and a small shoulder roll with the head on a gel donut (a strip of tape across the forehead stabilizes the head), or a radiolucent head-holder may be used. Place the *lateral* fluoro unit first, then the AP unit slides into the "C" of the lateral unit. Lateral fluoroscopy is used to assess reduction of the fracture fragment, and the head is repositioned to try and achieve reduction. If there is retrolisthesis of the odontoid, the neck may need to be slightly less extended. A radiolucent mouth gag is placed to hold the mouth open for AP transoral imaging (a small tape roll works well). Abort the procedure if AP and lateral fluoroscopic views do not adequately image the odontoid.

Once the patient is positioned, place a guidewire alongside the neck and verify with fluoro that the screw passing through C2 into the odontoid tip will project to a point that is not inside the patient's chest (if that cannot be corrected with repositioning of the head, then the operation should be aborted).

Approach

A Cloward-type of horizontal skin incision at $\approx$ C5–6 (the entry site can be localized by placing a guidewire adjacent to the patient's neck and taking a lateral fluoro as noted above) and approach identical to anterior cervical discectomy are used (all the way to exposing the longus colli muscles (p. 1284)). A Kittner is used to dissect superiorly anterior to the longus colli muscles in the loose areolar tissue up to C2. A self-retaining retractor (e.g., Caspar retractor, not distractor) with a superior retractor blade attached may be used (or a hand-held retractor, preferably radiolucent, may be used). Alternatively, a retractor tube system²³ (e.g., METRx® by Medtronic, which manufactures a specialized extra-angled bevelled tube for this procedure) may be used. The bovie is used to remove the soft tissue over the inferior front of C2.

Procedure

Localization: lateral fluoro is used to place the tip of an awl *as far anteriorly as possible* on the inferior endplate of C2 (► Fig. 108.1; common errors: 1) entering too far posterior along the inferior

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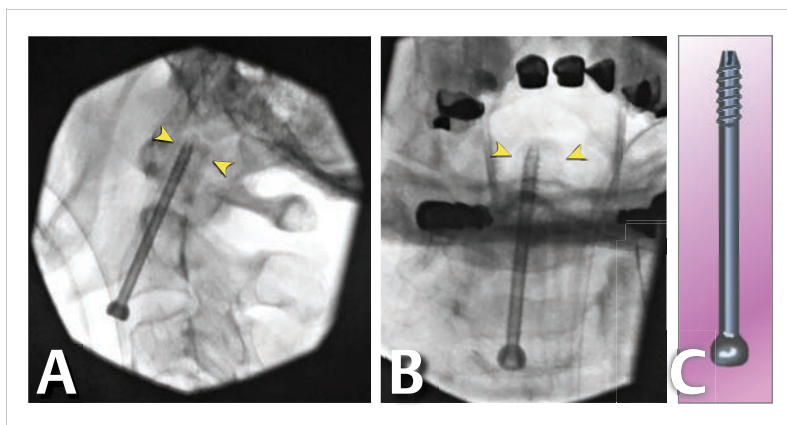


Fig. 108.1 Anterior odontoid screw.

Final radiographs, A: lateral, B: AP. The edges of the odontoid tip fragment are indicated by yellow arrowheads. C: illustration of a lag screw as used in this case.

margin of C2, which results in the guidewire ending up toward the back of the dens, 2) entering too far anterior which pre-disposes to the screw breaking out through the anterior cortex of C2). AP fluoro is used to place the awl in the exact center of the C2 body in the medio-lateral dimension. The awl is used to make a pilot hole at this location.

Guidewire placement, drilling, tapping, and ultimately screw placement are performed while monitoring the progress on frequent fluoro images, aiming for the exact middle of the dens on AP fluoro, and aiming toward the apex of the odontoid fracture fragment on lateral fluoro.

Drilling is performed under fluoro all the way *through* the apical cortex of the dens to avoid cracking the dens with the screw (the area just distal to the apex of the dens is safe).

Screw placement. If there is a minimal gap a fully threaded cannulated screw may be used. If there is a gap between the odontoid tip fragment and the body of the odontoid, a cannulated lag screw (a screw with an unthreaded proximal shaft, ► Fig. 108.1–C) is preferred. If a lag screw is not available, one can overdrill (i.e., using a larger diameter drill bit) the part of the path through the body of C2 up to the fracture and then use a fully threaded screw, which will slip through the overdrilled hole and still have a lag effect on the fracture fragment. If a second side-by-side screw is used, it may be fully threaded. In chronic nonunion cases, prior to advancing the screw, a bifaced curette may be inserted within the fracture space to freshen the fracture site. The screw(s) should be drawn up tightly to the inferior edge of C2. The challenge is to turn the screw enough to draw the fragment closer to the body to narrow the gap without turning it so much that it strips the bone in the fragment and the operation is for naught. ► Fig. 108.1 shows the final position of an anterior odontoid screw.

At the end of the procedure confirm integrity of the transverse ligament by carefully flexing the neck under lateral fluoro.

Postoperative immobilization

The immediate post-op strength of the odontoid + screw is only $\approx 50\%$ of the normal odontoid. Therefore, a cervical brace is recommended for 6 weeks¹⁹ (although some authors don't use one²¹). If the patient has significant osteoporosis, a halo brace is recommended.

Results

Healing takes ≈ 3 months (or longer with chronic nonunion). With fractures < 6 months old, the union rate was 95%. Chronic nonunions > 6 months old have a significant risk of hardware failure (screw breakage or pull-out), with a bony union rate of 31%, and 38% rate of presumed fibrous union.²¹ Thus, in cases of chronic nonunion > 6 months old, C1–2 arthrodesis is probably a better choice unless the need to maintain motion is worth the risk of needing a second operation if this one fails.

The average technical complication rate is $\approx 6\%$ (2% screw malposition, 1.5% screw breakout).

108.5 Atlantoaxial fusion (C1–2 arthrodesis)

108.5.1 Indications

NB: The patient will lose $\approx 50\%$ of head rotation with C1–2 fusion.

- instability of the C1–2 joints (usually with associated **atlantoaxial dislocation**), including:
 - incompetence of the transverse atlantal ligament (TAL) (most result from damage to the insertion point of the TAL on the medial tubercles of C1 (► Fig. 1.14):
 - rheumatoid arthritis (p. 1378): symptomatic patients, or asymptomatic patients with subluxation ≥ 8 mm
 - local infection
 - traumatic disruption of the TAL
 - Down syndrome (p. 1381) patients with laxity of the TAL
 - tumors
 - incompetence of the odontoid process
 - odontoid fractures meeting surgical criteria (p. 1174), including (in patients ≥ 7 years of age)
 - Type II fractures with ≥ 5 mm displacement
 - instability at the fracture site despite halo-vest traction
 - chronic nonunion of odontoid fractures
 - following transoral odontoidectomy
 - tumors destroying the odontoid process
- hangman's fractures that require surgical stabilization
- vertebrobasilar insufficiency with head turning (p. 1592) (bow hunter's sign)

108.5.2 Technical considerations

Some cases require incorporation of the occiput in addition to C1–2.

Surgical options include:

► Rigid instrumentation:

1. C1–2 fusion using polyaxial screws connected by rods:
 - a) C1: screws placed in lateral masses. May be used in cases where the posterior arch of C1 is compromised
 - b) C2 screw options:
 - screws may be placed in pedicles (pars)
 - screws may be placed in lateral masses
 - crossed C2 laminar screws²⁴
2. C1–2 posterior transarticular facet screws (TAS)^{25,26,27}

► **Posterior cervical wiring and fusion.** With the development of rigid fixation, these techniques are used less frequently. While they are poor in limiting rotation, they are effective in limiting flexion. And since the Dickman and Sonntag technique is effective in limiting extension, it has recently been used to offload C1 lateral mass screws which have a tendency to break at the bone interface (point of entry to the bone) of C1

1. interspinous fusion technique of Dickman and Sonntag (p. 1783)
2. not presented here:
 - a) Brooks fusion²⁸ (the Smith-Robinson technique as modified by Griswold²⁹): C1 to C2 sublaminar wires with 2 wedge bone grafts
 - b) Gallie fusion³⁰ (p. 1477–93) and its modifications: midline wire under the arch of C1 with an “H” bone graft

► **Halifax clamps with fusion.**³¹ These clamps are effective in minimizing movement in flexion, but are less stable in extension or with rotation

► **Odontoid compression screw fixation** (p. 1775). Essentially only for odontoid Type II fractures < 6 months old with intact transverse ligament (p. 1775).³² Preserves more mobility than C1–2 fusion

► **Combined anterolateral and posterior bone grafting.**³²

► **Combining anterior (transoral) decompression with posterior fusion.** Indicated when a significant anterior mass is present, causing neural compression and/or making passage of sublaminar wires at C1 unsafe

108.5.3 Techniques of atlantoaxial fusion

Positioning

The patient is placed in a halo ring (with a gap in the back and secured to the table using a Mayfield adapter) or Mayfield pin fixation and is then placed prone on the O.R. table on chest rolls. The table will usually need to be positioned in a maximal reverse-Trendelenburg position to bring up the surgical area. The patient's feet are allowed to rest on a padded footplate on the table to prevent the patient from sliding down. Lateral intraoperative X-rays are taken after patient positioning.

Incision and approach

A midline skin incision is made from just below theinion to the spinous process of C5 or C6.

C1–2 transarticular facet screws (TAS)

May be used as an adjunct to posterior C1–2 wiring and bone graft—e.g., technique of Dickman and Sonntag (p. 1783)—to achieve immediate stabilization without the need for postoperative external orthosis, or in cases where the posterior arch of C1 is fractured or absent. ✖ A major risk of the procedure is vertebral artery (VA) injury. Therefore many practitioners have adopted C1 lateral mass screws (p. 1780).

Selection of candidates

May be appropriate in elderly patients or those with rheumatoid arthritis, in whom there may be slow fusion, or for those who have failed a previous attempt at C1–2 wiring/fusion. Also in young individuals who have ligamentous laxity.

All patients must have thin-cut CT scans from the occipital condyles through C3 with sagittal reconstruction through the C1–2 facet on both sides to look for the presence of a vertebral artery in the intended path of the screw. Also, risk of VA injury can be reduced using CT scans reconstructed along the planned trajectory of the screw (aiming from a point 4 mm above the inferior C2 facet to a point in the anterior C1 button on CT³³).

Technique summary

A number of instrumentation sets are available for the procedure, and each has its own nuances. The following is intended to primarily cover the basic procedure common to most or all (see reference by Apfelbaum²¹ for details with that system).

► **Position.** Patient prone, with the head clamped in a Mayfield head-holder with a slight military tuck of the chin. Lateral C-arm fluoroscopy is used for the procedure, and some have advocated biplane fluoro.

► **Approach.** Utilize a standard midline posterior laminectomy approach from occiput to the C3 spinous process. The lamina of C2 and the posterior arch of C1 are exposed to the lateral aspect of the C2 inferior articular facet. The lateral extent of the spinal canal is defined using a small angled curette. The C1–2 facet is curetted to facilitate arthrodesis and permits observation of the drill as it crosses the joint.

► **Screw placement and fusion.** **ENTRY** 1–2 mm superior to the C2–3 facet on the midline axis of the pars interarticularis. The trajectory is determined fluoroscopically using a Kirschner wire (K-wire) placed on the side of the neck as a guide, aiming it through the C2 inferior articular process, pars interarticularis, superior articular process and across the C1–2 articulation into the lateral mass of C1. This helps establish the appropriate entry site for the drill guide through a separate stab wound, usually around the T1–2 level, 2–3 cm off the midline.

TRAJ A pilot hole is then drilled using visual guidance to maintain a straight parasagittal course (it helps to stand on 1 or 2 footstools to eliminate some of the parallax error) and fluoroscopic guidance to maintain the trajectory towards the C1 lateral mass. An assistant can reduce any atlanto-axial translational malalignment using a towel clip on C1 or C2 just prior to the drill crossing the C1–2 facet joint. To minimize the risk of VA injury, keep the drill as far dorsally as possible within the pars interarticularis. The pilot hole is then tapped and a fully threaded titanium screw is placed. If brisk arterial bleeding (not bone bleeding) occurs after drilling or tapping the first side, the VA may have been injured. The screw may still be placed but the contralateral hole and screw should *not* be placed. A post-op arteriogram is then performed to assess for propagating thrombus or dissection. Barring any contraindications, the procedure is repeated on the contralateral side. After screw placement, then posterior bone fusion—e.g., technique of Dickman and Sonntag (p. 1783)—is performed.

► **Post-operative care.** External immobilization is optional. Some surgeons do not feel it is mandatory as they consider the screws to supply adequate internal immobilization.

Results

A fusion rate of up to 99% with no complications has been reported.²⁵ Injury to the vertebral artery is the main potential complication.

C1–2 screw fixation

Placement of polyaxial mini screws in C1 (lateral mass, originated by Goel and Laheri³⁴ in 1994 and promulgated in 2001 by Harms and Melcher²⁵ or posterior arch lateral mass (PALM)^{36,37}) and C2 pedicle with rod fixation.

Advantages over C1–2 transarticular screws (see above):

1. the more superior and medial trajectory should reduce the risk of VA injury³⁵
2. may be used in the presence of C1–2 subluxation
3. may be usable in certain cases of aberrant VA course
4. in selected cases, this can be used for temporary fixation without fusion (since joint spaces remain intact) and the hardware may be removed after an appropriate time to reclaim motion in the C1–2 articulation

Booking the case: C1–2 lateral mass fusion

Also see defaults & disclaimers (p. 25) and pre-op assessment (see below).

1. position: prone, pin head-holder
2. anesthesia: intubation using video laryngoscopy, or second choice, asleep fiberoptic bronchoscopy (awake fiberoptic intubation is rarely used)⁶
3. equipment: C-arm or (optional) image-guided navigation system
4. implants:
 - a) mini-polyaxial screws (smooth shank screws needed if C1 lateral mass screws are a possibility)
 - b) cable for interspinous graft: C1 ring must be intact (optional, but recommended)
 - c) have occipital plates and instrumentation available in case of inability to place C1 screws, thereby enabling occipital-cervical fusion as a bail-out option
5. consent (in lay terms for the patient—not all-inclusive):
 - a) procedure: surgery to place screws & rods from the back of the neck to stabilize and usually to fuse the top 2 bones of the neck
 - b) alternatives: nonsurgical management in a collar, in some cases screws may be temporary and no fusion would be done
 - c) complications: screw breakage/pullout, failure to fuse which might require additional surgery, loss of some neck bending motion is expected ($\approx 20\%$ is typical)

Surgical technique (excerpted highlights)

See references.^{35,38}

Applied anatomy: there is no true neural foramen at C1–2, the C2 nerve root lies on the posterior surface of the capsule of the C1–2 articular joint.

Pre-op assessment

It is mandatory to know the position of the VA on both sides (and in particular, the location of both foramina transversarium of C1), and the following bony information (requires thin-cut CT scan):

1. craniocaudal thickness (height) of the posterior arch of C1 (in case the arch needs to be drilled to facilitate screw placement)
2. to determine screw length: distance from the planned entry point (see below) to the planned exit target (midposition of the anterior part of the superior C1 VB)
3. to estimate mediolateral angle for screws

C1 lateral mass screws

This is one option for C1 lateral mass screws. For the PALM technique, see below.

1. NB: if fusion is to accompany screw placement (i.e., permanent screw placement), strong consideration should be given to supplemental interspinous fusion, if not contraindicated (p. 1783) to prevent fatigue breakage of C1 screws where they penetrate the lateral mass
2. dissect over the superior surface of the C2 pars interarticularis to reach the posterior aspect of the C1–2 joint. Expose approximately the lateral half of the C1–2 joint (allows the C2 root to be retracted inferiorly and medially). Use a Penfield #4 dissector to palpate the medial aspect of the joint to accurately locate the entry point for the C1 lateral mass screws in the midpoint. Bleeding is controlled with bipolar cautery and/or Gelfoam-thrombin.
3. C1 lateral mass screws **ENTRY** visualization commonly requires caudal & medial retraction of the C2 dorsal root ganglion (occasionally this may not be feasible³⁸; sacrificing the C2 root may be required but this can lead to post-op pain and numbness³⁹; technique is to divide the *preganglionic* nerve fibers and to close the dural defect³⁸). The screw entry point is the midpoint of the inferior part of the C1 lateral mass (for both mediolateral and craniocaudal directions). An awl or a 1- to 2-mm high-speed drill is used to mark the position to prevent slippage while drilling the hole. Drilling a portion of the inferior arch of C1 is sometimes needed to allow screw placement (✗ CAUTION: the thickness of the arch in the craniocaudal dimension varies widely, and the horizontal segment of the VA lies immediately above—use pre-op CT for planning)
4. **TRAJ** averages $\approx 17^\circ$ medially, $\approx 22^\circ$ rostrally, **TARGET** the superior aspect of the anterior tubercle of C1 on lateral fluoro (see ► Fig. 108.2)
5. **C1 SCREWS** 3.5 or 4 mm diameter, length is determined from pre-op fine-cut CT to obtain bicortical purchase (✗ CAUTION: the ICA may be as close as 1 mm to the ideal exit site of the screw⁴⁰ ∴ some authors use only unicortical purchase). The screw needs to be proud to bring it up to the level of the C2 screw (it may actually be necessary to have the C1 screw protruding 1–2 mm more

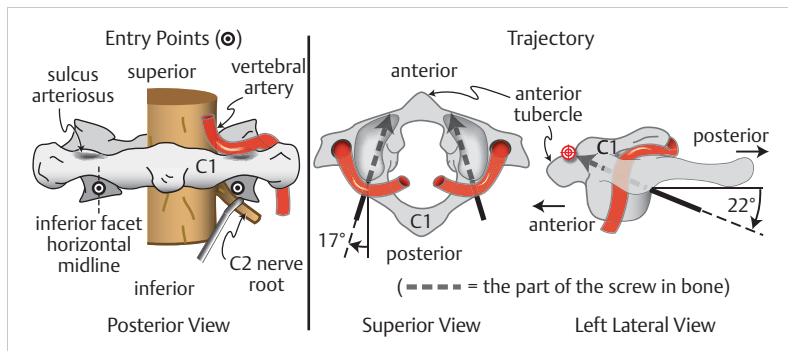


Fig. 108.2 Screw entry point and trajectory for C1 lateral mass screws.

than the C2 screw in order to allow rod attachment³⁸), and it should have an unthreaded portion (smooth shank) to minimize irritation of the C2 nerve, which could produce occipital neuralgia (unfortunately, most of the screws have only ≈ 10 mm smooth shank which is usually not long enough so some threads will contact the C2 nerve root)

6. C2 pedicle (pars) screws are placed as usual (see C2 pedicle (pars) screws (p. 1784))
7. if a fusion is to be performed: the posterior arch of C1 and the C2 lamina are decorticated with a drill. Onlay fusion substrate is then placed, taking care not to compress the dura. Optional adjunct: intra-articular decortication and packing bone within the C1–2 joint

Posterior arch lateral mass (PALM) C1 screws

C1 lateral mass polyaxial screws that enter through the arch as an alternative to entering the lateral mass directly. A slight variation results in “C1 pedicle screws”.⁴¹ May be used in patients who do not have a diminutive posterior C1 arch (the average C1 pedicle height ranges from 3.95–4.8 mm⁴¹; even then a small inferior breach should not be harmful). Advantages: reduces the amount of C1 lateral mass dissection needed with less bleeding, obviates the need to sacrifice the C2 nerve root which is required in some cases of direct lateral mass screws, avoids having exposed C1 screw that tends to break at the bone interface.^{36,37}

1. the position of the VA must be known. To protect it during drilling, it can be dissected subperiosteally from the sulcus arteriosus ($\gg$ Fig. 108.3) on the superior surface of C1 and lifted gently off the bone with a dissector³⁶
2. **ENTRY** the posterior arch of C1, various localizing options include
 - centered over C2 screw if already placed
 - in line with the center of the C2 facet^{36,41}
 - 2 cm from midline, 2 mm from inferior border of posterior arch⁴²

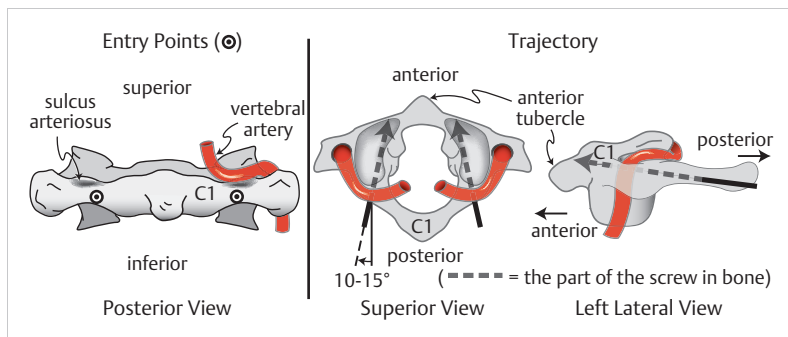


Fig. 108.3 Screw entry point and trajectory for C1 PALM screws.

3. **TRAJ** 10–15° medial (similar to above); craniocaudal direction is determined using lateral fluoroscopy to parallel the C1 arch⁴¹
4. **SCREWS** 3.5 mm diameter; length to bring the tip just behind the front of the C1 anterior tubercle
5. C2 pedicle (pars) screws are placed as usual if not already done (see C2 pedicle (pars) screws (p. 1784))

Post-op care

Optional: cervical collar (soft or rigid, as preferred) for 4–6 weeks.

Interspinous fusion technique of Dickman and Sonntag

A single bicortical graft is used, with multistranded cable passed sublaminar to C1 only. The bone graft is wedged between C1 and C2 (trapping it between loops of cable),^{43,44} see ► Fig. 108.4. Currently, this technique is infrequently used as the primary fixation for C1–2 fusion (unless technical difficulties prevent e.g., C1–2 lateral mass fusion). However, it may be most useful for limiting flexion and extension to offload C1 lateral mass screws to reduce the risk of screw breakage.⁴⁵

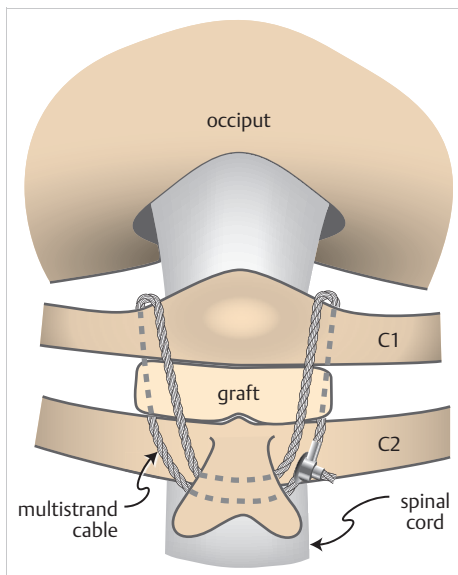


Fig. 108.4 Dickman and Sonntag C1–2 interspinous fusion. Illustrated with multistrand cable system, e.g., Medtronic ATLAS™ (formerly Songer Cable).

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Cannot be used if the posterior ring of C1 or C2 is not intact.

► **Bone graft.** Autologous bone is preferred. Bone is often taken from the posterior iliac crest (p. 1725). A tricortical graft of ≈ 4 cm length and > 1 cm height is obtained. The top edge is removed to create a bicortical graft of ≈ 1 cm height.

108.6 C2 screws

108.6.1 Options

1. pedicle screws (pars interarticularis screws): directed medially (see below)
2. lateral mass screws: directed laterally. Length is sized to fall short of foramen transversarium
3. C1–2 transarticular screws (p. 1779): associated with more risk of VA injury

- translaminar screws^{46,47}: 1 year stability appears to be less than C2 pedicle screws when used for subaxial fusions, but was $\approx$ as effective for axial fusions (C1–2 or C1–3).⁴⁸ May be useful as a “bail-out” for subaxial fusions when the C2 pars diameter is too small for pedicle screws⁴⁹

108.6.2 C2 pedicle (pars) screws

Check CT scan or MRI to rule out aberrant location of vertebral artery or unusual location of foramen transversarium before placing C2 pedicle screws. Some find image-guided navigation systems to be helpful.

Technique:

- ENTRY** palpate the medial and superior aspect of the pars with a Penfield 4 dissector ($\blacktriangleright$ Fig. 108.5). Enter at the estimated center of the surface projection of the C2 pars at the mid-point mediolaterally³⁵ in the superomedial quadrant of the surface of the C2 isthmus
- TRAJ** 20–30° medially (through the central axis of the C2 pedicle),⁵⁰ 25° superiorly (on lateral fluoroscopy, place the screw parallel to the pars) ($\blacktriangleright$ Fig. 108.6). To assist with trajectory, expose the proximal upper and medial border of the C2 pars interarticularis, and use a Penfield 4 to palpate during drilling ($\blacktriangleright$ Fig. 108.5)
- drill a shallow entry point, then drill with drill-stop set at 12 mm, monitoring progress at intervals under fluoro and palpating with probe, and if no breakout, then complete drilling by gradually increasing drilling depth by 2-mm increments either up to 15–20 mm to stay in the pedicles, or up to $\approx$ 30 mm depth to perform osteosynthesis for a hangman's fracture. $\times$ If withdrawal of the drill is followed by brisk bleeding, the screw should be inserted immediately to stop the bleeding. This bleeding may be from the vertebral artery; however, it is usually due to injury to the venous plexuses, and will not have any ill effects. In such cases it is best to not place the contralateral screw and to obtain an angiogram very soon post-op
- SCREWS** 3.5 mm dia. Screw length is not critical except when attempting to bridge a fracture gap (osteosynthesis) e.g., with a hangman's fracture in which case screws of 20–30 mm length are placed to avoid penetrating anterior C2 cortex (lag screws are used for this, or the proximal bone can be overdrilled); for most purposes screw lengths of 15–20 mm are used. Shorter screws (15–16 mm length) can still grip the pedicle with lower risk of VA injury

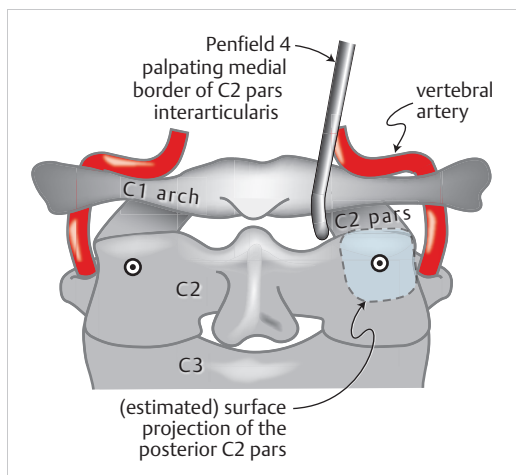


Fig. 108.5 Entry point for C2 pedicle screw placement (posterior view).

108.6.3 C3–6 fixation

Lateral mass screws

Generally applicable to C3–6. The lateral masses of the thoracic spine are usually too small and not strong enough⁵¹ for these screws. C7 is a transitional level, and lateral mass screws may sometimes be used. Occasionally even T1 may be amenable (see below).

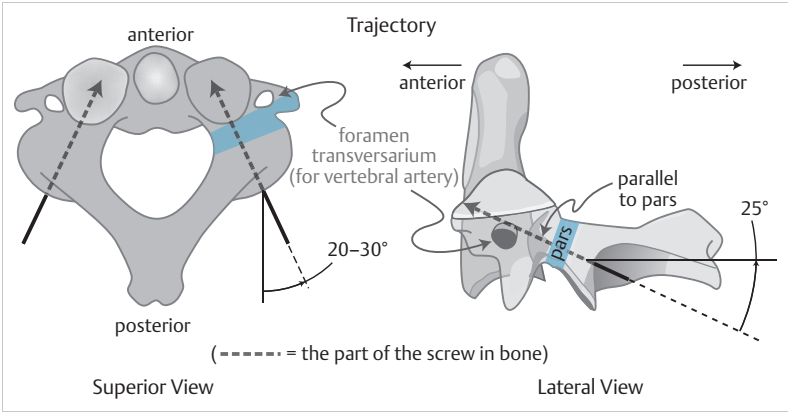


Fig. 108.6 Screw trajectory for C2 pedicle screws.

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Technique:

A number of methods have been promulgated with various screw entry points and trajectories (some are shown in ► Table 108.1). Comparing 3 techniques,⁵² there was a lower risk of nerve injury with the following (method of An⁵³):

1. **ENTRY**⁵³ 1 mm medial to the midpoint of the lateral mass (► Fig. 108.7). In the craniocaudal direction, the midpoint is used. A Penfield 4 may be used to palpate the medial wall of the pars to help determine entry point and trajectory

Table 108.1 Comparison of methods for lateral mass screw placement for C3–6

Method	Entry point		Trajectory angle	
	Mediolateral	Craniocaudal	Mediolateral	Craniocaudal
An	1 mm medial to midpoint	midpoint	30° lateral	15° cephalad
Magerl	2 mm medial to midpoint	2 mm cranial to midpoint	20–25° lateral	parallel to facet joint ^a
Roy-Camille	midpoint	midpoint	0–10° lateral	0°

^aangle can be determined by inserting probe into the joint

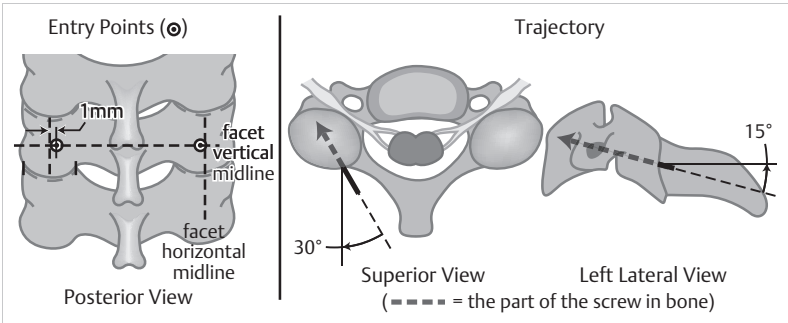


Fig. 108.7 Screw entry point and trajectory for C3–6 lateral mass screws (method of An).

2. **TRAJ** 30° laterally, 15° cephalad (editor's note: for upper cervical levels more cephalad trajectory is used, for caudal cervical levels 15° or less may be closer) (► Fig. 108.7).⁵³ To get the lateral angulation, the holes are best drilled from the contralateral side of the patient, holding the drill shaft almost up against the spinous processes (if they are still present)
 - a) **SCREWS** 3.5 mm diameter, 14–16 mm length (for C3–6)
 - b) **ROD** 3.5 mm diameter rods are usually used, and can be placed as far caudally as T3 as long as there is not gross instability (below T3, 5.5 mm diameter rods are used either via transitional rods or with rod connectors, e.g., “domino” connector)

Spinous process wiring may be used with intact spinous processes to help secure the bone graft.⁵¹

Transarticular screw fixation

An alternative to lateral mass fusion. First described in 1972 by Roy Camille. May be used alone or as an anchor point.

1. **PROS**:
 - a) screws cross 4 cortical surfaces for better purchase
 - b) compresses across the joint to promote fusion
 - c) useful at cervicothoracic junction where trajectory preserves facet capsule
 - d) lower implant profile
2. **CONS**: cannot correct deformity
3. **ENTRY** midpoint of lateral mass
4. **TRAJ** perpendicular to joint, neutral to 5° lateral (to avoid VA and exiting root)
5. biomechanics: stability equivalent to lateral mass screws⁵⁴
6. clinical: 25 patients (81 screws), 71 anchor, 10 fixation, 3.5 years F/U: solid fusion, no complications⁵⁵

Translaminar cervical screw fixation

May be used in cervical or thoracic spine.^{56,57}

1. indications: salvage technique when anatomy precludes pedicle screws
2. **PROS**:
 - a) avoids complications related to pedicle screws
 - b) no need for fluoroscopy (reduces radiation exposure)
3. **CONS**: requires intact posterior elements (cannot do with laminectomy)
4. **ENTRY** contralateral spinolaminar junction (at base of spinous process)
5. **TARGET** junction of the transverse process and the superior facet contralateral to the entry point
6. **SCREWS** 3.5–4.5 mm × 26 mm polyaxial screw
7. biomechanics: no data
8. clinical: 7 patients (C–T fixation), 14 months F/U, no hardware complications. Inconsequential ventral penetration in 5%⁵⁶

108.6.4 C7 screws

C7 is a transitional level, the lateral masses and/or pedicles may be relatively small and therefore difficult to instrument. For constructs spanning the cervicothoracic junction, C7 instrumentation is sometimes omitted which may actually facilitate the transition from cervical lateral mass screws to thoracic pedicle screws.

Screw fixation options:

1. pedicle screws (p. 1791): recommended especially when the C7 lateral mass is of inadequate size for lateral mass screws.⁵³ Placement with fluoroscopy may be difficult due to shoulder artifact on lateral fluoro, and direct visualization of the medial wall of the pedicle through a small laminotomy may be required as in the thoracic spine
2. lateral mass screws⁵⁰:
 - a) **ENTRY** as for C3–6 (see above)
 - b) **TRAJ** compared to C3–6 screws, slightly less lateral at ≈ 15° and a little less cephalad at ≈ 10°
 - c) **SCREWS** 3.5 mm diameter, 14 mm length
 - d) biomechanics: lab studies indicate that C7 lateral mass screws are biomechanically equivalent to C7 pedicle screws in constructs extending down to C7⁵⁸
3. C7 transfacet screw⁵⁹:
 - a) **PROS**: reduced risk to spinal cord and nerve roots

- b) CONS: disrupts C7–1 facet capsule, so T1 must be included in fusion. Short screws result in low pullout strength ∴ may be best used as an intermediate anchor point and not a construct endpoint
- c) **ENTRY** 1–2 mm medial and superior to center of facet
- d) **TRAJ** 30° inferiorly and 20° laterally, **TARGET** goal is bicortical purchase
- e) **SCREWS** 3.5 mm diameter × 8–10 mm polyaxial screws
- f) biomechanics: equivalent to C7–1 pedicle screws⁶⁰
- g) clinical: 10 patients, long cervicotoracic fixation, 6 months F/U, 3 patients with solid fusion

108.6.5 Cervical laminoplasty

General information

A procedure similar to laminectomy except that the decompressed posterior elements are not removed, and no fusion between vertebral levels is attempted. Pros & cons:

1. post-op range of motion is reduced 30–50% in extension, lateral bending, and rotation⁶¹ which is better than laminectomy + fusion, but not as good as with laminectomy alone
2. lower incidence of post-op kyphosis than with laminectomy
3. preserving the lamina might protect against some posterior trauma
4. contraindications: laminoplasty is not appropriate with
 - a) cervical instability
 - b) cervical kyphosis
5. disappointments
 - a) does not reduce the incidence of post-op C5 palsy (p. 1306)
 - b) does not mitigate against “bow-stringing” of the spinal cord in patients with high cervical lordosis (high Ishihara index)

Surgical techniques

The neck is positioned in slight flexion to open the interlaminar spaces.

A midline skin incision is used, dissection through this relatively avascular plane reduces blood loss. The interspinous ligament is preserved between levels and is divided above the uppermost and below the lowermost laminoplasty levels.

Methods for the laminoplasty include:

- unilateral (“open door”)⁶²: a partial-thickness trough is cut along one side at the junction between the lamina and lateral mass, and a complete cut is made on the other side. The fully cut side is pried open, creating a greenstick fracture on the partial-thickness cut side, and bone graft spacers (e.g., small discs sliced from iliac crest or cadaver rib) are inserted to maintain a gap on the open side (usually at every-other level). Plates (e.g., Centerpiece™ laminoplasty plates by Medtronic, or small cranial fixation plates) are placed (also at every other level)
- midline enlargement (“French door”): partial-thickness troughs are made on both sides, and a complete opening is made in the midline. The central opening is then pried apart creating greenstick fractures on both sides, and spacers are placed to maintain the opening. More symmetric decompression. Midline osteotomy is riskier than lateral troughs.

108.7 Anterior cervical vertebral body screw-plate fixation

Plate should be contoured to directly contact the front of the VBs being instrumented.

Typically, fixed-angle screws are used in the lowest level, and variable-angle screws are used at levels above. However, practices vary.

For the most rostral and most cranial screw holes, the plate should be sized such that the edge of the VB would just barely be visible when looking through the hole.

Screw length: Typical lengths vary from 12 mm (usually in women) to 16 mm. As a guide, if Caspar distraction pins (distributed by Aesculap Implant Systems) are used during the operation, the pin thread length can be used as a rough guide in estimating desired screw length based on how close the pins come to the posterior VB. Caspar pins are available in 12, 14, 16, & 18 mm tip lengths (14 mm is common, but you should verify the thread length on the pins you are using). Since the screws in the plate are usually angled, an additional 1–2 mm can be accommodated.

108.8 Zero profile interbody devices

These anteriorly placed devices incorporate holders for screw placement into an interbody cage without need for a separate anterior plate. Most often used in cervical spine.

1. PROS:

- are often easier to place adjacent to a previously placed anterior plate (because the plate will often cover too much of the VB to provide enough room for another plate on the same VB)
- avoids plates that are not parallel to the long axis of the spine
- posterior migration of the cage is prevented once the screws are placed

2. CONS: biomechanical stability is less than that with a plate (4-screw devices are more stable than 3-screw devices)

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109 Spine, Thoracic and Lumbar

109.1 Anterior access to the thoracic spine

109.1.1 General information

The thoracic spinal cord does not tolerate anything more than *minimal* manipulation. Therefore, unless the pathology (e.g., tumor or herniated disc) has already displaced the cord to one side, it is necessary to go around the spinal cord to gain access to the space anterior to the cord.

Indications: calcified thoracic disc herniation, spontaneous spinal cord herniation, tumors involving thoracic vertebral bodies.

109.1.2 Anterior access to the cervicothoracic junction/upper thoracic spine (T1, T2)

Depending on the patient's body habitus, anterior access below C7 may not be possible through a traditional anterior cervical approach. A sternal splitting procedure may be used. Permits access down to T3 (occasionally as far as T5) from an anterior midline approach (access to this region with a lateral (transthoracic) approach is poor due to the small volume of the pulmonary apices).

► **Sternal splitting procedure.** The neck and thorax are prepped down to the umbilicus. A hockey stick incision may be used, the horizontal portion is the usual for an ACDF. The vertical limb is centered over the sternum. In most cases, the services of a CV surgeon are employed to split the sternum and divide the sternocleidomastoid. This approach does not violate the pericardium or pleura, and a chest tube is not required (but is often used as a large-bore drain to prevent hemomediastinum, and also as a precaution in case the parietal pleura is cut during the exposure). Because of the depth of the approach, longer instruments than the routine 7-inch-length instruments used for an ACDF are required.

The exposed edges of the sternum may also be used to obtain cancellous bone for the graft.

109.1.3 Anterior access to mid- and lower thoracic spine

► **Transthoracic/retropleural approach.** If the pathology requires access that includes $\approx$ T10 and below, the attachment of the diaphragm increases the difficulty of a transthoracic approach, and a retrocoelmic approach (outside the pleural cavity) can facilitate surgery by allowing the diaphragm to be reflected off the VB.

Determining the level in the upper T-spine can be quite difficult intraoperatively. Counting up from the sacrum on an AP view using live fluoro will sometimes work when lateral spine X-rays cannot penetrate the upper T-spine due to the shoulders. If the patient does not have exactly 12 ribs and 5 lumbar vertebrae, this must be accounted for when counting levels. Image-guided navigation can also be helpful.

► **Laterality of approach.** If the pathology does not dictate use of one side over another:

1. advantages of right-sided thoracotomy: the heart, mediastinum, and brachiocephalic vein do not impede access
2. advantages of left-sided thoracotomy: aorta is easier to mobilize and retract than the vena cava

► **Position.** Lateral position on a bean-bag on the O.R. table with the break in the table under the level of pathology (remember to unbreak the table prior to instrumenting). Stabilize the patient using adhesive tape over surgical towels. An axillary roll is placed. Classically, a double-lumen endotracheal tube was recommended to permit deflating the lung on the side of the thoracotomy. The reality is that only minimal displacement of the lung is required to access the vertebral bodies, which permits continuous ventilation of both lungs with a normal single-lumen ET tube. The lung can be gently retracted with an "egg beater" retractor cushioned with a moistened sponge.

If a compression plate is going to be used, the goal is to be lateral on the VB; to achieve this, try to position it as far posterior as possible (rongeur off a little bit of the rib heads to facilitate this).

To increase the exposure, a rib may need to be resected. Generally, the level opened and the rib removed are one or two levels above the level of pathology (e.g., for T7 VB tumor, the T6 or T5 rib is removed).

109.1.4 Anterior access to thoracolumbar junction

► **Retroperitoneal approach.** Unless the pathology is predominantly right-sided, a left-sided approach is preferred because the spleen is easier to retract than the liver, and the aorta is easier to mobilize than the inferior vena cava. Flex the ipsilateral leg to relax the psoas muscle and thereby reduce traction on the ipsilateral lumbosacral plexus.

109.2 Thoracic pedicle screws

109.2.1 General information

Preferable to lateral mass screws because the transverse processes (which are analogous to the lateral masses in the cervical spine) of most thoracic vertebrae are not as strong.¹ Thoracic pedicles are typically very narrow in the lateral dimension (the width is usually a little larger at the cranial end) and are very tall in the cranio-rostral direction.

Accurate placement of thoracic pedicle screws is generally more challenging than the lumbar spine. There are at least 4 methods to place these screws, and a combination of them may be used (details follow this overview):

1. intraoperative fluoro: for parts of the thoracic spine, biplane fluoro may be used as in percutaneous lumbar spine pedicle screw (p. 1797) placement
 - a) PROS:
 - allows percutaneous screw placement
 - generally good accuracy in screw placement
 - b) CONS:
 - due to the dense bone of the shoulders, lateral fluoroscopy imaging from T1 to about T4 is usually difficult. For non-percutaneous cases, Steinman pins may be placed at the estimated entry points for the screws, and AP fluoro is used to fine-tune the position so that the screw enters the pedicle at the desired location
 - may increase radiation exposure to surgery team and patient
2. anatomic ("freehand" (p. 1792)) thoracic pedicle screw placement based on anatomic landmarks. Fluoroscopy is still usually performed after all screws are placed, and any screws with questionable position need to be reevaluated (usually removing the screw and re-probing) and possibly revised
 - a) PROS:
 - as the number of levels (and therefore, screws) placed in a patient increases, may save time over other methods
 - the small facetectomies that are needed to visualize the facet joint of the level below can facilitate correction of spinal curves and provides a good fusion surface as well as fusion material
 - likely reduces radiation to the surgical team and the patient
 - excels in complex scoliosis cases with varying degrees of rotation at different levels (which can be taxing when relying on fluoroscopy)
 - b) CONS: steep learning curve: this method probably takes the most practice to master
3. performing small laminotomies at each level where pedicles are not exposed by a laminectomy, and use position of pedicle either by visualization or by palpation of the medial and superior aspect of the pedicles with a dissector to get an approximation of the entry point and pedicle trajectory
 - a) PROS: can permit accurate placement at essentially any level with potentially less radiation (depending on how often the surgeon checks screw position)
 - b) CONS: takes a little time at each level, but overall this is comparable to other methods
4. image guidance using instruments that are fitted with specialized markers that are tracked real-time by "cameras" that project the drill and/or screw location on a CT or X-ray image viewed in the OR. Robotic screw placement simply automates the use of image guidance
 - a) PROS:
 - reduces intraoperative radiation to surgical team, and to a lesser extent to the patient
 - allows percutaneous screw placement
 - easily deals with spine rotation
 - b) CONS: accuracy may be compromised by movement of spinal segments relative to the registration array, or by technical errors. The surgeon must be vigilant for screw placement that is visually not appropriate based on anatomic knowledge

As in the lumbar spine, there are 2 options for craniocaudal trajectories (► Fig. 109.2:²

1. anatomic screw placement: parallel to the axis of the pedicle—provides a longer path for screw contact with bone but requires a multi-axial screw head
2. straightforward screw placement: parallel to the superior endplate. Allows fixed-head screws for spine derotation, therefore often used for pediatric scoliosis surgery. Provides 27% increase in pullout strength vs. the anatomic trajectory³

109.2.2 Fluoroscopy or laminotomy techniques for thoracic pedicle screw placement

1. **ENTRY:** several methods are used, including:
 - a) making a small laminotomy ► Fig. 109.1 (typically with high-speed drill) to allow palpation of the medial and cephalad edges of the pedicle e.g., with a Penfield #4 dissector. This works very well for very small pedicles where a 1–2-mm error in estimating the pedicle position may cause the pedicle to be missed
 - b) placing a small “dimple” and then a Steinman pin at the estimated entry points (see Freehand thoracic pedicle screw placement technique (p. 1792) for entry points using anatomic landmarks which can be useful here for first approximation) and then using AP fluoro to fine-tune the position of the pins (see ► Fig. 109.1).

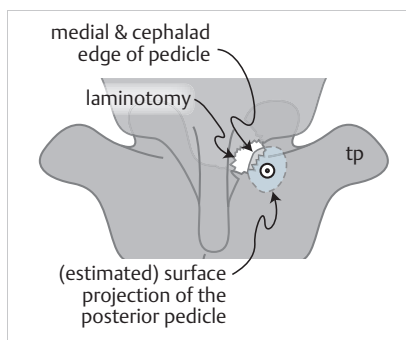


Fig. 109.1 Entry point for thoracic *pedicle* screws (posterior view). tp = transverse process. A small laminotomy is shown on the right side to allow localization by palpating the medial and cephalad edge of the pedicle with a dissector.

2. **TRAJ**
 - a) below T1: 5–10° *medially* and 10–20° *caudad*¹ (► Fig. 109.2). A thoracic Lenke probe may be used as a pedicle finder
 - b) T1: if a *lateral mass* screw is placed at T1 (instead of a pedicle screw), aim almost straight down at the floor (with patient positioned horizontally, i.e., without Trendelenberg or reverse-Trendelenberg)
3. **SCREWS** Smaller pedicles (usually T1–4, especially in females) usually require the smallest screw diameter (typically 4.5 mm). Others may accommodate 5.5 mm. Typical length: 20–25 mm
4. **ROD** When connecting to a cervical rod, down to ≈ T3 you can use a 3.5 mm diameter cervical rod throughout (here, the stiffer cobalt-chrome rod may be advantageous over titanium) with some systems using a 4.35–4.5 diameter rod. Below T3, ≈ 5.5 mm diameter rods (or 6.35 mm for scoliosis surgery) are usually used either via a transitional rod (that tapers from the larger diameter to the smaller one) or using a domino connector to mate the two rods

109.2.3 Anatomic (“freehand”) thoracic pedicle screw placement technique

Advantages and disadvantages

Possible advantages

- may speed up surgery especially when instrumenting a large number of levels
- reduces the amount of X-rays/fluoroscopy needed during surgery

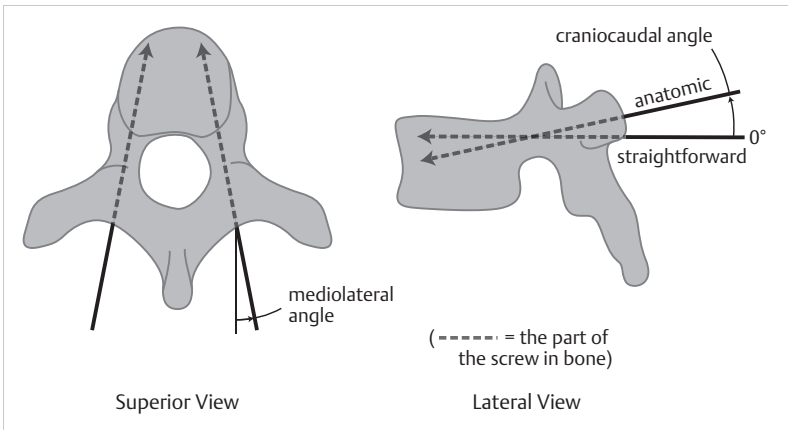


Fig. 109.2 Screw trajectory for thoracic *pedicle* screws. See text for mediolateral angle.

- avoids the challenges of getting fluoro aligned for each level in a scoliotic spine, especially when a rotational component is present
- not impeded in areas that are difficult to image on fluoroscopy (typically upper thoracic spine)
- accuracy as good as or better than other techniques.⁴ Pedicle breach rates were highest between T4 and T6⁴
- since facetectomies (removing part of the inferior facet) are usually employed, if an osteotomy or bone scalpel (as opposed to a drill) is used, it provides bone for use in bone graft (facetectomies can also be performed in conjunction with other screw placement techniques)
- exposes bone (of facet joints) to assist in fusion
- releases joints which facilitates reduction of scoliosis

Possible disadvantages

- cannot be used if anatomy is distorted by previous fusions, congenital anomalies...
- steep learning curve: proficiency requires knowing detailed level-specific anatomy and acquiring a feel for pedicle cannulation. Necessitates performing numerous screw placements, usually with a mentor
- possible increased blood loss from increased exposure and facetectomies
- individual variability from the "average" may produce sub-optimal screw placement

Procedure – freehand pedicle screws

A summary of key points/information⁵:

- complete exposure of entire posterior elements (lamina, pars interarticularis, facets, transverse process (tp)...) at every level (except the upper instrumented vertebra (UIV)) is mandatory
- intraoperative electrophysiologic monitoring: SSEP, MEP, triggered EMG
- inferior facetectomies: performed bilaterally at every level except the most superior (which is not being fused to the level above) to expose the superior facet of the level below. Make 2 cuts (► Fig. 109.3): vertical cut is parallel to spinal canal at the medial edge of facet joint (caution: in scoliosis, the spinal cord is at risk on the side of concavity), horizontal cut is at upper edge of inferior facet
- a series of steps are repeated without deviation with every screw placement:
 - make pilot hole at entry point (as indicated below)
 - insert curved pedicle finder with the tip directed laterally until 2 cm deep (do not force)
 - completely remove the pedicle finder, rotate it 180° and re-insert it down the same path just created, but now with the curve directed medially and insert it an additional centimeter, then remove it
 - palpate all 4 walls and the deep end with a thin ball probe, push it gently through the cancellous bone ("crunch through") until the tip strikes the cortical bone of the anterior vertebral body (sound and feel). Place a clamp where the probe enters the bone and...

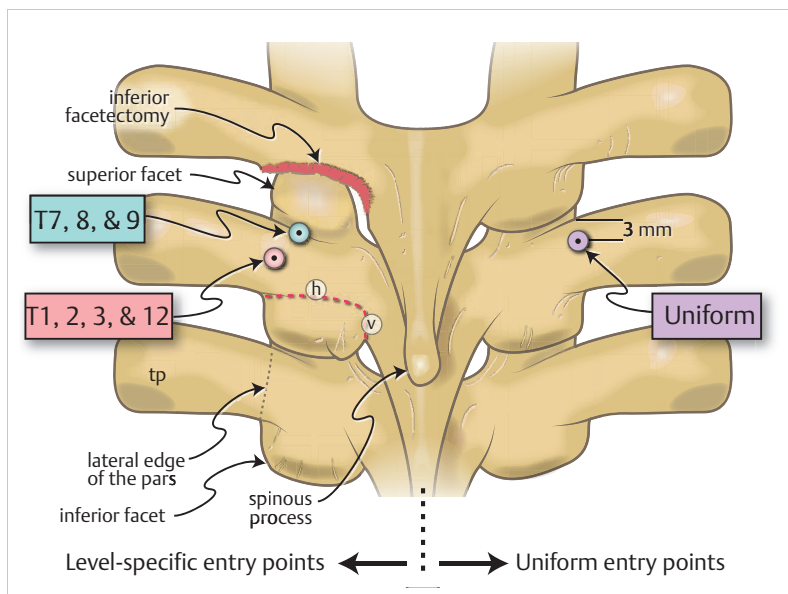


Fig. 109.3 Entry points for thoracic pedicle screws.

The left side illustrates level-specific entry points, and the right side shows uniform placement (see text).

Posterior view, stylized representation of generic thoracic level.

Mnemonic for level-specific entry points: "T1, 2, 3: mid tp (tp = transverse process). T7, 8, 9: top of the line (the "top of the line" being the superior border of the tp)."

Inferior facetectomy: red broken line shows cut lines for the (v) = vertical cut and (h) = horizontal cut. Facetectomy may also be used with the uniform entry point method.

- measure the depth that the probe was inserted to determine maximum screw length; if the length is much shorter than expected, the probe may be hitting the lateral VB wall and consideration should be given to redirecting the screw more medially
- tap the pedicle (i.e., cut the threads for the screw using a tap that is 0.5 mm smaller than the planned screw diameter)
- probe 2nd time with a thin ball probe and palpate all 4 walls and the deep
- place screw
- run a set of motor evoked potentials on electrophysiologic monitoring
- optional: after all screws are placed, stimulate each screw (e.g., with a Prass probe inserted into the screw head within the tulip of the pedicle screw) and determine the lowest current that produces an EMG response (triggered EMG): stimulation at <6 mA (some surgeons prefer 8 mA) or a threshold that is 65% or more decreased from the average of the other screws indicates a possible medial pedicle breach and should prompt a re-evaluation of the screw (remove screw & re-probe the hole or obtain a fluoro image)⁵

ENTRY⁵ (this describes "level-specific" entry points and trajectories. Alternatively, see uniform entry points & trajectories (p. 1795))

- craniocaudal (illustrated on the left side of ► Fig. 109.3)
 - T1, 2, 3, & 12: even with the middle of the transverse process (mnemonic: T1–2–3: "mid tp"); same for T12 (as with lumbar spine levels below it)
 - T7, 8, & 9: even with the top of the transverse process (mnemonic: T7–8–9: "top of the line")
 - for levels in between these 2 groups (viz. T4, 5, & 6), gradually move slightly superiorly at each level from mid-tp at T3 to the top of the tp at T7

- mediolateral (illustrated on the left side of ► Fig. 109.3). Tip: safe zone is *lateral* to the midpoint of the superior articular facet (SAF). Some use Lenke's "superior facet rule": enter 2–3 mm lateral to the SAF midline
 - T1, 2, & 3: even with the lateral edge of the pars
 - T7, 8, & 9: just lateral to midposition of the base of the superior articular facet (the most medial starting points)
 - T11, 12: at or just medial to lateral edge of pars
 - levels in between these groups: gradually transition the position

TRAJ: two methodologies

1. using anatomic landmarks: more commonly used than "angle technique" by most practitioners
 - advantage: angles are difficult to estimate, and using landmarks allows screws to be placed even in rotated, scoliotic spines
 - technique: the screw is inserted perpendicular to the surface of the superior articular facet (which is exposed during the facetectomies) while also "aiming" at the contralateral pedicle
2. angle technique: since estimating angles is difficult, this technique may be more helpful to conceptualize the general orientation
 - craniocaudal orientation (p. 1792)⁶:
 - for straightforward configuration: aim 0° relative to horizontal
 - for anatomic configuration: aim 10–15° caudal
 - mediolateral trajectory⁶: (► Fig. 109.2): the angle gradually becomes more medial as you progress from
 - T12 where the angle is slightly lateral ($\approx -5^\circ$) to
 - T11 where it is $\approx 0^\circ$ and as a rough approximation it increases by 2° per level above this to
 - T1 where the angle is $\approx 27^\circ$ medial

SCREW

- length: the depth to the anterior cortex varies from 40–45 mm when measured along the axis of the pedicle (or 30–42 mm when measured parallel to the sagittal plane).⁶ Typical thoracic screw length is 35–40 mm
- diameter: the narrowest pedicles in the mediolateral dimension are typically T4–7.⁶ Screw diameter should be approximately 80% of the pedicle diameter

109.3 Uniform entry points for freehand pedicle screws

Alternatively, a uniform entry point for thoracic pedicle screws (instead of modifying the entry point based on level as described above) may be used, e.g., the method of Baaj et al,⁷ illustrated on the right side of ► Fig. 109.3

ENTRY 3 mm inferior to the junction of the lateral edge of the superior facet and the transverse process.

TRAJ for a "straightforward" trajectory, the pedicle finder is oriented orthogonal to the curve of the spine in the sagittal plane.

109.4 Anterior access to the lumbar spine

109.4.1 Anterior lumbar interbody fusion (ALIF)

Relatively contraindicated in males because of risk of retrograde ejaculation in 1–2% (as high as 45% in some reviews). Other risks: injury to great vessels, especially with calcified arteries, in particular at L4–5.

Bowel prep the day before surgery may be used for complex cases.

Position: a general OR table is usually used. Place the iliac crests over the kidney rest or use a sacral bump to increase lordosis. Trendelenburg position allows gravity to retract the abdominal contents superiorly.

Approach: retroperitoneal usually through a Pfannenstiel's abdominal incision. A vertical incision is an alternative. Avoid injuring the ureters.

As a result of the bifurcation of the great vessels (aorta and inferior vena cava) which ranges from just above to just below the L4–5 disc space, this approach is best suited for access to L5–1.

At L5–1, the anterior sacral artery runs down the midline of the VB and has to be sacrificed to do an ALIF.

Dissection: At L5–S1, bovie cautery should be avoided when exposing the anterior longitudinal ligament because of risk of injury to the (superior) hypogastric plexus which can cause retrograde ejaculation in males. A vertical midline incision of the prevertebral soft tissues is made and the tissues are swept to their respective side using periosteal elevators.

Discectomy: a rectangular opening in the ALL and anterior anulus is incised with a scalpel and the discectomy is performed as in an ACD, using fluoroscopy to check the depth to avoid penetrating the PLL. Endplates are prepared using curettes and rasps to remove all cartilage from the endplates.

Implant: typical implant width is 35–38 mm for women, and 40–42 mm for men.

Complications: retrograde ejaculation in males occurs in 1–20% using a retroperitoneal approach. Venous thrombosis (usually of the common iliac veins) occurs in up to 10%.

109.5 Instrumentation/fusion pearls for the lumbar and lumbosacral spine

1. a lumbar fusion that includes L1 should not be terminated at L1 or T12
2. the taller the disc space the less likely that interbody grafts are well suited:
 - a) the disc may not be significantly degenerated to require discectomy
 - b) tall disc space means larger interbody implants which requires more retraction of nerve to insert (using a PLIF technique)
3. a long fusion should not be terminated at or near a vertebral level that is at the apex of scoliosis⁸ (p 382)
4. laminectomy without fusion should be avoided at the apex of scoliosis
5. posterior midline fusion: early experience with midline fusions resulted in lumbar spinal stenosis as a late complication. Therefore, current fusion techniques include posterolateral fusion, interbody fusion (from anterior or posterior approach), facet fusion...

109.6 Lumbosacral pedicle screws

109.6.1 General information

Pedicle screw pull-out strength is determined in part by the major screw diameter which should be 70–80% of pedicle diameter (in adults, larger screws can break through the pedicle wall or can burst the pedicle; in peds, larger screws may be used as the pedicle can expand). The minor diameter determines the strength of the screw and should be ≥ 5.5 mm in the adult lumbar spine.

109.6.2 Placement techniques

There are at least 4 screw placement techniques

1. intraoperative fluoro: biplane fluoro facilitates this technique
 - a) PROS:
 - allows percutaneous screw placement
 - generally good accuracy in screw placement
 - b) CONS:
 - imaging may be difficult in some parts of the lumbar spine, especially in larger patients. In these cases, the Steinman pin method below can be used to supplement
 - may increase radiation exposure to surgery team and patient
2. Steinman pin method: Steinman pins are placed at the estimated entry points for the screws, and AP (and often lateral) fluoro is used to fine-tune the position so that the screw enters the pedicle at the desired location
3. freehand placement based on anatomic landmarks. Usually with X-ray verification after all screws are placed. Greatly facilitated at levels where a laminectomy has been performed since the medial pedicle is exposed and is easily palpated
 - a) PROS: likely reduces radiation to the surgical team and the patient
 - b) CONS: requires somewhat more experience than the other methods; distortion of landmarks e.g., by previous surgery can preclude using this method; since the methodology uses averages in anatomy, it can be unreliable for patients whose anatomy differs from average
4. image guidance using instruments that are fitted with specialized markers that are tracked real-time by “cameras” that project the drill and/or screw location on a CT or X-ray image viewed in the O.R.
 - a) PROS:
 - reduces intraoperative radiation to surgical team, and to a lesser extent to the patient
 - allows percutaneous screw placement
 - b) CONS: accuracy may be compromised by movement of spinal segments relative to the registration array, or by technical errors. The surgeon must be vigilant for screw placement that does not look appropriate based on the anatomy

109.6.3 Open lumbar pedicle screw technique (see below for percutaneous placement)

- ENTRY** at the base of the transverse process, at the intersection of the center of the transverse process (in the rostral-caudal direction) and the sagittal plane through the lateral aspect of the superior facet. If a laminectomy has been performed at that level, the location of the pedicle is verified by palpation using a probe within the spinal canal, otherwise fluoroscopy is used. NB: degenerative changes include facet hypertrophy which can obscure these landmarks. At all but the superior-most level, it is often helpful to rongeur or drill off the hypertrophied part of the facet (avoid this at the superior level to reduce the risk of proximal junctional kyphosis)
- TRAJ**
 - a) the approximate mediolateral trajectory is shown in ► Table 109.1, and equals the lumbar vertebral number multiplied by 5° for each level from L1 to L5.⁹ The angle of the screw in the rostral-caudal direction is determined by fluoroscopy, maintaining a course that is either parallel to the vertebral endplate (for “straightforward” trajectory), or parallel to the axis of the pedicle (anatomic trajectory) (► Fig. 109.2). Image-guided navigation may be used
 - b) S1 craniocaudal trajectory: aim for the sacral promontory (the anterior superior edge of S1)
 - c) S2 screws may be oriented laterally and superiorly (into the ala) and can be as long as 60 mm. Alternatively, they can be oriented medially

Table 109.1 Medial angles for lumbar pedicle screw

Level	Medial angle
L1	5° medially
L2	10° medially
L3	15° medially
L4	20° medially
L5 & S1 ^a	25° medially
S2	40–45° laterally

^aaim for sacral promontory

- SCREWS** major screw diameter = 70–80% of pedicle diameter. Length should put tip 2/3 of the way across the VB (typical screw lengths: 40–55 mm) except for S1, which are usually only 35–40 mm long. Bicortical purchase or anterior VB penetration should be avoided (except for S1) to reduce the risk of injury to great vessels or abdominal viscera

- ROD** Typically 5–6.5 mm diameter

X-ray verification once pedicle screws are placed: on AP view if the screw tip crosses the midline to the contralateral side, there is likely to be a breach of the medial pedicle (sensitivity 0.87, specificity 0.97, accuracy 0.98).¹⁰ and if the screw does not pass medial to the medial pedicle wall there is likely to be lateral pedicle/VB violation (sensitivity 0.94, specificity 0.90, accuracy 0.96¹⁰).

109.6.4 Percutaneous pedicle screws

The principles here are also employed in accessing the pedicles for e.g., vertebroplasty/kyphoplasty, percutaneous biopsy of pathology in the pedicle and/or vertebral body.

Basic principles:

- requires AP and lateral fluoro, or image guidance (e.g., “O-arm”). With fluoro, biplane fluoro (1 C-arm dedicated for AP view, another for lateral) expedites the procedure
- this method can be employed essentially from T1 through S1 as long as adequate AP & lateral imaging of the involved level is possible. Using fluoro for upper thoracic placement (e.g., above ≈ T5) is challenging (small pedicles, and the shoulders interfere with lateral X-ray)
- the skin entry site is lateral to the lateral edge of the pedicle. This permits the needle to pass through the pedicle in a medial direction into the VB. The degree of angulation and therefore the distance off the midline for the entry site depends on the vertebral level being accessed (thoracic pedicles are oriented in a more AP direction, lumbar pedicles angulate medially inward) as well as the amount of overlying muscle/fat

Procedure:

- for an average size patient, the skin entry points will be about 4.5–5 cm lateral to midline; for larger patients this moves a little further laterally. Local anesthetic is injected using a spinal

needle which is adjusted under fluoro to determine an approximate entry site for the Jamshidi needle in the craniocaudal plane. The skin is incised just enough to accommodate the tower that is attached to the pedicle screw (usually ≈ 1.5 cm)

2. a Jamshidi needle (or equivalent) is inserted and the tip is positioned just short of entering the pedicle on the lateral fluoro (on left in ► Fig. 109.4)
3. at this point, on the AP view the needle should be at or just barely lateral to the lateral edge of the pedicle near the equator of the pedicle (“3:00” position on the right pedicle, “9:00” position on the left)

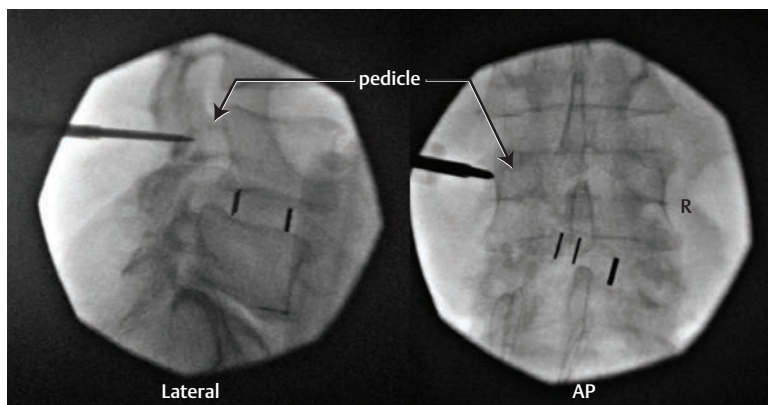


Fig. 109.4 Pedicle cannulation—entering pedicle.

4. the needle is advanced to just enter the pedicle on lateral fluoro; at this point it should be just within the pedicle margin on AP fluoro (as shown on the right in ► Fig. 109.4)
5. continue advancing the needle into the pedicle. Intermediate fluoro images can be obtained (e.g., to monitor trajectory on the lateral fluoro), but the next critical landmark is when the needle tip is just traversing the junction of the pedicle and the VB on the lateral fluoro (i.e., just entering the VB, as shown on the left in ► Fig. 109.5). It should be close to but never more medial than the medial border of the pedicle on the AP view (as shown on right in ► Fig. 109.5). If this criterion is

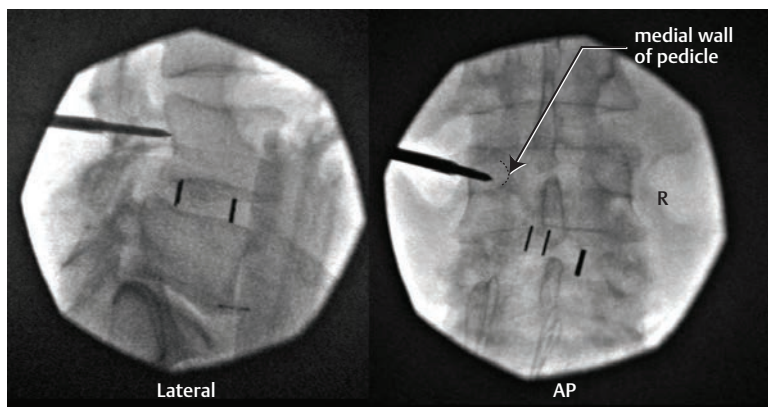


Fig. 109.5 Pedicle cannulation—entering the vertebral body.

maintained, the needle cannot breach the medial wall of the pedicle where it can threaten neural structures or compromise the purchase of the pedicle screw

6. after the needle tip is ≈ 1 cm into the VB, the stylet is removed and a guidewire is inserted through the needle for an additional 1–2 cm into the VB
7. the needle is removed taking great care not to dislodge the guidewire
8. in some cases, the cannulated screw with tower may be placed over the guidewire at this time, or the guidewire may be gently bent out of the way (without kinking it) for screw placement at a later step in the procedure

109.6.5 Pedicle-screw rod diameters

Approximate weight guidelines for pedicle-screw rod diameters are shown in ► Table 109.2.

Table 109.2 Minimum recommended titanium rod diameter for lumbar pedicle-screw fixation

Patient weight		Rod diameter (mm)
(lbs)	(kg)	
30–90	12–40	4.5
90–225	40–100	5.5
> 225	> 100	6.35 (1/4 inch)

109.6.6 Pedicle screw breach classification

CT is the diagnostic test of choice for evaluating pedicle screws for violation of bony margins (breach).¹¹ However, CT may distort the apparent screw location by as much as 25%.⁴

A number of breach grading scales have been proposed. Many are variations of 2 systems.

The Gertzbein system¹² measures *medial* pedicle breach from T8 to S1 and stratifies them as shown in ► Table 109.3. In the original study, medial violation > 4 mm was associated with neurologic deficit. There are nonvalidated opinions that 2 or 4 mm for medial breach, and 4 mm for lateral breach are “safe zones.”¹¹

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Table 109.3 Gertzbein classification of pedicle screw medial breach¹²

Grade	Breach distance (mm)
0	no breach
1	< 2
2	2–4
3	> 4

Table 109.4 Heary classification of pedicle screw breach¹³

Grade	Breach
1	no breach
2	lateral with screw tip within VB
3	anterior or lateral breach of screw tip
4	medial or inferior breach
5	proximity to sensitive structures necessitates immediate revision

The Heary system¹³ (► Table 109.4) takes into account the fact that sometimes a lateral breach is intentional (e.g., with small thoracic pedicles) and also that anterior breaches may have consequences. However, this system does not consider the degree of violation.

109.7 Lumbar cortical bone trajectory screw fixation

109.7.1 General information

AKA “Cortical Screws” (a trademarked name). Screws enter the bone of a lumbar vertebra posteriorly near the inferomedial border of the pedicle. They are directed superiorly and laterally, thereby

passing close to 3 cortical margins, which purportedly provides a pull-out strength close to that of traditionally oriented pedicle screws.

Cortical screw threads are closer together (higher pitch) than pedicle screws to better grip cortical bone. Typical screw diameter (4.5–5.5 mm) is smaller than pedicle screws.

Limitations: Use is generally restricted to lumbar spine and S1 (L1 to S1). May be precluded if entry points are compromised by previous laminectomy. Use is more controversial with high-grade spondylosis or scoliosis (especially if instrumentation needs to go higher than L1 or L2).

1. **PROS:**

- a) placement through a midline incision is somewhat easier compared to trying to aim medially from a more lateral entry point as with traditional pedicle screws (especially in obese patients)
- b) construct is medial to the segmental back muscles causing less injury to them
- c) may be used as rescue screws for failed pedicle screws
- d) may be used to extend a previous pedicle screw construct (may be placed in the same VB as traditional pedicle screw)

2. **CONS:**

- a) pull-out strength is probably slightly less than pedicle screws
- b) the bone that has to be left unviolated at the screw entry site may interfere with some of the decompression

109.7.2 Lumbar cortical bone trajectory screw technique

1. **ENTRY** (exceptions: the cranial-most screw and S1, see below)

- craniocaudal: the entry point is even with the cranial margin of the neural foramen (on lateral fluoro). Mediolateral: 3–5 mm medial to lateral edge of the pars interarticularis. On AP fluoro, this is at the inferomedial border of the pedicle
- there are 2 exceptions to this entry point
 - exception 1 – the cranial-most screw: enters 2–3 mm lower than the other screws (to avoid injuring the rostral facet joint that will not be part of the fusion)
 - exception 2 – S1 screw: enters midway between the L5–1 facet and the first dorsal foramen
- practical method: find the “corner” where the lateral pars and inferior TP (transverse process) meet (► Fig. 109.6), move the drill tip 3 mm medial to that, and make a dimple with the drill (at the cranial-most level, also move 2–3 mm inferior)
- pitfall: a severely hypertrophied facet of the level above may cover the entry point and may need to be partially removed

2. **TRAJ**

- overview: $\approx 20^\circ$ laterally, and $30\text{--}45^\circ$ cephalad
- practical method: with the drill tip in the entry dimple, bring the drill shaft medially to the spinous process (SP) and inferiorly to the bottom half of the SP (► Fig. 109.6). Aim for a point in the superior endplate approximately 1/3 of the way forward from the back of the VB on lateral fluoro
- drill with a pistoning technique. When first learning the technique, check that when you reach the midposition of the pedicle on lateral fluoro you should be in the center of the pedicle on AP fluoro. Drill the entire length of the screw (no pedicle finders!). Probe with a narrow probe to rule out breach and to measure screw length. Holes should be tapped to the actual screw diameter (not undertapped) for the entire screw length (tap “line-to-line”) to avoid fracturing the dense cortical bone
- 3. if needed, the laminectomy is done at this time, leaving a 3-mm cuff of bone around the screw entry points. Alternatively, modular screws (without the tulips attached) may be placed before the laminectomy

4. **SCREWS**

- use specialized cortical screws (not pedicle screws)
- typical screw diameter: 5 mm (range: 4.5 to 5.5).
- typical length: 30–35 mm.
- S1 screw: usually 6.6 mm diameter, length measured with a probe (usually ≈ 45 mm).
- leave the cranial-most screw 2–3 mm proud

5. **ROD** typically 5–6.5 mm diameter

109.8 Translaminar lumbar screw fixation

1. **indications:**

- a) short-segment lumbar fusion
- b) posterior component in a 360° fixation combined with interbody fusion

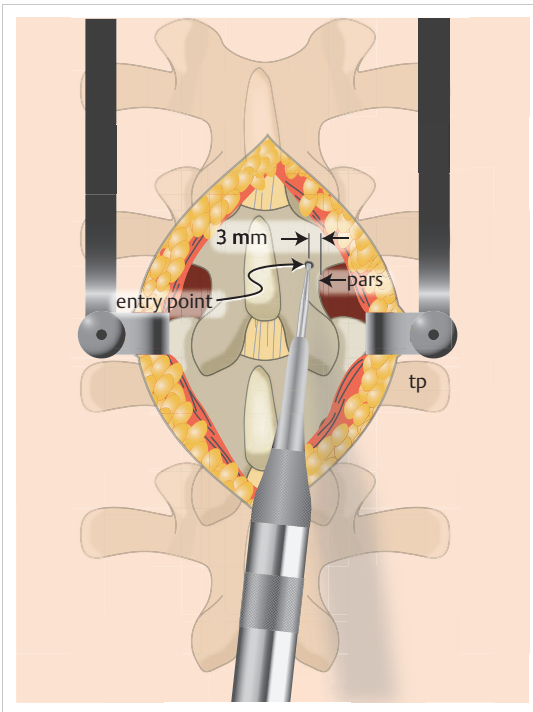


Fig. 109.6 Entry point for cortical bone trajectory screws (see text for differences for the most cranial level and S1).

The tip of the drill is positioned 3 mm medial to the lateral edge of the pars at the inferior edge of the tp.

Abbreviations: tp = transverse process; pars = pars interarticularis.

2. **PROS:**
 - a) small incision, minimal soft tissue disruption
 - b) decreased cost (fewer screws implanted)
 - c) decreased blood loss
 - d) adjacent facet joint spared
3. **CONS:**
 - a) requires intact posterior elements (cannot use with laminectomy)
 - b) cannot reduce
4. **ENTRY** skin incision 5–7 mm off midline, screw entry into bone in contralateral side of spinous process. Can be placed bilaterally
5. drill between the tables of the lamina across the center of the facet joint, terminating at the base of the transverse process
6. **SCREWS** 4.5 mm diameter fully threaded screws (no polyaxial head)
7. biomechanics: equivalent to bilateral pedicle screws.¹⁴ Limited in extension¹⁵
8. clinical: 476 patients, 10 years mean F/U, 74% good outcomes¹⁵

109.9 Posterior lumbar interbody fusion (PLIF and TLIF)

Originally developed by Cloward¹⁶ in 1943. Bilateral laminectomy and aggressive discectomy followed by the placement of bone grafts into the decorticated disc space. It has been advocated to reduce the movement in an abnormal “motion segment” (defined as the area between two vertebra). Relatively contraindicated with well-preserved disc-space height.

Many PLIFs when studied ≈ 1 year later show re-collapse of the disc space, which raises the question as to whether the PLIF has any benefit over simple discectomy. Areas of concern include: concerns regarding injury to nerves at the time of surgery or later due to retropulsion of bone graft.

Transforaminal lumbar interbody fusion (TLIF): a variation on a PLIF where the graft is placed from one side (via the “neural foramen”) after complete removal of the facet joint on that side.

Requires much less nerve root retraction than PLIF, and is often advantageous for reoperations with primarily unilateral pathology where going through the foramen avoids the scar tissue.

Stand-alone PLIFs or TLIFs may be associated with progressive spondylolisthesis at that level and are usually supplemented with pedicle screws/rods.

109.10 Minimally invasive lateral retroperitoneal transpsoas interbody fusion

109.10.1 General information

First introduced by Luiz Pimenta in 2001^{17,18} as an adaptation of an endoscopic lateral transpsoas approach to lumbar fusion described by Bergey et al.¹⁹ Trademarked names include “extreme-lateral” (XLIF™, NuVasive, San Diego, CA) or “direct-lateral” lumbar interbody fusion (DLIF™, Medtronic, Memphis, TN); the generic term lateral lumbar interbody fusion (LLIF) will be used here. Variants to the approach include the oblique lumbar interbody fusion (OLIF™, Medtronic, Memphis, TN) which utilizes a pre-psoas approach (at L5–1 the OLIF is halfway between an ALIF and an LLIF). A retroperitoneal approach indirectly decompresses nerves by distracting the disc space and fuses the spine with an interbody cage that has a large cross-sectional area. Access is best from L1–5. For L1–2, one can retract the 12th rib, or go between 11th & 12th rib, or excise the 12th rib. Iliac crest prevents access to L5–1 (axial-LIF may be used here) and occasionally to L4–5 (see below). A similar retropleural approach may be employed in the thoracic spine up to T4. ✖ With *thoracic* lateral interbody fusions, DO NOT penetrate the contralateral anulus. Intraoperative EMG monitoring is critical, so the anesthesiologist needs to use only *short-acting* neuromuscular blockade at beginning of case. In males, implants are typically 55–60 mm in length (oriented along patient's lateral axis) if placed in the midposition of the VB, or 45 mm in the anterior portion (lengths are 10% shorter in females).²⁰ Potential advantages include less tissue trauma, minimal blood loss, shorter operation time, fewer wound issues, placement of a larger cage, early patient mobilization,^{21,22} no risk of durotomy with CSF leak.

109.10.2 Indications

- central lumbar spinal stenosis (mild to moderate only) with neurogenic claudication
- foraminal stenosis (indirect decompression)
- spondylolisthesis grade 1 or 2
- axial low back pain associated with degenerative disc disease
- total disc replacement
- for correction of sagittal or coronal imbalance
- adjacent segment failure: LLIF is particularly attractive here because it obviates dealing with scar tissue and (often) hardware from previous surgery and which also reduces risk of durotomy
- burst fractures and tumors in the thoracolumbar area (corpectomy)
- to retrieve damaged or malpositioned lumbar disc replacement devices²³
- adult spinal deformity: can be used to correct scoliosis and to increase lumbar lordosis especially when combined with release of the anterior longitudinal ligament (ALL release)²²

109.10.3 Contraindications

1. cases requiring direct decompression: includes
 - a) pathology within the spinal canal e.g., herniated disc, synovial cyst, where simply distracting the disc space is less likely to correct the pathology
 - b) somewhat imprecisely defined “pinpoint” central canal stenosis (some cases may still respond)
2. tall disc spaces: disc space height > 12 mm usually implies that further distraction may be difficult to achieve. However, the interbody cage can still prevent compression at these levels when the patient stands up
3. prior retroperitoneal surgery on the planned side of LLIF (can still be done on contralateral side, sometimes may still be feasible on ipsilateral side) or extensive or complicated abdominal surgery (adhesions may preclude approach from either side)
4. pathology at L5–1: the technique usually cannot access L5–1 due to the interference from the ilium of the pelvis (an OLIF (oblique approach) or ALIF may be an alternative)
5. may not be able to access L4–5 if the iliac crest extends more than ≈ halfway up the L4 VB. Sometimes it is necessary to position the patient on the O.R. table with the table flexed and a bump under the hip to see if the space will “open up” and permit access. Angled instruments can

usually permit acceptable access to the space anterior to the lumbar plexus if it is not located too far forward on the VB

6. anomalous vascular anatomy interfering with approach: check position of great vessels on pre-op imaging (CR or MRI)
7. relative contraindications:
 - a) osteoporosis: may also be contraindication to lateral plates
 - b) active infection (relatively contraindicated with any fusion technique)

109.10.4 Surgical technique (MIS retroperitoneal transpsoas approach)

1. Position
 - a) lateral decubitus position with the top of the iliac crest just superior to the table break
 - b) choice of side: if there is no reason not to do so, the left side is usually placed up. Factors that might influence which side is to be up:
 - right side up if access is needed to L4–5 and the iliac crest is higher on left and would interfere (use AP X-rays, lateral X-rays, and/or lumbar CT to evaluate)
 - previous retroperitoneal surgery would cause one to consider placing the contralateral side up
 - if scoliosis is present and the intent is to correct this, the concave side is usually up: this usually provides better access to L4–5 if it is an operative level. Also usually allows access to multiple levels through potentially fewer and smaller incisions because the corridors to each disc space tend to converge
 - if ACR is planned, a posterior location of the great vessels and especially the lack of any soft tissue between the vertebral body/osteophyte and the vessel would speak for using the contralateral side if that is more favorable (if not, ACR may not be advisable)
 - c) true orthogonal position: the C-arm fluoroscopy is placed horizontal with 0° tilt, the patient position is then fine-tuned until the spinous process is exactly in the center between the pedicles on the AP view. If this is not possible due to rotation between levels, a relatively neutral level is chosen to start, and the O.R. table will need to be rotated slightly for each level as it is being worked on in order to make that level true lateral (centering spinous process).

Adhesive tape is applied to maintain the patient in this position

2. lateral fluoroscopy is used to mark the index disc space transversely and the posterior third of the disc space vertically. An exception is at L4–5, where the vertical mark is at the middle of the disc space based on the anatomic safe zones²⁴
3. retroperitoneal access through a single lateral skin incision and blunt dissection through abdominal muscles and fascia (external oblique, internal oblique, transversalis)
4. transpsoas approach and retractor placement is achieved using sequential tubular dilators that are placed under the guidance of fluoroscopic imaging (or navigation) and using directional EMG monitoring (Neurovision™, NuVasive, San Diego, CA), allowing the dilator to be placed anterior to the main lumbar plexus
5. discectomy and preparation of the disc space without violation of the endplates (to reduce the risk of subsidence) are performed. Work straight up-and-down in the ventral-dorsal plane (to avoid injuring the ALL anteriorly or entering the spinal canal posteriorly)
6. interbody spacer is placed usually with the posterior edge in the posterior third of the disc space (as the EMG monitoring allows)
7. for anterior column release (MIS-ACR), additional steps have to be performed after discectomy, which include dissection/section of the ALL and placement of hyperlordotic cages (20 or 30 degrees), usually more anteriorly in the disc space than a routine cage. This is a technique for the advanced lateral access surgeon

109.10.5 Instrumented augmentation (pedicle screws or lateral plate)

Standalone cage

A standalone cage (i.e., no additional instrumentation) may be feasible in the following circumstances:

- no osteoporosis
- no instability on pre-op lateral flexion/extension X-rays
- the ALL was not disrupted during LLIF surgery
- a cage width of at least 22 mm in the AP dimension has been placed: the large surface area reduces the risk of subsidence

When these conditions are not met, additional instrumentation should be considered.

Lateral plate and screws

May be applied through the same exposure. May not be optimal in patients with osteoporosis or age > 55 due to increased risk of subsidence from weaker bone. Not as practical for multilevel LLIFs.

Posterior instrumentation

Posterior instrumentation (e.g., pedicle screws, including percutaneous placement) provides the most stabilization. Also, may be indicated if laminectomy is needed for direct decompression.

109.10.6 Complications

1. thigh numbness: incidence is $\approx 10\text{--}12\%$.^{22,25} Due to injury to the genitofemoral nerve. Risk of direct injury in the quadrant anterior to the midline at L2–3 and in the anterior quadrant at L3–4 and L4–5.²⁴ A sensory nerve cannot be monitored with EMG. Usually transient, resolves in ≈ 2 weeks
2. thigh flexion weakness: due to psoas injury. Risk increases if > 2 levels are done. Usually transient, resolves in 1–8 weeks
3. femoral nerve or femoral/obturator nerve palsy as a result of nerve root/plexus injury.²⁶ 83% of cases show findings of axonotmesis (► Table 32.6) at 6 weeks.²⁷ Produces quadriceps weakness. Incidence is $\approx 2.6\%$.²⁷ Many cases are likely due to compression of the nerves within the psoas muscle when the muscle is retracted against the transverse processes; weakness is usually evident immediately post-op. Another possible etiology is psoas hematoma in which case the onset may be delayed 1–2 d. Prognosis: all 6 cases improved to 4/5 MRC strength by 12 months, some as early as 3 months (► Table 30.2).²⁷ Prior to recovery, patients may be able to ambulate with a brace
4. lumbar plexus injuries: the risk of direct injury to the lumbar plexus is reduced by staying at or anterior to “safe” working zones defined by Uribe et al²⁴ as follows (NB: it is often possible to work posterior to these zones using EMG monitoring to ascertain the proximity of the plexus, and conversely injuries can occur even in these safe zones).
Safe zones:
 - L1–2 through L3–4: middle of the posterior quarter of the VB
 - L4–5: midpoint of the VB
5. contralateral femoral nerve injury
6. genitofemoral neuralgia
7. abdominal viscera perforation
8. vascular injury²⁸ including common iliac artery (at the L4–5 level) or aortic injury (above L4–5), common iliac vein or inferior vena cava
9. kidney-ureter injury
10. graft subsidence
11. unintended rupture of anterior longitudinal ligament
12. psoas/retroperitoneal hematoma ipsilateral/contralateral
13. abdominal wall paresis or hernia²⁹
14. rhabdomyolysis (p. 136)
15. retrograde ejaculation (primarily with ACR and pre-psoas approaches)

109.10.7 Postoperative care

- for single-level lumbar LLIF: mobilize the patient in the immediate post-op period without a brace
- hip flexion pain on the approach side is anticipated during the immediate postoperative period
- transient hip flexion weakness related to psoas muscle manipulation is usually self-limited and improves by 8 weeks post-op
- in case of significant leg weakness (femoral nerve injury), a lumbar CT or MRI is indicated to rule out compression by a psoas hematoma, disc extrusion, or malposition of cage or screw. If compression is ruled out, patients can be followed by postoperative EMG at 6 weeks to define extent of injury (neuropraxia, axonotmesis, neuronotmesis), again at 3 months to evaluate if the expected interval improvement from neuropraxia has occurred and 5 months to follow up axonal growth²⁶

109.10.8 Outcomes

- fusion rates range from 91–100%³⁰
- outcomes scales (ODI and VAS leg and back) are significantly improved at follow-up³¹

Booking the case: Lateral interbody fusion

Also see defaults & disclaimers (p. 25) and pre-op orders (see below).

1. position: lateral decubitus, typically left-side up unless specified otherwise
2. equipment: C-arm
3. implants:
 - a) interbody graft
 - b) some stabilizing hardware is usually needed, especially if spondylolisthesis is present. Options:
 - pedicle screws: bilateral or unilateral
 - interspinous clamps
4. consent (in lay terms for the patient—not all-inclusive):
 - a) procedure: surgery through the side to place a spacer between two of the vertebrae (back-bones) to make more space for the nerves and to stop painful movement, screws/plates etc., will then need to be placed either from the side through the same opening or sometimes from the back. In case the procedure cannot be done from the side due to the position of the lumbar plexus (uncommon, when it occurs it is usually an issue at L4–5), determine if patient wants to have a posterior procedure (e.g., TLIF) and put that on the consent. Be sure to notify the vendor of this possibility
 - b) alternatives: nonsurgical management, open surgery through the back
 - c) complications: thigh weakness (usually temporary), knee weakness (uncommon), thigh numbness, graft subsidence/migration, failure to achieve desired relief

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109.11 Transfacet pedicle screws

109.11.1 General information

Screws placed directly across the lumbar facet joint into the pedicle of the level below. No rod is needed. Immobilizes only, does not provide any decompression, distraction, or fusion. Therefore not intended for use as a stand-alone. Can be placed percutaneously.

109.11.2 Indications

Placement is optimal for L3–4, L4–5, or L5–1. Difficulty increases in upper lumbar levels.

May be used as adjunct to:

1. ALIF
2. LLIF (when lateral plate not used)
3. contralateral to TLIF (pedicle screws could be used on the side of the TLIF, or a spinous process clamp could be used)
4. axial-LIF (Ax-LIF)

109.11.3 Contraindications

A transfacet pedicle screw cannot be used where the facet has been removed (e.g., for a TLIF) or with a pars defect in the upper of the two levels to be fused.

109.11.4 Technique

1. placed percutaneously or via an open procedure usually in the prone position
2. approximate skin incision site: a single midline ≈ 1.5-cm vertical incision is used
 - a) for L5–1 or L4–5: incision at L3 spinous process
 - b) for L3–4: incision at L2 spinous process
3. use AP & lateral fluoro to guide trajectory
 - a) AP fluoro: lay a guidewire on the patient's back and orient it to pass through the desired pedicle. Use a skin marker on the patient's back to mark the guidewire's trajectory
 - b) lateral fluoro: initial bony target is the midpoint of the inferior facet of the upper level. The tip of the guidewire should contact the bone directly posterior to the inferior endplate of the upper level

109.12 Facet fusion

A bone dowel (e.g., TruFUSE® by MinSURG) is placed into a predrilled opening in the facet joint to promote fusion across the joint. Marketed as a possible stand-alone.

109.13 S2 screws

May be directed medially (analogous to pedicle screws), or more commonly, directed laterally and superiorly into the ala (S2 alar screws) (see below for S2-alar-iliac screws). In either case, bicortical purchase is necessary.

The main pitfall to avoid is penetrating the sacroiliac (SI) joint with the screw.

109.14 Iliac fixation

109.14.1 General information

Stronger fixation than S1 or S2 alar screws (S2 alar screws not to be confused with S2-alar-iliac screws (p. 1807)).

Primary indications for extending fixation to ilium (i.e., pelvis)

1. long constructs (any construct from S1 that extends above L2)
2. high-grade spondylolisthesis (Meyerding grade (p. 1339) ≥ 3)
3. S1 screw failure
4. pseudarthrosis of prior attempted L5–S1 fusion
5. sacral insufficiency

In the absence of sacroiliac joint fusion (autofusion or surgical fusion), the iliac screws or rods connecting to them will eventually break. The function of the iliac screws here is to assist the S1 screws during the early stages of the fusion.

3 main options

1. iliac screw (AKA iliac bolt)
2. "percutaneous" iliac bolt
3. S2-alar-iliac screws

If the sacroiliac joint (SIJ) is not fused, the screw or the rod will usually eventually break because of continued movement across the SIJ.

109.14.2 Iliac screws

Wide exposure is needed. On the initial few cases, the surgeon may be better served by exposing all the way to the posterosuperior aspect of the sciatic notch so that the screw trajectory can be aimed using a palpating finger.

An offset adapter is often required to connect to rods passing through pedicle screws in the levels above.

ENTRY The classic entry point is the posterior superior iliac spine (PSIS) (► Fig. 109.7). A small amount of bone is removed just below and medial to the PSIS. This prevents the head of the screw from being too superficial, which may cause discomfort or skin breakdown.

TRAJ Overview: the screw trajectory aims towards the acetabulum to (ideally) pass approximately 1 cm superior to the sciatic notch. Placement is facilitated by 2 oblique fluoroscopy views:

1. obturator outlet view (AKA "teardrop view") (► Fig. 109.8 A): parallel to the iliac wing. On the teardrop view, stay within the hollow of the teardrop—the walls are cortical surfaces you don't want to violate. Also, avoid penetrating the cortex of the sciatic notch
2. iliac oblique view: perpendicular to the hemipelvis

SCREWS Length 50–80 mm (the screw should end just at or beyond the midpoint of the sciatic notch). Diameter: 6–8 mm.

109.14.3 Percutaneous iliac screw

Advantages

Compared to iliac screws

1. can line up with lumbar pedicle screws above without need for offset adapter

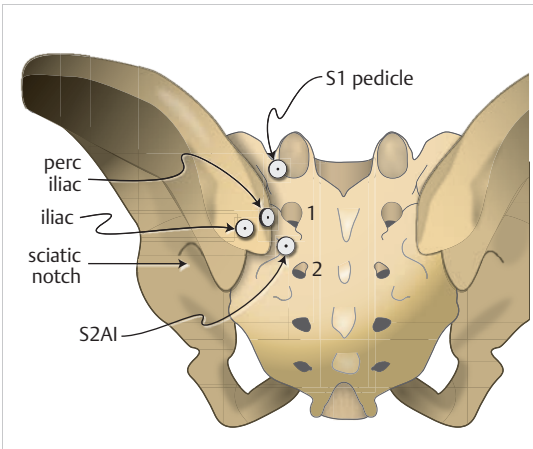


Fig. 109.7 Entry points for iliac fixation.

Entry points (white circles with black central dots) on the left side for S1 pedicle screws (depicted for reference, these are not iliac fixation screws), iliac screws (at the PSIS), percutaneous iliac screws, and S2-alar-iliac screws. Innominate bone and sacrum, dorsal view.

Key: 1 = 1st dorsal sacral foramen; 2 = 2nd dorsal sacral foramen; perc = percutaneous; S2Al = S2-alar-iliac.

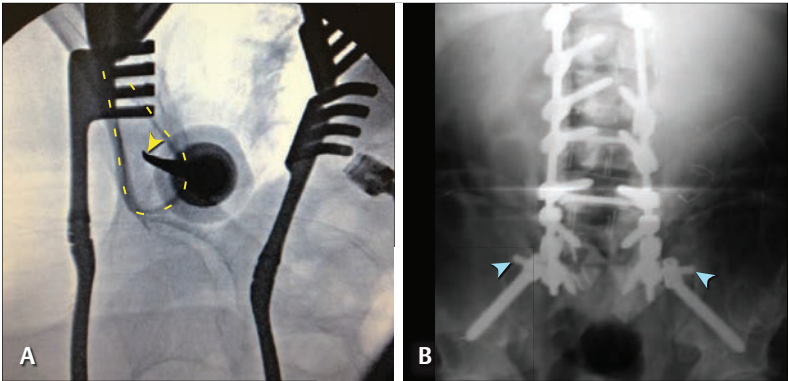


Fig. 109.8 Teardrop view and final AP X-ray for iliac screws.

A: Teardrop view (the teardrop is outlined as a broken yellow line). The pedicle finder (the tip is identified by the yellow arrowhead) is shown entering the teardrop.

B: A final AP X-ray. The offset connectors are identified with blue arrowheads.

2. less muscle dissection
3. less prominent on the surface of the skin
4. may be placed percutaneously

Technique

ENTRY The entry point is on the medial surface of the ilium (► Fig. 109.7).

TRAJ Same as for iliac screws.

109.14.4 S2-alar-iliac screws (S2Al screws)

Advantages

1. compared to iliac screws, same advantages as percutaneous iliac screws
 - a) can line up with lumbar pedicle screws without need for offset adapter
 - b) less muscle dissection
 - c) less prominent on the surface of the skin

- compared to percutaneous iliac screws: immobilizes the sacroiliac joint, which reduces screw/rod breakage (this joint may not be very mobile in patients with significant adult degenerative disc disease)

Technique

Options for placement include:

- fluoroscopy-assisted
- image navigation
- free-hand³²

ENTRY midway between the S1 and S2 neural foramen, at the lateral edge of the foramina, inline with the S1 pedicle screw (if used) (schematic shown in ► Fig. 109.7; the X-ray appearance is shown in ► Fig. 109.9 panel A).

SCREWS Length 80–90 mm (at or beyond the midpoint of the sciatic notch as with iliac screws). Diameter: 6–8 mm.

TRAJ Fluoroscopic method: initially aim for the anterior inferior iliac spine (AIIS), which is palpated through the drapes with the other hand. The curved pedicle finder is used with the curve directed posteriorly.³² The AIIS can be difficult to palpate in larger patients; in that case, palpate the greater trochanter and aim for a point 2–3 cm above that. A mallet is used to traverse the 2 cortical surfaces of the sacroiliac joint (SIJ) which are usually encountered at about 35–40 mm. Once the teardrop (on the obturator outlet fluoro view) is entered, follow the trajectory on fluoro and stay within the teardrop as with iliac screw and attempt to pass approximately 1 cm above the iliac notch. The hole is probed for breaches, tapped, and re-probed before the screw is placed.

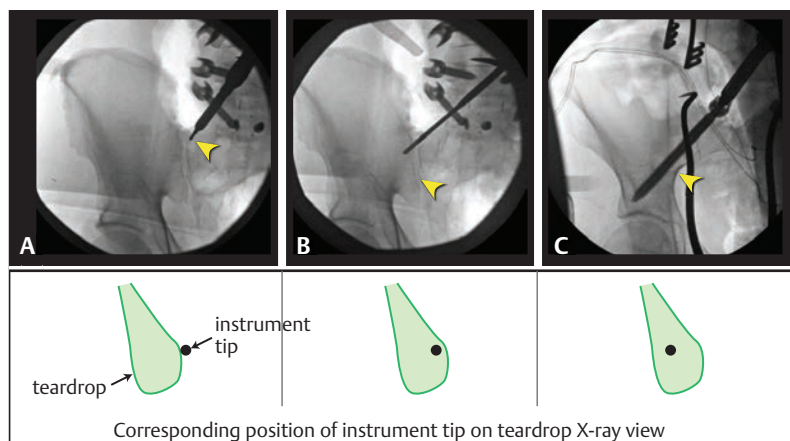


Fig. 109.9 S2-alar-iliac screws.

A: X-ray appearance of the entry point on AP view (arrowhead shows drill tip after drilling about 1 cm depth starting at the entry point shown in ► Fig. 109.7).

B: Pedicle finder crossing the sacroiliac joint (SIJ) (arrowhead).

C: Final screw placement on sciatic notch view. Note that the screw passes approximately 1 cm above the sciatic notch (arrowhead).

The lower panel shows corresponding location of the tip (black circle) of the pedicle finder on the teardrop view.

109.15 Post-op clinic visits—lumbar and/or thoracic spine fusion

109.15.1 Visit schedule

Patients are seen in the clinic at intervals depending on the preference of the surgeon. A typical follow-up schedule with studies routinely performed is shown in ► Table 109.5. For specific problems, additional investigations are usually needed.

Table 109.5 Sample post-op lumbar fusion clinic visit schedule^a

Time post-op	Agenda
7–10 d	wound check, D/C sutures/staples if used
6 wks	AP & lateral LS-spine X-ray in brace
10–12 wks	• AP & lateral LS-spine X-rays with flexion/extension views out of brace • if X-rays look good and patient is doing well, begin weaning brace
6 months	• AP & lateral LS-spine X-rays with flexion/extension views • some surgeons release patients at this time if they are doing well
1 year (optional)	• AP & lateral LS-spine X-rays with flexion/extension views • release patient if they are doing well
^a the same schedule can be used for thoracic fusions with the difference that standing AP & lateral X-rays are done in place of flexion/extension views	

109.15.2 Post-op X-rays

Items to check on post-op X-rays include:

1. alignment
2. position of grafts if used (e.g., interbody grafts)
3. integrity of hardware (look for screw or rod breakage, screw pull-out, rod disconnection)
4. lucencies around screws which may indicate motion and implies nonunion
5. any evidence of fusion (may be difficult, e.g., with synthetic interbody cages)
6. on flexion/extension films: look for motion across fused segments (sometimes absence of motion may be the only evidence of fusion on plain X-rays) and the development of abnormal motion at adjacent segments

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References

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110 Miscellaneous Surgical Procedures

110.1 Percutaneous ventricular puncture

110.1.1 Indications

In pediatrics, may be used to remove hemorrhagic ventricular fluid following intraventricular hemorrhage, or to obtain CSF specimen in cases of suspected ventriculitis. May be used emergently in pediatrics or adults as a temporizing measure in patients herniating from obstructive hydrocephalus.

110.1.2 Peds

Clip hair. 5-minute Betadine® prep.

The right side is preferred. Enter through coronal suture just lateral to anterior fontanelle (AF) using a 20–22 Ga spinal needle. If a CT scan has been done, it may be used to help judge angulation (usually varies between contra- and ipsilateral medial canthus and intersection with EAM).

110.1.3 Adult

See reference.¹

Only used emergently. Takes advantage of thin orbital roof in adult.

Prep conjunctiva and skin with antiseptic (e.g., ophthalmic Betadine). Elevate the eyelid and depress the globe. Using a 16–18 Ga spinal needle, penetrate the anterior third of orbital roof (1–2 cm behind orbital rim) with firm pressure (may need gentle tapping). Aim at coronal suture in the midline. The frontal horn should be about 3–4 cm deep.

110.2 Percutaneous subdural tap

110.2.1 Indications

Utilized in pediatrics. Used to be done for diagnostic purposes, but this has been supplanted by CT, MRI, & ultrasound. Currently, this procedure may be used emergently for decompression, to drain subdural collections, and to obtain fluid for diagnostic tests, such as culture (repeat taps may be used, but surgery should be considered after ≈ 5–6 taps).

110.2.2 Technique

Clip hair. Prep 5 minutes with povidone iodine (Betadine®). Using a short 20–21 Ga spinal needle (spinal needle is recommended because the stylet may reduce the risk of implanting epidermal cells into the CNS), penetrate the lateral margin of the anterior fontanelle (AF) or coronal suture at least 2 cm off midline. Remove the stylet and aspirate. With bilateral fluid collections, bilateral taps should be done.

110.3 Lumbar puncture

110.3.1 Contraindications

1. risk of tonsillar herniation (p.1815)
 - a) known or suspected intracranial mass (tumor, abscess, blood, cerebral edema...)
 - b) non-communicating (obstructive) hydrocephalus
 - c) relative contraindication: Chiari malformation, especially with a severe blockage of CSF circulation through the foramen magnum. This is less of a concern following satisfactory surgical decompression of Chiari malformation
2. infection in region desired for puncture: e.g skin ulceration. Choose another site if possible
3. coagulopathy
 - a) platelet count < 50,000/mm³ (p.161)
 - b) patient should not be on anticoagulants because of risk of epidural hematoma (p.1383) or subarachnoid hemorrhage² with secondary cord compression

4. use caution in suspected aneurysmal SAH: excessive lowering of the CSF pressure increases the transmural pressure (pressure across the aneurysm wall) and may precipitate rerupture
5. caution in patients with complete spinal block: 14% will deteriorate after LP³

Elevated ICP and/or papilledema by themselves are *NOT* contraindications (e.g., LP is actually used diagnostically and as a treatment in pseudotumor cerebri, see below).

110.3.2 Technique

Background and anatomy

The spinal cord and column are the same length in a 3-month fetus. After that, the spinal column grows faster than the cord. As a result, the conus medullaris (the lower terminus of the spinal cord) is located rostral to the termination of the thecal sac in the adult, situated between the middle thirds of the vertebral bodies (VB) of L1 and L2 in 51–68% of adults (the most common location), T12–1 in ≈ 30%, and L2–3 in ≈ 10% (with 94% of cords terminating within the territory of L1 and L2 VB).⁴ The thecal sac ends ≈ S2. The tips of the spinous processes as palpated on the surface are located caudal to the corresponding VB. The intercrystal line (connecting the superior border of the iliac crests) crosses the spine at the L4 spinous process or between the L4 and L5 spinous processes in most adults.

Procedure

Position: the procedure is usually performed in the lateral decubitus position. As the needle is advanced, it is helpful to have the patient bring the knees up and to flex the neck in order to open up the spaces between the posterior elements of the spine.

For diagnostic LP, a 20 Ga spinal needle is often selected. Larger needles (e.g., 18 Ga) may be used e.g., with pseudotumor cerebri to encourage post-procedure drainage of CSF into the soft tissues of the back.

The back is prepped and draped to create a sterile working area.

Entry point: in an adult, use the L4–5 interspace in most cases (located at or just below the intercrystal line) or 1 level higher (L3–4). Peds: L4–5 is preferred over L3–4.

The needle is always advanced with the *stylet in place* at least through the skin and some subcutaneous tissue to avoid introducing epidermal cells, which may cause iatrogenic *epidermoid tumors*; see Complications following LP (p. 1814). The needle is aimed slightly cranially (to parallel the spinous processes) and usually a little down towards the bed (aiming towards the umbilicus). If a Quincke LP (standard) needle is used, the bevel is turned parallel to the length of the spinal column to reduce the risk of post-LP H/A (p. 1817). In general, if bone is encountered it is more often due to deviation from a true midline trajectory rather than a failure to aim correctly in the rostral-caudal direction. The needle should be withdrawn to just below the skin surface before attempting a new trajectory.

If during insertion of the needle the patient experiences pain radiating down one LE, this usually indicates that a nerve root has been encountered. The needle should be withdrawn immediately and reinserted aiming more towards the side *contralateral* to the extremity that experienced the pain. The stylet is removed at intervals during the insertion to look for CSF (a distinct pop is sometimes felt as the needle penetrates the dura).

Once CSF flows, the needle is connected to a manometer through a 3-way stopcock, the pressure is measured and recorded (see below), and CSF is drained into sterile tubes (1–2 ml for each tube) for laboratory analysis (see below). The practitioner should also note the color of the fluid (clear, blood-tinged, xanthochromic...) and the clarity (clear, cloudy, purulent...).

At the end of the procedure, the stylet should be replaced before the needle is withdrawn (to reduce post-LP H/A, see below).

Opening pressure: The opening pressure (OP) should be measured and recorded for every LP. To be meaningful, the patient should be lying down and as relaxed as possible (should not be in forced fetal position), with the bed flat. The variation of pressure with respirations is usually a good indication of a communicating fluid column (the fluctuation is in-phase with the respiratory pressures in the inferior vena cava, rising with inspiration and falling with expiration⁵). Normal values: in the left lateral decubitus position, average OP = 12.2 ± 3.4 cm H₂O (8.8 ± 0.9 mm Hg).⁶ Also, see ► Table 23.1 for peds.

Queckenstedt's test: if a subarachnoid block is suspected (e.g., from spinal tumor), compress the jugular vein (JV) first on one side then on both sides (do not compress carotid arteries). If there is no

block, the pressure will rise to 10–20 cm of fluid, and will drop to the original level within 10 seconds of release of the JV.^{7(p 11)} Do *not* do JV compression if intracranial disease is suspected.

110.3.3 Laboratory analysis

Routinely, three tubes are sent for analysis as shown in ► Table 110.1. See ► Table 23.4 for interpreting the results of the laboratory analysis.

Table 110.1 Routine tests for CSF

Test	If there is <i>no</i> concern about possible traumatic tap	If there is concern about traumatic tap
cell count		Tube 1
gram stain + C & S (culture & sensitivity)	Tube 1	Tube 2
protein and glucose	Tube 2	Tube 3
cell count	Tube 3	Tube 4

If the tap is possibly traumatic (i.e., bloody), or if having an accurate cell count is essential (e.g., to R/O SAH) then 4 tubes are collected, and the first and last are sent for cell counts and are compared; see Traumatic tap (p. 1813).

If special cultures are required (e.g., acid-fast, fungal, viral) they are also specified on the tube for culture & sensitivity (C & S).

If CSF for cytology is desired (e.g., to R/O carcinomatous meningitis or CNS lymphoma), then at least 10 ml of CSF must be sent in one tube to pathology (where it is spun down and examined for cells).

110.3.4 Traumatic tap

General information

A traumatic tap (TT) (AKA bloody tap) occurs when the spinal needle damages a blood vessel with the result that (usually venous) blood either alone or admixed with CSF will be obtained.

TT has been defined as containing more than a certain number of RBCs per microliter, and different cutoffs have been used as a minimum: > 200, > 500, or > 1000 have been used.⁸

Incidence of TT varies depending on the study population and the cutoff used, and ranges from 8–30%.^{8,9}

Suggestions for “mining useful data” from a bloody tap for the purpose of diagnosing elevated WBCs (as in bacterial meningitis) or diagnosing SAH appear in the sections below.

Estimating true WBC count in CSF with a traumatic tap

When many RBCs and WBCs are present in the CSF due to a TT, it is difficult to know if there is a true leukocytosis on the CSF. It may help to determine if the WBCs are elevated or if they are present in the same ratio as in the peripheral blood. In non-anemic patients, there should be ≈ 1 –2 WBCs for every 1000 RBCs (as a correction^{10(p 176)}; subtract 1 WBC for every 700 RBCs^{10(p 176)}). In the presence of anemia or peripheral leukocytosis, use Fishman's formula^{10(p 176)} shown in Eq (110.1) to estimate the original WBC count in the CSF *before* the TT,

$$\text{WBC}_{\text{CSF Original}} = \text{WBC}_{\text{CSF}} - \frac{\text{WBC}_{\text{Blood}} \times \text{RBC}_{\text{CSF}}}{\text{RBC}_{\text{Blood}}} \quad (110.1)$$

where $\text{WBC}_{\text{CSF Original}}$ = WBC count in the CSF before the TT, WBC_{CSF} & RBC_{CSF} = WBC & RBC counts measured in the CSF, and $\text{WBC}_{\text{Blood}}$ & $\text{RBC}_{\text{Blood}}$ = WBC & RBC per mm^3 in the peripheral blood.

Estimating true total CSF protein content with a traumatic tap

If the hemogram and peripheral protein are normal, then have the cell count and protein content run on the *same tube*, and the correction is^{10(p 176)}:

- subtract 1 mg per 100 ml of protein for every 1000 RBC per mm^3

Differentiating SAH from traumatic tap

The sensitivity of detecting RBCs in CSF approaches 100%, and if it is assumed that 20% of LPs are traumatic (an approximation), then the specificity of a bloody tap for diagnosing the presence of blood in the CSF prior to the LP is only $\approx 80\%$.¹¹

See typical findings in SAH (p.1422). Some features helpful in differentiating SAH from TT are shown in ► Table 110.2.

Table 110.2 Features distinguishing traumatic tap from SAH

Feature	Traumatic tap (TT)	SAH
RBC count	< 2000 RBC/mm ³ ¹² (< 500 RBC/mm ³ had a 100% negative predictive value for SAH in one small study ¹³)	usually > 100,000 RBC/mm ³
clearing of RBCs from 1st tube to last tube	declines as CSF drains (controversial - see text)	changes little as CSF drains
ratio of WBC:RBC	similar to the ratio in peripheral blood (above)	usually promotes a leukocytosis (elevated WBC count)
supernatant	clear	xanthochromic (p. 1422) ³ (rarely in < 2 hrs, present in 70% by 6 hrs, and > 90% by 12 hrs after SAH)
clotting of fluid	usually clots if erythrocyte count > 200,000/mm ³	usually does not clot
protein concentration	fresh bleeding elevates CSF protein from normal by only ≈ 1 mg per 1000 RBC	blood breakdown products elevate this more than TT (measured protein exceeds the sum of normal protein + 1 mg protein/1000 RBC)
repeat LP at higher level	usually clear	remains bloody
opening pressure	usually normal	usually elevated

³NB: other conditions can cause xanthochromia

► **Clearing of RBCs with TT.** It is commonly held that the reduction of RBC count in the last tube compared to the first tube is indicative of a TT. An older study recommended a reduction > 30% as the requirement,¹⁴ and a smaller study (15 patients with SAH) recommended > 70% clearance as the cutoff.¹³

► **Xanthochromia.** Details and photo (p. 1422). Although it can occur earlier, a minimum of 4 hrs is usually required to develop xanthochromia (XTC). It is present in $\approx 100\%$ of bleeds at 12 hours, persists for ≈ 2 weeks. By 3 weeks it is found in about 70%, and may be detected as late as 4 weeks after the bleed.

► **Recommendations.** Conservative recommendation: clearing > 70% of RBCs from the first tube to the last with < 500 RBC/mm³ in the final tube is strongly supportive of a TT.¹³

The presence of XTC in the setting of bloody CSF virtually rules out a TT.

110.3.5 Complications following LP

General information

The overall risk of disabling or persistent symptoms (defined as severe H/A lasting > 7 days, cranial nerve palsies, major exacerbation of preexisting neurological disease, prolonged back pain, aseptic meningitis, and nerve root or peripheral nerve injuries) has been estimated at 0.1–0.5%.¹⁵ Severe side effects, which include brainstem herniation, infection, subdural hematoma or effusion, and SAH, are rare.¹⁶(p 171–2)

Possible complications

- tonsillar herniation
 - acute herniation in the presence of mass lesion (see below)
 - chronic tonsillar herniation (acquired Chiari 1 malformation): this has been reported after multiple traumatic LPs with presumed post-LP CSF leak¹⁷

2. infection (spinal meningitis)
3. "spinal headache": usually positional (diminishes with recumbency) (see below)
4. spinal epidural hematoma (p. 1383): usually seen only with coagulopathy
5. spinal epidural CSF collection: may be fairly common in patients with post-LP H/A. Usually resolves spontaneously
6. epidermoid tumor: risk may be increased by advancing LP needle without stylet (transplanting a core of epidermal tissue)^{18,19,20}
7. impinging nerve root with needle: usually causes transient radicular pain. May cause permanent radiculopathy in rare instances
8. intracranial subdural hygroma or hematoma^{21,22} (rare)
9. vestibulocochlear dysfunction²³:
 - a) subclinical (demonstrated on audiogram) or moderate reduction in hearing may occur, and seems to correlate with post-procedure CSF leakage. Most studies show reduction is at frequencies < 1000 Hz
 - b) sudden hearing loss may occur. Perform audiogram to quantify loss. Treat with bedrest for several days, prednisone 60 mg/d tapered over 2–3 weeks
 - c) pathogenesis: reduced CSF pressure may reduce perilymph pressure through the cochlear aqueduct (may be especially pronounced with a patent aqueduct),²⁴ producing endolymphatic hydrops
10. ocular abnormalities
 - a) abducens palsy: almost invariably unilateral. Often delayed 5–14 days post-LP, usually recovers after 4–6 wks²⁵
11. dural sinus thrombosis²⁶ (usually with underlying thrombophilia)

Risk of acute tonsillar herniation following lumbar puncture

The question of when to do LP first (to save time) and when to obtain CT scan first to R/O intracranial mass (for safety) before performing an LP is controversial.

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Issues

The time delay to initiating antibiotics is the most important variable in the outcome of meningitis. Mortality increase 13% per hour delay in treatment.²⁷ Time may be more crucial in community-acquired meningitis, where a virulent organism typically infects an immune-intact host (or immune-susceptible hosts such as children or the elderly), than in post-op neurosurgical meningitis (usually a low-virulence organism, e.g., *Staph. aureus*, in a host where the BBB has been disrupted but the patient is otherwise immune-intact).

The theoretical risk in performing an LP with elevated intracranial pressure (e.g., from intracranial mass, obstructive hydrocephalus, Chiari malformation...) is that the lowering of spinal CSF pressure resulting from removal of CSF during the procedure or the leakage of CSF through the puncture site after the procedure may precipitate tonsillar herniation due to the resultant increased pressure gradient.

Starting antibiotics without first having a CSF specimen from an LP may prevent cultures from growing in the lab which then risks the difficulties inherent in managing partially treated meningitis, or a suboptimal choice of antibiotic medication.

Clinical evaluation for possible contraindication to LP is unreliable. Papilledema is one possible indication of increased ICP. Papilledema takes a minimum of 6 hrs to develop after the onset of increased ICP, and in most cases it requires up to 24 hrs to develop. Therefore, its absence does not ensure normal ICP. Furthermore, papilledema may be seen in conditions where there is not a contraindication to LP, e.g., pseudotumor cerebri (p. 955), where LP is one of the accepted treatments.

The ready availability of CT scans, often within the emergency department itself, may involve a delay of only a few minutes, if qualified personnel to interpret the study are also immediately available. However, in practice, obtaining a CT scan delayed subsequent treatment by an average of 1.6 hours²⁷ and increased mortality.

Historical information

Herniation following LP was more common prior to ≈ 1950, long before CT scans were available, where the procedure was performed even when some patients had clear evidence of ↑ ICP, large-bore spinal needles (12–16 gauge) were more commonly employed, and large quantities of CSF were removed for therapeutic purposes. In a 1969 report of 30 patients who deteriorated after LP,²⁸ 73% had localizing signs (hemiparesis, anisocoria...) and 30% had papilledema. None of 5 patients with cerebral abscess deteriorated after the first of multiple LPs.

In a series of 129 patients with $\uparrow$ ICP,²⁹ the complication rate reported was 6%; however, some of these complications were probably unrelated to LP, and many of these patients were in extremis. In 7 series totalling 418 patients, a complication rate of 1.2% was calculated.²⁹ The risk of LP is small³⁰ especially with a 20 gauge or smaller needle and removing only a few ml of CSF.

Σ: LP in acute bacterial meningitis (ABM)

Herniation as a result of LP is consistently reported only in patients with severe non-infectious processes, often with accompanying signs of mass effect (localizing signs, papilledema...). Modified international guidelines²⁷ list these contraindications to LP in cases of suspected acute bacterial meningitis: papilledema, focal neurologic signs & immunocompromised state.

Antibiotics and corticosteroids should be started immediately (the goal is within 30–60 minutes, but this has been achieved in only 35%) in cases of suspected ABM. If a CT scan is indicated before LP, treatment can be started before the LP. If the CT scan does not show a space-occupying lesion, LP may be performed. In the unlikely event that there is acute deterioration associated with the withdrawal of a few ml of CSF, the (anecdotal) recommendation is to immediately replace the fluid through the LP needle.

Post-LP (myelogram) H/A

General information

AKA “postspinal headache” or “spinal headache.” May also follow procedures other than LP/myelogram, such as dural opening (p.1260). Can also occur with spontaneous intracranial hypotension (p.421) and following decompressive craniectomy.³¹

Clinical features

Important distinctive characteristic: H/A occurs when patient is erect, and is completely or partially (but significantly) relieved when recumbent. May be associated with nausea, vomiting, dizziness, or visual disturbances.

► **Time course.** Most post-LP headaches (PLPHA) have a delayed onset 24–48 hrs after the LP, and although they may occur weeks post-LP, most also develop within 3 days. The duration of PLPHA varies, with a mean of 4 days,³² and reports of duration of months³³ and even > 1 year.³⁴

Pathophysiology

Thought to be due to continued CSF leakage through the hole in the dura,³⁵ which reduces the CSF “cushion” of the brain. In the upright position, the pull of gravity on the brain produces traction on the blood vessels and any structures tethering the brain to the pain-sensitive dura. CSF may sometimes be demonstrable in the *epidural* space.

Epidemiology following LP

Reported incidence range is 2–40% (typically $\approx$ 20%), higher after diagnostic LP than for epidural anesthesia.³²

See also variables in LP that impact upon the risk of PLPHA (p.1816) (e.g., incidence is lower with smaller-gauge spinal needle).

1. risk factors for post-LP H/A outside the control of the physician:
 - a) age: incidence $\uparrow$ in younger patients
 - b) sex: incidence $\uparrow$ in females
 - c) prior headache history (including previous PLPHA)
 - d) body size: $\uparrow$ with small body mass index = weight/height² ³⁶
 - e) pregnancy
2. variables that have been shown to influence the incidence of PLPHA:
 - a) needle size: larger needles carry increased risk³⁷
 - b) bevel orientation: orienting the bevel parallel to the longitudinally running fibers of the dura reduces the risk of PLPHA³⁸
 - c) replacing the stylet prior to needle removal lowers the incidence³⁶
 - d) the number of dural punctures (may not be totally under the physician's control)

3. variables that may or may not influence the incidence of PLPHA: needle type:
 - a) Quincke needle: bevelled edge with cutting tip (the standard LP needle). Incidence of PLPHA with 20 and 22 gauge Quincke needles: 36%³⁹
 - b) atraumatic needles: a number of types are available (Sprotte, Whitacre...). Most are "pencil-pointed" and may produce a hole with a lower incidence of transdural leak.⁴⁰ Unproven³⁶
4. factors found *not* to affect the incidence of PLPHA:
 - a) the position of the patient after LP (does not seem to prevent PLPHA, but may delay the onset of symptoms^{41,42})
 - b) volume of fluid removed at the time of LP
 - c) hydration following LP³⁶

Treatment for H/A following LP

Initial "conservative" measures include:

1. flat in bed for at least 24 hrs
2. hydration (PO or IV)
3. analgesics for H/A
4. tight abdominal binder
5. deoxycortisone acetate 5 mg IM q 8 hrs³²
6. caffeine sodium benzoate 500 mg in 2 cc IV q 8 hrs up to 3 d max (70% of patients had relief with 1 or 2 injections)⁴³
7. high-dose steroids: report of success in a case of intracranial hypotension associated with spontaneous slit ventricles tapering down from a starting dose of dexamethasone 20 mg/day⁴⁴
8. blood patch if refractory

► **Epidural blood patch.** For refractory post-lumbar puncture or post-myelogram H/A. Works in one application in over 90% of cases, may be repeated if ineffective.³³ Theoretical risks: infection, cauda equina compression, failure to relieve H/A.

► **Technique.** Summary: 10 ml of non-heparinized autologous blood injected into *epidural* space.

Accessing epidural space (one of several techniques): proceed as routine LP. When ligaments are traversed, and needle tip is nearing spinal canal, stylet is removed. Then, either place drop of sterile saline in hub (hanging drop technique) and advance while watching for it to be drawn into needle as epidural space is entered, or gently try injecting air with small syringe (preferably glass, lower resistance) while advancing, when the epidural space is entered, resistance to injection disappears, but CSF cannot be aspirated.

A venipuncture site is prepared aseptically. 10 ml of the patient's blood is withdrawn. After verifying CSF cannot be aspirated through the spinal needle, the blood is injected into the epidural space. After 30 minutes supine, patient may ambulate ad lib.

110.4 Lumbar catheter CSF drainage

110.4.1 General information

Insertion of a catheter into the lumbar subarachnoid space for the purpose of draining CSF. Usually connected to a closed drainage system similar to that for an EVD. Generally used for periods of only a few days or so.

110.4.2 Indications

1. to reduce CSF pressure at a site of CSF leak/fistula. Examples:
 - a) dural breach during spine surgery or craniotomy (especially posterior fossa)
 - b) for spontaneous CSF fistula (p. 418) (rare)
2. to reduce intracranial pressure in cases of *communicating* hydrocephalus: e.g., drain test for NPH, or when an infected shunt has been removed
3. to reduce CSF pressure to attempt to increase perfusion of the spinal cord: e.g., during surgery for abdominal aortic aneurysm, or following spinal cord injury

110.4.3 Contraindications

As with lumbar puncture (see above).

110.4.4 Insertion technique

Positioning, entry site, and trajectory are all similar to lumbar puncture (see above). Placing the patient in slight reverse Trendelenburg helps distend the lumbar thecal sac. Instead of a spinal needle, a specialized needle designed for inserting a catheter is used. A Tuohy needle is commonly used, it is a spinal needle with a slightly curved beveled tip (► Fig. 110.1) and a stylet to avoid coring tissue and displacing it into the spinal canal. The needle with stylet in place is inserted with the bevel parallel to the fibers of the dura (rostral-caudal). As the needle is advanced, periodically the stylet is removed and if necessary the needle is aspirated with a syringe to see if CSF drains.⁴⁵ If not, the stylet is replaced and the needle is advanced. Once the tip is intrathecal (as evidenced by flow of CSF through the needle), the needle is rotated 90° (usually pointing rostrally – the direction of the needle curve is indicated by a small notch in the hub of the needle) and the catheter is threaded into the needle.

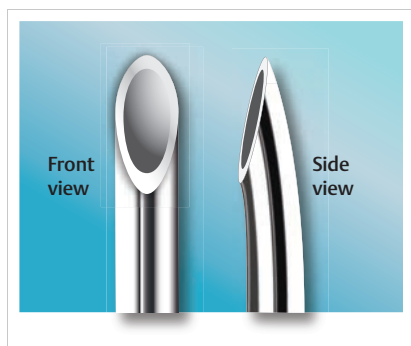


Fig. 110.1 Tuohy needle tip.
Diagram of the tip of a Tuohy needle.

✱ If the catheter does not thread, the needle must be withdrawn *together* with the catheter. Attempting to withdraw the catheter through the needle will shear off the catheter at the tip of the needle, leaving a portion of the catheter tube inside the patient.

110.4.5 Management

Nursing orders to maintain the catheter include:

1. instructions to regulate the CSF drainage. Most commonly, either:
 - a) by pressure: accomplished by specifying a height of the drip chamber, usually at the level of the tragus or shoulder
 - b) by withdrawing a specified amount of CSF per hour: usually 10–20 cc. This method reduces the risk of overdrainage if the drip chamber is too low
2. the risk of overdrainage (by either of the above methods) may be reduced by connecting the catheter to a volume-limiting drainage system (e.g., LimiTorr™ by Integra)
3. instructions for the exit site dressing: usually treated as an arterial-line

110.4.6 Complications

1. infection
2. overdrainage: usually as a result of the drainage bag being too low when using the pressure drainage method described above (either from falling to the floor, or not being raised when the patient sits or stands up) or from catheter disconnection. Can cause:
 - a) subdural hematoma from tearing of bridging veins from downward displacement of the brain
 - b) headache
3. pneumocephalus: usually from placing the drain height below the site of a fistula, and air is drawn in through the fistula tract
 - a) tension pneumocephalus: usually with a ball-valve effect at the fistula site
4. catheter pull-out: frequently occurs simply as a result of patient movement in bed or with patient transfers

110.5 C1–2 puncture and cisternal tap

110.5.1 Indications

Situations where CSF specimen is required but access via LP is difficult or contraindicated (lumbar arachnoiditis, superficial infection, marked obesity, patients who cannot be turned on their sides...), or to instill contrast to demonstrate the rostral extent of a block documented by dye injected via LP. Spinal headache is less common with these procedures than with LP. C1–2 puncture is safer than cisternal tap.

✖ **Contraindicated:** in patient with Chiari malformation (often present in myelomeningocele) due to low-lying cerebellar tonsils and medullary kink.

Normal CSF values for glucose and protein differ only slightly from CSF obtained by lumbar puncture. Opening pressures averaged 18 cm of fluid with lateral puncture.

110.5.2 C1–2 puncture

AKA lateral cervical puncture. Equipment: LP tray (useful for the specimen tubes, extension tube for contrast injection under fluoroscopy, lidocaine, and spinal needle) with a standard 20 Ga spinal needle, contrast if needed (e.g., Iohexol®). It is preferred to perform the procedure under fluoro, but it has also been described without fluoroscopic guidance with a completely cooperative patient.⁴⁶

Patient position: supine in bed without a pillow, with the head straight up. Avoid any head rotation which may bring the vertebral artery (VA) into the needle path.⁴⁷ Place head within lateral fluoroscopy unit (since this is rarely available, a C-arm fluoro positioned horizontally may be used).

If iodinated dye is being injected for myelography, the head should be elevated to prevent contrast from running into posterior fossa; in cases with cervical spine injury, one can put the entire bed in reverse Trendelenburg.

Entry point: 1 cm caudal and 1 cm posterior (dorsal) to the tip of the mastoid process. Needle insertion: use a 25 Ga needle to anesthetize the skin at the entry point. Under fluoro, advance a larger needle (e.g., 21 Ga) towards the C1–2 interspace while injecting local anesthetic: aim for a target in the middle of the posterior third of the bony spinal canal (or, alternatively, 2–3 mm anterior to the posterior margin of the bony canal) ("X" in ► Fig. 110.2). Leave this needle in as a marker.

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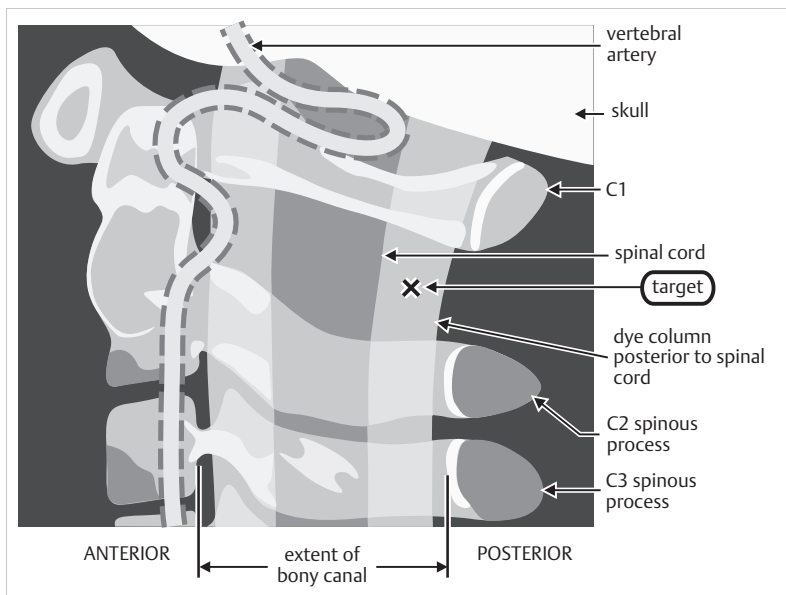


Fig. 110.2 C1–2 puncture target.*

*Left lateral view through upper C-spine: composite diagram of a myelogram and vertebral arteriogram illustrating the relative location of the spinal cord, CSF space, and VA. Only bony landmarks will be visible with fluoroscopy.

Insert the 20 Ga spinal needle parallel to the marker needle. Verify the course with fluoro. If fluoro is not used, insert the spinal needle at the entry point, and advance it parallel to the plane of the bed, perpendicular to the neck.⁴⁶ If the needle penetrates deeply without encountering bone or CSF, it is most likely that the tip is too far posterior. If bone is encountered, redirect the needle in the rostro-caudal plane.

Several “pops” may be felt, and the stylet should be removed after each to check for CSF return. The subarachnoid space is $\approx 5\text{--}6\text{ cm}$ deep to the skin surface in most adults.⁴⁸ The needle must be supported more than with a lumbar puncture.

To inject iodinated contrast, use e.g., $\approx 5\text{ ml}$ of 180 mgI/ml Iohexol® for cervical myelogram, watch dye on fluoro (should be able to see it in subarachnoid space).

Risks

Case report of a death from subdural hematoma due to puncture of an anomalous vertebral artery⁴⁹ (found in $\approx 0.4\%$ of population). If the VA is penetrated, the needle is withdrawn and local pressure is applied. Penetration of the upper spinal cord/lower medulla (risk of serious neurologic sequelae, even from this, is small). Herniation (as with LP) when there is increased ICP.

110.5.3 Cisternal tap

Suboccipital access to the cisterna magna. Usually done with patient sitting, with neck slightly flexed.⁵⁰ Overlying hair should be shaved. Local anesthetic is infiltrated. A 22 gauge spinal needle is inserted exactly in the midline between theinion and the C2 spinous process, directed superiorly toward the glabella until the needle strikes the occiput or enters the cisterna magna. If the occiput is encountered, the needle is withdrawn slightly and reinserted, directed slightly inferiorly, and the process is repeated (“walking down the occiput”) until the cisterna magna is entered (a “pop” will be felt).

The distance from the skin surface to the cisterna magna is 4–6 cm, and from the dura to the medulla is $\approx 2.5\text{ cm}$. However, due to tenting of the dura, the needle may be very close to the medulla before entering the subarachnoid space.

Risks

1. hemorrhage in the cisterna magna: may be due to perforation of a large vessel⁴⁶
2. piercing the medulla oblongata: may cause vomiting, respiratory arrest...
3. positioning may compromise blood flow in the vertebral artery in elderly patients

110.6 Ventricular catheterization

110.6.1 General information

Ventricular catheterization is a basic neurosurgical technique that is most commonly employed for one of the following uses:

1. external ventricular drain (EVD): drains CSF to an external collection system. Commonly performed at the bedside, usually in the ICU or E/R
2. CSF diversion (shunting): drains CSF internally to a distal site in the body. Typically performed in the OR
3. placement of ventriculoscope for endoscopic procedures. Typically performed in the OR

110.6.2 Indications

Indications range widely; however, ventricular catheterization is most often needed for:

1. intracranial pressure measurement (monitoring) (p. 1038) and management (p. 1046)
2. temporary CSF diversion
 - a) hydrocephalus: when definitive shunting is not immediately indicated or practical. Typically for acute hydrocephalus associated with obstruction by tumor or blood, or in a patient with an existing shunt that has failed
 - b) infection: management of infected shunt, occasionally in meningitis when repeated LPs or a lumbar drain are not suitable (primarily in cryptococcal meningitis (p. 409), but occasionally in bacterial meningitis)

- intrathecal drug administration: for long-term use, a ventricular access device (p. 1831) is often preferred. Typically for chemotherapy (e.g., for CNS lymphoma) or sometimes for antibiotics (for ventriculitis complicating meningitis)

110.6.3 Procedure risks

The primary risks of the procedure are:

- infection
- bleeding: including bleeding along catheter tract (intraparenchymal), within the subdural/epidural or intraventricular space, or SAH in patients with an aneurysm (sometimes as a result of lowering ICP which can increase the transmural pressure (p. 1427)). Risk of any bleeding = 7%, clinically significant bleeding 0.8%⁵¹
- malposition of catheter
- catheter failure: occlusion by blood, debris (includes brain tissue, infectious debris...), or ependymal lining of ventricles if the ventricles completely collapse on the catheter, damage to catheter, catheter pullout

110.6.4 Coagulation

To reduce the risk of bleeding, coagulopathies need to be identified and corrected prior to the procedure. Ideally, patients should have an INR ≤ 1.6 , platelet count $> 100K$, and no recent antiplatelet medication (can be screened by checking thromboelastogram (TEG)).

Elevated INR: for patients with an elevated INR (such as patients on warfarin), it has generally been recommended that placement of intraparenchymal catheters be delayed until the INR is ≤ 1.6 in order to reduce the risk of hemorrhage to an acceptable level.^{52,53} However, intraparenchymal ICP monitors were placed without hemorrhagic or thrombotic complications in 11 patients with grade III/IV hepatic encephalopathy associated with fulminant hepatic failure who had an average INR of 3 when it was performed within 15–120 minutes after receiving 36.7 micrograms/kg IV of recombinant activated factor VII (rFVIIa).⁵⁴ All of these studies showed that ICP monitors were inserted in the study patients without hemorrhagic complications – but bear in mind that absence of proof is not proof of absence.

110.6.5 Common ventricular catheter insertion sites

- ★ Kocher's point (coronal): localizes an entry point into the frontal horn of the lateral ventricle that passes anterior to the motor strip. The right side is usually used since that is more commonly the nondominant hemisphere. Often employed for ICP monitors, EVDs, shunts, ventriculoscopes.... Originally described as "about 2 cm from the central line and 3 cm from the precentral fissure."⁵⁵ A number of surface landmarks have been described to locate a point anterior to the motor strip that is in this general vicinity, and many of them refer to their target as "Kocher's point." Commonly cited landmarks:
 - entry site: (► Fig. 110.3)
 - 2–3 cm from midline, which is approximately the mid-pupillary line with forward gaze
 - 1 cm anterior to the coronal suture, which is approximately 11 cm back from the nasion in an adult (to avoid the motor strip (p.56))
 - trajectory: direct catheter perpendicular to surface of brain,⁵⁶ which can be approximated by aiming in coronal plane toward the medial canthus of the ipsilateral eye and in the AP plane toward the EAM
 - insertion length: advance catheter with stylet until CSF is obtained (should be $< 5-7$ cm depth; this may be 3–4 cm with markedly dilated ventricles). Advance catheter without stylet 1 cm deeper. ★ CAUTION: if CSF is not obtained until very long insertion length (e.g., ≥ 8 cm) the tip is probably in a cistern (e.g., prepontine cistern) which is undesirable
- ★ occipital-parietal region: commonly used for CSF shunt
 - entry site: a number of means have been described, including:
 - Frazier burr hole: placed prophylactically before p-fossa crani for emergency ventriculotomy in event of post-op swelling. Location: 3–4 cm from midline, 6–7 cm above inion⁵⁷ (p 520) (caution: an error in locating the inion could put the catheter in an undesirable location if this method alone is used)
 - parietal boss: flat portion of parietal bone
 - follow point from mid-pupillary line parallel to sagittal suture until it intersects line extending posteriorly from the top of the pinna
 - ≈ 3 cm above and ≈ 3 cm posterior to top of pinna

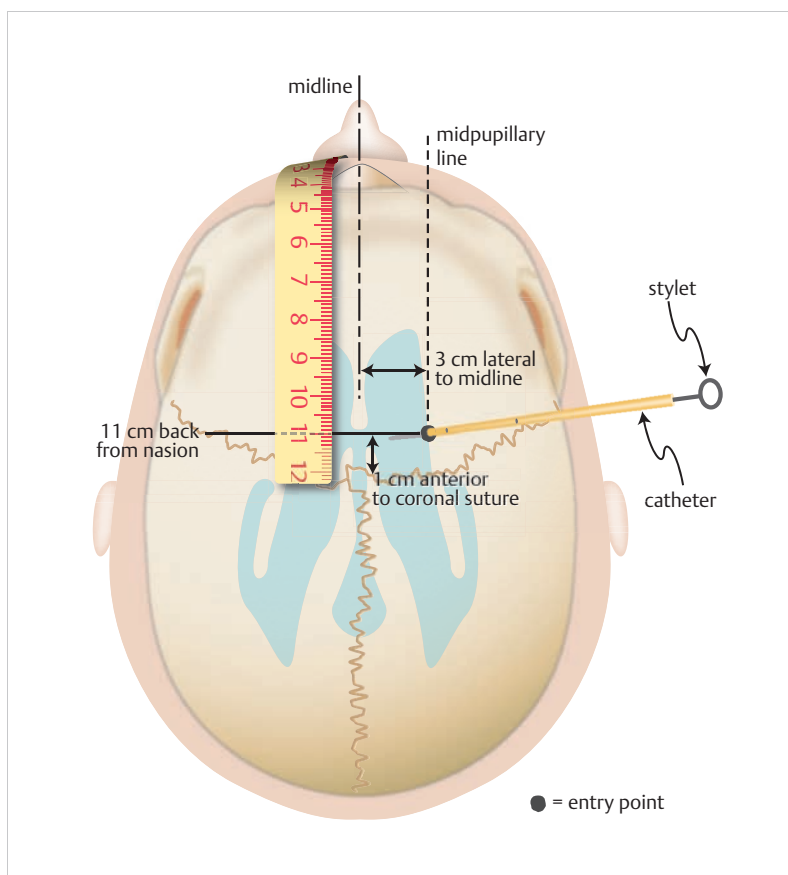


Fig. 110.3 Kocher's point. Ventriculostomy landmarks for Kocher's point. See text for details.

- b) trajectory: insert the catheter parallel to skull base:
 - initially aim for middle of forehead
 - if this fails, aim for ipsilateral medial canthus
- c) insertion length: ideally, the tip should be just anterior to the foramen of Monro in the frontal horn.⁵⁸ Ventriculoscopic guidance (if available) increases the accuracy to a significant degree. In the absence of this:
 - intracranial length should be $\approx$ two-thirds of the length of the skull (this is short enough to prevent penetration of frontal brain parenchyma, but long enough to take tip beyond the foramen of Monro to prevent catheter from ending up in the temporal horn where choroid plexus increases the chance of obstruction)
 - in adults without macrocrania the inserted length is usually $\approx$ 12 cm when the burr hole is in line with the axis of the lateral ventricle⁵⁹ (lengths > 12 cm are rarely required). In hydrocephalic infants usually $\approx$ 7–8 cm is required
 - use the stylet for the initial $\approx$ 6 cm of insertion, then remove it and insert the remaining length (keeps the catheter straight during penetration of occipital parenchyma and

prevents the tip from dropping into the temporal horn where there is choroid plexus; also, the temporal horn may collapse and occlude the catheter when the HCP is resolved)

3. Keen's point (posterior parietal): (placement in trigone) 2.5–3 cm posterior and 2.5–3 cm superior to pinna (was the usual site of occurrence of cerebral abscesses arising from otitis media, and was often used to tap these)
4. Dandy's point: 2 cm from midline, 3 cm aboveinion (may be more prone to damage visual pathways than above)

110.6.6 Ventriculostomy/ICP monitor – bedside insertion technique

General information

Intraventricular catheter (IVC) AKA external ventricular drain (EVD) placement at the bedside has a number of unique challenges that differ from ventricular catheterization in the OR where supplies, anesthesia, assistance (and possibly image navigation) are normally available.

★ 3 things that have been reproducibly shown to reduce the risk of infection⁶⁰:

1. tunneling the exit for the catheter > 2 cm
2. antibiotic impregnated ventricular catheter
3. shorter duration of presence of catheter

Informed consent

For procedures performed in the OR, see Booking the case (for shunts) (p.1831). The consent form for bedside ventricular catheterization should list the following which should be discussed with the patient or family when possible: infection, bleeding (stroke), and catheter malposition or failure. Often, patients requiring urgent placement of an EVD are unresponsive or unable to give informed consent, and an appropriate health care surrogate may not be available. In these situations, follow your specific institution's policy for emergency procedures.

Supplies

To enhance sterility and speed of the procedure, it is critical to have the required supplies immediately available. Many hospitals have an “EVD cart” with the supplies pre-stocked. A partial list is included here.

- ventricular catheter: an antibiotic impregnated catheter is recommended (see above)⁶⁰
- ventriculostomy kit: these kits usually contain a cranial access kit with scalpel, prep solution, drapes, razor, local anesthetic, hand drill with bit
- ventricular catheter drainage collection system: connects to the ventricular catheter. Usually consists of a drainage bag, a calibrated drip chamber with adjustable height, ports for sampling CSF and for connection to a pressure transducer
- sterile gowns, gloves, masks
- adhesive tape to stabilize the patient's head
- optional: marker to tape to middle of forehead that can be palpated through the drape to help the surgeon easily find this location. Items that work well: a “red-dot” EKG lead sticker, or a cap from a TB syringe turned so the “cup” faces forward

Insertion technique

Unless contraindicated (e.g., right ventricular bleed), the right (non-dominant) side is preferred.

1. clip the hair around the planned incision site and the exit site for the tunneled catheter (avoid shaving which compromises the skin barrier against infection)
2. mark the skin at Kocher's point (► Fig. 110.3): approximately 11 cm back from the nasion, and 2–3 cm (about 2 fingerbreadths) lateral to midline
3. stabilize the head with adhesive tape
4. don sterile gown and gloves, face mask, and eye protection
5. prep with an acceptable surgical prep and drape
6. infiltrate local anesthetic
7. a 2.5-cm skin incision is centered over Kocher's point. It is usually straight and sagittally oriented (in case it needs to be incorporated in a craniotomy flap in the future)
8. elevate the periosteum: the back end of the scalpel is convenient for this
9. place a self-retaining retractor

10. make a twist drill hole at Kocher's point (if you can see the coronal suture through the skin incision, drill the hole 1 cm anterior to it)
11. perforate the dura with the tip of a needle, moving it side-to-side (the dural opening must be large enough for the catheter or monitor probe)
12. inserting a ventricular catheter
 - the ventricular catheter is initially inserted with the stylet in place
 - insert the catheter *perpendicular* to the skull surface⁵⁶ to a maximum intracranial depth of 5–7 cm (most catheters are marked at 5 and 10 cm). If there is midline shift (MLS) on the pre-op CT, aim slightly towards the side of the MLS. When the catheter tip enters the ventricle, a subtle “pop” can be palpated, and if the CSF is under pressure some CSF may drip out through the proximal catheter around the stylet. The stylet may be removed if there is any question if the catheter tip is in the ventricle. It may be necessary to put the patient in reverse Trendelenburg to get CSF to flow if the pressure is low. See also other maneuvers (p. 1824). With any ventricular enlargement, the ventricle should be entered by 3–4 cm depth (with normal size ventricles, this may be 4–5 cm). If a pop is felt, the catheter is inserted an additional 1/2 cm with the stylet. Then an additional 1 cm is inserted without the stylet. Verify CSF flow
 - ✖ If no CSF is encountered by 7 cm, passing the catheter deeper will place it in an undesirable location (even if CSF is then obtained). At ≈ 9–11 cm the tip will often be in the pre-pontine cistern, a subarachnoid space, which is undesirable.
 - if unsuccessful after a maximum of three passes, then place a subarachnoid bolt or intraparenchymal monitor
 - the proximal end of the catheter is tunneled at least 2 cm⁶⁰ under the scalp using a sharp stylet before being brought out through the skin to reduce the risk of infection
 - luer lock adapter is attached to proximal end of catheter and is attached to collection system
13. for (Richmond) subarachnoid bolt: screw in until tip is flush with inner table
14. for intraparenchymal monitor, a path is “cleared” for a few cm into the brain parenchyma using the included blunt stylet. The monitor is then inserted and then backed out about ½ cm so that the tip does not butt up against brain tissue

► **Tricks of the trade: getting CSF with an IVC.** It is common to have difficulty obtaining CSF with ventricular catheterization. There are numerous maneuvers that may help. NB: the problem is almost never that you need to insert deeper! Possible problems and solutions include:

- malposition of catheter tip
 - a) again, the problem is almost never that you need to insert deeper. Never exceed 7 cm
 - b) mediolateral deviation: most of us tend to consistently aim too medial or too lateral. If you know your natural bias, reinsert the IVC aimed slightly in the opposite direction to compensate for this. If you don't know your bias, try reinserting the IVC slightly more medially or laterally
 - c) re-check the pre-op CT for midline shift and adjust aim accordingly
- low ICP:
 - lower the head of the bed (HOB) (reverse Trendelenburg)
 - gently compress the veins of the neck for no more than 1 minute
- try irrigating no more than 1–2 cc of preservative-free saline into the IVC and see if it comes back out. This can work for
 - plugging of the IVC with blood or debris or “air lock”
 - collapsed or near-collapsed ventricles

Removal

Patients receiving anticoagulants need to have normal coagulation and platelet function before discontinuing the catheter to reduce the risk of intracranial hemorrhage. For heparin and LMW heparin, stop the drug 24 hours prior to discontinuing the drain.

“Sump drainage”

The tip of a 25 gauge butterfly may be bent at a 90° angle, and inserted into a subcutaneous reservoir for (e.g., a ventricular access device (p. 1831), or the reservoir of a shunt with a *distal* occlusion) prolonged ventricular drainage.⁶¹ In one series of 34 patients, this was used for prolonged periods (up to 44 days) with acceptably low infection rate.⁶² The use of a one-way valve, continuous antibiotics (ampicillin and cloxacillin), and meticulous technique was credited for the lack of infection.

110.7 CSF diversionary procedures

110.7.1 Ventricular shunts

Booking the case: Ventricular shunt

Also see defaults & disclaimers (p. 25).

1. position: supine with shoulder roll
2. implants: need to specify shunt manufacturer and valve type (e.g., programmable, low-profile...) Uncommon components (e.g., ultra-low pressure, tumor filter) may need special order
3. equipment:
 - a) C-arm for ventriculo-atrial shunts
 - b) endoscopic display (e.g., if NeuroPen is used)
 - c) image-guided navigation system (infrequently used)
4. consent (in lay terms for the patient—not all-inclusive):
 - a) procedure: surgery to insert a permanent drainage tube from the brain to the abdomen, outside of the lungs, vein near the heart (as appropriate) to drain excess cerebrospinal fluid
 - b) alternatives: nonsurgical management (rarely effective for hydrocephalus), third ventriculostomy (for certain cases)
 - c) complications: infection, suboptimal position which might require reoperation, failure to relieve hydrocephalus/symptoms, subdural hematoma, bleeding in the brain, shunts are mechanical devices and will eventually fail (break, block up, move...) and need repair/replace-ment (sometimes sooner rather than later). Abdominal shunts: risk of bowel injury (which could require further surgery)

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Ventricular catheter

Kocher's point is currently used in most cases for insertion site of ventricular catheter; see Ventricular catheterization (p. 1820) for technique. An alternative is an occipital burr hole aiming toward the frontal horn of the lateral ventricle.

An inverted "J"-shaped incision is used to keep hardware from lying directly under the skin incision (minimizes risk of skin breakdown and also creates additional barrier to infection of subjacent hardware). CSF should be sent for culture at the time of insertion since it has been estimated that in $\approx 3\%$ of patients the CSF is already infected. 4 mg of preservative-free gentamicin may be instilled into the ventricular catheter by the technique of barbotage (a technique to administer a drug while reducing the amount of drug lost in the dead space of the catheter: a portion of the antibiotic solution is injected into the CSF, then a lesser amount of CSF is aspirated, a second portion is then injected and the process is repeated until all of the medication is administered).

If you think the catheter is in the ventricle, but you don't get CSF flow, it may be due to low pressure; you can compress the jugular veins or lower the head of the bed to try and induce CSF flow.

Connectors

If a connector must be used near the clavicle, place it rostral to (above) the clavicle. ✖ Avoid placing it caudal to the clavicle because this increases the risk of disconnection.

Distal catheter placement options

All things being equal, the general order of preference for distal catheter placement:

1. peritoneal cavity: intrinsic pressure $\approx 6\text{--}7$ cm H₂O. See below
2. pleural space (p. 454): not for age ≤ 7 years (approximate). For technique, see below
3. right atrium or superior vena cava (p. 1828): intrinsic pressure $\approx 4\text{--}5$ cm H₂O
4. infrequently used ("salvage" techniques) distal shunt sites
 - a) gallbladder (p. 1828): intrinsic pressure $\approx 10\text{--}20$ cm H₂O
 - b) internal jugular vein (with the catheter pointing "upstream")
 - c) superior sagittal sinus
 - d) urinary bladder
 - e) stomach (ventriculogastric shunt)

Peritoneal catheter placement

General information

Used for ventriculoperitoneal shunts, lumboperitoneal shunts...

For small children, use at least 30 cm length of intraperitoneal tubing to allow for continued growth (120 cm total length of peritoneal tubing was associated with a lower revision rate for growth without significant increase in other complications⁶³). A silver clip is placed at the point where the catheter enters the peritoneum so that the amount of residual intraperitoneal catheter can be determined on later films (more important in growing children).

Distal slits on the peritoneal catheter may increase the risk of distal obstruction,⁶⁴ and some authors recommend that they be trimmed off. Wire-reinforced catheters should not be used because of excessively high rate of viscus perforation, and this tubing was designed to prevent kinking which is not a problem with modern shunts.

Open technique

A vertical incision lateral and superior to the umbilicus is one of several choices. The following layers should be identified as they are traversed to avoid confusing preperitoneal fat with omentum and erroneously placing the tip in the preperitoneal space:

1. subcutaneous fat
2. anterior sheath of the abdominis rectus muscle (anterior rectus sheath)
3. abdominis rectus muscle fibers: should be split longitudinally
4. posterior rectus sheath
5. preperitoneal fat (may be very well developed in a few individuals, but is essentially nonexistent in most)
6. peritoneum (usually closely adherent to the posterior rectus sheath)

Trocar technique

A trocar (e.g., Codman #82-4095 disposable split trocar (► Fig. 110.4), which is designed to place catheters with outer diameters up to 3.0 mm) may be used. Also very helpful in conjunction with laparoscopic surgery.

✗ Contraindications: prior abdominal surgery, extremely overweight patients.

Technique:

1. place a Foley catheter to decompress the bladder prior to draping
2. 1 cm skin incision above and lateral to the umbilicus
3. tent-up the abdominal skin anteriorly (away from patient)
4. insert trocar with plastic stylet in place aiming toward the ipsilateral iliac crest
5. feel 2 "pops" of penetration: 1st = anterior rectus sheath, 2nd = posterior rectus sheath/peritoneum
6. remove the plastic stylet
7. feed the catheter through the central channel of the trocar. The peritoneal catheter should feed easily through trocar (if it doesn't, the tip of the trocar may be in the preperitoneal space, or pushing up against an organ, adhesion, etc.). You can check if a small amount of irrigation fluid runs into the peritoneum through the trocar (if not, placement may not be serviceable)
8. once the peritoneal catheter is in position, it is stabilized with bayonet pickups as the split trocar is withdrawn allowing the catheter to slide through the split opening in its side so that the catheter remains in place

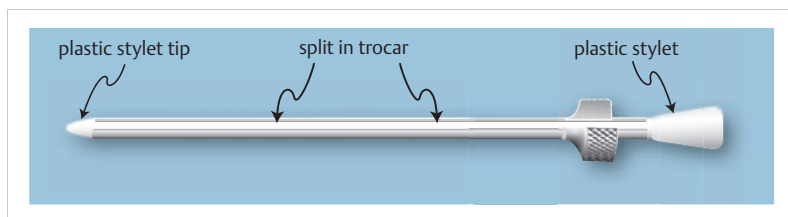


Fig. 110.4 Codman disposable split trocar.

VP shunt, post-op orders (adult)

1. flat in bed (to avoid overshunting and possible subdural hematoma) with gradual mobilization
2. if peritoneal end is new or revised, do not feed until bowel sounds resume (usually at least 24 hrs, due to ileus from manipulation of peritoneum)
3. shunt series (AP & lateral skull, and chest/abdominal X-ray) as baseline for future comparison (some surgeons obtain these films immediately post-op in case some immediate revision is indicated, e.g., ventricular catheter tip in temporal horn)

Pleural catheter placement

For ventriculopleural shunts, syringopleural shunts... see reference.⁶⁵ See also more details (p.454). In addition to the open technique described below, a trocar method (using the same trocar used for peritoneal catheters, ► Fig. 110.4) for introducing the pleural catheter has been described.⁶⁶

✱ Use under advisement in age ≤ 7 years⁶⁷ due to reduced surface area for resorption of CSF which may result in high pleural fluid levels.

A 3 cm horizontal incision is made just below the level of the breast either in the midclavicular line or in the anterior axillary line. Divide the subcutaneous tissue, deep fascia, and pectoralis muscle. The external and internal intercostal muscles are divided along the *superior* margin of the inferior of the two ribs exposed (to avoid the neurovascular bundle running along the inferior margin of each rib). A self-retaining retractor between the ribs aids the exposure. The parietal pleura is visualized with the visceral pleura sliding underneath with each respiration. The pleura is not opened until the catheter is brought out subcutaneously at this incision. Have the anesthesiologist hold respirations, and nick the parietal pleura (or use a blunt-tip hemostat to pop through) to admit the catheter. Allow the lung to drop away and insert 20–40 cm of tubing into the pleural cavity. If the pleural opening is lax around the catheter, it can be snugged with a 4–0 absorbable suture. Have the anesthesiologist provide a Valsalva maneuver before cinching down the pleural suture, and again before closing the deep muscle layer. A chest tube is usually not required. A maneuver that may sometimes be helpful is to place a red-rubber catheter next to the shunt tube at the same time (to permit the escape of air from the pleural space). Begin closing, but prior to placing the last deep suture, have the anesthesiologist perform a Valsalva maneuver and allow air to escape through the red-rubber catheter (you can place the end in saline to see the bubbles). Once the bubbles stop, pull the red-rubber catheter and close the last stitch. If the bubbles don't stop, there is an air leak in the visceral pleura and a pigtail catheter or a chest tube connected to a Pleur-evac® should be used.

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Right atrial catheter placement

General information

For ventriculo-atrial shunt...

Open method

The common facial vein (CFV) is located by making a diagonal cervical incision across the anterior border of the sternocleidomastoid at or just below the level of the angle of the mandible (the CFV may be as far as ≈ 2 cm below this point). The platysma is divided, and the CFV is located as it joins the internal jugular vein (IJV) at the level of the hyoid bone. The CFV is cannulated with the atrial tubing, and is secured with a snug ligature close to the junction with the IJV. If the CFV is not suitable, a purse-string suture is placed directly in the IJV, and the IJV is then opened in the center of the purse string and cannulated.

Percutaneous method

May be utilized in adults (and possibly peds). The IJV is catheterized using the Seldinger technique⁶⁸ with a guide-wire and needle through a stab incision at the anterior margin of the SCM. Fluoroscopy is used to place the tip of the wire at the desired location (see below). A No. 13 French peel-away introducer and dilator are then inserted over the wire, which is then bent at the skin edge and withdrawn⁶⁹ (for a pediatric case: may use a No. 7 French introducer with a 1.5 mm O.D. *lumboperitoneal* catheter for the distal atrial catheter). The atrial catheter is cut to the length of the wire distal to the bend, and the catheter is then threaded into the introducer. The position of the catheter tip should again be confirmed (e.g., with radio-opaque contrast under fluoroscopy). A short skin incision is then made starting at the point where the catheter penetrates the skin to permit subcutaneous tunneling of the tubing.

Location of distal tip

If the catheter repeatedly goes down the wrong vessel (e.g., the subclavian vein), a “J” guidewire may help. Also, rotating the head to a more neutral position sometimes works.

The ideal location of the distal tip is in the right atrium (unlike the location for central catheters in the superior vena cava (SVC)) so that the turbulent blood flow will reduce the risk of thrombus formation. The tip may enter the right atrium, but must not penetrate the tricuspid valve. A number of methods for optimal placement of the distal shunt tip may be employed, and include:

1. using an intraoperative chest X-ray to locate the tip between the level of T6–8 vertebra in an adult. In a growing child, initially insert to $\approx$ T10 level. This method is subject to error due to malalignment of the X-ray beam (parallax error)
2. place the tip near the level described above, then inject iodinated contrast, e.g., 20 ml of Omnipaque 180 (iohexol) (p. 230) under intraoperative fluoroscopy to locate the tip in right atrium
3. fill the catheter with normal or 3% saline and use the catheter as an EKG electrode. The P-wave changes from a downward to a biphasic morphology as the tip enters the atrium. A sharp upward deflection occurs as the tricuspid valve is approached.⁷⁰ Some recommend advancing the tip to maximal P-wave amplitude and then backing off a centimeter or two
4. fill the catheter with heparinized saline (1–5 U per cc NS) and measure the pressure as the tip is advanced,⁷¹ leave tip just short of where atrial pressure tracing occurs
5. utilizing intraoperative echocardiography⁷²

A growing patient is followed with annual CXRs. When the catheter tip is above $\approx$ T4, the catheter must be lengthened or converted to a VP shunt.

Gallbladder

Uncommonly used distal site. First described in 1958, there have been a number of publications,^{73,74} with a 63% success rate in one series.⁷⁴ Relevant physiology⁷⁵:

- gallbladder capacity: 30–60 cc. Can absorb up to 1500 cc of liquid per day
- pressure: resting internal pressure is 10–20 cm H₂O, which may counteract siphon effect of CSF. Post-prandial elevation of ICP up to 24 cm H₂O in children with ventriculogallbladder (VGB) shunts has been reported⁷⁶
- bile: lytic properties increase protein breakdown and may prevent adhesions. Normally sterile, but it has been purported that $\approx$ 50% of patients > 50 years of age have unsterile bile

Classically inserted via open laparotomy through the dome of the gallbladder using a purse-string suture to attain water-tight closure, which often necessitates using a metal connector inserted in the shunt tubing to prevent constriction.

Challenges:

1. contraindications:
 - prior gallbladder disease or procedures
 - gallstones
2. technical difficulties: optimal purse-string placement and tension
3. risks:
 - leakage of bile into abdomen due to disconnection or laxity in purse-string
 - occlusion of shunt tubing by too tight of a purse-string
 - resultant cholelithiasis
4. placement in young children may result in catheter pull-out of gallbladder as child grows

A novel option has been described of percutaneous ultrasound-guided placement of an 8-French pig-tail catheter into the gallbladder through the liver (transhepatic) – which is commonly done by interventional radiologists to drain bile or treat cholangitis – and connecting this to the ventricular catheter. This is less invasive, and decreases the risk of bile leakage, and if the catheter occludes, the tubing may be removed percutaneously without need for a laparotomy (which entails a 3% risk of major bile leak, and 3% risk of minor bile leak⁷⁷). Risk of percutaneous technique is 1.7% risk of minor bleeding and 0.4% risk of major bleeding.

110.7.2 Third ventriculostomy

General information

See also indications and complications (p. 453).

Older techniques include a subfrontal approach, opening the chiasmatic and lamina terminalis cisterns, and making a 5–10 mm opening in the lamina terminalis. Stereotactic third ventriculostomy (using contrast ventriculography⁷⁸ or CT-guided) has also been described. Current technique, endoscopic third ventriculostomy (ETV), often with assistance of image guidance, consists of fenestrating the floor of the third ventricle using a ventriculoscope.

Ventriculoscopic technique (endoscopic third ventriculostomy (ETV))

1. equipment: requires a rigid endoscope (does not work well with flexible)
2. image-guided stereotactic technology helps immensely with the trajectory, but once you've entered the third ventricle, you must navigate by visual landmarks and cannot rely on image guidance because of the limitations of the accuracy
3. burr hole: 2–3 cm lateral to the midline just anterior to the coronal suture (Kocher's point)
4. pass through the foramen of Monro and fixate the sheath just within the third ventricle
5. the floor of the third ventricle is inspected and must be thin enough and translucent enough to permit visualization of the basilar artery and mammillary bodies. If these structures cannot be visualized then the procedure should be aborted
6. the location of the opening is chosen:
 - a) in the midline (avoids PComA and PCA)
 - b) in the region of the tuber cinereum (prominence of the base of the hypothalamus, extending ventrally into the infundibulum and pituitary stalk)
 - c) posterior to the infundibular recess
 - d) anterior to the mammillary bodies
 - e) anterior to the tip of the basilar artery
7. an effective technique consists of "rubbing through" the floor of the third ventricle either with a probe or Decq forceps. Alternatively, hydrodissection or bipolar electrocautery may be used to thin down the lamina. ✖ Do not use laser due to possibility of injury to basilar artery!⁷⁹
8. the opening can be enlarged with the Decq forceps, or a 3 French Fogarty balloon or a double balloon (Fogarty or Neuroballoon™ catheter (Integra LifeSciences 7CBD10)). The balloon is inflated distal to the opening in the floor and is then withdrawn through the opening
9. the opening does not need to be large (unlike e.g., fenestrating an arachnoid cyst): ≈ 4–5 mm is usually adequate^{80,81}
10. after penetrating through the floor of the third ventricle, make certain that you can see vessels (sometimes the arachnoid is not perforated, or there is a second membrane or webs of membranes that need to be lysed)
11. consider injection of diluted iohexol or other intrathecal contrast agent into the lateral/third ventricle (see ventriculogram) prior to removal of scope. CT of head 1 hour after surgery will show diffuse subarachnoid contrast in cisterns and over convexity if ETV successful
12. sagittal T2 weighted, thin-slice sequence will show drop-out of T2 signal at stoma of ETV

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110.7.3 LP shunt placement

General information

Often placed without a mechanical valve (relies on the slits in the distal catheter ("slit valves") and the narrow inner diameter of the lumbar drain tubing to limit flow rate).

Commercially available systems include:

1. Integra Spetzler™ Lumbar Peritoneal Shunt Kit, a one-piece straight tube with slit valves on the peritoneal end. (NB: catheter length is only 80 cm, which may not be long enough for a morbidly obese adult. If more length is desired, a conventional peritoneal catheter can be connected to a lumbar drain using a step-down connector, as found in Edwards-Barbaro shunt kit (p. 1410)). If it is desired to inject contrast or to aspirate CSF to assess LP shunt patency post-op, one may optionally insert a Spetzler™ Lumbar Peritoneal Flushing Reservoir, a valveless reservoir (inlet and outlet tubing 0.7 mm inner diameter and 1.5 mm outer diameter). Separate in-line miter valves (low, medium, or high pressure) may optionally be inserted for added resistance.
2. Medtronic Lumboperitoneal Shunt Kit, #44420 contains a specialized programmable Strata® NSC valve with small-diameter connectors on the inlet and outlet ports, which accommodates the smaller lumbar catheter and the *specialized* small inner-diameter peritoneal/pleural catheter contained in the kit (both catheters are 0.7 mm ID, the distal catheter is 120 cm long). The valve has a tappable reservoir as well as inlet and outlet occluders.

Insertion technique

See reference.⁸²

1. position: lateral decubitus position, both knees flexed (right-side-up preferred)
2. prep back, flank, and abdomen
3. 1 cm skin incision over L4–5 or L5–1 (in obese patients, use larger skin incision carried down to fascia overlying spinous processes. This may also be superficially incised between spinous processes to aid insertion)
4. tilt table to 30° reverse-Trendelenburg to expand lumbar subarachnoid space
5. insert the 14 gauge Tuohy needle into subarachnoid space. When passing the needle, some surgeons orient it so that the opening in the needle tip is pointing laterally or medially in order that the tip penetrates the dura in line with the longitudinally running dural fibers. The needle is then rotated so that the opening is directed rostrally (caudal placement is also acceptable). Confirm there is CSF flow
6. remove the stylet, insert shunt tubing. For L4–5 insertion, insert a total of ≈ 17 cm of catheter into the needle. This will accommodate withdrawal of 9 cm when removing the needle, leaving about 8 cm within the canal, which minimizes conus medullaris irritation. The Spetzler one-piece shunt has 4 black tantalum-impregnated length markers: when the first marker is at the needle hub, the tip of the catheter is at the tip of the needle. The next 3 markers are at 5 cm intervals.
NB: the catheter pliability varies among manufacturers. Some are too flaccid to push through the Tuohy needle; for these, inserting the provided guidewire helps, but can be challenging itself. To facilitate guidewire placement, first wet the catheter and guidewire and pass the wire into the catheter while holding the catheter so that it is dangling vertically, such that the guidewire goes straight down into the catheter
7. ✖ never pull the catheter out through the needle—the catheter is prone to shear off on the angulated tip of the Tuohy needle
8. the Tuohy needle is then withdrawn until the tip of the needle comes out of the tissue (the standard Tuohy needle is 9 cm long), usually accompanied by the 9 cm of catheter inside the needle. Once the needle tip is visible, pinch the catheter there to hold it against further withdrawal and continue to remove the needle over the catheter
9. make a flank incision, pass a tunneler from the flank to the back incision. Pass the catheter from the back to the flank through the tunneler. Withdraw the tunneler, leaving the catheter
10. abdominal placement:
 - a) open: incision made through peritoneum. Place purse-string in peritoneal opening
 - b) trocar method (p.1826)
11. pass tunneler from abdominal incision to flank incision. Feed catheter from flank to abdomen. Withdraw tunneler over catheter
12. verify CSF flow. Place the catheter inside the peritoneum. For open technique: cinch and tie a purse-string snugly, but loose enough to allow the catheter to slide with gentle pushing
13. a snug-fitting retaining sleeve is placed around the catheter at all three incisions, and secured to subcutaneous tissue with non-absorbable suture

Insertion of programmable valve: in severely obese patients the valve will tend to twist within the subcutaneous tissue, which may make programming difficult or impossible. Some pointers regarding valve placement:

1. the valve must be fairly close to the surface of the skin to allow the programming magnet to work (Medtronic recommends ≤ 1 cm of thickness of tissue overlying the Strata NSC valve)
2. a technique to stabilize the valve is to use a linear skin the length of the valve with a dog-leg bend at both ends to allow the skin and a small amount of sub-Q tissue to be opened like a door. The valve is sutured *upside down* in all 4 corners to the underside of this flap so that when the flap is closed, the valve will be right-side-up under the skin
3. placing the valve over a rib (superficial to a rib, not directly on the rib) may provide some additional stability

Lumboperitoneal (LP) shunt evaluation

When problems develop, evaluation of function may be more difficult than with VP shunt. Evaluation may include:

1. abdominal X-rays: AP & lateral X-rays can rule out breakage or migration of a shunt component
2. noncontrast brain CT scan: can rule out complication such as subdural hematoma
3. LP: perform LP just above or below level of lumbar catheter. The pressure may be 0 or negative, and it may be necessary to aspirate CSF to confirm placement

- a) can give indirect evidence of shunt function by measuring the CSF pressure, which should be low if the shunt is working (only helpful in cases where the shunt was placed for elevated CSF pressure, e.g., pseudotumor cerebri; not helpful in NPH)
- b) “shunt-o-gram”: inject contrast into subarachnoid space through LP needle
 - radionuclide (p. 459): inject radio-isotope via LP and look for subsequent tracer activity in peritoneal cavity
 - with water-soluble contrast⁸³: inject 10 ml of iohexol and monitor the flow of contrast fluoroscopically as the patient is brought vertical. Coughing or Valsalva maneuver will accelerate the flow of contrast
- 4. shunt tap: if an antechamber has been installed, it is accessed after cleaning the skin with anti-septic using a 22 Ga or smaller needle placed perpendicular to the dome to prevent leakage. If there is no access chamber, it is sometimes possible to tap the tubing itself with a 27 gauge butterfly needle

110.8 Ventricular access device

110.8.1 General information

An indwelling ventricular catheter connected to a reservoir that is positioned under the scalp for the purpose of chronic access to the intrathecal space (usually the ventricular system), or sometimes other intracranial compartments such as tumor cysts. Sometimes referred to generically as an Ommaya® reservoir; however, this is actually a trade name.

110.8.2 Indications

1. administration of intrathecal (IT) antineoplastic chemotherapy:
 - a) for CNS neoplasms, including: carcinomatous meningitis, methotrexate for CNS lymphoma or leukemia (p. 844)
 - b) IT chemotherapy is often used for the following even in the absence of CNS involvement because of the high relapse rate in the CNS: acute lymphoblastic leukemia, lymphoblastic lymphoma, Burkitt's lymphoma
2. administration of intrathecal antibiotic for chronic meningitis
3. chronic removal of CSF from infants with intraventricular hemorrhage
4. for fluid aspiration from a chronic tumor cyst that is resistant to therapy (radiation or surgery)⁸⁴

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Booking the case: Ventricular access device

Also see defaults & disclaimers (p. 25).

1. position: supine
2. equipment
 - a) endoscopic display (e.g., if NeuroPen is used)
 - b) C-arm (optional) to verify position of ventricular catheter
 - c) image-guided navigation system (if used)
3. implants: need to specify reservoir manufacturer
4. consent (in lay terms for the patient—not all-inclusive):
 - a) procedure: surgery to insert a tube into the fluid space within the brain (ventricle) which is connected to a port under the skin so that fluids can be removed or injected (usually medication)
 - b) alternatives: sometimes fluid can be removed and medication can be injected using a lumbar puncture (spinal tap). The effectiveness of this may not be the same as the operation being discussed here
 - c) complications: infection, suboptimal position which might require reoperation, subdural hematoma, bleeding in the brain, this is a mechanical device and may eventually fail (break, block up...) and need repair/replacement

110.8.3 Technique of insertion

See reference.⁸⁵

Preferably placed in the right frontal region, unless indicated otherwise (e.g., for tumor cyst). Usually placed under endotracheal general anesthesia, although local anesthesia occasionally may be used (e.g., for patients too ill to tolerate general anesthesia).

Patient position: supine, head midline, neck flexed 5°.

Incision: inverted "U," slightly larger than the reservoir (the original Ommaya® reservoir is 3.4 cm diameter), with the center over the coronal suture approximately 3 cm from midline, roughly centered near Kocher's point (p. 1821). A circle of pericranium of diameter equal to that of the reservoir is excised and saved. Alternatively, the pericranium may be flapped separately in the opposite direction (i.e., a right-side-up "U"), and closed over the reservoir to help secure it in position.

Make a burr hole over the coronal suture 3 cm off midline. A cruciate incision in the dura is made large enough to visualize the cortical surface, minimal cortical bipolar coagulation is used, and a pial/cortical incision is made to avoid surface vessels.

One may inject 15–20 cc of filtered air into the ventricles with a ventricular needle prior to the catheter insertion to guide the tip of the catheter with intraoperative lateral skull X-rays (intraoperative pneumoencephalogram). Alternatively, image guidance may be used with the same tracking systems that may be used for ventricular shunts. The trajectory is towards a point intersecting a plane 2 cm anterior to the EAM aiming minimally towards the midline (1–2°). Alternatively, one may aim perpendicular to the surface of the skull.⁵⁶ A total length of ≈ 7.25 cm of catheter is fixed to the base of the reservoir, which allows the catheter to lie on the floor of the anterior horn of the lateral ventricle in most adults. This location can be verified with intraoperative pneumoencephalography⁸⁵ or with ventriculoscopic techniques.

The excised pericranium is placed over the dura, and the reservoir is sutured to the pericranium. Note: the dome of the original Ommaya® reservoir has a low resistance, and may be easily collapsed if too much tension is placed on the overlying scalp. If early use of the reservoir is desired (i.e., within 48 hrs post-op), the skin closure should be performed with a running nonabsorbable suture (e.g., nylon) and coated with cyanoacrylate skin adhesive, and the surgical site can then be left without a gauze dressing for easier access to the reservoir. A skin tattoo can be created over the center of the reservoir (to assist in localizing the reservoir for injection) using India ink and pricking the skin with a sterile needle.

110.8.4 Reservoir puncture

The scalp is prepped with antimicrobial scrub, and using sterile technique, a 25 gauge or smaller butterfly needle is introduced at an oblique angle, preferably with a non-coring needle. The original (Ommaya®) reservoir has firm plastic bottom surface which can be penetrated with the needle if too much force is applied.

110.9 Sural nerve biopsy

110.9.1 Indications & contraindications

Nerve biopsy plays an ever diminishing role in diagnosing peripheral neuropathies, to a significant degree as a result of increased sophistication of molecular biologic testing. In general, the yield of nerve biopsy is higher in cases of severe demyelinating, distal asymmetric, and multifocal neuropathy than in chronic, axonal, and symmetric types.^{86,87}

Nerve biopsy should not be performed if a diagnosis and etiology can be established by less invasive tests, which may include: clinical exam, electrodiagnostics (NCV & EMG), blood tests, CSF analysis, or skin biopsy for quantitative epidermal nerve fiber density.

Indications (only class IV evidence was found⁸⁶):

1. major indications: most helpful in
 - vasculitis (non-systemic vasculitic neuropathy)
 - diabetic neuropathy when superimposed chronic inflammatory demyelinating polyneuropathy (p. 195) (CIDP)⁸⁶ is suspected
2. minor indications (see reference for more extensive list⁸⁶)
 - infection: e.g., Hansen disease, HIV
 - may help distinguish between the two types of Charcot-Marie-Tooth syndrome (p. 568)
 - select hereditary neuropathies with negative appropriate genetic tests⁸⁷
 - may show demyelination in diabetic amyotrophy (p. 572)

Contraindications:

- absolute contraindication: infection or skin breakdown at the biopsy site
- relative contraindications: situations where yield of the test is low

- the nerve is normal clinically and on nerve conduction testing with no abnormality on MRI or ultrasonography (an exception to this may be when there is a strong suspicion of vasculitis with predominantly motor features and asymmetric involvement)⁸⁷
- evaluation of small-fiber neuropathy is rarely an indication for nerve biopsy. Skin biopsy can be done in this situation for quantification of intradermal nerve fiber density; and the test can be repeated in follow-up
- paraproteinemic neuropathies: IgM deposits in the nerve may predate serum IgM gammopathy
- if thorough and optimal nerve processing cannot be performed: in addition to fixed and frozen sections, the lab should be able to evaluate teased fibers, electron microscopy, and immunohistochemistry

110.9.2 Selection of nerve to be biopsied

Although a number of peripheral nerves may be biopsied, the sural nerve fulfills the criteria of being well studied, expendable with minimal deficit, easily accessible, and often involved in the pathologic process in question. However, being exclusively a sensory nerve, it would usually not be an appropriate choice for purely motor neuropathy.

Alternatively, the superficial peroneal nerve may be appropriate in select cases if a simultaneous muscle biopsy is indicated as the peroneus brevis is convenient.⁸⁷

110.9.3 Risks of procedure

1. sensory loss in the sural nerve distribution (lateral foot) is expected, often does not persist for more than several weeks (unless the underlying disease process prevents recovery)
2. pain and paresthesias: tend to recover more than sensory loss
3. problems with wound healing: the ankle is a notorious region for poor circulation and the loss of sensation (from the disease or biopsy) may render the area subject to repeated trauma without the patient being aware. Furthermore, many patients with an undiagnosed systemic disease requiring a sural nerve biopsy will have poor wound healing (a significant number are also diabetic)
4. failure to make a diagnosis: although biopsy may be able to exclude some contingencies, it often does not make a specific diagnosis

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110.9.4 Applied anatomy

The sural nerve is formed by the merging of the distal portion of the medial sural cutaneous nerve (one of the terminal branches of the tibial nerve) and the anastomotic ramus of the common peroneal nerve. It is entirely sensory except for some unmyelinated autonomic fibers. It supplies cutaneous sensation to the posterolateral third of the leg, the lateral heel and foot, and the little toe. At the level of the ankle it lies between the Achilles tendon and the lateral malleolus. This location is constant, superficial, and relatively protected from external trauma which might otherwise confuse the analysis.

110.9.5 Technique

Modified technique.⁸⁸ (p 771–2) Usually done under local anesthesia with sedation. Side to biopsy: if one leg shows more involvement, that is often the preferred side.

Position: under general anesthesia, patient prone. If patient is not under general anesthesia, use 3/4 oblique position (not prone) with a pillow between the legs. The leg to be biopsied is uppermost and is flexed 90° at the knee to relax tension on the nerve, the ankle is slightly everted. Compressing the calf (can be done using a sterile Penrose drain as a temporary tourniquet during surgery) dis-tends the lesser saphenous vein (LSV) at the lateral malleolus (LM), which (when visible) reliably locates the sural nerve usually deep and anterior to the vein.

After prepping, drape the limb with a sterile stockinette or similar drape, infiltrate local anesthetic subcutaneously just posterior to the LM and proximally, paralleling the Achilles tendon for ≈ 10 cm. A 7–10 cm incision is made overlying the course of the LSV beginning usually just posterior to and ≈ 1 cm proximal to the LM. The vein can be seen through the translucent Scarpa's fascia. The fascia is incised over the vein, which is gently retracted to reveal the nerve, usually deep to the vein. A common pitfall is to go too deep, but the nerve is fairly superficial; it is not necessary to go through the thick fascia. If at any time you see tendons to the toes, you have gone too deep.

To differentiate the sural nerve from the LSV (which may resemble the nerve in some cases): the nerve has many branches at acute angles, especially proximal to the LM, vs. the vein which has

right-angle branches. If in doubt, a frozen section may be helpful to verify that the biopsied structure is a nerve in order to avoid potentially embarrassing explanations and the possible need to repeat the procedure.

After exposing at least 3–5 cm of the nerve, anesthetize the proximal portion with 0.5% lidocaine using a 27 Ga needle and cut it sharply just distal to the infiltration site. Cut the nerve with slight tension on it to allow the ends to retract deep to the skin incision to prevent the formation of a scar neuroma. Some pathologists request that the proximal end of the nerve be marked, e.g., with a suture.

Although a full thickness nerve biopsy is most common, a fascicular biopsy (partial nerve biopsy) may be performed to minimize the sensory deficit (NB: if the disease is patchy, the chances of a positive biopsy is decreased) by opening the epineurium for the length of the exposure and teasing out a fascicle with minimal branching.

If it is desired to obtain a biopsy of the sural nerve higher up for comparison, it may be accessed in the mid-upper calf between the heads of the gastrocnemius muscle. Here it may be as deep as ≈ 2 cm. Gently tugging on the exposed nerve in the ankle may help in localization.

A dissolvable subcuticular closure may be used. Generous padding should be placed over the incision to protect it from bumps (since many patients are numb from the pathology and/or surgery, there is increased risk of inadvertently traumatizing the wound). An elastic pressure dressing is applied after closure.

110.9.6 Nerve handling

For light microscopy, which suffices in most cases, immerse the nerve in formalin. For electron microscopy, glutaraldehyde is used. For biochemical and immunofluorescence studies, use rapid freezing.

110.9.7 Post-op care

Pressure dressing should be worn for protection for two weeks. The patient is allowed to walk but should restrict their activity for 2–3 days. If nonabsorbable sutures are used instead of subcuticular closure, they should be left in place 10–14 days.

110.10 Nerve blocks

Also see Occipital nerve block (p.542).

110.10.1 Stellate ganglion block

General information

✱ Do not perform stellate ganglion block bilaterally (can cause bilateral laryngeal paralysis → respiratory compromise). The stellate ganglion is actually closer to C7 than C6, but risks at C7 are much higher (closer to pleura → pneumothorax, vertebral artery → arterial injection → seizures and/or hematoma, recurrent laryngeal nerve → unilateral vocal cord paralysis → hoarseness (common), brachial plexus → UE weakness). Other complications: intradural injection → spinal anesthesia, phrenic nerve block.

Technique

Position: patient supine with an interscapular roll, head tilted backward, mouth slightly open to relax the strap muscles.

Displace the SCM and carotid sheath laterally, and insert a 1.5 inch 22 Ga needle to contact Chassaignac's tubercle (anterior tubercle of transverse process of C6) AKA carotid tubercle (the most prominent in the C-spine), usually at the level of the cricoid cartilage, approximately 1.5–2 inches above clavicle.

Withdraw the needle 1–2 mm and aspirate to avoid intravascular injection. Inject small test dose, then full 10 ml of 0.5% bupivacaine (Marcaine®) or 20 ml of 1% lidocaine. Remove needle and elevate patient's head on pillow to facilitate spread.

Verify block by Horner syndrome, and anhidrosis and increased warmth of ipsilateral hand.

110.10.2 Lumbar sympathetic block

Technique

Patient prone on fluoro table. Use local anesthetic to allow insertion of 20–22 gauge spinal needles (10 to 12.5 cm long) at L2, L3, and L4 levels. Needle inserted 4.5–5 cm lateral to spinous process until transverse process contacted, then redirected caudally and inserted to a depth 3.5–4 cm deeper than transverse process. Final needle tip position should be just anterolateral to vertebral bodies. At each level, instill ≈ 8 ml of 1% lidocaine local after verifying that nothing can be aspirated.

Keep patient on bed rest for several hours, then ambulate with assist; watch for orthostatic hypotension due to vascular pooling in blocked lower extremity.

110.10.3 Intercostal nerve block

Indications

1. postthoracotomy pain
2. intercostal neuralgia
3. postherpetic neuralgia
4. pain from rib fractures

General principles

In order to obtain good anesthesia, the following should be noted:

1. a good site for injection is in the *posterior axillary line* (PAL) because
 - a) this is proximal to the origin of the lateral cutaneous nerve (which originates ≈ in the anterior axillary line)
 - b) this avoids the scapula for nerves above ≈ T7
 - c) this reduces the risk of pneumothorax from injecting closer to the spine (the latter requires a longer needle path and there is increased difficulty palpating landmarks)
2. due to overlap, at least 3 intercostal nerves usually need to be blocked to achieve at least some area of anesthesia; it is usually necessary to block 1–2 intercostal nerves both above and below the affected dermatome
3. the intercostal nerves lie on the undersurface of the corresponding rib in close proximity to the pleura; the order of structures from top down is: rib, vein, artery, nerve

Technique

1. after raising a skin wheal at the desired level in the PAL, insert a 22 Ga or smaller needle directly against the rib
2. walk the needle down the rib millimeter by millimeter until the needle just slips under the rib; to avoid pleural puncture, do not advance the needle more than one-eighth inch deep to the anterior surface of the rib
3. aspirate to be certain that there is no air (from lung penetration) or blood (from entering the intercostal artery or vein)
4. if no air or blood returns, inject 3–5 ml of local anesthetic
5. if there is any question about lung penetration, obtain a portable CXR to R/O pneumothorax

110.11 Multistranded cable for spine fusion

Multistranded cable (usually titanium alloy or stainless steel) has replaced twisted wire and has itself been largely supplanted by rigid immobilization (spinal screws/rods) for internal immobilization of the spine. There are still some situations where cable has a role (most commonly in the cervical spine) usually supplementing rigid immobilization, such as C1–2 lateral mass fusion (p. 1780). Systems include Medtronic ATLAS™ cable (formerly Songer cable), Synthes cable system.

ATLAS™ cable system: maximal cable tension for titanium alloy cables is 30 lbs, for stainless steel it is 60 lbs (these maximums are for the cables; actual tension depends on the quality of the bone).

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111 Functional Neurosurgery and Stereotactic Neurosurgery

111.1 Introduction

Functional neurosurgery (FN) involves the use of electrical stimulation, ablative therapy, or therapeutic infusions to mask or relieve symptoms of aberrant neurophysiology. Functional procedures may be performed in the brain, spinal cord, or on cranial or peripheral nerves and may be used in the treatment of:

1. movement disorders
2. neurovascular compression syndromes
3. autonomic dysfunction
4. psychiatric disease
5. pain (p.518)
6. epilepsy (p.1889)

Deep brain techniques (lesioning or stimulation) rely on stereotactic methods discussed below.

111.2 Stereotactic surgery

111.2.1 General information

The techniques of stereotactic surgery are utilized in some functional procedures (e.g., DBS) as well as for biopsies, cyst drainage, etc. The term stereotactic (Greek: stereo = 3-dimensional, tactic = to touch) surgery was initially used in animals, and was based on atlases of three-dimensional coordinates compiled from dissections. The term was then used for surgery performed in humans, usually for thalamic lesioning to treat parkinsonism (see Surgical treatment of Parkinson's disease (p.1840)), where the target site to be lesioned was located relative to landmarks with intraoperative pneumoencephalography or contrast ventriculography. Use of this procedure fell off dramatically in the late 1960s with the introduction of L-dopa for parkinsonism.¹

Current techniques would be more appropriately termed *image-guided* stereotactic surgery. In the first part of the procedure, a CT scan or MRI (or occasionally, angiogram) is performed. For increased precision, "fiducial" markers or a stereotactic frame is attached to the patient's head during this image acquisition phase. Acceptable accuracy for biopsies can often be obtained using high-resolution thin-cut imaging slices (usually with a 0 angle of the gantry), and then surface-matching algorithms in the guidance system will match the pre-op CT/MRI to the patient's head. This is not accurate enough for lesion generation or electrode placement.

The second part of the procedure usually takes place in an operating room. The patient is "registered" with the pre-op images, and then tracking cameras follow the movement of instruments with appropriate attachments to show in "real-time" the location of the instrument with respect to the pre-op image. An important limitation to be aware of is the fact that the pre-op images are "historical" and are not updated as the surgical procedure alters the anatomy of the patient. Even the administration of mannitol can cause brain shifts that may cause the target of the surgery to move away from its pre-op location by several millimeters.²

111.2.2 Indications for stereotactic surgery

1. biopsy (also, see below)
 - a) deeply located cerebral lesions: especially near eloquent brain
 - b) brainstem lesions: may be approached through the cerebral hemisphere³
 - c) multiple small lesions (p.357) (e.g., in some AIDS patients)
 - d) patient medically unable to tolerate general anesthesia for open biopsy
2. catheter placement
 - a) drainage of deep lesions: colloid cyst, abscess
 - b) indwelling catheter placement for intratumoral chemotherapy
 - c) radioactive implants for interstitial radiation brachytherapy⁴
 - d) shunt placement: for hydrocephalus (rarely used) or to drain cyst
3. electrode placement
 - a) depth electrodes for epilepsy
 - b) "deep brain stimulation" for chronic pain (requires electrophysiologic stimulation)

4. lesion generation
 - a) movement disorders: parkinsonism (p. 1841), dystonia, hemiballismus
 - b) treatment of chronic pain
 - c) treatment of epilepsy (rarely used)
5. evacuation of intracerebral hemorrhage
 - a) using an Archimedes' screw device^{5,6}
 - b) with adjunctive urokinase^{7,8} or recombinant tissue-plasminogen activator (p. 1626)⁹
6. stereotactic "radiosurgery"—see Stereotactic radiosurgery & radiotherapy (p. 1902)
7. to localize a lesion for open craniotomy (e.g., AVM,¹⁰ deep tumor)
 - a) using a ventricular-type catheter
 - b) using a blunt biopsy needle or introducer¹¹
 - c) systems using visible light laser beam for guidance
8. transoral biopsy of C2 (axis) vertebral body lesions¹²
9. "experimental" or unconventional applications
 - a) stereotactic clipping of aneurysms¹³
 - b) stereotactic laser surgery
 - c) CNS transplantation¹⁴; e.g., for parkinsonism (p. 184)
 - d) foreign body removal¹⁵

111.2.3 Stereotactic biopsy

General information

This section presents information regarding stereotactic brain biopsy (SBB) in general. For SBB in specific conditions, see the index entry for that condition. Biopsy may be through a small "cookie cutter" craniotomy, or through a smaller burr hole with a needle. Although it may be performed under local or general anesthesia, it is more commonly done under general.

Contraindications

1. coagulation disorders
 - a) coagulopathies: bleeding diatheses, iatrogenic (heparin or coumadin)
 - b) low platelet count (PC): PC < 50,000/ml is an absolute contraindication, it is desirable to get the PC ≥ 100,000
2. inability to tolerate general anesthesia and to cooperate for local anesthesia

Yield

The yield rate (i.e., the ability to make a diagnosis from an SBB) reported in large series in the literature ranges from 82–99% in nonimmunocompromised (NIC) patients, and is slightly lower in AIDS patients at 56–96%. Higher yield rates in AIDS may result from improved surgical technique and histologic evaluation.¹⁶

The yield rate is higher for lesions that enhance with contrast on CT or MRI (99% in NIC patients) than with lesions that do not enhance (74%).¹⁷

Complications

The most frequent complication is hemorrhage, although most are too small to have clinical impact. The risk of a major complication (mostly due to hemorrhage) in NIC patients ranges from 0–3% (with most < 1%), and 0–12% in AIDS.¹⁷ Higher complication rates seen in AIDS patients in some series may be due to reduced platelet count or function, and to vessel fragility in primary CNS lymphoma. In NIC patients, multifocal high-grade gliomas had the highest complication rate.

Infection is an infrequent complication with needle biopsy.

111.3 Deep brain stimulation

111.3.1 General information

Deep brain stimulation (DBS): the use of electrical stimulation through surgically implanted electrodes to produce neuromodulation of electrical signals for the purpose of symptom improvement. For many indications, DBS has supplanted ablative procedures in the brain. A variety of conditions may be treated including:

1. movement disorders
 - a) Parkinson's disease: STN stimulation may be superior to best medical management^{18,19} because of similar efficacy to levodopa with fewer side effects (primarily dyskinesias) (see below)
 - b) dystonia (p. 1842)
 - c) tremor (p. 1841)
2. epilepsy (p. 1891)
3. pain (p. 1886): response is variable, typically only 25–60% respond
4. potential uses
 - a) psychiatric disorders (p. 1841): mainly
 - Tourette syndrome (p. 1842): GPI, STN, ALIC
 - obsessive-compulsive disorders (p. 1842): anterior capsule and STN stimulation²⁰ and, recently, targets more posterior & rostral²¹
 - depression (p. 1842): subgenual cingulate gyrus²² and anterior capsule stimulation²³
 - b) obesity²⁴
 - c) drug addiction²⁵
 - d) hypertension (case report of lowering BP in a patient treated for pain²⁶)

111.3.2 Surgical treatment of Parkinson's disease

Historical background

Prior to the availability of effective medications, surgical procedures were developed to treat Parkinson's disease (PD). An early procedure was ligation of the anterior choroidal artery. Due to variability in distribution, destruction often extended beyond the desired confines of the pallidum and the results were too unpredictable (p. 1539). Anterodorsal pallidotomy became an accepted procedure in the 1950's, but long-term improvement was mainly in rigidity, while tremor and bradykinesia did not improve.²⁷ The ventrolateral thalamus subsequently became the preferred target. Lesions there were most effective in diminishing tremor. In actuality, the tremor was often not the most debilitating symptom, particularly since it is a resting tremor at first (it may become more pervasive later). Bradykinesia and rigidity were frequently more problematic for the patient. Furthermore, the procedure only reduces tremor in the contralateral half of the body, and bilateral thalamotomies were not recommended due to an unacceptably high risk of post-op dysarthria and gait disturbance. Use of thalamotomy fell off dramatically in the late 1960's with the introduction of L-dopa.¹

Current trends

At some point most patients will experience problematic side effects and/or resistance to treatment with antiparkinsonian drugs. This, together with recent refinements in surgical techniques resulting in improved outcomes, has produced a resurgence of interest in the operative treatment of Parkinson's disease. Tissue transplantation (e.g., with adrenal medullary tissue) has been all but abandoned in the U.S.

Since 1987, a shift has taken place from lesioning to stimulation techniques. The subthalamic nucleus (STN) was an early target in PD. Deep brain stimulation (DBS) in the area of the globus pallidus pars interna (GPI)²⁸ or the ventral intermediate nucleus of the thalamus (Vim) can also relieve parkinsonian symptoms²⁹ without irreversibly destroying tissue. A randomized study showed similar efficacy between thalamotomy and DBS, but fewer side effects with DBS.³⁰

Indications for surgical treatment of PD

1. patients refractory to medical therapy (including multiple agents). However, some investigators feel the response to surgery might be better if done early
2. primary indication (based on an opinion survey³¹): patients with levodopa-induced dyskinesias (especially those with associated painful muscle spasms). Initial results indicate that these are very responsive to pallidotomy
3. gait and postural instability³² as well as falls and freezing (non-human primate data)³³ may respond to DBS of the pedunculopontine nucleus (PPN)
4. patients primarily with rigidity or bradykinesia (unilateral or bilateral), on-off fluctuations, or dystonia. Tremor may be present, but if it is the predominant symptom, then using the ventralis intermedius (VIM) nucleus of the thalamus as the target (for ablation (thalamotomy) or stimulation)³⁴ is a better procedure. VIM stimulation is also used to treat essential tremor³⁵

Contraindications

1. patients with significant dementia: further cognitive impairment has been noted primarily in patients with cognitive deficits prior to treatment
2. patients with risk of intracerebral hemorrhage: those with coagulopathy, poorly controlled hypertension, those on anti-platelet drugs that cannot be withheld (may consider stereotactic radiosurgery lesions for these rare patients, see below)
3. patients with ipsilateral hemianopsia: due to the risk of post-op contralateral hemianopsia from optic tract injury, which would make the patient blind
4. age ≥ 85 yrs
5. patients with secondary parkinsonism (p.185), i.e., *not* idiopathic Parkinson's disease: respond poorly, presumably due to different pathophysiology. Look for:
 - a) signs of autonomic nervous system dysfunction (suggests Shy-Drager)
 - b) EOM abnormalities (may occur in progressive supranuclear palsy (PSNP))
 - c) long-tract signs
 - d) cerebellar findings (as in olivo-ponto-cerebellar atrophy (OPCA))
 - e) failure to improve with levodopa
 - f) MRI: lacunar infarcts in basal ganglia (as in arteriosclerotic parkinsonism), or tumor in region of substantia nigra
 - g) PET scanning (if available): decreased striatal metabolism detected by deoxyglucose PET scan (suggests striato-nigral degeneration (SND))
6. patients with normal dopamine transport (DaT) scan which may rule out PD as the cause of a tremor

Results

At present, the major focus of therapy has been on improvement of motor symptoms. Although 97% of patients showed at least some improvement (some poor results may derive from inclusion of some patients with secondary parkinsonism), in 17% the degree of improvement was graded as mild.

Significant reduction of levodopa-induced dyskinesias occurred in 90%. Bradykinesia improved in 85%, rigidity in 75%, and tremor in 57%. Other areas of improvement include: speech, gait, posture, and reduction of on-off phenomenon and freezing. Although symptoms may be ameliorated, overall functional improvement may not be remarkable.³⁶

Although dosages of antiparkinsonian medication are often reduced, continued medical therapy is usually required, and no change is made for at least 2 months.

Indications are that beneficial surgical effects can last ≥ 5 years, with early failures possibly due to production of too small of a lesion, and late failures possibly due to progression of the disease.

Ongoing studies are investigating longer-term results, microelectrode recording, alternate lesioning targets, the role of early surgery... Until more information is available, one cannot make any statements about the optimal target, localizing method, etc.

111.3.3 Tremor

Tremor is the most common movement disorder (intention > postural > resting), with essential tremor affecting 5–10 million persons in the U.S. Thalamic stimulation may be useful for tremors that are refractory to medical treatment, including tremor-dominant Parkinson's disease, essential tremor,^{37,38} cerebellar and posttraumatic tremor.³⁴

Prior to being considered for surgical intervention, it should be determined that the patient has failed maximal medical therapy.

Side effects of Vim stimulation include: paresthesias, H/A, dysequilibrium, dysarthria, dystonia, and localized pain.

111.3.4 Psychiatric disorders

General principles/disorders

Surgical treatment of psychiatric disorders has a long and storied history. The first documented surgeries aimed at treating psychiatric disease occurred in the late 1800's. Psychosurgery fell out of favor due to negative public perception, the advent of effective anti-psychotic medications, and skepticism regarding surgical outcomes.

Most current applications of functional neurosurgery in psychiatric disease remain investigational. Several ongoing trials have shown promising preliminary results for severe refractive obsessive-compulsive disorder (OCD),³⁹ depression,⁴⁰ & Tourette syndrome⁴¹ (see below).

Tourette syndrome (TS)

A disorder characterized by random, repeated, and stereotyped motor or vocal tics for > 1 year,⁴² usually in several “bouts” per day. Onset is before age 18 years (mean age: 5 years). Male:female ratio is 4:1. The tics may be socially inappropriate, as such, are disabling. TS is often associated with OCD, ADHD, and other personality disorders.

Medical therapy has focused on blocking dopamine receptors, central adrenergic receptors, or on catecholamine-depleting agents.⁴³

Surgical targets for TS have included the frontal lobes, the cingulate gyrus, the anterior limb of the internal capsule (ALIC), the limbic system, and the subthalamic zona incerta.⁴⁴ Current targets of interest for DBS include: GPi, STN, ALIC, and thalamus. Early results have been promising.⁴⁵

Obsessive-compulsive disorder (OCD)

A debilitating psychiatric disorder characterized by anxiety-provoking intrusive thoughts (obsessions) that lead to stereotyped motor, cognitive acts, or rituals that are performed (compulsions) to relieve the associated anxiety.⁴⁶ Affects up to 2% of the population, and is the 10th leading cause of disability worldwide.^{47,48} 12–20% of cases remain refractory to standard behavioral psychotherapy and medical treatment with serotonin reuptake inhibitors (SSRI) or atypical antipsychotics.⁴⁹

DBS targets explored for OCD include the anterior limb of the internal capsule (ALIC), nucleus accumbens (NAc), ventral capsule/ventral striatum (VC/VS), subthalamic nucleus (STN), and inferior thalamic peduncle (ITP).^{50,51} The FDA recently granted a humanitarian device exemption approving the use of VC/VS DBS for medically intractable OCD.

Major depressive disorder (MDD)

Major depressive disorder (MDD) is the fifth leading cause of non-fatal disability worldwide.⁵² First-line treatment includes pharmacotherapy, psychotherapy, and forms of non-invasive brain stimulation such as transcranial magnetic stimulation and electroconvulsive therapy. Patients who fail conservative therapy may be candidates for functional neuromodulatory procedures.

Initial attempts at surgical treatment for chronic depression or MDD included nonspecific destructive surgeries such as the prefrontal leucotomy (lobotomy). After these procedures fell out of favor, more selective ablative procedures targeting the orbitofrontal cortex and the medial prefrontal cortex were attempted. Recently, DBS targets investigated for treatment of medically intractable depression include: the subcallosal cingulate gyrus, inferior thalamic peduncle, nucleus accumbens, and ventral capsule/striatum.^{53,54,55} Several of these targets gained interest after patients undergoing DBS for OCD noted improved mood as a side effect.⁵⁶ Vagus nerve stimulation (VNS) has also been explored as an option after patients receiving VNS for seizures were noted to have improvement in depressive symptoms independent of seizure control.⁵⁷

111.3.5 Dystonia

Pallidal stimulation is the primary surgical treatment for dystonia.⁵⁸ Response is better for primary dystonias, e.g., tardive dystonias than for secondary dystonias such as postanoxic, postencephalitic, perinatal, and poststroke dystonias⁵⁸ (other targets need to be assessed).

For primary dystonias, the globus pallidus internus (GPi) is the most common primary target (► Fig. 111.1). Good results have also been reported with STN DBS. Dyskinetic cerebral palsy in children may also be treated with pallidal stimulation.⁵⁹

111.3.6 Deep brain stimulation technique

Overview

1. medications (anti-tremor or anti-parkinsonian depending on the condition being treated) are withheld the morning of the procedure to bring out symptoms
2. a stereotactic frame is applied under local anesthetic parallel to the orbitomeatal line (which aligns somewhat with the AC-PC line (p. 58))

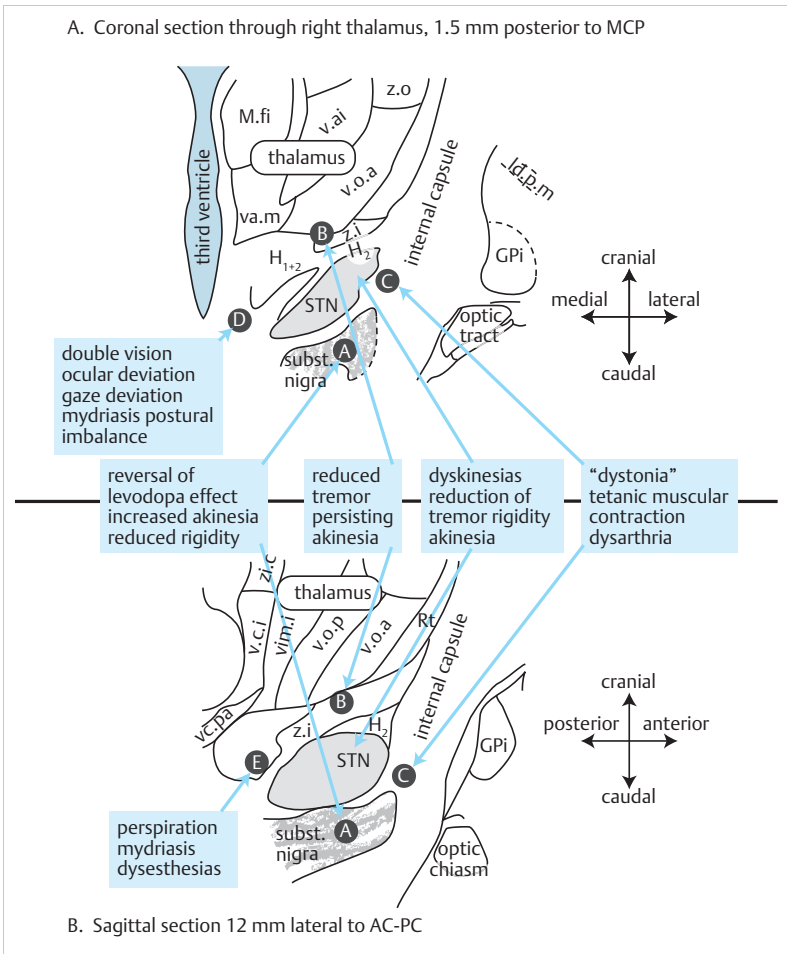


Fig. 111.1 Illustration of some targets for functional brain surgery.

Abbreviations: AC = anterior commissure; GPi = globus pallidus interna; H₁ = Forel's H₁ field; MCP = midcommissural point (halfway between AC & PC); PC = posterior commissure; STN = subthalamic nucleus; subst. nigra = substantia nigra; z.i = zona incerta.

3. radiologic target localization: (depends on the condition being treated)
 - a) may utilize MRI or CT. MRI is the most common imaging modality, and may best demonstrate the desired anatomy, but is susceptible to geometric distortion. Therefore many centers also utilize CT to supplement MRI. T1WI images are commonly employed; however, some feel that optimal MRI imaging may be performed with gadolinium-enhanced axial and coronal projections using 1-mm slice intervals and an STIR or gradient echo acquisition protocol
 - b) identify the AC-PC line on imaging
4. for some situations (e.g., GPi or STN targets for Parkinson's disease), there is a choice between awake and asleep placement of DBS electrodes, and the results appear fairly equivalent.⁶⁰ For some conditions, the patient must be awake to assess the response to stimulation

5. electrophysiologic target localization (for awake operations)
 - a) stimulation is required to verify the target which varies between individuals. For patients with dyskinesias that occur only following a dose of medication, the normal dose of the medicine is given after the imaging to bring out the symptoms
 - b) impedance typically drops when a white matter tract is encountered
 - c) pallidum stimulation usually increases (but occasionally decreases) contralateral muscle tone. Also look for reduction of tremor or dyskinesia. Contralateral weakness or hypotonia indicates proximity to the internal capsule (medial to GPi). Visual scotomata suggest stimulation of the optic tract (inferior to GPi)

Typical targets used in deep brain stimulation

► Fig. 111.1 depicts relationships of typical targets to other structures and effects of DBS (or lesioning). This figure is intended for illustrative purposes only, and is not presented for purposes of performing surgical procedures.

111.4 Torticollis

111.4.1 General information

AKA wry neck. A form of dystonia resulting in a failure to control head position (if shoulders or trunk are also involved, dystonia is a more proper label).

111.4.2 Etiologies

A symptom of diverse causes. Differential diagnosis includes:

1. congenital torticollis (may be the initial presentation of dystonia musculorum deformans)
2. spasmodic torticollis, AKA wry neck: a specific subtype of torticollis that is idiopathic by definition. The shortened sternocleidomastoid (SCM) muscle is usually in spasm
3. extrapyramidal lesions (including degenerative): often alleviated by lying down; EMG shows abnormal grouped activity
4. psychogenic (often mentioned, seldom verified)
5. torticollis from atlantoaxial rotatory subluxation (p. 1158): the *elongated* SCM may be in spasm (opposite of that in spasmodic torticollis)
6. neurovascular compression of the 11th nerve (see below)
7. hemorrhage into sternocleidomastoid muscle (with subsequent contracture)
8. infection of the cervical spine
9. cervical adenitis
10. syringomyelia
11. cerebellar tumors in children
12. bulbar palsies
13. "pseudotorticollis" may develop as an unconscious correction to reduce diplopia that occurs with imbalance of extraocular eye musculature

111.4.3 Non-surgical treatment of torticollis

Should be attempted first, and includes:

1. relaxation training, including biofeedback
2. thorough neuropsychiatric evaluation
3. trans-epidermal neuro-stimulation (TENS) to the neck

111.4.4 Surgical procedures

Reserved for disabling, refractory cases. Includes:

1. dorsal cord stimulation
2. local injection of botulinum toxin: may work for retrocollis, is poor for lateral torticollis (must inject posterior cervicals and both SCM, and may cause temporary pharyngeal muscle dysfunction resulting in dysphagia), and is totally ineffective for anterocollis
3. selective rhizotomy and spinal accessory nerve section

111.4.5 Torticollis of 11th nerve origin

- usually a horizontal type (manifests as horizontal head movement) which may be exacerbated when supine (unlike extrapyramidal torticollis)
- contraction of SCM is usually accompanied by activity in contralateral agonist muscles
- may be treated surgically. Procedures include
 - sectioning of the anastomotic branches between the 11th nerve and the upper cervical posterior root (C1 anastomotic branch is sensory only)
 - microvascular decompression of the 11th nerve (most cases caused by vertebral artery, but PICA compression is also described⁶¹). Relief takes several weeks post-op

111.5 Spasticity

111.5.1 General information

Spasticity results from lesions in upper motor neuron pathways, causing absence of inhibitory influence on alpha motor neurons (α MN) (alpha spasticity) as well as on gamma motor neurons (intrafusal fibers) (gamma spasticity). Result: uninhibited reflex arc between α MN and Ia afferents from muscle spindles, resulting in a hypertonic state of muscles with clonus, and sometimes with involuntary movements. Etiologies include: injury to cerebrum (e.g., stroke) or spinal cord (spasticity is an expected sequela of spinal cord injury rostral to the conus medullaris), multiple sclerosis, and congenital abnormalities (e.g., cerebral palsy, spinal dysraphism).

111.5.2 Clinical

General information

Increased resistance to passive movement, hyperactive muscle stretch reflexes, simultaneous activation of antagonistic muscle groups, may occur spontaneously or in response to minimal stimuli. Characteristic postures include scissoring of legs or hyperflexion of thighs. May be painful, or may disrupt patient's ability to sit in wheelchair, lie in bed, drive modified vehicles, sleep, etc. May also promote development of decubitus ulcers. A spastic bladder will have low capacity and will empty spontaneously.

Spasticity is often exacerbated by same type of stimuli that aggravate autonomic hyperreflexia; see Autonomic hyperreflexia (p. 1222).

The onset of spasticity following spinal cord injury may be delayed for several days to months (the latency period is attributed to “**spinal shock**” (p. 1119), during which time there is decreased tone and reflexes).⁶² Onset of spasticity following spinal shock starts with increasing flexor synergistic activity over 3–6 mos, with more gradual increases of extensor synergy which ultimately predominates in most cases.

Some “beneficial” aspects of mild spasticity:

- maintains muscle tone and therefore bulk; provides support for patient when sitting in wheelchair, helps prevent decubitus ulcers over bony prominences
- muscle contractions may help prevent DVTs
- may be useful in bracing

Grading spasticity

The Ashworth scale (► Table 111.1) is commonly used for the clinical grading of the severity of spasticity. Assessment should be performed with patient supine and relaxed. Many attempts have been made to quantitate spasticity electrodiagnostically, the most reliable has been H-reflex measurement.

Table 111.1 Ashworth scores⁶³

Ashworth score	Degree of muscle tone
1	no increase in tone (normal)
2	slight increase, a “catch” with flexion or extension
3	more marked increase, passive movements easy
4	considerable increase, passive movements difficult
5	affected part rigid in flexion or extension

111.5.3 Treatment

General information

Depends on extent of useful function (or potential for same) present in areas at and below the level where spasticity starts (complete spinal cord injuries usually have little function, whereas patients with MS may have significant function).

Non-surgical treatment of spasticity

1. "prevention": measures to decrease inciting stimuli (physical therapy to prevent joint damage, good skin & bladder care... see Autonomic hyperreflexia (p. 1222))
2. prolonged stretching (more than just range of motion): not only prevents joint and muscle contractures, but modulates spasticity
3. oral medications⁶⁴; (see Surgical treatment (p. 1846) for intrathecal medications): few drugs are effective without significant undesirable side effects
 - a) **diazepam** (Valium®): activates GABA-A receptors, increases presynaptic inhibition of α MN. Most useful in patients with complete spinal cord injuries.
R start with 2 mg PO BID-TID, increase by 2 mg per day q 3 days up to a max of 20 mg TID.
Side effects: may cause sedation, weakness, decreased stamina (most of which may be minimized by gradual increases in dosage). Abrupt discontinuation may cause depression, seizures, withdrawal syndrome
 - b) **baclofen** (Lioresal®): a GABA derivative that activates presynaptic GABA-B receptors of Ia muscle spindle afferents, causes presynaptic inhibition of α MN and decreases nociception. May be most useful in patients with spinal cord lesions (complete or incomplete).
R start with 5 mg PO BID-TID, increase in 5 mg increments q 3 days up to max of 20 mg QID.
Side effects: sedation, lowers seizure threshold. Must be tapered to discontinue (abrupt discontinuation may result in seizures, rebound hyper-spasticity or hallucinations)
 - c) **dantrolene** (Dantrium®): reduces depolarization induced Ca^{++} influx into sarcoplasmic reticulum of skeletal muscle; acts on all skeletal muscle (with no preferential effect on spasmogenic reflex arc).
R start with 25 mg PO q d, increase q 4–7 days to BID, TID, then QID, then by 25 mg per day up to max of ≈ 100 mg QID (may take 1 week at new steady state to see effect);
Side effects: muscle weakness (may make ambulation impossible), sedation, idiosyncratic hepatitis (may be fatal; more common in patients on > 300 mg/d $\times > 2$ mos) that is often preceded by anorexia, abdominal pain, N/V; D/C if no benefit is seen by ≈ 45 days; follow LFTs (SGPT or SGOT)
 - d) **progabide**: activates both GABA-A and GABA-B receptors. Useful for patients with severe flexor spasms
 - e) other agents that have not been successful include⁶²: phenothiazines (reduce gamma spasticity, but only at high PO doses or parenterally), clonidine, tetrahydrocannabinol...
4. intramuscular injections: botulinum toxin (Botox®), phenol

Surgical treatment of spasticity – overview

Reserved for spasticity refractory to medical treatment, or where side effects of medications are intolerable. Generally either orthopedic (e.g., tendon release operations (tenotomies) of heel cord (Achilles tendon lengthening) or hamstrings, iliopsoas myotomies, adductor lengthening, derotational osteotomies...) or neurosurgical (e.g., nerve blocks, neurectomies, myelotomy, rhizotomies...).

Neurosurgical treatment options:

1. nonablative procedures
 - a) intrathecal (IT) baclofen: delivered via an implantable pump (see below)
 - b) intrathecal morphine (tolerance and dependence may develop)
 - c) electrical stimulation via percutaneously placed epidural electrodes⁶⁵
2. ablative procedures, with *preservation* of potential for ambulation
 - a) selective dorsal rhizotomy (p. 1851): divides *sensory* rootlets involved in spastic reflex arc
 - b) motor point block⁶² (intramuscular phenol neurolysis): preserves sensation and existing voluntary function. Especially useful in patients with incomplete myelopathies; time-consuming
 - c) phenol nerve block: similar to motor point block, but used when spasticity more severe and complete block of muscle desired. Open phenol block used instead of percutaneous when nerve is mixed and sensory preservation is desired (also reduces post-block dysesthesias)⁶⁶
 - d) selective neurectomies⁶²
 - sciatic neurectomy (may be done with RF lesion)⁶⁷

- obturator neurectomy: useful if strong hip adductor spasticity that causes scissoring and wasted energy expenditure in ambulating
 - pudendal neurectomy: useful if excessive detrusor dyssynergy interferes with bladder retraining
- e) percutaneous radiofrequency foraminal rhizotomy⁶⁷: small unmyelinated sensory fibers are more sensitive to RF lesions than larger myelinated A-alpha fibers of motor units. Technique: start at S1 on one side, and work up to T12, then repeat on other side. At each level: verify needle position by stimulating with 0.1–0.5 V and watch for movement in appropriate myotome (tip should be extradural, avoid subarachnoid placement), ablate with 70–80 °C × 2 mins for S1, and 70 °C × 2 mins for L5 to T12 (to preserve motor function). If symptoms recur, may repeat with lesions at 90 °C × 2 mins
- f) myelotomies⁶⁸
- Bischof's myelotomy: divides anterior and posterior horns via laterally placed incision, disrupts reflex arc. No effect on α -spasticity
 - midline "T" myelotomy: interrupts reflex arc from sensory to motor units without disrupting connections from corticospinal tract to anterior motor neurons. Slightly higher risk of losing motor function.
Technique: laminectomy from T11 to L1. Mobilize midline dorsal longitudinal vein and incise cord in midline from T12 at a depth of 3 mm to S1 at a depth of 4 mm (preserving S2–4 maintains bladder reflex pathways. Unilateral extension up to conus medullaris reduces bladder spasticity and increases capacity before reflex emptying occurs)
- g) stereotactic thalamotomy or dentatotomy: may be useful in cerebral palsy.⁶⁹ Useful for unilateral dystonia, but cannot be used for bilateral dystonia as bilateral lesions would be required which jeopardizes speech, cognition... Effective only for dystonia *distal* to shoulders or hips, and should not be used if the condition is rapidly progressive
3. ablative procedures, with *sacrifice* of potential for ambulation (in complete cord injuries, nonablative procedures are not indicated because there is no motor function to recover). Used after failure of percutaneous rhizotomy (see above) and "T" myelotomy (see above)
- a) intrathecal injection of 6 ml of 10% phenol (by weight) in glycerin mixed with 4 ml of iohexol (Omnipaque® 300) (p. 230) for a final concentration of 6% phenol and $\approx$ 120 mg iodine/ml. Administered via LP at L2–3 interspace with patient in lateral decubitus position (most symptomatic side down) under fluoro until T12–1 nerve root sleeves are filled (sparing S2–4 for bladder function). Patient is maintained in this position × 20–30 mins and then kept sitting upright × 4 hrs⁷⁰ (absolute alcohol provides more permanent blocks, but is hypobaric and more difficult to control)
- b) selective anterior rhizotomy: results in flaccid paralysis with denervation atrophy of muscles
- c) neurectomies, often combined with tenotomies⁶⁷
- d) intramuscular neurolysis by phenol injection⁶⁷
- e) cordectomy⁷¹: most drastic measure, reserved for patients who do not respond to any other measure. Results in total flaccidity with loss of benefits from mild spasticity. Converts bladder from UMN to LMN control. Works well for progressive deficit from syringomyelia and for spasticity, but poor for "phantom" leg pain⁷²
- f) cordotomy: rarely used

Intrathecal baclofen

See references.^{73,74,75,76}

Selection criteria used in one study⁷⁵ are shown in ► Table 111.2. Other indications include: stroke,⁷⁷ cerebral palsy, TBI, dystonia, stiff-man syndrome.

Test doses: Incremental test doses of 50, 75, and then 100 mcg intrathecal baclofen (ITB) via lumbar puncture or temporary catheter were used,⁷⁵ randomly alternated with placebo, with dose escalation halted if a response to active drug occurred. The following parameters were evaluated at 0.5, 1, 2, 4, 8, & 24 hrs post injection: pulse and respiratory rate, BP, hypertonia (Ashworth score, ► Table 111.1), reflexes, spasm score, voluntary muscle movement, and adverse effects (if any, including seizures). Pump implantation was offered if there was a 2-point reduction in the Ashworth score and muscle spasm score for $\geq$ 4 hrs after bolus injection of active drug without intolerable side effects. Usual daily dose for ITB is twice the test dose, typically 200 micrograms/d.

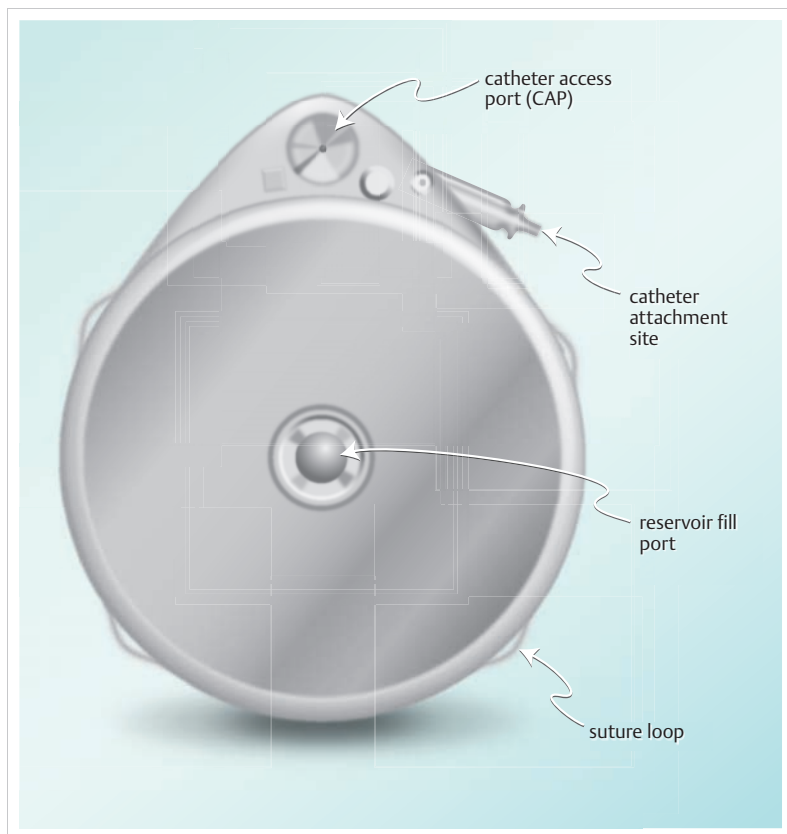
Alternatively, give 25 mcg IT in the O.R., and if the patient improves, insert subcutaneous pump.⁶⁴

Pumps: available programmable systems include those manufactured by Medtronic, Inc., Minneapolis, MN: the SynchroMed II Pump (► Fig. 111.2) with the N'Vision programmer, or the somewhat similarly shaped SynchroMed EL which has a screen over the CAP to prevent inadvertent access.

Table 111.2 Selection criteria for baclofen pump

- age 18–65 yrs (older patients treated on compassionate use basis)
- able to give informed consent
- severe, chronic spasticity (≥ 12 mos duration) due to spinal cord lesion or MS
- spasticity refractory to oral drugs (including baclofen), or unacceptable side effects
- no CSF block (e.g., on myelography)
- positive response to IT baclofen at test dose ≤ 100 mcg and no response to placebo
- no implanted programmable device such as cardiac pacemaker^a
- females of childbearing potential: not pregnant & using adequate contraception
- no hypersensitivity (allergy) to baclofen
- no history of stroke, impaired renal function, or severe hepatic or GI disease

^athis study used a programmable IT pump

**Fig. 111.2** Medtronic Synchron II.

Insertion technique: IT catheter is typically inserted $\approx$ L2–3, and is threaded rostrally $\approx$ 3 levels, but should be no higher than T10 (risk of rostral progression of hypotonia). Record the catheter length used (this is programmed into the Synchron II and is important when injecting the port or withdrawing drug).

Post-op orders: Guidelines for postoperative orders following baclofen pump insertion

1. admit PACU, transfer to:
 - a) floor if insertion follows test dosing or if patient has just been transitioned from stable PO dose
 - b) ICU if there has been a hiatus in baclofen therapy
2. neuro checks q 2 hrs for 1st 24 hours
3. baclofen:
 - a) for patients on oral or IV baclofen: continue baclofen at the previous dose via the same route (oral or IV) until the ITB takes effect (usually 2–4 hrs, full effect is delayed up to 24 hrs). The IV/PO drug is then tapered
 - b) if there has been a hiatus in baclofen therapy: baclofen 20 mg PO QID
4. have 2 vials of IV physostigmine available and labeled "FOR EMERGENCY USE ONLY" for possible baclofen overdose

► **Device-related complications.** Shown in ► Table 111.3. The frequency of most is $\approx 1\%$, except catheter-related problems which had a rate of $\approx 30\%$ ⁷⁵

Table 111.3 ITB pump complications^a

1. mechanical problems
 - a) pump underinfusion
 - b) catheter problems: occlusion, kink, dislodgment, cut, break (including microleaks), or disconnection
2. wound complications
 - a) pocket erosion
 - b) incisional pain
 - c) infection:
 1. infection at the pump site can be introduced at the time of insertion, or when accessing any of the ports, or from hematogenous spread from unrelated bacteremia
 2. infection of the catheter may be associated with meningitis
 - d) seroma (may require aspiration)
 - e) CSF collection

^adevice-related complications requiring a secondary invasive procedure

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Pump site infection: If the patient is still receiving ITB (i.e., not going into baclofen withdrawal (p.1849)) and not showing signs of sepsis or meningitis, start antibiotics and attempt to taper the IT baclofen over a few weeks if tolerated, supplementing as necessary with PO medication. Once the pump-delivered dose is negligible, the pump and catheter can be removed and antibiotics continued until the infection is convincingly cleared (typically 3–6 months), and then a new system can be implanted. For patients that are septic, the pump may need to be removed emergently and the ITB is managed as described below.

► **Baclofen overdose.** Etiologies: wrong concentration of drug (standard concentrations: 500, 1000 or 2000 mcg/cc, compounding pharmacies may make custom doses as high as 3000–5000 mcg/cc), programming error, not aspirating catheter access port (CAP) prior to injections.

Presentation: rostral progression of hypotonia, respiratory depression, coma, and seizures.

Attend to ABCs (airway/breathing/circulation). Intubate if necessary.

Pump-related measures:

1. empty pump reservoir to stop drug flow (record amount withdrawn)
2. administer physostigmine if not contraindicated:
 - a) **R** adult: 0.5–1.0 mg IM or IV @ rate ≤ 1 mg/min
(may repeat q 10–30 minutes PRN)
 - b) **R** peds: 0.02 mg/kg IM or IV @ rate ≤ 0.5 mg/min
(may repeat q 5–10 min up to 2 mg max)
3. withdraw 30–40 ml CSF either via LP or through catheter access port
4. notify the pump manufacturer

► **Intrathecal baclofen withdrawal.** ITB results in down-regulation of GABA-B receptors that cannot be compensated by PO baclofen or small doses of ITB.

Interruption of ITB therapy may occur as a result of: empty pump reservoir, pump battery failure, end of pump life (Synchroned II pumps shut off automatically after 5 years, audible alarms give ample warning to replace the pump), catheter migration/breakage/kinking/disconnection/occlusion, programming error, catheter tip granuloma (p.1389) (less common with ITB than with opioid, see that section for management) or stiff-man syndrome (antibodies to glutamic acid decarboxylase which reduces available endogenous GABA).

If it is necessary to electively (or semi-electively) remove a pump system (e.g., for infection), the optimal scenario is to gradually taper the drug by reprogramming the pump and/or by filling the reservoir with solution of decreased baclofen concentration while supplementing with PO baclofen as needed and as tolerated.

The severity of withdrawal syndrome depends on dose of drug used (increased with higher dose) and duration of therapy (increased with longer therapy).

Syndromes arising from abrupt discontinuation of ITB:

1. mild withdrawal symptoms: return of spasticity and rigidity, tachycardia, piloerection (goose-bumps) & pruritus (pruritus without a rash is very suggestive of ITB withdrawal)
2. moderate withdrawal symptoms: seizures & hallucinations
3. severe withdrawal symptoms (estimated incidence: 3–5%)⁷⁸: increased rebound spasticity, rigidity, fever, labile BP, and reduced level of consciousness. If untreated, the severe syndrome can progress over 24–72 hours to rhabdomyolysis (with elevated creatine phosphokinase (CPK) and transaminase), hepatic and renal failure, DIC, and occasionally death. Differential diagnosis:
 - malignant hyperthermia (p. 112)
 - autonomic dysreflexia (p. 1222): exaggerated sympathetic response, classically presents with HTN + bradycardia in a patient with a spinal cord injury at T6 or above
 - neuroleptic malignant syndrome: caused by dopamine (DA) receptor blocking drugs or sudden cessation of DA agonists. Treat with DA agonists and IV dantrolene
 - sepsis
 - status epilepticus

Management of *abrupt* ITB withdrawal syndrome (adapted^{78,79}):

1. ABCs (airway/breathing/circulation). Intubate if necessary
2. assess for noxious stimulation, infection, other drugs, progression of underlying disease
3. labs: CPK
4. primary goal is to reestablish ITB therapy at the same dose as soon as possible
5. assess pump/system as outlined in ► Table 111.4

Table 111.4 Assessing ITB infusion systems (adapted⁷⁹)

- interrogate the pump (with device programmer): alarm log for low reservoir, low battery, or end of service, check rotor log (rotor stops temporarily in MRI scanners)
- verify reservoir volume: aspirate entire volume of drug and compare to volume calculated by the programmer; discrepancies > 25% may signal mechanical faults. Refill reservoir with drug if empty (by experienced ITB practitioner)
- obtain AP & lateral X-rays to assess location of catheter tip and to look for breaks^a, kinks, or migration
- program the pump to administer a prescribed bolus, observe for decreased spasticity (takes up to 4 hours)
- aspirate 3–4 ml thru the catheter aspiration port (CAP) (see ► Fig. 111.2). Difficulty aspirating or bubbles are abnormal
- contrast study (either of the following can miss a microleak):
 - if abrupt onset: aspirate the CAP (if unsuccessful, do NOT inject dye because of danger of overdose), then inject iodinated contrast into the CAP either under fluoroscopy or while the patient is in the CT scanner (do not move the patient between injection & imaging). High rate of false negatives
 - insidious onset: an indium study can be done, it takes ≈ 24 hrs to get the radio-isotope. DPTA radio-isotope is injected into the reservoir fill port (see ► Fig. 111.2) - not into the CAP! (OK for it to mix with baclofen). Image with nuclear medicine camera (length of study depends on programmed flow rate). This test is expensive, takes a long time to complete (2–3 days from the time it is ordered) and has low sensitivity

^aNB: in 2012 Medtronic replaced the radio-opaque catheter with a stronger Ascenda catheter which is radiolucent except for a small spot at the distal tip

6. early use of high-dose oral/enteral baclofen: ≥ 120 mg/d in 6–8 divided doses if the patient's condition permits (NB: PO baclofen is not reliable as the lone treatment for ITB withdrawal, and safety not established for age < 12 yrs)
7. attempt to restore ITB therapy at or near pre-withdrawal dosage by experienced physician by one of the following:
 - a) using a programmed bolus through the pump
 - b) injection via the catheter access port
 - c) via LP
 - d) via new externalized catheter
8. if restoration of ITB therapy is delayed and if symptoms persist
 - a) move patient to ICU if not already there

- b) parenteral benzodiazepine (BDZ) infusion: diazepam or midazolam. Titrate dose to reduce muscle rigidity, hyperthermia, BP lability, seizures. BDZs do not prevent baclofen withdrawal syndrome, but they may buy some time
 - c) cyproheptadine⁸⁰: a serotonin antagonist. Start with 4 mg PO q 6 hrs
 - d) diphenhydramine 50 mg PO or IM, may repeat q 6 hrs for pruritus
 - e) dantrolene: may *not* be as effective as it is for malignant hyperthermia
 - f) rectal baclofen
9. surgical repair/replacement of defective components

Selective dorsal rhizotomy

General information

Key concepts: Selective dorsal rhizotomy (SDR)

- an ablative technique primarily used to reduce LE spasticity in children with cerebral palsy
- uses intraoperative stimulation and EMG to identify and divide a portion of the *sensory* lower-extremity rootlets involved in the spastic reflex arc (no motor roots are cut)
- optimal candidates: no consensus; consider for children > 2 yrs old with spastic diplegia (LEs > UEs) that hampers gait, premie at birth or full term + brain MRI → no major abnormalities
- avoid pts with: dystonia, nonambulatory or unable to maintain antigravity posture, significant scoliosis or hip dislocation, orthopedic procedure within last 1 year or botox in last 3 months
- results in gait improvement in ambulatory patients at 5, 10, & 15 yrs. Physiotherapy is a must
- risks: scoliosis, urinary retention, CSF leak

Selective dorsal rhizotomy (SDR): uses intraoperative electrophysiological stimulation and EMG to eliminate a portion of sensory rootlets involved in “handcapping spasticity” (leaves all motor rootlets and those sensory rootlets subserving “useful spasticity” intact). Decreases the afferent limb of the pathologic reflex arc. No effect on α -spasticity. Ambulatory children with cerebral palsy have improved gait, has been used in nonambulatory children with improvement of spasticity (helps facilitate care) but they are still not able to ambulate afterwards.

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Selection criteria

There is currently no consensus on the optimal candidate.⁸¹ Factors to consider:

1. most often used in children > 2 years of age
2. some surgeons reserve SDR for patients who have failed IT baclofen
3. mild-to-moderate LE weakness with the ability to walk or maintain antigravity posture.
On the GMFCS (► Table 111.5), most are Level II or III, some Level IV are candidates
4. spasticity hampers gait and/or development of motor skills
5. legs worse than arms
6. gestation (to be certain that you're dealing with CP and not some other entity)
 - a) premies < 33 weeks gestation at birth with spastic diplegia
 - b) full-term children with spastic diplegia and brain MRI with no major structural abnormality (periventricular leucomalacia is common and is acceptable)

Table 111.5 Gross Motor Function Classification System – Expanded and Revised (GMFCS – E&R) (see reference⁸² for full description and website⁸³ for helpful illustrations)

Level	Brief description
I	able to walk at home, school, & outdoors; climbs stairs without railing; speed, balance, and coordination are limited
II	able to walk in most settings & climb stairs holding railing; difficulty on long distances or uneven terrain, inclines, & crowded spaces
III	able to walk with hand-held mobility device in most indoor settings; climb stairs with railing + assist/supervision, use wheeled mobility for long distances
IV	assistance or powered mobility in most situations, may walk short distances with assist at home
V	wheeled mobility in all settings; limited ability to maintain antigravity postures and control leg & arm movements

7. avoid patients with dystonia (becomes evident by age 5 years, SDR does not work for this), significant scoliosis (relative contraindication – may be candidates with small conus exposure), hip dislocation, orthopedic procedure (e.g., tendon release) within the last year or botox injection within the last month
8. willing and able to undergo intense physical therapy post-op

Technique

1. EMG monitoring leads: muscles of LEs, pudendal, bladder, and anal sphincter
2. anesthesia techniques (p. 112) to maintain EMG
3. position: prone, in slight Trendelenburg to reduce loss of CSF
4. laminectomy or laminoplasty: more commonly used options (check pre-op MRI for the level of the conus)
 - a) wide opening to expose cauda equina: e.g., L1–5. May be higher risk for developing scoliosis
 - b) small opening over conus. In children < 10 years: ultrasound can be used through the skin to localize the conus in the OR. For age > 10 years, the laminectomy is made based on the location of the conus gleaned from the pre-op MRI (usually ≈ L1–2). May be harder to identify lower sacral roots (the ones you don't want to cut) which tend to be thinner and more medially situated
5. open the dura, identify sensory rootlets (dorsal to the dentate ligament), and place a thin barrier underneath (e.g., strip cut from a sterile glove) to keep them separated from the motor roots
6. L2 is identified by locating the root exiting the neural foramen between L2 & L3
7. stimulation: each dorsal root from L2 through S1 is stimulated before subdividing into separate rootlets. The root is elevated with hooked stimulator probes, the cathode probe is positioned cranial to the anode probe. Starting at 0.5 mA, progressively larger single-pulsed stimuli are applied (up to 4 mA max) until there is a response. Then this amplitude is used in a 1-second 50 Hz train and the response is graded as shown in ► Table 111.6

Table 111.6 EMG response grading to a 1-second train of stimuli for SDR⁸⁴

Grade	EMG response
0	unsustained or single discharge isolated to that root
1+	sustained discharges from segmentally innervated muscles
2+	sustained discharges from segmentally innervated muscles and muscles from immediately adjacent segments
3+	sustained discharges from segmentally innervated muscles & muscles from distant segments
4+	sustained discharges from contralateral muscles ± sustained discharges from ipsilateral muscles

8. selective sectioning (adapted from Park,⁸⁴ see reference for details):
 - a) roots with a 0 response are left intact
 - b) roots with 3+ or 4+ response: separate into rootlets along natural planes, divide all but 1 rootlet (1 is left to avoid complete sensory loss)
 - c) roots with 1+ or 2+ response: separate into rootlets, divide 50–65% of rootlets from that nerve – targeting the ones with the most activity if any (e.g., if there are 6 rootlets, cut 3 (50%) or 4 (66%)). Move the rootlets to be spared beneath the barrier to protect them
 - d) spare sacral roots (to avoid bladder problems): in general, do not cut the S2 root (which has a single fascicle), or at most, cut ≤ 35% of S2.⁸⁵ Do not cut S3 or lower
 - e) the L1 root is identified and 50% is cut without testing (Park feels L1 testing is unreliable⁸⁴)

► **Post-op care.** Most institutions have their own protocol. Patients are generally kept flat on bed rest for 48–72 hours.

Patients are started very early with PT.

NB: with reduction in spasticity, the patient's metabolic requirements often decrease substantially. The patient and caregivers must be aware of this and decrease calorie intake accordingly to avoid increases in body weight that could impair mobility because the UE & LE muscles may not be strong enough to compensate.

Outcome

Benefits are not immediately evident, it may take ≈ 2–3 months.

Ashworth scale (► Table 111.1) scores improve by 0.5–1.5.⁸⁶

Benefits may be temporary, but functional gains of improved mobility, self-care, ambulation and quality of life (for patient and caregivers) can be shown at 5, 10, & 15 years.^{87,88,89}

Patients with cerebral palsy plateau in motor development. Patients with SDR plateau later.

For inexplicable reasons, L1–5 selective dorsal rhizotomies often reduces spasticity in the UEs.

► Complications

1. sensory loss or dysesthesia: transient (typically lasting ≤ 3 months): 2.5–40%⁸⁵; permanent but not disabling sensory loss: 0–6%
2. transient urinary retention (1.25–24%⁸⁵) or urinary incontinence (4.4%)
3. CSF leak: 3%
4. infection: 2%
5. motor weakness: must differentiate from underlying weakness unmasked from SDR. Cutting less of L4 to preserve more quadriceps tone may help (this does not increase strength, just the spasticity)
6. spinal deformity: scoliosis, spondylolisthesis. True incidence is unknown, patients with spastic CP have higher rates in general even without SDR.⁹⁰ Possible lower risk with limited laminectomy is uncertain

111.6 Sympathectomy

111.6.1 Cardiac sympathectomy (angina pectoris)

With the advances in percutaneous coronary artery techniques, cardiovascular surgery and drugs, cardiac sympathectomy for angina pectoris has found less application. However, it may still be useful in patients who have no further treatment options. Bilateral sympathectomy from the stellate ganglion through the T7 ganglia is required. Newer thoroscopic techniques may revive some interest in this.

111.6.2 Upper extremity sympathectomy

General information

Various pathologies that may be indications for upper extremity sympathectomy are shown in ► Table 111.7.

Table 111.7 Indications for UE sympathectomy

- essential hyperhidrosis
- primary Raynaud's disease
- shoulder-hand syndrome
- intractable angina
- $\pm$ causalgia major (p. 525)

Hyperhidrosis

General information

Either essential (primary, or idiopathic) or secondary (etiologies include: hyperthyroidism, diabetes mellitus, pheochromocytoma, acromegaly, parkinsonism, CNS trauma, syringomyelia, hypothalamic tumors, menopause).⁹¹

Due to overactivity of eccrine sweat glands (found over entire body, highest concentration in palms and soles of feet). They produce a hypotonic secretion with saline as the primary constituent. These glands are under control of the sympathetic nervous system; however, the neurotransmitter is paradoxically acetylcholine (i.e., they are cholinergic, unlike most sympathetic end organs, which are adrenergic). Most eccrine sweat glands serve a thermoregulatory function; however, those on the palms and soles respond primarily to emotional stress.⁹¹

Essential hyperhidrosis is a generalized condition that usually manifests mostly in the palms. The prevalence is approximately 2.8% in the United States.⁹²

Treatment

Mild cases are treated medically with

1. topical agents: astringents (potassium permanganate, tannic acid...) or antiperspirants (contact dermatitis usually limits use of these agents)
2. or systemically with anticholinergics: including atropine, propantheline bromide... (side effects of dry mouth and blurred vision usually limits use of these)
3. tap water iontophoresis: may produce keratinization of palmar epithelium

Severe cases refractory to medical therapy may be candidates for surgical sympathectomy (see below).

Surgical technique

Removal of only the second thoracic ganglion is probably adequate, and avoids a Horner syndrome in most. Techniques used include: anterior transthoracic, thoracic endoscopic,⁹³ percutaneous radiofrequency, and supraclavicular. An approach via a midline posterior incision with a T3 costotransversectomy allows bilateral access.^{91,94} The risk of significant complications is $\approx 5\%$, and they include pneumothorax, intercostal neuralgia, spinal cord injury, and Horner syndrome.

111.6.3 Upper thoracic sympathectomy

Approaches include

1. posterior paravertebral approach
2. axillary thoracotomy with transthoracic exposure of the sympathetic chain
3. supraclavicular, retropleural exposure
4. percutaneous radiofrequency technique^{95,96}
5. video endoscopic approach⁹⁷

111.6.4 Lumbar sympathectomy

Primary indication is for causalgia major of the lower extremity. Preoperative lumbar sympathetic blocks may be utilized to evaluate patient for response.

Removal of the L2 and L3 sympathetic ganglion is usually adequate to remove sympathetic tone from the lower extremities (occasionally L1 and sometimes T12 are also removed for causalgia of the thigh).

The most common approach is a retroperitoneal approach through a flank incision. The patient is placed in a lateral oblique position, and the skin incision is made from the anterior superior iliac spine to the tip of the 12th rib. The peritoneum is dissected from the muscular wall and is retracted anteriorly. The kidney and ureter are retracted anteriorly; injury to the ureter being a major risk of the operation. The sympathetic chain is identified on the lateral aspect of the vertebral bodies. The vena cava makes a right-sided approach more difficult as the aorta is easier to deal with on left-sided approaches.

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112 Neurovascular Compression Syndromes

112.1 General information

Syndromes due to compression of cranial nerves at the root entry zone (REZ) (or in the case of motor nerves, root exit zone) typically by an adjacent artery. The REZ (AKA Obersteiner-Redlich zone) is the point where central myelin (from oligodendroglial cells) changes to peripheral myelin (from Schwann cells).

A neurovascular compression syndrome has been described for cranial nerves V, VII, VIII, IX, X, and XI as shown in ► Table 112.1.

Table 112.1 Neurovascular compression syndromes

Cranial nerve	Syndrome	Typical offending vessel
V	trigeminal neuralgia	superior cerebellar artery (SCA)
VII (facial nerve proper)	hemifacial spasm (p. 1870)	anterior inferior cerebellar artery (AICA)
VII (nervus intermedius)	geniculate neuralgia (p. 1873)	
VIII	disabling positional vertigo (p. 1873)	posterior inferior cerebellar artery (PICA)
IX	glossopharyngeal neuralgia (p. 1873)	
X	superior laryngeal neuralgia (p. 1874)	PICA or VA
XI	torticollis of XI nerve origin (p. 1845)	VA or (rarely) PICA

112.2 Trigeminal neuralgia

112.2.1 General information

112

Key concepts

- sharp, electric shock-like paroxysmal lancinating pain in the distribution of one or more branches of the trigeminal nerve on one side
- characterized by periods of remission and initial response to carbamazepine
- neurologic exam must be intact (only exception: mild sensory loss)
- 80–90% of cases are caused by compression of the trigeminal nerve at the root entry zone by the superior cerebellar artery. In MS patients, may be due to MS plaque (MS patients are usually less responsive to procedures)
- 75% will ultimately fail medical therapy and require a procedure (main options: microvascular decompression, percutaneous rhizotomy, or radiosurgery). Choice of modality depends on patient age, location of symptoms, prior treatment and the side effect profile of the treatment modality

Trigeminal neuralgia (TGN) (AKA tic douloureux): paroxysmal lancinating electric-like pain lasting a few seconds, often triggered by sensory stimuli, confined to the distribution of one or more branches of the trigeminal nerve (► Fig. 112.1) on one side of the face, with no neurologic deficit.

Atypical facial pain (AFP): this term is sometimes used to describe any other type of facial pain. Some of these conditions may mimic TGN, but have features inconsistent with TGN, such as: constant pain, hyperactive autonomic dysfunction, hypesthesia to pinprick. The term “atypical trigeminal neuralgia” should be avoided.

Rarely, TGN manifests as status trigeminus, a rapid succession of tic-like spasms triggered by seemingly any stimulus. IV carbamazepine (where available) or phenytoin may be effective for this.

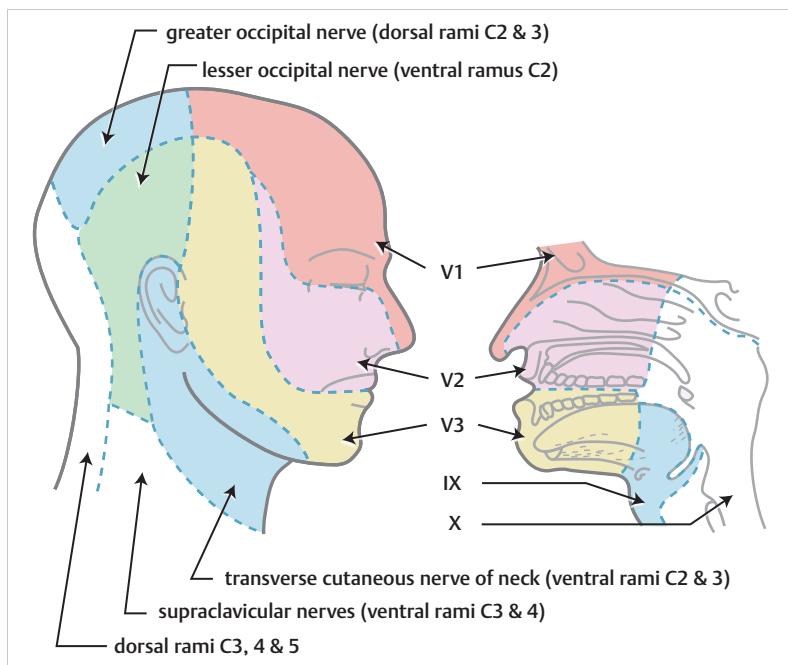


Fig. 112.1 Pain/temperature innervation of the head*.

* KEY: V = branches of the trigeminal nerve (V1 = ophthalmic nerve; V2 = maxillary nerve; V3 = mandibular nerve); IX = glossopharyngeal nerve; X = vagus nerve.

112.2.2 Applied anatomy

The trigeminal nerve is a mixed motor and sensory cranial nerve. The major branch (portio major) penetrates the pons and conveys pain, touch, and temperature sensation (but not taste) of cutaneous and deep facial structures (the dermatomal distribution of sensation is shown in ► Fig. 112.1) as well as special visceral afferents (*not* motor fibers) subserving proprioception of mastication muscles.

The motor branch (portio minor) exits the pons immediately adjacent to the portio major, and controls 4 of the muscles of mastication and the tensor tympani, tensor palatini, mylohyoid, and the anterior belly of the digastric.

The nerve traverses the cerebellopontine angle cistern (where it can come into proximity of the superior cerebellar artery which can be a source of trigeminal neuralgia) as it crosses between the pons and Meckel's cave.

Meckel's cave is a space between two layers of dura (dura propria and periosteal layers) located in the anterior petrous apex (shown on MRI in ► Fig. 99.7) that normally measures $\approx 4 \times 9$ mm wide $\times 15$ mm long.¹ It rests in a depression of the petrous bone (impressio trigemini) and houses the trigeminal cistern, a subarachnoid CSF containing space, which in turn contains the Gasserian ganglion (AKA semilunar ganglion) wherein the cell bodies of the sensory portion of the trigeminal nerve reside. The entrance to the cave is the porus trigeminus. Distal to the Gasserian ganglion, the trigeminal nerve trifurcates into the ophthalmic division (V1), mandibular division (V3), and the maxillary division (V2). V1 & V2 pass through the cavernous sinus to exit the skull through the superior orbital fissure (V1) and foramen rotundum (V2). V3 does not enter the cavernous sinus, and exits the skull through the foramen ovale. See abnormalities within Meckel's cave (p.1660), some of which can rarely cause trigeminal neuralgia.

112.2.3 Epidemiology

See ► Table 112.2. Annual incidence 4/100,000. There is no correlation with herpes simplex infection.² There is a tendency for spontaneous remission, with pain-free intervals of weeks or months being characteristic, regardless of treatment. 2% of patients with MS have TGN,³ whereas ≈ 18% of patients with bilateral trigeminal neuralgia have MS.⁴

Table 112.2 Epidemiology of trigeminal neuralgia^{5,6}

age (years)	typically > 50 (average 63)
female:male	1.8:1
Laterality	
right	60%
left	39%
both	1%
Division involved	
V1 only	2%
V2 only	20%
V3 only	17%
V1 & V2	14%
V2 & V3	42%
all three	5%

112.2.4 Pathophysiology

Probably due to ephaptic transmission in trigeminal nerve from large-diameter partially demyelinated A fibers to thinly myelinated A-delta and C (nociceptive) fibers. Pathogenesis may be due to:

1. vascular compression of the trigeminal nerve at the root entry zone (NB: compression may be seen in up to 50% of autopsies in patients without TGN⁷):
 - a) most commonly (80%) by the SCA; see Neurovascular compression syndromes (p.1857) for more details
 - b) persistent primitive trigeminal artery (p.83)⁸
 - c) dolichoectatic basilar artery⁹ (p.1108)
2. posterior fossa tumor (see below)
3. in MS, plaque within brainstem may cause TGN that is often poorly responsive to microvascular decompression

In addition to the sensory division of the trigeminal nerve, other possible pain pathways include¹⁰: the motor branch of the 5th nerve (portio minor), or the 7th or 8th nerve.

112.2.5 Tumors and trigeminal neuralgia

In >2000 patients with facial pain seen over 10 yrs, only 16 harbored tumor (<0.8% incidence).¹¹ 3 tumors outside cranial vault included nasal carcinoma and skull base mets; all had hypalgesia and atypical facial pain (AFP). 6 middle fossa tumors included 2 meningiomas, 2 schwannomas (1 primary tumor of Gasserian ganglion), and 1 PitNET/adenoma. Posterior fossa tumors are the most likely to cause symptoms that most closely resemble true TGN; of these, vestibular schwannoma (VS) is most common. 2 of 7 VSs had tumors contralateral to the neuralgia (presumably due to brainstem shift). Patients with true TGN initially responded to carbamazepine, none with AFP did.

When facial pain is caused by tumor, especially with peripheral tumors, the pain is frequently atypical (usually constant), neurologic abnormalities are often present (usually sensory loss, although some are neurologically normal at first), and the age is often younger than typical TGN.

112.2.6 Differential diagnosis

See Craniofacial pain syndromes (p.519).

112.2.7 Evaluation

History and physical (in addition to routine):

1. history
 - a) accurate description of pain localization to determine which divisions of trigeminal nerve need to be treated
 - b) determine time of onset of TGN, trigger mechanisms
 - c) ascertain presence and length of pain-free intervals (lack of any pain-free interval is atypical for TGN)
 - d) determine duration, side effects, dosages, and responses to medications tried
 - e) inquire about symptoms that may indicate the presence of conditions other than TGN: e.g., history of herpetic vesicles, excessive tearing of the eye (may indicate SUNCT (p.520)), facial twitching (tic convulsif), tongue pain (glossopharyngeal neuralgia), sensory loss (tumor...), progressive relentless pain (tumor, herpes...), symptoms that suggest MS
2. physical exam: the exam should be normal in TGN, any neurologic deficit (except very mild sensory loss) in previously unoperated patient should prompt search for structural cause, e.g., tumor (see below). This exam also serves as a baseline for post-op comparison
 - a) assess sensation in all 3 divisions of trigeminal nerve bilaterally (include corneal reflexes)
 - b) assess masseter function (bite) and pterygoid function (on opening mouth, chin deviates to weak side)
 - c) assess EOM function

112.2.8 Imaging

MRI is often used to evaluate these patients for possible intracranial tumors or MS plaques, especially in cases with atypical features. The yield in typical cases is low.

112.2.9 Medical therapy for trigeminal neuralgia

Drug info: Carbamazepine (Tegretol®)

Complete or acceptable relief in 69% (if 600–800 mg/d are tolerated and give no relief, diagnosis of TGN is suspect³). **Side effects:** Drowsiness. Rash in 5–10%. Possible Stevens-Johnson syndrome. Relative leukopenia is common (usually does not require discontinuing drug). See precautions under carbamazepine (CBZ, Tegretol®) (p.491).

R 100 mg PO BID, increase by 200 mg/d up to maximum of 1200 mg/d divided TID. **Supplied:** See supply information (p.491).

Drug info: Oxcarbazepine (Trileptal®)

Rapidly metabolized to carbamazepine, similar efficacy, often tolerated at higher doses than carbamazepine. **Side effects:** symptomatic hyponatremia.

R for trigeminal neuralgia: 300 mg PO BID, increase by 600 mg/d q week. Usual dose: 450–1200 mg. Maximum of 2400 mg/d. **Supplied:** 150, 300, 600 mg tablets; 500 mg/5-ml suspension.

Drug info: Baclofen (Lioresal®)

2nd DOC (not as effective as carbamazepine, but fewer side effects). Caution: teratogenic in rats. Avoid abrupt withdrawal (can cause hallucinations and seizures). May be more effective if used in conjunction with low-dose carbamazepine.

R Start low, 5 mg PO TID, increase q 3 d by 5 mg/dose; not to exceed 20 mg QID (80 mg/d); use smallest effective dose.

Drug info: Gabapentin (Neurontin®)

An antiseizure medication, may act synergistically with carbamazepine and baclofen. **Side effects:** include ataxia, sedation and rash.

R start with 100 mg PO BID, titrate to 5–7 mg/kg/day (3600 mg/d max).

112.2.10 Miscellaneous drugs

Also possibly effective:

1. phenytoin (Dilantin®): may be useful IV in patients in too much pain to open their mouths to take carbamazepine orally
2. capsaicin (Zostrix®): 1 gm applied TID for several days resulted in remission of symptoms in 10 of 12 patients (4 relapsed in < 4 mos, but remained pain-free for 1 yr after 2nd course)¹²
3. clonazepam (Klonopin®) (p. 495): works in 25%
4. lamotrigine (Lamictal®)
5. amitriptyline (Elavil®): more commonly used for atypical facial pain
6. botulinum toxin (Botox®): reduces transmission of CGRP, producing a direct effect on the sensory nerve fibers
7. intranasal lidocaine spray (2 metered sprays of 0.1 ml of 8%, or 4 sprays of 4%) into the nostril on the affected side with the patient positioned supine with HOB 30–45° (and remaining in that position for 30 seconds) may provide temporary relief (approximately 4 hours) for V2 TGN by blocking V2 as it passes through the sphenopalatine ganglion (SPG) posterior to the middle turbinate¹³

112.2.11 Surgical therapy for trigeminal neuralgia

Indications for surgery

Reserved for cases refractory to medical management, or when side effects of medications exceed risks and drawbacks of surgery.

Surgical options

1. peripheral trigeminal nerve branch procedures to block or ablate the division involved with *pain*, or can be used to block the trigger¹⁴:
 - a) means of blocking
 - local blocks (phenol, alcohol)
 - neurectomy of trigeminal branch involved
 - b) nerve branches:
 - V1 (ophthalmic division) at the supraorbital, supratrochlear, or infraorbital nerves
 - V2 (maxillary division) at the foramen rotundum
 - V3 (mandibular division) block at the foramen ovale, or neurectomy of inferior dental nerve
2. blocking the *trigger*: either via percutaneous rhizotomy or alcohol block
3. percutaneous trigeminal rhizotomy (PTR): AKA percutaneous (stereotactic) rhizotomy (PSR) of trigeminal (Gasserian) ganglion (see below) (not truly a stereotactic procedure in the current sense of the word, therefore the term percutaneous trigeminal rhizotomy is preferred). Objective is to selectively destroy A-delta and C fibers (nociceptive) while preserving A-alpha and beta fibers (touch). Ideally, a retrogasserian lesion (not a ganglionic lesion). May also be used to block trigger. Lesioning techniques include (see below for comparison of techniques):
 - a) *radiofrequency* rhizotomy (RFR) (originated by Sweet and Wespis¹⁵). Uses radiofrequency energy to thermocoagulate the pain fibers. Requires the patient to be awake at intervals during the procedure
 - b) *glycerol* injection into Meckel's cave^{16,17}: possibly lower incidence of sensory loss and anesthesia dolorosa than with radiofrequency lesion.¹⁸ Water-soluble contrast cisternography was recommended in original description, may not be essential¹⁹
 - c) *mechanotrauma* (percutaneous microcompression (PMC) rhizolysis): via inflation of No. 4 Fogarty catheter balloon.^{20,21,22} Does not require the patient to be awake
 - d) injection of sterile boiling water

4. Spiller-Frazier subtemporal extradural approach with retrogasserian rhizotomy (rarely used today). Original Spiller-Frazier technique involved avulsion of the nerve, which had an unacceptably high risk of bleeding. Subsequently, the same exposure was used with partial sectioning of the nerve or just mildly traumatizing the ganglion
5. intradural retrogasserian trigeminal nerve section (sensory portion $\pm$ motor root, see below): may be performed during MVD if no vascular compression is identified
6. cutting descending trigeminal tract in lower medulla (99.5% success): rarely used
7. microvascular decompression (MVD)²³: (see below) microsurgical exploration of root entry zone, usually via posterior fossa craniectomy, and displacement of vessel impinging on nerve (if such a vessel is found). Usually with the placement of a non-absorbable "insulator" (Ivalon® sponge or shredded Teflon felt; see relative merits of Ivalon® vs. Teflon felt (p. 1872)
8. complete section of the nerve proximal to the ganglion via a p-fossa crani
9. stereotactic radiosurgery (p. 1864)
10. motor cortex stimulation²⁴: (somewhat analogous to spinal cord stimulation for spinal or extremity pain). Better for neuropathic trigeminal pain (as distinct from trigeminal neuralgia)

Selection of surgical option

Some pearls that influence treatment option choices (expert opinion²⁵):

1. V3 neuralgia: RF. Can selectively treat V3 without involving other divisions
2. V1 or V2: balloon compression. Causes numbness in all 3 divisions, but unlike the lesion with RF the corneal numbness is better tolerated and the corneal reflex is often preserved
3. bilateral pain: glycerol. It has the shortest duration of effect, which is an advantage if you think you may need to treat the other side at some point
4. SRS: due to latency until pain relief, suboptimal for patients who need immediate pain relief

Peripheral nerve ablation and neurectomies

Limited to pain or trigger points in territory of supraorbital/supratrochlear, infraorbital, or inferior dental nerves. Neurectomy may be a consideration especially for elderly patients who are not candidates for MVD (neurectomy may be done under local anesthesia) with pain in the forehead (to avoid anesthesia of the eye, as could occur with RFR). Disadvantages include sensory loss in the distribution of the nerve and a high rate of pain recurrence due to nerve regeneration (usually in 18–36 months) which often responds to repeat neurectomy.²⁶ May also be used following PTR.

Supraorbital and supratrochlear nerve: See also information on supraorbital neuralgia (SON) or supratrochlear neuralgia (STN) (p.521). SON may be treated with rhizotomy (e.g., with alcohol or radiofrequency) or with neurectomy. Alcohol injection is used with caution for STN because of risk of injury to the superior oblique muscle. For neurectomy, these nerves are exposed through a 2 cm incision parallel to and just above the medial portion of the eyebrow (never through the eyebrow as this can create an unsightly "bi-brow"; shaving the eyebrow is also discouraged since it occasionally does not grow back). The incision is carried down to the bone and the periosteum is elevated caudally towards the supraorbital foramen or notch. The nerves will be visible on the undersurface of the periosteal flap. The supraorbital nerve is freed in its foramen/notch, and is then avulsed by grasping it with a mosquito hemostat and twisting the clamp. The nerve avulses "like pulling a worm out of a hole." The distal portion of the nerve should be located at the site where the periosteum was incised and it, too, should be avulsed. The process can be repeated for the more medially situated supratrochlear nerve.

Other nerves: Not covered here, other nerve branches that may be cut or avulsed include: infra-trochlear, lacrimal (the branch of V1 at the lateral edge of the orbit), infraorbital nerve, inferior alveolar, lingual and mental nerves.²⁷ (p 290)

Percutaneous trigeminal rhizotomy (PTR)

Recommended for patients who: are poor risk for general anesthesia (elderly or those with increased risk for general anesthesia), wish to avoid "major" surgery, have unresectable intracranial tumors, have MS, have impaired hearing on the other side, or have limited life expectancy (<5 yrs).¹⁸ For "atypical facial pain," denervating the painful region of the face benefits <20% of patients, and worsens 20%.²⁸ Recurrences are easily treated by repeat procedures. May be used to treat failures of peripheral nerve ablation.

► **Choice of lesion technique.** Recurrence rates and incidence of dysesthesias are comparable among the various lesioning techniques. Incidence of intraoperative hypertension is less with

percutaneous microcompression (PMC) than with radiofrequency rhizotomy (RFR) lesion²² (no reports of intracerebral hemorrhage).

RFR requires a patient who is able to cooperate; PMC can be done with the patient asleep. Paralysis of ipsilateral trigeminal motor root (e.g., pterygoids) is more common after PMC (usually temporary) than RFR, and so PMC should not be done if there is already contralateral paralysis from a previous procedure. See also description of technique (p. 1865).

Bradycardia may occur with any of the percutaneous rhizotomy techniques. It is usually not harmful, however, preparation for possible symptomatic bradycardia should routinely include:

1. place defibrillator leads on chest prior to beginning procedure
2. have atropine available (some surgeons administer atropine prophylactically²⁹)
3. have crash cart available

Complications with percutaneous radiofrequency (note: some “numbness” is actually expected in most successful PTRs and occurs in 98% of cases,⁶ and is not considered a complication here):

1. mortality: only 17 deaths in over 22,000 procedures (includes less experienced neurosurgeons and patients often considered poor surgical risks)³
2. dysesthesias⁶ (sometimes called “annoying paresthesias”): higher rate in more complete lesions
 - a) minor: 9%
 - b) major (requiring medical treatment): 2%
 - c) anesthesia dolorosa (severe, constant, burning aching pain that is refractory to all treatment): 0.2–4%
3. meningitis⁵: 0.3%
4. alterations in salivation³⁰: 20% (increased in 17%, decreased in 3%)
5. partial masseter weakness (usually not perceived by patient, ► Table 112.3)
6. oculomotor paresis (usually temporary, ► Table 112.3)
7. reduced hearing (secondary to paresis of tensor tympani, ► Table 112.3)
8. **neuroparalytic keratitis** (keratitis due to fifth nerve deficit which impairs sensation, ► Table 112.3)

Table 112.3 Complications with percutaneous radiofrequency

Complication	850 cases ⁴⁰		315 cases ³⁰
	straight electrode (N = 700)	curved electrode (N = 150)	
partial masseter weakness (usually not perceived by patient)	15–24%	7%	50%
oculomotor paresis (usually temporary)	2%	0	
reduced hearing (secondary to paresis of tensor tympani)	0	0	27%
neuroparalytic keratitis (keratitis due to fifth nerve deficit which impairs sensation)	4%	2%	0

9. intracranial hemorrhage: personal report of 7 cases (6 fatal) in > 14,000 procedures, probably due to transient HTN (SBP up to 300 torr)
10. alterations in lacrimation³⁰: 20% (increased in 17%, decreased in 3%)
11. herpes simplex eruption: prescribe antiherpetic drug if patient develops symptoms, e.g., Acyclovir® (p. 399).
12. bradycardia and hypotension: 1% with RFR compared to up to 15% with glycerol injection
13. rare^{31,32}:
 - a) temporal lobe abscess
 - b) aseptic meningitis
 - c) intracerebral abscess: 0.1%
 - d) trigeminal trophic syndrome (TTS)³³: triad of unilateral crescentic nasal alar ulceration with anesthesia and paresthesia of the trigeminal dermatome (may present with severe pruritus and self-induced skin lesions from scratching). A result of trigeminal nerve injury. Treatment has included: carbamazepine, diazepam, amitriptyline, chlorpromazine, clonazepam or pimizide³⁴
 - e) complications related to needle placement³⁵:
 - carotid-cavernous fistula (CCF): may occur with any percutaneous technique³⁶ (including balloon microcompression³⁷)

- blindness: from penetration of inferior orbital fissure³⁸
- injury to other cranial nerves: II, III, IV, VI³⁹
- f) subarachnoid hemorrhage
- g) seizures

Microvascular decompression (MVD)

See also more detailed information (p.1867).

Recommended for patients with inadequate medical control of pain with >5 years anticipated survival and able to tolerate a small craniotomy¹⁸ (surgical morbidity increases with age). Relief is often long lived, persevering 10 yrs in 70%. Incidence of facial anesthesia is much less than with PTR, and anesthesia dolorosa does not occur. Mortality: <1%. Incidence of aseptic meningitis (AKA hemogenic meningitis): 20%. 1–10% major neurologic morbidity. Failure rate: 20–25%.

1–2% of patients with MS will have a demyelinating plaque at the root entry zone; this usually does not respond to MVD, and one should attempt a PTR.

Stereotactic radiosurgery (SRS)

The first use of SRS by Leksell was for the treatment of TGN. Initially, this was reserved for refractory cases following multiple operations,⁴¹ now becoming more widely practiced. SRS is the least invasive procedure. Generally recommended for patients with co-morbidities, high-risk medical illness, pain refractory to prior surgical procedures, or those on anticoagulants (anticoagulation does not have to be reversed to have SRS).

Treatment plan: 4–5 mm isocenter in the trigeminal nerve root entry zone identified on MRI. Use 70–80 Gy at the center, keeping the 80% isodose curve outside of the brainstem.

Results: Significant pain reduction after initial SRS: 80–96%,^{42,43,44,45} but only ≈ 65% become pain-free. Median latency to pain relief: 3 months (range: 1 d–13 months).⁴⁶ Recurrent pain occurs within three years in 10–25%. Patients with TN and multiple sclerosis are less likely to respond to SRS than those without MS. SRS can be repeated, but only after four months following the original procedure.

Favorable prognosticators: higher radiation doses, previously unoperated patient, absence of atypical pain component, normal pre-treatment sensory function.⁴⁷

Side effects: Hypesthesia occurred in 20% after initial SRS, and in 32% of those requiring repeat treatment⁴⁶ (higher rates associated with higher radiation doses⁴³).

Management of surgical treatment failures

90% of recurrences are in distribution of previously involved divisions; 10% are in new division and may represent progression of the underlying process. Some treatment failures are not persistent TN, but rather represent trigeminal neuropathic pain (AKA trigeminal deafferentation pain).

PTR may be repeated in patients who have a recurrence with some preservation of facial sensation. Attempted repeat PTR is often productive, and failures can be managed as below.

MVD may be performed in patients failing PTR, but the success rate may be reduced⁴⁸ (91% for patients undergoing MVD first, vs. 43% for those having MVD following PTR; note: 91% may be an unrealistically high success rate, and taking patients that fail PTR may select for a more difficult subgroup). Repeat MVD may also be performed, with attention given to possible slippage of the insulating sponge, or the fact that the true offending vessel may be “artificially” moved away from the nerve secondary to the surgical positioning.

SRS can be repeated, using the same dose, with reported significant reduction in pain in 89%, and complete relief in 58%.⁴⁶

Intradural retrogasserian trigeminal nerve section

May be used as a measure of last resort in patients who have recurrent TGN following one or more PTRs in the presence of total facial anesthesia, or in patients undergoing posterior-fossa craniectomy for the purpose of MVD when no impinging vessel can be identified. In the latter case, a partial rhizotomy is performed by sectioning 2/3 of the nerve, with resulting partial anesthesia. In the case of patients with facial anesthesia pre-op, consideration should be given to sectioning the motor division (portio minor) as an alternate pain pathway.¹⁰

112.2.12 Percutaneous trigeminal rhizotomy (PTR)

Due to concerns about hemorrhage, check coagulation profile (PT/PTT, consider bleeding time), and discontinue ASA and NSAIDs, preferably 10 days pre-op. Procedure may be performed in OR with fluoro, or in angiography suite in X-ray department.

Booking the case: Percutaneous trigeminal rhizotomy

(For any of the percutaneous methods: balloon, glycerol, RFR)

Also see defaults & disclaimers (p.25) and pre-op orders (see below).

1. position: supine
2. anesthesia: MAC with sedation
3. equipment:
 - a) lesion generator and needle kit for radiofrequency rhizotomy
 - b) C-arm fluoroscopy (2 C-arms for balloon compression)
 - c) calibrated inflatable balloons (as in kyphoplasty) for balloon rhizotomy
4. consent (in lay terms for the patient—not all-inclusive):
 - a) procedure: put a needle into the cheek to numb the nerve to the face
 - b) alternatives: medical treatment, surgery through the back of the skull (microvascular decompression), radiation (stereotactic radiosurgery)
 - c) complications: facial numbness is anticipated, rarely: stroke, bleeding, blindness

Pre-op orders (RFR)

1. NPO after MN except meds
2. continue Tegretol® & other meds PO with sips of water
3. AM of procedure: IV NS @ KVO in arm contralateral to neuralgia
4. atropine 0.4 mg IM PRN (✗ contraindications include rapid a-fib)
5. non-disposable LP tray to accompany patient

Technique: Percutaneous trigeminal radiofrequency rhizotomy (RFR)

Adapted technique.⁴⁹

NB: needle insertion and/or lesioning may cause HTN, consider monitoring BP. Use either a straight electrode (bare 5 mm for 1 division, 7.5 mm for 2 divisions, or 10 mm for total lesions) or a curved electrode.⁴⁰

Electrode insertion

1. attach ground electrode to patient's upper arm
2. prep the cheek on the involved side with Betadine
3. entry point: under short-acting anesthetic agent—e.g., propofol (Diprivan®) (p.140) or methohexital (Brevital®) (p.139)—insert electrode-needle 2.5–3 cm lateral to oral commissure
4. trajectory:
 - a) palpate the buccal mucosa with a gloved finger inside the mouth (lateral to the teeth) and with the other hand pass the electrode medial to the coronoid process of the mandible (keeping the needle deep to the oral mucosa, i.e., outside the oral cavity), initially aiming towards the plane intersecting a point 3 cm anterior to EAM and the medial aspect of the pupil when the eye is directed forward. Be careful not to contaminate the field with the hand that was in the patient's mouth
 - b) as insertion progresses, use fluoroscopy to direct the tip towards the intersection of the top of the petrous bone with the clivus (5–10 mm below floor of sella along clivus)
 - c) upon entering foramen ovale the masseter often contracts, causing the jaw to briefly close. Remove the stylet, look for CSF to verify location (may not occur in re-do cases), and insert electrode through needle

In difficult cases, intraoperative fluoroscopy may assist in localizing the needle to Meckel's cave and to R/O e.g., entry into superior orbital fissure (which can cause blindness after lesioning), or entry into foramen spinosum (middle meningeal artery). If necessary to visualize (e.g., when there is difficulty entering), the foramen ovale is optimally seen on a submental X-ray by hyperextending neck 20° and rotating head 15–20° away from side of pain.⁵⁰

Impedance measurements: from the tip of the electrode, when available, may help indicate location of needle tip. Impedance: CSF (or any fluid) low ($\approx 40\text{--}120\ \Omega$); connective tissue, muscle, or nerve is usually $200\text{--}300\ \Omega$ (may be up to $400\ \Omega$); if $>400\ \Omega$ this likely indicates electrode is contacting periosteum or bone. After starting the lesion, impedance often goes down by $30\ \Omega$ transiently, and then as the lesioning continues it gradually returns to baseline or $\approx 20\ \Omega$ above it. If char develops on the electrode tip, the impedance will read higher than where it started.

Stimulation and repositioning

Once the foramen ovale is entered, the needle is positioned with the following guidelines: for V3 division lesion the curved electrode should be just short of the clivus and pointing down, for V2 it is at the clivus and directed up, for V1 it is 5 mm beyond clivus and pointing up. ✖ At no time should the needle tip extend > 8 mm beyond clival line (to avoid Cr. N. III or VI complications).

The patient is allowed to wake up and is stimulated through the electrode with the following settings: frequency = 50–75 Hz, 1 mS duration, start at 0.1 V amplitude and slowly increase (usually 0.2–0.5 V is adequate, higher voltages may indicate that the needle is not near the target and that stimulation is due to far-field currents; however, in previously lesioned patients up to 4 V may sometimes be necessary). If stimulation does not reproduce pain in the distribution of the patient's TGN, then the amplitude is returned to 0, the electrode is repositioned (straight electrode: advance needle < 5 mm at a time, until the tip is in the vicinity of the clival line; curved tip electrode: advance and/or rotate); then slowly elevate the voltage again from 0 and repeat the repositioning-stimulating process until stimulation reproduces the distribution of tic pain. If previous lesions have produced analgesia and the patient cannot feel the stimulating current, one may stimulate at 2 Hz and watch for masseter twitch (requires preserved motor root).

Lesioning

When stimulation produces pain in the involved distribution of the TGN, perform the first lesion under short-acting anesthesia at 60–70 °C × 90 sec. A facial flush may be noted.⁵⁰ After every lesion, perform a post-lesion assessment (see below). The goal is analgesia (but not anesthesia) in the areas of tic pain and hypalgesia in areas of trigger points. An average of three lesions are necessary at the first sitting, each ≈ 5 °C higher than the previous for 90 seconds. Anesthetic may not be needed after the first lesion if moderate analgesia has been produced by previous lesions.

Post-lesion assessment

After each lesion and at completion of procedure, assess:

1. sensitivity to pinprick and light touch in all three divisions of trigeminal nerve (grading: normal, hypalgesic, analgesic, anesthetic)
2. corneal reflex bilaterally
3. EOM function
4. masseter muscle strength (patient clenches teeth, palpate cheeks for contraction)
5. pterygoid muscle strength (ask patient to open mouth, chin deviates towards side of pterygoid weakness)

Post-op care (PTR)

Include in post-op orders:

1. ice pack to face on side of procedure for 4 hrs
2. soft diet
3. routine activity when alert
4. avoid narcotics (usually not necessary)
5. if corneal reflex impaired: risk of neuroparalytic keratitis. Natural tears 2 gtt q 2 hrs while awake to eye on affected side. Lacrilube® to eye & tape eye shut q hs

Prior to discharge from hospital, repeat post-lesion assessment (see above). Patients are then weaned off of carbamazepine as tolerated.

112.2.13 Percutaneous microcompression rhizolysis balloon (PMC)

Via inflation of No. 4 Fogarty catheter balloon.

Technique:

1. the needle is placed as with RFR (p. 1865).
2. aim for balloon placement in the medial foramen ovale (to avoid entering the middle fossa). After placing the balloon, insert the stylet to visualize where the balloon will go. Use Omnipaque 240 to fill the balloon
3. inflate to 1.4 atmospheres of pressure

112.2.14 Results

Results of various PTR techniques compared to microvascular decompression (MVD) are shown in

► Table 112.4. Recurrence rate is higher in patients with multiple sclerosis (50% at 3 yrs mean F/U).⁵¹

Table 112.4 Comparison of outcomes of percutaneous techniques to MVD

Parameter	Percutaneous techniques (PTR)			MVD
	RFR ^a	Glycerol	Balloon	
initial success rate ^{6,10}	91–99%	91%	93%	85–98%
medium-term recurrence rate	19% at 6 yrs ⁵	54% at 4 yrs	21% at 2 yrs	15% in 5 yrs
long-term recurrence rate	80% at 12 yrs ^{30b}			30% at 10 yrs
facial numbness ⁶	98%	60%	72%	2%

^aabbreviations: RFR = radiofrequency rhizotomy; MVD = microvascular decompression; balloon = balloon microcompression
^bthis author included initial failures to PTR requiring repeat procedures during same hospitalization

112.2.15 Microvascular decompression (MVD) for trigeminal neuralgia

Indications

1. patients unable to achieve adequate medical control of trigeminal neuralgia with ≥ 5 yrs anticipated survival, without significant medical or surgical risk factors¹⁸ (although a small p-fossa exploration is usually well-tolerated, surgical morbidity increases with age)
2. may be used in patients who do not fit the above criteria, but have intractable pain and fail PTR
3. patient with tic involving V1 for whom the risk of exposure keratitis due to corneal anesthesia would be unacceptable (e.g., already blind in contralateral eye) or patient wishing to avoid facial anesthesia for any reason
4. ✕ patients with MS are usually not considered candidates for MVD due to low response rate

Booking the case: Microvascular decompression

Also see defaults & disclaimers (p. 25) and pre-op preparation (see below).

1. position: park bench
2. equipment: microscope
3. implants: Ivalon sponge or shredded Teflon
4. intra-op monitoring: (optional) BAER, facial EMG (monitors VII and portio minor (motor) of VII), VIII (CNAP (compound nerve action potential) using Cueva electrode placed directly on VIII nerve referenced to ipsilateral earlobe)
5. consent (in lay terms for the patient—not all-inclusive):
 - a) procedure: surgery behind the ear to move a blood vessel from the sensory nerve of the face, if no offending vessel can be identified then possible partial sectioning of the appropriate part of the trigeminal nerve with associated numbness)
 - b) alternatives: needle procedures through the cheek (percutaneous rhizotomy), radiation (stereotactic radiosurgery)
 - c) complications: (in addition to usual craniotomy complications (p. 25)), CSF leak, hearing loss ($\approx 10\%$), facial numbness, pain near incision (occipital neuralgia or lesser occipital neuralgia), rarely: diplopia, facial paralysis, failure of the procedure

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Technique

Also see Paramedian suboccipital craniectomy (p. 1740) for important pointers, including use of armored endotracheal tube.

Preoperative preparation

An MRI is recommended (with FIESTA sequence or equivalent, if available) to rule out mass lesion or vascular abnormality. Baseline BAER are performed by some⁵² (see below for intra-op monitoring).

O.R. setup

Setup for lateral oblique suboccipital (posterior-fossa) craniotomy (p. 1738). Microscope: observer's eyepiece is placed on the side opposite to that of the tic.

Positioning

See reference.⁵³

1. lateral oblique position (p. 1738), symptomatic side *up*, axillary roll
2. thorax elevated 10–15° to reduce venous pressure
3. 3-pin skull fixation. Head position:
 - a) head rotation: head rotated 10–15° away from the affected side. Do not exceed 30° rotation
 - b) lateral head tilt
 - for trigeminal neuralgia or VIII nerve approach: the head is parallel to the floor (if it is lower, nerves VII & VIII will obscure view of V)
 - for VII nerve or lower, the vertex is tilted 15° *down* from the horizontal
 - c) flex neck: leave 2 fingerbreadths room between the chin and the sternum
4. upper shoulder retracted caudally with adhesive tape
5. option: lumbar spinal drain. Drain 20–30 cc during craniotomy, then drain off small amounts from time-to-time during the case to keep the field mostly dry, but occasionally letting CSF build up to bathe cranial nerves

Intraoperative monitoring

Option: intraoperative monitoring of facial EMG and BAER (assesses acoustic nerve).⁵²

Approach

1. skin incision⁵³: vertical incision 3–5 cm in length, 5 mm medial to mastoid notch, a small “5–6–4” incision (p. 1740); in thick or short-necked patients, a slightly longer incision that angles infero-medially is used. 75% of the incision is inferior to the transverse sinus, 25% superior
2. burr hole:
 - a) 1 cm inferior and 1 cm medial to the asterion⁵⁴ (p. 60)
 - b) if the asterion is not easily identified or if there are concerns about the reliability of the asterion as a landmark for the junction of the transverse and sigmoid sinuses,⁵⁵ place the burr hole directly over the mastoid emissary vein which drains superolaterally into the sigmoid sinus
3. craniotomy: top of bone opening as close as possible to transverse sinus. The position of the transverse sinus can be approximated by a line drawn from the posterior base of the zygomatic process to theinion, or roughly ≈ 2 finger-breadths above the upper end of the mastoid notch. Lateral limit of bone opening is sigmoid sinus. A triangular bony opening with a leg along each sinus works well. Craniectomy diameter needs to be only ≈ 3 cm. Apply bone wax liberally (blocks off any possible opening into the mastoid air cells)
4. dural opening: either a curvilinear with each end at a sinus and the convexity *away* from the junction (Jannetta) or an inverted “T” (with one incision towards each sinus and the third towards junction of sinuses, lets you get as close as possible to the sinuses)
5. minimal or no retraction of cerebellum is usually required
6. allow CSF to drain before proceeding: this may require gentle advancement of a cottonoid in the CPA. A lumbar drain should be placed if CSF cannot be drained
7. follow the junction of tentorium with temporal bone deep. Place a retractor that both medially displaces the cerebellum *and* slightly “lifts” the cerebellum towards the surgeon (medial displacement alone is not as effective)
8. superior petrosal vein (SPV): drains to the superior petrosal sinus within the tentorial dura and often blocks access to Cr. N V (sometimes there is a venous complex of 2–3 veins). Coagulating and dividing the SPV is controversial,^{56,57,58} with risks including cerebellar infarction, midbrain and pontine infarction (0–5%) and thus should be avoided if possible.^{57,58} If the vein is torn, the dural side is tamponaded (sometimes up to 30 minutes is needed) while the free end is coagulated
9. V is deeper than the VII/VIII complex, which should not even be seen with this approach. If VII/VIII are seen, move the retractor superiorly as even gentle traction may cause hearing loss (► Fig. 1.11). In some cases the suprameatal tubercle (a small hillock of bone just posterior to Meckel’s cave) may be large enough to obscure the site where the fifth nerve enters the cave and may need to be reduced with a diamond burr

Decompression of nerve

1. arachnoid overlying the fifth nerve is sharply divided (caution re Cr. N. IV, which follows the tentorial opening in the arachnoid rostral to the fifth nerve). Intra-op changes in BAER are often attributed to retraction of arachnoid that is tethered to the VII/VIII complex
2. the fifth nerve may be markedly atrophic if previous PTRs have been done

3. identify the smaller motor root (portio minor) of the fifth nerve
4. arteries and/or veins compressing V should be dissected off the nerve. NB: vessels located proximally are the most likely offenders; however, the dorsal root entry zone (which is the sensitive part of the nerve) may be variable in location and peripheral vessels may be culpable. The nerve should be inspected and freed of vessels from its origin at the brainstem all the way to its entrance into Meckel's cave.⁵³ Veins may be coagulated and then should be divided (to prevent recanalization)
5. the most common cause of compression is the superior cerebellar artery (SCA)
6. check the nerve at the junction with brainstem for any residual compression prior to the next step
7. insulating material is interposed between nerve and vessel to prevent re-compression. Options include:
 - a) e.g., Ivalon® (polyvinyl formyl alcohol) sponge (Ivalon Surgical Products, 1040 OCL Parkway, Eudora, KS, 66025, U.S.A. distributed by Fabco in the U.S.A. (860) 536-8499, toll free: (888) 813-8214, <http://fabco.net/catalog/ivalon-ophthalmic/>) cut in a saddle shape. Note: if an Ivalon block is used instead of pre-packaged sterile pads, it must be rinsed thoroughly to remove formalin, then autoclaved. Ivalon should be hydrated in NS for 10 minutes prior to cutting it
 - b) shredded Teflon felt; see merits of Ivalon® vs. Teflon or muscle (p. 1872)
8. Wilson recommends performing a partial sensory rhizotomy of the inferior one-half to two-thirds of the portio major for the following: cases where no vascular contact with the nerve or no deformity of the nerve is identified, in most cases of patients undergoing a repeat MVD, or for cases with duration of symptoms > 8–9 yrs, as this latter group tends to have a lower success rate with MVD alone⁵⁹
9. if the procedure is for a failed MVD and it is desired to partially divide the nerve, the nerve is organized somatotopically with V1 fibers superiorly, and V3 inferiorly. If the goal is total elimination of pain pathways and there is concern about pain conduction through ancillary pathways, consider also dividing the motor root (portio minor)

Closure

1. bone wax should be applied liberally to the exposed lateral bone edges (to paraphrase Dr. Jan-netta⁵³ and Mr. Miyagi,⁶⁰ "Wax in, wax out.")
2. irrigate gently with warm saline (avoid "jet" irrigation which can damage the VIII nerve)
3. intra-op BAER decline may occur on dural closure and should prompt re-opening of the dura and checking for tension on the VIII nerve from a vessel or Telfa
4. perform several Valsalva maneuvers to ensure watertight closure of dura
5. the bone defect should be covered e.g., with burr hole cover to reduce chance of pain associated with uncovered craniectomy
6. after fascial closure, Valsalva maneuver is performed again to ensure watertight closure
7. use 4–0 running locked nylon to approximate skin in watertight fashion (avoid excessive tension)

Post-op care following MVD

Include in post-op orders

1. admit to ICU
2. arterial-line for continuous BP monitoring
3. analgesics (e.g., codeine 30–60 mg IM q 3 hrs)
4. anti-emetics (e.g., ondansetron 4 mg IV q 6 hrs)
5. medication to aggressively treat HTN (viz. SBP > 160 mm Hg)

Post-op H/A, nausea, and pain

Patients routinely have H/A and nausea for 2–3 days (there tends to be less intracranial air and less "pneumoencephalogram sickness" if the park-bench position is used instead of the sitting position). However, severe H/A should prompt a STAT CT to R/O bleeding. If the CT is negative, severe H/A may be due to transient elevation of CSF pressure that occurs in some, and which usually responds to 1, or at most 2, LPs to halve the pressure. Aseptic meningitis usually responds to steroids. Some patients have continued but lessened tic douloureux pain for several days post-op; this usually subsides.⁵³

Complications

The short list:

1. cerebellar injury
2. hearing loss
3. CSF leak

The long list:

1. mortality: 0.22–2% in experienced hands (> 900 procedures)^{61,62}
2. meningitis
 - a) aseptic meningitis (AKA hemogenic meningitis): H/A, meningismus, mild fever, culture-negative CSF, pleocytosis. Incidence: $\approx$ 2% (up to 20% has been reported). Usually occurs 3–7 days post-op. Responds to LP + steroids
 - b) bacterial meningitis: 0.9%
3. major neurologic morbidity: 1–10% (higher rates with less experienced surgeons), including:
 - a) deafness: 1%
 - b) vestibular nerve dysfunction
 - c) facial nerve dysfunction
4. mild facial sensory loss: 25%
5. cranial nerve palsies⁶³:
 - a) fourth nerve (diplopia): 4.3% (only $\approx$ 0.1% are permanent)
 - b) facial nerve: 1.6% (most are transient)
 - c) eighth nerve (hearing loss): 3%
6. postoperative hemorrhage⁶⁴: subdural, intracerebral (1%⁶), subarachnoid
7. seizures: including status epilepticus⁶⁴
8. infarction⁶⁴: including posterior cerebral artery distribution, brainstem
9. CSF leak: resolves with lumbar drainage in most cases
10. pneumonia: 0.6%

Outcome

1. success rate: 75–80% (rates may be lower in patients having prior destructive procedure); good but not total relief in an additional $\approx$ 10%
2. recurrence rate in large series is difficult to ascertain from literature; in a series of 40 patients followed 8.5 yrs mean⁶²:
 - a) major recurrence (recurrent tic not controlled by medications) rate: 31%
 - b) minor recurrence (mild or controlled by medications) rate: 17%
 - c) using Kaplan-Meier curve, expect 70% to be either pain-free or have minor recurrence by 8.5 years (or $\approx$ 80% at 5 years)
 - the risk for a *major* recurrence after MVD is 3.5% annually
 - the risk for a *minor* recurrence after MVD is 1.5% annually
 - d) major recurrence rate is lower for patients having major arterial cross-compression of the nerve discovered at the time of surgery (patients with venous compression had much higher rate)
 - e) this study found no correlation between previous destructive surgery and major recurrence rate (in 11 patients)

Some feel that the longer one waits before performing an MVD, the lower the success rate.

112.3 Hemifacial spasm

112.3.1 General information

Key concepts

- intermittent unilateral painless contractions of facial muscles
- typically caused by compression of VII nerve by AICA
- along with palatal myoclonus: the only movement disorder that persists in sleep
- responds well to microvascular decompression, but risk of hearing loss is $\approx$ 20%

Hemifacial spasm (HFS) is a condition of intermittent, painless, involuntary, spasmodic contractions of muscles innervated by the facial nerve in one side of the face only. May be limited to the upper or lower half of the face only, and excess lacrimation may be present. HFS usually begins with rare contractions of the orbicularis oculi, and slowly progresses to involve the entire half of the face and increases in frequency until the ability to see out of the affected eye is impaired.

HFS may be associated with trigeminal neuralgia, geniculate neuralgia; see **Tic convulsif** (p. 1873), or vestibular and/or cochlear⁶⁵ nerve dysfunction.

HFS is more common in women, is seen more often on the left, and usually presents after the teenage years. Auditory function testing reveals abnormal acoustic middle ear reflex in almost half of patients, indicating some degree of VIII compromise.⁶⁵

Meige's syndrome: hemifacial spasm with oral movements.

► **Note.** HFS and palatal myoclonus are the only involuntary movement disorders that persist during sleep.⁶⁶

112.3.2 Etiologies

1. vascular compression syndrome (see below): the most common etiology (much more common than with trigeminal neuralgia)
2. idiopathic
3. tumor compressing the nerve
4. can follow some cases of Bell's palsy
5. conditions that can mimic HFS
 - a) blepharospasm (bilateral spasmodic closure of the orbicularis oculi muscles) which is more common in the elderly, and may be associated with organic brain syndrome. Blepharospasm is notorious for disappearing when the patient presents for medical evaluation (an effect of alerting), but may be elicited by asking the patient to gently close the eyes and then rapidly open them, following which a blepharospasm may occur. HFS usually involves more than the ocular muscles
 - b) facial myokymia: continuous facial spasm which may be a manifestation of an intrinsic brain-stem glioma or of multiple sclerosis. Often associated with other findings

112.3.3 Vascular compression

HFS is usually caused by compression of the facial nerve at the root exit zone (REZ) by a vessel, which is most often an artery (most commonly AICA⁶⁷ (either pre- or postmeatal⁶⁸), but other vascular possibilities include an elongated PICA, SCA, a tortuous VA, the cochlear artery, a dolichoectatic basilar artery, AICA branches...), aneurysm, a vascular malformation, and rarely, veins have been implicated. In typical HFS (onset in the orbicularis oculi, and progressing downward over the face), the vessel impinges on the antero-caudal aspect of the VII/VIII nerve complex; in atypical HFS (beginning in the buccal muscles and progressing upward over the face), the compression is rostral or posterior to VII.⁶⁹

Vessels contacting the REZ of the vestibular nerve may cause vertigo, whereas tinnitus or hearing loss may result from cochlear nerve REZ compression.

Infrequently, benign tumors or a cyst in the cerebellopontine angle, multiple sclerosis, adhesions, or osseous skull deformities will be the cause of HFS.

Evidence indicates that there is not cross (ephaptic) conduction at the compressed REZ, but that the facial motonucleus is involved secondarily as a result of the REZ compression, via a phenomenon similar to kindling.⁷⁰ In addition to the spasm, a 2nd electrophysiological phenomenon associated with HFS is synkinesis, where stimulation of one branch of the facial nerve results in delayed discharges through another branch (average latency: 11 mSec⁷¹).

112.3.4 Evaluation

In typical cases of HFS, the diagnostic work-up is negative.

Most patients should have MRI of the posterior fossa (CT scan is less sensitive here) to R/O tumors or AVMs.

Vertebral angiography is usually not performed if imaging is normal. The neurovascular compression responsible for HFS usually cannot be identified on angiography.

112.3.5 Treatment

Medical management

HFS is generally a surgical condition. Early, mild cases may be managed expectantly. Carbamazepine and phenytoin are generally ineffective, unlike the situation with the causally similar condition of trigeminal neuralgia. Local injection of botulinum toxin (Oculinum®) may be effective in treating HFS and/or blepharospasm.^{72,73} Baclofen has been advocated but is not very effective.

Surgical management

General information

Many ablative procedures are effective for HFS (including sectioning of divisions of the facial nerve); however, this leaves the patient with some degree of facial paresis. The current procedure of choice for HFS is microvascular decompression (MVD), wherein the offending vessel is physically moved off of the nerve and a sponge (e.g., Ivalon®, polyvinyl formyl alcohol foam) is interposed as a cushion. Other cushions may not prove to be as satisfactory (muscle may disappear, and Teflon felt may thin⁷⁴).

Most often, the offending vessel approaches the nerve at a right angle, and causes grooving in the nerve. Compression must occur at the root exit zone; decompression of vessels impinging distal to this area is usually ineffective.

Operative risks: see below.

Postoperatively, there may be episodes of mild HFS, but they usually begin to diminish 2–3 days following MVD. Severe spasm that does not abate suggests failure to achieve adequate decompression, and reoperation should be considered.

Surgical results of MVD depends on the duration of symptoms (shorter duration has better prognosis) as well as on the age of the patient (elderly patients do less well). Complete resolution of HFS occurred in 44 (81%) of 54 patients undergoing MVD; however, 6 of these patients had relapse.⁷⁵ 5 patients (9%) had partial improvement, and 5 (9%) had no relief.

112.3.6 Technique of MVD

Intraoperative brainstem auditory evoked potential (BAER),⁷⁶ or more applicable, direct VIII nerve monitoring,⁷⁷ may help prevent hearing loss during MVD for 7th or 8th nerve dysfunction. Furthermore, monitoring for the disappearance of the (delayed) synkinetic response may aid in determining when adequate decompression has been achieved (generally reserved for teaching institutions).⁷⁰

For a diagram of the normal anatomy of the CPA, see ► Fig. 1.11. The facial nerve should not be manipulated, and one should avoid dissection around the VII and VIII nerves near the IAC.⁷⁸ Vessels must be preserved, especially the cochlear artery and small perforators.

Place gentle medial traction on the cerebellum (< 1 cm is recommended⁷⁸), and incise the arachnoid membrane between the flocculus and the eighth nerve (to avoid tension on nerves that could cause post-op deficit). The IX nerve may be followed medially from the jugular foramen to locate the origin of the VII nerve (the origin of VII is 4 mm cephalad and 2 mm anterior to that of the IX nerve⁷⁹).

112.3.7 Surgical results

Complete resolution of spasm occurs in $\approx 85\text{--}93\%$.^{74,80,81,82,83} Spasm is diminished in 9%, and unchanged in 6%.⁸³ Of 29 patients with complete relief, 25 (86%) had immediate post-op resolution, and the remaining 4 patients took from 3 mos to 3 yrs to attain quiescence.

Recurrence

Return of symptoms after a period of complete resolution of HFS occurs in up to 10% of patients, 86% of recurrences happen within 2 yrs of surgery, and the risk of developing recurrence after 2 yrs of post-op relief is only $\approx 1\%$.⁸³

Surgical complications

1. ipsilateral hearing loss: may occur from traction injury or a vasospasm
 - a) total hearing loss occurs in $\approx 13\%$ (range: 1.6–15%) (2.8% in one series,⁶⁵ 15% in another series⁷⁵)
 - b) partial hearing loss: 6%
2. facial weakness
 - a) transient: 18%
 - b) permanent facial weakness: 6%⁸¹
3. ataxia in 1–6%
4. other complications that are minor or temporary include:
 - a) aseptic meningitis (AKA hemogenic meningitis) in 8.2%
 - b) hoarseness or dysphagia in 14%
 - c) CSF rhinorrhea in 0.3%
 - d) perioral herpes in 3%⁷⁸

112.4 Geniculate neuralgia

112.4.1 General information

Geniculate neuralgia (GeN) AKA Hunt's neuralgia AKA nervus intermedius neuralgia: a very rare neuralgia affecting the nervus intermedius (the somatic sensory branch of the facial nerve primarily innervating mechanoreceptors of the hair follicles on the inner surface of the pinna and deep mechanoreceptors of nasal and buccal cavities and chemoreceptors in the taste buds on the anterior 2/3 of the tongue).

Symptoms: unilateral paroxysmal otalgia (lancinating pain experienced deep within the ear, often described as an "ice pick in the ear") radiating to the auricle, with occasional burning sensations around the ipsilateral eye and cheek, and prosopalgia (pain referred to deep facial structures, including orbit, posterior nasal, and palatal regions). During pain attacks, some patients have: salivation, bitter taste, tinnitus, or vertigo.

GeN occasionally has cutaneous trigger points in the anterior EAC and tragus, and pain may also be triggered by cold, noise, or swallowing.

Work-up includes neuro-otologic evaluation with audiometry and ENG. Some patients may require imaging (MRI or high-resolution CT) and angio (to R/O aneurysm).

112.4.2 Variants

Tic convulsif (AKA convulsive tic): GeN combined with hemifacial spasm, usually due to neurovascular compression of both the sensory and motor roots of the facial nerve,⁸⁴ most often by AICA. First described by Cushing in 1920.

GeN may be associated with herpetic infections of the geniculate ganglion (AKA herpetic ganglionitis, AKA Ramsay Hunt syndrome (RHS)) in which case herpetic lesions appear on pinna, in EAC, and possibly on TM. May include facial palsy, decreased auditory acuity, tinnitus or vertigo. Unlike idiopathic GeN, RHS is more chronic and less paroxysmal, tends to remit with time, and is usually refractory to carbamazepine. Idiopathic GeN tends to be more painful than RHS, and does not remit spontaneously.

112.4.3 Treatment

1. medical therapy
 - a) mild cases may respond to carbamazepine, sometimes in combination with phenytoin
 - b) may respond to valproate (Depakote®) 250 mg PO BID
 - c) topical antibiotics for secondary infections of herpetic lesions
 - d) local anesthetic to EAC
2. surgery: for severe cases where medical treatment fails or is not tolerated
 - a) microvascular decompression together with division of the nervus intermedius (nerve of Wrisberg).⁸⁵ To find the nerve, a micro nerve hook is hooked around the VII nerve, the hook is rotated 90°, and the nerve is fished out from in front of VII. Operating under local anesthesia allows verification by stimulating nerve
 - b) geniculate ganglion section⁸⁶

112.5 Disabling positional vertigo

As described by Jannetta et al.⁸⁷ constant disabling positional vertigo or dysequilibrium, causing ≈ constant nausea. No vestibular dysfunction nor hearing loss (tinnitus may be present). One possible cause is vascular compression of the vestibular nerve which may respond to microvascular decompression

112.6 Glossopharyngeal neuralgia

112.6.1 Epidemiology

Incidence: 1 case for every 70 of trigeminal neuralgia.⁸⁸ (p 3604–5)

112.6.2 Clinical

Severe, lancinating pain in the distribution of the glossopharyngeal and vagus nerves (throat & base of tongue most commonly involved, radiates to ear (otalgia), occasionally to neck), occasionally with salivation and coughing. Rarely: hypotension,⁸⁹ syncope,⁹⁰ cardiac arrest and convulsions may accompany. May be triggered by swallowing, talking, chewing. Trigger zones are rare.

112.6.3 Treatment

Pain may be reduced by application of locan anesthetic to tonsillar pillars and fossa. Usually, the persistence and severity of pain requires surgical intervention. One may either perform microvascular decompression, or nerve division via extra- or intra-cranial approach (latter may be required for permanent relief).

Intracranial approach: Section of preganglionic glossopharyngeal nerve (IX) and upper one-third or two fibers (whichever is larger) of vagus (X). IX is readily identified at its dural exit zone where it is separated from X by a dural septum. The upper third of X is usually composed of a single rootlet, or less commonly, multiple small rootlets. Initial post-op dysphagia usually resolves. Cardiovascular complications following vagal section have been reported, warrants close monitoring × 24 hrs.

112.7 Superior laryngeal neuralgia

A rare disorder characterized by severe pain in the distribution of the superior laryngeal nerve. Patients experience pain along the anterior cervical triangle with extension to the ipsilateral ear and eyes, intermittent hoarseness, and paralysis of the ipsilateral cricothyroid muscle on laryngoscopy.⁹¹ Pain is usually worsened by swallowing, shouting, or by head turning. The etiology of this disorder may vary: can occur after trauma, iatrogenic injury (after surgery in the anterior neck such as carotid endarterectomy), or as a result of vascular compression by PICA or vertebral artery. It can be treated with superior laryngeal nerve blocks, nerve sectioning, or less commonly by microvascular decompression.

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113 Pain Procedures

113.1 Prerequisites for pain procedures

Medical therapy must be maximized before a patient is a candidate for a pain procedure. Usually this requires escalating the dose of oral narcotic pain medications until the point that the pain is relieved or the side effects (usually somnolence or hallucinations) are intolerable (e.g., up to 300–400 mg/day of MS Contin may sometimes be necessary).

113.2 Choice of pain procedure

► Table 113.1 shows some pain procedures that may be used for various indications (the list is not intended to be all-inclusive, but is rather to serve as a starting point for organizing pain procedures). In general, nonablative procedures are exhausted before resorting to ablative procedures.

Table 113.1 Choice of pain procedures^a

Unilateral pain		Bilateral or midline pain	
Head, face, neck, UE	Pain at or below C5 dermatome	Below diaphragm	Above diaphragm
DBS	cordotomy ^b	spinal IT narcotics	intraventricular narcotics
stereotactic mesencephalotomy		↓ commissural myelotomy	

^aabbreviations: IT = intrathecal; UE or LE = upper or lower extremity

^bcordotomy (open or percutaneous) if pain is unresponsive to or too high for spinal IT narcotics

113.3 Types of pain procedures

See pain procedures particular to trigeminal neuralgia (p. 1861).

Techniques for other conditions include:

- electrical stimulation
 - deep brain stimulation (p. 1839)¹: targets include thalamus and periaqueductal or periventricular gray matter
 - spinal cord stimulation (p. 1883)
- direct drug administration into the CNS:
 - different routes: spinal (p. 1881), epidural or intrathecal, intraventricular (p. 1882)
 - different agents: local anesthetics, narcotics (without motor, sensory, or sympathetic impairment seen with local anesthetics) (p. 1881)
- intracranial ablative procedures:
 - cingulotomy: theoretically reduces the unpleasant effect of pain without eliminating the pain. Must be done bilaterally, recently with MRI. Intolerable pain usually recurs after ≈ 3 mos. 10–30% develop flattened affect
 - medial thalamotomy: no longer used (presented for historical reasons). Controversial. Was used for some for nociceptive cancer pain. Performed stereotactically
 - stereotactic mesencephalotomy**²: for unilateral head, neck, face, and/or UE pain. Use MRI to create lesion 5 mm lateral to Sylvian aqueduct at the level of the inferior colliculus. Unlike spinal cordotomy, the lesion is not near any motor tracts. Main complication is diplopia due to interference with vertical eye movement, often transient
- spinal ablative surgical procedures
 - cordotomy: see below
 - open
 - percutaneous
 - cordectomy
 - commissural myelotomy: for bilateral pain (p. 1880)
 - punctate midline myelotomy: for relief of visceral cancer pain
 - dorsal root entry zone lesion (p. 1886)
 - dorsal rhizotomy: not useful for large areas of involvement
 - dorsal root ganglionectomy (an extraspinal procedure)

- h) sacral cordotomy: for patients with pelvic pain who have colostomy and ileostomy. A ligature is tied around the dural sac below S1 nerve roots
5. sympathectomy: possibly for causalgia major; see Sympathectomy (p. 1853) and Complex regional pain syndrome (CRPS) (p. 525)
6. peripheral nerve procedures
 - a) nerve block³:
 - neurolytic: injection of neurodestructive agents (e.g., phenol or absolute alcohol) on or near the target nerve
 - nonneurolytic: using local anesthetics, sometimes in combination with corticosteroids
 - b) neurectomy: (e.g., intercostal neurectomy for pain due to infiltration of chest wall by malignancy). Performed open or percutaneously with radiofrequency lesion. May sacrifice motor function with mixed nerves
 - c) peripheral nerve stimulators: rarely discussed

113.4 Cordotomy

113.4.1 General information

Interruption of the lateral spinothalamic tract fibers in the spinal cord. Cordotomy is the procedure of choice for *unilateral* pain below the C5 dermatomal level ($\approx$ nipple; occasionally pain as high up as the mandible may be treated), in a terminally ill patient. Better for aching pain, poor for central pain, dysesthesias, causalgia (deafferentation pain), midline visceral pain. May be performed as an open procedure, but is more easily performed percutaneously at the C1–2 interspace (which limits the procedure to the cervical region). If there is any contralateral pain, it will tend to be magnified following the procedure and often leads to dissatisfaction with cordotomy. If there is any bladder dysfunction, it will usually be worse following cordotomy. Bilateral cervical cordotomies carry a risk of the loss of automaticity of breathing⁴ (one form of sleep apnea, so-called Ondine's curse⁵). Therefore, if bilateral cordotomies are desired, the second should be staged after normal respiratory function and CO₂ responsiveness are verified following the first procedure, or the second stage may be done as an open procedure in the thoracic region.

Review the cross-sectional spinal cord anatomy for relationships of the critical tracts (spinothalamic and corticospinal) to the dentate ligament, the anterior spinal artery, respiratory ($\gg$ Fig. 1.15), and bladder areas ($\gg$ Fig. 3.2).

113.4.2 Pre-op evaluation

Spirometric measurement of minute volume before and after breathing a mixture of 5% CO₂ and 95% O₂ for 5 minutes. If the MV decreases, these patients are at increased risk of having sleep apnea (usually transient), no increased risk if MV increases or stays the same. Also, patients with <50% of predicted values on PFTs are not candidates.

In patients with pulmonary cancer contralateral to the planned side of cordotomy, check that the contralateral diaphragm is functioning with fluoroscopy; otherwise, if the ipsilateral diaphragm is lost due to cordotomy, the patient may be hypopneic.

113.4.3 Percutaneous cordotomy

General information

Indicated for unilateral pain below $\approx$ C4–5 in a terminally ill patient. Radiofrequency current is used to lesion the lateral spinothalamic tract.

Technique

Patient does not need to be NPO. Usual pain medications should be given. The patient must be awake and cooperative (any movement with the needle in the cord may lacerate the cord); however, one may give e.g., hydroxyzine (Vistaril®) 50 mg IM on call to procedure for relaxation.

The procedure is performed in the X-ray department with either fluoroscopic or CT guidance. For fluoroscopy, the head is placed in a Rosomoff head-holder with the height adjusted to keep the mastoid process in the same horizontal plane as the acromioclavicular joint. Working on the side *contralateral* to the pain, local anesthetic without epinephrine is infiltrated 1 cm caudal to the mastoid tip. An 18 gauge lumbar puncture needle is inserted perfectly horizontal aiming halfway between

the posterior margin of the body of C2 and the anterior portion of the C2 spinous process. Stay rostral to the C2 lamina to avoid the nerve (which is painful).

The dura will be penetrated at about the time that the tip of the needle is approximately even with the midline of the odontoid process on AP fluoro. A few ml of CSF are aspirated and shaken in a syringe together with a few ml of Pantopaque®, and several ml of the mixture are injected into the subarachnoid space under lateral fluoro guidance (**note:** Pantopaque is no longer available, and water-soluble agents are less effective). A needle endoscopic technique may be able to localize the spinal cord anterior to the dentate ligaments. Some dye will layer on the anterior cord, some on the dentate ligament, and most in the posterior thecal space. The dye will only stay momentarily on the dentate ligament, thus be ready to immediately advance the needle just barely anterior to this while monitoring the tip impedance, which will jump from $\approx 300\text{--}500\ \Omega$ (ohms) in the CSF to $\approx 1200\text{--}1500\ \Omega$ as the spinal cord is penetrated.

Stimulation at 100 Hz should produce contralateral tingling at a threshold of ≤ 1 volt. No motor response should be elicited with 100 Hz in the spinothalamic tract, and if muscle tetany occurs, lesioning must *not* be performed. If tingling is in the arm, lesioning will usually render from the arm and below analgesic. If tingling is in the lower extremity it will render only that limb analgesic. Stimulation at 2 Hz should produce ipsilateral twitching of the arm or neck at $\approx 1\text{--}3$ volts.

Radiofrequency lesioning is performed for 30 seconds while the patient sustains contraction of the *ipsilateral* hand and the voltage is gradually increased from zero. Any twitching of the hand is an indication to back down on the voltage. A second lesion is performed in the same region and is usually less painful. The appropriate body area is then checked for analgesia to pinprick.

If the procedure is performed satisfactorily, an ipsilateral Horner syndrome usually occurs.

Complications

For complications, see ► Table 113.2.

Table 113.2 Post-cordotomy complications

Complication	Frequency
ataxia	20%
ipsilateral paresis	5% total 3% permanent
bladder dysfunction	10% total 2% permanent
postcordotomy dyesthesia	8%
sleep-induced apnea	0.3% unilateral cordotomy 3% bilateral cordotomy
death (respiratory failure)	0.3% unilateral cordotomy 1.6% bilateral cordotomy

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Outcome

In experienced hands, 94% will achieve at least significant pain relief at the time of hospital discharge. The level of analgesia falls with time. At 1 year 60% will be pain-free, and at 2 years this will be only 40%.

Post-procedure management

CSF leakage will cease spontaneously. Patient is kept supine for 24 hrs to prevent “spinal” (post-LP) headache. Pain medication appropriate to postoperative management is prescribed. If successful, one can rapidly stop the narcotics for the primary pain; withdrawal syndromes occur only rarely.

113.4.4 Open cervical cordotomy (Schwartz technique)

General information

A relatively quick method for open cervical cordotomy.⁶ Can theoretically be done under local for patients who cannot tolerate general anesthesia.

Technique

Position: prone; face carefully placed on padded horseshoe headrest, neck slightly flexed to open the interlaminar spaces and to lower the head to prevent accumulation of intracranial air.

Skin incision: midline from occiput to C3. Working only on the side *contralateral* to the pain, muscles are stripped off the posterior lip of the foramen magnum, and from the lamina of C1 and C2. A Schwartz or Gelpi retractor is engaged between the occiput and C2. To increase exposure, the inferior half of C1 and superior half of C2 lamina are removed with a punch.

Dural incision: the ligamentum flavum is thin between C1–2, and can usually be opened with the dura in a linear incision from the lamina of C1 to C2 placed in the lateral third of the exposure, taking care to avoid bleeding from epidural veins. An angle is cut in the incision at either end to allow increased dural retraction. Tack-up sutures are placed in the dura, the arachnoid is opened, the dentate ligament is located and is gripped with a hemostat and divided between the hemostat and the dura.

Cordotomy: the dentate ligament is used to slightly rotate the spinal cord. A cordotomy knife (or 11 blade) with bone wax placed at 5 mm, is inserted into the cord in an avascular area just anterior to the dentate ligament, sharp side down. The anterolateral quadrant of the cord is cut with the following caveats:

- do not go posterior to the dentate ligament (to avoid corticospinal tract)
- do not cross the midline of the spinal cord
- do not injure the anterior spinal artery
- for patients with lower extremity pain, be sure to start exactly at the dentate ligament (to avoid missing lumbar and sacral fibers).

113.5 Commissural myelotomy

113.5.1 General information

AKA mediolongitudinal myelotomy. Interrupts pain fibers crossing in the anterior commissure on their way to the lateral spinothalamic tract.

113.5.2 Indications

Bilateral or midline pain, primarily below the thoracic levels (including abdomen, pelvis, perineum, and lower extremities).

113.5.3 Technique

Laminectomy must extend at least 3 levels above the highest dermatome involved in pain. The dura is opened longitudinally and the operating microscope is then used to identify the midline sulcus (this is usually very difficult to see, and is then estimated as being halfway between where the dorsal roots enter the cord). Veins in the midline are sacrificed for the length of the proposed incision. A number 11 scalpel blade is then placed in a hemostat with 6–7 mm of the tip exposed. The blade is inserted in the midline at the upper end of the desired incision and is then passed caudally for the length of the planned incision (usually 3–4 cm).

113.5.4 Outcome

60% of patients have complete pain relief, 28% have partial, and 8% have none.

113.5.5 Complications

Weakness in the lower extremities occurs in $\approx 8\%$ (usually lower motor neuron, presumably due to injury to anterior horn motor neurons). Dysesthesias occur in almost all patients, but persist $>$ a few days in $\approx 16\%$ (these patients also have impaired joint position sense, all of which are presumably due to posterior column injury). Bladder dysfunction is seen in $\approx 12\%$. Sexual dysfunction may also occur. There is a risk of injury to the anterior spinal artery (rare).

113.6 CNS narcotic administration

113.6.1 Intraspinal narcotics

General information

Spinal narcotics may be administered epidurally or intrathecally for pain relief. Satisfactory pain control can usually be achieved for pain below the neck, although for pain above the diaphragm/umbilicus some recommend intraventricular morphine (p.1882).⁷ May also be performed on a “one-time” basis e.g., injection into epidural space following a lumbar laminectomy. Or, it may be given on a short-term continuous basis, via an external epidural or intrathecal catheter. It may also be performed on an intermediate-term basis (<60 days) with the use of a subcutaneous reservoir⁸ or on a long-term basis with an implantable drug infusion pump⁹ (e.g., Infusaid® or Medtronic® pump). Advantages over systemic narcotics include less sedation and/or confusion, less interference with GI motility (constipation), and possibly less N/V. The effectiveness is usually limited to ~ 1 year and is thus not indicated for chronic benign pain. With time, increased doses are required because of the development of tolerance and/or progression of disease¹⁰ with the concomitant development of the usual narcotic side effects.

Agents to use

Must be preservative-free (for either intrathecal or epidural use). This may be prepared by a pharmacist (e.g., add enough preservative-free 0.9% saline to 1 or 3 gm morphine sulfate powder to yield a total of 100 ml produces 10 or 30 mg/ml solution respectively, and then filter this through a 0.22 mcm filter¹¹). Alternatively, commercially available preparations include Duramorph® (available as 0.5 or 1 mg/ml) and Infumorph® (available in 20 ml ampules of 10 or 25 mg/ml), any of which may be diluted to a lower strength with preservative-free diluent (normal saline). Cross-tolerance to systemic narcotics does occur, and spinal narcotics are more effective in patients who have not been on continuous high-dose IV opiates (patients on high-dose IV narcotics need higher initial intraspinal narcotic doses).

► **Side effects.** Include pruritus (often diffuse, and may be experienced most intensely in the nose), respiratory depression (the respiratory depression with spinal narcotics is usually very gradual, and is often easily detected by monitoring respiratory rate q 1 hr and taking action if the rate decreases), urinary retention, and N/V.

Trial injection

Before implanting a permanent delivery system, a test injection should be performed to verify pain relief and tolerance for medication. Administered via a percutaneously inserted epidural or intrathecal catheter connected to an external pump. Doses required for intrathecal catheters are usually ~ 5–10 times lower than those for epidural catheters.

Sample post-injection orders after a one-time injection:

1. use no other narcotics for ~ 24 hrs (with a continuous infusion additional narcotics should be withheld until the effect of the spinal narcotics has been determined)
2. 2 ampules (0.4 mg each) of naloxone (Narcan®) and syringe taped to patient's bed (for the first 24 hrs after a single injection; at all times with continuous infusion)
3. head of bed elevated $\geq 10^\circ$ for 24 hrs
4. record respiratory rate q 1 hr for 24 hrs; if asleep and respiratory rate < 10 breaths/min, awaken patient. If unable to awaken, administer naloxone 0.4 mg IV and notify physician. Repeat naloxone 0.4 mg IV q 2 min PRN
5. **optional:** pulse oximeter for 24 hrs
6. diphenhydramine (Benadryl®) 25 mg IV q 1 hr PRN itching
7. droperidol (Inapsine®) 0.625 mg (which is 0.25 ml of the 2.5 mg/ml standard concentration available) IV q 30–60 mins PRN nausea
8. PRN supplemental pain medication:
 - a) narcotic agonist/antagonist: e.g., nalbuphine (Nubain®) 1–4 mg IV q 3 hrs,
OR
 - b) ketorolac tromethamine (Toradol®) 15 mg IV or IM or 30 mg IM q 6 hrs (use lower dose for weight < 50 kg, age > 65 yrs, or reduced renal function)

Implantable drug delivery pumps

Although satisfactory pain control can be achieved with either epidural or intrathecal narcotics (morphine diffuses easily through the dura to the CSF where it gains access to pain receptors), epidural catheters commonly develop problems with scarring and may become less effective sooner than intrathecal catheters. Pumps should only be implanted if patients have successful pain control with test injection of spinal epidural (5–10 mg) or intrathecal (0.5–2 mg) morphine. A life expectancy of > 3 months is recommended for implantable pumps (if shorter longevity is anticipated, an external pump may be used).

One such series of commonly used implantable drug delivery pumps is manufactured by Infusaid Inc. The only needle that should be used with their devices are special 22 gauge Huber (non-coring) needles. Delivery rates increase with body temperature 10–13% per °C above 37 °C, they decrease by the same amount for every °C below 37 °C, and they also become inaccurate at ≤4 ml of reservoir fluid. These pumps should never be allowed to run until empty, as this may permanently affect accuracy and reliability of drug delivery. In addition to the pump reservoir port, most models have one or more side “bolus” ports that deliver injected fluid directly to the outlet tubing. One should not aspirate when accessing either port.

Medtronic produces a programmable pump.

Surgical insertion

Similar to the insertion of a lumbar-peritoneal shunt (p.1829). The patient is placed in the lateral position, such as on a bean-bag device. The pump is inserted into a subcutaneous pocket, created with a slightly curved 8–10 cm skin incision. The pump may be sutured to the fascia of the abdomen (in obese patients, it may be sutured to the subcutaneous tissue). Excess tubing should be coiled underneath the pump to prevent inadvertent puncture when accessing either reservoir.

The spinal catheter is inserted through a Tuohy needle inserted between lumbar spinous processes either percutaneously or via a small incision 2–3 mm lateral to the spinous processes. Alternatively, it may be inserted directly via a hemilaminectomy. Fluoroscopy may be used intraoperatively to verify rostral placement of the catheter; radiographic visualization of the catheter may be aided by filling it with iodinated contrast, e.g., Omnipaque-300 (p.230). All bends in the tubing should be very gradual to avoid kinking.

Post-op pain management

Although the pump will be infusing when the patient leaves the operating room, unless they have been on intraspinal narcotics up until the time of surgery, it will usually take several days for the drug to reach equilibrium in the CSF before the level of pain control will be adequate. This can be mitigated by a bolus infusion (3–4 mg morphine for epidural catheters, or 0.2–0.4 mg for intrathecal catheters).

Complications

Meningitis and respiratory failure are rare complications. CSF fistula and spinal H/A may occur. Disconnection or dislodgment of catheter tip may result in failure to control pain, but can usually be surgically corrected.

Outcome

Cancer pain is significantly improved in up to 90%. Success rate for neuropathic pain (e.g., postherpetic neuralgia, painful diabetic sensory neuropathy): 25–50%.

113.6.2 Intraventricular narcotics

Indications

May be used for cancer pain (especially head and neck)¹² unresponsive to other methods in patients with a life expectancy < 6 mos.

Technique

An intraventricular catheter is connected to a ventricular access device (p.1831). 0.5–1 mg of intrathecal morphine is injected via the VAD and usually provides ~ 24 hrs of analgesia.

Complications

► **Side effects.** Common ones include dizziness, N/V. The risk of respiratory depression is minimized by using correct dosing. Complications in a series of 52 patients¹²: bacterial colonization of reservoir (4%), dislodged catheter (2%), blocked catheter (6%), postoperative meningitis (2%).

Outcome

Pain is successfully controlled in 70% at 2 mos, but thereafter the effectiveness diminishes as a result of tolerance to the narcotics.

113.7 Spinal cord stimulation (SCS)

113.7.1 General information

Spinal cord stimulation (SCS) is an invasive pain management technique that is useful in treatment of selected cases of chronic limb or axial pain. Has superseded the technique known as dorsal column stimulation (DCS). Pain relief with SCS can be achieved with ventral stimulation and is not reliant on stimulation-induced paresthesias as with DCS. Pain relief can persist even after the cessation of stimulations, and is not reversed by naloxone. The exact mechanism of action is undetermined.

Consultation

If seeing a consultation for a new SCS, for best results:

1. failure of multidisciplinary approach to adequately control pain
2. screen for appropriate indications (e.g., neuropathic pain, FBSS etc.)
3. chronic neuropathic limb pain without high-yield structural pathology that can be remedied with definitive surgery
4. permanent implant only if patient undergoes a successful trial

If a patient already has an SCS system and they have experienced a change in stimulation:

1. obtain radiographs to see if the lead has changed position or fractured
2. interrogate the generator and/or leads for functionality
3. if system functionality confirmed, re-program
4. if re-programming unsuccessful, consider revision

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113.7.2 Indications/contraindications

The most common SCS application is for relief of neuropathic leg pain.

FDA-approved indications include:

1. failed back surgery syndrome (p. 1884) (FBSS)¹³: especially if LE pain > back pain
2. complex regional pain syndrome (p. 1885) (CRPS), formerly “reflex sympathetic dystrophy”

Possible indications include:

1. diabetic neuropathy (p. 1886)
2. refractory angina pectoris (p. 1886)
3. postthoracotomy pain (intercostal neuralgia)
4. postherpetic neuralgia
5. painful limb ischemia from inoperable peripheral vascular disease (p. 1885)
6. functional: spastic hemiparesis, dystonia, bladder dysfunction

Relative contraindications include:

1. cancer pain
2. patients with limited life expectancy
3. patients with untreated depression
4. primarily nociceptive pain (p. 518)
5. patients with polysubstance abuse
6. patients with psychosis

113.7.3 Technique

The implant trial procedure is performed with the patient awake, under local anesthetic, to confirm appropriate stimulation coverage.¹⁴

Two techniques are used to place electrodes in the *epidural* space:

1. wire-like electrodes placed percutaneously with a Tuohy needle.
2. "paddle electrodes" placed via laminotomy

SCS leads have been placed anywhere from C1–5:

- leads at C1–2 usually provide coverage for neck, shoulder, or arm pain
- midline truncal stimulation between the neck and low back for pain is often difficult to achieve
- posterior column stimulation evokes tingling paresthesias targeted to overlay the area of pain and electrodes are always positioned in the dorsal epidural space
- leads are usually inserted under fluoroscopic guidance in the midline; however, if the symptoms are lateralized, leads can be placed eccentrically toward the affected side
- thoracic percutaneous leads are introduced into the epidural space via a Tuohy needle puncture at some level below the conus. They are threaded rostrally until the desired coverage is achieved. Wide distribution can be attained
- thoracic paddle leads are placed by performing a laminotomy at T9–10 with the lead passed rostrally to the T8–9 vertebral body level¹⁴

Following electrode placement, a trial with an external generator over several days is used to determine if there is a response. A trial is considered successful if there is greater than 50% pain reduction on visual analog scale (VAS). Responders undergo subsequent conversion to a permanent system with internalization of an implanted pulse generator (IPG). IPGs may be primary cell devices or rechargeable and are placed in a subcutaneous pocket in the flank, buttock, or lower abdominal quadrant. The IPG can be externally reprogrammed to "fine-tune" stimulation.

113.7.4 Complications

Overall risk of device-related complications is 32%.^{14,15,16} Potential complications include:

1. lead migration: 13%. More common in percutaneous and cervical stimulators
2. lead breakage: 9.1%
3. infection: 3.5% incidence. Treated with electrode and/or IPG removal and antibiotics
4. hardware malfunction: 2.9%
5. unwanted stimulation: 2.4%
6. less common complications: CSF leak, radicular pain, intermittent interference with cardiac pacemakers, and neurologic deficit.

113.7.5 Outcome

The primary measure of success is a sustained reduction in pain > 50%.

Secondary goals may include: improved health-related quality of life, reduction in pain medication requirement, increased functional capacity, and return to work.

In a retrospective report on 410 patients with SCS implantation for a variety of indications followed for a mean of 96 months, successful therapy thus defined was achieved in 74%.¹⁷

113.7.6 Specific syndromes treated

Failed back surgery syndrome (FBSS)

Σ: Effectiveness of spinal cord stimulators for failed back surgery syndrome

The addition of SCS improves pain control over either PT or medical management alone for FBSS. At 24 months, SCS is as effective as reoperation in treating radicular pain, with no difference in ADLs or work status.

PROCESS trial¹⁸ (Prospective Randomized Controlled Multicenter Trial of the Effectiveness of Spinal Cord Stimulation): 100 FBSS patients were randomized to SCS placement plus conventional medical management (CMM) (52 patients) vs. CMM alone (48 patients). The health-related quality of life, measured using EuroQol-5D questionnaire, was greater in the SCS group despite a higher total health care cost at 6 months.¹⁵ In 24-month follow-up, the primary outcome was achieved in 37% of patients that were randomized to SCS plus CMM and 2% of patients that were randomized to CMM only. Patients were allowed to cross over. After crossover, the primary outcome was achieved in 47% of patients (34 of 72) who had SCS plus medical management as final treatment versus 7% of patients (1 of 15) in the other group ($P=0.02$).¹⁹

In another randomized prospective study, patients with persistent or recurrent radicular pain after lumbosacral surgery were randomized to reoperation vs. SCS. On an average of 3-year follow-up, the SCS group required fewer opiate analgesics. 9 of 19 patients in the SCS group compared to only 3 of 26 patients reoperated had self-reported pain relief and satisfaction ($P<0.01$), and there was no difference in ADLs and work status. Patients in the SCS group were less likely to crossover to undergo reoperation (5 of 24 patients from the SCS group versus 14 of 26 patients in the reoperation group, $P=0.02$).²⁰

In a study evaluating the cost-effectiveness of SCS, the initial cost of SCS is offset by future reductions in healthcare resource utilization. The medium- to long-term costs of SCS seem to be economically favorable compared to other therapies for FBSS.²¹

Complex regional pain syndrome (CRPS)

Σ: Effectiveness of SCS for CRPS

SCS may be effective for treating CRPS during the first couple of years, but no significant benefit was evident at 5-year follow-up.

CRPS is a chronic pain condition marked by continuous disabling intense aching or burning pain. Type I is nonspecific soft-tissue pain with no known nerve injury. Type II (p.525) follows a nerve injury. Spontaneous pain, hyperpathia, and allodynia disproportionate to the inciting stimulus are hallmarks of CRPS. The pathophysiology leading to disproportional pain response is unknown and treatment options are limited. In a randomized clinical trial,²² patients with CRPS Type I were randomized to receive SCS plus physical therapy (PT) (36 patients) or PT alone (18 patients). 24 of 36 patients had successful SCS trial and underwent implantation. At 6 months, in the group that received SCS plus PT, pain intensity reduced by 2.4 cm on the visual-analogue scale as opposed to an increase of 0.2 cm in the PT-only group ($P<0.001$). In addition, 39% of patients in the SCS group had "much improved" globally perceived effect vs. 6% ($P=0.01$). The health-related quality of life only improved in the SCS group. At 2-year follow-up, pain intensity in the SCS group reduced by 2.1 vs. 0.0 cm in the PT group compared to baseline ($P<0.001$) and global perceived effect was "much improved" in 43% vs. 6% ($P=0.001$).²³ However, these benefits were no longer significant in 5 years.²⁴

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Peripheral vascular disease

Σ: SCS for peripheral vascular disease

SCS does help with pain due to inoperable limb ischemia. It may or may not improve healing of pressure ulcers.

Peripheral vascular disease may result in clinical limb pain or ischemic ulcers.

A review of six controlled studies of nearly 450 patients, SCS + medical treatment was compared to medical treatment alone. Although there was no significant difference in ulcer healing, the use of analgesics was less and limb salvage after 12 months was significantly higher in the SCS group (relative risk = 0.71).²⁵

Angina pectoris

Σ: SCS for angina pectoris

SCS was as effective as CABG in controlling refractory angina and protecting against MIs. SCS improves exercise capacity by an unknown mechanism.

In select patients SCS can reduce anginal pain and improve exercise capacity.

In a prospective study of 104 patients who underwent SCS placement for refractory angina pectoris (average follow-up $\approx$ 13 months), 73% had $>50\%$ reduction of weekly anginal episodes compared to baseline.²⁶

Diabetic neuropathy

Σ: SCS for diabetic neuropathy

Available data is limited, but SCS may be a viable modality for refractory pain from diabetic neuropathy. Further study is needed.

No good clinical data are available. A few studies with small numbers of patients suggest SCS can provide significant pain relief in most patients with diabetic neuropathy who have failed conservative management.^{27,28,29}

A small prospective, open-label study reported 9 of 11 patients with diabetic neuropathy that failed conservative treatment had significant pain relief after SCS implantation at 6 months. Pain score on visual analogue scale decreased from 77 to 34. Microcirculatory perfusion did not change significantly from baseline.²⁹

113.8 Deep brain stimulation (DBS)

Deafferentation pain syndromes (anesthesia dolorosa, pain from spinal cord injury, or thalamic pain syndromes) may benefit from stimulation of sensory thalamus (ventral posteromedial (VPM) or ventral posterolateral (VPL)). DBS for chronic neuropathic pain produces a 40–50% reduction in pain in about 25–60% of patients.³⁰

Nociceptive pain syndromes are more likely to benefit from stimulation of periventricular gray matter (PVG) or periaqueductal gray matter (PAG), although PAG stimulation is rarely used because it often produces unpleasant side effects. Still, response rate has been only $\approx 20\%$,³¹ resulting in failure of the FDA to approve these devices for pain.

Cluster headaches: may respond to hypothalamic stimulation, but larger trials with longer follow-up are needed.³⁰

113.9 Dorsal root entry zone (DREZ) lesions

113.9.1 General information

Although use has been reported for a variety of indications, DREZ lesions appear to be most effective in treating the following:

1. deafferentation pain resulting from nerve root avulsion.^{32,33,34} This most commonly occurs in motorcycle accidents
2. spinal cord injuries (SCI) with pain around the lowest spared dermatome with caudal extension of pain restricted to a few dermatomes (SCI with diffuse pain involving the entire body and limbs below the injury is less responsive)
3. postherpetic neuralgia (p. 522): usually good initial response, but early recurrence in $\leq$ few months is common, and only 25% have long-term relief of pain
4. postamputation phantom limb pain: there is some support for this in the literature, but others feel this is not a good indication³⁵
5. ✖ generally not used for cancer pain

113.9.2 Technique

A laminectomy is performed over the involved segment(s) using radiographic localization. The dura is opened, and the DREZ is identified under microscope magnification using intact posterior rootlets above or below for orientation (contralateral rootlets may also be used to estimate the mirror-image location). Lesions are created ipsilateral to the avulsed nerve roots by radiofrequency current (approximately 50–60 lesions are required for several segments, each lesion is done at 75° for $\approx$ 15 seconds) or selective incisions extending from the last completely normal rootlet at the rostral end to the first normal rootlet caudally. The lesioning needle or knife blade is angled 30–45° medially and inserted to a depth of 2–3 mm. DREZ lesions may be combined with a cordectomy at the level of anatomic cord disruption in paraplegic patients.³⁵

113.9.3 Post-op management

Bed rest for 3 days may reduce the risk of CSF leakage. Analgesics appropriate for a multilevel laminectomy are administered.

113.9.4 Complications

Ipsilateral weakness (related to corticospinal tract) or loss of proprioception (dorsal columns) occurs in 10% of patients, and is permanent in $\approx$ half (i.e., 5%).

113.9.5 Outcome

In pain related to brachial plexus avulsion, 80–90% long-term significant improvement can be expected. Paraplegics with pain limited to the region of injury have an 80% rate of improvement, compared to 30% for those with pain involving the entire body below the lesion.

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114 Seizure Surgery

114.1 Indications for seizure surgery

114.1.1 General information

20-36% of patients continue to have seizures despite rigorous attempt at medical management with ASMs. Many of these patients may be candidates for surgical procedures to control their seizures.^{1,2}

Epilepsy treatment should be aggressive to reduce: brain damage, development of secondary seizure foci, psychosocial distress and mortality (p.480). Seizure surgery is associated with a reduction in mortality rate in medically refractory epilepsy, and while the reduction was highest when seizures are abolished, mortality is also reduced with significant reduction of tonic-clonic seizure frequency.³ Unfortunately, many patients who would benefit from surgical treatment are not referred or are referred late for surgery due to the misconception surgery should be a last resort.⁴

The seizure disorder must be severe, debilitating, or socially disruptive to the patient.

114.1.2 Types of seizures amenable to surgery

The three general categories of seizure types suitable for seizure surgery⁵:

1. focal onset seizures (see below for criteria)
 - a) temporal origin: the largest group of surgical candidates (especially mesial temporal lobe epilepsy (MTLE), ► Table 26.1, which is often medically refractory)
 - b) extratemporal origin
2. symptomatic generalized seizures: e.g., for Lennox-Gastaut (p.484)syndrome, corpus callosotomy may reduce the incidence of drop attacks
3. unilateral, multifocal epilepsy associated with infantile hemiplegia syndrome

114.1.3 Surgical candidates for focal onset seizures

There is no standard, widely accepted criteria for resective/ablative seizure surgery. Therefore patient selection differs between centers.⁶

A literature review found that most epilepsy centers define surgical candidacy by seizure frequency and medically refractory seizures, but that the definition of these terms varies.⁶ The most common criteria found were: seizure frequency ≥ 1 seizure per month, and failure of ≥ 2 ASMs. They also point out that even "infrequent seizures may still pose significant risks and impair quality-of-life".⁶

Medically refractory epilepsy: most centers rely upon failure of at least 2 adequate trials of 2 simultaneous tolerable ASMs (the probability of control after 2 failures is 9.6%²). The ILAE recommends seizure freedom (Category 1 outcome) for whichever of the following is *longer*⁷:

- 12 months or
- 3 times the longest pre-intervention time between seizures during the past 12 months

114.2 Definitions

In addition to the classification of seizure (p.480), the following terms & concepts are particularly relevant to seizure surgery.⁸

Epileptogenic lesion: a structural lesion believed to be producing the seizure, e.g., area of abnormal cortical development, or cavernous malformation.

Ictal onset zone (AKA seizure onset zone): the area that actually generates the seizure. Typically identified depth electrode recording (scalp electrodes can often give an approximation).

Symptomatogenic zone: region of cortex that produces the observed symptoms when activated by a seizure.

Irritative zone: region of cortex that produces interictal spikes detected on scalp or invasive EEG, magnetoencephalography (MEG), or triggered fMRI. Isolated spikes generally do not produce clinical symptoms.

Epileptogenic zone: region of cortex essential for the generation of a clinical seizure. Unlike the irritative zone, it can also be detected by single photon emission CT (SPECT).

Functional deficit zone: area of cortex that is impaired in the interictal period, either by destruction by a lesion or by functional inhibition.

114.3 Pre-surgical evaluation for seizure surgery

114.3.1 General information

All patients should undergo high-resolution MRI study to rule out e.g., neoplasm, AVM, cavernous malformations, mesial temporal sclerosis, or hippocampal lesion. Noninvasive techniques allow localization in the majority of cases.

25% of patients referred to specialized epilepsy clinics do not have epilepsy.^{9,10} See ► Table 28.2 for some etiologies.

114.3.2 Noninvasive evaluation techniques

► **Video-EEG monitoring.** Preoperative long-term inpatient video-EEG monitoring (surface electrodes) to correlate the clinically disabling seizure with appropriate electrical abnormalities and possibly to identify the seizure focus is required.

► **High-resolution MRI.** The imaging modality of choice. Extremely good for detecting hippocampal asymmetry of mesial temporal sclerosis (MTS), and neuronal developmental abnormalities (e.g., cortical dysplasia) that may produce complex partial seizures (CPS).¹¹

► **CAT scan.** A seizure focus may enhance with IV contrast shortly following a seizure. Subtle enhancement may be present on the side of the focus on interictal CT scan.¹²

► **PET scan (positron emission tomography).** Interictal PET scan using 18-fluorodeoxyglucose (18FDG) shows hypometabolism lateralized to the side of temporal lobe focus in 70% of patients with medically refractory CPS (does not show actual *site* of origin). Useful when MRI and EEG cannot localize.

► **SPECT scan (single-photon emission computed tomography).** Used to demonstrate increased blood flow during a seizure to help localize site of onset. [99m]Technetium (Tc) hexamethyl-propylene-amine-oxime (HMPAO) is usually administered immediately after onset of seizure, and the scan may be obtained within several hours.¹³

► **MEG (magnetoencephalography).** Functional imaging technique for mapping brain activity by recording magnetic fields created by neuronal activity (electrical current).¹⁴ Synchronized neuronal currents induce a weak magnetic field. Clinical uses include detecting and localizing pathological activity in patients with epilepsy and in localizing eloquent cortex for preoperative surgical planning. Requires a magnetically shielded room.

114.3.3 Mildly invasive techniques

► **Wada test.**¹⁵ AKA intracarotid amytal test. Localizes dominant hemisphere (side of language function) and assesses ability of hemisphere without lesion to maintain memory when isolated. Usually reserved for candidates for large resections.¹⁶ Each cerebral hemisphere is individually anesthetized via selective carotid catheterization (usually by a neurointerventionalist) and injection of short-acting barbiturate.

Start with angiogram to assess cross-flow and to R/O persistent trigeminal artery (p.83). Significant cross-flow is a relative contraindication to anesthetizing the side of dominant supply (patient goes to sleep).

Wada test may be grossly inaccurate with high-flow AVM. Also, portions of hippocampus may be supplied by posterior circulation (not anesthetized by ICA injection).

EEG monitoring is usually performed during the test when it is being done for seizure surgery. Patient will show delta waves during deepest level of anesthesia.

Technique:

- instruct patient as to what is expected
- catheterize ICA: usually start on side of lesion
- have patient hold contralateral arm in air, and instruct them to hold it there
- inject 100–125 mg sodium *amobarbital* (Amytal®) rapidly into internal carotid artery (effect starts almost instantaneously, begins to subside after ≈ 8 minutes; may subside in ≈ 2 minutes with AVM where flow rates are high)
- determine adequacy of injection by assessing motor function in elevated arm (should be ≈ flaccid)
- assess language skills by showing patient pictures of objects and asking them to name each one out loud and remember each one

- assess memory function by asking patient to name as many of the pictures as they can ≈ 15 minutes after test: if they have difficulty, ask them to pick out pictures from a group that contains additional ones not shown to patient
- repeat procedure on other side (use lower Amytal doses with each subsequent injection)

114.3.4 Evaluation techniques requiring surgery

- ▶ **EEG obtained with invasive electrodes.** Indications: Lack of lateralizing or localizing electrophysiology in preoperative evaluation requires invasive electrodes for better definition of seizure focus. Surgical options:
 - depth electrodes
 - electrodes are placed stereotactically
 - stereoencephalography (sEEG): popularized in Europe by J. Talairach and J. Bancaud during the 1950s for invasive mapping of refractory focal epilepsy. The technique requires the placement of multiple depth electrodes in an orthogonal orientation to localize seizure onset.^{17,18,19}
 - 2–3% risk of intracerebral hemorrhage.¹⁶ Risk of infection with depth electrodes¹⁶: 2–10%.
 - subdural grids or strips
 - grids are frequently used for extraoperative functional mapping (helpful in children or in the mentally retarded). Subdural grid electrodes are placed with a craniotomy.
 - surface strip electrodes may be placed through a burr hole
 - useful technique for intraoperative functional mapping

114.4 Surgical techniques

114.4.1 Basic procedures

Three basic types of procedure: resections, disconnections, and stimulation.¹ The chances of being seizure free after surgery are two to three times higher when a lesion is identified on histopathology or MRI.²⁰

1. resections

- a) resection of epileptic focus: higher chance of completely controlling seizures. Performed in noneloquent brain. Seizures must have focal onset (resection is discouraged if multifocal onset). Includes:
 - anterior temporal lobectomy or selective amygdalo-hippocampectomy for MTLE (p. 1893)
 - neocortical resections: especially with neuronal migration abnormalities
- b) resection of lesion in secondary epilepsy (lesional epilepsy e.g., tumor, AVM, cavernous malformation²¹...). In most cases the seizure focus is in or near the lesion, but some structural lesions are not responsible for the seizures. For seizure foci within the mesial temporal lobe, seizure control is better when lesionectomy is accompanied by amygdalo-hippocampectomy²²

2. disconnections: used when eloquent brain is involved, or to separate the electrical activity of the two cerebral hemispheres

- a) section of corpus callosum (callosotomy): when drop attacks are the most disabling seizure type or for multiple bilateral foci (see below)
- b) hemispherectomy: for unilateral seizures with widespread hemispheric lesions and profound contralateral neurologic deficit. If any cortex is left, must make sure it is functionally deafferented (disconnected)
 - anatomic hemispherectomy
 - functional hemispherectomy: preservation of the basal ganglia isolates the abnormal side with ≈ 80% seizure control rate (similar to anatomic hemispherectomy, but with lower complication rate)
- c) multiple subpial transections^{23,24}: for partial seizure originating in eloquent cortical areas (e.g., sensorimotor cortex). The cortex is transected at 5 mm intervals leaving the pia and surface vessels intact, thus interrupting the horizontal spread of the seizure while sparing the vertically oriented functional fibers. Produced seizure freedom in 55% when combined with resection and in 24% when used alone

3. stimulation: a reversible and modifiable (via programming changes) mode of therapy which may be particularly advantageous for poorly localized seizure foci or for seizure foci in areas of eloquent brain. 30–40% seizure reduction for most patients. Risks include hardware failure, hemorrhage (not typically seen with VNS), infection, and stimulation-induced side effects

- a) open-loop stimulation: blind stimulation continuously or intermittently
 1. vagus nerve stimulation (p. 1895) (VNS): side effects include voice changes
 2. deep brain stimulation (DBS). Targets:

centromedian nucleus of the thalamus²⁵: better for generalized tonic-clonic seizures
 bilateral anterior nucleus of the thalamus: for partial seizures (SANTE trial).²⁶ At 5 years 68% had ≥ 60% fewer seizures, 16% of 110 patients were seizure free for 6 months
 hippocampus²⁵: for partial seizures

- b) closed-loop stimulation: stimulation in response to detection of a seizure, e.g., responsive neurostimulation (RNS®)^{27,28} (NeuroPace). Shortcoming: detection of seizure can be difficult and imprecise, and may require extra (detection) leads. In 111 patients with MTLE, median seizure reduction was 70%, and 29% of patients had at least 6 months seizure free, and 15% had no seizures for ≥ 1 year.²⁸ This modality may be best for patient with unilateral or bilateral MTE who are not eligible for resection or have failed previous resection. Risks include soft-tissue infections with a rate of 0.03 per implant year
- c) chronic subthreshold cortical stimulation (CSCS or CS²)²⁹: continuous (or intermittent) cortical stimulation at low levels to suppress a seizure focus

114.4.2 Anesthetic considerations

If intraoperative electrocorticography is to be performed:

- under local anesthesia: the only anesthetic agents that may be used are narcotics (usually fentanyl) and droperidol
- under general anesthesia: *avoid* benzodiazepines and barbiturates

114.4.3 Intraoperative electrocorticography (ECoG)

Subdural strips and/or grids are useful for ECoG and motor/speech mapping.

Methohexital (Brevital®) may be given to try to provoke a seizure: observe for ↓ fast activity in suspected focus.

114.4.4 Intraoperative cortical mapping

See techniques of cortical mapping (p. 1732).

114.5 Surgical procedures

114.5.1 Corpus callosotomy

Indications and contraindications

Partial (typically anterior 2/3) or total section may be most effective for generalized major motor seizures. Of little benefit for simple or complex seizures. Benefit has been supported for:

1. frequent episodes of *atonic seizures* (“drop attacks”) where loss of postural tone → falls and injuries³⁰ (70% reduction with callosotomy), typically seen in Lennox-Gastaut syndrome
2. possibly for generalized seizure disorder with unilateral hemisphere damage (e.g., infantile hemiplegia syndrome); hemispheric resection may be better for this type, whereas callosotomy may promote partial seizures.

Note: a “functional hemispherectomy” is recommended over “anatomically complete” hemispherectomy to reduce morbidity and mortality⁵

3. some patients with generalized seizures without identifiable, resectable focus
4. ✖ contraindication: major behavioral and/or language deficits may occur even with partial division in patients with speech and dominant handedness located in *opposite* hemispheres (“crossed dominance”). Thus, Wada test is recommended in all left-handed patients

Technical details

Division of the anterior two-thirds of the corpus callosum (CC) (minimizes the risk of disconnection syndrome, see below) may be advantageous over complete callosotomy (controversial). Some advocate sectioning the CC with intraoperative EEG until the typical bisynchronous discharges that are usually seen become asynchronous.³¹ No need to section anterior commissure. Can usually be performed via a bifrontal craniotomy utilizing a bicoronal skin incision.

May produce post-op ↓ verbalization or akinetic mutism that usually resolves in weeks.

Callosotomy has been reported in a small number of patients using laser interstitial thermal therapy (LITT) alone, or as a technique to extend a previous surgical callosotomy.^{32,33}

MRI sagittal cuts are ideal for assessing extent of division of the CC.³⁴

Disconnection syndrome

In a patient with a dominant left hemisphere, consists of left tactile anomia, left-sided dyspraxia (may resemble hemiparesis), pseudohemianopsia, right-sided anomia for smell, impaired spatial synthesis of right hand resulting in difficulty copying complex figures, decreased spontaneity of speech, incontinence.

More common with larger surgical sections of the CC. Risk is less if the anterior commissure is spared. Patients usually adapt after 2–3 months, with final function normal for most daily activities (deficits may show up on neuropsychological testing).

114.5.2 Mesial temporal lobe epilepsy (MTLE)

General information

80% of patients with medically intractable temporal seizures have a demonstrable focus in the anterior-mesial temporal lobe. Most patients have neuronal loss and gliosis of mesial temporal structures (mesial temporal sclerosis (p.482) (MTS)). For this, a standard resection of temporal tip (with amygdalo-hippocampectomy) may be performed. In a randomized trial, anterior temporal lobectomy (ATL) was shown to be superior to medical management for treatment of medically resistant epilepsy. Results at one year show seizure reduction and improved quality of life in those patients who underwent ATL versus medical management alone for intractable epilepsy.³⁵

Limits of resection (without significant neurologic deficit) with ATL

Note that these values are generally considered safe; however, variations occur from patient to patient and only intraoperative mapping can reliably determine the location of language centers.³⁶ Most centers spare the superior temporal gyrus.³⁷ The following measurements are made along the middle temporal gyrus:

- *dominant* temporal lobe: up to 4–5 cm may be removed. Over-resection may injure speech centers, which cannot be reliably localized visually
- *non-dominant* temporal lobe: 6–7 cm may be resected. Slight over-resection may → partial contralateral upper quadrant homonymous hemianopsia; resection of 8–9 cm → complete quadrantanopsia

Alternatively, intraoperative electrocorticography may be used to guide resection of electrically abnormal areas.

Resection should be performed in subpial plane to prevent injury to vascular branches.

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114.5.3 Selective amygdalo-hippocampectomy (SAH)

Anterior temporal lobectomy was the initial surgery for temporal lobe epilepsy described by Penfield and Baldwin. In 1958, Niemeyer described a more selective approach to the hippocampus and amygdala through the middle temporal gyrus.³⁸ It would be almost thirty years until selective amygdalo-hippocampectomy (SAH) was later modified. The goal of SAH is to remove the epileptogenic focus while minimizing disruption of nearby neurovascular structures and white matter tracts. Image guidance is very helpful for these techniques. More recent studies have compared standard ATL to SAH and found that both techniques have similar outcome with regard to seizure freedom but suggest an improved neuropsychological outcome with SAH versus ATL.^{39,40}

Three basic approaches:

1. transcortical: inferior temporal gyrus (ITG) approach. This technique uses a minimal-access approach via a trephine craniotomy for SAH⁴¹
2. transsylvian: approach requires a pterional craniotomy. More restrictive and greater risk of injury to M1 portion of MCA within Sylvian fissure⁴²
3. subtemporal: uses a temporal fossa approach to access mesial structures^{43,44}

114.6 Risks of seizure surgery

Major risks are related to⁴⁵:

1. removal of essential areas of cortex

- injury to medullary core underlying cortical resection (projection fibers, association fibers, and/or commissural fibers): the most common deficit after temporal lobectomy is a contralateral (homonymous) superior quadrantanopsia (so-called “pie-in-the-sky” defect, due to an injury to Meyer’s loop wherein the fibers for the superior visual field of the optic radiation take a slight rostral “detour” towards the temporal tip; see ► Fig. 33.2)
- injury to vessels in area of resection → ischemic damage to areas supplied: especially Sylvian branches during temporal lobectomy, anterior choroidal artery resulting in hemiparesis during mesial temporal lobe resection, or ACA branches with corpus callosotomy
- injury to nearby cranial nerves: especially third nerve during hippocampectomy where it lies medial to tentorium

114.7 Other ablation techniques

114.7.1 Radiofrequency ablation (RF ablation)

Electric current is passed through depth electrodes to create small localized lesions approximately 4 mm diameter with the patient awake. Can be performed at the patient’s bedside.

114.7.2 MRI-guided laser interstitial thermal therapy (MRGLITT)

Laser-induced thermal therapy uses thermal energy transmitted through a fiberoptic probe to induce cell death by damaging DNA and causing protein denaturation. The current therapy is performed with simultaneous MRI stereotactic guidance and real-time feedback from the ablated lesion.^{46,47} It is considered less invasive than microsurgery. Main advantage is a shorter postoperative recovery period. Technique has been used for lesional (e.g., cavernous malformation) and nonlesional epilepsy. 82% of 17 patients > 1 year follow-up (mean 30 months) were seizure-free at 1 year (59% Engel class 1A ► Table 114.1, 31% off all ASMs),⁴⁸ and ablations targeting more anterior, medial and inferior temporal lobe structures were more likely to be associated with Engel I outcomes.⁴⁹ Longer-term data is pending. LITT has been used to perform corpus callosotomy (see above).

114.7.3 Stereotactic radiosurgery (SRS) of mesial temporal lobe

The Radiosurgery or Open Surgery for Epilepsy (ROSE) trial found better seizure control with open anterior temporal lobectomy (78%) than with 24 Gy SRS (52%).⁵⁰ SRS had about a year latency before benefits were realized. The safety profile was similar. The conclusion was that SRS could be an option for patient who are not operative candidates or who choose not to have surgery.⁵⁰

114.8 Postoperative management for seizure surgery (epilepsy surgery)

- ICU for observation (24 hrs)
- for seizures in the immediate post-op period (“honeymoon seizures”), not necessary to treat only one brief generalized seizure, otherwise load appropriately with IV Keppra or phenytoin
- 10 mg dexamethasone (Decadron®) IV before surgery followed by q 8 hrs as necessary (longer taper for laser ablation and radiosurgery)
- antiseizure medications are continued × 1–2 years even if no post-op seizures occur
- after discharge: neuropsychiatric evaluation 6–12 months after surgery

114.9 Outcome

The modified Engel classification and the International League Against Epilepsy (ILAE) of outcome following seizure surgery are shown in ► Table 114.1.^{51,52}

114.9.1 Outcome with resection of seizure focus

The greatest effect of seizure surgery is *reduction of seizure frequency*³⁷; however, any surgical procedure may fail to have a beneficial effect.

Seizure control is usually assessed at 1, 3, & 6 mos post-op, and then annually. A post-op MRI is usually obtained at 3 mos post-op to assess extent of surgical resection. Most patients take anti-epileptic drugs (ASMs) for 2 years post-op, and then may be discontinued in those free of seizures.

Table 114.1 Modified Engel & ILAE classification of seizure surgery outcome⁵²

Engel Class	General Description	Detailed Engel Description	ILAE Classification
I	seizure-free or residual auras	I-A: completely seizure free since surgery	Class 1. Completely seizure free; no auras. Class 1a. Completely seizure free since surgery; no auras.
		I-B: nondisabling simple partial seizures only since surgery	
		I-C: some disabling seizures after seizures surgery, but free from disabling seizures for 2 yrs	Class 2. only auras; no other seizures
		I-D: generalized convulsions with ASM discontinuation only	
II	rare disabling seizures (<3 complex partial seizures per year)	II-A: initially free from disabling seizures, but still has rare seizures	Class 3. 1-3 seizure days/yr; $\pm$ auras
		II-B: rare disabling seizures since surgery	
		II-C: occasional disabling seizures since surgery, but rare seizures for the last 2 yrs	
		II-D: nocturnal seizures only	
III	worthwhile improvement	III-A: worthwhile seizure reduction	Class 4. 4 seizure days/yr - 50% reduction in baseline number of seizure days; $\pm$ auras
		III-B: prolonged seizure-free intervals of seizure days; auras amounting to 50% of follow-up period, but not 2 yrs	
IV	no worthwhile seizure improvement	IV-A: significant seizure reduction	Class 5. <50% reduction in baseline number of seizure days - 100% of baseline number of seizure days; $\pm$ auras
		IV-B: no appreciable change	
		IV-C: seizures worse	Class 6. >100% increase in baseline number of seizure days; $\pm$ auras

Recurrent seizures: although late seizures may occur, 90% of seizures that recur do so within 2 years.

2 years post-op in patients maintained on ASMs: 50% are seizure-free, and 80% have over 50% reduction of seizure frequency.

For temporal lobectomies in the dominant hemisphere without intraoperative monitoring, there is a 6% risk of mild dysphasia. Significant memory deficits occur in $\approx$ 2%.

114.9.2 Radiosurgery for epilepsy

Stereotactic radiosurgery has been suggested to be an effective treatment for pharmacoresistant epilepsy with the potential for less morbidity than resection.^{53,54} Seizure-free outcomes $\approx$ 65% for MTL (delayed response to therapy $\approx$ 6–12 months). Potential for long-term complications remains a concern (radionecrosis).⁵⁵

114.9.3 Vagus nerve stimulation (VNS)

Electrodes wrapped around the *left* vagus nerve in the neck are connected to an implanted programmable generator to stimulate the nerve to reduce seizure frequency. As is also true with many ASMs, the mechanism of action is not well understood.

Indications: Although it has been used (off-label) for treatment-resistant depression and other psychiatric conditions, the FDA-approved indication is for adjunctive therapy for patients > 12 years old with partial-onset seizures refractory to medical treatment.

Complications: Main risk of surgery is transient or permanent vocal cord paralysis. Infection necessitating removal of the device occurred in 0.6%.⁵⁶

Outcome: In an 11-year retrospective review of VNS in 400 patients, seizure control $\geq$ 90% occurred in 22% of patients, control $\geq$ 75% in 40%, control $\geq$ 50% in 64%.⁵⁶

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115 Radiation Therapy (XRT)

115.1 Introduction

Ionizing radiation comprises a portion of the electromagnetic spectrum and includes X-rays and gamma rays (both of which are electromagnetic radiation and transmit their energy via photons) and particulate radiation (e.g., protons). The goal of radiation therapy (XRT) in treating tumors is to cause cell death or to stop cell replication. Photons impart critical energy to achieve this result by the photoelectric effect (at lower energies, <0.05 MeV), by Compton scattering (at higher energies of 0.1–10 MeV, e.g., in linear accelerators and Gamma knives), or by pair-production (at the highest energies).¹ In the Compton effect, the initial collision of the photon with an atom creates a free electron which then ionizes other atoms and breaks chemical bonds. The absorption of radiation by indirect ionization in the presence of water produces free radicals (containing an unpaired electron), which causes cellular injury (usually by damaging DNA) within the tumor.

See discussion of radiation dosage and units (p.234).

115.2 Conventional external beam radiation

115.2.1 Fractionation

The practice in which the total radiation dose is delivered in a series of smaller brief applications. This is one means of increasing the therapeutic ratio (the ratio of the effectiveness of XRT on tumor cells to that of normal cells). Radiation injury is a function of the dose, the exposure time, and the area exposed. Radiation oncologists refer to the four "R's" of radiobiology²:

1. Repair of sublethal damage
2. Reoxygenation of tumor cells that were hypoxic before XRT: oxygenated cells are more sensitive than hypoxic cells because oxygen combines with unpaired electrons to form peroxides, which are more stable and lethal than free radicals
3. Repopulation of tumor cells following treatment
4. Redistribution (or reassortment) of cells within the cell cycle: cells in the mitotic phase are the most sensitive

115.2.2 Dosing

The biologically effective dose of fractionated radiation is often modeled by the linear-quadratic equation (LQ-model) shown in Eq (115.1), where n = the number of doses, d = dose per fraction, and the factors α & β are used to describe the cell response to radiation. A high α/β ratio ≥ 10 is designated as early-responding tissue such as tumor cells, and a ratio ≤ 3 is considered late-responding tissue (mitotically quiescent), such as normal brain and also AVMs.

Linear quadratic equation:

$$\text{biologically effective dose (BED) (Gy)} = n \times d \times \left[1 + \frac{d}{\alpha/\beta} \right] \quad (115.1)$$

115.3 Cranial radiation

Following surgery for tumor (craniotomy or spinal surgery), most surgeons wait ≈ 7 –10 days before instituting XRT to the surgical site (allows initiation of healing from surgery).

CNS tumors that are very radiosensitive include:

1. lymphoma – they “melt away” with XRT but tend to recur later
2. germ cell tumors
3. metastatic small cell lung cancer

For specific indications and doses, see the appropriate section (e.g., GBM, metastasis...).

For side effects, see Radiation injury and necrosis (p.1899).

115.4 Spinal radiation

115.4.1 General information

Most spine tumors are metastatic cancer. There is no proof that any treatment for spinal metastases will prolong survival. Treatment goals, regardless of the treatment modality, are pain relief and preservation of function.

Radiation therapy (XRT) is the main treatment modality for radiosensitive spinal metastases. Even tumors that are considered “radioresistant” can respond to XRT.

115.4.2 Typical spinal radiation

For most metastatic spine tumors treated with conventional radiation therapy (i.e., not stereotactic radiosurgery), the usual fractionation is 30 Gy administered over 10 fractions.

115.4.3 Emergency spinal radiation

For acute spinal cord paralysis from tumor, if emergency surgery is not a consideration, 8 Gy can be administered in the 1st fraction for lymphoma, multiple myeloma, and (although this is not standard) it could be considered for small-cell (neuroendocrine) carcinoma since it is so radiosensitive.

115.4.4 Side effects

1. radiation myelopathy (p. 1901)
2. due to overlap, radiation injury to GI tract: N/V, diarrhea
3. bone marrow suppression
4. growth retardation in children³
5. risk of developing cavernous malformations of the spinal cord (p. 1525)

115.5 Radiation injury and necrosis

115.5.1 General information

Radiation necrosis (RN) may mimic recurrent (or denovo) tumor both clinically and radiographically. Differences in prognosis and treatment make it important to distinguish between tumor and RN.

115.5.2 Pathophysiology

As radiation is selectively toxic to more rapidly dividing cells, the two normal cell types within the CNS most vulnerable to RN are vascular endothelium (which has a turnover time of $\approx 6\text{--}10$ mos) and oligodendroglial cells. Vascular injury may be the primary limiting factor to the tolerance of cranial XRT.⁴ Injury from XRT occurs at lower doses when given concurrently with chemotherapy (especially methotrexate).

115.5.3 Etiology of side effects

The mechanism(s) by which XRT causes side effects is not known with certainty, but may be due to:

1. damage to vascular endothelium: effects on cerebral vasculature may differ substantially from effects on systemic vessels⁴
2. glial injury
3. immune system effect

Radiation effects are divided into 3 phases⁵:

1. *acute*: occur during treatment. Rare. Usually an exacerbation of symptoms already present. Probably secondary to edema. Treat with $\uparrow$ steroids
2. *early delayed*: few weeks to 2–3 mos following completion of XRT. In spinal cord $\rightarrow$ Lhermitte's sign. In brain $\rightarrow$ post-irradiation lethargy & memory difficulties
3. *late delayed*: 3 mos–12 yrs (most within 3 years). Due to small artery injury $\rightarrow$ thrombotic occlusion $\rightarrow$ white matter atrophy or frank coagulative necrosis

Manifestations of radiation effects:

1. decreased cognition
 - a) dementia may develop following XRT⁶ in as little as 1 year post-XRT. Incidence was higher when doses of 25–39 Gy were given in fractions > 300 cGy⁷
 - b) children: may attain lower IQ by $\approx$ 25 points, especially with > 40 Gy whole-brain XRT. Measurable IQ differences occur in children radiated before age 7, but more subtle deficits occur even in older children⁸
2. radiation necrosis
3. injury to anterior optic pathways
4. injury to hypothalamic-pituitary axis $\rightarrow$ hypopituitarism $\rightarrow$ growth retardation in children; see radiation injury to pituitary (p. 893)
5. primary hypothyroidism (especially in children)
6. may induce formation of new tumor: tumors most commonly identified as having increased incidence following radiation treatment are gliomas (including glioblastoma⁹), meningiomas,¹⁰ and nerve sheath tumors.¹¹ Skull base tumors have been reported following EBRT¹²
7. malignant transformation: e.g., after SRS for vestibular schwannomas (p. 799)
8. leukoencephalopathy: profound demyelinating/necrotizing reaction 4–12 mos after combined RTX and methotrexate, especially in children with acute lymphoblastic leukemia (ALL) and adults with primary CNS tumors

115.5.4 Evaluation (differentiating RN from recurrent tumor)

Greenberg IMHO

Over the years many methods have been championed to differentiate radiation necrosis from recurrent high-grade glioma. Some are listed below. None have proven adequately reliable, and this may not even be a useful exercise. Tumor cells are frequently found on biopsy. The decision whether to reoperate is usually based on whether there is progressive mass effect (regardless of whether it is necrosis or tumor), taking into consideration the patient's neurologic condition, projected longevity, patient desires...

CT and MRI

Cannot reliably differentiate some cases of RN from tumor (especially astrocytoma; RN occasionally resembles glioblastoma).

MR spectroscopy (p. 244) was reliable in distinguishing pure tumor (elevated choline) from pure RN (low choline), but was less definitive with mixed tumor/necrosis.¹³

DWI: mean ADCs were lower with recurrence ($1.18 \pm 0.13 \times 10^{-3}$ mm/s) vs. necrosis ($1.4 \pm 0.17 \times 10^{-3}$ mm/s)¹⁴ (not all cases biopsy proven).

Nuclear brain scan

Some reports of success with thallium-201 and technetium-99 m brain scans.

Computerized radionuclide studies

PET (positron emission tomography) scan: because positron-emitting isotopes have short half-lives, PET scanning requires a nearby cyclotron to generate the radiopharmaceuticals at great expense. Utilizing 18-fluorodeoxyglucose (18FDG), regional glucose metabolism is imaged and is generally increased with recurrent tumor, and is decreased with RN. Specificity for distinguishing RN from tumor recurrence is > 90%, but sensitivity may be too low to make it reliable.¹⁵ Amino acid tracers such as [11C]methionine and [18F]tyrosine are taken up by most brain tumors,¹⁶ especially gliomas, and may also be used to help differentiate tumor from necrosis. Accuracy may be increased by fusing PET scan with MRI.¹⁷

SPECT (single-photon emission computed tomography): “poor man's PET scan.” Uses radio-labeled amphetamine. Uptake depends on presence of intact neurons and the condition of cerebral blood vessels (including blood-brain barrier). Decreased radionuclide uptake indicates necrosis, whereas tumor recurrence has no decreased uptake.

115.5.5 Treatment

1. steroids: symptoms from any form of radiation toxicity often respond initially to steroids (glucocorticoids) or to an increased dose of steroids. Some patients may be "steroid-dependent" in that they manifest symptoms when attempts are made to wean steroids. Most symptoms can be mitigated with dexamethasone 2 mg PO TID to QID
2. bevacizumab (Avastin®): may be considered for refractory cases. May work by decreasing capillary leakage.
The "MD Anderson protocol"¹⁸: bevacizumab 7.5 mg/kg IV q 3 weeks
3. reoperation and excision: appropriate if there is deterioration from mass effect, regardless of whether the mass effect is from recurrent tumor or RN. The decision to reoperate should be based on the patient's Karnofsky rating (p. 1640). Although some benefit has been shown, most reoperation studies are biased because they often select the patients who are doing better
4. other contentious therapies include: hyperbaric oxygen and anticoagulation
5. patients with documented tumor recurrence (as opposed to RN) may also be considered for additional radiation (external beam, interstitial brachytherapy, or stereotactic radiosurgery (SRS)) or chemotherapy

115.5.6 Prevention

Injury is dependent on total radiation dose, number of treatments or fractions (less damage occurs with more frequent small treatments), and volume treated.

Various studies to determine the tolerance of *normal* brain to XRT have estimated that 65–75 Gy given over 6.5–8 wks in 5 fractions/week is usually tolerated (radiation necrosis will occur in $\approx$ 5% after 60 Gy fractionated in 30 treatments over 6 weeks). Other studies have shown tolerance to 45 Gy for 10 fractions, 60 Gy for 35 fractions, and 70 Gy for 60 fractions.⁵

115.5.7 Radiation myelopathy

Radiation myelopathy (RM) typically occurs in patients with spinal cord included in radiation therapy (XRT) ports used to treat cancer outside the spinal cord. This includes breast, lung, thyroid, and epidural mets. Radiation (brachial) plexopathy (p. 571) may occur with irradiation in the region of the axilla for carcinoma of the breast. In the lower extremities, XRT for pelvic or bone tumors (e.g., of the femur) may produce lumbar plexopathy. In addition to permanent changes, radiation therapy may also produce spinal cord edema, which may resolve after completion of radiation therapy.

Epidemiology

Incidence difficult to estimate due to the fact that the onset is typically delayed, together with the poor survival of patients with malignant disease requiring XRT.

Most cases reported involve the cervical cord in spite of the higher frequency with which the thoracic cord is exposed to XRT (perhaps due to higher XRT doses to the head and neck and longer survival than with lung Ca).¹⁹ Delay between completion of XRT and onset of symptoms is usually $\approx$ 1 yr (reported range: 1 mo–5 yrs).

Important factors relating to the occurrence of RM include¹⁹:

1. rate of application (probably the most important factor)
2. total radiation dose
3. extent of cord shielding
4. individual susceptibility and variability
5. amount of tissue radiated
6. vascular supply to the region radiated
7. source of radiation

Pathophysiology

Effects of XRT on the spinal cord that lead to RM are:

1. direct injury to cells (including neurons)
2. vascular changes, including endothelial cell proliferation $\rightarrow$ thrombosis
3. hyalinization of collagen fibers

Clinical

Clinical types of radiation myelopathy

Four clinical types have been described and are shown in ► Table 115.1.

Onset is usually insidious, but abruptness has also been described; the presentation often mimics epidural mets. First symptoms: usually paresthesias and hypesthesia of LEs, and Lhermitte's sign. Then spastic weakness of LEs with hyperreflexia develops. A Brown-Séquard syndrome is not uncommon.

Approximately 50% of patients developing RM also have dysphagia from esophageal strictures requiring dilatations (the dysphagia often predates the myelopathy).

Table 115.1 Types of radiation myelopathy

Type	Description
1	benign form; commonly several mos following XRT (reported as late as 1 yr). Usually resolves completely within several mos. Mild sensory symptoms (frequently limited to a Lhermitte's sign) without objective neurological findings
2	injury to anterior horn cells → lower motor neuron signs in arms or legs
3	described only in experimental animals after doses larger than normal XRT. Complete cord lesion within hours due to injury to blood vessels
4	the type commonly reported. Chronic, progressive myelopathy (see text)

Evaluation

Essentially a diagnosis of exclusion. Radiographic imaging (CT, myelography) will be normal. MRI may show spinal cord infarction. The history of previous radiation is key. The differential diagnosis is included in Acute paraplegia or quadriplegia (p.1702).

Prognosis

Prognosis for Type 4 RM is poor. Usually progresses to complete (or near-complete) cord lesion. Paraplegia and/or sphincter involvement are poor signs.

Prevention

Maximum recommended cord radiation dose depends on size of port, and varies with investigator. With large-field techniques (> 10 cm of cord), the risk of RM is negligible with ≤ 3.3 Gy in 42 days (0.55 Gy/wk), and with small-field techniques ≤ 4.3 Gy in 42 days (0.717 Gy/wk). Larger doses may possibly be given safely if fractionated over longer periods. Recommended upper limit: 0.2 Gy/fraction.

115.6 Stereotactic radiosurgery and radiotherapy

115.6.1 General information

Key concepts

- stereotactic radiosurgery (SRS): the application of a single large dose of radiation to a stereotactically localized target usually ≤ 3 cm diameter with minimal radiation delivered to surrounding tissue. May be a single treatment or up to 5 treatment fractions
- stereotactic radiotherapy (SRT) is similar but employs hypofractionated dosing (2–5 treatment fractions) e.g., with a positioning mask, which can allow for larger targets
- both SRS and SRT may be performed with properly equipped linear accelerators (LINAC Scalpel, CyberKnife®...), collimated beams from multiple radioactive sources (Gamma Knife®), or less commonly, protons and heavy charged particle beams

Stereotactic radiosurgery (SRS)

Lars Leksell coined the term “radiosurgery” in 1951.²⁰ His concept was to replace the use of electrodes or “knives” (scalpels) with multiple intersecting beams of radiation to converge on an intracranial target through an intact skull. The dose of radiation at the point of intersection (the “isocenter”) is higher than outside of the isocenter where the dose falls off sharply such that adjacent tissues receive only minimal radiation from each individual incident beam. When combined with a reliable method for aiming the beams at an intracranial target (e.g., using a stereotactic frame system and three-dimensional imaging) the technique came to be known as stereotactic radiosurgery.

Initially developed to create a necrotic lesion in specific nuclei or pathways for functional disorders, it has subsequently been proven that subnecrotic doses could trigger cellular reactions in tumors and vasculature that would lead to tumor shrinkage or control and obliteration of vascular malformations.

With conventional radiation treatments, the “R’s” of radiobiology (p.1898) are exploited. In contrast, with SRS, precision and accuracy are used to affect the target (e.g., damage tumor or thrombose AVM) and spare normal tissue.

Stereotactic radiotherapy (SRT)

Methods of immobilization and point-of-delivery imaging technologies have since made it practical to deliver stereotactic radiation spread out over several individual treatment sessions (known as fractions) when desirable. Many authors refer to this as stereotactic radiation therapy (SRT). The precise definition of SRT has evolved over time, with some publications describing it as using a conventional fractionation scheme (1.8–2 Gy/fraction). However the majority of authors refer to SRT in terms of a hypofractionated approach, generally limited to five treatment fractions (see below).

Fractionation capitalizes on the differential response of normal tissue from tumors to radiation insult; see the Four “R’s” of radiobiology (p.1898). The value of fractionation is more apparent for tissues with high rates of proliferation and less capability to repair sublethal DNA damage (high α/β ratio in Eq (115.1)).²¹ However these models apply more directly in conventional fractionation schemes; their use in single- or hypo-fractionated schemes is a subject of current research.

Treatment over multiple fractions renders impractical the traditional stereotactic frame. SRT therefore employs various techniques for patient immobilization, including thermoplastic masks, dental-impression-based bite-blocks, and other relocatable frame systems. Displacement errors can be as high as 2–8 mm with mask systems; however, these uncertainties can be reduced with in-room imaging systems such as cone-beam CT (CBCT) which can assist in patient localization, as well as intra-fraction motion monitoring techniques such as orthogonal kilovoltage X-rays, surface tracking systems, and infrared marker tracking systems.

Blurred lines between SRS and SRT

Although some purists insist that SRS is performed in a single session, the current definition was expanded by AANS/CNS/ASTRO in 2007 to include radiosurgical procedures “using a rigidly attached stereotactic guiding device, other immobilization device, and/or a stereotactic image-guidance system ... performed in a limited number of sessions, up to a maximum of five.”²²

To further muddy the waters, for billing purposes in the U.S., the cpt codes used by Medicare for SRS (for brain and spine)^{23,24} describe treatment in one session, whereas SBRT (stereotactic body radiation therapy) allows treatment delivery not to exceed 5 fractions within the body. For 5 or more fractions, Medicare considers it to be intensity-modulated radiation therapy (IMRT).

Comparison of SRS technologies

Various methods for delivering SRS/SRT are clinically available. Three main categories (based on sources of radiation) are Gamma Knife, linear accelerator-based, and heavy charged particle radiosurgery. Fundamentally, there is no difference between a photon created by radioactive decay (gamma ray) and a photon created using electrical energy in a linear accelerator (X-ray).

► **Gamma knife radiosurgery.** In the original Gamma Knife (GK), the source of radiation is gamma decay of 201 cobalt-60 sources that align with an inner collimator to direct the resulting photon beams. A treatment couch includes a mount for an external collimation “helmet,” each of which has either a 4-, 8-, 14-, or 18-mm diameter beam aperture for each source. The stereotactic frame affixed to the patient’s head is positioned within the collimator helmet so the area to be treated is at the focus point of treatment unit. Several dwell positions (also called “shots” or “isocenters”) may be defined to match dose distributions to irregularly shaped targets.

In the newer GK model Perfexion®, 192 cobalt-60 sources are distributed on 8 sectors attached to sector drive motors to move the sources along a fully internalized tungsten collimator. This allows each sector to move from “home,” 4-, 8-, 16-mm collimators or shielded positions. The design makes possible composite use of different beam diameters to help optimize dose distribution. The Gamma Knife is specifically designed for cranial and upper cervical lesions, and is best suited for smaller lesions (<3 cm diameter).

As the gamma sources age, output declines, and treatment times necessarily become longer. Eventually, the sources must be replaced, which is a time-consuming and expensive process.

► **Linear accelerator-based radiosurgery.** Linear accelerators (linacs) generate X-rays by accelerating electrons and directing them to strike a target of a substance with a high atomic number. The X-ray source and beam collimation is mounted on a rotating gantry, creating a fixed isocenter in space. Beam convergence is achieved via rotating arcs with the isocenter fixated on the target. The target is aligned to the isocenter using a treatment table that is adjustable in up to 6 degrees of freedom (6 dof, 3 translations, 3 rotations). Beams are collimated using either narrow-aperture cones (SRS cones) or multileaf collimators (MLCs), the latter of which use banks of computerized leaves to shape the treatment field and may be modulated to achieve specific dose distributions. Linacs are more versatile than GK in that they can treat cranial and extracranial targets, they often have built-in CBCT imaging to assist with patient setup and target localization, and they can have a higher dose rate (and thus potentially faster treatment). However, they are generally technically more complicated than GK and require substantially more quality assurance to maintain technical confidence. The Cyberknife is an SRS-specific linac that uses a robotic arm rather than an isocentrically-mounted linac to achieve 6 dof targeting.

► **Heavy charged particle radiosurgery.** Heavy charged particles (protons or helium ions) from a cyclotron can be used for radiosurgery.²⁵ Unlike high-energy photons (gamma and X-rays), which deposit the majority of their energy upon entrance into tissue and continue to deposit decreasing amounts of energy as they travel through the body, heavy charged particle beams have a shorter, bounded range of penetration wherein particles sharply increase energy deposition near the terminal depth of penetration (Bragg peak effect). Particle radiosurgery achieves a well-localized volume of high-dose radiation by taking advantage of cross-firing of a number of beams as well as the Bragg peak. Due to the expense and increased complexity of heavy charged particle SRS, this therapy is only available in a few centers in the world.

115.6.2 Indications

In general, SRS is useful for well-circumscribed lesions less than approximately 3 cm diameter. For larger lesions, the radiation dose must be reduced because of anatomic and radiobiological constraints.

Published uses of SRS include:

1. vascular lesions
 - a) AVMs (including dural arteriovenous fistulas)
 - b) cavernous malformations
2. tumors
 - a) metastases
 1. as primary treatment (p. 1907)
 2. to treat resection cavity after surgery (p. 1908)
 - b) vestibular schwannomas
 - c) meningiomas
 - d) PitNET/adenomas
 - e) gliomas
 - f) others: craniopharyngioma, pineal tumors, etc.
3. functional disorders
 - a) trigeminal neuralgia^{26,27}
 - b) intractable chronic pain: thalamotomy²⁸
 - c) movement disorders: pallidotomy for Parkinson's disease or thalamotomy for tremor (usually not a technique of choice because of inability to perform physiologic stimulation prior to lesioning. May be a consideration for the rare patients who cannot undergo placement of a stimulator/lesioning needle)
 - d) psychiatric diseases (e.g., obsessive-compulsive disorder)
 - e) epilepsy²⁹

115.6.3 Contraindications

Compressive tumors of the spinal cord, brainstem or optic structures: even with the sharp falloff of radiation dose, there remains radiation delivered within a few millimeters of the margins of the isocenter. This, together with post-radiation swelling, might create significant risk of neurologic injury. Surgical removal should be considered in these situations, especially for benign lesions in young individuals.

115.6.4 Treatment procedure

The treatment procedure includes placement of stereotactic frame (in framed-based SRS), obtaining stereotactic images, target definition, treatment planning and execution of treatment.

Target localization

MRI is the predominant imaging modality for SRS procedures given its superior soft tissue and tumor contrast. Typical MRI protocols include T1-weighted pre- and post-contrast images using 3D pulse sequences. If visualization of structures in surrounding CSF (trigeminal nerves, CPA tumors) is required, special sequences such as constructive interference in steady state (CISS) is useful. Fat-saturation sequences are used in cases with previously resected skull base tumors. NB: there is a 1–2 mm shift due to spatial distortion artifact from the MRI magnet. This effect is more prominent in high field strength MRI.

CT accuracy is never better than 0.6 mm, which is the pixel size. Usually is used when MRI is contraindicated or in cases where distortion in MRI may be a concern (dental brace, shunt). CT does not suffer susceptibility distortion as MRI and one can use a stereotactic CT to merge with nonstereotactic MRI for treatment planning.

Stereotactic angiography remains the best method to define the AVM nidus and their arterial supply and venous drainage. Given that angiograms provide only two sets of orthogonal images (usually AP and lateral images), CTA and/or MRI/MRA can be used as an adjunct to provide vascular anatomy on axial plane.

Treatment planning

Treatment planning is the process by which a dose distribution is created with the goal of adequately treating the target areas while largely sparing normal surrounding structures. In GK treatment, a dose distribution is created by defining one or more isocenters (shots). Each isocenter can make use of full complement beams of various diameters; however, in certain cases one or more of the beams can be blocked to shape the dose distribution for irregular targets and protect adjacent critical structures. For linac-based SRS, treatment planning is accomplished using computer simulation programs to help select the number of arcs or beams with certain orientation. Also, static and dynamic collimators have been developed. Intensity modulation is also a means of delivering the desired dose to a target while decreasing the dose given to surrounding structures.

Lesions that are not round or ellipsoid in shape are not a problem when using linac, but with older GKs, multiple isocenters must be used to conform to an irregular surface. This results in multiple "hotspots." This problem is eliminated with the GK Perfexion, which can create a single isocenter using different beam diameters. In the case of multiple metastases, it can generate individually shaped isocenters for each tumor and thus avoid having to change the helmet or add isocenters during a treatment.³⁰

► **Normal tissue tolerance.** Cranial nerves: Damage to small nutrient vessels and Schwann cells or oligodendroglia are the possible mechanisms of radiation injury to cranial nerves. Special sensory nerves (optic, vestibulocochlear) are the most radiosensitive. The precise dose tolerance of cranial nerve is unclear; the optic nerve can probably tolerate doses lower than 8–10 Gy. Nerves in the parasellar region, the facial nerve and lower cranial nerves tend to tolerate higher doses.

SRS treatment may also have a deleterious effect in structures sensitive to swelling, such as brainstem. However, the structures at highest risk are those within the higher isodose lines immediately adjacent to the lesion.

► Table 115.2 shows maximum recommended doses of various organs for a single fraction. In the brain, critical radiation-sensitive structures include: optic vitreous, nerve, and chiasm, brainstem, pituitary gland, and cochlea.

► **Dose.** Dose is usually prescribed as a particular dose (in units of Gray) to the periphery of target. The periphery of the target is usually defined as the isodose curve, which covers the substantial (usually 95–100%) of the target. The isodose curve is the curve of equal dose usually defined as a

Table 115.2 Maximum recommended radiation dose of critical organs (delivered in a single fraction)

Structure	Maximum dose (cGy)	% of maximum (at a prescribed dose of 50 Gy)
eye lens (cataract induction begins at 500 cGy)	100	2%
optic nerve ³¹	100	2%
skin in beam	50	1%
thyroid	10	0.2%

percentage of the maximum dose point. Traditionally, treatment planning for GK uses the 50% isodose line because for a single isocenter this is the location of steepest dose gradient. Linac-based SRS used higher isodose line (70–90%) to increase homogeneity of dose distribution.

Dose-volume relation: The dose of radiation that can be tolerated is highly dependent on the volume being treated (larger treatment volumes require lower doses to avoid radiation injury). Dose selection is made based on known information or is estimated from dose-volume relationship. If uncertain, err on the side of a lower dose or slightly underdose the margin of tumor. Previous radiotherapy must also be taken into account by the treatment team, as local structures are more sensitive.

115.6.5 Lesion-specific issues

Arteriovenous malformations

SRS is best accepted for the treatment of small to moderate-sized (<3 cm) AVMs that are deep or border on eloquent brain and have a “compact” (i.e., sharply demarcated) nidus, or for patients who are too high-risk for surgery due to medical comorbidities. The radiation induces endothelial cell damage, smooth muscle cell proliferation, thickening of the vascular wall, and ultimately obliteration of the lumen over a period of 1–3 years (latency period).³² During the latency period, the annual risk of hemorrhage is ≈ 1–3% which is the same as the natural history.³³

► **Dose.** Early study from Karlsson et al demonstrated increased obliteration rate with increased margin dose.³⁴ The effect reached the plateau at 25 Gy and a higher dose produced more complications without extra benefits. The optimal dose for AVMs generally ranges between 23–25 Gy. The dose might be reduced for nidi at critical locations or with a large volume. At McGill with linac SRS, they use 25–50 Gy delivered to the 90% isodose curve at the edge of the nidus. With Bragg-peak, complications occurred less frequently with doses <19.2 Gy compared to doses above that (this may reduce the obliteration rate or increase the latency period).³⁵

► **Results.** SRS has an overall obliteration rate of 70–80% across all cases of AVMs treated.³⁶ At 1 year, 46–61% of AVMs were completely obliterated on angiography, and at 2 years 86% were obliterated. There was no reduction in size in <2% of cases. Smaller lesions have higher obliteration rates (with Bragg-peak in AVMs <2 cm diameter, 94% thrombosed at 2 years, and 100% at 3 years).³⁵ AVMs >25 mm in diameter have only ≈ 50% chance of obliteration with 1 SRS treatment. Angiographic features associated with high-flow AVMs predict a lower obliteration rate with SRS.³⁷

► **Grading system.** SRS-based grading systems have been developed to predict patient outcomes since scales developed for surgical resection (e.g., the Spetzler-Martin scale) have not proven applicable for AVM radiosurgery. Long-term outcomes of 1012 patients treated with GK at UVA were analyzed to create the Virginia Radiosurgery AVM Scale (VRAS) (► Table 115.3 and ► Table 115.4).³⁶

Table 115.3 Virginia Radiosurgery AVM Scale (VRAS)³⁶: variables and points

Variable	Points		Score
AVM volume (cm ³)	<2	0	(0 - 2)
	2–4	1	
	>4	2	
Eloquent location	no	0	(0 - 1)
	yes	1	
History of hemorrhage	no	0	(0 - 1)
	yes	1	
VRAS → TOTAL			(0 - 4)

Table 115.4 Virginia Radiosurgery AVM Scale (VRAS)³⁶: total points and % favorable outcome

Total points	Favorable outcome (%)
0	83
1	79
2	70
3	48
4	39

Multivariate analysis showed that AVM volume, non-eloquent location, and no history of hemorrhage were independent predictors of AVM obliteration without post-treatment hemorrhage or permanent neurologic deficits.

► **Embolization.** Controversy exists as to whether embolization is helpful or harmful prior to SRS. Some experts find that target definition after embolization is extraordinarily difficult because of multiple small residual nidi. Experimental studies have shown attenuation of radiation effect by embolic material.³⁸ However, for large AVMs, embolization remains an effective adjunct to reduce the size of nidi to make AVMs amenable to SRS. Additionally, pre-SRS embolization should be considered for high-flow fistulas, which are more radioresistant, and intranidal/perinidal aneurysms, which are at risk for rupture.

► **Large AVMs.** Large AVMs (> 10 cm³) remain a significant challenge to treat with any modality. Single-stage SRS has resulted in low obliteration and high complication rates. To maximize the dose/volume response, SRS can be performed in a volume-staged fashion. The Pittsburgh group reported overall rates of total obliteration of 18%, 45%, and 56% at 5, 7, and 10 years, respectively, in a group of 47 patients with large AVMs. Ten patients had hemorrhage after SRS, and 5 of these patients died.³⁹

► **Residual lesions after SRS.** Factors associated with treatment failures include: incomplete angiographic definition of the nidus (the most frequent factor, responsible for 57% of cases), recanalization of the nidus (7%), masking of nidus by hematoma, and a theorized “radiobiological resistance.”⁴⁰ In some, no discernible reason for failure could be identified. In this series, the complete obliteration rate was = 64%. If AVMs persist, retreatment with SRS 2–3 years later is an option.⁴⁰

The ultimate goal of radiosurgery for AVMs is elimination of risk of hemorrhage. An imperfect surrogate marker for this goal has been total obliteration of the nidus on imaging. Reduced, unchanged, or increased hemorrhagic rates during the latency period have been reported. Proton beam treatment of AVMs affords no protection against hemorrhage in the first 12–14 months following treatment²⁵; this is similar to the 12–24 month latency for photon radiation.⁴¹ Hemorrhages may occur during the latency period even in AVMs that had never bled before,³⁵ and the question has been raised whether a partially thrombosed AVM is more likely to bleed because of increased outflow resistance. Yen et al reviewed a large group of AVM patients undergoing radiosurgery and reported that the hemorrhagic rates reduced from 6.6% before SRS to 2.5% after SRS. The protective effect is more significant in those AVMs that had a prior hemorrhage (10.4% to 2.8%).⁴²

Vascular lesions other than AVMs

Dural arteriovenous fistulas (AVFs) have also shown promising response to SRS.⁴³ However, dural AVFs with cortical drainage should not be treated with SRS because these AVFs pose a high risk of hemorrhage. SRS is of no benefit for venous angiomas.⁴⁴ SRS for cavernous malformations remains controversial, as the lesions could not be precisely evaluated with MRI or angiography. However, published retrospective studies have demonstrated reduction of hemorrhage rates following SRS.^{45,46}

Metastases, primary treatment

The gold standard and level 1 recommendation for a single brain metastasis amenable to surgical resection is surgery followed by WBRT.^{47,48} This does not apply to extremely radiosensitive tumors such as lymphoma, small cell lung cancer, germ cell tumors, and multiple myeloma. There has not been a randomized study to compare surgery alone to SRS alone. A prospective randomized study by Muacevic et al showed comparable survival and local tumor control between SRS and surgery plus WBRT in patients with a single brain metastasis but a higher incidence of distant recurrence in the SRS arm.⁴⁹ When tissue is required for diagnosis, surgery should be considered. Tumors with

significant mass effect should also be resected if the patients are fit for surgery. In patients with ≤ 3 mets, single-dose SRS provides superior survival advantage as compared to WBRT (level III recommendation). Radiographic local control rate of $\approx 88\%$ (reported range: 82–100%) has been cited.⁵⁰

No significant difference has been found with SRS between tumors considered “radiosensitive” and those that are “radioresistant” as defined by standards developed for EBRT (see ► Table 55.7; however, histology may affect the rate of response). The lack of significance of “radioresistance” may be due in part to the fact that the sharp dose drop-off with SRS allows higher doses to be delivered to tumors than would be used with EBRT.

General guidelines for when to consider SRS for brain metastases are:

- total tumor number ≤ 10
- total tumor volume $\leq 15 \text{ cm}^3$
- single tumor volume $< 10 \text{ cm}^3$, and
- no leptomeningeal disease present.

Recent studies on SRS for brain metastases have shown that total tumor volume is a better predictor of overall survival, local control, and even distant failure than tumor number.^{51,52,53}

► **Dose.** RTOG 90-05 study recommended 24 Gy as the maximum tolerated doses of single-fraction SRS for tumors $\leq 20 \text{ mm}$ in maximum diameter; 18 Gy for 21–30 mm; and 15 Gy for 31–40 mm.⁵⁴

Metastases, post-resection treatment

SRS may be used for cavity treatment following removal of a metastasis. In analyzing treatment failures of post-op SRS, 25% of local failures occurred at the margin, and per-op dural involvement was very common.⁵⁵ For treatment planning, the following contouring guidelines are suggested:

1. the circumference of the contour should extend 2 mm beyond the cavity walls⁵⁶
2. the surgical tract should be included in the treatment plan (even for deep lesions)⁵⁵
3. for tumors that were in contact with the dura pre-operatively, the contour should extend a few mm along the dura⁵⁵
4. for tumors that were in contact a dural venous sinus per-op, the contour should extend a few mm beyond the initial sinus contact⁵⁵

Vestibular schwannomas (VS)

Possible indications of SRS for VS are: poor surgical candidates (due to poor medical condition and/or advanced age, some use > 65 or 70 years as a cutoff), patient refusing surgery, bilateral VS, post-operative treatment of incompletely removed VS that continue to grow on serial imaging, or recurrences following surgical removal. See the Management algorithm (p.787) under Vestibular schwannomas for strategies that take into account the natural history.

► **Dose.** As there is little difference in radiographic control at different doses, single-dose SRS $< 13 \text{ Gy}$ is recommended for hearing preservation and to minimize new onset or worsening of cranial nerve deficits (Level III⁵⁷)

Meningiomas

In a large series of meningiomas treated with SRS by the Pittsburgh group,⁵⁸ among 942 patients with 1045 tumors treated, the control rate for patients with surgically verified WHO grade 1 meningiomas was 93%. Control rate for presumed meningiomas based on imaging (no prior histological confirmation) was 97%. The control rates for WHO grades II and III tumors were 50% and 17%, respectively.

► **Dose.** The authors used a mean dose of 14 Gy to the tumor margin.⁵⁸

PitNET/adenomas

Surgery is the mainstay of management for the majority of pituitary tumors (major exception: prolactinomas (p. 888)), especially for symptomatic nonsecretory (nonfunctioning) tumors (p. 887) and secretory tumors that fail medical treatment. For residual/recurrent secretory or nonsecretory tumors, SRS is an appropriate tool to arrest tumor growth and/or normalize hormonal function. In a large series of 418 patients with residual/recurrence PitNET/adenomas treated with GK at UVA,⁵⁹ the

overall tumor control rate was 90% in patients with imaging follow-up available. The endocrine remission rates are 53% for acromegaly, 54% for Cushing's disease, and 26% for prolactinomas.

► **Dose.** The usual dose for nonsecretory tumors is ≈ 16 –18 Gy, and a higher dose is required for secretory tumor ≈ 25 Gy.

► **Results.** Early studies of SRS showed limited numbers of hypopituitarism and cranial nerve injuries. Available long-term follow-up studies now demonstrate a 20–30% rate of new endocrinopathy after SRS and a low but not negligible risk of cranial nerve damage.⁵⁹

Infiltrating tumors

SRS is generally not indicated as a primary treatment for infiltrating tumors, e.g., gliomas, due to the lack of a definable capsule and very close relationship between target volume and tolerated radiation dose. SRS has been used for recurrent lesions following standard treatment (surgical excision and adjuvant EBRT 60 Gy and temozolomide). One of the arguments for SRS in these tumors is the fact that 90% of recurrences are within the original radiographic solid tumor volume.⁶⁰ However, support for SRS boost declined after a 2004 RTOG trial 9305 showing no survival benefit in newly diagnosed GBM patients with upfront use of SRS followed by EBRT and BCNU chemotherapy. SRS can be used as a salvage treatment for small ($< 10 \text{ cm}^3$), recurrent GBM after standard treatment. Kong et al showed SRS salvage treatment in these cases prolonged OS to 23 from 12 months.⁶¹

115.6.6 Treatment morbidity and mortality

Immediate morbidity and mortality

Immediate mortality from the actual treatments themselves is probably zero. Morbidity: all but $\approx 2.5\%$ of patients were discharged home within 24 hours. Many centers do not admit patients overnight. Some immediate adverse reactions include⁶²:

- 16% of patients require analgesics for post-procedural headaches and antiemetics for nausea/vomiting
- at least 10% of patients with subcortical AVMs had focal or generalized seizures within 24 hrs of treatment (only one was on subtherapeutic ASMs. All were controllable with additional ASMs.)

Premedication

The Pittsburgh Gamma Knife group gives methylprednisolone 40 mg IV and phenobarbital 90 mg IV immediately after the radiation dose to patients with tumors or AVMs to reduce these adverse effects.⁶² For small lesions and patients without prior history of seizures, premedication with steroids or antiepileptics are probably not necessary.

Delayed morbidity

Long-term morbidity directly related to the radiation may occur, and just as with conventional XRT, is more frequent with larger doses and treatment volumes. Radiation complications include:

- radiation-induced changes, high intensity on MRI T2WI, or low density on CT: usually occur ≈ 13 months after SRS for AVMs. The incidence is 34% with 8.6% having imaging changes associated with neurological symptoms (focal deficits, seizures, or headache) and 1.8% developing radiation necrosis and permanent deficits.⁶³ The possible mechanisms of this side effect include glial cell damage, breakdown of blood-brain barrier, or early venous thrombosis. Premature venous thrombosis or occlusion before obliteration of AVM nidus can produce venous hyperemia or intracranial hemorrhage⁶⁴
- vasculopathy: diagnosed by narrowing seen on angiography or by ischemic changes on imaging in $\approx 5\%$ of cases
- cranial nerve deficits: occur in $\approx 1\%$ of all cases. Incidence is higher with tumors of CPA or skull base
- radiation-induced tumors: only a few case reports of newly formed malignant tumors (glioblastomas) or malignant transformation of a benign tumor (vestibular schwannoma). Loeffler reported 6 cases over 80,000 radiosurgical procedures for benign diseases.⁶⁵ Radiation-induced meningioma is a well-known complication of radiation therapy.⁶⁶ In a large series of AVM patients treated with SRS, the incidence of radiation-induced meningioma was estimated to be 0.7%.⁶⁷

115.7 Interstitial brachytherapy

115.7.1 General information

Technique whereby radioactive implants are used to deliver locally high doses of radiation directly to tumors while exposing nearby normal brain to less toxic doses. At present, the numbers are too small and the follow-up too short to determine the efficacy of interstitial brachytherapy.⁶⁸

Interstitial brachytherapy (IB) may reduce the rate of tumor growth, but it rarely produces clinical improvement. Patients are generally not considered for IB unless their Karnofsky score is ≥ 70 .

115.7.2 Techniques

Techniques include:

1. insertion of high-activity iodine-125 pellets which remain in place (either by conventional open surgery or by stereotactic technique)
2. insertion of catheters (so-called afterloading catheters) containing radioactive source (such as gold or I-125) by stereotactic technique, which are then removed at a predetermined time (usually 1–7 days)
3. instillation of radioactive liquids (e.g., phosphorous isotope) into a cyst cavity

I-125 has several characteristics that favor its use: it emits low-energy gamma rays, which are absorbed by surrounding tissues minimizing radiation exposure of the normal brain, medical personnel, and visitors. It is available as low-activity (< 5 mCi) or high-activity (5–40 mCi) seeds.

Treatment planning is devised to deliver 60 Gy to the edge of a volume that extends 1 cm beyond the contrast-enhancing tumor, with variations included to spare radiosensitive structures (e.g., optic chiasm). Usual delivery rates are 40–50 cGy/hr to the tumor margin (30 cGy/hr is the critical dose for cessation of human tumor growth) requiring that the seeds stay in the afterloading catheter ≈ 6 days.

115.7.3 Radiation necrosis

Symptomatic radiation necrosis (RN) occurs in $\approx 40\%$ of cases, and may occur as early as several months after IB. It may be impossible to differentiate from recurrent tumor in many cases. Symptomatic treatment is often achieved with increased corticosteroid dosages. Continued neurologic deterioration may require craniotomy.

115.7.4 Outcome

IB is often used as a “last-ditch” effort in a patient with a recurrent malignant tumor who has received maximal external beam irradiation and who is not a candidate for reoperation (as expected, the results in patients with such poor prognoses are not good). However, patients eligible for IB are usually better than those who are not candidates, and this may bias the results towards a better outcome.⁶⁹ Some studies with early (primary treatment) use have shown possible benefit.⁷⁰

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116 Endovascular Intervention

116.1 Introduction

AKA endovascular neurosurgery, neuroendovascular surgery, endovascular and surgical neuroradiology (ESNR), interventional neuroradiology (INR) or interventional neurology (IN), combines catheter-based techniques and imaging for the diagnosis and treatment of specific cerebral and spine conditions. Training is certified by the Committee on Advanced Subspecialty Training (CAST).

116.2 Indications/conditions treated

Endovascular intervention includes diagnosis as well as treatment of the following:

- aneurysms: coiling ($\pm$ stent or balloon assistance), flow-diverting stents e.g., Pipeline, parent vessel sacrifice
- arteriovenous malformations (AVMs): embolization (preoperative or curative)
- dural arteriovenous fistula (DAVF): curative or palliative embolization
- spinal AVMs: embolization
- arteriovenous fistulas: e.g., carotid-cavernous fistulas (CCF)
- acute embolic stroke: intra-arterial thrombectomy, angioplasty, stenting, chemical thrombolysis
- cerebral venous thrombosis (CVT): thrombolysis or mechanical thrombectomy
- cerebrovascular arterial dissections: stenting or parent artery sacrifice
- common and internal carotid artery stenosis: angioplasty/stenting
- tumors: embolization. Primarily used before surgery as an adjunct to decrease vascularity e.g., with some meningiomas and hemangioblastomas
- intracranial atherosclerosis
- vasospasm
- chronic subdural hematoma: embolization of middle meningeal artery
- transverse sinus stenosis: stenting as may be indicated in pseudotumor cerebri (p.955)
- inferior petrosal sinus sampling for localizing pituitary macroadenomas
- iatrogenic vascular injuries: stenting or embolization to achieve hemostasis
- refractory epistaxis: embolization to achieve hemostasis
- Wada testing: evaluation for language and memory localization (e.g., epilepsy patients being considered for surgery)
- intra-arterial chemotherapy: e.g., retinoblastoma
- intraoperative angiography (p. 1466): typically used in aneurysm surgery to confirm exclusion of the aneurysm and patency of parent vessels, and during AVM surgery to confirm elimination of nidus

116.3 Contraindications

- uncorrected (life-threatening) bleeding disorders
- relative contraindications:
 - poor renal function (due to iodine dye load): hemodialysis can be arranged in emergent situations; alternatively, patient may be hydrated and monitored for return to baseline creatinine
 - connective tissue disorder that predisposes to vessel dissection
 - severe allergy to iodine contrast: requires prior administration of "dye allergy prep" (p.232)
 - major atheroma of aortic arch or plaque/atherosclerosis of great vessels (innominate, subclavian, common carotid artery) due to high risk for thromboembolic complications
 - for spinal angiography: thoracic aortic aneurysm (relative contraindication)

116.4 Risks of cerebral angiography

Risk varies with the nature of the pathology being investigated and with the experience of the angiography team. Overall risk of a complication resulting in a permanent neurologic deficit^{1,2}: 0.1%. In ACAS, there was a 1.2% complication rate (p. 1549).

Risks include:

1. arterial dissection
2. perforation of artery or aneurysm
3. embolic thrombus

4. air embolism (p.1738) : extremely rare (incidence of 0.08% in one series³). This incidence is likely higher with endovascular procedures (which may use pressurized arterial lines) than with diagnostic angio.³ Treatment recommendations: elevate BP, ventilate with 100% O₂, consider hyperbaric oxygen³
5. puncture site complications
 - a) hematoma at the puncture site
 - b) embolism distal to the puncture site, e.g., that may compromise blood supply to the lower extremity

116.5 Miscellaneous angiography

► **Tumors.** While angiography is no longer used diagnostically for tumors, there are a few general principles that may be useful. Typically, non-vascular deep lesions cause changes in venous structures, whereas superficial lesions affect arterial structures. Malignant neoplasm (e.g., glioblastoma): the classic feature on angiography is an early draining vein. Meningiomas: the stain (contrast) “arrives early, stays late” (appears early in arterial phase, blush persists beyond venous phase); see also other angiographic findings with meningiomas (p.812).

► **Allcock test.** Evaluates flow through the posterior communicating arteries by vertebral injection with simultaneous common carotid artery compression in the neck.

116.6 Pharmacologic agents

116.6.1 General information

This section presents drugs as they are used in neuroendovascular procedures.⁴ The indications cited are specific to endovascular intervention.

116.6.2 Abciximab (ReoPro)

General information

The Fab fragment of an antibody. Prevents binding of fibrinogen to platelet GP IIb/IIIa receptors. Platelet inhibition lasts up to 48 hours.

Indications and case selection

- acute endoarterial thrombus during endovascular intervention
- dissection with thrombus adherent to intimal flap
- prophylaxis for intracranial or extracranial stent implantation

Contraindications

Do not administer concurrently with tPA (p.1564).

Dosing

R: bolus with 0.25 mg/kg IV over 10–60 minutes (shorter duration for the acute complications during intervention), followed by infusion of 0.125 mcg/kg/min (max. 10 mcg/min) for 12 hours.

Reversal

Discontinue abciximab infusion. Allow 10–30 minutes for clearance of the drug from plasma, followed by platelet transfusion. Surgical intervention should be delayed for 12–24 hours after discontinuation.

116.6.3 Aspirin

General information

Irreversibly inactivates cyclooxygenase (COX), resulting in platelet inhibition by preventing formation of prostaglandins from arachidonic acid that persists through the lifespan of the exposed platelets.

Indications and case selection

- Intra- (short-term) and postprocedural (short- + long-term) prophylaxis of thromboembolic events, e.g., during
 - diagnostic cerebral angiography
 - coil embolization of aneurysms
 - stent implantation (typically with a second antiplatelet agent)
 - balloon test occlusions
 - therapeutic occlusion of large arteries
- Subacute management of procedural complications, e.g.
 - parent artery coil herniations
 - thrombus or clot on coil phenomena
 - in-stent thrombus (alone or in combination with a second agent)

Dosing

R: typical dose does not exceed 325 mg PO daily

Uncoated Aspirin (ASA) achieves peak plasma concentrations within 30–40 minutes.^{5,6} Enteric-coated ASA achieves peak plasma concentrations in up to 6 hours.⁷

Data on the prevalence of ASA resistance in neurointerventional procedures is scarce and the assessment is highly assay-dependent. A dose-related effect (i.e., improved response to increasing the dose) indicates that ASA also exerts antiplatelet effects through non-cyclooxygenase pathways.⁸

24–28% of patients are classified as aspirin-resistant.^{9,10} However, results are conflicting on whether aspirin (or clopidogrel) resistance results in higher rates of thrombotic complications in patients undergoing neuroendovascular procedures.^{11,12,13}

Reversal

ASA inhibition of COX is irreversible. Therefore, reversal is achieved by platelet transfusion.

116.6.4 Clopidogrel (Plavix™)

General information

A platelet ADP receptor antagonist.

Indications and case selection

- prevention of intraprocedural and short-term postprocedural (4–12 weeks) thromboembolic events related to endovascular procedures including
 - coil embolization of wide-neck cerebral aneurysms where stent will be used
 - stent implantation (with a second antiplatelet agent)
 - therapeutic occlusion of large arteries (often with a second antiplatelet agent)
- subacute management of procedural complications (alone or in combination with a second agent)
 - parent artery coil prolapse/herniation
 - thrombus or clot on coil phenomena
 - in-stent thrombus (may be more effective than other agents)

Dosing

R: 75 mg PO daily. Start 7–10 days prior to the actual procedure because there is a 3–7-day latency period to full therapeutic effect

LD: 300–600 mg PO, if there was no time to achieve therapeutic effect over a course of days. A therapeutic effect can usually be achieved within 2 to 3 hours of LD.

Reversal

Platelet transfusion.

116.6.5 Eptifibatide (Integrilin®)

General information

A reversible inhibitor of platelet aggregation by preventing the binding of fibrinogen, von Willebrand factor, and other adhesive ligands to GP IIb/IIIa. Platelet aggregation inhibition appears dose- and

concentration-dependent and is reversible following discontinuation of eptifibatide. It causes a 5-fold increase in bleeding time and has no measurable effect on PT or aPTT.

Indications and case selection

These are the same as for abciximab (p. 1914).

Dosing

R: bolus 180 mcg/kg IV (max 22.6 mg) over 1–2 minutes followed by infusion of 2 mcg/kg/min.

Reversal

- discontinue drug. Significant reduction of anti-platelet effects occurs in 2–4 hours¹⁴
- for clinical signs or imaging evidence of hemorrhage:
 - for ICH, hemodynamic compromise, or decrease in Hb > 5 g/dl or decrease in HCT > 15%, give platelet transfusion
 - for a decrease in Hb < 5 g/dl or decrease in HCT < 15%, give desmopressin **R:** 0.3 mcg/kg × 1.

116.6.6 Heparin

General information

A glycosaminoglycan that indirectly inhibits thrombin by modulating antithrombin III (AT III) activity. It also indirectly inactivates factors IXa, Xa, XIa, and XIIa and restores electronegativity to endothelial surfaces. It prevents thrombin-platelet aggregation and inhibits von Willebrand factor. The anticoagulant effects are immediate. The half-life of IV heparin is approximately 1.5 hours.^{15,16}

Indications and case selection

- prophylaxis during diagnostic angiography (used in flush solutions only)^{4,17}
- almost all neuroendovascular procedures (except for diagnostic angiograms) including:
 - embolization of intracranial aneurysms (coil or Pipeline stent)
 - therapeutic occlusion of carotid, vertebral, or other large cerebral artery
 - embolization of brain AVM or dural AVF
 - cervical and intracranial angioplasty
 - intra- or extracranial stent implantation
 - balloon test occlusion of carotid, vertebral, or other large cerebral artery

Dosing

R: **Flush systems:** 4000–6000 i.u. per liter of 0.9% normal saline (typically 5000, yielding 5 i.u. per cc) for flush systems used in neuroendovascular procedures. For intraoperative angiography: 2500 i.u. per 1 liter of 0.9% normal saline (2.5 i.u. per cc) (lower dosage during surgery is a precaution against excessive bleeding from operative sites, e.g., during craniotomy, but intra-op angios generally use very little flush).

Endovascular interventions:

Aneurysm (coil, stent-assisted coil, flow diversion) embolization for unruptured aneurysm: bolus 60–70 units per kg of patient weight. Check the activated clotting time (ACT) 3–5 minutes after bolus and then hourly. Administer heparin 0–5000 units hourly PRN to maintain ACT between 250 and 300.

Coil embolization for ruptured aneurysm: several protocols exist.

- place the first (framing coil) prior to heparin bolus
- if balloon-assisted coiling is anticipated, IV heparin may be given sooner
- some interventionalists give 2000–3000 units as a bolus, followed by additional heparin bolus after placement of 1st coil

Stroke thrombectomy: because most thrombectomy procedures with modern devices can be performed rather quickly, many interventionalists no longer administer heparin bolus prior to thrombectomy.

Carotid artery stenting: similar to unruptured aneurysm embolization.

Balloon-test occlusion or prolonged stent-assisted coiling: for procedures with significant stasis of blood flow and thus increased risk of thrombus formation, higher ACT range of 300–350 may be utilized.

► **Post-procedure heparinization.** Generally not done unless acute clot formed. For aneurysm embolization, one of the following regimens are recommended:

Heparin dosing for post coiling neurosurgery patients

- University of Cincinnati Protocol¹⁸: devised to administer a safe dose of heparin that does not require repeat blood draws and titration to ACT or aPTT
 - weight-based dosing
 - weight ≤ 75 kg: 900 units/hour, no bolus doses, for 12 hrs
 - weight > 75 kg: 1300 units/hour, no bolus doses, for 12 hrs
 - labs: no ACT or other laboratory draws required.
- *alternative*: the most commonly used heparin weight-based protocol (NB: this differs from dosing, e.g., for coronary indications or treatment of DVT or PE¹⁹): bolus dose of 60–70 units/kg followed by a maintenance IV infusion rate of 18 units/kg/hour

Reversal

IV protamine sulfate at 1 mg per 100 units of circulating heparin (not to exceed 50 mg total).

A preloaded syringe of 50 mg should be available at all times. Normally protamine is administered as an IV infusion over 10–30 minutes, to prevent idiosyncratic hypotension and anaphylactoid symptoms. In an emergency, e.g., vessel or intracranial aneurysm perforation, anticoagulation must be immediately reversed by rapid IV bolus of 10 mg protamine over 1–3 minutes.

116.6.7 Nitroglycerine

General information

Produces strong and immediate vasodilatation by stimulation of cGMP, which results in vascular smooth muscle relaxation.

Indications and case selection

Vessel spasm during catheterization.

Dosing

R: 100–300 mcg through the catheter.

116.6.8 Papaverine

General information

A benzyloisoquinolone alkaloid that vasodilates by inhibition of cAMP and cGMP phosphodiesterases in smooth muscle, leading to increased intracellular levels of cAMP and cGMP. It may also inhibit the release of calcium from the intracellular space by blocking calcium ion channels in the cell membrane. Papaverine is short-acting, with half-life of less than 1 hour.

Indications and case selection

Pre-treatment for angioplasty. The resulting vasodilatation will assist with balloon catheter placement. Due to its short duration of action necessitating repeated administration, other agents e.g., verapamil, are generally preferred.

Dosing

R: 300 mg of 3% papaverine (30 mg/ml) at pH 3.3, is diluted in 100 ml of normal saline to obtain a 0.3% concentration. It is administered intra-arterially through the microcatheter, which is positioned just proximal to the affected vascular segment at a rate of 3 ml/minute.

✗ Do not mix Papaverine with contrast agents or heparin, which may result in precipitation of crystals.

116.6.9 Sodium amytal

General information

A barbiturate derivative that activates GABA_A receptors. When it is administered into cerebral vasculature during Wada test, it temporarily anesthetizes the perfused region, resulting in isolation of the contralateral hemisphere, permitting assessment of cortical functions including language and memory. It blocks neuronal activity, and in conjunction with lidocaine, which blocks axonal activity, it is administered as a test injection in procedures such as spinal AVM embolization, or embolization of spinal tumors, prior to the injection of the embolic agent.²⁰

Indications and case selection

Wada test

Test injection prior to embolization of feeder vessel of AVM. In order to suppress both neuronal and axonal activity, amytal injection is followed by xylocaine injection.

Dosing

R: 50–100 mg per test injection via catheter.

For administration, dilute 500 mg sodium amytal in 20 cc 0.9% NS to yield a concentration of 25 mg/cc of sodium amytal.

For Wada test, after catheterization of vessel of interest, 100 mg of sodium amytal (4 cc of the preparation described above) are injected through the catheter. Additional boluses of 25 mg (1 cc) or modification of the original bolus may be required, per the neurologist's request.

116.6.10 Tissue plasminogen activator (tPA, alteplase)

General information

A fibrin-specific thrombolytic protease that converts plasminogen to plasmin.

Indications for intravenous (IV) thrombolysis with tPA

The use of IV tPA has decreased due to superior results of mechanical clot removal (p.1932) in situations when it is available. When IV tPA is employed, the following information is provided.²¹

Inclusion criteria:

- diagnosis of ischemic stroke causing measurable neurological deficit
- onset of symptoms ≤ 3 hrs before treatment begins (up to 4.5 hrs is supported by data with slightly lower level of evidence)
- age ≥ 18 years

✗ Exclusion criteria:

- arterial puncture at noncompressible site in previous 7 days
- history of previous intracranial hemorrhage
- intracranial neoplasm, AVM, or aneurysm
- recent intracranial or intraspinal surgery
- active internal bleeding
- elevated blood pressure (systolic > 185 mm Hg or diastolic > 110 mm Hg)
- acute bleeding diathesis including but not limited to:
 - platelet count $< 100 \times 10^3/\text{mm}^3$
 - heparin received within 48 hours resulting in an aPTT above normal
 - current use of anticoagulant with an elevated PT (> 15 sec) or INR (> 1.7)
 - current use of direct thrombin inhibitors (desirudin, bivalirudin, argatroban)²² or direct factor Xa inhibitors (rivaroxaban, edoxaban, betrixaban)²³ with elevated sensitive laboratory tests (e.g., aPTT, INR, platelet count, Ecarin Clotting Time (ECT), TT, or appropriate factor Xa activity assays)
- blood glucose concentration < 50 mg/dL (2.7 mmol/L)
- CT demonstrates multilobar infarction (hypodensity $> 1/3$ cerebral hemisphere)

Relative exclusion criteria:

- only minor or rapidly improving symptoms of stroke
- pregnancy

- seizure at onset with postictal residual neurological impairments
- major surgery or serious trauma within 14 days
- history of GI or urinary tract hemorrhage within 21 days
- recent acute myocardial infarction (within previous 3 months)

Dosing

R: Intravenous: 0.9 mg/kg (max. 90 mg). The first 10% of the calculated dose is administered as an IV bolus over 1 minute and the remainder is infused starting *immediately* afterwards over an hour as long as the initial bolus is given within the 4.5-hour window.

R: Intra-arterial thrombolysis with tPA: The dose is independent of any previously administered intravenous dose if IV tPA was given first. Rarely used as primary endovascular treatment nowadays given its low efficacy and high adverse event risk in comparison to other methods of thrombectomy. IA tPA is mainly used as adjunct treatment after thrombectomy if more distal emboli are noted angiographically.

First, 1–2 mg tPA is administered manually proximal to or within the clot, then an infusion of 0.5 mg/ml at 20 ml/hr (10 mg/hr). If not effective in lysing the clot, the infusion is prepared by mixing 10 mg of tPA in 20 ml in normal saline, resulting in a concentration of 1 mg tPA per 2 ml saline (or 0.5 mg/ml). An infusion pump may be used for more precise administration.

Frequent angiography is performed to determine angiographic response.

Discontinue tPA if

- adequate recanalization is achieved
- extravasation of contrast material is noted on angiography
- the maximum dose has been administered, or the administered dose approaches the maximum dose without clinical or angiographic improvement

R Cerebral venous thrombosis (CVT)

Similar to arterial stroke thrombectomy, chemical thrombolysis is rarely used as a sole treatment but rather as an adjunct to mechanical clot extraction. The bolus is usually 2–5 mg administered locally via a microcatheter into the thrombus. If unsuccessful, the microcatheter is left inside the affected sinus and then an infusion is started at a rate of 1 mg/hr, usually for 12–24 hours with repeat daily angiograms to determine the degree of response to thrombolysis. If a clot burden is still present on angiography, a longer duration of administration until the clot resolves is a consideration; however, after 24 hours bleeding complications become more frequent.

Reversal

If symptomatic intracranial bleeding is suspected²¹:

- immediately stop tPA infusion
- diagnostics:
 - bloodwork: CBC, PT (INR), aPTT, fibrinogen level, and type- and cross-match
 - STAT non-contrast head CT
- cryoprecipitate (includes factor VIII): 10 U infused over 10–30 min (onset: 1 hr, peak: 12 hrs), additional dose for fibrinogen level of <200 mg/dL
- tranexamic acid (TXA) 1000 mg IV infused over 10 min OR ε-aminocaproic acid 4–5 g over 1 hour, followed by 1 g IV until bleeding is controlled. There is a risk of seizures with TXA, especially in renal failure
- hematology and neurosurgery consult as appropriate

116.6.11 Verapamil

General information

A nondihydropyridine calcium channel blocker that reduces the influx of calcium through L-type calcium channels in smooth-muscle cells, enabling vasodilation. Half-life is about 3–7 hours.

Indications and case selection

- before balloon angioplasty: chemical vasodilatation prior to mechanical vasodilatation may enable a smoother and safer angioplasty
- mild vasospasm, that does not warrant angioplasty
- moderate vasospasm, that cannot be safely treated with angioplasty

Dosing

R: 5–10 mg IA. It is slowly infiltrated (over 2–10 minutes) into the vasospastic vessel via a microcatheter into the intracranial vessel, and/or via diagnostic or guide catheter into larger vessels, e.g., ICA or VA.

Reversal

For clinically significant hypotension or high degree AV block: treat with vasopressors and cardiac pacing, e.g., epinephrine, norepinephrine, and vasopressin infusions. Calcium chloride infusion is given in large doses, e.g., 1 gm/hr for more than 24 hours. 20% intralipid infusion (100 ml bolus followed by continuous infusion at 0.5 ml/kg/hr) may also be useful.²⁴ Atropine may be administered for bradycardia. Hemodialysis is ineffective.

116.6.12 Xylocaine

General information

Blocks the fast voltage-gated Na⁺ channels in neuronal cell membranes. There may be inhibition of postsynaptic neurons and, consequently, action potentials.

Indications and case selection

As a local anesthetic prior to arteriotomy. Preparations with or without epinephrine may be used.

Cardiac xylocaine is used to test axonal function. It may be used alone or in conjunction with amobarbital.

Used for functional testing in case of spinal vascular disorders.

Provocative functional testing in cerebral AVMs.

Dosing

R: for local anesthesia: Approximately 5 ml of 2% lidocaine (max. 4 mg/ml to 280 mg; 14 ml)

R: for neurophysiological testing (Wada test): 10–40 mg i.a. of cardiac lidocaine.

Reversal

Lidocaine overdose²⁵: intravenous bolus of 20% lipid emulsion 1.5 ml/kg over 1 min and start IV infusion at 15 ml/kg/hr. The bolus may be repeated twice at 5 min intervals, if cardiovascular stability is not restored. Additionally, the infusion may be doubled to 30 ml/kg/hr if instability persists after 5 minutes. Lipid emulsion is continued until cardiovascular stability is restored or maximum dose administered.²⁵ Do not exceed the maximum cumulative dose of 12 ml/kg.

Concurrent supportive care includes following ACLS protocols, e.g., securing airway, hemodynamic support, etc.

✖ Propofol is not a substitute for lipid emulsion in such cases. It has only 10% lipids, which is too low to be of benefit, and the cardio-depressant properties of propofol may be counterproductive in such situations.

116.7 Neuroendovascular procedure basics

116.7.1 Vascular access

General information

Most commonly, vascular access is obtained through femoral artery. If femoral access is not possible, then the radial artery, brachial artery, or carotid artery (least common) may be used.

Femoral artery access

After prepping and draping the groin, place the left little finger on anterior superior iliac spine and span with thumb to the pubic symphysis. This approximately demarcates the ilioinguinal ligament. Bisect this line with the other hand and palpate the femoral pulse. The site of access is 3 fingerbreadths below the point of bisection, to ensure the vessel puncture site is *below* the ilioinguinal ligament and therefore compressible. A marker (e.g., hemostat) can be placed over the pulse and a fluoro shot taken to confirm that the planned access site is situated over the middle of the femoral head.

A small, superficial stab incision in the skin is made at the selected point, after infiltration of local anesthesia. In an elective case, a micropuncture set with a 7-cm 21- or 23-gauge needle may be used for vessel puncture. In an emergency (e.g., stroke) a larger 18 G single wall needle may be used to minimize the amount of time required to obtain femoral access. The artery is palpated and immobilized between the index and middle finger of one hand, while the needle is introduced through the stab at 45°. Once the artery is accessed, blood will emanate through the needle hub. An exchange over wires is made using modified Seldinger technique to place the sheath of desired size. Angiography is performed to confirm successful positioning of the sheath and ensure no local vasospasm, dissection, or active contrast extravasation is seen. The sheath is connected to a continuously running flush of heparinized saline and secured with adhesive or suture to prevent dislodgement.

Radial artery access

First perform Allen's test with pulse oximetry to ensure the hand has satisfactory vascular supply, in case the procedure results in radial artery occlusion.

Allen's test: Palpate the radial and ulnar arteries and place a pulse oximeter on the thumb or index finger. Make the patient flex and extend their fingers repeatedly. With digital pressure, compress both the radial and ulnar arteries during finger extension and maintain the compression until the oximetry pulse is lost. The wrist is maintained in approximately 20° flexion in order to avoid false positive test when the wrist is in hyperextension. Release the pressure on the ulnar artery. Measure the time taken to achieve visual capillary refill in finger pads and at least 92% oxygen saturation. Normal capillary refill time is < 5 sec, refill times of 5–15 secs are considered equivocal. A refill time longer than 15 sec is abnormal. Allen's test can also be performed using ultrasonography.

Reverse Allen's test: It should be performed when the radial artery is being subjected to a repeat procedure. Compress both the radial and ulnar arteries with digital pressure during finger extension and maintain the compression. Release the pressure on the *radial* artery. Measure the time taken to achieve visual capillary refill in finger pads and at least 92% oxygen saturation, as indicated above.

During access, the following modifications are made from the technique described above. A shorter 21 G needle (e.g., 3 cm instead of 7 cm) is used. Avoid entering the artery at a steep angle, as this may cause difficulties in threading the wire through the artery.

Advance a 0.018" wire through the needle hub into the radial artery and remove the needle. Make a nick in the skin over the wire, to aid in smoother insertion of larger sheaths. A 4 or 5 Fr (5 Fr because most diagnostic catheters for neuro procedures are still 5 Fr) micropuncture sheath (with dilator) is advanced over the wire and then the wire and dilator are removed. A cocktail of heparin (5000 IU/ml), verapamil (2.5 mg), 2% lidocaine (1.0 ml), and nitroglycerin (0.1 mg) are administered through the introducer sheath to relieve and/or prevent vasospasm. The patient should be forewarned about a transient but uncomfortable sensation of severe burning, as the cocktail is injected into the artery. The standard 0.035" wire is advanced through the sheath. The desired sheath size is then placed.

Note: After completion of the procedure, a closure device is *not* used for radial artery. Only manual compression for 15–20 minutes is applied. Alternatively, compression devices such as TR Band (Terumo) can be used instead of manual compression to achieve hemostasis.

116.7.2 Sheath management

Once placed, the sheath is connected to a continuously running heparinized saline solution at a rate of 30 ml/hr. The flush consists of 5000 units of heparin in 1000 ml of 0.9% saline bag (5 units/ml). The saline bag is placed in a pressure infuser which is inflated to 300 mm Hg. It is important that the sheath be continuously irrigated and the saline bag be under a pressure greater than the patient's own arterial pressure. The patient's leg on the side of the sheath is kept straight to prevent kinking of the sheath. If the intention is to maintain the sheath for several hours or a few days, it should be secured by suturing to the patient's skin.

116.7.3 Arteriotomy closure

After completion of the procedure, if vascular access will not be required within next few days, the sheath is removed. To do this, the following options are available:

- pressure
 - *manual pressure:* palpate the artery proximal to site of arteriotomy, remove sheath and apply manual pressure for 15–30 minutes and gradually decreasing pressure every 5 minutes
 - Femostop™: prior to application, ensure ACT is < 150 and BP is under control. The inner circle of the Femostop dome should be positioned 1 cm superior and 1 cm medial to the actual puncture

site and over the femoral artery. Inflate the Femostop to 20–30 mm Hg above the patient's systolic pressure. If this does not result in hemostasis, inflate to higher pressures until distal pulses are occluded. Maintain distal pulse occlusion for 5–7 minutes, then readjust the manometer pressure until good pedal pulse and good color of extremities is achieved.

Continue to progressively decrease the applied pressure over the course of several hours, until the device can be discontinued entirely. Usually, a Femostop is maintained for 6–12 hours.

- percutaneous closure devices
 - ✖ neither of the following should be used below the common femoral artery bifurcation if the vessel diameter is too small (typically < 4–5 mm) or if significant atherosclerosis is present
 - Angioseal™: effects closure by deposition of fibrin plug in the vessel which is then pulled back into the arteriotomy defect of the vessel wall. Angioseal is available in 6 and 8 Fr sizes. Deployed by exchanging it with the sheath over a wire. After deployment, the patient remains supine for 2 hours (with pillow under the head) with the leg on the side of Angioseal kept straight. Patient is mobilized thereafter
 - Mynx™: a bio-absorbable intravascular sealant that is deposited in the arteriotomy defect. Available in 5, 6, 7 Fr sizes. Tends to be the least painful
 - Starclose™: Also exchanged with the sheath over a wire. Deployment results in stitch closure of the arteriotomy defect. Tends to be more painful

116.8 Diagnostic angiography for cerebral subarachnoid hemorrhage

116.8.1 General information

Objective: diagnose source of hemorrhage, assess supplying blood vessels and collateral vessels (may require additional provocative maneuvers), assess vessels for possible bypass, assess for vasospasm.

The most common cause of nontraumatic SAH is ruptured cerebral aneurysms. Other causes include AVM (cerebral or spinal), vasculitis, pre-truncal nonaneurysmal subarachnoid hemorrhage, arterial dissection, dural sinus thrombosis, pituitary apoplexy, bleeding dyscrasias, sickle cell disease, and cocaine abuse. In 14–22%, a cause for SAH is not found on angiography.²⁶

116.8.2 Setup

A sheath is placed in the femoral artery, connected to a continuously running flush of heparinized saline. A diagnostic catheter is attached to a continuously running flush of heparinized saline. It is introduced into the vasculature via the sheath and advanced over the guidewire through the aorta to the target vessel. The catheter is advanced with the guidewire leading and pulled back with the guide wire completely retracted into the catheter. Once the catheter is in the desired position, the guidewire is withdrawn completely and angiography performed, by using manual injections or the auto-injector.

116.8.3 Planning

Pre-planning the procedure will save time, amount of contrast administered, and radiation exposure. Standard views (half Towne's and lateral for anterior circulation; Towne's and lateral for posterior circulation) are obtained. Additionally, the images in appropriate projections are obtained, keeping in mind the patient's pathology. Typically, a six-vessel angiogram, including bilateral internal, external carotid, and vertebral arteries, is performed to determine the source of the SAH. In certain cases, a spinal angiogram may also be needed. As needed, rotational angiography is performed and the 3D reconstruction manipulated for further examination, as well as selection of further projections for angiography.

116.9 Disease-specific intervention

116.9.1 Aneurysms

General information

Endovascular therapy (coiling, and more recently, flow diversion) vs. clip debate is a moving target, especially given the fact that as long-term data about endovascular treatment is becoming available, the technology continues to evolve, which renders the data obsolete. While there will always be a

role for surgical clipping, endovascular therapy has emerged as a first line therapy for most aneurysms (particularly ruptured aneurysms or in poor surgical candidates). Surgery still remains a strong option for MCA aneurysms and PICA aneurysms.

Indications

The selection of aneurysm for treatment depends upon the following considerations:

- ▶ **Ruptured vs. unruptured.** A ruptured aneurysm needs to be treated urgently, as the risk or re-rupture is 2–3%/day in the first few days and 20% in 2 weeks. The risk of morbidity and mortality of an untreated ruptured aneurysm is 45–50%.^{27,28}
 - ▶ **Symptoms other than rupture.** Aneurysms that present with symptoms, e.g., cranial nerve palsy, loss of vision or ischemia, may be at a higher risk of rupture than asymptomatic aneurysms (some symptoms may be due to acute expansion).^{29,30,31,32,33,34}
 - ▶ **Size.** Large aneurysms (>7–10 mm) are more likely to rupture than small (<7 mm) aneurysms.^{34,35}
 - ▶ **Shape.** An irregularly shaped aneurysm may be at a greater risk of rupture than a spherical saccular aneurysm. Irregular shape includes morphological characteristics such as daughter blebs, or irregular borders.³⁶
 - ▶ **Aspect ratio.** In addition to size, the aspect ratio (aneurysm depth/neck width) may predict the aneurysms at risk of rupture. An aspect ratio greater than 1.6 may create low flow conditions in the dome of the aneurysm that lead to stasis, thrombosis, and a fibrinolytic cascade that results in breakdown of the intima.³⁷ A retrospective evaluation of 75 ruptured and 107 unruptured aneurysms demonstrated the mean aspect ratio to be 2.7 for ruptured and 1.8 for unruptured aneurysms ($p < 0.001$). The mean depth of aneurysm was also greater in ruptured aneurysms (7.7 ± 4.9 mm vs. 5.1 ± 4.5 mm). 75% of the ruptured aneurysms were <10 mm, and 62% of these had aspect ratio > 1.6.³⁸
 - ▶ **Location.** The ISUIA studies demonstrated posterior circulation aneurysms had a higher risk of rupture compared to anterior circulation.^{29,39} Conversely, cavernous segment aneurysms had a 0% risk of rupture until they reached a size of 13–24 mm, when the 5-year cumulative risk became 3%.
 - ▶ **Choice of treatment.** The choice of treatment should take all of the above in addition to individual patient considerations, e.g., age and overall health.
- Wide necked aneurysms were previously thought better suited for clipping, but the availability of stents that may act as scaffolding for coils, e.g., Enterprise™ stent, or act as vascular reconstruction devices, e.g., Pipeline, has greatly increased the spectrum of aneurysms amenable to endovascular treatment. Additionally, small aneurysms (<4 mm in size) may be less favorable for coiling.

Endovascular options

- ▶ **Coiling.** This is the treatment of first choice for most narrow-necked aneurysms. The first one or two coils (framing coils) used are equal to the average size of aneurysm (calculated as height + width/2). The coils are progressively downsized and changed in softness as embolization progresses. The goal is to maximally pack the aneurysm such that no contrast is seen entering it and without causing any loop to herniate into the vessel lumen. Even after maximal packing as seen angiographically, the actual packing is about 20–30%.⁴⁰
- ▶ **Coiling with stenting.** For wide-necked aneurysms, a stent may be used to prevent the coils from herniating out of the aneurysm into the blood vessel. When a stent is used, the patient is required to be on ASA (most commonly indefinitely, although certain centers stop all antiplatelet drugs after 1 year) and clopidogrel or alternative agents such as ticagrelor or prasugrel (typically for 3–6 months). Therefore, stent assisted coiling is generally avoided in ruptured aneurysms, in part due to the fact that if an EVD, ventricular shunt, or craniotomy is needed it may require temporary reversal of antiplatelet medication, which increases the risk of acute in-stent thrombosis. However, it has also been undertaken successfully in ruptured cases, with 93% technical success, clinically significant ICH in 8% (including 10% known to have EVDs), and significant thromboembolic events in 6%.⁴¹
- ▶ **Balloon-assisted coiling.** This technique may be used for wide necked aneurysms where stenting is deemed less desirable, e.g., ruptured aneurysm, since it bypasses the need for dual anti-platelet

therapy. A balloon catheter is selected based on the diameter of the parent artery and the width of the aneurysm neck. The balloon segment of the catheter (indicated by radiopaque markers) is positioned across the neck and kept inflated during coil deposition into the aneurysm. It is deflated prior to coil detachment and the stability of the deposited coil assessed. If the coil appears stable, it is detached. This inflation-deflation technique is continued as required, until the aneurysm has been completely coiled. Close attention should be paid to ACT as prolonged balloon inflation times and local blood stasis may trigger local thrombus formation. It is typically not recommended to keep the balloon inflated for more than 3–5 minutes.

► **Dual catheter technique.** In this case two microcatheters are placed in the aneurysm. Coil loops are deposited alternatively from each. The technique is proposed to decrease the risk of coil loop prolapse/herniation in wide-neck aneurysms.

► **Coil types. Bare platinum coils:** detachable coils that come in various diameters (mm), lengths (cm), shapes (3D, box coils, helical), and degrees of softness. Framing coils are typically more “stiff” and will provide a solid frame, whereas finishing coils are very “soft” and will find vacant compartments within the coil mass.

Hydrocoils: platinum coils coated with hydrogel, which expands upon contact with blood filling the residual space between loops. These are typically used as “finishing” coils and are believed to reduce the risk of aneurysm recurrence/residual.

► **Pipeline Embolization Device (PED).** The woven design of this particular stent makes it low-porosity, which attenuates blood entry into the aneurysm and therefore encourages stasis. If needed, two or more of these may be deployed within each other across the neck of the aneurysm to cause adequate flow stasis in the aneurysm. Angiography immediately post-deployment demonstrates contrast stasis in the aneurysm. Six-month follow-up angiography usually demonstrates complete obliteration of the aneurysm, such that it is no longer visualized. Once an aneurysm is successfully treated with PED, there is a 0% recurrence rate. The most recent iteration of the device known as Pipeline Flex™ is a 48-braid device that is somewhat easier to deploy than the earlier version. It is available in diameters from 2.5–5 mm. Sometimes, coil deployment in the aneurysm may also be required to encourage aneurysm thrombosis.

A strategy used in case of ruptured aneurysms is to “undercoil” the aneurysm to address the risk of re-rupture and then complete treatment with PED when it becomes safe to use dual antiplatelet therapy

In case of ruptured blister aneurysms, Pipeline device has been used with success by administering an abciximab bolus (0.125 mcg/kg) IV, approx. 10 minutes prior to device deployment.^{42,43}

Indications:

- Large or giant wide-necked ICA aneurysms, from petrous to the superior hypophyseal segments
- Presently, patients must be 22 years of age or older
- PED has been used in cases outside the established indications, e.g., in MCA, and vertebral and basilar arteries^{44,45,46}

Contraindications:

- Ruptured aneurysm (due to requirement of pre-embolization dual anti-platelet therapy). However, it has been used in ruptured blister aneurysms^{42,43}
- Patients in whom dual anti-platelet therapy is contraindicated
- Patients who have not received dual anti-platelet therapy (ASA and clopidogrel) prior to procedure
- Active bacterial infection
- Metal allergy to cobalt, chromium, platinum or tungsten
- Patients with pre-existing stent in the parent artery at the target aneurysm location (relative contraindication)

Treatment of aneurysmal rupture during coiling

- notify anesthesia: for assistance with critical care management and in case you need to go to the O.R.
- immediately lower the blood pressure
- inflate balloon if balloon-assisted coiling
- immediately reverse anticoagulation. Give 50 mg of protamine (protamine should always be available during the procedure)

- do not remove the coil that caused the perforation; continue deploying it and follow with additional coils in rapid succession
- insert an extraventricular drain (EVD)

116.9.2 Endovascular management of vasospasm

General information

A ruptured aneurysm needs to be secured by coiling or clipping prior to hyperdynamic therapy or endovascular intervention for vasospasm.

Indications

Failure of resolution, or worsening of symptoms of vasospasm with hyperdynamic therapy for 12–24 hours.

Patients with conditions such as congestive heart failure, cardiac ischemia, or pulmonary edema that limit the institution of hyperdynamic therapy.⁴⁷

Treatment options

In addition to pharmacological (hyperdynamic or triple-H) therapy, endovascular options for management of vasospasm include intra-arterial chemical spasmolysis, utilizing selective catheterization of the involved segment, and mechanical spasmolysis using angioplasty.

► **Chemical spasmolysis.** Verapamil: Drug of first choice for spasmolysis in many centers. Its advantage is a relatively long half-life (6–12 hours).

Indications:

- Mild to moderate vasospasm that does not warrant angioplasty. Or, when angioplasty cannot be used safely
- Vasospasm consequent to manipulation during endovascular intervention
- Prior to performing angioplasty, so that the dilation is performed on the relaxed dilated artery rather than a relatively rigid vasoconstricted one

Dose: 5–10 mg injected gradually (over 2–10 minutes) via microcatheter, as the microcatheter is withdrawn through the spasmotic segment. Up to 20 mg may be given into each arterial tree. It is injected gradually to prevent a significant drop in BP or bradycardia.

Other drugs used for chemical spasmolysis include nicardipine, papaverine and nitroglycerine. Similar to verapamil, nicardipine has a relatively long half-life (9 hours). Papaverine (<1 hour) and nitroglycerine (minutes) are comparatively short acting.

► **Angioplasty.** A balloon mounted catheter is positioned across the arterial segment in spasm. The balloon is gradually inflated under fluoroscopic visualization to the desired width (equal to or less than the caliber of the adjacent non-spastic segment. The rate of balloon inflation is at a rate ≤ 1 atm/15 sec (to “stretch,” not “crack” the vessel).

A combination of chemical and mechanical spasmolysis may also be used such that initially the spastic segment is dilated using verapamil infiltration, followed by balloon angioplasty.

Balloon dilatation produces durable results. However, even verapamil induced spasmolysis has been noted to persist on angiographies performed 48–72 hours post treatment.

Angioplasty is performed under direct fluoroscopic visualization. To this effect, contrast media is admixed with normal saline (50:50 to 2/3–1/3 ratio) that is used to inflate the balloon.

Precautions during angioplasty:

- Avoid inflating the balloon beyond the normal caliber of the vessel
- Inflate gradually at a rate ≤ 1 atm/15 sec (to “stretch,” not “crack” the vessel)
- Do not inflate the balloon beyond its indicated “burst pressure.” This can also be avoided by selecting a balloon catheter with the appropriate width of the particular vessel and length of the segment in spasm

► **Complications and management.** Vessel rupture. This can be avoided by not inflating the balloon beyond the normal caliber of the vessel. Following rupture, reverse therapeutic heparinization by administration of protamine 1 mg per 100 units of heparin (max. 50 mg). In such an emergency, rapid IV bolus of protamine (10 mg over 1–3 minutes) is delivered rather than over 10–30 minutes, which is usually done to prevent idiosyncratic hypotension and anaphylactoid symptoms.

Maintain wire (0.014") access across the injured vessel. Exchange for a balloon catheter 1 mm smaller in size and inflate it for about 20 minutes. Then deflate and perform angiography to see if bleeding has been controlled. If necessary, sacrifice the involved vessel using coils. With this approach, e.g., in the M2 segment of MCA, the complication of stroke is accepted with the aim of saving the patient's life. Additionally, decompressive craniotomy and evacuation of the clot may be required.

► **Dissection.** If it is minor, non-flow limiting and without a raised intimal flap, no intervention may be necessary. The dissection may be re-examined after an interval to verify resolution. If managed conservatively, place the patient on aspirin, with or without Plavix.

In case of a significant dissection, place a stent across the affected segment and start patient on ASA and plavix. Plavix 75 mg PO daily is usually administered for a month while ASA is continued indefinitely.

► **Thromboembolic complications.** A thrombus consequent to hardware, vessel injury blood stagnation or inadequate heparinization is apparent as a filling defect on control angiography. This can usually be readily addressed by administration of abciximab bolus of 0.25 mg/kg IV over 10–15 minutes, followed by infusion of 0.125 mcg/kg/min (max. 10 mcg/min) for 12 hours. Repeat angiography 15 minutes following the initiation of abciximab. If the thrombus persists, an angioplasty may be performed to flatten the thrombus against the wall of the vessel. Once blood flow is restored, its lytic properties, as well as the abciximab, may resolve the thrombus. Another consideration is to deploy a stent to restore lumen and blood flow.

► **ICH.** This may happen due to causes including vessel injury, bleeding into a previous area of infarct, hyperperfusion in a previously compromised area, or hypertension. When detected, discontinue and reverse the heparinization using protamine. Monitor patient closely. A decompressive craniotomy or craniectomy may be required if the hemorrhage is significant.

► **Surveillance and follow-up.** In addition to diligent monitoring in NICU, the progress or resolution of vasospasm can also be assessed by performing serial TCDs. Depending upon extent of availability, these can be daily, every other day or at least bi-weekly (e.g., q Monday and Thursday). The TCDs may be discontinued when resolution of vasospasm is apparent clinically and radiologically.

Angiography may be repeated approximately 3 days following initial intervention, or when there is significant change indicating worsening vasospasm. It need not be repeated, if the patient is obviously improving. Alternatively, CTA may be used for surveillance. At the time of surveillance angiography, additional treatments with chemical spasmolysis and/or balloon angioplasty may be considered. Therefore, start off with a larger sheath (usu. 6 Fr) to enable intervention.

Usually, the patient will remain in the NSICU for approximately 10–14 days, considering that vasospasm is maximal by 7–8 days and usually resolves by 14 days.^{48,49,50,51}

116.9.3 Arteriovenous malformation

Indications for endovascular intervention

- the most common form of endovascular intervention for AVM is preoperative embolization to facilitate surgical AVM resection
- presence of associated lesions, e.g., aneurysm or pseudoaneurysm on the feeding pedicle or nidus, venous thrombosis, venous outflow restriction, venous pouches, or dilations
- a small surgically inaccessible AVM, or where surgery carries a high risk of morbidity and mortality. Curative AVM embolization is rare and limited to small lesions with simple angioarchitecture. The small surgically inaccessible AVM can also be treated with radiosurgery, which has a better track record than curative AVM embolization attempts
- as a palliative treatment in an AVM that is not completely treatable by any approach or their combination, due to location and/or diffuse morphology, but is symptomatic. Use with caution: data suggests that partial embolization of complex AVMs may increase rupture rate and worsen outcome

AVM embolization

This can be performed using numerous agents (or their combination).

The agents include coils, onyx, NBCA, and PVA.

► **Coils.** These may be used to close down a vessel supplying the AVM, an AVM pouch, or aneurysms on the arteries associated with AVM. However, it cannot be relied upon to completely obliterate the AVM nidus or effect a cure.

► **Onyx™.** A “lava-like” liquid embolic agent—Ethylene vinyl alcohol (EVOH) copolymer; ethylene and vinyl alcohol dissolved in dimethyl sulfoxide (DMSO) with micronize tantalum (for radioopacity)—that solidifies through the process of precipitation, which is initiated when it comes into contact with an aqueous solution (e.g., blood, body fluids, normal saline, water) to form a cast. Not an adhesive. Amongst all agents currently available, it has the best and most controlled penetration of the AVM nidus. Therefore, Onyx has the greatest likelihood of achieving complete cure. Complete cure rates with onyx alone are possible in 20–51% of highly selected patients.^{52,53,54} Supplied in pre-mixed ready-to-use vials of Onyx-18, Onyx-34 and Onyx-500 in which the number (e.g., 18 for Onyx-18) corresponds to the nominal viscosity (as measured in centistokes), which the manufacturer controls by altering the EVOH concentration. Prior to use, the product must be shaken on a mixer for at least 20 minutes. Higher numbers indicate greater viscosity. Onyx 18 is most commonly used, Onyx 34 is used for very high flow AVMs.

Onyx is used with DMSO compatible microcatheters (marathon, echelon, XT-17, Duo) or balloon microcatheter (Scepter C) that have been primed with approximately 0.3 to 0.8 ml of DMSO injected slowly through the microcatheter (depending on the deadspace of the microcatheter) prior to injecting Onyx itself. This is done to eliminate any contrast, saline, or blood within the microcatheter that will cause the onyx to solidify in the catheter. The microcatheter tip is wedged in an arterial branch supplying the AVM, as close to the nidus as possible. It is ensured that the branch is exclusively supplying the AVM, by performing angiography through the microcatheter. The microcatheter is then primed with DMSO and Onyx embolization commenced under fluoroscopy. The onyx is visible due to admixed tantalum powder. It is injected slowly and continuously, avoiding too much force, as this could cause the Onyx to reflux instead of flowing forward, and prematurely close off access to the nidus. No more than 1 cm of Onyx reflux on the microcatheter should usually be allowed, as it may make the removal of catheter difficult later and may well cause disastrous complications (this is obviated with detachable 1-cm and 3-cm tip Apollo microcatheters which tolerate greater Onyx reflux). If reflux is noted, wait for 1–2 minutes and then continue with the injection. The already deposited Onyx will form a plug around the microcatheter causing the new deposit of Onyx to flow forward. The injection is stopped once Onyx is noted only to reflux and not flow into the nidus. Care is taken not to allow the Onyx to flow into the major draining vein or sinus, especially before the entire arterial supply is interrupted. The concept is the same as in surgical removal of AVMs, where major draining veins are occluded only after eliminating the arterial supply of AVM. Otherwise, the result is the same with ensuing venous hypertension resulting in AVM rupture and catastrophic ICH. In case of a large AVM, it is sometimes best to stage the procedure, addressing one major arterial tree at a time.

Intravenous or combined IV, and IA embolization of AVMs has been performed in a select group of small, deep seated AVMs, with single deep drainage.⁵⁵

► **NBCA.** N-butyl cyanoacrylate is an embolic agent that is a glue which rapidly solidifies when it comes in contact with blood. Since the introduction of Onyx, its use has declined considerably due to the drawbacks of very short working time, greater risk of embolization into venous sinuses, and adhesion to catheters, rendering their extraction following embolization difficult. Such adhesion may cause intracranial bleeds or inadvertent catheter fractures with retained foreign body.

► **PVA.** Polyvinyl alcohol (PVA) particles are available in sizes ranging from 50 to 1000 μ m. While it may be useful in temporarily devascularizing the AVM, e.g., in preparation of craniotomy and surgical resection of AVM, the endovascular treatment with PVA alone is not durable.

Postoperative management

Post-op orders:

- admit the patient to NSICU
- keep leg on side used for procedure straight X 2 hrs (in case of Angioseal closure) or 6–8 hours (in case manual compression was applied), with HOB elevated 15°
- postprocedure heparinization is not required with AVMs because most ischemic events occur during the procedure and are related to the passage of embolic materials into normal blood vessels¹⁵
- check groins, DPs, vitals and neuro checks q 15 min X 4 hr, q 30 min X 4 hr, then q hr
- maintain mild hypotension for 12–72 hours, especially in case of larger AVMs, and monitor the patient for perfusion pressure break through bleeding, seizures, and other possible complications
- review/resume preprocedure medications (hold metformin for 48 hours after intervention; hold all oral hypoglycemics until good PO intake is established)⁵⁶

Follow-up:

- outpatient appointment in 4 weeks

- follow-up angiography at 3 months. In case of staged AVM embolization, the interval between sessions is at the discretion of the interventionalist, e.g., 1–4 weeks apart

116.9.4 Dural arteriovenous fistulae (DAVF)

General information

These are abnormal direct arteriovenous shunts within the leaflets of the dura.

Classification

There are numerous DAVF classifications. Borden (p.1517) and Cognard (p.1517) are amongst the more widely used.

Indications for endovascular intervention

- DAVF with “aggressive” features (► Table 116.1) are always considered for treatment. Due to the high annual mortality rate (10.4%) and annual hemorrhage rate (8.1%), the treatment should be expeditious.⁵⁷
- DAVF with angiographic findings including:
 - selective contrast injection into ICA or VA demonstrating delayed cerebral circulation time. This is indicative of venous congestive encephalopathy.
 - pseudophlebitic pattern: Brain surface demonstrating tortuous, dilated collateral veins in the venous phase of the angiogram. This finding is associated with a greater risk of hemorrhage or non-hemorrhagic neurological deficit.
 - cortical venous reflux (CVR): To ensure that this is not missed, always perform selective (rather than global, non-selective) angiography when assessing for DAVF. Venous stenosis or obstruction is commonly found in patients with CVR.

If a DAVF is detected, look for additional fistulae, as they are multiple in up to 8%.

Those with “benign” features (see ► Table 116.1), may be considered for treatment if the symptoms are causing considerable discomfort to the patient, or are angiographically progressive. In many cases the treatment is palliative, i.e., the symptoms are reduced but the fistula is not completely obliterated. The benign type of DAVF should continue to be followed, even when treatment is not indicated because of the risk of conversion to a more aggressive form with CVR. The follow-up may be clinical, with radiological assessment for any change in symptoms. A suggested protocol is to perform MRA with gadolinium annually, with follow-up conventional catheter angiography at 3 years. If there is any change in the patient’s clinical condition, whether it is worsening, improvement, or resolution of the symptoms, perform standard angiography to assess for CVR.

Asymptomatic DAVF can usually be followed.

Table 116.1 Differentiating aggressive from benign DAVF

“Aggressive” symptoms	“Benign” symptoms
• Cortical venous reflux (CVR): the hallmark of an aggressive DAVF • Intracerebral hemorrhage • Focal neurological deficit • Dementia • Papilledema • Increased intraocular pressure	• Pulsatile bruit • Orbital congestion (without increased intraocular pressure) • Cranial nerve palsy • Chronic headaches

Contraindications to endovascular intervention

Most contraindications are relative, and a risk-benefit assessment is performed on a case-by-case basis:

- NBCA should not be used in those with allergy to cyanoacrylates, ethiodol, or iodine. Premedication in those with iodine allergies is a consideration.
- PVA should not be used as a therapeutic option (other than in cases of epistaxis). It is usually indicated for pre-surgical devascularization of the lesion.

DAVF embolization

The approach can be transarterial, transvenous, or a combination. When feasible, a transvenous approach is preferred, as the probability of fistula obliteration is greater via the venous route. Very

infrequently, it is possible to access the venous side from the arterial route because of a large connection between the dural artery and adjacent vein, e.g., in traumatic DAVF.

For transvenous route, consider the potential outcome of venous occlusion, e.g., venous infarct, if the sinus being occluded is also the main source of drainage for normal veins. In such a situation, consider highly selective occlusion that will spare the normal drainage. Alternatively, instead of performing a complete occlusion, consider partial treatment only such that CVR is eliminated, converting the fistula into the benign Borden type I.

When using the venous route, ascertain that the venous channel is not tenuous (e.g., acute DAVF), rendering it prone to rupture during manipulation. The venous walls become sturdier when the fistula has been present for a while.

► **Coils.** Measure the maximum width of the fistulous site to be occluded and select the appropriate sized coils. Deposit as many coils as necessary to occlude the fistula.

Sometimes a “combined” strategy may be used, in which coils are initially deposited to slow the rapid blood flow through the fistula, followed by occlusion using a liquid embolic agent. If this strategy is used, start off with or exchange for a microcatheter compatible with the liquid embolic agent.

► **Onyx.** During transarterial embolization with liquid embolic agents, wedge the catheter as close as possible to the fistula. The technique of deposition is the same as for AVM (see AVM section (p.1927)). It is important to disrupt the fistulous connection to achieve cure. Therefore, Onyx must penetrate into the venous aspect of the DAVF.

► **NBCA.** It is used less frequently since the availability of Onyx, for reasons described above.

► **PVA.** It really has no significant role in the treatment of DAVF.

116.9.5 Carotid-cavernous fistulae (CCF)

General information

CCF are categorized as direct and indirect. The direct CCF are usually posttraumatic and are a high-flow single shunt between the ICA and the cavernous sinus. Indirect CCF are low-flow from meningeal branches. The section below describes management of the direct type.

Indications for endovascular intervention

Direct fistulae usually require treatment as they frequently do not resolve spontaneously. Other indications: corneal exposure, diplopia, proptosis, intolerable bruits, or headaches.

Timing of treatment

If the patient is stable, treatment may usually be performed within a couple of days of the diagnosis (i.e., treatment does not have to be emergent).

Indications for urgent treatment: ICH, epistaxis, increased IOP, decreased visual acuity, rapidly progressive proptosis, cerebral ischemia, and enlargement of traumatic aneurysm beyond the cavernous sinus.

CCF embolization

The treatment goal is to eliminate the fistula.

Angiography is performed to identify the exact location and size of the fistula and its venous drainage. To address the high flow, consider angiography at 7.5 fps, instead of the usual 2–4 fps. In addition to the CCF, also look for other vascular injuries/anomalies.

Selective catheterization of both ECAs and ICAs is performed to assess their contribution to the CCF. Angiography is also performed after manual compression of the CCA on the side of fistula, to better assess cross-flow from the contralateral side. The digital compression will attenuate the high blood flow to the fistula, enabling its visualization. Do not compress both carotids simultaneously.

Rotational angiography with 3D reformatting may be performed to study the fistula and select appropriate working views for intervention. It is important to be cognizant of the venous involvement, including cavernous sinuses, superior and inferior ophthalmic veins, sphenoparietal sinus, superior and inferior petrosal sinuses, and the pterygoid plexi.

The following routes may be utilized for treating CCF: transarterial, transvenous, and via the superior ophthalmic vein (if conventional routes are not available). Other indications: corneal exposure, diplopia, proptosis, intolerable bruits, or headaches.⁵⁸

► **Coils.** Currently, the method of choice is transarterial coil embolization of the CCF.

Using roadmapping images, the microcatheter is advanced over microwire into the cavernous sinus via the fistula. This may take some effort. Coils are then deployed and detached, as previously described for aneurysms. Periodic angiography is performed and further coils are placed until the cavernous sinus (CS) is completely occluded. Complete occlusion is indicated by no further contrast entering the CS.

► **Onyx.** Refer to AVM section (p.1927) for details regarding usage. In case of a high-flow fistula, prior to Onyx deposition, it may be advisable to initially deposit coils into the CS to slow the blood flow. This will prevent the inadvertent embolization of Onyx into the draining veins and sinuses. A balloon can be inflated within the parent ICA in order to protect it as Onyx is being injected into the cavernous sinus.

► **NBCA.** NBCA should be used with great caution, preferably after having slowed the flow through the CCF in order to prevent untoward deposition in the venous sinuses.

There is also potential for reflux into the carotid artery, which could cause a stroke. This may particularly occur when the CCF closure is near completion and the pressure gradient between the carotid artery and the CS is lowered. As with Onyx, a balloon can be inflated in the parent artery to protect it.

► **Coil occlusion of the ICA.** The desirable treatment for CCF is the occlusion of the fistula itself, essentially resulting in ICA reconstruction. This is frequently not possible. If the CCF is not amenable to treatment by any other route, sacrifice of the involved ICA is an option, especially if satisfactory supply has been confirmed from the contralateral ICA via anterior communicating and/or supply via the posterior communicating arteries.

Postoperative management and follow-up

Post-op orders:

- admit the patient to NSICU for overnight observation. Further ICU stay will depend upon the patient's clinical condition
- consult or arrange follow-up with ophthalmic surgery
- 0.9% NS + 20 meq KCl @ 150 cc/hr X 2 hrs, then decrease to 100 cc/hr if patient is NPO overnight, then advance diet as tolerated
- keep the leg that was used for procedure straight for 2 hrs in case of Angioseal closure, or 6–8 hours when manual compression was applied. May elevate 15° HOB, e.g., a pillow under the head
- check groin, DPs, vitals, and neuro checks q 15 min X 4 hrs, q 30 min X 4 hrs, then q hr
- review/resume preprocedure medications (hold metformin for 48 hours after intervention; hold all oral hypoglycemics until good PO intake is established)

Follow-up:

- D/C home the next morning after mobilizing, if there are no complications/other ongoing medical concerns requiring hospitalization
- follow-up on outpatient basis in 4 weeks
- follow-up angiography at 3 months

116.9.6 Vertebrojugular fistulae (VJF)

Etiologies

- Iatrogenic, e.g., during spine surgery or angiography, chiropractic manipulation, nerve block injection, or radiation therapy^{59,60}
- Trauma, e.g., penetrating injury or GSW
- Vasculitis

Endovascular management of VJF

► **Stenting.** A polytetrafluoroethylene (PTFE) covered stent, e.g., Jostent, may be used to cover the ostia of the fistula.⁵⁹

► **Coil occlusion.** In the presence of adequate blood flow through contralateral healthy vertebral artery, the fistulous artery may be occluded with coils.⁶¹ Verify that the arterial wall with the fistulous connection is part of the occluded segment.

► **NBCA occlusion.** Rarely, NBCA occlusion has been performed when stenting or coil occlusion were not possible.⁶² Onyx may also be used similarly.

116.9.7 Carotid dissection

General information

See General information on arterial dissections (p. 1576).

Angiographic features

Luminal stenosis (65%), occlusion (28%), pseudoaneurysm (28%), luminal irregularity (13%), embolic distal branch occlusion (13%), intimal flap (12%), and slow ICA-MCA flow (11%).⁶³

Management

The CADISS randomized trial showed no difference in efficacy of antiplatelet agents versus anticoagulation at preventing stroke and death in patients with symptomatic carotid and vertebral artery dissection.⁶⁴ Because the management with heparin and Coumadin is more complex, most physicians now favor the use of antiplatelets such as Aspirin. If anticoagulation is indicated, for example based on comorbid conditions such as atrial fibrillation or the presence of flow limiting thrombus, the following protocol may be considered.

The initial management in the absence of ICH is intravenous heparin for 12 hrs–7 days followed by warfarin.⁶⁵ The goal aPTT with heparin is 1.5–2.0 times the control value (50–80 sec). Warfarin is continued for 3–6 months with target INR range of 2.0–3.0.

Indications for endovascular intervention

- persistent, worsening, or recurrent ischemic symptoms despite anticoagulation therapy
- flow-limiting lesion with hemodynamic compromise

Stenting with/without coiling

The endovascular treatment for carotid dissection is stenting, including the use of flow diversion stents if needed. In case of intimal flap, the stent will appose the flap back to the arterial wall. Pseudoaneurysms have also been successfully occluded with stenting. Both uncovered and covered stents have been used successfully.⁶⁶ JoStent is a PTFE covered stent that is available in US. A vein covered stent has also been used.⁶⁷

In case of a pseudoaneurysm that continues to show significant residual filling after stenting, coiling of the pseudoaneurysm will cause occlusion.⁶⁶

After stenting, the patient remains on dual antiplatelet therapy (ASA + Plavix) for at least a month and ASA alone indefinitely.

Follow-up

Follow-up should be arranged for patients on warfarin (e.g., “Coumadin clinic”).

Follow-up study in 3–6 months, which could be CTA, Doppler ultrasonography, or catheter angiogram.

116.9.8 Subclavian artery stenosis

General information

Radiologically demonstrable stenosis of the subclavian or innominate artery is present in approximately 17%. Of these, 2.5% have angiographic flow reversal in vertebral artery. Only 5.3% of those with angiographic steal have neurologic symptoms.⁶⁸

► **Symptoms.** The 5 D's of VBI (i.e., diplopia, dysarthria, defective vision, dizziness, and drop attacks).

Other symptoms include headache, nystagmus, hearing loss, and focal seizures.^{69,70}

The arterial stenosis is proximal to the origin of VA. Symptoms are induced by exercise or exertion using the arm ipsilateral to the stenosis. The increased flow demand due to the exertion results in retrograde blood flow through the VA. The neurological symptoms may be because of continuous brainstem ischemia or more commonly, ischemia due to ipsilateral arm exercise or exertion.⁷¹

Indications for endovascular intervention

Symptomatic subclavian artery stenosis (i.e., stenosis resulting in subclavian steal syndrome).

Endovascular intervention

This includes angioplasty and stenting. A balloon-mounted stent may be used, as the stent is deployed concurrently with angioplasty.⁷² However, if the stenosis is particularly severe (e.g., <2 mm), pre-dilatation may be performed by a smaller balloon to achieve a caliber of 4 mm at site of stenosis. Normal antegrade blood flow is restored following successful angioplasty and stenting. Radial or brachial access is frequently used for such cases.

Postoperative management

The patient is monitored at least overnight in NSICU.

After stenting, the patient remains on dual antiplatelet therapy (ASA + Plavix) for at least 1 month, and ASA alone indefinitely.

Follow-up study in 3–6 months, which could be CTA, Doppler ultrasonography, or catheter angiogram.

Complications of angioplasty and stenting

The frequency of complications is 17.8% (of 73 procedures) for innominate and VA angioplasty and stenting. These include access-site bleeding and distal embolization.⁷³

116.9.9 Ischemic stroke

IV tPA

The target for IV tPA administration (“door-to-needle time”) is to administer within 60 minutes from time of arrival to hospital in $\geq 50\%$ of patients.²¹ However, IV tPA can be administered up to 4.5 hours after the onset of symptoms if the patient does not have any contraindications.⁷⁴ Candidates for IV tPA should receive it even if endovascular therapy is being considered (Level I²¹).

Dose: 0.9 mg/kg (max 90 mg) with 10% of the dose administered as a bolus over 1 minute and the rest infused over 60 minutes.

Do not wait for a patient with suspected stroke from large vessel occlusion to “improve” after IV tPA initiation (possible harm²¹). Once the IV tPA drip is started, the patient should be immediately taken to the angiography suite or transported to a center with neuroendovascular capabilities, preferably a comprehensive stroke center. An ongoing debate is whether IV tPA is “futile” in cases with large vessel occlusion given the low chance of large clot lysis, increased risk profile with the use of systemic tPA, and time delay to the initiation of thrombectomy. Trials to address this issue are currently underway in Europe.

Endovascular intervention

General information

Endovascular intervention is extremely safe and effective when performed up to 24 hours from stroke symptom onset, including “wake-up” strokes. The number needed to treat (NNT) (the number of patients to which a treatment must be administered before either a single patient achieves a benefit or one adverse event is prevented) with thrombectomy is 2–4, compared to 3–19 for IV tPA in acute ischemic stroke (AIS). Trials favor rapid endovascular intervention in AIS with anterior circulation proximal vessel occlusion, small infarct core, and moderate-to-good collateral circulation.^{75,76,77,78,79,80} Meta-analysis of these randomized trials shows that even large core infarcts or more distal occlusions still benefit from thrombectomy.⁸¹

For AHA guidelines for mechanical thrombectomy, see Practice guideline: Mechanical thrombectomy for acute ischemic stroke (AIS) (p. 1562).

Indications and case selection for intra-arterial tPA (IA-tPA)

- not recommended as primary therapy given widespread use of mechanical thrombectomy and data supporting its high safety and efficacy. A clinically beneficial dose has not been established (Level I²¹)
- as adjunct treatment after mechanical thrombectomy for more distal clots

- primary endovascular treatment for very distal clots or for use within the first 6 hours of major ischemic stroke due to MCA occlusion (Level 1²¹), although it is frequently used up to 24 hours as long as imaging did not show a large core infarct

Contraindications to intervention

Most contraindications are relative and have to be weighed against the risk of not intervening. These contraindications include:

- hemorrhagic infarct or ICH
- CT demonstrating hypodensity or mass effect consistent with evolving infarct of more than one-third of middle cerebral artery territory
- recent major surgery
- pregnancy
- when considering stenting, contraindication to anticoagulants and/or thrombolytics

Preprocedural management

This may be under the supervision of a stroke neurologist, or the neurosurgeon. Ensure the following:

- rapid transfer of patient to a facility with endovascular capabilities such as a comprehensive stroke center (CSC). Thrombectomy-capable stroke centers (TSC) have been recently introduced by the AHA/Joint Commission and are the subject of heated debate regarding its validity
- ABC's take precedence
- ensure patient has two intravenous lines, preferably 18G or larger. Start monitoring BP, pulse oximetry, ECG, O₂ saturation, cardiac rate and rhythm, respiratory rate
- insert a Foley catheter
- verify laboratory values including: platelet count, BUN, CR, APTT, PT/INR, β -HCG for females of reproductive age group
- maintain MAP $\geq$ 90 mm Hg
- non-contrast head CT scan: To rule out ICH and to determine the Alberta Stroke Program Early CT Score (ASPECTS) (p. 1559).⁸² The score ranges from 0 to 10. A higher number corresponds to smaller stroke core burden. Score 6–10 is considered a "favorable" profile for endovascular therapy
- CTA: to assess location of the clot (hyperdense artery sign (p. 1560)), collaterals & vascular tortuosity
- MRI of the head is rarely indicated (may be obtained in select cases)
- if available and can be done without delay, then perfusion studies, e.g., CTP or MRP, especially in cases of wake-up strokes or when symptom onset is over 6 hours prior. These perfusion studies will demonstrate viable brain (penumbra) vs. completed stroke to help in selecting patients for thrombectomy
- in centers where available, CT, CTA, and CTP all are performed during the same session on CT scanner
- be cognizant of renal insufficiency, diabetes, congestive heart failure, etc., in which case consider diluted non-ionic contrast agent and carefully pre-plan to minimize contrast load
- the goal of intervention is to re-establish successful recanalization, defined as mTICI score⁸³ 2b/3 ($\gg$ Table 116.2), as soon as possible

Table 116.2 Modified Treatment in Cerebral Ischemia (mTICI) scale⁸³

Grade	Description
0	no perfusion
1	antegrade perfusion past the initial occlusion, but limited distal branch filling with little or slow distal reperfusion
2a	antegrade perfusion of <50% of the ischemic territory of the occluded target artery (e.g., in 1 major division of the MCA & its territory)
2b	antegrade perfusion of >50% of the ischemic territory of the occluded target artery (e.g., in 2 major divisions of the MCA & their territories)
3	complete antegrade reperfusion of the ischemic territory of the occluded target artery, with absence of visualized occlusion in all distal branches

Abbreviations: MCA = middle cerebral artery.

Techniques

► **Stent retrievers.** Due to a high success rate, stent retrievers became the method of first choice for clot removal in embolic stroke. The successful recanalization rate is 70–85%.^{84,85,86,87} The two devices most commonly employed currently in the U.S. are Solitaire and Trevo. Also available in the U.S.: Penumbra System® 3D™ Revascularization Device and EmboTrap®.

A 7 or 8 Fr sheath is placed in the femoral artery, through which a guide sheath or a balloon-guide catheter is positioned in the cervical ICA (for anterior circulation strokes). The balloon-guide catheter increases the rate of successful recanalization, decreases the number of thrombectomy passes, and reduces the risk of distal embolic formation with thrombectomy. Angiography is performed to identify the site of occlusion. Using fluoroscopy and road mapping, a microcatheter is advanced over a microwire, across the site of occlusion. The microwire is removed and the stent retriever is advanced through the microcatheter such that it extends proximal and distal to the clot. The stent retriever is unsheathed by retracting the microcatheter as the retriever is maintained stationary. The stent retriever expands to its actual size and this results in restoration of flow in the occluded artery. After five minutes, if a balloon catheter is used, the balloon on the guide catheter is inflated to arrest blood flow. Maintaining gentle aspiration on the guide catheter, the stent retriever and microcatheter are retracted simultaneously. Once both the microcatheter and retriever are within the guide catheter, vigorous aspiration is applied as the two devices are concurrently retracted and removed from the patient. Angiography is performed to confirm reconstitution of circulation. Some operators utilize a distal aspiration catheter to capture the clot proximally and apply aspiration while removing the stent retriever. This approach is called “Solumbra” or “Trenumbra”.⁸⁸

Vessel perforation during stent retriever withdrawal has been reported.⁸⁹ The risk of perforation is believed to increase with treatment of more distal small occlusions.⁹⁰ Distal aspiration alone (discussed in the next paragraph) may be a preferable approach in such cases.

► **Primary aspiration.** Primary aspiration has emerged as a faster and cheaper alternative to primary stent retriever thrombectomy. This approach was first introduced as the ADAPT technique (A Direct Aspiration first Pass Technique). A large-bore catheter is placed in direct contact with the thrombus and aspiration is applied using a pump or manually with a syringe to extract the clot.⁹¹ If unsuccessful alone, a stent retriever is introduced through the aspiration catheter as a rescue treatment to remove the clot. This approach is now also called “direct aspiration,” “contact aspiration,” and “primary aspiration” with a variety of catheters of different length and diameter available on the market to accommodate vessels of various sizes and locations ranging from the ICA terminus to the middle cerebral artery M2 and 3 segments. 3 recent randomized trials (ASTER, 3D, and COMPASS) showed equal safety and efficacy of both approaches in achieving successful recanalization.^{92,93,94} The COMPASS trial also demonstrated significant cost-effectiveness and shorter time to recanalization with primary aspiration vs. stent retriever use.

► **Intra-arterial tPA.** This may be the simplest endovascular technique; however, on its own, recanalization rates are inferior compared to the above-mentioned mechanical techniques^{95,96} (although it may be better than IV tPA). Currently, IA tPA is used primarily in conjunction with other techniques in ischemic stroke.

► **Other techniques.** Other techniques to extract thrombus have also been employed with mixed results including: aspiration with a simple syringe attached to a microcatheter, snares, angioplasty at the site of thrombus, stenting, etc.

Stent retrievers are preferred to the MERCI device (Level I²¹), as well as to other devices which may be reasonable in some circumstances (Level II²¹).

116.9.10 Cerebral venous thrombosis (CVT)

General information

Also see Cerebral venous thrombosis (p. 1594).

Hydration with IV fluids and IV anticoagulation with heparin or low-molecular-weight heparins (LMWH) are part of the initial treatment for cerebral venous thrombosis (CVT), even in the face of parenchymal venous hemorrhagic infarcts. Prior to initiation of treatment, blood for hypercoagulopathy tests is drawn.

Indications for endovascular intervention

1. clinical deterioration despite anticoagulation

2. contraindications to anticoagulation: bleeding diathesis, thrombocytopenia (< 100 K), or recent GI hemorrhage
3. coma
4. deep CVT
5. intracerebral hemorrhage (ICH)

Endovascular treatment

Chemical thrombolysis (tPA): A catheter may be advanced to the involved sinus or close to it, through the femoral vein. The advantage of local administration is that a larger amount of tPA actually reaches the clot vs. systemic administration through a peripheral vein. Usually, 2–5 mg are administered through the thrombus and then an infusion started at a rate of 1–2 mg/hr, usually for 12–24 hours. Bleeding complications increase after ≈ 24 hours. Repeat angio in 24 hours. If clot burden is still present on angiography, the infusion may be continued for longer, until the clot resolves.

For CVT, the infusion may be prepared in a concentration of 1 mg/10 ml (0.1 mg/ml), for a rate of 10 ml/hr.

Mechanical thrombectomy: Similar to arterial embolic stroke, devices such as Stentriever or Penumbra may be used for clot extraction. Additionally, devices intended for other sites, e.g., clot extraction from dialysis fistula, have also been used in cranial sinuses.³⁷ The challenge during endovascular intervention is negotiating the sigmoid-transverse sinus junction, especially when using bulkier catheters, e.g., AngioJet (very stiff).

116.9.11 Chronic subdural hematoma

Middle meningeal artery (MMA) embolization is a minimally invasive potential option for treating chronic SDH (CSDH). Membranes of the CSDH contain friable neovasculature⁹⁸ that may be responsible for repeated rebleeding into the subdural space and may contribute to recurrence after surgical evacuation. It has been shown to be safe and effective in small series as treatment for recurrent CSDH^{99,100} as well as for primary treatment,¹⁰¹ and larger trials are anticipated.

Technique (summary): angiography of the common and external carotid arteries is performed on the affected side. The distal MMA vasculature may exhibit irregular wispieness.¹⁰² The MMA is selectively catheterized using a microcatheter, and 150–250-micron diameter polyvinyl alcohol particles¹⁰¹ or 15–20% n-butyl-2-cyanoacrylate⁹⁹ is injected under fluoro roadmap until there is flow stasis. Post-injection angio confirms effective MMA embolization.

Branches to avoid: potential ophthalmic artery collaterals and branches supplying cranial nerves.

116.9.12 Tumor embolization

Indications

Preoperative devascularization of vascular tumors, including

- meningiomas: embolization for meningiomas is location-specific, size-dependent, institution-dependent, and potentially controversial. May be best reserved for large hypervascular convexity meningiomas. Usually performed 24 hours to 1 week preoperatively. The devascularization causes attenuation in intraoperative blood loss and the resultant necrosis frequently renders the tumor softer and easier to remove. However, tumor swelling may occur and occasionally an emergency craniotomy may be required
- hemangiopericytomas
- juvenile nasopharyngeal angiofibromas
- glomus jugulare tumors
- hemangioblastomas
- vascular metastases

Technique

A sheath is placed in the femoral artery and a guide catheter is positioned as close as possible to the vessels of interest, e.g., in case of a meningioma, the guide catheter tip is positioned in the proximal ECA. Angiography and roadmapping are performed through the guide catheter. Using fluoroscopy and road mapping, a microcatheter is advanced over wire into the branches supplying the tumor. Angiography is performed through the microcatheter to ascertain which branch supplies the tumor and that no concerning collaterals with intracranial circulation exist. A blank road map is obtained and embolization commenced. PVA particles or Onyx may be used for embolization. In case of Onyx, a DMSO-compatible catheter must be used. PVA may be cheaper and quicker to use for tumor

embolization. However, the devascularization is not durable and the occluded vessels may recanalize; therefore, with PVA the surgery should be performed within a few days of the embolization.

116.9.13 Refractory epistaxis

Indications

Epistaxis that has not responded to treatment including manual compression, nasal packing, local vasoconstrictors, endoscopic cauterization, or surgical ligation of sphenopalatine arteries.

Preoperative management

Verify lab values, including platelet count, BUN, CR, APTT, PT/INR, and β -HCG for females of reproductive age group. In renal insufficiency, diabetes, CHF, etc., use diluted non-ionic contrast agent and pre-plan carefully to maintain contrast load to minimum.

Liquids only on morning of procedure. NPO (for $\approx$ 6 hours) when procedure performed under general anesthesia.

Obtain informed consent for angiography and embolization of ECA branches.

Ensure two IV lines inserted. Insert Foley. The patient will be more comfortable and cooperative with an empty bladder, if the procedure becomes prolonged.

► **Technique.** Position patient on the neuroangiography table. Attach pulse oximetry and ECG leads for monitoring O₂ saturation, HR, cardiac rhythm respiratory rate, and BP.

A sheath is placed in the femoral artery. A guide catheter is positioned in the proximal ECA on the side of bleeding or pathology. Angiography and roadmapping are performed through the guide catheter. Using fluoroscopy and road mapping, a microcatheter is advanced over the wire into the sphenopalatine branches. Angiography is performed through the microcatheter to ascertain appropriate positioning and to ensure no concerning collaterals with intracranial circulation exist. Contrast extravasation, tumor blush, or pseudoaneurysms may be detected. A blank road map is obtained and embolization of the offending vessel commenced. PVA particles (250–300 mcg) or Onyx (18 or 34) may be used. In case of Onyx, a DMSO-compatible catheter is used. PVA may be cheaper and quicker to use. Monitor for dangerous anastomosis with the ICA, and especially the ophthalmic artery as Onyx may cause blindness via penetrating even the smallest collaterals.

Postoperative management

Post-op orders:

- admit to ICU for overnight observation. If nasal packing is used, typically removed for inspection for bleeding the next day, or managed by ENT if they are involved
- IV: 0.9% NS + 20 meq KCl @ 150 cc/hr X 2 hrs, then decrease to 100 cc/hr, if patient is NPO
- activity: keep right/left leg (whichever side was used for procedure) straight for 2 hrs (in case of Angioseal closure), or 6–8 hours (in case manual compression was applied), with HOB elevated 15°. This is achieved by placing a pillow under the patient's head. There should be no flexion in the femoral region. If more head elevation is required, place bed in reverse-Trendelenburg position
- check groin, DPs, vitals, and neuro checks q 15 min X 4 hrs, q 30 min X 4 hrs, then q hr
- advance diet as tolerated. Review/resume preprocedure medications (except oral hypoglycemics, until good PO intake is established)

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